

Statistics for Insurance

Theory and methods for life and non-life insurance data

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Preface

The practice of actuarial science has undergone a profound transformation in recent decades. Where the field once primarily relied on more deterministic and aggregate methods, it now commonly deals with the complexity of stochastic processes and the sheer volume of granular data that describes them. The fundamental challenge, however, remains the same: to manage the risk of processes arising in insurance and financial operations. The discipline of statistics for insurance is our primary scientific tool for this task. It offers the required methods to describe the behaviour of the underlying risks, which is the first and arguably the most fundamental step in designing any effective risk mitigation strategy.

Modern insurance data comes in a great variety of shapes and sizes, and the statistical methods required to analyse it must be equally varied and sophisticated. In this monograph, we study this complexity from different angles. We are interested in data that can give rise to a claim, which may possess both discrete and continuous components, such as in the interplay between claim frequency and severity. We consider processes that evolve through time, requiring pathwise analysis, and we address the practical challenges of varying exposure periods and reporting delays. We also face rare or catastrophic events that manifest as heavy-tails, the influence of different sources of randomness, and the common issues of truncated or censored data. The main aim of these notes is to provide a representative and central exposition of statistical tools that can tackle these characteristics. While the classical distinction between life and non-life insurance has its place, a further goal of this work is to close the gap between these two perspectives, presenting a more unified approach, striving to use the same set of mathematical tools to treat both cases.

In selecting the material, a deliberate choice was made to emphasise not primarily the application of statistical methods, but rather their mathematical foundations. This aligns with my conviction that a deep understanding of the mathematical structure is what drives intuition the most, even when practical application is the ultimate goal. The development of novel methods to solve the emerging problems in insurance often builds upon established theories. For this reason, we strive to document and understand how mathematical statistics quantifies risk, and in particular embarking on the understanding of the detailed proofs of precise statements. Practical examples, however, are also amply provided in both the main text and the exercises.

The order of the presented chapters creates a path through the core of modern statistics for insurance, though clearly only one of many. The material has

grown from a shorter set of lecture notes used to teach a semester course at the masters level; in its current length, a selection of about half of the material is recommended. We begin with preliminaries on stochastic convergence, before moving to exploratory tools, the study of heavy-tailed data, and parametric models. We then explore selected topics of machine learning, with an emphasis on model selection and explainability, before turning to specialised topics like stochastic claims reserving, survival analysis, and nonparametric inference for Markov processes. The later chapters cover more advanced topics on multi-state modelling, including both parametric and nonparametric approaches. A Bayesian chapter finalises our treatment. A notable omission is a systematic treatment of parametric dependence modelling and time-series analysis. However, the reader may find that the mathematical tools presented can often be adapted to cover such cases. Numerous references have been pivotal to the development of this monograph, and they are purposefully collected in an organised manner at the end of each chapter, albeit merely representative and far from being comprehensive.

This work would not have been possible without the support and inspiration of many people, but I prominently want to thank my masters and PhD students at the University of Copenhagen, as well as my collaborators. They have been a constant source of motivation, and their intellectual curiosity, dedication to learning and teaching the material, and their help with earlier versions of the manuscript have been essential. Above all, their enthusiasm for the subject has sharpened my own understanding in many ways.

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Chapter 1

Probabilistic Foundations

The mathematical foundations of statistics rest on results in probability theory, such as laws of large numbers, central limit theorems, and general principles of stochastic convergence. These results provide the framework for analysing the behaviour of statistical procedures and for establishing their validity in large samples. A broad range of methods in insurance statistics, developed in later chapters, build directly on these tools. We begin with finite-dimensional convergence and limit theorems that are used repeatedly later on, then move to empirical summaries and process-level convergence.

1.1 Stochastic convergence in finite dimensions

We now recall some of the main results used to analyse independent and identically distributed (iid) random vectors. Here, we interpret such vectors as real-valued risks, e.g. claim sizes from a non-life insurance portfolio. We start by fixing notation for deterministic and stochastic orders, and then list standard convergence tools. From real analysis we recall the following notation, which makes precise the notion of speed of convergence.

Definition 1.1 (*o*-notation) Let a_n, b_n be sequences of real numbers. We write $a_n = o(b_n)$ if $\lim_{n \rightarrow \infty} a_n/b_n = 0$. We write $a_n = O(b_n)$ if there exists $K < \infty$ such that $|a_n/b_n| \leq K$ for all $n \geq 1$.

When working with stochastic convergence, this notation can be ambiguous because several modes of convergence exist. Nevertheless, it can be adapted as a convenient shorthand that proves useful in practice.

Definition 1.2 (Types of convergence) Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables defined on some probability space. We say that Z_n converges almost surely (a.s.) to Z , denoted by $Z_n \xrightarrow{a.s.} Z$, if for the metric d of interest

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} d(Z_n, Z) = 0\right) = 1.$$

We say that Z_n converges in probability to Z , denoted by $Z_n \xrightarrow{\mathbb{P}} Z$, if for all

$\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(d(Z_n, Z) > \varepsilon) = 0.$$

Finally, we say that Z_n converges in distribution to Z , denoted by $Z_n \xrightarrow{d} Z$, if for all bounded continuous (equivalently, bounded Lipschitz) functions $f : \mathbb{R}^p \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{E}[f(Z_n)] = \mathbb{E}[f(Z)].$$

Convergence in distribution is also characterised by convergence of cumulative distribution functions at continuity points, or by convergence of characteristic functions.

Almost sure statements are often proved by controlling whether a sequence of exceptional events happens infinitely many times. The next definition and lemma formalise this idea and are used repeatedly in this chapter.

Definition 1.3 (Infinitely often) For events (A_n) , define

$$\{A_n \text{ i.o.}\} := \bigcap_{n=1}^{\infty} \bigcup_{k \geq n} A_k.$$

This is the event that infinitely many of the events A_n occur.

Lemma 1.1 (Borel–Cantelli) *Let $(A_n)_{n \geq 1}$ be events.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then*

$$\mathbb{P}(A_n \text{ i.o.}) = 0.$$

2. *If, instead, the events A_n are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then*

$$\mathbb{P}(A_n \text{ i.o.}) = 1.$$

In particular, under independence,

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty \quad \Leftrightarrow \quad \mathbb{P}(A_n \text{ i.o.}) = 0.$$

Proof. If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then by monotone convergence,

$$\mathbb{E}\left[\sum_{n=1}^{\infty} 1_{A_n}\right] = \sum_{n=1}^{\infty} \mathbb{E}[1_{A_n}] = \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty.$$

Hence $\sum_{n=1}^{\infty} 1_{A_n} < \infty$ almost surely, which means only finitely many A_n occur, i.e. $\mathbb{P}(A_n \text{ i.o.}) = 0$. Now assume independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. For $m \geq n$,

$$\mathbb{P}\left(\bigcup_{k=n}^m A_k\right) = 1 - \prod_{k=n}^m \mathbb{P}(A_k^c) = 1 - \prod_{k=n}^m (1 - \mathbb{P}(A_k)) \geq 1 - \exp\left(-\sum_{k=n}^m \mathbb{P}(A_k)\right),$$

where we used $1 - x \leq e^{-x}$. Letting $m \rightarrow \infty$ and by continuity from below, $\mathbb{P}(\cup_{k \geq n} A_k) = 1$ for every n . Since these events decrease in n , we get by continuity from above $\mathbb{P}(A_n \text{ i.o.}) = 1$. $\square$

The above theorem is often specialised to a criterion of almost sure convergence.

Corollary 1.1 (Borel–Cantelli criterion for almost sure convergence) *Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables. If for every $\varepsilon > 0$,*

$$\sum_{n=1}^{\infty} \mathbb{P}(\|Z_n - Z\| > \varepsilon) < \infty,$$

then $Z_n \xrightarrow{a.s.} Z$.

Proof. For $\varepsilon > 0$, define $A_n^\varepsilon = \{\|Z_n - Z\| > \varepsilon\}$. By Lemma 1.1, $\mathbb{P}(A_n^\varepsilon \text{ i.o.}) = 0$ for every $\varepsilon > 0$. Hence, for each $m \geq 1$, with probability one, only finitely many of the events $A_n^{1/m}$ occur. Intersecting over $m \geq 1$, we obtain, with probability one, that for every $m \geq 1$ there exists $N_m < \infty$ such that $\|Z_n - Z\| \leq 1/m$ for all $n \geq N_m$. $\square$

Example 1.1 (Laws of large numbers under fourth moment condition) Let (Z_n) be iid real-valued random variables with $\mathbb{E}[Z_1] = 0$, $\mathbb{E}[Z_1^4] < \infty$, and define

$$S_n = \frac{1}{n} \sum_{i=1}^n Z_i, \quad n \geq 1.$$

For any $\varepsilon > 0$, Markov's inequality gives $\mathbb{P}(|S_n| > \varepsilon) \leq \mathbb{E}[S_n^4]/\varepsilon^4$. Using

$$\mathbb{E}\left[\left(\sum_{i=1}^n Z_i\right)^4\right] = n\mathbb{E}[Z_1^4] + 3n(n-1)(\mathbb{E}[Z_1^2])^2,$$

it follows that there exists $C < \infty$ such that $\mathbb{P}(|S_n| > \varepsilon) \leq C/n^2\varepsilon^4$, which immediately yields $S_n \xrightarrow{\mathbb{P}} 0$. Moreover, since $\sum_{n=1}^{\infty} n^{-2} = \pi^2/6 < \infty$, Corollary 1.1 strengthens the statement to $S_n \xrightarrow{a.s.} 0$. $\diamond$

A complementary tool for weak convergence is a set-wise characterisation. One can test convergence in distribution through probabilities of open or closed sets. This is also a main tool used below.

Theorem 1.1 (Portmanteau theorem) *Let Z and (Z_n) be a random element, and sequence of random elements, respectively, in a common metric space (S, d) . The following are equivalent:*

1. $Z_n \xrightarrow{d} Z$.
2. $\liminf_{n \rightarrow \infty} \mathbb{P}(Z_n \in G) \geq \mathbb{P}(Z \in G)$ for every open set $G \subset S$.
3. $\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Z \in F)$ for every closed set $F \subset S$.

Proof. Assume convergence in distribution, and let $G \subset S$ be open. Define

$$f_m(s) = \min\{1, m d(s, G^c)\}, \quad s \in S, \quad m \geq 1.$$

Then each f_m is continuous, $0 \leq f_m \leq 1_G$, and $f_m \uparrow 1_G$ pointwise. Hence

$$\mathbb{E}[f_m(Z_n)] \leq \mathbb{P}(Z_n \in G).$$

Taking $\liminf_{n \rightarrow \infty}$ and using the assumption, $\mathbb{E}[f_m(Z)] \leq \liminf_{n \rightarrow \infty} \mathbb{P}(Z_n \in G)$. Now let $m \rightarrow \infty$ and apply monotone convergence to obtain the second condition. The equivalence between the second and third conditions follows by complements. Finally, assume the second condition, and let $f \geq 0$ be continuous on S . Since $\{f > t\}$ is open for every $t > 0$, the assumption gives

$$\mathbb{P}(f(Z) > t) \leq \liminf_{n \rightarrow \infty} \mathbb{P}(f(Z_n) > t).$$

Using Fatou's lemma,

$$\begin{aligned} \mathbb{E}[f(Z)] &= \int_0^\infty \mathbb{P}(f(Z) > t) \, dt \leq \int_0^\infty \liminf_{n \rightarrow \infty} \mathbb{P}(f(Z_n) > t) \, dt \\ &\leq \liminf_{n \rightarrow \infty} \int_0^\infty \mathbb{P}(f(Z_n) > t) \, dt = \liminf_{n \rightarrow \infty} \mathbb{E}[f(Z_n)]. \end{aligned}$$

Now let f be bounded continuous, with $|f| \leq c < \infty$. Applying the previous inequality to $c \pm f$ yields

$$\mathbb{E}[f(Z)] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[f(Z_n)] \quad \text{and} \quad \limsup_{n \rightarrow \infty} \mathbb{E}[f(Z_n)] \leq \mathbb{E}[f(Z)].$$

Hence $\mathbb{E}[f(Z_n)] \rightarrow \mathbb{E}[f(Z)]$ for every bounded continuous f , so convergence in distribution holds. $\square$

The above types of convergence are the three most commonly encountered in statistics, and they are intrinsically related in several ways. The next lemma collects the standard implications:

Lemma 1.2 (Relation between types of convergence) *Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables. Then*

1. $Z_n \xrightarrow{a.s.} Z$ implies $Z_n \xrightarrow{\mathbb{P}} Z$.
2. $Z_n \xrightarrow{\mathbb{P}} Z$ implies $Z_n \xrightarrow{d} Z$.
3. Let c be a constant. Then $Z_n \xrightarrow{\mathbb{P}} c$ if and only if $Z_n \xrightarrow{d} c$.

Proof. Assume first that $Z_n \xrightarrow{a.s.} Z$. Fix $\varepsilon > 0$ and set

$$A_n = \{\|Z_n - Z\| > \varepsilon\}.$$

Then $1_{A_n} \rightarrow 0$ almost surely and $0 \leq 1_{A_n} \leq 1$, so dominated convergence yields

$$\mathbb{P}(A_n) = \mathbb{E}[1_{A_n}] \rightarrow 0,$$

which is exactly $Z_n \xrightarrow{\mathbb{P}} Z$. Next assume $Z_n \xrightarrow{\mathbb{P}} Z$, and let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ be bounded and continuous. Take an arbitrary subsequence (Z_{n_k}) . Since $Z_{n_k} \xrightarrow{\mathbb{P}} Z$, we can choose a further subsequence $(Z_{n_{k_j}})$ such that for all $j \geq 1$

$$\mathbb{P}(\|Z_{n_{k_j}} - Z\| > 2^{-j}) < 2^{-j}$$

By Lemma 1.1 (first part), $\|Z_{n_{k_j}} - Z\| \xrightarrow{a.s.} 0$, hence $f(Z_{n_{k_j}}) \xrightarrow{a.s.} f(Z)$. Because f is bounded, the dominated convergence theorem gives

$$\mathbb{E}[f(Z_{n_{k_j}})] \rightarrow \mathbb{E}[f(Z)].$$

Thus every subsequence of $\mathbb{E}[f(Z_n)]$ has a further subsequence converging to $\mathbb{E}[f(Z)]$, so $\mathbb{E}[f(Z_n)] \rightarrow \mathbb{E}[f(Z)]$, and this proves $Z_n \xrightarrow{d} Z$. Finally, assume $Z_n \xrightarrow{d} c$ and fix $\varepsilon > 0$. The set

$$F_\varepsilon = \{x \in \mathbb{R}^p : \|x - c\| \geq \varepsilon\}$$

is closed, so Theorem 1.1 yields

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\|Z_n - c\| \geq \varepsilon) = \limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F_\varepsilon) \leq \mathbb{P}(c \in F_\varepsilon) = 0.$$

Hence $\mathbb{P}(\|Z_n - c\| > \varepsilon) \rightarrow 0$, i.e. $Z_n \xrightarrow{\mathbb{P}} c$. □

It is interesting to note that the converse implications in general fail.

Example 1.2 (Converse implications fail) The implications in Lemma 1.2 are, in general, not reversible.

1. *Convergence in distribution does not imply convergence in probability.* Let Y satisfy $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = 1/2$, and define

$$Z_n = (-1)^n Y, \quad n \geq 1.$$

Since Y and $-Y$ have the same law, each Z_n has the same distribution as Y , hence $Z_n \xrightarrow{d} Y$. However,

$$\|Z_{n+1} - Z_n\| = 2|Y|,$$

so $\mathbb{P}(\|Z_{n+1} - Z_n\| > 1) = 1$ for all n . If (Z_n) converged in probability to some random variable Z , then

$$\mathbb{P}(\|Z_{n+1} - Z_n\| > 2\varepsilon) \leq \mathbb{P}(\|Z_{n+1} - Z\| > \varepsilon) + \mathbb{P}(\|Z_n - Z\| > \varepsilon) \rightarrow 0$$

for each $\varepsilon > 0$, a contradiction.

2. *Convergence in probability does not imply almost sure convergence.* Let $U \sim \text{Unif}(0, 1)$. For $n = 2^m + j$ with $m \geq 0$ and $j = 0, \dots, 2^m - 1$, set

$$A_n = [j2^{-m}, (j+1)2^{-m}), \quad Z_n = 1_{A_n}(U).$$

Equivalently, (A_n) is the typewriter sequence

$$[0, 1), [0, 1/2), [1/2, 1), [0, 1/4), [1/4, 1/2), [1/2, 3/4), [3/4, 1), \dots$$

Then $\mathbb{P}(Z_n = 1) = |A_n| = 2^{-m} \rightarrow 0$, hence $Z_n \xrightarrow{\mathbb{P}} 0$. On the other hand, fix $u \in (0, 1)$ and define the m -th block of indices

$$B_m = \{2^m, \dots, 2^{m+1} - 1\}.$$

In each block B_m , exactly one interval among

$$[0, 2^{-m}), [2^{-m}, 2 \cdot 2^{-m}), \dots, [(2^m - 1)2^{-m}, 1)$$

contains u . Hence, for exactly one $n \in B_m$ we have $Z_n(u) = 1$, and for all other $n \in B_m$ we have $Z_n(u) = 0$. Therefore

$$\limsup_{n \rightarrow \infty} Z_n(u) = 1, \quad \liminf_{n \rightarrow \infty} Z_n(u) = 0,$$

so $(Z_n(u))$ has no limit for any $u \in (0, 1)$, and in particular Z_n does not converge almost surely.

◇

Remark 1.1 (Verifying convergences) There are several useful ways of verifying convergence in probability and in distribution apart from the definition. For instance *Chebyshev's inequality* states that for any increasing function $\varphi : [0, \infty) \rightarrow [0, \infty)$ one has, for any $\varepsilon > 0$,

$$\mathbb{P}(\|Z_n - Z\| > \varepsilon) \leq \frac{\mathbb{E}[\varphi(\|Z_n - Z\|)]}{\varphi(\varepsilon)}, \quad (1.1)$$

so if the numerator on the right-hand side tends to zero, convergence in probability follows. Likewise, the *Cramér–Wold device* states that

$$Z_n \xrightarrow{d} Z \Leftrightarrow a^\top Z_n \xrightarrow{d} a^\top Z, \quad \forall a \in \mathbb{R}^p,$$

linking one-dimensional convergence in distribution, which can be verified using standard results such as convergence of distribution functions at continuity points, and multidimensional convergence. Characteristic functions provide yet another standard route. ○

Theorem 1.2 (Strong law of large numbers) *Let (Z_n) be iid random variables with $\mathbb{E}[|Z_1|] < \infty$. Then*

$$\frac{1}{n} \sum_{i=1}^n Z_i \xrightarrow{a.s.} \mathbb{E}[Z_1].$$

Proof. It is enough to treat $\mathbb{E}[Z_1] = 0$. Write $S_n = Z_1 + \dots + Z_n$. For any $\varepsilon > 0$ define

$$A_\varepsilon = \left\{ \limsup_{n \rightarrow \infty} S_n/n > \varepsilon \right\}, \quad Z_i^\varepsilon = (Z_i - \varepsilon)1_{A_\varepsilon}, \quad S_n^\varepsilon = \sum_{i=1}^n Z_i^\varepsilon.$$

Let $M_n = S_1 \vee \dots \vee S_n$ and $M'_n = \max_{1 \leq i \leq n} (Z_2 + \dots + Z_{i+1})$, we have for $i \leq n$, $S_i = Z_1 + (Z_2 + \dots + Z_i) \leq Z_1 + M'_n$, hence $M_n \leq Z_1 + M'_n$. By stationarity, M'_n has the same distribution as M_n , so

$$\mathbb{E}[Z_1; M_n > 0] \geq \mathbb{E}[M_n - (M'_n)_+; M_n > 0] \geq \mathbb{E}[(M_n)_+ - (M'_n)_+] = 0,$$

and letting $n \rightarrow \infty$ gives $\mathbb{E}[Z_1; \sup_n S_n > 0] \geq 0$, which holds for any stationary sequence.

Now, A_ε is measurable with respect to $\mathcal{T} := \cap_n \sigma(Z_i : i > n)$, and since $Z_1, \dots, Z_n$ are independent of $\mathcal{T}$, a generating-class argument yields $\sigma(Z_n : n \geq 1) \perp \mathcal{T}$, so $\mathcal{T} \perp \mathcal{T}$, and hence $\mathbb{P}(A_\varepsilon) = 1$ or 0 . Moreover $\{\sup_n S_n^\varepsilon > 0\} = \{\sup_n S_n/n > \varepsilon\} \cap A_\varepsilon = A_\varepsilon$. Further, since A_ε is shift-invariant, (Z_n^ε) is stationary. Then, by the above inequality,

$$0 \leq \mathbb{E}[Z_1^\varepsilon; \sup_n S_n^\varepsilon > 0] = \mathbb{E}[Z_1 - \varepsilon; A_\varepsilon] = -\varepsilon \mathbb{P}(A_\varepsilon),$$

and hence $\mathbb{P}(A_\varepsilon) = 0$. Thus $\limsup_n S_n/n \leq 0$ a.s. Applying the same argument to $-S_n$ yields $\liminf_n S_n/n \geq 0$ a.s., so $S_n/n \xrightarrow{a.s.} 0$. $\square$

A further useful inequality, which helps bound the speed of weak convergence from above, is the following. It can also be paired with the Borel–Cantelli Lemma 1.1 when proving almost sure convergence.

Theorem 1.3 (Bernstein’s inequality) *Let (Z_n) be independent random variables with values in a measurable space $\mathcal{Z}$ and let γ be a real-valued function on $\mathcal{Z}$ such that $\mathbb{E}[\gamma(Z_n)] = 0$ for all $n \geq 1$. Assume that for some constant $\kappa > 0$,*

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}[|\gamma(Z_i)|^m] \leq \frac{m!}{2} \kappa^{m-2}, \quad m = 2, 3, \dots$$

Then for any $t > 0$,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n \gamma(Z_i) \geq \kappa t + \sqrt{2t}\right) \leq e^{-nt}.$$

Proof. For any centred real-valued random variable X with $\mathbb{E}[X] = 0$ and $c > 0$,

$$\exp[X - c] - 1 \leq \frac{\exp[X]}{1 + c} - 1 = \frac{e^X - 1 - X + X - c}{1 + c} \leq \frac{e^{|X|} - 1 - |X| + X - c}{1 + c}.$$

Choosing $c = \mathbb{E}[e^{|X|}] - 1 - \mathbb{E}[|X|]$ and taking expectations gives $\mathbb{E}[e^{X-c}] \leq 1$, or

$$\log \mathbb{E}[e^X] \leq c = \mathbb{E}[e^{|X|}] - 1 - \mathbb{E}[|X|].$$

Applying this to $X_i = \gamma(Z_i)/L$, expanding the exponential, and using the moment condition we obtain

$$\begin{aligned} \log \mathbb{E}\left[\exp\left(\frac{S_n}{L}\right)\right] &\leq \sum_{i=1}^n (\mathbb{E}[e^{\gamma(Z_i)/L}] - 1 - \mathbb{E}[\gamma(Z_i)/L]) \\ &\leq \sum_{m=2}^{\infty} \sum_{i=1}^n \frac{\mathbb{E}[|\gamma(Z_i)|^m]}{L^m m!} \\ &\leq \sum_{m=2}^{\infty} \frac{1}{L^m m!} \left(\frac{nm!}{2} \kappa^{m-2}\right) = \frac{n}{2L^2(1 - \kappa/L)}. \end{aligned}$$

Let $S_n = \sum_{i=1}^n \gamma(Z_i)$. Fix $a > 0$ and let $L > \kappa$. By Markov’s inequality

$$\mathbb{P}(S_n \geq a) = \mathbb{P}(e^{S_n/L} \geq e^{a/L}) \leq \exp\left(-\frac{a}{L} + \frac{n}{2(L^2 - L\kappa)}\right).$$

Optimising over L by taking $\kappa/L = 1 - (1 + 2a\kappa/n)^{-1/2}$ and then reparametrising by taking $a/n = \kappa t + \sqrt{2t}$ yields the result. $\square$

Remark 1.2 In Monte Carlo estimation, $n^{-1} \sum_{i=1}^n \gamma(Z_i)$ is an average of iid simulation outputs. Bernstein's inequality provides a non-asymptotic error bound for

$$\mathbb{P}\left(\left|n^{-1} \sum_{i=1}^n \gamma(Z_i) - \mathbb{E}[\gamma(Z_1)]\right| \geq \varepsilon\right),$$

under moment/boundedness conditions, which can be used to pick a simulation size n for a desired accuracy and confidence. $\circ$

Numeric example 1.1 (SLLN paths and Bernstein concentration for bounded risks) Let (Z_n) be iid with $Z_i \sim \text{Unif}(-1, 1)$, and define $\bar{Z}_n = n^{-1} \sum_{i=1}^n Z_i$. Then $\mathbb{E}[Z_i] = 0$ and $\text{Var}(Z_i) = 1/3$. Taking $\gamma(z) = z$ in Theorem 1.3, the moment condition is satisfied (for instance) with $\kappa = 1/3$, so that for $t > 0$,

$$\mathbb{P}(\bar{Z}_n \geq \kappa t + \sqrt{2t}) \leq e^{-nt}.$$

By symmetry, for any fixed $\varepsilon > 0$ and t_ε defined by $\varepsilon = \kappa t_\varepsilon + \sqrt{2t_\varepsilon}$, we obtain

$$\mathbb{P}(|\bar{Z}_n| \geq \varepsilon) \leq 2e^{-nt_\varepsilon}.$$

For comparison, Chebyshev gives

$$\mathbb{P}(|\bar{Z}_n| \geq \varepsilon) \leq \frac{\text{Var}(Z_1)}{n\varepsilon^2} = \frac{1}{3n\varepsilon^2}.$$

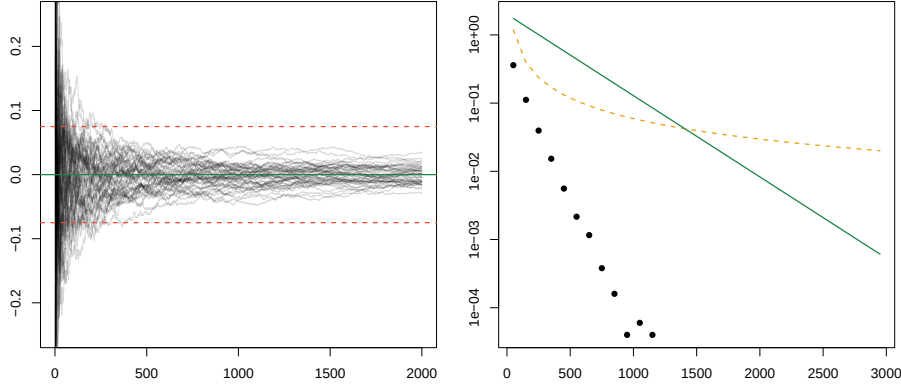


Figure 1.1: Left: 50 simulated trajectories of $\bar{Z}_n$ (black) with horizontal reference lines at 0 (green) and $\pm\varepsilon$ (red dashed), with $\varepsilon = 0.075$. Right (log scale): Monte Carlo empirical probabilities $\mathbb{P}(|\bar{Z}_n| \geq \varepsilon)$ (black points), Chebyshev bound $1/(3n\varepsilon^2)$ (orange dashed), and Bernstein bound $2e^{-nt_\varepsilon}$ (green solid), where t_ε solves $\varepsilon = \kappa t + \sqrt{2t}$ with $\kappa = 1/3$.

The left panel of Figure 1.1 visualises the pathwise statement behind Theorem 1.2, with visible early fluctuations, and trajectories stabilising and eventually remaining inside the $\pm\varepsilon$ region. The right panel highlights the rate contrast on a logarithmic scale, and although both inequalities are not tight, Chebyshev's inequality decays polynomially in n , while Bernstein's inequality decays exponentially in n . $\diamond$

Example 1.3 (Law of large numbers for non-iid risks) Let (Z_n) be independent (not necessarily identically distributed) random variables. Write $\mu_i = \mathbb{E}[Z_i]$ and $Y_i = Z_i - \mu_i$. Assume there exist constants $v > 0$ and $\kappa > 0$ such that for all i and all $m \geq 2$,

$$\mathbb{E}[|Y_i|^m] \leq \frac{m!}{2} v^2 \kappa^{m-2}.$$

Applying Theorem 1.3 to the centred variables $Y_1, \dots, Y_n$ with the scaling function $\gamma(y) = y/v$, the moment condition holds with parameter κ/v . Hence,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n Y_i \geq \kappa t + v\sqrt{2t}\right) \leq e^{-nt}, \quad t > 0,$$

and the same bound applies to $-Y_i$. For any $\varepsilon > 0$, set $t = \kappa^{-2}(\sqrt{v^2 + 2\kappa\varepsilon} - v)^2/2$, so that $\kappa t + v\sqrt{2t} = \varepsilon$. Therefore

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n (Z_i - \mu_i)\right| > \varepsilon\right) \leq 2e^{-nt},$$

which is clearly summable in n . Therefore, Lemma 1.1 yields

$$\frac{1}{n} \sum_{i=1}^n (Z_i - \mu_i) \xrightarrow{\text{a.s.}} 0.$$

◇

Two results that are often combined to analyse the asymptotic behaviour of complex expressions by reducing to simpler components are the following.

Theorem 1.4 (Continuous mapping theorem) *Let Z and (Z_n) be an $\mathbb{R}^p$ -valued random variables and sequence, respectively, and let $g : \mathbb{R}^p \rightarrow \mathbb{R}^q$ be continuous. Then*

1. $Z_n \xrightarrow{\text{a.s.}} Z$ implies $g(Z_n) \xrightarrow{\text{a.s.}} g(Z)$.
2. $Z_n \xrightarrow{\mathbb{P}} Z$ implies $g(Z_n) \xrightarrow{\mathbb{P}} g(Z)$.
3. $Z_n \xrightarrow{d} Z$ implies $g(Z_n) \xrightarrow{d} g(Z)$.

Proof. For the first statement, let $\Omega_0 = \{\omega : Z_n(\omega) \rightarrow Z(\omega)\}$. Then $\mathbb{P}(\Omega_0) = 1$. For $\omega \in \Omega_0$, continuity of g gives $g(Z_n(\omega)) \rightarrow g(Z(\omega))$. Hence $g(Z_n) \xrightarrow{\text{a.s.}} g(Z)$. For the second statement, fix $\varepsilon, \eta > 0$. Choose M such that $\mathbb{P}(\|Z\| > M) \leq \eta$ and define

$$K = \{x \in \mathbb{R}^p : \|x\| \leq M + 1\}.$$

Since g is continuous on the compact set K , it is uniformly continuous on K , so there exists $\delta \in (0, 1)$ such that

$$\|x - y\| \leq \delta, \quad x, y \in K \quad \Rightarrow \quad \|g(x) - g(y)\| \leq \varepsilon.$$

On the event $\{\|Z\| \leq M, \|Z_n - Z\| \leq \delta\}$ we have $\|Z_n\| \leq M + 1$, so $Z, Z_n \in K$, hence $\|g(Z_n) - g(Z)\| \leq \varepsilon$. Therefore

$$\mathbb{P}(\|g(Z_n) - g(Z)\| > \varepsilon) \leq \mathbb{P}(\|Z_n - Z\| > \delta) + \mathbb{P}(\|Z\| > M).$$

Using $Z_n \xrightarrow{\mathbb{P}} Z$ now gives

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\|g(Z_n) - g(Z)\| > \varepsilon) \leq \eta.$$

Since $\eta > 0$ is arbitrary, $g(Z_n) \xrightarrow{\mathbb{P}} g(Z)$. Finally, for the third statement, let $F \subset \mathbb{R}^q$ be closed. Then $g^{-1}(F) \subset \mathbb{R}^p$ is closed, and by Theorem 1.1 (iii),

$$\limsup_{n \rightarrow \infty} \mathbb{P}(g(Z_n) \in F) = \limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in g^{-1}(F)) \leq \mathbb{P}(Z \in g^{-1}(F)) = \mathbb{P}(g(Z) \in F).$$

Applying Theorem 1.1 again, with the reverse implication, yields $g(Z_n) \xrightarrow{d} g(Z)$. $\square$

Lemma 1.3 (Slutsky's lemma for random variables) *Let (Z_n) and Z be $\mathbb{R}^p$ -valued random variables. If $Z_n \xrightarrow{d} Z$ and $W_n \xrightarrow{d} c$ with c a constant, then*

$$(Z_n, W_n) \xrightarrow{d} (Z, c).$$

Proof. We first compare (Z_n, W_n) to (Z_n, c) in probability, and then use convergence in distribution for the latter. We have that $\|(Z_n, W_n) - (Z_n, c)\| = \|W_n - c\| \xrightarrow{\mathbb{P}} 0$ by the third point of Lemma 1.2. We also have $(Z_n, c) \xrightarrow{d} (Z, c)$ by the definition of convergence in distribution. Indeed, for any bounded continuous function f , the map $g(x) = f(x, c)$ is again bounded and continuous, so

$$\mathbb{E}[f(Z_n, c)] = \mathbb{E}[g(Z_n)] \rightarrow \mathbb{E}[g(Z)] = \mathbb{E}[f(Z, c)].$$

The two pieces together imply the result, as per Exercise 6. $\square$

We now state a useful approximation theorem which helps to prove $Z_n \xrightarrow{d} Z$ by choosing approximations $Y_n^{(k)}$ of Z_n and $Y^{(k)}$ of Z such that $Y_n^{(k)} \xrightarrow{d} Y^{(k)}$ for fixed k , when the approximation errors are uniformly small.

Theorem 1.5 (Weak convergence by approximation) *Let (S, d) be a metric space and let $Z, Z_n, Y^{(k)}, Y_n^{(k)}$ be S -valued random elements. Assume that for each fixed k , $Y_n^{(k)} \xrightarrow{d} Y^{(k)}$ as $n \rightarrow \infty$, and that $Y^{(k)} \xrightarrow{d} Z$ as $k \rightarrow \infty$. If in addition, for any given $\varepsilon > 0$,*

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(d(Y_n^{(k)}, Z_n) > \varepsilon) = 0, \quad (1.2)$$

then $Z_n \xrightarrow{d} Z$.

Proof. Let $F \subset S$ be closed and fix $\varepsilon > 0$. Write $F^\varepsilon = \{s \in S : d(s, F) \leq \varepsilon\}$. Then

$$\mathbb{P}(Z_n \in F) \leq \mathbb{P}(Y_n^{(k)} \in F^\varepsilon) + \mathbb{P}(d(Y_n^{(k)}, Z_n) > \varepsilon).$$

For fixed k , Theorem 1.1 (iii) yields

$$\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Y^{(k)} \in F^\varepsilon) + \limsup_{n \rightarrow \infty} \mathbb{P}(d(Y_n^{(k)}, Z_n) > \varepsilon).$$

By (1.2) the second term vanishes as $k \rightarrow \infty$. Since F^ε is closed and $Y^{(k)} \xrightarrow{d} Z$, another application of Theorem 1.1 gives

$$\limsup_{k \rightarrow \infty} \mathbb{P}(Y^{(k)} \in F^\varepsilon) \leq \mathbb{P}(Z \in F^\varepsilon).$$

Hence

$$\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Z \in F^\varepsilon).$$

Letting $\varepsilon \downarrow 0$ and using continuity from above for the closed sets $F^\varepsilon \downarrow F$ yields $\limsup_{n \rightarrow \infty} \mathbb{P}(Z_n \in F) \leq \mathbb{P}(Z \in F)$. Since F was arbitrary, Theorem 1.1 implies $Z_n \xrightarrow{d} Z$. $\square$

Then we may make the following definitions, which are the stochastic analogues of Definition 1.1.

Definition 1.4 (Stochastic o -notation) Let (Z_n) be $\mathbb{R}^p$ -valued random variables and r_n be positive random variables. We say that Z_n is bounded in probability, denoted by $Z_n = O_{\mathbb{P}}(1)$ if

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(\|Z_n\| > M) = 0.$$

Likewise, we say that Z_n converges to zero in probability, denoted by $Z_n = o_{\mathbb{P}}(1)$ if $Z_n \xrightarrow{\mathbb{P}} 0$. We also define $Z_n = O_{\mathbb{P}}(r_n)$ if $Z_n/r_n = O_{\mathbb{P}}(1)$ and similarly $Z_n = o_{\mathbb{P}}(r_n)$ if $Z_n/r_n = o_{\mathbb{P}}(1)$.

An immediate consequence is that bounded in probability follows from distributional convergence, a useful fact which is repeatedly used in the remaining chapters.

Example 1.4 (Convergence in distribution implies bounded in probability) Let (Z_n) be a real-valued random sequence converging in distribution to Z with distribution function F . Then at continuity points $M, -M$ we have

$$\mathbb{P}(Z_n > M) \rightarrow 1 - F(M), \quad \mathbb{P}(Z_n \leq -M) \rightarrow F(-M).$$

Since both of the expressions tend to zero as $M \rightarrow \infty$ it follows that

$$Z_n \xrightarrow{d} Z \quad \Rightarrow \quad Z_n = O_{\mathbb{P}}(1),$$

which is the desired implication. $\diamond$

A companion transfer rule lets us propagate stochastic bounds across approximations, simplifying notation.

Example 1.5 (Convergence in distribution of an approximating sequence) We show that if $Z_n = O_{\mathbb{P}}(a_n)$ and $W_n = o_{\mathbb{P}}(\|Z_n\|)$, then $W_n = o_{\mathbb{P}}(a_n)$.

The assumptions that $Z_n = O_{\mathbb{P}}(a_n)$ and $W_n = o_{\mathbb{P}}(\|Z_n\|)$ mean that

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(\|Z_n\|/a_n > M) = 0, \quad \lim_{n \rightarrow \infty} \mathbb{P}(\|W_n\|/\|Z_n\| > \varepsilon) = 0$$

for all $\varepsilon > 0$. We want to show that for all $\varepsilon, \delta > 0$ there exists L such that for $n \geq L$, we have $\mathbb{P}(\|W_n/a_n\| \leq \varepsilon) > 1 - \delta$. For $\varepsilon_1, \varepsilon > 0$ we get

$$\begin{aligned} \mathbb{P}(\|W_n\|/a_n \leq \varepsilon) &\geq \mathbb{P}(\{\|W_n\|/\|Z_n\| < \varepsilon_1\} \cap \{\|Z_n\|/a_n < \varepsilon/\varepsilon_1\}) \\ &\geq \mathbb{P}(\|W_n\|/\|Z_n\| < \varepsilon_1) + \mathbb{P}(\|Z_n\|/a_n < \varepsilon/\varepsilon_1) - 1. \end{aligned}$$

By assumption we can find L such that for $n \geq L$ we have $\mathbb{P}(\|Z_n\|/a_n < \varepsilon/\varepsilon_1) > 1 - \delta/2$ and $\mathbb{P}(\|W_n\|/\|Z_n\| < \varepsilon_1) > 1 - \delta/2$. The result follows. $\diamond$

1.2 Fundamental theorems in finite dimensions

In insurance, we need not only point estimates of risks but also uncertainty quantification. The main results we encounter in asymptotic statistics are limit theorems describing the distribution of suitably normalised estimators. These results fall under the umbrella term “central limit theorems”, or CLTs, and play a central role in describing the *large sample behaviour* of estimators.

Theorem 1.6 (Laplace’s multivariate CLT) *Let (Z_n) be a sequence of iid random vectors in $\mathbb{R}^p$. Assume that $\mathbb{E}[\|Z_1\|^2] < \infty$. It holds that*

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n Z_i - \mathbb{E}[Z_1] \right) \xrightarrow{d} N(0, \text{Var}(Z_1)),$$

where N denotes the p -dimensional normal distribution.

Proof. Let $\mu = \mathbb{E}[Z_1]$ and fix $a \in \mathbb{R}^p$. Put $\xi_k = a^\top (Z_k - \mu)$ so that $\mathbb{E}[\xi_1] = 0$ and $\mathbb{E}[\xi_1^2] = a^\top \text{Var}(Z_1)a$. Let $\varphi(t) = \mathbb{E}[e^{it\xi_1}]$ be the characteristic function of ξ_1 . Since $\mathbb{E}[\xi_1^2] < \infty$ and $|e^{it} - 1| \leq |t|$, dominated convergence yields

$$\varphi'(t) = \mathbb{E}[(i\xi_1)e^{it\xi_1}], \quad \varphi''(t) = \mathbb{E}[(i\xi_1)^2 e^{it\xi_1}], \quad t \in \mathbb{R}.$$

Hence $\varphi'(0) = i\mathbb{E}[\xi_1] = 0$ and $\varphi''(0) = -\mathbb{E}[\xi_1^2]$, and Taylor’s formula gives

$$\varphi(t) = 1 - \frac{t^2}{2} \mathbb{E}[\xi_1^2] + o(t^2), \quad t \rightarrow 0.$$

Hence the characteristic function of $n^{-1/2} \sum_{k=1}^n \xi_k$ is

$$\varphi\left(\frac{t}{\sqrt{n}}\right)^n = \left(1 - \frac{t^2}{2n} \mathbb{E}[\xi_1^2] + o(n^{-1})\right)^n \rightarrow \exp\left(-\frac{t^2}{2} \mathbb{E}[\xi_1^2]\right),$$

which is the characteristic function of $N(0, a^\top \text{Var}(Z_1)a)$. By the Cramér–Wold device this implies

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n Z_i - \mu \right) \xrightarrow{d} N(0, \text{Var}(Z_1)),$$

as claimed. $\square$

Sometimes the distribution of the random variables may depend on the sample size, or the risks may be transformed using an n -dependent function. In that case, it results convenient to work with the following notion.

Definition 1.5 (Triangular array) A triangular array of random variables (or in general of stochastic elements) is a sequence with a representation on the form

$$Z_{nm}, \quad n = 1, 2, \dots, \quad m = 1, \dots, n.$$

We require mutual independence for elements within the same row but do not make any assumptions on the distribution, nor on the dependence between rows. A triangular array can be visualised as

$$\begin{array}{cccc} Z_{11} & & & \\ Z_{21} & Z_{22} & & \\ Z_{31} & Z_{32} & Z_{33} & \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

Theorem 1.7 (Lindeberg's multivariate CLT) *Let (Z_{nm}) be a triangular array of random variables with values in $\mathbb{R}^p$ with $\mathbb{E}[Z_{nm}] = 0$ and $\mathbb{E}[\|Z_{nm}\|^2] < \infty$ for all $m \leq n$. Assume that*

$$\sum_{m=1}^n \text{Var}(Z_{nm}) \rightarrow \Sigma,$$

for a fixed $p \times p$ positive semi-definite matrix Σ , and that the associated real-valued array $(\|Z_{nm}\|)$ satisfies Lindeberg's condition, i.e. for all $c > 0$:

$$\sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^2; \|Z_{nm}\| > c] \rightarrow 0.$$

Then

$$\sum_{m=1}^n Z_{nm} \xrightarrow{d} N(0, \Sigma),$$

where N denotes the p -dimensional normal distribution.

Proof. Fix $a \in \mathbb{R}^p$ and set $X_{nm} = a^\top Z_{nm}$. If $a = 0$ there is nothing to prove, so assume $a \neq 0$. Then $\mathbb{E}[X_{nm}] = 0$ and

$$c_{nm} = \mathbb{E}[X_{nm}^2] = a^\top \text{Var}(Z_{nm})a, \quad c_n = \sum_{m=1}^n c_{nm} \rightarrow a^\top \Sigma a.$$

Moreover, for any $\varepsilon > 0$,

$$\sum_{m=1}^n \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon] \leq \|a\|^2 \sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^2; \|Z_{nm}\| > \varepsilon/\|a\|] \rightarrow 0,$$

so the (one-dimensional) Lindeberg condition holds for (X_{nm}) . It therefore suffices, by the Cramér–Wold device, to prove that $\sum_{m=1}^n X_{nm} \xrightarrow{d} N(0, a^\top \Sigma a)$.

We first note that for any $\varepsilon > 0$,

$$\sup_m c_{nm} \leq \varepsilon^2 + \sup_m \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon] \leq \varepsilon^2 + \sum_{m=1}^n \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon],$$

so $\sup_m c_{nm} \rightarrow 0$ by the Lindeberg condition. Let $\varphi_{nm}(t) = \mathbb{E}[e^{itX_{nm}}]$. Introduce independent $\zeta_{nm} \sim N(0, c_{nm})$ with characteristic functions $\psi_{nm}(t) = \exp(-t^2 c_{nm}/2)$, so $\sum_{m=1}^n \zeta_{nm} \sim N(0, c_n)$. Using the elementary bound (proved by induction)

$$\left| \prod_{m=1}^n z_m - \prod_{m=1}^n z'_m \right| \leq \sum_{m=1}^n |z_m - z'_m|, \quad |z_m| \leq 1, \quad |z'_m| \leq 1,$$

To control the remainders, set $h_m(x) = e^{ix} - \sum_{k=0}^m (ix)^k/k!$. Then $h_m(x) = i \int_0^x h_{m-1}(s) ds$, $|h_{-1}| \equiv 1$, and $|h_0| \leq 2$, so by induction

$$\left| e^{ix} - \sum_{k=0}^m \frac{(ix)^k}{k!} \right| \leq \frac{2|x|^m}{m!} \wedge \frac{|x|^{m+1}}{(m+1)!}.$$

With $m = 2$ this gives $|e^{itx} - 1 - itx + \frac{1}{2}t^2x^2| \leq |tx|^2(1 \wedge |tx|)$, so

$$\begin{aligned} |\varphi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| &\leq t^2 \mathbb{E}[X_{nm}^2(1 \wedge |tX_{nm}|)], \\ |\psi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| &\leq t^2 \mathbb{E}[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)]. \end{aligned}$$

Combining this with the product bound and the triangle inequality yields

$$\begin{aligned} &\left| \prod_{m=1}^n \varphi_{nm}(t) - \prod_{m=1}^n \psi_{nm}(t) \right| \\ &\leq \sum_{m=1}^n |\varphi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| + \sum_{m=1}^n |\psi_{nm}(t) - 1 + \tfrac{1}{2}t^2c_{nm}| \\ &\leq |t|^2 \sum_{m=1}^n \mathbb{E}[X_{nm}^2(1 \wedge |tX_{nm}|)] + |t|^2 \sum_{m=1}^n \mathbb{E}[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)]. \end{aligned}$$

It remains to note that

$$\sum_{m=1}^n \mathbb{E}[X_{nm}^2(1 \wedge |tX_{nm}|)] \leq \varepsilon |t| \sum_{m=1}^n c_{nm} + \sum_{m=1}^n \mathbb{E}[X_{nm}^2; |X_{nm}| > \varepsilon],$$

vanishes as $n \rightarrow \infty$ and $\varepsilon \downarrow 0$, by the Lindeberg condition and $c_n \rightarrow a^\top \Sigma a$. Likewise,

$$\sum_{m=1}^n \mathbb{E}[\zeta_{nm}^2(1 \wedge |t\zeta_{nm}|)] \leq |t|^3 \mathbb{E}[|\xi|^3] \sum_{m=1}^n c_{nm}^{3/2} \leq |t|^3 \mathbb{E}[|\xi|^3] c_n \sup_m c_{nm}^{1/2} \rightarrow 0,$$

where $\xi \sim N(0, 1)$. □

A simple and effective way of checking Lindeberg's condition is through a moment condition, as given by the following result.

Lemma 1.4 (Lyapounov's condition) *If a triangular array (Z_{nm}) satisfies Lyapounov's condition of order $\alpha > 2$, i.e.*

$$\sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^\alpha] \rightarrow 0,$$

then Lindeberg's condition is fulfilled.

Proof. Fix $c > 0$. On the event $\{\|Z_{nm}\| > c\}$ we have $\|Z_{nm}\|^2 \leq c^{2-\alpha}\|Z_{nm}\|^\alpha$ since $\alpha > 2$. Hence

$$\sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^2; \|Z_{nm}\| > c] \leq c^{2-\alpha} \sum_{m=1}^n \mathbb{E}[\|Z_{nm}\|^\alpha] \rightarrow 0,$$

which is exactly Lindeberg's condition. $\square$

To see the Lindeberg condition in action, we consider a portfolio with non-identical risks.

Example 1.6 (Portfolio with heterogeneous risks) Let (Z_n) represent $\mathbb{R}$ -valued independent but not identically distributed risks. By subtracting their means we can assume w.l.o.g. that they have mean zero. Let

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i, \quad \sigma_i = \text{Var}(Z_i)/n, \quad s_n = \left(\sum_{i=1}^n \sigma_i \right)^{1/2}.$$

We are interested in conditions under which $S_n/s_n \xrightarrow{d} N(0, 1)$. One can achieve this by applying Lindeberg's condition to the array defined by $Z_{nm} = n^{-1/2}Z_m/s_n$, assuming s_n has a positive finite limit $\bar{\sigma}$. For instance, if there exists a constant M such that $|Z_m| \leq M$ for all m , then since $n^{1/2}s_n \rightarrow \infty$, there exists N such that for $n \geq N$ we have $n^{1/2}s_n > M/c$, and so

$$\frac{1}{ns_n^2} \sum_{m=1}^n \mathbb{E}[|Z_m|^2 1_{(|Z_m| > cn^{1/2}s_n)}] = 0.$$

for every $n \geq N$, i.e. Lindeberg's condition is fulfilled. $\diamond$

We turn our attention to uniform consistency. By the law of large numbers, we know that the empirical distribution function

$$\mathbb{F}^{(n)}(t) = \frac{1}{n} \sum_{i=1}^n 1_{(Z_i \leq t)}$$

converges almost surely to $F(t) = \mathbb{P}(Z \leq t)$ for any fixed t . Sometimes, we may be interested in providing a pathwise result of the convergence, for instance guaranteeing that the latter convergence is uniform in t . That is exactly what the following celebrated theorem states.

Theorem 1.8 (Glivenko–Cantelli) *Let (Z_n) be a sequence of iid real-valued random variables with distribution function F . It holds that*

$$\sup_{t \in \mathbb{R}} |\mathbb{F}^{(n)}(t) - F(t)| = \sup_{t \in \mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^n 1_{(Z_i \leq t)} - F(t) \right| \xrightarrow{\text{a.s.}} 0.$$

Using the same proof technique, we prove a slightly more general result which is useful when generalising to other estimators that are not bounded but still non-decreasing and càdlàg (right-continuous with left limits); then the above result follows as a corollary. Crucially, the limit must be bounded, but the requirement does not extend to the approximating sequence.

Lemma 1.5 (Abstract Glivenko–Cantelli) *Assume that $T: \mathbb{R} \rightarrow \mathbb{R}$ is a non-decreasing, bounded and càdlàg function. Assume that $(T^{(n)})$ is a sequence of random non-decreasing càdlàg maps $T^{(n)}: \mathbb{R} \rightarrow \mathbb{R}$ such that $T^{(n)}(x) \xrightarrow{\text{a.s.}} T(x)$ and $T^{(n)}(x-) \xrightarrow{\text{a.s.}} T(x-)$ for all $x \in \mathbb{R}$. It then holds that*

$$\sup_{x \in \mathbb{R}} |T^{(n)}(x) - T(x)| \xrightarrow{\text{a.s.}} 0.$$

Proof. The idea is to control oscillations between jumps by a finite partition and use pointwise convergence on the grid. Given $\varepsilon > 0$ choose some $k \in \mathbb{N}$ and a sequence $-\infty = x_0 < \dots < x_k = \infty$ satisfying $T(x_j-) - T(x_{j-1}) < \varepsilon$. We can do this by including possible jumps in the sequence. By assumption it holds for each finite point in the partition x_j that

$$|T^{(n)}(x_j) - T(x_j)| \xrightarrow{\text{a.s.}} 0, \quad |T^{(n)}(x_j-) - T(x_j-)| \xrightarrow{\text{a.s.}} 0.$$

Let

$$\Delta^{(n)} = \max_{j=1, \dots, k-1} \{|T^{(n)}(x_j) - T(x_j)|, |T^{(n)}(x_j-) - T(x_j-)|\}$$

Let $x \in \mathbb{R}$ be arbitrary. By construction there is a j such that $x \in [x_{j-1}, x_j)$ and we have

$$T^{(n)}(x_{j-1}) - T(x_{j-1}) - \varepsilon \leq T^{(n)}(x_{j-1}) - T(x_j-) \leq T^{(n)}(x) - T(x)$$

and also

$$T^{(n)}(x) - T(x) \leq T^{(n)}(x_j-) - T(x_{j-1}) \leq T^{(n)}(x_j-) - T(x_j-) + \varepsilon.$$

Then

$$\begin{aligned} T^{(n)}(x_{j-1}) - T(x_{j-1}) - \varepsilon &\leq T^{(n)}(x) - T(x) \leq T^{(n)}(x_j-) - T(x_j-) + \varepsilon \\ \Rightarrow |T^{(n)}(x) - T(x)| &\leq \Delta^{(n)} + \varepsilon. \end{aligned}$$

Since x was arbitrary, it holds that

$$\sup_{x \in \mathbb{R}} |T^{(n)}(x) - T(x)| \leq \Delta^{(n)} + \varepsilon.$$

Note that $\Delta^{(n)}$ is the maximum of finitely many terms converging to zero almost surely, so $\Delta^{(n)} \xrightarrow{\text{a.s.}} 0$, and in turn

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} |T^{(n)}(x) - T(x)| \leq \varepsilon, \quad \text{a.s.}$$

Since $\varepsilon > 0$ was arbitrary, the result follows. $\square$

A similar result can be established for arrays, though the proof is not available through standard techniques, see Theorem 3.2.1 in [Shorack and Wellner \[2009\]](#).

Theorem 1.9 (Extended Glivenko–Cantelli) *Let (Z_{nm}) be a triangular array of real-valued random variables with arbitrary distribution functions F_{nm} and empirical distribution function*

$$\mathbb{F}^{(n)}(t) = \frac{1}{n} \sum_{m=1}^n 1_{(Z_{nm} \leq t)}$$

for every n . Let $F_n = \frac{1}{n} \sum_{m=1}^n F_{nm}$. Then it holds that

$$\sup_{t \in \mathbb{R}} |\mathbb{F}^{(n)}(t) - F_n(t)| \xrightarrow{a.s.} 0.$$

We now introduce an important tool for transferring asymptotic distributions through transformations of estimators. It is to be distinguished from the continuous mapping theorem in that the transformation happens before normalisation.

Theorem 1.10 (δ -method) *Let (T_n) and Z be $\mathbb{R}^p$ -valued random variables and (a_n) be a positive deterministic sequence. Let $h : \mathbb{R}^p \rightarrow \mathbb{R}$ be a differentiable function at c , with derivative $\nabla h(c)$. If $a_n(T_n - c) \xrightarrow{d} Z$ then*

$$a_n(h(T_n) - h(c)) \xrightarrow{d} \nabla h(c)^\top Z.$$

Proof. We linearise h around c and then show the remainder is negligible at the scale a_n . By Slutsky’s Lemma [1.3](#) (with the constant sequence) and the continuous mapping Theorem [1.4](#) (for the inner product operation) we have that

$$\nabla h(c)^\top a_n(T_n - c) \xrightarrow{d} \nabla h(c)^\top Z.$$

Now, from Example [1.4](#) we know that $a_n(T_n - c) = O_{\mathbb{P}}(1)$, or in other words $(T_n - c) = O_{\mathbb{P}}(a_n^{-1})$. By Taylor expansion and Example [1.5](#) we get

$$h(T_n) - h(c) = \nabla h(c)^\top (T_n - c) + o(\|T_n - c\|) = \nabla h(c)^\top (T_n - c) + o_{\mathbb{P}}(a_n^{-1}),$$

so that by multiplication with a_n we may piece the two results together to obtain

$$a_n(h(T_n) - h(c)) = \nabla h(c)^\top a_n(T_n - c) + o_{\mathbb{P}}(1) \xrightarrow{d} \nabla h(c)^\top Z$$

by Exercise [6](#). □

1.3 Basic exploratory tools

In anticipation of specialised tools in the subsequent chapters, we now introduce some basic functionals for exploring the distribution of a dataset and indicate how the above results can be used to quantify their uncertainty. The focus is on functionals that later appear in tail modelling and dependence analysis, and we emphasise their large-sample behaviour.

1.3.1 Some concrete estimators

We start with variance as a baseline measure of dispersion and then turn to tail-oriented diagnostics. For measuring the spread of a distribution, it is common to use the empirical variance, which is defined as

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}_n)^2, \quad \bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i.$$

Proposition 1.1 (Asymptotics of the empirical variance) *Let (Z_n) be an iid sequence with finite fourth moment, and central moments $\mu^{(k)} = \mathbb{E}[(Z_1 - 1_{(k>1)}\mathbb{E}[Z_1])^k]$, $k = 1, \dots, 4$. Then we have*

$$\begin{aligned} \hat{\sigma}_n^2 &\xrightarrow{a.s.} \mu^{(2)}, \\ \sqrt{n}(\hat{\sigma}_n^2 - \mu^{(2)}) &\xrightarrow{d} N(0, \mu^{(4)} - (\mu^{(2)})^2). \end{aligned}$$

Proof. First note that

$$\hat{\sigma}_n^2 = h(T_n), \quad T_n = \left(\frac{1}{n} \sum_{i=1}^n Z_i, \frac{1}{n} \sum_{i=1}^n Z_i^2 \right)^\top, \quad h(x, y) = y - x^2.$$

From this decomposition, the first part follows from the law of large numbers and the continuous mapping Theorem 1.4. Let the raw moments be given by μ_k , $k = 1, \dots, 4$. By Laplace's CLT, Theorem 1.6, $\sqrt{n}(T_n - (\mu_1, \mu_2)^\top) \xrightarrow{d} N(0, \Sigma)$ where

$$\Sigma = \begin{pmatrix} \mu_2 - \mu_1^2 & \mu_3 - \mu_1\mu_2 \\ \mu_3 - \mu_1\mu_2 & \mu_4 - \mu_2^2 \end{pmatrix}$$

We then apply the δ -method, Theorem 1.10, to the function h , with gradient $\nabla h(x, y) = (-2x, 1)^\top$. The resulting variance is calculated by

$$\nabla h(\mu_1, \mu_2)^\top \Sigma \nabla h(\mu_1, \mu_2) = \mu^{(4)} - (\mu^{(2)})^2.$$

□

As an illustration of the multivariate delta method, consider the sample correlation.

Example 1.7 (Sample correlation and Fisher's z -transform) Let (X_n, Y_n) be iid bivariate random variables with $\mathbb{E}[X_i] = \mathbb{E}[Y_i] = 0$, $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2] = 1$, and $\text{Corr}(X_i, Y_i) = \rho \in (-1, 1)$. Define

$$S_{XX} = \frac{1}{n} \sum_{i=1}^n X_i^2, \quad S_{YY} = \frac{1}{n} \sum_{i=1}^n Y_i^2, \quad S_{XY} = \frac{1}{n} \sum_{i=1}^n X_i Y_i,$$

and the sample correlation

$$r_n = \frac{S_{XY}}{\sqrt{S_{XX}S_{YY}}}.$$

By Laplace's CLT, Theorem 1.6,

$$\sqrt{n}((S_{XY}, S_{XX}, S_{YY}) - (\rho, 1, 1)) \xrightarrow{d} N(0, \Sigma),$$

where direct moment calculations for the bivariate normal yield

$$\Sigma = \begin{pmatrix} 1 + \rho^2 & 2\rho & 2\rho \\ 2\rho & 2 & 2\rho^2 \\ 2\rho & 2\rho^2 & 2 \end{pmatrix}.$$

Let $g(a, b, c) = a/(bc)^{1/2}$. Then $\nabla g(\rho, 1, 1) = (1, -\rho/2, -\rho/2)^\top$, and the δ -method, Theorem 1.10, gives

$$\sqrt{n}(r_n - \rho) \xrightarrow{d} N(0, (1 - \rho^2)^2).$$

Define the Fisher z -transform $z_n = \frac{1}{2} \log((1 + r_n)/(1 - r_n))$. Since $h(r) = \frac{1}{2} \log((1 + r)/(1 - r))$ has derivative $h'(\rho) = 1/(1 - \rho^2)$, another application of the δ -method yields

$$\sqrt{n}(z_n - \frac{1}{2} \log(\frac{1+\rho}{1-\rho})) \xrightarrow{d} N(0, 1).$$

◇

The tail of a distribution refers to its survival behaviour at very large values. Such values are sometimes deemed “catastrophic” from a non-life insurance perspective, and their accurate estimation is crucial for risk management in both insurance and reinsurance. For determining or classifying tail behaviour, one can look at the following function.

Definition 1.6 (Empirical mean excess function) Let u be a threshold. Then

$$\mathfrak{e}_n(u) = \frac{\sum_{i=1}^n Z_i 1(Z_i > u)}{\sum_{i=1}^n 1(Z_i > u)} - u,$$

is called the empirical mean excess function.

Clearly, the analysis of $\mathfrak{e}_n(u)$ is equivalent to that of $\mathfrak{e}_n(u) + u$. The following result establishes convergence to the true mean excess function, defined as $e_F(u) = \mathbb{E}[Z_1 - u | Z_1 > u]$. Define also $e_F^{(2)}(u) = \mathbb{E}[(Z_1 - u)^2 | Z_1 > u]$. A list of mean excess functions of several parametric families may be found in [Embrechts et al., 1997, p. 161].

Theorem 1.11 (Asymptotics of the empirical mean excess function) Let (Z_n) be an iid sequence with finite second moment and with distribution function F . For any u such that $F(u) < 1$ we have

$$\begin{aligned} \mathfrak{e}_n(u) &\xrightarrow{a.s.} e_F(u), \\ \sqrt{n}(\mathfrak{e}_n(u) - e_F(u)) &\xrightarrow{d} N\left(0, \frac{e_F^{(2)}(u) - e_F(u)^2}{1 - F(u)}\right). \end{aligned}$$

Proof. We express $e_n(u)$ as a smooth function of two sample averages and apply a bivariate CLT. The first part follows directly from the law of large numbers and the first part of the continuous mapping Theorem 1.4. By Laplace's CLT, Theorem 1.6,

$$T_n = \left(\frac{1}{n} \sum_{i=1}^n Z_i 1_{(Z_i > u)}, \frac{1}{n} \sum_{i=1}^n 1_{(Z_i > u)} \right)^\top$$

converges, when normalised, to a centred Gaussian vector with covariance matrix Σ with entries

$$\begin{aligned} \sigma_1^2 &= \mathbb{E}[Z_1^2 1_{(Z_1 > u)}] - \mathbb{E}[Z_1 1_{(Z_1 > u)}]^2 \\ \sigma_2^2 &= \mathbb{P}(Z_1 > u) \mathbb{P}(Z_1 \leq u) \\ \sigma_{12} &= \mathbb{E}[Z_1 1_{(Z_1 > u)}] \mathbb{P}(Z_1 \leq u). \end{aligned}$$

We then apply the δ -method, Theorem 1.10, to the function $h(x, y) = x/y$, with gradient $\nabla h(x, y) = (1/y, -x/y^2)^\top$. The resulting variance is calculated straightforwardly through

$$\nabla h(\mathbb{E}[Z_1 1_{(Z_1 > u)}], \mathbb{P}(Z_1 > u))^\top \Sigma \nabla h(\mathbb{E}[Z_1 1_{(Z_1 > u)}], \mathbb{P}(Z_1 > u)).$$

□

Notice, however, that the above result is pointwise in u and does not describe the behaviour of the process $u \mapsto e_n(u)$ across thresholds.

Definition 1.7 (Empirical cumulative hazard without product integral) Let (Z_n) be an iid sequence. We define its empirical cumulative hazard by

$$\Lambda^{(n)}(t) = \log(n) - \log \left(\sum_{i=1}^n 1_{(Z_i > t)} \right).$$

The following result establishes convergence to the cumulative hazard function

$$\Lambda(t) = -\log(1 - F(t)),$$

which for absolutely continuous F coincides with the usual integral representation

$$\int_{-\infty}^t \frac{dF(s)}{ds} \frac{1}{1 - F(s)} ds = \int_{-\infty}^t \frac{1}{1 - F(s)} dF(s).$$

Theorem 1.12 (Asymptotics of the cumulative hazard) Let (Z_n) be an iid sequence with distribution function F . For any t such that $F(t) < 1$ we have

$$\begin{aligned} \Lambda^{(n)}(t) &\xrightarrow{a.s.} \Lambda(t), \\ \sqrt{n}(\Lambda^{(n)}(t) - \Lambda(t)) &\xrightarrow{d} N\left(0, \frac{F(t)}{1 - F(t)}\right). \end{aligned}$$

Proof. This is another direct δ -method application, now for the log transform of the empirical CDF. The first part follows directly from the law of large numbers

and the first part of the continuous mapping Theorem 1.4. By Laplace's CLT, Theorem 1.6,

$$\sqrt{n}(\mathbb{F}^{(n)}(t) - F(t)) \xrightarrow{d} N(0, F(t)(1 - F(t))).$$

Notice that

$$\Lambda^{(n)}(t) = -\log(1 - \mathbb{F}^{(n)}(t)).$$

Thus, we apply the δ -method, Theorem 1.10, to the function $h(x) = -\log(1 - x)$, with gradient $h'(x) = 1/(1 - x)$. The resulting variance is given by

$$h'(F(t))^2 F(t)(1 - F(t)) = \frac{F(t)}{1 - F(t)}.$$

□

1.3.2 Finite sample behaviour

Recall that given any estimator T_n with $\sqrt{n}(T_n - \eta) \xrightarrow{d} N(0, \sigma^2)$, one can approximate its distribution by considering

$$T_n \overset{d}{\approx} N(\eta, \sigma^2/n) \quad (1.3)$$

and obtain quantiles on the right-hand side expression to match the ones on the left-hand side. Of course, the variance is in general unknown, so that a common strategy is to approximate it by a suitable estimator $\hat{\sigma}_n^2$ and plug it into the relevant quantile calculation.

It should be noted that the quality of the approximation (1.3) can vary substantially. For instance, Figure 1.2 shows that for the given sample sizes, the cumulative hazard performs better visually than the mean excess function. The behaviour of an estimator before its limit is well approximated is often referred to as *finite sample behaviour*.

We see from the above theorems that we are able to capture the behaviour of several of the exploratory tools for a fixed time (or threshold). Similar results hold when considering finitely many time points. For instance,

$$\sqrt{n}(\mathbb{F}^{(n)}(t) - F(t), \mathbb{F}^{(n)}(s) - F(s)) \xrightarrow{d} N(0, \Sigma(s, t)). \quad (1.4)$$

where $\sigma_{12}(s, t) = \mathbb{E}[1_{(Z_1 \leq t)} 1_{(Z_1 \leq s)}] - \mathbb{E}[1_{(Z_1 \leq t)}] \mathbb{E}[1_{(Z_1 \leq s)}] = F(s \wedge t) - F(s)F(t)$, and the variances are unchanged from the unidimensional case. This raises the question of whether one-dimensional central limit theorems suffice to describe the full behaviour of $\mathbb{F}^{(n)}$, especially for hypotheses that depend on the entire curve. The answer is negative. For instance, in Figure 1.3 we plot empirical versus true cumulative distribution functions for two sample sizes, where we wish to test whether F belongs to one of three given distributions. The probability that an empirical curve jumps at any fixed time $t_0 < \dots < t_m$ is zero, so the maximal distance between curves requires a genuinely continuous inspection of the paths.

In fact, although finite-dimensional distributions are enough to characterise the distribution of a stochastic process, their convergence is not enough to establish *pathwise* (see next section) convergence between e.g. $t \mapsto \mathbb{F}^{(n)}(t)$ and a

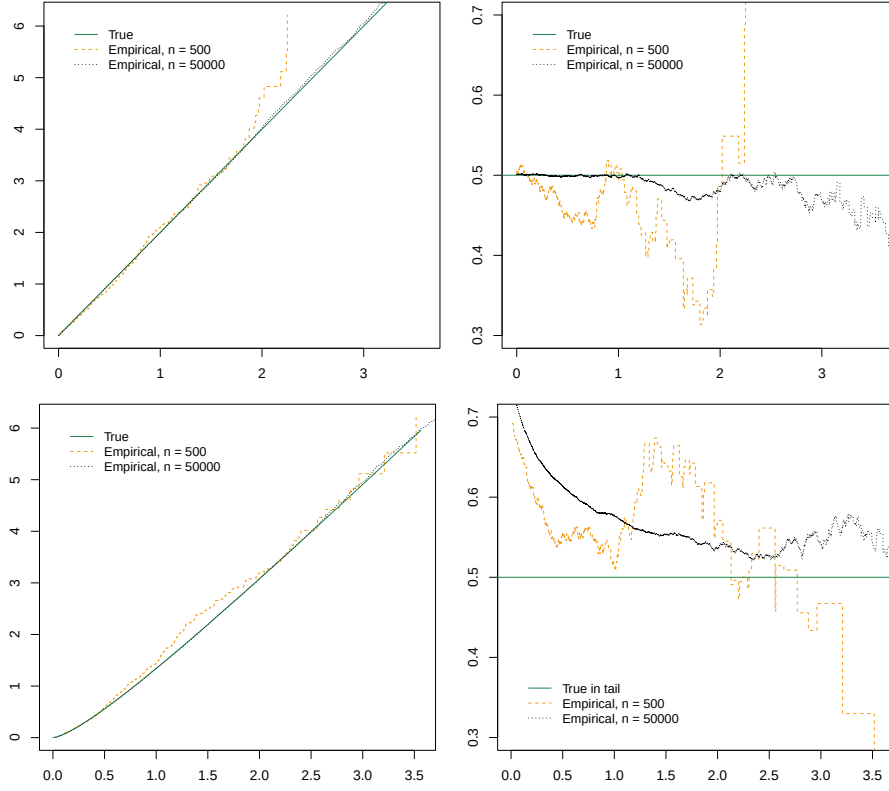


Figure 1.2: Empirical versus theoretical cumulative hazard (left, as a function of t) and mean excess functions (right, as a function of threshold u) for Exponential (top) and Gamma (bottom) distributed random variables.

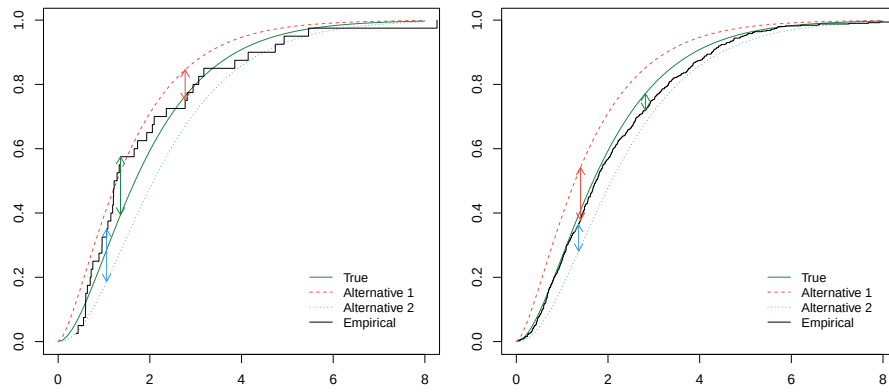


Figure 1.3: Empirical versus theoretical cumulative distribution functions $F(t)$ for Gamma distributed samples of size $n = 40$ (left) and $n = 500$ (right). Vertical arrows provide the maximal distance between curves.

suitable limit process. Therefore, our next task is to define pathwise convergence and provide additional conditions under which we may formulate *functional central limit theorems*.

1.4 Convergence of stochastic processes

In this section, we provide an overview of the relevant theory on stochastic processes and stochastic convergence, along with a couple of preliminary examples. The main application of these methods lies in nonparametric statistical procedures. The notation introduced here is used extensively in the sequel, particularly when establishing limit results for models in parametric models, extremes, survival analysis, and multi-state models. Additional results on the convergence of stochastic processes and the functional δ -method may be found in references such as [van der Vaart and Wellner \[1996\]](#) and [van der Vaart \[1998\]](#).

1.4.1 Introduction to stochastic processes

A stochastic process in continuous time is a collection of random variables indexed by some real subset, that is $X = (X_t)$, where typically $t \in I \subseteq \mathbb{R}$. Equivalently, from a pathwise point of view, given an underlying probability space $(\Omega, \mathcal{A}, \mathbb{P})$, the maps

$$\omega \mapsto X(\omega) = (X_t(\omega))_{t \in I}$$

define a random element in a suitable metric space of functions $(\mathbb{D}, d)$, where d is a suitable metric. Let $\mathcal{D}$ be the Borel σ -field generated by d . We usually refer to the space as $(\mathbb{D}, \mathcal{D})$ when no ambiguity arises. A schematic representation is provided in Figure 1.4.

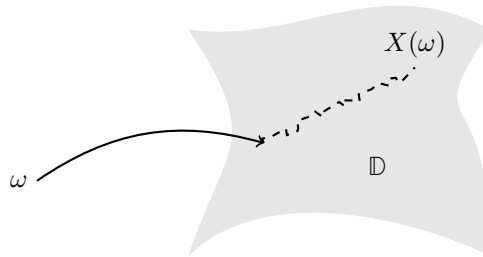


Figure 1.4: A schematic representation of a trajectory of a stochastic process X in the space $\mathbb{D}$.

Common choices for this metric space of functions are:

1. A finite-dimensional Euclidean space $\mathbb{D} = \mathbb{R}^p$, equipped with any norm.
2. $\mathbb{D} = C[a, b]$, the set of all continuous functions $z: [a, b] \rightarrow \mathbb{R}$ for $a, b \in \bar{\mathbb{R}}$, equipped with the uniform norm $\|z\|_{[a, b]} = \sup_{s \in [a, b]} |z(s)|$.

3. $\mathbb{D} = D[a, b]$, the set of all càdlàg functions $z: [a, b] \rightarrow \mathbb{R}$ for $a, b \in \bar{\mathbb{R}}$, equipped with the uniform norm $\|z\|_{[a, b]} = \sup_{s \in [a, b]} |z(s)|$. Another common choice is the Skorokhod norm.
4. $\mathbb{D} = \ell^\infty(\mathcal{F})$, the set of all uniformly bounded real functions on an arbitrary set $\mathcal{F}$: all functions $z: \mathcal{F} \rightarrow \mathbb{R}$ such that $\|z\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} |z(f)| < \infty$, equipped with the uniform norm $\|z\|_{\mathcal{F}}$. For instance, $\mathcal{F}$ could be a Euclidean subset.

Note that the convergence theory sketched below reduces to the classical one in finite dimensions for the first case above. The key observation in that case is that all norms in finite-dimensional spaces are equivalent.

In any of the four cases, to speak about the distribution of X , and hence about convergence of distributions, we define $C_b(\mathbb{D})$, the set of all bounded continuous functions on $\mathbb{D}$, and consider expectations of the kind

$$\mathbb{E}[f(X)], \quad f \in C_b(\mathbb{D}). \quad (1.5)$$

The spaces in items 1 and 2 are usually measurable and thus the above expectation is well-defined. Unfortunately, when dealing with processes in 3 or 4, subtle but important measurability issues arise when using the supremum norm. For instance, for $\mathbb{F}^{(n)}(t) = n^{-1} \sum_{i=1}^n 1_{(Z_i \leq t)}$ the process

$$t \mapsto X_n(t) = \sqrt{n}(\mathbb{F}^{(n)}(t) - F(t)), \quad t \in [a, b],$$

has paths in $\mathbb{D} = D[a, b]$ but the Borel σ -field $\mathcal{D}$ is too large for the pre-image to satisfy $X_n^{-1}(\mathcal{D}) \subseteq \mathcal{B}^n$, where $\mathcal{A} = \mathcal{B}^n$ is the underlying product space supporting $Z_1, \dots, Z_n$.

Equipping the target space with the sup-norm is natural, since it allows us to study uniform convergence of paths. However, as mentioned, it also causes measurability issues. We address these by working with outer expectations, which coincide with the usual expectation when measurability holds, and allow us to extend standard convergence results.

1.4.2 Convergence of processes

We introduce the concept of *weak convergence* for stochastic processes. Note, however, that thanks to the wealth of theorems available for this extended notion, we seldom have to deal with the foundational definitions directly. First, we introduce the so-called *outer integrals* and *outer probabilities*. They play a fundamental role in the rigorous construction of the theory, though their practical role is minor in terms of applications to statistics for insurance.

Definition 1.8 (Inner and outer integral and probability) Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $T: \Omega \rightarrow \bar{\mathbb{R}}$ be an arbitrary map. The outer integral of T with respect to $\mathbb{P}$ is defined as

$$\mathbb{E}^*[T] = \inf \{ \mathbb{E}[U] \text{ such that } U \geq T, U: \Omega \rightarrow \bar{\mathbb{R}} \text{ measurable with } \mathbb{E}[U] < \infty \}.$$

The outer probability of a subset $B \subseteq \Omega$ is

$$\mathbb{P}^*(B) = \inf \{ \mathbb{P}(A) : B \subseteq A, A \in \mathcal{A} \}.$$

The inner integral and inner probability are respectively defined as

$$\mathbb{E}_*[T] = -\mathbb{E}^*[-T], \quad \mathbb{P}_*(B) = 1 - \mathbb{P}^*(\Omega \setminus B).$$

Outer probabilities enter the picture because sup-norm functionals over uncountable index sets need not be measurable. In most applications we work with separable or càdlàg versions, in which case the outer and ordinary notions coincide for sets of interest (more precisely on sets generated by balls in $\mathbb{D}$; see [van der Vaart and Wellner, 1996, Section 1.7]), so the machinery is primarily technical rather than conceptual. Nonetheless, it is important to remember that there is a subtle distinction.

Convergence in distribution of a stochastic process evaluated at fixed indices is referred to as convergence in finite dimensions. The latter notion is implied by the following stronger condition.

Definition 1.9 (Weak convergence) Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Let $(\mathbb{D}, d)$ be a metric space and let $\mathcal{D}$ be the Borel σ -field on $\mathbb{D}$. Let (X_n) be a sequence of arbitrary maps $X_n: \Omega \rightarrow \mathbb{D}$. The sequence (X_n) converges weakly to a Borel probability measure¹ L if

$$\mathbb{E}^*[f(X_n)] \rightarrow \int f \, dL \quad \text{for every } f \in C_b(\mathbb{D}).$$

This is denoted $X_n \xrightarrow{d} L$ or $X_n \xrightarrow{d} X$ if $X \sim L$.

Notice that the definition subsumes the distributional convergence in finite dimensions, and therefore the use of the same symbol is justified.

Remark 1.3 The limit X is usually a continuous stochastic process, which is measurable with the uniform metric, and so we can write $X_n \xrightarrow{d} X$ whenever, equivalently,

$$\mathbb{E}^*[f(X_n)] \rightarrow \mathbb{E}[f(X)] \quad \text{for every } f \in C_b(\mathbb{D}).$$

Also note that the metric on $\mathbb{D}$ does not play a role in the existence within this definition, since the outer expectation always exists (we have $\inf \emptyset = \infty$). $\circ$

The following lemma shows that establishing weak convergence in a larger space than required is not a problem, as it is equivalent to showing it in the possibly smaller space where the processes take values.

Lemma 1.6 (Subspaces) Let $\mathbb{D}_0 \subseteq \mathbb{D}$ be arbitrary metric spaces equipped with the same metric. If X and every X_n take their values in $\mathbb{D}_0$, then $X_n \xrightarrow{d} X$ as maps in $\mathbb{D}_0$ if and only if $X_n \xrightarrow{d} X$ as maps in $\mathbb{D}$.

For stochastic processes we may also define convergence in probability and almost surely using the outer probability notion. Again, the definitions coincide in finite dimensions and therefore the same notation is justified.

Definition 1.10 (Convergence in outer probability and outer almost surely) Let (X_n) be a sequence of arbitrary maps $X_n: \Omega \rightarrow \mathbb{D}$ and let $X: \Omega \rightarrow \mathbb{D}$.

¹A Borel probability measure L is a probability measure defined on the Borel σ -field $\mathcal{D}$.

- (i) (X_n) converges in *outer probability* to X if $\mathbb{P}^*(d(X_n, X) > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$. This is denoted $X_n \xrightarrow{\mathbb{P}} X$.
- (ii) (X_n) converges *outer almost surely* to X if $\mathbb{P}^*(\lim_{n \rightarrow \infty} d(X_n, X) = 0) = 1$ for d the metric of interest. This is denoted $X_n \xrightarrow{a.s.} X$.

The following result generalises Lemma 1.2.

Lemma 1.7 (Relation between types of convergence) *Let (X_n) be a sequence of arbitrary maps and X Borel measurable. Then*

1. $X_n \xrightarrow{a.s.} X$ implies $X_n \xrightarrow{\mathbb{P}} X$.
2. $X_n \xrightarrow{\mathbb{P}} X$ implies $X_n \xrightarrow{d} X$.
3. Let $c \in \mathbb{D}$. $X_n \xrightarrow{\mathbb{P}} c$ if and only if $X_n \xrightarrow{d} c$.

Sometimes it can be convenient to deal with stochastic processes and their distributional representations using results from real analysis directly; in that connection the following result allows for such treatment in an appropriate probability space, sometimes referred to as a Skorokhod construction.

Theorem 1.13 (A.s. representation) *Let $X_n : \Omega \rightarrow \mathbb{D}$ be an arbitrary sequence of maps, and let X be Borel measurable and separable. If $X_n \xrightarrow{d} X$, then there exists a probability space $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{\mathbb{P}})$ and maps $\tilde{X}_n : \tilde{\Omega} \rightarrow \mathbb{D}$ with*

1. $\tilde{X}_n \xrightarrow{a.s.} \tilde{X}$;
2. $\mathbb{E}^*[f(\tilde{X}_n)] = \mathbb{E}^*[f(X_n)]$, for every bounded $f : \mathbb{D} \rightarrow \mathbb{R}$ and every n .

The following definition is quite technical, but required for full understanding of weak convergence of stochastic processes. All limit processes considered in the sequel are tight, as a consequence of some general-purpose central limit theorems given below.

Definition 1.11 (Tightness) A Borel probability measure L is *tight* if for every $\varepsilon > 0$ there exists a compact set K with $L(K) \geq 1 - \varepsilon$. Likewise we say the process $X \sim L$ is tight if L is tight.

The following lemma allows us to establish the joint weak convergence of two stochastic processes from that of the individual processes and the cross-finite-dimensional distributions. This is useful since the latter condition usually follows easily by elementary means.

Lemma 1.8 (From marginal to joint weak convergence) *Let $X_n, X : \Omega \rightarrow \ell^\infty(T)$ and $Y_n, Y : \Omega \rightarrow \ell^\infty(S)$, satisfy $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ with tight X and Y . If additionally*

$$(X_n(t_1), \dots, X_n(t_k), Y_n(s_1), \dots, Y_n(s_l)) \xrightarrow{d} (X(t_1), \dots, X(t_k), Y(s_1), \dots, Y(s_l)),$$

for any $t_1, \dots, t_k \in T$ and $s_1, \dots, s_l \in S$, then there exist versions of X and Y with bounded sample paths and $(X_n, Y_n) \xrightarrow{d} (X, Y)$ in $\ell^\infty(T) \times \ell^\infty(S)$.

We state a general version of the continuous mapping theorem. Slutsky's lemma in combination with the continuous mapping theorem is a strong tool used to conclude weak convergence for different kinds of processes in what follows.

Theorem 1.14 (Continuous mapping) *Let $\mathbb{D}_n \subseteq \mathbb{D}$ and $g_n: \mathbb{D}_n \rightarrow \mathbb{E}$ satisfy the following: if $x_n \rightarrow x$ with $x_n \in \mathbb{D}_n$ for every n and $x \in \mathbb{D}_0$, then $g_n(x_n) \rightarrow g(x)$, where $\mathbb{D}_0 \subseteq \mathbb{D}$ and $g: \mathbb{D}_0 \rightarrow \mathbb{E}$. Let (X_n) be a sequence of maps with values in $\mathbb{D}_n$, let X be Borel measurable and tight, and take values in $\mathbb{D}_0$. Then*

1. $X_n \xrightarrow{a.s.} X$ implies $g_n(X_n) \xrightarrow{a.s.} g(X)$.
2. $X_n \xrightarrow{\mathbb{P}} X$ implies $g_n(X_n) \xrightarrow{\mathbb{P}} g(X)$.
3. $X_n \xrightarrow{d} X$ implies $g_n(X_n) \xrightarrow{d} g(X)$.

Lemma 1.9 (Slutsky's lemma for general random elements) *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$ with X tight and c a constant then $(X_n, Y_n) \xrightarrow{d} (X, c)$.*

Notice that the following theorem is akin to the Glivenko–Cantelli result, and the proof is also very similar.

Theorem 1.15 (Dini's theorem, for monotone processes) *Let (X_n) be a sequence of monotone stochastic processes indexed by (some subset of) $\mathbb{R}$ converging marginally in probability to a continuous deterministic function f . Then the convergence is uniform in probability on compact subsets of the index set.*

Proof. Monotonicity lets us reduce uniform deviations to a finite grid. Assume w.l.o.g. that the processes are non-decreasing. Let $\eta > 0$ and consider a partition $a = s_0 < s_1 < \dots < s_K = b$ such that $f(s_i) - f(s_{i-1}) \leq \eta/2$. For $s \in [s_i, s_{i+1})$ we have

$$\begin{aligned} X_n(s) - f(s) &\leq X_n(s_{i+1}) - f(s_{i+1}) + f(s_{i+1}) - f(s) \\ &\leq X_n(s_{i+1}) - f(s_{i+1}) + \eta/2 \leq \max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| + \eta/2. \end{aligned}$$

Similarly $X_n(s) - f(s) \geq -\max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| - \eta/2$, which gives

$$\sup_{s \in [a, b]} |X_n(s) - f(s)| \leq \max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| + \eta/2.$$

Using the convergence in probability of the K finite terms on the right-hand side, we obtain

$$\begin{aligned} &\limsup_{n \rightarrow \infty} \mathbb{P} \left(\sup_{s \in [a, b]} |X_n(s) - f(s)| > \eta \right) \\ &\leq \limsup_{n \rightarrow \infty} \mathbb{P} \left(\max_{0 \leq k \leq K} |X_n(s_k) - f(s_k)| > \eta/2 \right) = 0. \end{aligned}$$

This finishes the proof. □

1.4.3 The functional δ -method

When dealing with stochastic processes, the nature of transformations that we can apply to their paths can be rather complex, since they may vary in a pathwise manner. The *functional δ -method* is a tool that preserves weak convergence of stochastic processes when transformed under pathwise maps that satisfy the natural but important requirement of *Hadamard differentiability*.

Definition 1.12 (Hadamard differentiability) Let $\mathbb{E}$ and $\mathbb{D}$ be two normed spaces. A map $\phi: \mathbb{D}_\phi \subseteq \mathbb{D} \rightarrow \mathbb{E}$ is called Hadamard differentiable at $\theta \in \mathbb{D}_\phi$ if there is a continuous linear map $\phi'_\theta: \mathbb{D}_\phi \rightarrow \mathbb{E}$ such that

$$\frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n} \rightarrow \phi'_\theta(h)$$

for all converging scalar sequences $t_n \rightarrow 0$ and functions $h_n \rightarrow h$ such that $\theta + t_n h_n \in \mathbb{D}_\phi$ for every n . We say that ϕ is *Hadamard differentiable tangentially* to a set $\mathbb{D}_0 \subseteq \mathbb{D}$ by requiring that every $h_n \rightarrow h$ in the definition has $h \in \mathbb{D}_0$ and the derivative need only be defined on $\mathbb{D}_0$. For a specific h , we speak of Hadamard differentiability in the direction of h .

Remark 1.4 (Hadamard differentiability implies continuity) Hadamard differentiability implies continuity at θ in the norm of $\mathbb{E}$ (e.g. the supremum norm). Indeed, let $x_n \rightarrow \theta$ in $\mathbb{D}$ and set $r_n = x_n - \theta$. Define $t_n = \|r_n\|^{1/2}$ and $h_n = r_n/t_n$, so that $t_n \rightarrow 0$, $h_n \rightarrow 0$, and $\theta + t_n h_n = x_n$. The definition of Hadamard differentiability yields

$$\frac{\phi(\theta + r_n) - \phi(\theta)}{t_n} \rightarrow \phi'_\theta(0),$$

hence $\|\phi(x_n) - \phi(\theta)\| = o(t_n) \rightarrow 0$. Thus ϕ is continuous at θ for whichever norm is placed on $\mathbb{E}$, as long as Hadamard differentiability holds in that norm. $\circ$

Notice that the above defines a pathwise derivative as a map of functions, so one needs to know how it transforms functions to be able to evaluate it. An effective tool is the following.

Theorem 1.16 (Chain rule) *Let the following maps be Hadamard differentiable:*

1. $\phi: \mathbb{D}_\phi \subseteq \mathbb{D} \rightarrow \mathbb{E}_\phi$ at $\theta \in \mathbb{D}_\phi$ tangentially to $\mathbb{D}_0$, and
2. $\psi: \mathbb{E}_\psi \supseteq \mathbb{E}_\phi \rightarrow \mathbb{F}$ at $\phi(\theta)$ tangentially to $\phi'_\theta(\mathbb{D}_0)$.

Then $\psi \circ \phi$ is Hadamard differentiable at θ tangentially to $\mathbb{D}_0$ with derivative $\psi'_{\phi(\theta)} \circ \phi'_\theta$.

Proof. We may simply rewrite

$$\frac{\psi \circ \phi(\theta + t_n h_n) - \psi \circ \phi(\theta)}{t_n} = \frac{\psi(\phi(\theta) + t_n \{ \frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n} \}) - \psi(\phi(\theta))}{t_n},$$

and obtain the limit by the composition rule for limits in normed spaces, using the Hadamard differentiability of ϕ with derivative ϕ'_θ for the expression in brackets, and then the Hadamard differentiability of ψ . $\square$

Before providing more specialised Hadamard differentiability results of intricate functions, we provide a few elementary examples, all with respect to the uniform norm.

Proposition 1.2 (Elementary Hadamard differentiability) *Fix $\tau > 0$ and equip $\mathbb{D} = \ell^\infty([0, \tau])$ with the sup norm $\|\cdot\|_\infty$.*

1. *Weighted integral functional. Let w be bounded and measurable on $[0, \tau]$, and define*

$$\phi(F) = \int_0^\tau F(t)w(t) \, dt, \quad F \in \mathbb{D}.$$

Then $\phi : \mathbb{D} \rightarrow \mathbb{R}$ is Hadamard differentiable at every F tangentially to $\mathbb{D}$ with derivative $\phi'_F(h) = \int_0^\tau h(t)w(t) \, dt$.

2. *Cumulative integral map. Define*

$$\Phi(F)(t) = \int_0^t F(s) \, ds, \quad 0 \leq t \leq \tau.$$

Then $\Phi : \mathbb{D} \rightarrow \mathbb{D}$ is Hadamard differentiable at every F tangentially to $\mathbb{D}$ with derivative $\Phi'_F(h)(t) = \int_0^t h(s) \, ds$.

3. *Ratio of integrals. Let w be bounded and measurable on $[0, \tau]$, and set*

$$\psi(F) = \frac{\int_0^\tau F(t) \, dt}{\int_0^\tau F(t)w(t) \, dt}, \quad F \in \mathbb{D}_\psi := \left\{ F \in \mathbb{D} : \int_0^\tau F(t)w(t) \, dt \neq 0 \right\}.$$

Then ψ is Hadamard differentiable at any $F \in \mathbb{D}_\psi$, with

$$\psi'_F(h) = \frac{\int_0^\tau h(t) \, dt}{\int_0^\tau F(t)w(t) \, dt} - \frac{\int_0^\tau F(t) \, dt}{\left(\int_0^\tau F(t)w(t) \, dt\right)^2} \int_0^\tau h(t)w(t) \, dt.$$

Proof. For the first statement, let $t_n \rightarrow 0$ and $h_n \rightarrow h$ in $\|\cdot\|_\infty$, and simply note that

$$\frac{\phi(F + t_n h_n) - \phi(F)}{t_n} - \phi'_F(h) = \int_0^\tau (h_n(t) - h(t))w(t) \, dt.$$

Hence

$$\left| \int_0^\tau (h_n - h)w \, dt \right| \leq \|h_n - h\|_\infty \int_0^\tau |w(t)| \, dt \leq \tau \|w\|_\infty \|h_n - h\|_\infty \rightarrow 0.$$

For the second statement, for each t , $\{\Phi(F + t_n h_n) - \Phi(F)\}(t)/t_n = \int_0^t h_n(s) \, ds$. Thus

$$\sup_{0 \leq t \leq \tau} \left| \int_0^t (h_n - h)(s) \, ds \right| \leq \tau \|h_n - h\|_\infty \rightarrow 0,$$

so the convergence holds in $\|\cdot\|_\infty$. For the third statement, let $L_1(F) = \int_0^\tau F(t) \, dt$ and $L_2(F) = \int_0^\tau F(t)w(t) \, dt$. By the previous statements and Remark 1.4 both

are continuous linear functionals on $\mathbb{D}$. For $F \in \mathbb{D}_\psi$, we have $L_2(F) \neq 0$. Hence, by continuity, there exists N such that for $n \geq N$,

$$|L_2(F + t_n h_n) - L_2(F)| < \frac{1}{2} |L_2(F)|,$$

so $|L_2(F + t_n h_n)| \geq \frac{1}{2} |L_2(F)|$ and $F + t_n h_n \in \mathbb{D}_\psi$. Using linearity,

$$\frac{\psi(F + t_n h_n) - \psi(F)}{t_n} = \frac{L_1(h_n)L_2(F) - L_1(F)L_2(h_n)}{L_2(F + t_n h_n)L_2(F)}.$$

Since $L_1(h_n) \rightarrow L_1(h)$ and $L_2(h_n) \rightarrow L_2(h)$, and $L_2(F + t_n h_n) \rightarrow L_2(F)$, the limit is the stated derivative. $\square$

We now provide some more complicated maps, which are used sporadically in the remaining chapters.

Lemma 1.10 (Hadamard differentiability of inverses and compositions) *The following hold.*

1. *We consider the inverse map with different support assumptions.*

(i) *Let $0 < u_1 < u_2 < 1$ and let F be continuously differentiable on $[a, b] = [F^{-1}(u_1) - \varepsilon, F^{-1}(u_2) + \varepsilon]$ with strictly positive derivative f . Then the inverse $\phi(h) = h^{-1}$ as a map from the set of restrictions of distribution functions to $[a, b]$, to $\ell^\infty([u_1, u_2])$ is Hadamard differentiable at F tangentially to $C[a, b]$. The derivative is*

$$\phi'_F(h)(\cdot) = -(h/f) \circ F^{-1}(\cdot).$$

(ii) *Let F have compact support $[a, b]$ and be continuously differentiable on $[a, b]$ with strictly positive derivative f . Then the inverse $\phi(h) = h^{-1}$ as a map from the set of distribution functions supported on $[a, b]$, to $\ell^\infty(0, 1)$ is Hadamard differentiable at F tangentially to $C[a, b]$. The derivative is again*

$$\phi'_F(h)(\cdot) = -(h/f) \circ F^{-1}(\cdot).$$

2. *Let $\varphi : (a, b) \rightarrow \mathbb{R}$ be a function with uniformly continuous and bounded derivative. Let $\phi(F) = \varphi(F)$ be a map from the subset of functions in $\ell^\infty(\mathcal{F})$ with image in (a, b) , to $\ell^\infty(\mathcal{F})$. The derivative of ϕ for F in such subset is*

$$\phi'_F(h)(\cdot) = \varphi'(F(\cdot))h(\cdot).$$

3. *Suppose $F_1 : \mathcal{X} \rightarrow \mathcal{Y}$ is a map, and $F_2 : \mathcal{Y} \rightarrow \mathcal{Z}$ is Fréchet differentiable² uniformly in y in the range of F_1 , with derivatives $F'_{2,y}$ such that $y \mapsto F'_{2,y}$ is uniformly norm-bounded. Then the composition $\phi(F_1, F_2)(x) = F_2 \circ F_1(x)$ is Hadamard differentiable at (F_1, F_2) tangentially to the set $\ell^\infty(\mathcal{X}, \mathcal{Y}) \times UC(\mathcal{Y}, \mathcal{Z})$ ³ with derivative*

$$\phi'_{(F_1, F_2)}(h_1, h_2)(x) = h_2 \circ F_1(x) + F'_{2, F_1(x)}(h_1(x)).$$

²A technical strengthening of Hadamard differentiability, where the limit should be continuous and linear, and the convergence uniform in h on bounded subsets of $\mathbb{D}_\phi$.

³The space of uniformly continuous functions from $\mathcal{Y} \rightarrow \mathcal{Z}$.

Proof. We treat the three items in turn.

1. Let $t_n \rightarrow 0$ and let $h_n \rightarrow h$ uniformly on $[a, b]$, with $h \in C[a, b]$, such that $F + t_n h_n$ lies in the stated domain for all n . For $u \in (0, 1)$ set $q_u = F^{-1}(u)$ and $q_{u,n} = (F + t_n h_n)^{-1}(u)$.

We first prove a generic pointwise expansion for fixed $u \in (0, 1)$. Choose $\varepsilon_n > 0$ with $\varepsilon_n = o(t_n)$. By the definition of the inverse,

$$(F + t_n h_n)(q_{u,n} - \varepsilon_n) \leq u \leq (F + t_n h_n)(q_{u,n}).$$

Since (h_n) is uniformly bounded, this yields

$$F(q_{u,n} - \varepsilon_n) + O(t_n) \leq u \leq F(q_{u,n}) + O(t_n).$$

Since F is monotone and continuous with $F(q_u) = u$ and strictly positive derivative at q_u , the function F is bounded away from u outside any neighbourhood of q_u . Hence the preceding inequalities imply $q_{u,n} \rightarrow q_u$. Moreover, uniform convergence of h_n and continuity of h give $h_n(q_{u,n} - \varepsilon_n) \rightarrow h(q_u)$. By Taylor's theorem at q_u we obtain

$$\begin{aligned} u + (q_{u,n} - q_u)f(q_u) - o(|q_{u,n} - q_u|) + O(\varepsilon_n) + t_n h(q_u) - o(t_n) &\leq u \\ &\leq u + (q_{u,n} - q_u)f(q_u) + o(|q_{u,n} - q_u|) + O(\varepsilon_n) + t_n h(q_u) + o(t_n), \end{aligned}$$

and hence $q_{u,n} - q_u = O(t_n)$ and

$$\frac{q_{u,n} - q_u}{t_n} \rightarrow -\frac{h(q_u)}{f(q_u)}.$$

- (i) The continuity and strict positivity of f on $[a, b]$ imply uniform continuity and a positive lower bound. The Taylor remainder above is therefore uniform in $u \in [u_1, u_2]$, and the same argument yields

$$\sup_{u \in [u_1, u_2]} \left| \frac{q_{u,n} - q_u}{t_n} + \frac{h(q_u)}{f(q_u)} \right| \rightarrow 0,$$

which proves Hadamard differentiability as a map into $\ell^\infty([u_1, u_2])$ with the stated derivative.

- (ii) Set $\varepsilon_{u,n} = t_n^2 \wedge (q_{u,n} - a) > 0$ for each $u \in (0, 1)$. Because $F + t_n h_n$ lies in the stated domain, it is supported on $[a, b]$, so $a < q_{u,n} \leq b$ and thus

$$(F + t_n h_n)(q_{u,n} - \varepsilon_{u,n}) \leq u \leq (F + t_n h_n)(q_{u,n}).$$

The smoothness of F on $[a, b]$ implies that

$$F(q_{u,n} - \varepsilon_{u,n}) = F(q_{u,n}) + O(\varepsilon_{u,n}),$$

uniformly in $u \in (0, 1)$. Since $h_n \rightarrow h$ uniformly, we may replace h_n by h at the cost of an $o(t_n)$ term. Hence

$$-t_n h(q_{u,n}) + o(t_n) \leq F(q_{u,n}) - F(q_u) \leq -t_n h(q_{u,n} - \varepsilon_{u,n}) + o(t_n),$$

with $o(t_n)$ uniform in u . Since f is continuous and strictly positive on $[a, b]$, the middle term is bounded above and below by a constant times $|q_{u,n} - q_u|$. This yields $|q_{u,n} - q_u| = O(t_n)$ uniformly in u . Using uniform convergence of h_n and uniform continuity of h , the preceding Taylor expansion is uniform in $u \in (0, 1)$, and we obtain

$$\sup_{u \in (0,1)} \left| \frac{q_{u,n} - q_u}{t_n} + \frac{h(q_u)}{f(q_u)} \right| \rightarrow 0.$$

This proves Hadamard differentiability as a map into $\ell^\infty(0, 1)$ with the stated derivative.

2. Let $t_n \rightarrow 0$ and let $h_n \rightarrow h$ in $\ell^\infty(\mathcal{F})$ such that $F + t_n h_n$ remains in the stated domain for all n . For each $x \in \mathcal{F}$, the mean value theorem yields a $\xi_{n,x} \in (0, 1)$ with

$$\varphi(F(x) + t_n h_n(x)) = \varphi(F(x)) + \varphi'(F(x) + \xi_{n,x} t_n h_n(x)) t_n h_n(x).$$

Therefore

$$\begin{aligned} & \left\| \frac{\phi(F + t_n h_n) - \phi(F)}{t_n} - \varphi'(F)h \right\|_\infty \\ & \leq \left\| (\varphi'(F + \xi_n t_n h_n) - \varphi'(F))h_n \right\|_\infty + \left\| \varphi'(F)(h_n - h) \right\|_\infty, \end{aligned}$$

where ξ_n denotes the pointwise selection $x \mapsto \xi_{n,x}$. Since $h_n \rightarrow h$ in the sup norm, the sequence (h_n) is bounded and $t_n \|h_n\|_\infty \rightarrow 0$. By uniform continuity of φ' on (a, b) the first term vanishes, and the second term tends to 0 because φ' is bounded. Hence ϕ is Hadamard differentiable at F with derivative $\phi'_F(h)(\cdot) = \varphi'(F(\cdot))h(\cdot)$.

3. Let $t_n \rightarrow 0$ and let $h_{1n} \rightarrow h_1$ and $h_{2n} \rightarrow h_2$ in $\ell^\infty(\mathcal{X}, \mathcal{Y})$ and $\ell^\infty(\mathcal{Y}, \mathcal{Z})$, respectively, with $h_2 \in UC(\mathcal{Y}, \mathcal{Z})$, and such that $F_1 + t_n h_{1n}$ and $F_2 + t_n h_{2n}$ remain in the stated domain for all n . Write

$$F'_{2,F_1}(h_1)(x) = F'_{2,F_1(x)}(h_1(x)), \quad x \in \mathcal{X},$$

and similarly for $F'_{2,F_1}(h_{1n})$. Then

$$\begin{aligned} & \frac{(F_2 + t_n h_{2n}) \circ (F_1 + t_n h_{1n}) - F_2 \circ F_1}{t_n} - h_2 \circ F_1 - F'_{2,F_1}(h_1) \\ & = (h_{2n} - h_2) \circ (F_1 + t_n h_{1n}) + h_2(F_1 + t_n h_{1n}) - h_2(F_1) \\ & \quad + \frac{F_2(F_1 + t_n h_{1n}) - F_2(F_1)}{t_n} - F'_{2,F_1}(h_{1n}) + F'_{2,F_1}(h_{1n} - h_1). \end{aligned}$$

The first term converges to zero in $\ell^\infty(\mathcal{X}, \mathcal{Z})$ because

$$\|(h_{2n} - h_2) \circ (F_1 + t_n h_{1n})\|_\infty \leq \|h_{2n} - h_2\|_\infty \rightarrow 0.$$

The second term converges to zero since h_2 is uniformly continuous and $\|t_n h_{1n}\|_\infty \rightarrow 0$. For the third term, uniform Fréchet differentiability of F_2 in the range of F_1 yields

$$\sup_{x \in \mathcal{X}} \left\| \frac{F_2(F_1(x) + t_n h_{1n}(x)) - F_2(F_1(x))}{t_n} - F'_{2,F_1(x)}(h_{1n}(x)) \right\|_\infty \rightarrow 0.$$

Finally, by uniform boundedness of $y \mapsto F'_{2,y}$,

$$\|F'_{2,F_1}(h_{1n} - h_1)\|_\infty \leq \sup_y \|F'_{2,y}\| \|h_{1n} - h_1\|_\infty \rightarrow 0.$$

Thus the difference quotient converges to $h_2 \circ F_1 + F'_{2,F_1}(h_1)$, and the stated derivative follows. $\square$

The two maps presented below are useful because of their link to the cumulative hazard function and to the distribution function and because they are Hadamard differentiable. They are used extensively in the final three chapters, where censoring plays an important role.

Lemma 1.11 (Hadamard differentiability of integrals) *Let $\psi: [0, M] \rightarrow \mathbb{R}$ be a C^2 -function. Then the function*

$$g(F_1, F_2)(\cdot) = \int_{(-\infty, \cdot]} \psi(F_1(s)) \, dF_2(s)$$

*is Hadamard differentiable as a map from the domain*⁴

$$\mathbb{D}_g = \text{BV}_M[-\infty, \infty] \times \text{BV}_M[-\infty, \infty] \subseteq D[-\infty, \infty] \times D[-\infty, \infty] = \mathbb{D}$$

to $\mathbb{E} = D[-\infty, \infty]$ where we equip the spaces with the supremum norm. The derivative is given by

$$\begin{aligned} g'_{(F_1, F_2)}(h_1, h_2)(\cdot) &= h_2(s)\psi(F_1(s)) \Big|_{s=-\infty}^{s=\cdot} - \int_{(-\infty, \cdot]} h_2(s-) \, d\psi(F_1(s)) \\ &\quad + \int_{(-\infty, \cdot]} \psi'(F_1(s))h_1(s) \, dF_2(s). \end{aligned}$$

Proof. Let $t_n \downarrow 0$ and let $h_{1n} \rightarrow h_1$, $h_{2n} \rightarrow h_2$ in $D[-\infty, \infty]$ be such that $F_{2n} = F_2 + t_n h_{2n} \in \text{BV}_M$ for all n . Then each h_{2n} has bounded variation. Set $F_{1n} = F_1 + t_n h_{1n}$ and fix $x \in [-\infty, \infty]$. By adding and subtracting $\psi'(F_1)h_1$ we obtain

$$\begin{aligned} &\frac{g(F_{1n}, F_{2n})(x) - g(F_1, F_2)(x)}{t_n} \\ &= \int_{(-\infty, x]} \left(\frac{\psi(F_{1n}) - \psi(F_1)}{t_n} - \psi'(F_1)h_1 \right) dF_{2n} \\ &\quad + \int_{(-\infty, x]} \psi(F_1) \, dh_{2n} + \int_{(-\infty, x]} \psi'(F_1)h_1 \, dF_{2n}. \end{aligned}$$

For the first term, Taylor's theorem and boundedness of (h_{1n}) in the supremum norm yield

$$\|(\psi(F_{1n}) - \psi(F_1))/t_n - \psi'(F_1)h_1\|_\infty \leq \|\psi'\|_\infty \|h_{1n} - h_1\|_\infty + \frac{1}{2} \|\psi''\|_\infty t_n \|h_{1n}\|_\infty^2,$$

⁴Here, we use BV_M to denote the space of càdlàg functions of bounded variation with total variation at most M .

and therefore

$$\begin{aligned} \sup_x \left| \int_{(-\infty, x]} \left(\frac{\psi(F_{1n}) - \psi(F_1)}{t_n} - \psi'(F_1)h_1 \right) dF_{2n} \right| \\ \leq (\|\psi'\|_\infty \|h_{1n} - h_1\|_\infty + \tfrac{1}{2}\|\psi''\|_\infty t_n \|h_{1n}\|_\infty^2) M, \end{aligned}$$

since $F_{2n} \in \text{BV}_M$ and as $t_n \rightarrow 0$, this term converges to zero uniformly in x .

By integration by parts,

$$\int_{(-\infty, x]} \psi(F_1) dh_{2n} = (\psi(F_1(s)) h_{2n}(s)) \Big|_{s=-\infty}^{s=x} - \int_{(-\infty, x]} h_{2n}(s-) d\psi(F_1(s)),$$

and the right-hand side converges uniformly in x to the corresponding expression with h_2 , because $h_{2n} \rightarrow h_2$ in supremum norm and $\psi(F_1)$ is of bounded variation.

Finally, write

$$\int_{(-\infty, x]} \psi'(F_1)h_1 dF_{2n} = \int_{(-\infty, x]} \psi'(F_1)h_1 dF_2 + \int_{(-\infty, x]} \psi'(F_1)h_1 d(F_{2n} - F_2).$$

Set $r = \psi'(F_1)h_1$, which is bounded and càdlàg on $[-\infty, \infty]$. Given $\varepsilon > 0$, choose a finite partition $-\infty = u_0 < \dots < u_m = \infty$ such that the oscillation of r on each $[u_{i-1}, u_i]$ is at most ε . Then, uniformly in x ,

$$\left| \int_{(-\infty, x]} r d(F_{2n} - F_2) \right| \leq 2M\varepsilon + \sum_{i=1}^m |r(u_{i-1})| |(F_{2n} - F_2)[u_{i-1}, u_i]|.$$

For fixed partition the sum tends to zero because $F_{2n} \rightarrow F_2$ uniformly, and the first term is bounded by $2M\varepsilon$. Hence the second integral vanishes uniformly in x as $n \rightarrow \infty$.

Collecting terms shows that the difference quotient converges uniformly in x to $g'_{(F_1, F_2)}(h_1, h_2)$, proving the required Hadamard differentiability. $\square$

Definition 1.13 (Product integral) Let $F \in D(0, b]$ and let $\Delta F(t) = F(t) - F(t-)$ and $F^c(t) = F(t) - \sum_{0 < s \leq t} \Delta F(s)$. The product integral is defined as

$$\phi(F)(t) = \prod_{0 < s \leq t} (1 + dF(s)) = \exp(F^c(t)) \prod_{0 < s \leq t} (1 + \Delta F(s)).$$

As additional notation we have

$$\phi(F)(s, t] = \prod_{s < u \leq t} (1 + dF(u)) = \frac{\phi(F)(t)}{\phi(F)(s)}, \quad \text{for } s < t.$$

Let X be a finite signed $(k \times k)$ matrix-valued measure defined on the Borel sets $((0, b], \mathcal{B}(0, b])$. By this we mean that each entry $X(i, j)$ is a finite signed measure on $((0, b], \mathcal{B}(0, b])$. The following definition does not require that X is commutative:

$$\prod_{0 < s \leq t} (\text{Id} + dX(s)) = \lim_{\max |t_j - t_{j-1}| \rightarrow 0} \prod_j (\text{Id} + X(t_{j-1}, t_j]),$$

for partitions $0 = t_0 < \dots < t_n = t$ of $(0, t]$.

Remark 1.5 The product integral can be defined in several ways, and more generally than Definition 1.13. For some equivalent definitions, see Gill and Johansen [1990]. $\circ$

Lemma 1.12 (Hadamard differentiability of the product integral) *For fixed, finite, positive constants b and M , the product integral map $\phi: \text{BV}_M(0, b] \subseteq D(0, b] \rightarrow D(0, b]$ is Hadamard differentiable with derivative*

$$\phi'_F(h)(\cdot) = \int_{(0, \cdot]} \phi(F)(0, u-) \phi(F)(u, \cdot] \, dh(u).$$

The scalar result above is the template for product-integral arguments throughout the book. In multi-state settings we need the matrix analogue, so we provide it next before invoking the general functional δ -method.

Lemma 1.13 (Hadamard differentiability of the product integral - matrix version) *Let $\mathbb{D}$ be a space of matrices with function entries each of bounded variation, indexed by $[0, \theta]$, and equip it with the uniform norm. For $F \in \mathbb{D}$, define*

$$\phi(F)(t, s) = \prod_{(t, s]} (\text{Id} + dF), \quad 0 \leq t < s \leq \theta.$$

The map ϕ is Hadamard differentiable at each $F \in \mathbb{D}$ w.r.t. the uniform norm on $[0, \theta]$ and has derivative

$$\phi'_F(h)(t, s) = \int_{(t, s]} \prod_t^{u-} (\text{Id} + dF) \, dh(u) \prod_u^s (\text{Id} + dF), \quad 0 \leq t < s \leq \theta,$$

where the integral w.r.t. h is defined by the integration by parts formula.

Theorem 1.17 (Functional δ -method) *Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces. Let $\phi: \mathbb{D}_\phi \subseteq \mathbb{D} \rightarrow \mathbb{E}$ be Hadamard differentiable at θ tangentially to $\mathbb{D}_0$. Let $X_n: \Omega \rightarrow \mathbb{D}_\phi$ be maps with $r_n(X_n - \theta) \xrightarrow{d} X$ for some sequence of constants $r_n \rightarrow \infty$, where X is tight and takes its values in $\mathbb{D}_0$. Then $r_n(\phi(X_n) - \phi(\theta)) \xrightarrow{d} \phi'_\theta(X)$.*

Proof. We simply rescale via a clever function g_n and then apply the continuous mapping theorem to the weak limit. Indeed, define the function $g_n(h) = r_n(\phi(\theta + r_n^{-1}h) - \phi(\theta))$ on $\mathbb{D}_n = \{h: \theta + r_n^{-1}h \in \mathbb{D}_\phi\}$. By definition of Hadamard differentiability $g_n(h_n) \rightarrow \phi'_\theta(h)$ for any $h_n \rightarrow h \in \mathbb{D}_0$. Then the continuous mapping Theorem 1.14 gives

$$r_n(\phi(X_n) - \phi(\theta)) = g_n(r_n(X_n - \theta)) \xrightarrow{d} \phi'_\theta(X).$$

$\square$

Remark 1.6 The phrase *tangentially to $\mathbb{D}_0$* in Theorem 1.17 highlights that ϕ is defined on the full domain $\mathbb{D}_\phi$, but differentiability is only required along directions in $\mathbb{D}_0$. In the theorem, the perturbations are precisely $h = r_n(X_n - \theta)$, and the assumption $X \in \mathbb{D}_0$ means these directions live in $\mathbb{D}_0$. For example, if X_n are càdlàg and $\sqrt{n}(X_n - \theta) \xrightarrow{d} X$ with X a continuous Gaussian process on $[0, \tau]$, then $\theta \in \mathbb{D}[0, \tau]$ while $h = X \in C[0, \tau]$, so it suffices that ϕ be differentiable along continuous directions even if the derivative may or may not be defined on the whole of $\mathbb{D}[0, \tau]$. $\circ$

1.4.4 Entropy theorems

There exist similar central limit theorems for stochastic processes that turn out very useful in the sequel. Many times, the estimators we deal with are sums of iid terms. Therefore, before introducing these results, it is useful to define the empirical distribution function indexed by a function class. Thus, we introduce the so-called *empirical process*.

The empirical measure $\mathbb{P}^{(n)}$ of a sample of iid random variables $Z_1, \dots, Z_n \sim L$ on a measurable space $(\mathcal{Z}, \mathcal{B})$ is the discrete measure given by

$$\mathbb{P}^{(n)}(C) = \frac{1}{n} \sum_{i=1}^n \delta_{Z_i}(C),$$

for a measurable set $C \in \mathcal{B}$. Let $\mathcal{F}$ be a collection of measurable functions $f: \mathcal{Z} \rightarrow \mathbb{R}$. Then the empirical measure induces the following map from $\mathcal{F}$ to $\mathbb{R}$

$$f \mapsto \mathbb{P}^{(n)} f = \int f \, d\mathbb{P}^{(n)} = \frac{1}{n} \sum_{i=1}^n f(Z_i).$$

We scale and centre the above map to obtain the $\mathcal{F}$ -indexed empirical process $\mathbb{G}^{(n)} = \sqrt{n}(\mathbb{P}^{(n)} - L)$ given by

$$f \mapsto \mathbb{G}^{(n)} f = \int f \, d\mathbb{G}^{(n)} = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n f(Z_i) - \int f \, dL \right).$$

If $\sup_{f \in \mathcal{F}} |f(z) - \int f \, dL| < \infty$, for almost every $z \in \mathcal{Z}$, then we view the empirical process $(\mathbb{G}^{(n)} f)_{f \in \mathcal{F}}$ as a well-defined stochastic process with paths in $\ell^\infty(\mathcal{F})$.

The complexity of $\mathcal{F}$ is later quantified via covering or bracketing numbers. For instance, the class of indicator half-lines $\{1_{(-\infty, t]} : t \in \mathbb{R}\}$ has small bracketing numbers because each function can be bracketed by neighbouring indicators with close thresholds, which is why this class has favourable asymptotic properties, as shown below.

Definition 1.14 (Glivenko–Cantelli class) Given an empirical measure $\mathbb{P}^{(n)}$ of a random sample (Z_n) defined on a measurable space $(\mathcal{Z}, \mathcal{B})$ where every Z_i has law L , a class $\mathcal{F}$ of measurable functions $f: \mathcal{Z} \rightarrow \mathbb{R}$, is called a $\mathbb{P}$ -Glivenko–Cantelli class if

$$\|\mathbb{P}^{(n)} - L\|_{\mathcal{F}} \rightarrow 0,$$

where the convergence is in outer probability or outer almost surely.

Definition 1.15 (Donsker class) A collection $\mathcal{F}$ of measurable functions $f: \mathcal{Z} \rightarrow \mathbb{R}$ is a $\mathbb{P}$ -Donsker class if

$$\mathbb{G}^{(n)} = \sqrt{n}(\mathbb{P}^{(n)} - L) \xrightarrow{d} \mathbb{G} \quad \text{in } \ell^\infty(\mathcal{F}),$$

for a tight Gaussian limit process $\mathbb{G} \in \ell^\infty(\mathcal{F})$.

Remark 1.7 Sometimes we may treat empirical processes $\mathbb{G}^{(n)}$ as stochastic processes indexed exchangeably by either: a) classes of functions $\mathcal{F}$; or b) intervals. For instance, if we consider sets on the form

$$\mathcal{F} = \{f: [0, \infty) \rightarrow \{0, 1\} \text{ s.t. } f(z) = 1_{[0, t]}(z) \text{ for } t \in [0, T]\},$$

implying that $\mathbb{G}^{(n)}f = \sqrt{n}(\frac{1}{n} \sum_{i=1}^n 1_{(Z_i \leq t)} - F(t))$, for $f = 1_{[0, t]}$, and we may rewrite weak convergence as

$$\mathbb{G}^{(n)} = \sqrt{n}(\mathbb{P}^{(n)} - L) \xrightarrow{d} \mathbb{G} \quad \text{in } \ell^\infty([0, T]),$$

meaning that the t -indexed stochastic process $(\mathbb{G}^{(n)}(t))_{t \in [0, T]}$ converges weakly to a pathwise bounded process. $\circ$

Definition 1.16 (Bracketing number) Given two functions l and u the bracket $[l, u]$ is the set of all functions f with $l \leq f \leq u$. An ε -bracket is a bracket with $\|u - l\| < \varepsilon$ where $\|\cdot\|$ is some norm. The bracketing number $N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)$ is the minimum number of ε -brackets needed to cover $\mathcal{F}$. The entropy (with bracketing) is defined as $\log N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)$.

The bracketing number increases as ε decreases. Intuitively speaking, the larger the entropy, the more the class of functions wiggles, and the more difficult it becomes to establish limit theorems. The precise size is quantified in the theorems below, collectively referred to as entropy theorems.

The next three results form a progression. Finite L^1 -bracketing controls uniform laws, finite L^2 entropy integral yields Donsker limits, and Theorem 1.20 extends the same logic to changing classes, where n -dependent functions allow us to extend from the iid case to arrays.

Theorem 1.18 (Glivenko–Cantelli) *A class $\mathcal{F}$ is $\mathbb{P}$ -Glivenko–Cantelli if*

$$N_{[]}(\varepsilon, \mathcal{F}, L^1(\mathbb{P})) < \infty$$

for every $\varepsilon > 0$.

Theorem 1.19 (Bracketing CLT, Donsker’s theorem) *A class $\mathcal{F}$ is $\mathbb{P}$ -Donsker if*

$$J_{[]} (1, \mathcal{F}, L^2(\mathbb{P})) = \int_{(0,1)} \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L^2(\mathbb{P}))} \, d\varepsilon < \infty.$$

The theorem below is stated for wide classes of separable processes. For practical purposes, we only apply this theorem in the following chapters for càdlàg processes indexed by euclidean subsets, which are separable. It is also worth remarking that even separability can be relaxed.

Theorem 1.20 (Bracketing CLT for changing functions) *Let (X_n) be a sequence of iid separable stochastic processes. Define the classes $\mathcal{F}_n = \{f_{n,t} : t \in I\}$ indexed by a Euclidean subspace I . We require a specific norm, so redefine the bracketing number $N_{[]}(\varepsilon \|F_n(X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P}))$ as the minimal number of*

sets N_ε in a partition $\mathcal{F}_n = \bigcup_{j=1}^{N_\varepsilon} \mathcal{F}_{\varepsilon_j}^n$ of the index set into sets $\mathcal{F}_{\varepsilon_j}^n$ such that for every one of these sets it holds that

$$\mathbb{E} \left[\sup_{f_{n,t}, f_{n,s} \in \mathcal{F}_{\varepsilon_j}^n} |f_{n,t}(X_1) - f_{n,s}(X_1)|^2 \right] \leq \varepsilon^2 \|F_n(X_1)\|_{L^2}^2,$$

where $F_n = \sup_t |f_{n,t}|$ is an envelope function of $\mathcal{F}^n$. Suppose

$$\mathbb{E} [F_n(X_1)^2] = O(1) \quad (1.6)$$

$$\mathbb{E} [F_n(X_1)^2 1_{(F_n(X_1) > \eta\sqrt{n})}] \rightarrow 0 \quad \text{for every } \eta > 0 \quad (1.7)$$

$$\sup_{\|s-t\| \leq \delta_n} \mathbb{E} [(f_{n,t}(X_1) - f_{n,s}(X_1))^2] \rightarrow 0 \quad \text{for every } \delta_n \rightarrow 0 \quad (1.8)$$

$$\int_{[0, \delta_n]} \sqrt{\log(N_{[]}(\varepsilon \|F_n(X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})))} \, d\varepsilon \rightarrow 0 \quad \text{for every } \delta_n \rightarrow 0. \quad (1.9)$$

Then the sequence $(\mathbb{G}^{(n)} f_{n,t})_{f_{n,t} \in \mathcal{F}_n}$ converges in distribution in $\ell^\infty(I)$ to a tight Gaussian process (with a continuous version) provided the sequence of covariance functions $\mathbb{E} [f_{n,t}(X_1) f_{n,s}(X_1)] - \mathbb{E} [f_{n,s}(X_1)] \mathbb{E} [f_{n,t}(X_1)]$ converge pointwise for each $(s, t) \in I^2$. If for each n the partition points (in t) can be chosen independently of n , then the condition in (1.8) is unnecessary.

The following examples illustrate the basic idea behind the entropy theorems. For low-dimensional parametric classes, a simple grid in the parameter space often yields a polynomial bound on the bracketing number.

Example 1.8 (A one-parameter linear class) Let (Z_n) be iid real-valued random variables with law L and assume $\mathbb{E}[Z_1^2] < \infty$. Consider

$$\mathcal{F}_1 = \{f_\theta : \mathbb{R} \rightarrow \mathbb{R} : f_\theta(z) = \theta z, \theta \in [-1, 1]\}.$$

Fix $\varepsilon > 0$ and choose a grid $-1 = \theta_0 < \theta_1 < \dots < \theta_m = 1$ with mesh size at most $\varepsilon / \|Z_1\|_{L^2}$. For $j = 0, \dots, m-1$, define

$$l_j(z) = \min\{f_{\theta_j}(z), f_{\theta_{j+1}}(z)\}, \quad u_j(z) = \max\{f_{\theta_j}(z), f_{\theta_{j+1}}(z)\}.$$

Then every f_θ with $\theta \in [\theta_j, \theta_{j+1}]$ belongs to $[l_j, u_j]$, and

$$\|u_j - l_j\|_{L^2} = (\theta_{j+1} - \theta_j) \|Z_1\|_{L^2} \leq \varepsilon.$$

Hence

$$N_{[]}(\varepsilon, \mathcal{F}_1, L^2(\mathbb{P})) \leq \left\lceil \frac{2\|Z_1\|_{L^2}}{\varepsilon} \right\rceil.$$

Since $\|g\|_{L^1} \leq \|g\|_{L^2}$, we also have $N_{[]}(\varepsilon, \mathcal{F}_1, L^1(\mathbb{P})) < \infty$ for every $\varepsilon > 0$, so $\mathcal{F}_1$ is $\mathbb{P}$ -Glivenko–Cantelli by Theorem 1.18. Moreover, a simple calculation shows $J_{[]}(\mathcal{F}_1, L^2(\mathbb{P})) < \infty$ and thus $\mathcal{F}_1$ is even $\mathbb{P}$ -Donsker by Theorem 1.19. $\diamond$

Example 1.9 (A two-parameter affine class) Under the same moment assumption as the previous example, consider

$$\mathcal{F}_2 = \{f_{a,b} : \mathbb{R} \rightarrow \mathbb{R} : f_{a,b}(z) = a + bz, (a, b) \in [-1, 1]^2\}.$$

Fix $\varepsilon > 0$ and choose grids in $[-1, 1]$ with mesh sizes $\Delta_a = \varepsilon/2$ and $\Delta_b = \varepsilon/(2\|Z_1\|_{L^2})$. This partitions $[-1, 1]^2$ into rectangles $R_{jk} = [a_j, a_{j+1}] \times [b_k, b_{k+1}]$. For each rectangle, define

$$l_{jk}(z) = \inf_{(a,b) \in R_{jk}} (a + bz), \quad u_{jk}(z) = \sup_{(a,b) \in R_{jk}} (a + bz).$$

Then every $f_{a,b}$ with $(a, b) \in R_{jk}$ belongs to $[l_{jk}, u_{jk}]$. Since $a + bz$ is linear in (a, b) , the extrema occur at the corners, and therefore

$$u_{jk}(z) - l_{jk}(z) \leq \Delta_a + \Delta_b |z|.$$

Consequently,

$$\|u_{jk} - l_{jk}\|_{L^2} \leq \Delta_a + \Delta_b \|Z_1\|_{L^2} \leq \varepsilon.$$

The number of rectangles is at most $\lceil 2/\Delta_a \rceil \lceil 2/\Delta_b \rceil$, and hence

$$N_{[]}(\varepsilon, \mathcal{F}_2, L^2(\mathbb{P})) \leq \frac{C}{\varepsilon^2}$$

for some constant $C < \infty$. As in the previous example, this implies finite $L^1(\mathbb{P})$ -bracketing numbers for every $\varepsilon > 0$, so $\mathcal{F}_2$ is $\mathbb{P}$ -Glivenko–Cantelli by Theorem 1.18, and the entropy integral in Theorem 1.19 is again finite, hence $\mathcal{F}_2$ is $\mathbb{P}$ -Donsker. $\diamond$

1.4.5 k -nearest neighbours empirical processes

The tools above allow the study of fixed-bandwidth localisation, as is conveniently exploited in the following chapter. However, one can sometimes instead use a k -nearest-neighbour (k -NN) approach, where the neighbourhood is selected by spatial search and therefore adapts automatically to the covariate density. The corresponding local empirical-process CLT can be viewed as a corollary of the bracketing CLT for changing classes, Theorem 1.20, but the proof is technically involved; we therefore refer to the detailed arguments in Portier [2025]. Equivalently, the k -NN rule may be interpreted as a kernel method with a data-dependent bandwidth, when choosing a uniform kernel on the random ball centred at x whose radius is the k -NN distance.

Definition 1.17 (Local k -nearest-neighbour empirical measure) Let $(S, \mathcal{S})$ be a measurable space and let (X_n, Y_n) be iid random elements in $(\mathbb{R}^d \times S, \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{S})$, with $x \in \mathbb{R}^d$ fixed. Let $\mathcal{K}_{n,\kappa}(x)$ be the index set of the κ nearest covariates X_i to x (ties are broken lexicographically). For $A \in \mathcal{S}$, define

$$\hat{\mu}_{n,\kappa,x}(A) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i \in A)}.$$

For measurable $g: S \rightarrow \mathbb{R}$, write $\hat{\mu}_{n,\kappa,x}(g) = \int g \, d\hat{\mu}_{n,\kappa,x}$.

The next theorem is the local k -NN counterpart of Theorem 1.20, adapted from Portier [2025].

Theorem 1.21 (Bracketing CLT for local k -NN empirical processes) *Let $(S, \mathcal{S})$ be a measurable space. Let (X_n, Y_n) be iid random elements in $(\mathbb{R}^d \times S, \mathcal{B}(\mathbb{R}^d) \otimes \mathcal{S})$, let μ_z denote the regular conditional law of Y given $X = z$, and fix $x \in \mathbb{R}^d$. Assume X has density f_X with $f_X(x) > 0$ and f_X continuous at x . Let $\mathcal{F}$ be a class of measurable functions on S with envelope F , and for $u > 0$ define*

$$\mu_{x,u}(A) = \frac{\mathbb{E}[\mu_X(A)1_{(X \in B(x, u^{1/d}))}]}{\mathbb{E}[1_{(X \in B(x, u^{1/d}))}]}, \quad A \in \mathcal{S}.$$

Here $B(x, r) = \{z \in \mathbb{R}^d : \|z - x\| \leq r\}$. Suppose there exists $\delta > 0$ such that with

$$\rho_\delta(f_1, f_2) = \sup_{|u| \leq \delta} \|f_1 - f_2\|_{L^2(\mu_{x,u})}$$

the following holds:

$$\sup_{|u| \leq \delta} \int_{[0, \delta_n]} \sqrt{\log N_{[]}(\varepsilon \|F\|_{L^2(\mu_{x,u})}, \mathcal{F}, L^2(\mu_{x,u}))} \, d\varepsilon \rightarrow 0 \quad \text{for every } \delta_n \rightarrow 0, \quad (1.10)$$

$$\sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F > M)}) \rightarrow 0 \quad \text{as } M \rightarrow \infty, \quad (1.11)$$

$$\sup_{f \in \mathcal{F}} \min_{1 \leq j \leq m_\eta} \rho_\delta(f, f_j^\eta) \leq \eta \quad \text{for every } \eta > 0 \text{ and some } f_1^\eta, \dots, f_{m_\eta}^\eta \in \mathcal{F}, \quad (1.12)$$

$$z \mapsto (\mu_z(f_1 f_2), \mu_z(f_1), \mu_z(F^2)) \text{ is continuous at } x, \text{ for every } f_1, f_2 \in \mathcal{F}. \quad (1.13)$$

If $\kappa = \kappa_n$ satisfies $\kappa \rightarrow \infty$ and $\kappa/n \rightarrow 0$, then

$$\mathbb{U}_{\kappa,x}^{(n)}(f) := \kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} (f(Y_i) - \mu_{X_i}(f)), \quad f \in \mathcal{F},$$

converges weakly in $\ell^\infty(\mathcal{F})$ to a tight centred Gaussian process $\mathbb{U}_x$ with covariance

$$\text{Cov}(\mathbb{U}_x(f_1), \mathbb{U}_x(f_2)) = \mu_x(f_1 f_2) - \mu_x(f_1)\mu_x(f_2).$$

The next corollary adds a deterministic localisation-bias bound and an explicit rate condition to pass from the centred process to the full local empirical measure.

Corollary 1.2 (From centred to full k -NN empirical-process CLT) *Assume the conditions of Theorem 1.21. Suppose additionally that there are constants $\tau_x, L_x > 0$ such that*

$$\sup_{f \in \mathcal{F}} |(\mu_z - \mu_x)(f)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x), \quad (1.14)$$

$$\kappa^{(d+2)/2}/n \rightarrow 0. \quad (1.15)$$

Then

$$\{\sqrt{\kappa}(\hat{\mu}_{n,\kappa,x}(f) - \mu_x(f))\}_{f \in \mathcal{F}} \xrightarrow{d} \mathbb{U}_x \quad \text{in } \ell^\infty(\mathcal{F}).$$

As a first elementary example we show how the previous results recover asymptotics for a k -NN regression from a singleton class. It also highlights that, when the response is bounded, the local Lindeberg requirement is automatic.

Example 1.10 (k -NN local regression at a fixed covariate) Let (X_n, Y_n) be iid with values in $\mathbb{R}^d \times \mathbb{R}$, and fix $x \in \mathbb{R}^d$. Write

$$m(z) = \mathbb{E}[Y_1 | X_1 = z], \quad \hat{m}_{n,\kappa}(x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} Y_i.$$

Assume first that $|Y_1| \leq B$ a.s. for some $B < \infty$, that $f_X(x) > 0$ with f_X continuous at x , and that $z \mapsto \mathbb{E}[Y_1 | X_1 = z]$ and $z \mapsto \mathbb{E}[Y_1^2 | X_1 = z]$ are continuous at x . Assume also that there exist $\tau_x, L_x > 0$ such that

$$|m(z) - m(x)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x),$$

and let $\kappa = \kappa_n$ satisfy $\kappa \rightarrow \infty$, and $\kappa^{(d+2)/2}/n \rightarrow 0$. Set $\mathcal{F} = \{g\}$ with $g(y) = y$. Then $\hat{m}_{n,\kappa}(x) = \hat{\mu}_{n,\kappa,x}(g)$, and:

$$\begin{aligned} N_{[]}(\varepsilon \|F\|_{L^2(\mu_{x,u})}, \mathcal{F}, L^2(\mu_{x,u})) &= 1, \quad \varepsilon > 0, \\ \sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F>M)}) &= 0 \quad \text{for } M > B, \end{aligned}$$

with envelope $F \equiv B$. Hence (1.10)–(1.12) are immediate; condition (1.13) is exactly continuity of the conditional first and second moments at x . The boundedness assumption can be weakened; it is enough that for some $\delta_0 > 0$ and $\tau'_x > 0$,

$$\sup_{\|z-x\| \leq \tau'_x} \mathbb{E}[|Y_1|^{2+\delta_0} | X_1 = z] < \infty.$$

Indeed, taking $F(y) = |y|$ and $\delta < (\tau'_x)^d$ gives

$$\sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F>M)}) \leq M^{-\delta_0} \sup_{|u| \leq \delta} \mu_{x,u}(F^{2+\delta_0}) \rightarrow 0,$$

so (1.11) still holds. Therefore Theorem 1.21 gives

$$\kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} (Y_i - m(X_i)) \xrightarrow{d} N(0, \sigma^2(x)),$$

where $\sigma^2(x) = \text{Var}(Y_1 | X_1 = x)$. By (1.14) and (1.15), Corollary 1.2 further yields

$$\sqrt{\kappa}(\hat{m}_{n,\kappa}(x) - m(x)) \xrightarrow{d} N(0, \sigma^2(x)).$$

This is the k -NN analogue of the fixed-point Nadaraya–Watson asymptotic normality, to be treated in the following chapter, with normalisation $\sqrt{\kappa}$ and without explicit bandwidth. $\diamond$

The next example treats discrete responses through a finite class of indicator functions. This gives, in one step, binary, categorical and multinomial asymptotics together with the familiar conditional multinomial covariance structure.

Example 1.11 (Binary, categorical and multinomial k -NN local probabilities) Let (X_n, Y_n) be iid and fix $x \in \mathbb{R}^d$. Assume $f_X(x) > 0$, that f_X is continuous at x , and let $\kappa = \kappa_n$ satisfy $\kappa \rightarrow \infty$ and $\kappa/n \rightarrow 0$.

For the categorical case, suppose $Y_1 \in \{1, \dots, J\}$. Define

$$p_j(z) = \mathbb{P}(Y_1 = j \mid X_1 = z), \quad \hat{p}_{j,\kappa}(x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i=j)}, \quad j = 1, \dots, J.$$

Take $\mathcal{F} = \{g_1, \dots, g_J\}$ with $g_j(y) = 1_{(y=j)}$ and envelope $F \equiv 1$. Then, for every $\varepsilon \in (0, 1)$ and every probability measure Q ,

$$N_{[\cdot]}(\varepsilon \|F\|_{L^2(Q)}, \mathcal{F}, L^2(Q)) \leq J,$$

so (1.10) holds; condition (1.11) is immediate since $F \equiv 1$, and condition (1.12) holds because $\mathcal{F}$ is finite. If $z \mapsto p_j(z)$ is continuous at x for each j , then (1.13) holds as well, since

$$\mu_z(g_j) = p_j(z), \quad \mu_z(g_j g_r) = \mu_z(1_{(Y=j)} 1_{(Y=r)}).$$

Hence, if $\kappa \rightarrow \infty$ and $\kappa/n \rightarrow 0$,

$$\left(\kappa^{-1/2} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} (g_j(Y_i) - \mu_{X_i}(g_j)) \right)_{j=1, \dots, J} \xrightarrow{d} N_J(0, \Sigma_x),$$

with

$$[\Sigma_x]_{jr} = p_j(x) 1_{(j=r)} - p_j(x) p_r(x), \quad 1 \leq j, r \leq J.$$

If, in addition, there exist $\tau_x, L_x > 0$ such that

$$\max_{1 \leq j \leq J} |p_j(z) - p_j(x)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x),$$

and $\kappa^{(d+2)/2}/n \rightarrow 0$, then Corollary 1.2 gives

$$\sqrt{\kappa}(\hat{p}_{\cdot,\kappa}(x) - p(x)) \xrightarrow{d} N_J(0, \Sigma_x),$$

where $\hat{p}_{\cdot,\kappa}(x) = (\hat{p}_{1,\kappa}(x), \dots, \hat{p}_{J,\kappa}(x))^\top$ and $p(x) = (p_1(x), \dots, p_J(x))^\top$.

The binary case is $J = 2$, yielding

$$\sqrt{\kappa}(\hat{p}_{1,\kappa}(x) - p_1(x)) \xrightarrow{d} N(0, p_1(x)(1 - p_1(x))).$$

For multinomial responses with a fixed number m of trials, write $Y = (Y_1, \dots, Y_J)$ with $\sum_{j=1}^J Y_j = m$, and apply the same argument to $g_j(y) = y_j/m \in [0, 1]$, which is valid because $g_j(Y_1)$ is the within-observation category- j proportion and satisfies $\mathbb{E}[g_j(Y_1) \mid X_1 = x] = p_j(x)$. The statistical similarity with the categorical case means they estimate the same conditional probability vector, with multinomial observations carrying within-observation replication so the covariance is divided by m (the categorical case corresponding to $m = 1$). One obtains the same limit covariance form, scaled by $1/m$:

$$[\Sigma_x]_{jr} = \frac{1}{m} (p_j(x) 1_{(j=r)} - p_j(x) p_r(x)), \quad 1 \leq j, r \leq J.$$

◇

1.5 Empirical distributions, quantiles and copulas

With the functional theorems in place, we can now transfer them to quantile- and dependence-based summaries that are central to risk measurement. We consider once again the process $t \mapsto \mathbb{F}^{(n)}(t) = n^{-1} \sum_{i=1}^n 1_{(Z_i \leq t)}$. We may extend the proof of Theorem 1.12 to recognise that its finite-dimensional distributions converge to a centred Gaussian process $\mathbb{G}$ with covariance function

$$\text{Cov}(\mathbb{G}(t), \mathbb{G}(s)) = F(s \wedge t) - F(s)F(t),$$

or in other words, $\mathbb{G} = \mathbb{B} \circ F$ a.s. where $\mathbb{B}$ is a standard Brownian bridge⁵. We can now assert the full pathwise convergence.

Theorem 1.22 (Pathwise convergence of the empirical distribution function) *Let (Z_n) be an iid sequence of positive random variables with distribution function F . Then*

$$\sqrt{n}(\mathbb{F}^{(n)} - F) \xrightarrow{d} \mathbb{B} \circ F \quad \text{in } \ell^\infty([0, \infty)).$$

Moreover, consider another sample (W_n) and denote the joint empirical and true distribution functions for the pair (Z_n, W_n) respectively by $\mathbb{H}^{(n)}$ and H . Then

$$\sqrt{n}(\mathbb{H}^{(n)} - H) \xrightarrow{d} \mathbb{G}_H \quad \text{in } \ell^\infty([0, \infty)^2)$$

with $\mathbb{G}_H$ a tight Gaussian process satisfying

$$\text{Cov}(\mathbb{G}_H(s, t), \mathbb{G}_H(u, v)) = H(s \wedge u, t \wedge v) - H(s, t)H(u, v).$$

Proof. Given that we already have recognised the limit process, we simply apply the Bracketing CLT, Theorem 1.19. Define $\mathcal{F} = \{f: [0, \infty) \rightarrow \{0, 1\} \text{ s.t. } f(z) = 1_{[0, t]}(z) \text{ for } t \in [0, \infty)\}$. We calculate the bracketing number. Consider for $j = 1, \dots, k$ brackets on the form $l_j(x) = 1_{[0, t_{j-1})}(x)$, $u_j(x) = 1_{[0, t_j]}(x)$ for sequences on the form $0 = t_0 < t_1 < \dots < t_k = \infty$. Let $\mathcal{F}_j = \{f \in \mathcal{F} : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F} = \bigcup_{j=1}^k \mathcal{F}_j$. Let $\varepsilon > 0$ be given, and pick a sequence for $k \leq \lceil 2/\varepsilon^2 \rceil$ such that

$$F(t_j -) - F(t_{j-1}) < \varepsilon^2 \quad \text{for } 1 \leq j \leq k,$$

which is possible because F is a distribution function, in particular, it is càdlàg, non-decreasing and bounded by 1. Then this sequence covers all of $\mathcal{F}$ with brackets satisfying

$$\|u_j(Z_1) - l_j(Z_1)\|_{L^2} = (F(t_j -) - F(t_{j-1}))^{1/2} < \varepsilon.$$

Consequently $N_{[]}(\varepsilon, \mathcal{F}, L^2(\mathbb{P})) \leq k \leq \lceil 2/\varepsilon^2 \rceil$. We then obtain

$$\int_{(0,1)} \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L^2(\mathbb{P}))} \, d\varepsilon \leq \int_{(0,1)} \sqrt{\log(2/\varepsilon^2 + 1)} \, d\varepsilon < \infty,$$

which shows that $\mathcal{F}$ is $\mathbb{P}$ -Donsker. The second part is similar and left for the reader. $\square$

⁵A Brownian motion and Brownian bridge are centred stochastic processes on $\mathbb{R}$ and $[0, 1]$, respectively. Their covariance functions are given by $(s, t) \mapsto t \wedge s$ and $(s, t) \mapsto t \wedge s - ts$, respectively.

Quantiles are crucial in insurance for assessing risk, and they are related to the concept of Value at Risk (VaR). Indeed the VaR at a confidence level $\alpha \in (0, 1)$ is defined simply as the α -quantile of the loss distribution. Formally, for a loss random variable Z , the VaR at level α is given by:

$$\text{VaR}_\alpha(Z) = \inf\{z \in \mathbb{R} \mid \mathbb{P}(Z \leq z) \geq \alpha\} = F^{-1}(\alpha).$$

In other words, $\text{VaR}_\alpha(Z)$ represents the maximum loss not exceeded with a confidence level of α . As a risk measure VaR possesses several important properties:

1. *Monotonicity*: if $Z_1 \leq Z_2$ almost surely, then $\text{VaR}_\alpha(Z_1) \leq \text{VaR}_\alpha(Z_2)$.
2. *Translation invariance*: for any constant $c \in \mathbb{R}$, $\text{VaR}_\alpha(Z+c) = \text{VaR}_\alpha(Z) + c$.
3. *Homogeneity*: for any positive scalar $\lambda > 0$, $\text{VaR}_\alpha(\lambda Z) = \lambda \text{VaR}_\alpha(Z)$.
4. *Subadditivity for elliptical distributions*⁶: for loss variables Z_1 and Z_2 , and $\alpha \geq 1/2$ we have $\text{VaR}_\alpha(Z_1 + Z_2) \leq \text{VaR}_\alpha(Z_1) + \text{VaR}_\alpha(Z_2)$.

For instance, Solvency II requires that pensions fulfil the Solvency Capital Requirement that enough capital is held to withstand a “200-year event”. The loss of such 200-year event can be estimated as the 99.5% upper quantile of the total loss.

Given the widespread use of VaR in the insurance industry, in particular as one of the gold standard risk measures, finding limit results for nonparametric estimators for the quantile process, which considers all quantiles simultaneously, is of considerable interest. Generally speaking, nonparametric methods provide a flexible and robust approach to estimating risk without relying on specific distributional assumptions. This is true for medium and large quantiles using the method below. However, it should also be noted that for estimating *very high quantiles*, other nonparametric tools from Extreme Value Theory are better suited.

Theorem 1.23 (Pathwise convergence of the quantile process) *Let $0 < p < q < 1$. Let (Z_n) be an iid sequence of real-valued random variables such that their distribution function F is continuously differentiable on $[a, b] = [F^{-1}(p) - \varepsilon, F^{-1}(q) + \varepsilon]$ for some $\varepsilon > 0$ (which may span the support of F if compact) with strictly positive derivative f . Then*

$$\sqrt{n}([F^{(n)}]^{-1} - F^{-1}) \xrightarrow{d} -\frac{\mathbb{B}}{f \circ F^{-1}} \quad \text{in } \ell^\infty([p, q]).$$

Proof. By the first part of Lemma 1.10 we know that the inverse map is Hadamard differentiable tangentially to the space of functions that are continuous on $[a, b]$,

⁶Distributions with $\mathbb{E}[e^{i\langle t, (Z_1, \dots, Z_r) - \mu \rangle}] = \psi(t\Sigma t^\top)$ for every $t \in \mathbb{R}^r$, for some location parameter μ , some nonnegative-definite matrix Σ and some scalar function ψ . For instance, Gaussian random vectors.

and hence we may apply the functional δ -method, Theorem 1.17, to the already established convergence from Theorem 1.22. This yields that

$$\sqrt{n}([\mathbb{F}^{(n)}]^{-1} - F^{-1}) \xrightarrow{d} -\frac{(\mathbb{B} \circ F)}{f} \circ F^{-1} \quad \text{in } \ell^\infty([p, q]),$$

as desired. $\square$

Quantile limits immediately deliver distribution theory for robust scale; the interquartile range is the prototypical case.

Example 1.12 (Asymptotics of the interquartile range) Let $q_p = F^{-1}(p)$ and $\hat{q}_p = [\mathbb{F}^{(n)}]^{-1}(p)$. Under the assumptions of Theorem 1.23 with $p = 1/4$ and $q = 3/4$, we have

$$\sqrt{n}(\hat{q}_p - q_p) \xrightarrow{d} -\frac{\mathbb{B}(p)}{f(q_p)},$$

so the vector $\sqrt{n}(\hat{q}_{1/4} - q_{1/4}, \hat{q}_{3/4} - q_{3/4})$ is asymptotically normal with covariance matrix

$$\Sigma = \begin{pmatrix} \frac{(1/4)(3/4)}{f(q_{1/4})^2} & \frac{(1/4)(1-3/4)}{f(q_{1/4})f(q_{3/4})} \\ \frac{(1/4)(1-3/4)}{f(q_{1/4})f(q_{3/4})} & \frac{(1/4)(3/4)}{f(q_{3/4})^2} \end{pmatrix},$$

since $\text{Cov}(\mathbb{B}(p_1), \mathbb{B}(p_2)) = p_1 \wedge p_2 - p_1 p_2$. The interquartile range is $\text{IQR} = q_{3/4} - q_{1/4}$ and $\widehat{\text{IQR}} = \hat{q}_{3/4} - \hat{q}_{1/4}$, so by the continuous mapping theorem,

$$\sqrt{n}(\widehat{\text{IQR}} - \text{IQR}) \xrightarrow{d} N(0, \sigma_{\text{IQR}}^2),$$

with

$$\sigma_{\text{IQR}}^2 = \frac{3}{16} \left(\frac{1}{f(q_{3/4})^2} + \frac{1}{f(q_{1/4})^2} \right) - \frac{1}{8} \frac{1}{f(q_{1/4})f(q_{3/4})}.$$

$\diamond$

We now examine the Lorenz curve, Gini coefficient, and expected shortfall, which are defined as integral functionals of the quantile process.

Example 1.13 (Lorenz curve and Gini coefficient) Let $Z \geq 0$ with mean $\mu = \mathbb{E}[Z] \in (0, \infty)$ and distribution function F with strictly positive density f . Define the quantile function $q_p = F^{-1}(p)$. Fix $0 < \delta < 1/2$ and assume that f is continuous and strictly positive on $[q_\delta, q_{1-\delta}]$. The *trimmed* Lorenz curve is

$$L_\delta(p) = \frac{1}{\mu_\delta} \int_\delta^p q_u \, du, \quad p \in [\delta, 1-\delta], \quad \mu_\delta = \int_\delta^{1-\delta} q_u \, du.$$

Let $\hat{q}_p$ be the empirical quantile and $\hat{\mu}_\delta = \int_\delta^{1-\delta} \hat{q}_u \, du$. By Theorem 1.23,

$$\sqrt{n}(\hat{q} - q) \xrightarrow{d} G \quad \text{in } \ell^\infty([\delta, 1-\delta]), \quad G(p) = -\frac{\mathbb{B}(p)}{f(q_p)}.$$

The map $q \mapsto \int_{\delta}^{(\cdot)} q_u \, du$ is Hadamard differentiable by Lemma 1.11, and the quotient map is Hadamard differentiable on $(0, \infty)^2$. Hence, by the functional δ -method,

$$\sqrt{n}(\widehat{L}_{\delta}(p) - L_{\delta}(p)) \xrightarrow{d} \frac{1}{\mu_{\delta}} \int_{\delta}^p G(u) \, du - \frac{L_{\delta}(p)}{\mu_{\delta}} \int_{\delta}^{1-\delta} G(u) \, du, \quad p \in [\delta, 1-\delta].$$

Define the trimmed Gini coefficient by

$$G_{\delta} = 1 - \frac{2}{1-2\delta} \int_{\delta}^{1-\delta} L_{\delta}(p) \, dp, \quad \widehat{G}_{\delta} = 1 - \frac{2}{1-2\delta} \int_{\delta}^{1-\delta} \widehat{L}_{\delta}(p) \, dp.$$

Another application of Lemma 1.11 yields

$$\sqrt{n}(\widehat{G}_{\delta} - G_{\delta}) \xrightarrow{d} -\frac{2}{1-2\delta} \int_{\delta}^{1-\delta} \left(\frac{1}{\mu_{\delta}} \int_{\delta}^p G(u) \, du - \frac{L_{\delta}(p)}{\mu_{\delta}} \int_{\delta}^{1-\delta} G(u) \, du \right) dp,$$

which is mean-zero normal. If additional regularity holds near $p = 0$ and $p = 1$ (for instance f is bounded away from 0 on a compact support), one may let $\delta \downarrow 0$ and recover limit results for the untruncated Lorenz and Gini definitions. $\diamond$

Remark 1.8 The quantile process limit in Theorem 1.23 is only guaranteed on compact subintervals of $(0, 1)$. Near the endpoints, the factor $1/f(q_p)$ may blow up when $f(q_p)$ is small or vanishes (as often happens for distributions on $[0, \infty)$). The trimmed versions avoid this problem; as mentioned above, extending to $\delta = 0$ requires extra endpoint regularity. $\circ$

Example 1.14 (Expected shortfall) Let $0 < \alpha < 1$ and define the *trimmed* expected shortfall

$$\text{ES}_{\alpha, \delta}(Z) = \frac{1}{1-\alpha-\delta} \int_{\alpha}^{1-\delta} q_u \, du, \quad 0 < \delta < 1-\alpha,$$

with estimator $\widehat{\text{ES}}_{\alpha, \delta} = (1-\alpha-\delta)^{-1} \int_{\alpha}^{1-\delta} \widehat{q}_u \, du$. Under the assumptions of Theorem 1.23 on $[\alpha, 1-\delta]$, the map $q \mapsto \int_{\alpha}^{(\cdot)} q_u \, du$ is Hadamard differentiable by Lemma 1.11, and thus

$$\sqrt{n}(\widehat{\text{ES}}_{\alpha, \delta} - \text{ES}_{\alpha, \delta}) \xrightarrow{d} \frac{1}{1-\alpha-\delta} \int_{\alpha}^{1-\delta} G(u) \, du, \quad G(u) = -\frac{\mathbb{B}(u)}{f(q_u)}.$$

If additional endpoint regularity holds so that the quantile process extends to $[\alpha, 1)$ and $f(q_u)^{-1}$ is integrable near $u = 1$, one may let $\delta \downarrow 0$ and recover the usual expected shortfall ES_{α} . $\diamond$

A *copula* is a mathematical function that allows us to study and model the dependence structure between multiple random variables independently of their marginal distributions. Formally, a copula is a multivariate distribution function with uniform marginals on the interval $[0, 1]$. According to Sklar's Theorem, for any multivariate distribution function H with continuous marginals $F_1, F_2, \dots, F_r$, there exists a copula $C : [0, 1]^r \rightarrow [0, 1]$ such that

$$H(x_1, x_2, \dots, x_r) = C(F_1(x_1), F_2(x_2), \dots, F_r(x_r)).$$

Conversely, if C is a copula and $F_1, F_2, \dots, F_r$ are distribution functions, then H defined as above is a multivariate distribution function with marginals $F_1, \dots, F_r$.

In risk management and insurance, copulas are commonly used for their ability to model dependencies between different risk factors. Some fundamental properties of copulas include:

1. *Uniform marginals:* for a copula $C(u_1, u_2, \dots, u_r)$, the marginal distribution of each variable is uniform on $[0, 1]$.
2. *Grounded and monotone:* a copula is grounded, meaning $C(u_1, \dots, u_r) = 0$ if any $u_i = 0$. It is also r -increasing, that is C is non-decreasing in each argument.
3. *Invariant under strictly increasing transformations:* the copula remains unchanged under strictly increasing transformations of the marginal distributions.
4. *Bounds:* copulas are bounded by the Fréchet–Hoeffding bounds, where the lower bound is $\max(\sum_{i=1}^r u_i - r + 1, 0)$ and the upper bound is given by $\min(u_1, u_2, \dots, u_r)$.

Many parametric copula functions exist, and we refer the reader to [Nelsen \[2006\]](#) for general methods on how to construct parametric copulas. We now define a nonparametric estimator of a bivariate copula, i.e. for $r = 2$ and with marginals F_1 and F_2 , referred to as the *empirical copula* function:

$$\mathbb{C}^{(n)}(u, v) = \mathbb{H}^{(n)}([\mathbb{F}_1^{(n)}]^{-1}(u), [\mathbb{F}_2^{(n)}]^{-1}(v)).$$

Theorem 1.24 (Convergence of the empirical copula process) *Let $0 < p < q < 1$. Let (Z_n, W_n) be iid vectors with joint distribution function H and marginal distribution functions F_1 and F_2 . Assume the latter are continuously differentiable on $[a, b] = [F_1^{-1}(p) - \varepsilon, F_1^{-1}(q) + \varepsilon]$ and $[c, d] = [F_2^{-1}(p) - \varepsilon, F_2^{-1}(q) + \varepsilon]$ with positive derivatives f_1 and f_2 , respectively, for some $\varepsilon > 0$. Additionally, assume that $\partial H / \partial x, \partial H / \partial y$ are continuous on the product of the intervals. Then*

$$\sqrt{n}(\mathbb{C}^{(n)} - C) \xrightarrow{d} \mathbb{G}_C \quad \text{in } \ell^\infty([a, b] \times [c, d]).$$

where with $\mathbb{G}_H$ as in Theorem 1.22 we have

$$\begin{aligned} \mathbb{G}_C &= \mathbb{G}_H(F_1^{-1}, F_2^{-1}) \\ &- \frac{\partial H}{\partial x}(F_1^{-1}, F_2^{-1}) \frac{\mathbb{G}_H(F_1^{-1}, \infty)}{f_1(F_1^{-1})} - \frac{\partial H}{\partial y}(F_1^{-1}, F_2^{-1}) \frac{\mathbb{G}_H(F_2^{-1}, \infty)}{f_2(F_2^{-1})}. \end{aligned}$$

Proof. Notice that we may construct the copula map as a composition of Hadamard differentiable maps. Indeed, define the maps

$$\begin{aligned} \phi(H)(x, y) &= (H(x, y), H(x, \infty), H(\infty, y)) \\ \psi(H, F_1, F_2)(x, y) &= (H(x, y), F_1^{-1}(x), F_2^{-1}(y)) \\ \nu(H, F_1^{-1}, F_2^{-1})(x, y) &= H \circ (F_1^{-1}(x), F_2^{-1}(y)), \end{aligned}$$

with Hadamard derivatives given by

$$\begin{aligned}\phi'(a)(x, y) &= (a(x, y), a(x, \infty), a(\infty, y)) \\ \psi'(a, b, c)(x, y) &= (a(x, y), -(b/f_1) \circ F_1^{-1}(x), -(c/f_2) \circ F_2^{-1}(y)) \\ \nu'(a, b, c)(x, y) &= a \circ (F_1^{-1}(x), F_2^{-1}(y)) \\ &\quad + (\partial H/\partial x, \partial H/\partial y) \circ (F_1^{-1}, F_2^{-1})(x, y)(b, c)^\top,\end{aligned}$$

where we have used the second, first, third, and second statements in Lemma 1.10 for the first, second, and third maps, respectively (one can also apply the definition directly to see that these are indeed the derivatives). Thus, by the chain rule, Theorem 1.16, we obtain that $\nu \circ \psi \circ \phi$ is Hadamard differentiable at H with derivative

$$\begin{aligned}\nu'_{\psi \circ \phi(H)} \circ \psi'_{\phi(H)} \circ \phi'_H(h)(u, v) &= h(F_1^{-1}(u), F_2^{-1}(v)) \\ &\quad + \left\langle (\partial H/\partial x, \partial H/\partial y) \circ (F_1^{-1}, F_2^{-1})(u, v), \left(-\frac{h(F_1^{-1}(u), \infty)}{f_1(F_1^{-1}(u))}, -\frac{h(F_2^{-1}(v), \infty)}{f_2(F_2^{-1}(v))} \right) \right\rangle,\end{aligned}$$

so that we may apply the functional δ -method, Theorem 1.17, to Theorem 1.22 to obtain the desired result. $\square$

Numeric example 1.2 (Empirical copula approximation) In Figure 1.5 we illustrate how sample size improves both the empirical copula and the associated empirical copula process. The copula considered is the Gaussian copula, constructed as per $C(u, v) = H(F_1^{-1}(u), F_2^{-1}(v))$. More precisely, the right panels display the centred and scaled field

$$(u, v) \mapsto \sqrt{n}(\mathbb{C}^{(n)}(u, v) - C(u, v)),$$

that is, the process appearing in Theorem 1.24. For $n = 100$, the empirical copula surface still exhibits visible sampling roughness, and the empirical-process surface does not yet resemble a typical realisation of the Gaussian limit particularly well. For $n = 10,000$, both visual approximations are substantially more stable. The true copula is not depicted, since it is virtually indistinguishable from the $n = 10,000$ approximation to the eye. $\diamond$

1.6 Stochastic simulation

We have already encountered some simulation studies, without providing details on *how* to simulate. Although stochastic simulation is not, strictly speaking, a topic of statistics for insurance, it is an indispensable tool throughout modern actuarial work, and throughout this book. It is the foundation for parametric bootstrap methods, enables scenario generation for both random variables and stochastic processes, and makes it possible to approximate integrals and probabilities when closed-form expressions are unavailable or intractable. Below, we give a very basic treatment, and leave more sophisticated methods to the chapters which require them.

At the root of essentially every simulation method lies the ability to generate values that behave like independent draws from the standard uniform distribution. In practice these numbers are produced by deterministic pseudorandom

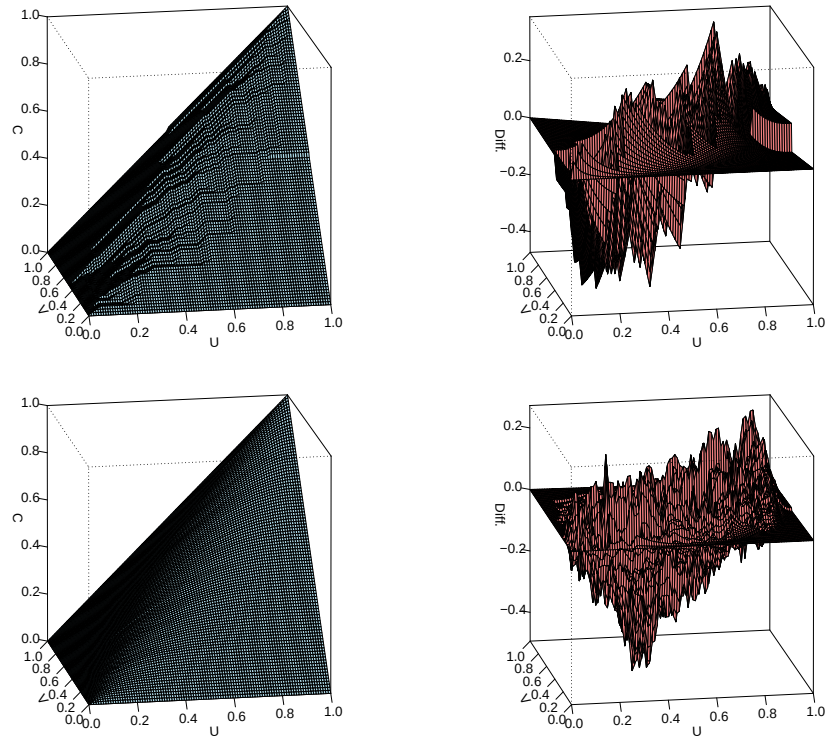


Figure 1.5: Empirical copulas (left) and their associated empirical processes (right) for simulated Gaussian vectors (correlation $\rho = 0.8$) of sample size $n = 100$ (top) and $n = 10,000$ (bottom).

number generators, designed so that their output has good statistical properties and appears random for computational purposes. Here we adopt the standard working assumption that the underlying random number generator provides independent and identically distributed samples from $\text{Unif}(0, 1)$, and we build subsequent simulation algorithms from that starting point.

Discrete random variables are then immediate to simulate. If Z is discrete and takes values $x_1, \dots, x_N$ with probabilities $p_1, \dots, p_N$, define cumulative sums $c_j = \sum_{i=1}^j p_i$ with $c_0 = 0$ and $c_N = 1$. Draw $U \sim \text{Unif}(0, 1)$ and return $Z = x_j$ whenever $c_{j-1} < U \leq c_j$. In other words, we split $(0, 1)$ into intervals of lengths $p_1, \dots, p_N$ and map each interval to its corresponding value.

For continuous random variables Z with density f , this direct interval partition is no longer available, and three general-purpose methods are common in practice: inversion, acceptance-rejection, and Fourier inversion. Beyond these, many specialised simulation routines exploit specific structural properties of the distribution of Z . We present theory for univariate laws. For multivariate laws, the most general direct construction is to factor the joint law into successive conditional laws and simulate componentwise (known as the Rosenblatt construction); when these conditional laws are not tractable, one typically turns to more specialised case-by-case techniques.

1.6.1 Inversion method

The inversion method is the most simple and robust and general-purpose simulation technique. Its caveat is that computing the inverse of a function might be numerically costly.

Definition 1.18 (Generalised inverse) Let F be a distribution function on $\mathbb{R}$. Its generalised inverse is

$$F^{-1}(u) = \inf\{y \in \mathbb{R} : F(y) \geq u\}, \quad u \in (0, 1).$$

Proposition 1.3 (Generalised inverse and inversion) Let F be a distribution function and let F^{-1} be its generalised inverse.

1. For $u \in (0, 1)$ and $z \in \mathbb{R}$,

$$u \leq F(z) \quad \Leftrightarrow \quad F^{-1}(u) \leq z.$$

2. If $U \sim \text{Unif}(0, 1)$, then $F^{-1}(U)$ has distribution function F .
3. If Z has distribution function F and F is continuous, then $F(Z) \sim \text{Unif}(0, 1)$.

Proof. The equivalence in the first statement follows directly from the definition

$$F^{-1}(u) = \inf\{y \in \mathbb{R} : F(y) \geq u\},$$

since $\{y : F(y) \geq u\} = [F^{-1}(u), \infty)$. Therefore, for any $z \in \mathbb{R}$,

$$\mathbb{P}(F^{-1}(U) \leq z) = \mathbb{P}(U \leq F(z)) = F(z),$$

which proves that $F^{-1}(U)$ has cdf F . Now assume $Z \sim F$ and that F is continuous. For $u \in (0, 1)$,

$$\mathbb{P}(F(Z) \geq u) = \mathbb{P}(Z \geq F^{-1}(u)).$$

By continuity of F , $\mathbb{P}(Z \geq a) = \mathbb{P}(Z > a) = 1 - F(a)$ for every a , hence

$$\mathbb{P}(F(Z) \geq u) = 1 - F(F^{-1}(u)).$$

Now $F(F^{-1}(u)) \geq u$ by definition of F^{-1} . If $y_n < F^{-1}(u)$ and $y_n \uparrow F^{-1}(u)$, then $F(y_n) < u$ for all n , and continuity gives $F(F^{-1}(u)) \leq u$. Therefore $F(F^{-1}(u)) = u$, so

$$\mathbb{P}(F(Z) \geq u) = 1 - u, \quad u \in (0, 1),$$

which is equivalent to $F(Z) \sim \text{Unif}(0, 1)$. $\square$

In simulation, the key statement is the second one: a random variable with cdf F is obtained by generating $U \sim \text{Unif}(0, 1)$ and setting $Z = F^{-1}(U)$.

Example 1.15 (Exponential building blocks and inversion) Let $F(z) = 1 - \exp(-\lambda z)$, $z \geq 0$, be the exponential cdf with rate $\lambda > 0$. Its inverse is

$$F^{-1}(u) = -\frac{\log(1-u)}{\lambda}, \quad u \in (0, 1),$$

so if $U \sim \text{Unif}(0, 1)$ then

$$Z = -\frac{\log(1-U)}{\lambda} \sim \text{Exp}(\lambda).$$

Since $1 - U \stackrel{d}{=} U$, one may equivalently use $Z = -\log(U)/\lambda$. This basic construction is often exploited to generate richer laws. For instance, if $E_1, \dots, E_p$ are independent $\text{Exp}(\lambda)$ random variables and $p \in \mathbb{N}$, then

$$G_p = \sum_{i=1}^p E_i \sim \text{Gamma}(p, \lambda).$$

Likewise, if (E_n) are iid $\text{Exp}(\lambda)$ and

$$N = \max \left\{ n \geq 0 : \sum_{i=1}^n E_i \leq 1 \right\},$$

then $N \sim \text{Poisson}(\lambda)$. Indeed, this N is the unit-time count of the process with iid exponential inter-arrival times, hence of a Poisson process with rate λ . Writing $E_i = -\log(U_i)/\lambda$ with iid $U_i \sim \text{Unif}(0, 1)$ gives the equivalent product representation

$$N = \max \left\{ n \geq 0 : \prod_{i=1}^n U_i > e^{-\lambda} \right\}.$$

At the same time, direct inversion is available for both families. One may generate

$$G_p = F_\Gamma^{-1}(U), \quad N = F_{\text{Pois}}^{-1}(U),$$

where F_Γ and F_{Pois} denote the Gamma and Poisson cdf, respectively. For Gamma this means numerical inversion of the Gamma cdf, while for Poisson it is the smallest $m \in \mathbb{N}_0$ such that

$$\sum_{k=0}^m e^{-\lambda} \frac{\lambda^k}{k!} \geq U.$$

The broader message is that simulation can proceed either by direct inversion of the target cdf or by exploiting probabilistic identities that transform simpler random inputs into the target law. $\diamond$

1.6.2 Acceptance-rejection

We now turn to the second general-purpose method. Let f be a target density and let g be a proposal density that is easy to simulate, with

$$f(z) \leq Cg(z), \quad z \in \mathbb{R},$$

for some finite constant $C \geq 1$. The method repeatedly draws $Y \sim g$ and $U \sim \text{Unif}(0, 1)$, independently, and accepts the proposal whenever

$$U \leq \frac{f(Y)}{Cg(Y)}.$$

If the proposal is rejected, one draws a new pair and repeats until acceptance. The next result guarantees that such a method correctly samples from f .

Proposition 1.4 (Acceptance-rejection) *Let (Y_n, U_n) be iid with $Y_n \sim g$, $U_n \sim \text{Unif}(0, 1)$, and $Y_n \perp U_n$. Define*

$$A_n = \left\{ U_n \leq \frac{f(Y_n)}{Cg(Y_n)} \right\}, \quad T = \inf\{n \geq 1 : A_n \text{ occurs}\}, \quad Z = Y_T.$$

Then Z has density f .

Proof. Set $p = \mathbb{P}(A_1)$. Since U_1 is independent of Y_1 ,

$$p = \mathbb{E}\left[\mathbb{P}\left(U_1 \leq \frac{f(Y_1)}{Cg(Y_1)} \mid Y_1\right)\right] = \mathbb{E}\left[\frac{f(Y_1)}{Cg(Y_1)}\right] = \frac{1}{C} \int f(z) \, dz = \frac{1}{C}.$$

Now let $B \subseteq \mathbb{R}$ be measurable. Then

$$\begin{aligned} \mathbb{P}(Z \in B) &= \sum_{n=1}^{\infty} \mathbb{P}(T = n, Y_n \in B) = \sum_{n=1}^{\infty} \mathbb{P}(A_1^c, \dots, A_{n-1}^c, A_n, Y_n \in B) \\ &= \sum_{n=1}^{\infty} (1-p)^{n-1} \mathbb{P}(A_1, Y_1 \in B), \end{aligned}$$

by independence of the iid trials. Furthermore,

$$\mathbb{P}(A_1, Y_1 \in B) = \int_B g(z) \frac{f(z)}{Cg(z)} dz = \frac{1}{C} \int_B f(z) dz = p \int_B f(z) dz.$$

Hence

$$\mathbb{P}(Z \in B) = \sum_{n=1}^{\infty} (1-p)^{n-1} p \int_B f(z) dz = \int_B f(z) dz,$$

which proves that Z has density f . $\square$

Remark 1.9 (Efficiency) The acceptance probability is

$$\mathbb{P}(A_1) = \mathbb{E} \left[\mathbb{P} \left(U_1 \leq \frac{f(Y_1)}{Cg(Y_1)} \mid Y_1 \right) \right] = \frac{1}{C},$$

so the expected number of proposals per accepted draw equals C . Therefore the method is efficient when C is close to 1, which in turn requires a proposal g that resembles f closely on both body and tail. $\circ$

This following simple construction illustrates the bounded-support case.

Example 1.16 (Beta simulation) Suppose the target is $\text{Beta}(\alpha, \beta)$ on $(0, 1)$ with $\alpha, \beta \geq 1$, so its density

$$f(z) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} z^{\alpha-1} (1-z)^{\beta-1}, \quad 0 < z < 1,$$

is bounded. Let

$$M = \sup_{0 < z < 1} f(z) < \infty, \quad g(z) = 1_{(0,1)}(z),$$

so g is the uniform density on $(0, 1)$ and $f(z) \leq Mg(z)$ on $\mathbb{R}$. Taking $C = M$ gives a valid acceptance-rejection scheme with acceptance probability $1/M$. $\diamond$

On unbounded domains, or when densities are sharply peaked, good proposals are typically chosen by matching concavity and tail behaviour so that C remains moderate. When f itself is also costly to evaluate, the following technique is useful.

Remark 1.10 (Squeezing) When f is expensive to evaluate, one can reduce calls to f by adding simple bounds. Suppose there are simple functions l and u such that

$$0 \leq l(z) \leq \frac{f(z)}{Cg(z)} \leq u(z) \leq 1.$$

Given a proposal Y and a uniform U , accept immediately if $U \leq l(Y)$ and reject immediately if $U > u(Y)$. Only on the intermediate event $l(Y) < U \leq u(Y)$ is the exact ratio $f(Y)/(Cg(Y))$ evaluated. This leaves the target law unchanged while reducing expensive evaluations. $\circ$

1.6.3 Fourier inversion

In some models the distribution function F or density f is not available in explicit form, while a transform is. Let $Z \sim F$ and define the characteristic function

$$\psi(s) = \widehat{F}[is] = \mathbb{E}[e^{isZ}], \quad s \in \mathbb{R}.$$

This permits simulation constructions based directly on Fourier inversion. The first result gives the density representation directly from ψ .

Proposition 1.5 (Fourier inversion) *If $\psi \in L^1(\mathbb{R})$, then Z has a bounded continuous density*

$$f(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \psi(s) e^{-isz} \, ds, \quad z \in \mathbb{R}.$$

Proof. Define f by the displayed integral. Since $\psi \in L^1(\mathbb{R})$, dominated convergence implies that f is bounded and continuous. Moreover, for $a < b$,

$$\int_a^b f(z) \, dz = \lim_{M \rightarrow \infty} \frac{1}{2\pi} \int_{-M}^M \psi(s) \frac{e^{-isa} - e^{-isb}}{is} \, ds.$$

By the Lévy inversion formula, the right-hand side equals $F(b) - F(a)$ at continuity points a, b of F . Hence the finite measure with density f agrees with F on intervals $(a, b]$ and therefore coincides with F . $\square$

The next example gives the practical approximation route for simulation, and shows how the equidistant-grid case leads to an FFT implementation.

Example 1.17 (Grid approximation and FFT implementation) Assume the density f of Z is continuous and supported on $[a, b]$. For any fixed integer $n \geq 2$, we define an equidistant grid of step size $\epsilon = (b - a)/(n - 1)$ with points

$$z_m = a + (m - 1)\epsilon, \quad m = 1, \dots, n.$$

We construct a discrete random variable $\tilde{Z}_n$ taking values in $\{z_1, \dots, z_n\}$ with probabilities proportional to the density:

$$p_m = \mathbb{P}(\tilde{Z}_n = z_m) = \frac{f(z_m)}{\sum_{j=1}^n f(z_j)}.$$

This discrete variable $\tilde{Z}_n$ approximates Z as $n \rightarrow \infty$. Indeed, for every bounded continuous function h on $[a, b]$, Riemann-sum convergence yields

$$\mathbb{E}[h(\tilde{Z}_n)] = \frac{\sum_{m=1}^n h(z_m) f(z_m) \epsilon}{\sum_{m=1}^n f(z_m) \epsilon} \rightarrow \int_a^b h(z) f(z) \, dz = \mathbb{E}[h(Z)],$$

hence $\tilde{Z}_n \xrightarrow{d} Z$. Typically, such an approximation requires the point values $f(z_m)$ to be evaluated via numerical quadrature (e.g. using Proposition 1.5).

However, because the grid is explicitly equidistant, we can significantly accelerate this computation using the Fast Fourier Transform (FFT) applied directly

to the characteristic function $\psi(s) = \mathbb{E}[e^{isZ}]$. Let $w = e^{2\pi i/n}$ and define the discrete frequencies $s_r = 2\pi(r-1)/(n\epsilon)$ for $r = 1, \dots, n$. The Discrete Fourier Transform (DFT) of the probability vector (p_m) is given by

$$\hat{p}_r = \frac{1}{n} \sum_{m=1}^n p_m w^{(r-1)(m-1)} = \frac{e^{-is_r z_1}}{n} \sum_{m=1}^n p_m e^{is_r z_m}.$$

Recognising the sum as a discrete approximation to the characteristic function $\psi(s_r)$, we can bypass the direct computation of $f(z_m)$ by defining proxy Fourier coefficients:

$$\hat{q}_r = \frac{e^{-is_r z_1}}{n} \psi(s_r).$$

Applying the inverse transform to $(\hat{q}_r)$ yields candidate probability masses $\tilde{p}_m$:

$$\tilde{p}_m = \sum_{r=1}^n \hat{q}_r w^{-(r-1)(m-1)}.$$

To correct for minor numerical artifacts and ensure a valid distribution, we take the absolute values and re-normalise:

$$p_m^* = |\tilde{p}_m|, \quad p_m = \frac{p_m^*}{\sum_{j=1}^n p_j^*}.$$

The resulting probability vector (p_m) over the grid (z_m) allows the approximate simulation of Z to be reduced to standard discrete sampling. $\diamond$

The final result gives an exact transform-based acceptance-rejection route, avoiding grid discretisation.

Proposition 1.6 (Transform bounds for acceptance-rejection) *Assume $\psi, \psi', \psi'' \in L^1(\mathbb{R})$. Let f be the density of Z , and define*

$$c = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi(s)| \, ds, \quad k = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi''(s)| \, ds.$$

Then

$$f(z) \leq c \wedge \frac{k}{z^2}, \quad z \neq 0.$$

Consequently, with $b = \sqrt{k/c}$ and $A = 4\sqrt{kc}$, the density

$$g(z) = \frac{1}{A} (c 1_{(|z| \leq b)} + k z^{-2} 1_{(|z| > b)})$$

yields the valid acceptance-rejection envelope $f(z) \leq Ag(z)$.

Proof. From Proposition 1.5,

$$|f(z)| \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi(s)| \, ds = c.$$

Because $\psi, \psi', \psi'' \in L^1$ we have $\psi(s) \rightarrow 0$ and $\psi'(s) \rightarrow 0$ as $|s| \rightarrow \infty$. Hence for $z \neq 0$, integration by parts gives

$$f(z) = \frac{1}{2\pi iz} \int_{-\infty}^{\infty} \psi'(s) e^{-isz} \, ds,$$

and integrating by parts again gives

$$f(z) = -\frac{1}{2\pi z^2} \int_{-\infty}^{\infty} \psi''(s) e^{-isz} \, ds.$$

Hence

$$|f(z)| \leq \frac{1}{2\pi z^2} \int_{-\infty}^{\infty} |\psi''(s)| \, ds = \frac{k}{z^2}.$$

Therefore $f(z) \leq c \wedge kz^{-2}$ for $z \neq 0$, while $f(0) \leq c$. Now,

$$A = \int_{-\infty}^{\infty} (c \wedge kz^{-2}) \, dz = 2 \int_0^b c \, dz + 2 \int_b^{\infty} kz^{-2} \, dz = 2cb + \frac{2k}{b} = 4\sqrt{kc}.$$

So g is a density. Also, on $|z| \leq b$, one has $Ag(z) = c \geq f(z)$; on $|z| > b$, one has $Ag(z) = kz^{-2} \geq f(z)$. Hence $f \leq Ag$. $\square$

Simulation from g is straightforward. With $b = \sqrt{k/c}$, the density g is a mixture of a uniform law on $(-b, b)$ and a symmetric shifted Pareto component with density $(b/2)z^{-2}1_{(|z|>b)}$. A convenient representation for the tail component is Sb/U , where $S \in \{-1, 1\}$ is symmetric and $U \sim \text{Unif}(0, 1)$. The acceptance-rejection step is then to accept Z with probability

$$\frac{f(Z)}{Ag(Z)},$$

where $A = 4\sqrt{kc}$ and $f(Y)$ is evaluated from Proposition 1.5.

1.7 Notes and Comments

The classical limit results and tools on stochastic convergence have long histories that can be traced back to the development of measure-theoretic probability in the early twentieth century. In [Kolmogoroff \[1933\]](#) the modern axiomatic framework was laid out, within which the principal convergence modes are formulated. The Law of Large Numbers and the Central Limit Theorem both have eighteenth-century origins; for the CLT under broad moment conditions, key proofs are due to [Lyapunov \[1901\]](#) and [Lindeberg \[1922\]](#). The technical tools of asymptotic statistics were developed in parallel, with Slutsky's lemma [[Slutsky, 1925](#)], the Continuous Mapping Theorem [[Mann and Wald, 1943](#)], and the Cramér–Wold device [[Cramér and Wold, 1936](#)] all entering the literature in this period, together with the Glivenko–Cantelli theorem on uniform convergence of the empirical distribution function [[Glivenko, 1933](#), [Cantelli, 1933](#)]. Stochastic order notation was formalised in [Mann and Wald \[1943\]](#), and [Cramér \[1946\]](#) gave a systematic asymptotic treatment that helped make this notation and delta-method arguments standard tools. The classical monograph [Feller \[1971\]](#) offers a thorough probabilistic treatment of these results, while modern references that present these results systematically include [van der Vaart \[1998\]](#), [Billingsley \[1999\]](#), [Kallenberg \[2002\]](#). At the finite-dimensional level, standard references for estimator examples and finite-sample behaviour include [Serfling \[1980\]](#), together with sharp nonasymptotic inequalities for empirical distributions such as [Dvoretzky et al. \[1956\]](#), [Massart \[1990\]](#).

Convergence in law for stochastic processes required further structural tools. Tightness and relative compactness criteria were given by [Prokhorov \[1956\]](#), and almost-sure representations were introduced by [Skorokhod \[1956\]](#), forming the early foundations of weak convergence theory in metric spaces. A complementary development was martingale theory, as provided in [Doob \[1953\]](#), where maximal inequalities, optional-sampling arguments, and martingale convergence theorems became standard tools in stochastic-process limit proofs. The invariance principle of [Chung and Fuchs \[1951\]](#), establishing convergence of rescaled partial-sum processes to Brownian motion, marks the early beginning of functional central limit theorems. Various of these results were unified by [Billingsley \[1999\]](#), whose work largely systematised weak convergence of probability measures on function spaces.

Building on the foundational work, the development of empirical process theory is treated systematically in [Dudley \[1984\]](#), [Pollard \[1984\]](#), [Shorack and Wellner \[2009\]](#), [van der Vaart and Wellner \[1996\]](#). The functional delta method, formulated in terms of Hadamard differentiability, connects these weak convergence results to asymptotic expansions for estimators and test statistics [[van der Vaart, 1998](#), [van der Vaart and Wellner, 1996](#)]. The formulation of weak convergence using outer probability and outer expectation, which elegantly deals with measurability issues, follows Hoffmann–Jørgensen's work [[Hoffmann-Jørgensen, 1974](#), [Hoffmann-Jørgensen and Pisier, 1976](#)] and has become standard in the empirical-process literature. The k -nearest neighbours empirical process treatment is taken from [Portier \[2025\]](#) and builds on the same empirical-process framework as [van der Vaart and Wellner \[1996\]](#). The entropy-theorem block, based on bracketing arguments, fits directly into this same empirical-process line of work, as does the section on empirical distributions and quantiles fol-

lows standard weak-convergence treatments from [Shorack and Wellner \[2009\]](#), [van der Vaart and Wellner \[1996\]](#).

Finally, the simulation subsections are classically covered in random-variate monographs such as [Devroye \[1986\]](#), [Hörmann et al. \[2004\]](#), [Ripley \[1987\]](#), [Robert and Casella \[2004\]](#), with the material being inspired by the treatment of [Asmussen and Glynn \[2007\]](#). Inversion and acceptance-rejection are treated there in depth, while acceptance-rejection traces back to [von Neumann \[1951\]](#), and transform-based inversion dates back to the inversion identity of [Gil-Pelaez \[1951\]](#) and practical numerical inversion methods such as [Abate and Whitt \[1995\]](#).

Beyond the provided material, these theoretical results have found systematic application in statistical methodology, as expanded in subsequent chapters; here we only mention a few directions. Survival analysis and event-history modelling rely heavily on martingale limit theorems and counting-process techniques, as developed in [Andersen et al. \[1993\]](#), [Fleming and Harrington \[2005\]](#). Extreme value theory rests on weak convergence arguments for maxima, with standard references in [Resnick \[1987\]](#), [de Haan and Ferreira \[2006\]](#). Dependence modelling via copulas has led to the study of empirical copula processes; foundational copula references include [Sklar \[1959\]](#), [Nelsen \[2006\]](#), [Joe \[2014\]](#), [Durante and Sempi \[2015\]](#). Other specialized developments include treatment of stochastic-process limits in [Whitt \[2002\]](#), on Markov processes in [Ethier and Kurtz \[1986\]](#), and a semimartingale framework in [Jacod and Shiryaev \[2003\]](#). The rigorous treatment of stochastic simulation methods of [Asmussen and Glynn \[2007\]](#) is used throughout the book.

1.8 Exercises

1. By applying Theorem 1.3 to $-\gamma$, show that we also obtain

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n\gamma(Z_i)\right|\geq\kappa t+\sqrt{2t}\right)\leq 2e^{-nt}.$$

Equivalently, show that there exists a constant $C > 0$ such that for all $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n\gamma(Z_i)\right|\geq\varepsilon\right)\leq 2\exp\left(-\frac{Cn\varepsilon^2}{1+\kappa\varepsilon}\right).$$

2. Let (Z_n) be iid with $\mathbb{P}(Z_1 > z) = z^{-\alpha}$ for $z \geq 1$ and $\alpha \in (0, 1)$.

(a) Show that $\mathbb{E}[|Z_1|] = \infty$.

(b) Show that $\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n Z_i = \infty$, and hence Theorem 1.2 fails without the moment assumption.

Hint: Use Borel–Cantelli with the events $\{Z_n > Mn\}$ for any $M > 0$.

3. Let (Z_n) be iid symmetric with $\mathbb{P}(|Z_1| > z) = z^{-\alpha}$ for $z \geq 1$ and $\alpha \in (1, 2)$. Show that the Lindeberg condition for the normalisation $\sqrt{n}$ fails, by proving that for any fixed $\varepsilon > 0$,

$$n \mathbb{E}[Z_1^2; |Z_1| > \varepsilon\sqrt{n}] \rightarrow \infty.$$

Show that $n^{-1/2} \sum_{i=1}^n Z_i$ cannot converge in distribution to a non-degenerate normal law. Conclude that Theorem 1.6 fails when $\mathbb{E}[Z_1^2] = \infty$.

4. Let $Z_n = Z_1$ almost surely for all n , where Z_1 has a continuous distribution function F . Show that $\mathbb{F}^{(n)}(t) = 1_{(Z_1 \leq t)}$ for all n and hence

$$\sup_{t \in \mathbb{R}} |\mathbb{F}^{(n)}(t) - F(t)| = \max\{F(Z_1), 1 - F(Z_1)\} \geq \frac{1}{2} \quad \text{a.s.}$$

Conclude that the Glivenko–Cantelli theorem fails without independence.

5. Explain why the extended Glivenko–Cantelli Theorem 1.9 cannot be proven using a combination of Theorem 1.7 and Lemma 1.5.

6. Let (Z_n) and (W_n) be $\mathbb{R}^p$ -valued random variables. Show that if $\|Z_n - W_n\| = o_{\mathbb{P}}(1)$ and $Z_n \xrightarrow{d} Z$, then also $W_n \xrightarrow{d} Z$.

Hint: Use the definition of convergence in distribution.

7. Go through the details of Example 1.5, and discuss the practical implications of this result.

8. Prove a multidimensional version of (1.4). That is, for $-\infty < t_1 < t_2 < \dots < t_M < \infty$, $M \in \mathbb{N}$, show that

$$\sqrt{n}(\mathbb{F}^{(n)}(t_1) - F(t_1), \dots, \mathbb{F}^{(n)}(t_M) - F(t_M)) \xrightarrow{d} N(0, \Sigma(t_1, \dots, t_M)),$$

where

$$(\Sigma(t_1, \dots, t_M))_{i,j} = \sigma(t_i, t_j) = F(t_i \wedge t_j) - F(t_i)F(t_j).$$

9. Derive the mean excess function for each of the following distributions:

- (a) If Z is a Pareto-distributed random variable with scale parameter $\kappa > 0$ and shape parameter $\alpha > 0$, then $\bar{F}_Z(z) = (\kappa/z)^\alpha$, $z \geq \kappa$. Show that if $\alpha > 1$ then $e_F(u) = u/(\alpha - 1)$, $u \geq \kappa$, where $e_F(u) = \mathbb{E}[Z - u | Z > u]$ is the mean excess function.
- (b) If Z is log-normally distributed with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, then the distribution function of Z is given by

$$G_Z(z) = \Phi\left(\frac{\log z - \mu}{\sigma}\right), \quad z > 0,$$

where Φ denotes the distribution function for a standard normal random variable. Straightforward calculations yield that

$$e_G(u) = e^{\mu + \frac{\sigma^2}{2}} \frac{\Phi\left(\frac{(\mu + \sigma^2) - \log u}{\sigma}\right)}{\Phi\left(\frac{\mu - \log u}{\sigma}\right)} - u.$$

Show that as $u \rightarrow \infty$,

$$e_G(u) = \frac{\sigma^2 u}{\log u - (\mu + \sigma^2)} (1 + o(1)),$$

where $o(1)$ is understood as $u \rightarrow \infty$.

Hint: Recall Mills' ratio, which states that for $z \rightarrow -\infty$ we have that $\Phi(z) \sim -\phi(z)/z$, where ϕ denotes the density of a standard normal distribution.

- 10. Generalise Proposition 1.1 to independent but not identically distributed risks, using a uniformly bounded condition as in Example 1.6.
- 11. In Proposition 1.1 we saw that the empirical variance is asymptotically Gaussian. While empirical variance is commonly used for measuring the spread of a distribution, it is also common to use the empirical skewness to measure the asymmetry of a distribution.

$$\hat{\xi}_n = \frac{\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}_n)^3}{\left(\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}_n)^2\right)^{\frac{3}{2}}}, \quad \bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i,$$

which targets the standardised central third moment

$$\xi := \frac{\mathbb{E}[(Z - \mathbb{E}[Z])^3]}{(\mathbb{E}[(Z - \mathbb{E}[Z])^2])^{\frac{3}{2}}}.$$

Let (Z_n) be iid $\mathbb{R}$ -valued random variables with finite sixth moment.

- (a) Show that $\hat{\xi}_n \xrightarrow{\mathbb{P}} \xi$.
- (b) State and prove a central limit theorem for $\sqrt{n}(\hat{\xi}_n - \xi)$.

12. Show that $\int_{(0,1]} \sqrt{\log(2/\varepsilon^2 + 1)} \, d\varepsilon < \infty$.

13. Prove Lemma 1.9 by applying Lemma 1.8.

14. Go through the details of the proof of Theorem 1.22 in the univariate case, and provide the details in the bivariate case.
15. In Chapter 1 we proved that the empirical process (scaled difference) of the cumulative hazard is asymptotically Gaussian for each t with $F(t) < 1$. The goal of this exercise is to prove a functional (pathwise) CLT for the empirical process of the cumulative hazard. A useful route is to first identify the functional map and its derivative, and then apply the functional δ -method.

(a) Consider the map ϕ , given by

$$\phi(h)(t) = -\log(1 - h(t)),$$

such that $h(t) < 1$ for all t . Show that the Hadamard derivative of ϕ at a function f in the direction of h is given by

$$\phi'_f(h)(t) = \frac{h(t)}{1 - f(t)}.$$

Hint: For small x , $-\log(1 - x)$ is approximated by the identity function.

- (b) State and prove a functional limit theorem of $\sqrt{n}(\mathbb{A}^{(n)} - \Lambda)$ and compute the corresponding covariance function.
16. State and prove a functional CLT for the mean excess function.
17. Go through the details for obtaining the derivatives of Theorem 1.24.
18. Prove the following:

Lemma 1.14 (Duhamel equation) *Let $B_1 = \phi(A_1)$ and $B_2 = \phi(A_2)$ be two product integrals. They satisfy the Duhamel equation*

$$B_2(s, t] - B_1(s, t] = \int_{(s, t]} B_1(s, u) B_2(u, t] (A_2 - A_1) \, du.$$

19. Prove the following:

Lemma 1.15 (Integration by parts for the Duhamel equation) *Suppose that $A, B \in D(0, \infty)$ and that $h \in D(0, \infty)$ is of bounded variation. Then it holds that*

$$\begin{aligned} & \int_{(0, t]} \prod (1 + dB)(0, u) \prod (1 + dA)(u, t] \, dh(u) \\ &= h(t) + \int_{(0, t]} \prod (1 + dB)(0, s) [h(t) - h(s)] \, dB(s) \\ & \quad + \int_{(0, t]} h(r-) \prod (1 + dA)(r, t] \, dA(r) \\ & \quad + \int_{(0, t]} \int_{(s, t]} \prod (1 + dB)(0, s) [h(r-) - h(s)] \prod (1 + dA)(r, t] \, dA(r) \, dB(s). \end{aligned}$$

20. Let (Z_n) be iid with characteristic function $\varphi(t) = \mathbb{E}[e^{itZ_1}]$ and define the empirical characteristic function

$$\varphi_n(t) = \frac{1}{n} \sum_{i=1}^n e^{itZ_i}, \quad t \in [-T, T].$$

- (a) Show that the class $\{e^{it\cdot} : t \in [-T, T]\}$ is Donsker and conclude that $\sqrt{n}(\varphi_n - \varphi) \xrightarrow{d} \mathbb{G}$ in $\ell^\infty([-T, T])$ for a centred Gaussian process with covariance

$$\text{Cov}(\mathbb{G}(s), \mathbb{G}(t)) = \varphi(s - t) - \varphi(s)\varphi(-t).$$

- (b) For fixed t_0 , use the δ -method (viewing $\varphi_n(t_0)$ as a complex-valued random element) to derive the asymptotic distribution of $|\varphi_n(t_0)|$.

21. Let $w: [0, 1] \rightarrow \mathbb{R}$ be Lipschitz and define the L -statistic functional

$$T(F) = \int_0^1 w(u) F^{\leftarrow}(u) \, du.$$

Assume F is continuous with strictly positive density on an interval containing the support of w . Show that T is Hadamard differentiable at F and derive the asymptotic distribution of $T(\mathbb{F}^{(n)})$ by the functional δ -method.

22. Let H be a bivariate distribution function with continuous marginals F and G , and copula C .

- (a) Show that if a and b are strictly increasing, then the copula of $(a(X), b(Y))$ equals C .
 (b) Assume that H has a density h with marginal densities f and g . Show that the copula density exists and is given by

$$c(u, v) = \frac{h(F^{\leftarrow}(u), G^{\leftarrow}(v))}{f(F^{\leftarrow}(u))g(G^{\leftarrow}(v))}, \quad (u, v) \in (0, 1)^2.$$

23. Establish consistency of the Lorenz curve and Gini coefficient. Comment on the truncated versus untruncated case.

24. Let Z be a discrete random variable with values $x_1, \dots, x_N$ and probabilities $p_1, \dots, p_N$, and define $c_j = \sum_{i=1}^j p_i$ with $c_0 = 0$ and $c_N = 1$. Let F be the distribution function of Z and F^{-1} its generalised inverse.

Use Proposition 1.3 to show that if $U \sim \text{Unif}(0, 1)$ and $\tilde{Z} = F^{-1}(U)$, then

$$\tilde{Z} = x_j \quad \Leftrightarrow \quad c_{j-1} < U \leq c_j,$$

and therefore

$$\mathbb{P}(\tilde{Z} = x_j) = c_j - c_{j-1} = p_j, \quad j = 1, \dots, N.$$

Conclude that the standard discrete simulation algorithm (partitioning $(0, 1)$ into intervals of lengths p_j and mapping each interval to x_j) is exactly the inversion method applied to a discrete target law.

25. Prove that the squeezing variant of acceptance-rejection method of Remark 1.10 produces a random variable with density f .

Practical exercises

1. Verify through simulation the variance of the approximations in Theorems 1.1 and 1.11.
 - (a) Let (Z_n) be a sequence of iid random variables where $Z_n \sim N(0, 1)$ for all n . Verify through simulation the asymptotic distribution $\sqrt{n}(\hat{\sigma}_n^2 - \mu^{(2)})$, where $\mu^{(2)} = \mathbb{E}[(Z_1 - \mathbb{E}[Z_1])^2]$.
 - (b) Let (Z_n) be a sequence of iid random variables where Z_n is Pareto distributed with shape parameter $\alpha = 3$ and scale parameter $\kappa = 100$ for all n . Verify through simulation the asymptotic distribution for $u = 200$ of $\sqrt{n}(\hat{e}_n(u) - e_F(u))$.
Hint: In R you can simulate Pareto distributed variables using the function `rPareto()` from the R package `Pareto`.
2. In the R package `CASdatasets` Dutang and Charpentier [2024] one may find the dataset `frecomfire`, within which there is a column called `ClaimCost2007`, a vector of claim costs in thousand euros from commercial French fire insurance, normalised to 2007. In this exercise we assume that the data in `ClaimCost2007` are iid and examine the variance, mean excess, and cumulative hazard.
 - (a) Plot a histogram of the column `ClaimCost2007`,
 - (b) Construct a 95% confidence interval for the variance using Theorem 1.1,
 - (c) Construct a 95% confidence interval for the mean excess function in $u = 5000$, using Theorem 1.11,
 - (d) Construct a 95% confidence interval for the cumulative hazard in $t = 2000$, using Theorem 1.12.
3. Consider the following simulation studies:
 - (a) For $Z \sim \text{Exp}(\lambda)$, $\lambda > 0$ we have $F_Z(z) = 1 - e^{-\lambda z}$. In R simulate $n \in \{10, 100, 1000\}$ iid $\text{Exp}(1/2)$ -variables, construct and plot $\mathbb{F}^{(n)}(z)$ for $z \in (0, 10]$ with 95% confidence bounds using Theorem 1.22.
 - (b) Denote $D_n = \sup_x |\mathbb{F}^{(n)}(x) - F(x)|$. State the limit distribution for $\sqrt{n}D_n$. With the same setup as in (a), simulate 100 realisations of $\sqrt{n}D_n$ for each n . Plot the histograms and compare with the limit distribution of $\sqrt{n}D_n$ either through the exact limit density or through simulations; for the latter, the function `rbridge()` from the R package `e1071` can be useful.
Hint: The supremum of a continuously time-changed Brownian bridge is the same as a non-time-changed one.
4. If $Y \sim N(\mu, \Sigma)$ is an r -dimensional multivariate random vector then its corresponding copula is the so-called *Gaussian copula*. The marginal distributions can be standardised through strictly increasing transformations to obtain a new vector $X \sim N(0, \rho_\Sigma)$, where $\rho_\Sigma = \rho$ is the correlation

matrix of Y . Since copulas are invariant under strictly increasing transformations it follows that Y and X have the same copula. It is given by

$$\begin{aligned} C_\rho^{\text{Ga}} &= \mathbb{P}(\Phi(X_1) \leq u_1, \dots, \Phi(X_r) \leq u_r) \\ &= \Phi_\rho(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_r)), \end{aligned}$$

where Φ_ρ is the cdf of an r -dimensional zero-mean Gaussian vector with covariance matrix ρ . We consider $r = 2$ and X a centred Gaussian vector with $\sigma_1^2 = \sigma_2^2 = 1$ and $\sigma_{12} = 0.7$.

- (a) Simulate $n = 1000$ observations from $N(0, \rho)$ and set $U_i = \Phi(X_{i1})$, $V_i = \Phi(X_{i2})$. Construct the empirical copula

$$\mathbb{C}^{(n)}(u, v) = \frac{1}{n} \sum_{i=1}^n 1_{(U_i \leq u, V_i \leq v)}$$

on a grid in $(0, 1)^2$ and compare it visually with the true Gaussian copula $C(u, v) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v))$. (Either numerically evaluate or approximate C by Monte Carlo simulation using a very large sample).

- (b) To investigate the scaled differences $\sqrt{n}(\mathbb{C}^{(n)} - C)$, simulate 1000 realisations of sample size $n = 1000$ and then compute $\sup_{(u,v) \in [0,1]^2} |\sqrt{n}(\mathbb{C}^{(n)}(u, v) - C(u, v))|$ for each simulation. Plot a histogram of these values. Compare with $n = 100, 000$.
5. Implement simulation of $\text{Gamma}(p, \lambda)$ (with integer p) and $\text{Poisson}(\lambda)$ by two conceptually different methods. Use the distributional identities from the chapter to construct $\text{Gamma}(p, \lambda)$ as a sum of p iid exponentials with rate λ , and to construct $\text{Poisson}(\lambda)$ from exponential inter-arrival times (or equivalently from the product representation involving uniforms). Then implement direct inversion, i.e. $F_\Gamma^{-1}(U)$ and $F_{\text{Pois}}^{-1}(U)$ with $U \sim \text{Unif}(0, 1)$. Using a large sample, compare the two routes through empirical means and variances against the theoretical values, compare computation times, and discuss briefly which method is preferable in your implementation and why.
6. Consider the three distribution functions

$$F_P(z) = 1 - z^{-1/\gamma}, \quad z \geq 1, \quad F_{Fr}(z) = \exp(-z^{-1/\gamma}), \quad z > 0,$$

and

$$F_B(z) = 1 - (1 + (z/\theta)^\eta)^{-\beta}, \quad z \geq 0, \quad \theta, \eta, \beta > 0,$$

which are the Pareto, Fréchet and Burr parameterisations used in Chapter 3. Derive their inverse distribution functions and implement inversion-based simulation for all three laws from a uniform sample.

7. Consider the standard normal target law, for which $\psi(s) = \exp(-s^2/2)$. Use Example 1.17 and Proposition 1.6 to compare Fourier simulation by FFT approximation and by exact acceptance-rejection.

- (a) Compute explicitly

$$c = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi(s)| \, ds, \quad k = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\psi''(s)| \, ds,$$

and deduce $b = \sqrt{k/c}$ and $A = 4\sqrt{kc}$ for the acceptance-rejection envelope.

- (b) Implement the FFT-based method of Example 1.17 on a fixed equidistant grid.
- (c) Implement the exact transform-based acceptance-rejection method from Proposition 1.6.
- (d) Using large samples, compare approximation quality (moments, selected quantiles, cdf values), computational cost, and, for acceptance-rejection, empirical acceptance probability.
- (e) For the FFT method, study sensitivity with respect to truncation window and mesh size.

Conclude which method is preferable in your implementation for a given accuracy target.

Chapter 2

Smoothing covariates

In this chapter, we introduce nonparametric regression estimators. They are a cornerstone of statistical methods for investigating the distribution of a risk variable Z and its dependence on covariates X . However, kernel methods work best when the data, including the covariates, are not sparse. Other nonparametric methods from machine learning are better suited for high-dimensional data, but their statistical properties are generally less transparent than those of kernel methods.

Kernel nonparametric methods are valuable in insurance, as they are well suited for estimating skewed, multimodal, or heteroscedastic claim distributions and regression functions, which can be crucial for accurate risk assessment and premium pricing, particularly for medium (or attritional) claims. One important limitation of the methods in this chapter is the accurate modelling of large claims, i.e. heavy-tailed distributions, which is treated in the next chapter. Other actuarial applications of kernel regression include smoothing irregular development patterns or risk classification. We first develop the basic kernel estimators and their asymptotics, then turn to distributional regression, discuss practical bandwidth selection, and finally study k-nearest neighbour distributional regression.

2.1 Kernel estimators

Despite their drawbacks in higher dimensions, we first define the estimators for covariates in $\mathbb{R}^k$; for simplicity we provide proofs for the $k = 1$ case, which is also where kernel methods are most applicable. We refer the interested reader to Section 1.4 in [Ghosh \[2018\]](#) and [Stute \[1986\]](#) for general treatments. We begin by defining the kernel and the two core estimators that are analysed throughout the chapter.

Consider a kernel, defined as a bounded function $K: \mathbb{R}^k \rightarrow \mathbb{R}_+ \cup \{0\}$ with

bounded support¹ and satisfying

$$\int K(u) \, du = 1, \quad \int u_i K(u) \, du = 0,$$

for all $i \in \{1, \dots, k\}$. The first condition guarantees that the kernel is a density function, and the second is a balance property. We assume that

$$\int u_i^2 K(u) \, du > 0,$$

which guarantees that there are no dimensions where the kernel is degenerate. Since the kernel is bounded and of bounded support, we also have

$$\int |u_i|^m K(u) \, du < \infty, \quad m \in \mathbb{N}.$$

Henceforth we assume the kernel is symmetric around the origin, so the first-moment conditions are automatically satisfied.

Definition 2.1 (Parzen–Rosenblatt density estimator) Let (X_n) be a sequence of iid random vectors in $\mathbb{R}^k$ with density f . Let $K: \mathbb{R}^k \rightarrow \mathbb{R}$ be a k -dimensional kernel. Let b_n be a sequence of $k \times k$ non-singular matrices with non-negative entries. The *Parzen–Rosenblatt density estimator* of f is given by

$$\mathbb{f}^{(n)}(x) = \frac{1}{n|b_n|} \sum_{i=1}^n K(b_n^{-1}(x - X_i)),$$

where $|b_n|$ is the determinant of b_n .

The next definition builds directly on this construction. The estimator $\mathfrak{m}^{(n)}$ targets the product $m(x) = e(x)f(x)$, while $\mathfrak{e}^{(n)}$ is obtained by normalising with $\mathbb{f}^{(n)}$.

Definition 2.2 (Priestley–Chao and Nadaraya–Watson estimators) Let (Y_n, X_n) be a sequence of iid random variables in $\mathbb{R} \times \mathbb{R}^k$ with density f for X_1 and conditional expectation $e(x) = \mathbb{E}[Y_1 | X_1 = x]$. Let $K: \mathbb{R}^k \rightarrow \mathbb{R}$ be a k -dimensional kernel. Let b_n be a sequence of $k \times k$ non-singular matrices with non-negative entries. The *Priestley–Chao estimator* of $m(x) = e(x)f(x)$ is given by

$$\mathfrak{m}^{(n)}(x) = \frac{1}{n|b_n|} \sum_{i=1}^n Y_i K(b_n^{-1}(x - X_i)).$$

Similarly, the *Nadaraya–Watson regression estimator* of $e(x)$ is given by

$$\mathfrak{e}^{(n)}(x) = \frac{\mathfrak{m}^{(n)}(x)}{\mathbb{f}^{(n)}(x)} = \sum_{i=1}^n \frac{Y_i K(b_n^{-1}(x - X_i))}{\sum_{i=1}^n K(b_n^{-1}(x - X_i))}.$$

Diagonal bandwidth matrices correspond to variable-wise smoothing (often with a common scalar bandwidth), whereas non-diagonal b_n allow for rotations

¹Kernels of unbounded support are common in the literature; here we stick with the bounded support assumption for mathematical convenience.

and rescalings of the covariate space but are less common in practice. As k grows, the amount of data falling in a local neighbourhood shrinks rapidly, which is why we focus on the univariate case for proofs and applications.

For simplicity, we assume for the remainder of the chapter that the conditional distribution of Y_1 has the first three moments, that is $m^{(j)}(x) = \mathbb{E}[|Y_1|^j | X_1 = x]f(x) < \infty$ for $j = 1, 2, 3$, and we let $x \mapsto m^{(j)}(x)$ and $f(x)$ be C^2 . When necessary, we also assume existence of the joint density. A common choice sets b_n to be diagonal with identical entries h_n , and uses product kernels. In this case, the Parzen–Rosenblatt estimator reduces to

$$\mathbb{f}^{(n)}(x) = \frac{1}{nh_n^k} \sum_{i=1}^n \prod_{j=1}^k K_0\left(\frac{X_{ij} - x_j}{h_n}\right)$$

where X_{ij} is the j –th entry of the vector X_i . The Priestley–Chao and Nadaraya–Watson estimators collapse similarly. We now concentrate on the case $k = 1$.

In practice, Gaussian kernels are a common choice (despite not satisfying the bounded support assumption). Uniform kernels are another simple option, given by

$$K_0(u) = \begin{cases} \frac{1}{2} & u \in [-1, 1] \\ 0 & \text{otherwise.} \end{cases}$$

2.1.1 Density estimator

For coherence, we redefine the estimator for univariate covariates. We write b_n for matrix bandwidths and h_n (or simply h) for scalar bandwidths. Let K be a univariate kernel function. For univariate X , the Parzen–Rosenblatt density estimator is given as

$$\mathbb{f}^{(n)}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right). \quad (2.1)$$

Theorem 2.1 (Weak consistency of the Parzen–Rosenblatt density estimator)
 Let $h = h_n$ be a sequence of bandwidths with $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$. Then

$$\mathbb{f}^{(n)}(x) \xrightarrow{\mathbb{P}} f(x).$$

Proof. We expand the bias by Taylor’s theorem and show the variance vanishes under the bandwidth conditions. Notice that by the change of variables $u = (x - y)/h$ and Taylor expansion of f around x , the bias of the estimator is given by

$$\begin{aligned} \mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x) &= \int \frac{1}{h} K((x - y)/h) \{f(y) - f(x)\} dy \\ &= \int K(u) \{f(x - hu) - f(x)\} du \\ &= \int K(u) \{f'(x)(-hu) + \frac{1}{2}f''(x)(hu)^2 + o(h^2)\} du \\ &= \frac{h^2}{2} f''(x) \int u^2 K(u) du + o(h^2), \end{aligned}$$

which vanishes when $h \rightarrow 0$. Notice that under the integral sign the remainder is $o(u^2 h^2)$, and by boundedness of the kernel, $\int u^2 K(u) du < \infty$. A similar calculation for the variance yields

$$\begin{aligned} \text{Var}(\mathbb{f}^{(n)}(x)) &= \frac{1}{nh} \int K^2(u) f(x - hu) du - \frac{1}{n} \left(\int K(u) f(x - hu) du \right)^2 \\ &= \frac{f(x)}{nh} \int K^2(u) du + O(1/n) - O(1/n), \end{aligned} \quad (2.2)$$

which vanishes as $nh \rightarrow \infty$ and $h \rightarrow 0$. We now apply Chebyshev's inequality (1.1) with the second moment to conclude that $\mathbb{f}^{(n)}(x) - f(x) \xrightarrow{\mathbb{P}} 0$. $\square$

Notice the above theorem holds even for $f(x) = 0$. Keeping the bandwidth fixed, however, removes consistency; the next example makes this explicit.

Example 2.1 (Inconsistency if the bandwidth does not shrink) Assume $h_n \equiv h_0 > 0$ is fixed. Then

$$\mathbb{E}[\mathbb{f}^{(n)}(x)] = \int \frac{1}{h_0} K\left(\frac{x-y}{h_0}\right) f(y) dy = (K_{h_0} * f)(x), \quad K_{h_0}(u) = h_0^{-1} K(u/h_0).$$

Moreover $\text{Var}(\mathbb{f}^{(n)}(x)) = O(n^{-1})$, so $\mathbb{f}^{(n)}(x) \xrightarrow{\mathbb{P}} (K_{h_0} * f)(x)$. Unless f is a fixed point of this convolution, the estimator is inconsistent. For instance, if f is the standard normal density and K is the uniform kernel on $[-1, 1]$, then

$$(K_{h_0} * f)(0) = \frac{1}{2h_0} \int_{-h_0}^{h_0} f(u) du < f(0)$$

since f is strictly concave at 0. $\diamond$

Theorem 2.2 (Weak convergence of the kernel density estimator) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Then*

$$\sqrt{nh}(\mathbb{f}^{(n)}(x) - f(x)) \xrightarrow{d} N\left(0, f(x) \int K^2(u) du\right).$$

Proof. We split the error into a stochastic term and a bias term, and apply a CLT to the former. Consider the decomposition

$$\begin{aligned} \sqrt{nh}(\mathbb{f}^{(n)}(x) - f(x)) &= \sqrt{nh}(\mathbb{f}^{(n)}(x) - \mathbb{E}[\mathbb{f}^{(n)}(x)]) + \sqrt{nh}(\mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x)) \\ &=: A_n + B_n. \end{aligned}$$

Recall that $\mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x) = O(h^2)$. Thus B_n is a bias term of order $O(\sqrt{nh^5})$, which vanishes by assumption.

The remaining term A_n can be handled using Lindeberg's CLT for arrays, Theorem 1.7 via Lyapounov's condition as per Lemma 1.4. Namely, we need to verify that the row sums of the variances of the array

$$Z_{ni} = (nh)^{-1/2} \{K(h^{-1}(x - X_i)) - \mathbb{E}[K(h^{-1}(x - X_i))]\}$$

converge, and that the row sums of the absolute central moments of order $\alpha > 2$ vanish. We now make a computation similar to (2.2) to obtain

$$\begin{aligned} \sum_{i=1}^n \sigma_{n,i}^2 &= \sum_{i=1}^n \text{Var}(Z_{ni}) = \int K^2(u) f(x - hu) \, du - h \left(\int K(u) f(x - hu) \, du \right)^2 \\ &= f(x) \int K^2(u) \, du + O(h) - O(h), \end{aligned}$$

which converges to the desired variance. Now note that for the nonnegative random variable $\tilde{Z}_{ni} = (nh)^{-1/2} K(h^{-1}(x - X_i))$, by Minkowski's inequality, positivity, and the fact that L^p norms are increasing in p , we have

$$\mathbb{E}[|Z_{ni}|^3]^{1/3} \leq \mathbb{E}[\tilde{Z}_{ni}^3]^{1/3} + |\mathbb{E}[\tilde{Z}_{ni}]| = \mathbb{E}[\tilde{Z}_{ni}^3]^{1/3} + \mathbb{E}[|\tilde{Z}_{ni}|] \leq 2\mathbb{E}[\tilde{Z}_{ni}^3]^{1/3}.$$

Thus, the sum of third absolute central moments gives

$$\sum_{i=1}^n \mathbb{E}[|Z_{ni}|^3] \leq 2^3 \sum_{i=1}^n \mathbb{E}[\tilde{Z}_{ni}^3] = 8f(x) \frac{1}{\sqrt{nh}} \int K^3(u) \, du + O(h/\sqrt{nh}),$$

which vanishes as $nh \rightarrow \infty$ and $h \rightarrow 0$. This concludes the proof. $\square$

Since we have calculated the bias in the above proof, we may actually examine an asymptotic mean integrated square error (AMISE) calculation, which makes the bias–variance tradeoff explicit and suggests a concrete bandwidth selection rule.

Example 2.2 (AMISE and optimal bandwidth) Assume f is twice continuously differentiable and K is symmetric with $\mu_2(K) = \int u^2 K(u) \, du$ and $R(K) = \int K^2(u) \, du$. A Taylor expansion as above gives the pointwise bias

$$\mathbb{E}[\mathbb{f}^{(n)}(x)] - f(x) = \frac{h^2}{2} \mu_2(K) f''(x) + o(h^2),$$

while Theorem 2.2 yields the variance term $\text{Var}(\mathbb{f}^{(n)}(x)) = \frac{1}{nh} f(x) R(K) + o((nh)^{-1})$. Hence the mean squared error satisfies

$$\text{MSE}(x) = \frac{h^4}{4} \mu_2(K)^2 f''(x)^2 + \frac{1}{nh} f(x) R(K) + o(h^4) + o((nh)^{-1}).$$

Integrating over x gives the asymptotic mean integrated squared error

$$\text{AMISE}(h) = \frac{h^4}{4} \mu_2(K)^2 \int (f''(x))^2 \, dx + \frac{1}{nh} R(K).$$

Minimising yields the optimal bandwidth

$$h_{\text{AMISE}} = \left(\frac{R(K)}{\mu_2(K)^2 \int (f''(x))^2 \, dx} \right)^{1/5} n^{-1/5}.$$

A natural issue with this procedure is that we require an estimate of f'' to compute the optimal bandwidth, which itself is aimed at estimating f . This circular difficulty is in practice solved by inputting some parametric estimate for f'' . $\diamond$

Remark 2.1 Kernel density smoothing on the positive line encounters finite-sample boundary issues, since symmetric kernels place mass outside the support. When estimating near zero, such kernels extend into the negative region, where no data exist, leading to underestimation. Similar issues arise near the edges of compact supports. These effects are negligible asymptotically, but important in finite samples. $\circ$

Example 2.3 (Boundary correction by reflection) Suppose X has support on $[0, \infty)$ and f is C^2 in a neighbourhood of 0. For a symmetric kernel K with bounded support, define the reflected estimator for $x \geq 0$ by

$$\mathbb{f}_R^{(n)}(x) = \frac{1}{nh} \sum_{i=1}^n \left\{ K\left(\frac{x - X_i}{h}\right) + K\left(\frac{x + X_i}{h}\right) \right\}.$$

Let $f_{\text{ref}}(y) = f(|y|)$ denote the even reflection of f . Then a change of variables shows

$$\mathbb{E}[\mathbb{f}_R^{(n)}(x)] = \int K(u) f_{\text{ref}}(x - hu) \, du,$$

while the uncorrected estimator $\mathbb{f}^{(n)}(x)$ satisfies, at $x = 0$,

$$\mathbb{E}[\mathbb{f}^{(n)}(0)] = \int_{-\infty}^0 K(u) f(-hu) \, du = \frac{1}{2}f(0) + hf'(0) \int_0^{\infty} uK(u) \, du + o(h),$$

where $f'(0)$ denotes the right derivative. Thus the boundary bias of the uncorrected estimator equals $-\frac{1}{2}f(0) + O(h)$. For $x = 0$, using the symmetry of K , we have $\mathbb{E}[\mathbb{f}_R^{(n)}(0)] = 2 \int_0^{\infty} K(u) f(hu) \, du$. A Taylor expansion yields

$$\mathbb{E}[\mathbb{f}_R^{(n)}(0)] = f(0) + 2hf'(0) \int_0^{\infty} uK(u) \, du + o(h).$$

Thus the reflected estimator removes the leading boundary loss of order $O(1)$. Generally, a bias of order $O(h)$ remains, involving the first moment of the kernel on the half-line. However, if $f'(0) = 0$ (e.g. if the density has a shoulder at the boundary), the linear term vanishes and the bias becomes of order h^2 . $\diamond$

Numeric example 2.1 (Boundary reflection on a compact support) To illustrate the boundary effect and its correction, consider iid observations (X_n) with density f from the Beta(1, 0.8) density. We take $n = 1000$ and estimate f with the uniform kernel. The reflection idea from the previous example is applied at both boundaries. This yields a reflected estimator of the same form as $\mathbb{f}_R^{(n)}$, now correcting, to some degree, the boundary problems at both 0 and 1.

The resulting comparison is shown in Figure 2.1, where the correction is clearly visible. This specific density has the feature that it diverges to infinity at the right endpoint of the support, which unsurprisingly is not fully corrected by the reflected estimator. $\diamond$

2.1.2 Nonparametric regression

The Nadaraya–Watson estimator is a fundamental tool in nonparametric statistics for estimating the conditional expectation of a dependent risk variable Y

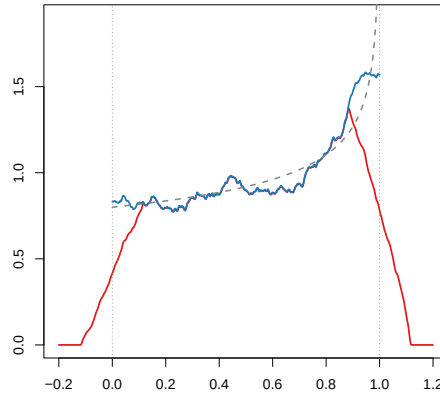


Figure 2.1: Boundary correction by reflection for Beta(1, 0.8) data on $[0, 1]$, using a uniform kernel. Red solid: uncorrected estimator $\mathbb{f}^{(n)}$; blue solid: reflected estimator; grey dashed: true density f . Vertical grey dashed lines delimit the support, and the vertical range is truncated since the true density is unbounded at 1.

given a covariate X . It can be viewed as a regression model that makes no parametric assumptions on the conditional distribution, which offers flexibility for modelling nonlinear relationships. It is therefore often used as a visual or exploratory tool to assess whether parametric models fit adequately. We now study its asymptotics in detail, and notice that for $k = 1$ the estimator is given by

$$\mathfrak{e}^{(n)}(x) = \sum_{i=1}^n \frac{Y_i K(h_n^{-1}(x - X_i))}{\sum_{i=1}^n K(h_n^{-1}(x - X_i))}.$$

Theorem 2.3 (Consistency of the Nadaraya–Watson estimator) *Let $h = h_n$ be a sequence of bandwidths with $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$. Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\mathfrak{e}^{(n)}(x) \xrightarrow{\mathbb{P}} \mathbb{E}[Y | X = x].$$

If in addition, $nh_n/\log n \rightarrow \infty$ and for some $\delta \in (0, 1]$

$$\sum_{n=1}^{\infty} \left(\frac{\log n}{nh_n} \right)^{\frac{1+\delta}{2}} < \infty,$$

then the convergence strengthens to

$$\mathfrak{e}^{(n)}(x) \xrightarrow{a.s.} \mathbb{E}[Y | X = x].$$

Proof. Recall that $\mathfrak{e}^{(n)}(x) = \mathfrak{m}^{(n)}(x)/\mathbb{f}^{(n)}(x)$ where $\mathfrak{m}^{(n)}(x)$ is the Priestley–Chao estimator. The calculations below only use $m(x) = \mathbb{E}[Y | X = x]f(x)$ and $m^{(2)}(x) = \mathbb{E}[Y^2 | X = x]f(x)$, so the argument applies equally to discrete responses; we keep the density notation for brevity. The latter estimator has

bias given by

$$\begin{aligned}
& \mathbb{E}[\mathfrak{m}^{(n)}(x)] - m(x) \\
&= \int \int K(u) \{yf(y, x - hu) - yf(y, x)\} \, du \, dy \\
&= \int \int K(u) y \left\{ \frac{\partial}{\partial x} f(y, x)(-hu) + \frac{1}{2} \frac{\partial^2}{\partial x^2} f(y, x)(hu)^2 + o(h^2) \right\} \, du \, dy \\
&= \frac{h^2}{2} m''(x) \int u^2 K(u) \, du + o(h^2),
\end{aligned}$$

which goes to zero as $h \rightarrow 0$. The variance is given by

$$\begin{aligned}
\text{Var}(\mathfrak{m}^{(n)}(x)) &= \frac{1}{nh} \int \int y^2 K^2(u) f(y, x - hu) \, du \, dy \\
&\quad - \frac{1}{n} \left(\int \int y K(u) f(y, x - hu) \, du \, dy \right)^2 \\
&= \frac{\mathbb{E}[Y^2 | X = x] f(x)}{nh} \int K^2(u) \, du + O(1/n) - O(1/n),
\end{aligned}$$

which vanishes as $nh \rightarrow \infty$. Thus we may apply Chebyshev's inequality with the second moment to conclude that $\mathfrak{m}^{(n)}(x) \xrightarrow{P} m(x) = e(x)f(x)$. Division with the consistent estimator $\mathfrak{f}^{(n)}(x)$ of $f(x) > 0$ now finishes the proof of consistency by Theorem 1.4.

For almost sure convergence, let $M_n := \sqrt{nh/\log n}$ and decompose the estimator into a truncated part and a remainder:

$$Y_i = Y_i^{(n)} + R_i^{(n)}, \quad Y_i^{(n)} := Y_i 1_{(|Y_i| \leq M_n)}, \quad R_i^{(n)} := Y_i 1_{(|Y_i| > M_n)}.$$

Accordingly, let $\mathfrak{m}^{(n)} = \mathfrak{m}_T^{(n)} + \mathfrak{m}_R^{(n)}$. Since $|Y_i^{(n)}| \leq M_n$ and K is bounded with compact support, the terms in $\mathfrak{m}_T^{(n)}$ are bounded. Bernstein's inequality, Theorem 1.3, implies that for each fixed $\varepsilon > 0$, there exists a constant $C > 0$ such that

$$\mathbb{P}\left(|\mathfrak{m}_T^{(n)}(x) - \mathbb{E}[\mathfrak{m}_T^{(n)}(x)]| > \varepsilon\right) \leq 2 \exp \left\{ -\frac{C\varepsilon^2 nh}{1 + \varepsilon M_n} \right\}.$$

Since $M_n = \sqrt{nh/\log n} \rightarrow \infty$, the exponent is dominated by $-C'\sqrt{nh \log n}$, which ensures that the series $\sum_{n=1}^{\infty} \mathbb{P}(|\mathfrak{m}_T^{(n)}(x) - \mathbb{E}[\mathfrak{m}_T^{(n)}(x)]| > \varepsilon)$ is summable. By the Borel–Cantelli Lemma, $\mathfrak{m}_T^{(n)}(x) - \mathbb{E}[\mathfrak{m}_T^{(n)}(x)] \xrightarrow{a.s.} 0$.

Now, for the remainder term, we use the bound $|y| 1_{(|y| > b)} \leq |y|^{2+\delta} b^{-(1+\delta)}$. Under the assumed $(2 + \delta)$ -moment bounds, we have

$$\mathbb{E}[|\mathfrak{m}_R^{(n)}(x)|] \leq C M_n^{-(1+\delta)} = C \left(\frac{\log n}{nh} \right)^{\frac{1+\delta}{2}}.$$

By Markov's inequality and the assumed growth rate of nh relative to $\log n$, we also obtain $\mathfrak{m}_R^{(n)}(x) \xrightarrow{a.s.} 0$ by summability. Finally, since $\mathbb{E}[\mathfrak{m}^{(n)}(x)] \rightarrow m(x)$ and $\mathbb{E}[\mathfrak{m}_R^{(n)}(x)] \rightarrow 0$, it follows that $\mathbb{E}[\mathfrak{m}_T^{(n)}(x)] \rightarrow m(x)$. Combining these results, we obtain $\mathfrak{m}^{(n)}(x) \xrightarrow{a.s.} m(x)$. The same argument for $\mathfrak{f}^{(n)}(x)$ yields $\mathfrak{f}^{(n)}(x) \xrightarrow{a.s.} f(x)$, and the continuous mapping Theorem 1.4 completes the proof. $\square$

Theorem 2.4 (Weak convergence of the Nadaraya–Watson estimator) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\sqrt{nh}(\mathfrak{e}^{(n)}(x) - \mathbb{E}[Y | X = x]) \xrightarrow{d} N\left(0, \frac{\sigma(x)^2}{f(x)} \int K^2(u) \, du\right),$$

where $\sigma(x)^2 = \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2$.

Proof. First consider the following decomposition:

$$\begin{aligned} \sqrt{nh_n}(\mathfrak{e}^{(n)}(x) - \mathbb{E}[Y | X = x]) &= \sqrt{nh_n}(\mathfrak{e}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)]/\mathbb{E}[\mathfrak{f}^{(n)}(x)]) \\ &\quad + \sqrt{nh_n}(\mathbb{E}[\mathfrak{m}^{(n)}(x)]/\mathbb{E}[\mathfrak{f}^{(n)}(x)] - m(x)/f(x)) \\ &=: A_n + B_n. \end{aligned}$$

Recall that both $\mathbb{E}[\mathfrak{f}^{(n)}(x)] - f(x) = O(h^2)$ and $\mathbb{E}[\mathfrak{m}^{(n)}(x)] - m(x) = O(h^2)$. Thus the B_n term is $O(\sqrt{nh^5})$, which vanishes by assumption.

We can decompose the A_n term further as

$$\begin{aligned} A_n &= \frac{1}{\mathbb{E}[\mathfrak{f}^{(n)}(x)]\mathfrak{f}^{(n)}(x)} \sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) \mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathbb{E}[\mathfrak{m}^{(n)}(x)] \mathfrak{f}^{(n)}(x) \} \\ &= \frac{1}{\mathbb{E}[\mathfrak{f}^{(n)}(x)]\mathfrak{f}^{(n)}(x)} \sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) (\mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathfrak{f}^{(n)}(x)) \} \\ &\quad + \frac{1}{\mathbb{E}[\mathfrak{f}^{(n)}(x)]\mathfrak{f}^{(n)}(x)} \sqrt{nh_n} \{ (\mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)]) \mathfrak{f}^{(n)}(x) \} \\ &= \frac{m(x)}{f^2(x)} \sqrt{nh_n} \{ \mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathfrak{f}^{(n)}(x) \} + \frac{1}{f(x)} \sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)] \} \\ &\quad + o_{\mathbb{P}}(\sqrt{nh_n} \{ \mathbb{E}[\mathfrak{f}^{(n)}(x)] - \mathfrak{f}^{(n)}(x) \}) + o_{\mathbb{P}}(\sqrt{nh_n} \{ \mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)] \}). \end{aligned}$$

Thus it suffices to verify that $\mathfrak{f}^{(n)}(x)$ and $\mathfrak{m}^{(n)}(x)$ are asymptotically *jointly* normal, so we can evaluate the sum of the two leading terms of A_n , which by Exercise 6 of Chapter 1 establishes the weak convergence of A_n and hence the theorem.

We handle this joint convergence using Lindeberg's CLT for arrays, Theorem 1.7, via Lyapounov's condition in Lemma 1.4. We show that the array

$$Z_{ni} = (nh)^{-1/2} \{ K(h^{-1}(x - X_i)) \{ \lambda_1 + \lambda_2 Y_i \} - \mathbb{E}[K(h^{-1}(x - X_i)) \{ \lambda_1 + \lambda_2 Y_i \}] \}$$

for arbitrary constants λ_1, λ_2 satisfies Lyapounov's condition. We first check the variance convergence in Theorem 1.7 by computing

$$\begin{aligned} \sum_{i=1}^n \sigma_{n,i}^2 &= \sum_{i=1}^n \text{Var}(Z_{ni}) = \int \int (\lambda_1 + \lambda_2 y)^2 K^2(u) f(y, x - hu) \, du \, dy \\ &\quad - h \left(\int \int (\lambda_1 + \lambda_2 y) K(u) f(y, x - hu) \, du \, dy \right)^2 \\ &= f(x) \mathbb{E}[(\lambda_1 + \lambda_2 Y)^2 | X = x] \int K^2(u) \, du + O(h^2) - O(h). \end{aligned}$$

Similarly, Lyapounov's condition with $\alpha = 3$ follows by defining $\tilde{Z}_{ni} = (nh)^{-1/2}K(h^{-1}(x - X_i))\{\lambda_1 + \lambda_2 Y\}$ and noting that

$$\begin{aligned} \sum_{i=1}^n \mathbb{E}[|Z_{ni}|^3] &\leq 2^3 \sum_{i=1}^n \mathbb{E}[|\tilde{Z}_{ni}|^3] \\ &= 8 \frac{f(x) \mathbb{E}[|\lambda_1 + \lambda_2 Y|^3 | X = x]}{\sqrt{nh}} \int K^3(u) \, du + O(h/\sqrt{nh}), \end{aligned}$$

which vanishes as $nh \rightarrow \infty$ and $h \rightarrow 0$. Thus, by the Cramér–Wold device, joint asymptotic normality holds.

Returning to A_n , choose $\lambda_1 = -\mathbb{E}[Y | X = x]/f(x)$ and $\lambda_2 = 1/f(x)$ to obtain convergence to a normal random variable with mean zero and variance given by

$$\begin{aligned} &f(x) \mathbb{E}[(\lambda_1 + \lambda_2 Y)^2 | X = x] \int K^2(u) \, du \\ &= \frac{\mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2}{f(x)} \int K^2(u) \, du, \end{aligned}$$

concluding the proof. $\square$

As a corollary, we consider local linear regression, a popular alternative for estimating the regression function $e(x) = \mathbb{E}[Y | X = x]$. The approach improves upon kernel smoothing by fitting a straight line, rather than just a constant, to the data within a local neighbourhood of a target point x .

This is accomplished by solving a weighted least squares problem for coefficients $(\hat{\beta}_0, \hat{\beta}_1)$ that minimise the sum of squared residuals, with weights determined by a kernel function K and bandwidth h :

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1(X_i - x))^2 h^{-1} K(h^{-1}(X_i - x))$$

The resulting local linear estimator, $\mathfrak{r}^{(n)}(x)$, is the intercept of this locally fitted line, $\hat{\beta}_0$. (Actually, generalised versions exist in the literature where higher order terms are subtracted). The following theorem describes its asymptotic properties.

Theorem 2.5 (Asymptotic distribution of the local linear regression estimator) *Let $h = h_n$ be a sequence of bandwidths with $h_n \rightarrow 0$, $nh_n \rightarrow \infty$, and $nh_n^5 \rightarrow 0$. Then for fixed x such that $f(x) > 0$, the Local Linear Regression estimator $\mathfrak{r}^{(n)}(x)$ satisfies:*

$$\sqrt{nh}(\mathfrak{r}^{(n)}(x) - \mathbb{E}[Y | X = x]) \xrightarrow{d} N\left(0, \frac{\sigma(x)^2}{f(x)} \int K^2(u) \, du\right),$$

where $\sigma(x)^2 = \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2$.

Proof. The estimator $\mathfrak{r}^{(n)}(x)$ admits the explicit form

$$\mathfrak{r}^{(n)}(x) = \frac{T_{n,0}S_{n,2} - T_{n,1}S_{n,1}}{S_{n,0}S_{n,2} - S_{n,1}^2},$$

where $S_{n,j} := \sum_{i=1}^n h^{-1} K(h^{-1}(X_i - x))(X_i - x)^j$ and $T_{n,j} := \sum_{i=1}^n h^{-1} K(h^{-1}(X_i - x))(X_i - x)^j Y_i$. Notice that $\mathfrak{e}^{(n)}(x) = T_{n,0}/S_{n,0}$. The difference is

$$\mathfrak{r}^{(n)}(x) - \mathfrak{e}^{(n)}(x) = \frac{S_{n,1}(T_{n,0}S_{n,1} - T_{n,1}S_{n,0})}{S_{n,0}(S_{n,0}S_{n,2} - S_{n,1}^2)}.$$

Each individual term in both numerator and denominator can be expanded into bias and stochastic terms. Lengthy but straightforward calculations yield a total rate of $O(h^2) + O_{\mathbb{P}}(\sqrt{h/n})$. Thus, $\sqrt{nh}(\mathfrak{r}^{(n)}(x) - \mathfrak{e}^{(n)}(x)) \xrightarrow{\mathbb{P}} 0$, and the estimators are asymptotically equivalent. By Theorem 2.4, the result follows. $\square$

The consistency of several kernel regression estimators follows directly from the univariate case. For asymptotic normality, a bit more work is required. In this case we have two sequences (Y_n) and (Z_n) of dependent variables and (X_n) of covariates, and we specialise notation to

$$m_Y(x) = \mathbb{E}[Y | X = x]f(x), \quad m_Z(x) = \mathbb{E}[Z | X = x]f(x)$$

which should satisfy the same assumptions as $m(x)$. We may also denote the associated Nadaraya–Watson estimators by $\mathfrak{e}_Y^{(n)}$ and $\mathfrak{e}_Z^{(n)}$. It is also clear how to generalise the following theorem beyond two estimators.

Theorem 2.6 (Weak convergence of two Nadaraya–Watson estimators) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let Z satisfy the same conditions as Y . Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\sqrt{nh}(\mathfrak{e}_Y^{(n)}(x) - \mathbb{E}[Y | X = x], \mathfrak{e}_Z^{(n)}(x) - \mathbb{E}[Z | X = x]) \xrightarrow{d} N\left(0, \frac{\int K^2(u) du}{f(x)} \Sigma(x)\right),$$

where $\Sigma(x)$ is the conditional covariance matrix:

$$\begin{aligned} \sigma_1^2(x) &= \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2 \\ \sigma_2^2(x) &= \mathbb{E}[Z^2 | X = x] - \mathbb{E}[Z | X = x]^2 \\ \sigma_{12}(x) &= \mathbb{E}[ZY | X = x] - \mathbb{E}[Y | X = x]\mathbb{E}[Z | X = x]. \end{aligned}$$

Proof. Joint normality can be established as in Theorem 2.4, now considering the array

$$\begin{aligned} Z_{ni} &= (nh)^{-1/2} K(h^{-1}(x - X_i))\{\lambda_1 + \lambda_2 Y_i + \lambda_3 Z_i\} \\ &\quad - (nh)^{-1/2} \mathbb{E}[K(h^{-1}(x - X_i))\{\lambda_1 + \lambda_2 Y_i + \lambda_3 Z_i\}] \end{aligned}$$

and invoking the Cramér–Wold device once again. It remains to compute the covariance. Using the same method as for the individual variances, for $A_i = (nh)^{-1/2} K(h^{-1}(x - X_i))\{\alpha_1 + \alpha_2 Y_i\}$ and $B_i = (nh)^{-1/2} K(h^{-1}(x - X_i))\{\beta_1 + \beta_2 Z_i\}$ we obtain

$$\sum_{i=1}^n \text{Cov}(A_i, B_i) = f(x) \mathbb{E}[\{\alpha_1 + \alpha_2 Y\}\{\beta_1 + \beta_2 Z\} | X = x] \int K^2(u) du + O(h^2).$$

It then remains to plug in $\alpha_1 = -\mathbb{E}[Y | X = x]/f(x)$, $\beta_1 = -\mathbb{E}[Z | X = x]/f(x)$ and $\alpha_2 = \beta_2 = 1/f(x)$ and the result follows. $\square$

For finite-dimensional weak convergence in the covariate argument, it is also natural to compare the same Nadaraya–Watson estimator at different target covariate points. This is an easy task, as when the target points are distinct, the kernel windows eventually do not overlap, and the asymptotic fluctuations become independent. The next theorem makes this precise for two fixed covariate values.

Theorem 2.7 (Weak convergence at two distinct covariate points) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let $x_1 \neq x_2$ be such that the density of X satisfies $f(x_1), f(x_2) > 0$. Then*

$$(\sqrt{nh}(\mathfrak{e}^{(n)}(x_1) - \mathbb{E}[Y | X = x_1]), \sqrt{nh}(\mathfrak{e}^{(n)}(x_2) - \mathbb{E}[Y | X = x_2])) \xrightarrow{d} (G_1, G_2),$$

where G_1 and G_2 are independent and

$$G_j \sim N\left(0, \frac{\sigma(x_j)^2}{f(x_j)} \int K^2(u) \, du\right), \quad j = 1, 2,$$

with $\sigma(x)^2 = \mathbb{E}[Y^2 | X = x] - \mathbb{E}[Y | X = x]^2$.

Proof. By the same decomposition as in the proof of Theorem 2.4, for each $j \in \{1, 2\}$ we have

$$\sqrt{nh}(\mathfrak{e}^{(n)}(x_j) - \mathbb{E}[Y | X = x_j]) = W_n(x_j) + o_{\mathbb{P}}(1),$$

where

$$W_n(x_j) = \frac{1}{f(x_j)\sqrt{nh}} \sum_{i=1}^n \left(K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j]) - \mathbb{E}[K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j])] \right).$$

Hence, by Slutsky's theorem, it suffices to establish joint weak convergence of $(W_n(x_1), W_n(x_2))$. The Cramér–Wold and Lindeberg arguments are the same as in Theorem 2.6, so it only remains to identify the limiting covariance structure.

The marginal variances are exactly those in Theorem 2.4. Thus it suffices to show that the cross-covariance tends to zero. Let

$$\begin{aligned} \xi_{n,i}^{(j)} &= (nh)^{-1/2} \frac{1}{f(x_j)} \left(K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j]) \right. \\ &\quad \left. - \mathbb{E}[K(h^{-1}(x_j - X_i))(Y_i - \mathbb{E}[Y | X = x_j])] \right), \quad j = 1, 2. \end{aligned}$$

Then $\text{Cov}(W_n(x_1), W_n(x_2)) = \sum_{i=1}^n \text{Cov}(\xi_{n,i}^{(1)}, \xi_{n,i}^{(2)})$. Let $M > 0$ be such that $K(u) = 0$ for $|u| > M$. Since $x_1 \neq x_2$ and $h \rightarrow 0$, for all sufficiently large n we have $2Mh < |x_1 - x_2|$, and therefore

$$K(h^{-1}(x_1 - z))K(h^{-1}(x_2 - z)) = 0, \quad z \in \mathbb{R}.$$

Hence the product term in $\mathbb{E}[\xi_{n,i}^{(1)}\xi_{n,i}^{(2)}]$ is zero for large n . The remaining centring term is negligible because, as in the proof of Theorem 2.4,

$$\begin{aligned} &\mathbb{E}[K(h^{-1}(x_j - X))(Y - \mathbb{E}[Y | X = x_j])] \\ &= h(\mathbb{E}[\mathfrak{m}^{(n)}(x_j)] - \mathbb{E}[Y | X = x_j]\mathbb{E}[\mathfrak{f}^{(n)}(x_j)]) = O(h^3), \quad j = 1, 2. \end{aligned}$$

Therefore $\text{Cov}(W_n(x_1), W_n(x_2)) = O(h^5) \rightarrow 0$, and consequently we get that $(W_n(x_1), W_n(x_2)) \xrightarrow{d} (G_1, G_2)$. $\square$

2.2 Nonparametric distributional regression

Distributional regression, unlike traditional regression which predicts the mean of the dependent variable given the covariates, models the entire conditional distribution. This provides a fuller picture of the distribution and serves not only as a predictive model but also as a goodness-of-fit tool for checking whether conditional distributions satisfy a prescribed parametric structure. The starting point is the conditional empirical distribution function, whose asymptotics we establish first.

Definition 2.3 (Conditional empirical distribution function) The conditional empirical distribution function (cdf) is given for $t \in [0, \infty)$, $x \in \mathbb{R}$ by

$$\mathbb{F}^{(n)}(t | x) = \sum_{i=1}^n \frac{1_{(Y_i \leq t)} K(h^{-1}(x - X_i))}{\sum_{i=1}^n K(h^{-1}(x - X_i))}.$$

Theorem 2.8 (Pathwise weak convergence of the cdf) *Let $h = h_n$ be a sequence of bandwidths with $nh_n^5 \rightarrow 0$ and $nh_n \rightarrow \infty$. Let x be such that the density of X satisfies $f(x) > 0$. Then*

$$\sqrt{nh}(\mathbb{F}^{(n)}(\cdot | x) - F(\cdot | x)) \xrightarrow{d} \left(\frac{\int K^2(u) du}{f(x)} \right)^{1/2} \mathbb{B} \circ F(\cdot | x)$$

in $\ell^\infty([0, \infty))$, where $\mathbb{B}$ is a standard Brownian bridge.

Proof. We establish finite-dimensional convergence to identify the covariance structure, and then apply a bracketing CLT to control the process in t . For a fixed t , we have a Nadaraya–Watson estimator. Thus, finite-dimensional convergence follows as Theorem 2.6 (for binary responses). Next, we apply the Bracketing CLT for changing functions, Theorem 1.20. However, we know from the proof of Theorem 2.4 that $(nh)^{-1} \sum_{i=1}^n 1_{(Y_i \leq t)} K(h^{-1}(x - X_i))$ and $(nh)^{-1} \sum_{i=1}^n K(h^{-1}(x - X_i))$ are jointly weakly convergent, and this can be extended to finite-dimensional distributions too. Consequently, since the latter term is not a process in t , by Theorem 1.8 it suffices to establish pathwise weak convergence of the first process $\tilde{X}(t) = (nh)^{-1} \sum_{i=1}^n 1_{(Y_i \leq t)} K(h^{-1}(x - X_i))$.

Define

$$\begin{aligned} \mathcal{F}_n &= \{f_{n,t}: [0, \infty) \times \mathbb{R} \rightarrow [0, \infty), \\ &\text{s.t. } f_{n,t}(y, z) = h^{-1/2} 1_{[0,t)}(y) K(h^{-1}(x - z)) \text{ for } t \in [0, \infty)\}. \end{aligned}$$

Clearly the following random variable is an envelope

$$F_n(Y_1, X_1) = h^{-1/2} K(h^{-1}(x - X_1)).$$

It satisfies by Taylor expansion (as in the proof of Theorem 2.2)

$$\mathbb{E}[F_n(Y_1, X_1)^2] = f(x) \int K^2(u) du + O(h) = O(1),$$

showing condition (1.6). We also have

$$\begin{aligned} \mathbb{E}[F_n(Y_1, X_1)^2 1_{(F_n(Y_1, X_1) > c\sqrt{nh})}] &\leq \frac{1}{c\sqrt{nh}} \mathbb{E}[F_n(Y_1, X_1)^3] \\ &= f(x) \frac{1}{c\sqrt{nh}} \int K^3(u) du + O(h/\sqrt{nh}), \end{aligned}$$

which vanishes as $nh \rightarrow \infty$, showing (1.7). The bracketing number is calculated as in Theorem 1.22, but with slight adaptation. We provide the details.

Consider for $j = 1, \dots, k$ brackets on the form

$$l_j(y, z) = h^{-1/2} 1_{[0, t_{j-1}]}(y) K(h^{-1}(x-z)), u_j(y, z) = h^{-1/2} 1_{[0, t_j)}(y) K(h^{-1}(x-z))$$

for sequences on the form $0 = t_0 < t_1 < \dots < t_k = \infty$. Let $\mathcal{F}_j^n = \{f \in \mathcal{F}_n : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F}_n = \bigcup_{j=1}^k \mathcal{F}_j^n$. Let $\varepsilon > 0$ be given, and pick a sequence for $k \leq \lceil 4/\varepsilon^2 \rceil$ such that

$$F(t_j - |x) - F(t_{j-1} |x) \leq \varepsilon^2/2 \quad \text{for } 1 \leq j \leq k,$$

which is possible because F is a cumulative distribution function, in particular, it is càdlàg, non-decreasing and bounded by 1. Then this sequence covers all of $\mathcal{F}_n$ with brackets satisfying

$$\begin{aligned} & \|u_j(Y_1, X_1) - l_j(Y_1, X_1)\|_{L^2} \\ &= (F(t_j - |x) - F(t_{j-1} |x))^{1/2} \left\{ f(x) \int K^2(u) \, du + O(h) \right\}^{1/2} \\ &= (F(t_j - |x) - F(t_{j-1} |x))^{1/2} \|F_n(Y_1, X_1)\|_{L^2} (1 + o(1)) \\ &\leq \varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \end{aligned}$$

Consequently $N_{[\cdot]}(\varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})) \leq k \leq \lceil 2/\varepsilon^2 \rceil$. We then obtain

$$\int_{(0,1)} \sqrt{\log N_{[\cdot]}(\varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P}))} \, d\varepsilon \leq \int_{(0,1)} \sqrt{\log(4/\varepsilon^2 + 1)} \, d\varepsilon < \infty,$$

which shows (1.9). Finally, since the partitions were chosen independently of n , (1.8) does not need to be verified. The bias term is handled as in finite dimensions and the proof is done. $\square$

2.3 Bandwidth selection

All the above models crucially depend on an adequate choice of bandwidth. There exist many approaches in the literature for selecting an appropriate bandwidth, most of them minimising certain criterion functions, as in the AMISE example at the beginning of the chapter, or in a data-driven way. This section anticipates one of the fundamental concepts used in machine learning to provide a bandwidth selection strategy based on predictive performance. A broader and more detailed discussion pertains to later chapters. We first illustrate the approach for density estimation, and then note how it extends to conditional mean and distribution estimation.

The bandwidth selection strategy that we present is the *leave-one-out cross-validation* technique and is applicable in similar ways for density, conditional mean (local regression), and conditional distribution estimation.

2.3.1 Density estimator

For density estimators we evaluate the quality of the estimator $\mathbb{f}^{(n)}$ in terms of a specific risk measure called the *mean integrated squared error*, defined as

$$R(h) = \mathbb{E}[L(h)] = \mathbb{E} \int (\mathbb{f}^{(n)}(x) - f(x))^2 dx.$$

However, the true f is not available, which appears not only inside the square term, but also through integration. Note that we may rewrite

$$L(h) = \int \mathbb{f}^{(n)}(x)^2 dx - 2 \int \mathbb{f}^{(n)}(x)f(x) dx + \int f^2(x) dx,$$

and the last term does not depend on the bandwidth. Thus, we are interested in minimising the expected value of

$$\begin{aligned} J(h) &= \int \mathbb{f}^{(n)}(x)^2 dx - 2 \int \mathbb{f}^{(n)}(x)f(x) dx \\ &= \int \mathbb{f}^{(n)}(x)^2 dx - 2\mathbb{E}\mathbb{f}^{(n)}(X). \end{aligned}$$

The expectation in the second term could in principle be approximated by an empirical average of $\mathbb{f}^{(n)}$ at each observed X_i :

$$\frac{1}{n} \sum_{i=1}^n \mathbb{f}^{(n)}(X_i).$$

The problem, however, is that the maximisation of such quantity would tend to drastically overfit the data, since it is optimal to take a very spiky $\mathbb{f}^{(n)}$ at each datapoint. The fundamental technique to overcome this issue is to fit the estimator with all the data except X_i , call it $\mathbb{f}_{-i}^{(n)}$, so that it does not use the value X_i , and then the empirical average can reflect a true loss.

Definition 2.4 (Cross-validation risk estimator and bandwidth, for the density estimator) We define the cross-validation risk estimator as

$$\hat{J}(h) = \int \mathbb{f}^{(n)}(x)^2 dx - \frac{2}{n} \sum_{i=1}^n \mathbb{f}_{-i}^{(n)}(X_i).$$

The minimiser of the above function provides the cross-validation bandwidth, that is

$$\hat{h} = \operatorname{argmin}_h \hat{J}(h).$$

The latter bandwidth selection strategy, along with several other alternatives and refinements are usually implemented in standard software packages².

²For instance via the R function `density()`.

2.3.2 Local regression

The risk in the local regression case is the *mean squared error*, given by

$$R(h) = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\mathfrak{e}^{(n)}(X_i) - e(X_i))^2 \right],$$

where we once again encounter a problem of a term being unknown, in this case, the $e(\cdot)$ function. Similarly, the natural guess for estimating the mean squared error is simply

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \mathfrak{e}^{(n)}(X_i))^2,$$

which turns out to be a poor estimator and drastically overfits (or under-smooths) the data for the same reason as in the density. The correct approach is to use cross-validation now with the estimator fitted without the i -th pair (Y_i, X_i) , denoted by $\mathfrak{e}_{-i}^{(n)}$, as the function to test Y_i against.

Definition 2.5 (Cross-validation risk estimator and bandwidth, for the regression estimator) We define the cross-validation risk estimator as

$$\hat{R}(h) = \frac{1}{n} \sum_{i=1}^n (Y_i - \mathfrak{e}_{-i}^{(n)}(X_i))^2.$$

The minimiser of the above function provides the cross-validation bandwidth, that is

$$\hat{h} = \underset{h}{\operatorname{argmin}} \hat{R}(h).$$

A good rule of thumb is to *always* plot $\hat{R}(h)$ to verify that the minimum is well-behaved. Other approaches may be found for instance in Chapter 5 of [Wasserman \[2006\]](#).

2.3.3 Distributional regression

In the case of distributional regression, things can be a bit trickier to define. Any bandwidth criterion must aggregate over t , for instance by integrating over t or by focusing on a set of quantiles of interest. A naive choice can over-emphasise the central part of the distribution, so tail-sensitive goals may require reweighting. We would like to minimise the following risk function

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (\mathbb{F}^{(n)}(t | X_i) - F(t | X_i))^2 \right]$$

for all t at the same time. Several approaches come directly to mind:

$$\begin{aligned} R_1(h) &= \frac{1}{n} \sum_{i=1}^n \int (\mathbb{F}^{(n)}(t | X_i) - F(t | X_i))^2 dt \\ R_2(h) &= \frac{1}{n} \sum_{i=1}^n \int (\mathbb{F}^{(n)}(t | X_i) - F(t | X_i))^2 dF(t | X_i). \end{aligned}$$

However – and quite surprisingly – this *does not work*. Several workarounds are possible, though not entirely straightforward. The issue is that the asymptotic behaviour of the leave-one-out cross-validation in this case is equivalent to not leaving any observation out (cf. Altman and Léger [1995]), in stark contrast to density estimation and even local regression where this asymptotic equivalence does not hold. A simple practical solution which works reasonably well is to find the optimal $\hat{h}$ for density estimation of X as per Definition 2.4 and plug it into $\mathbb{F}^{(n)}(t | X_i)$.

Numeric example 2.2 (Conditional distributional regression) One convenient aspect of distributional regression is that, apart from a global bandwidth selection, no model choice is required. For instance, consider risks Y with conditional Weibull distribution given a covariate X , with constant (unit) scale parameter and shape parameter specified by $1/X$. Let X be uniformly distributed, which in practice can (and should) be attained by transforming the covariates X through the probability integral transform (PIT), that is by applying the empirical distribution function $\mathbb{F}_X^{(n)}(\cdot)$ of the data (X_n) to the data itself. We simulate $n = 5000$ joint observations. By applying the naïve plug-in strategy to find $\hat{h}$ delineated above, we get a satisfactory performance as shown in Figure 2.2, where we have also provided two sets of confidence bounds constructed by: a) not considering the pathwise properties (separately for each t); and b) considering the pathwise properties and using the result of Theorem 2.8 and the fact that the distribution of

$$\text{KS} = \sup_t |\mathbb{B}(t)|$$

is the well-known Kolmogorov–Smirnov distribution, which can explicitly be computed through

$$\mathbb{P}(\text{KS} \leq x) = 1 - 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 x^2} = \frac{\sqrt{2\pi}}{x} \sum_{k=1}^{\infty} e^{-(2k-1)^2 \pi^2 / (8x^2)}.$$

The difference between confidence bands at their thickest is not negligible. Notice also the simplicity of the approach, which automatically captures the nonlinear dependence structure of the vectors (Y, X) , which can serve as a particularly useful tool to verify the adequacy of other models. $\diamond$

2.4 k -nearest neighbours distributional regression

We now parallel the kernel-based distributional-regression analysis with a k -nearest-neighbour localisation. The strategy is the same as before, where we aim to establish pathwise weak convergence for the conditional cdf estimator, then obtain the conditional quantile process by the functional δ -method.

Definition 2.6 (k -nearest-neighbour conditional empirical distribution function) For $t \in [0, \infty)$ and fixed $x \in \mathbb{R}$, define

$$\mathbb{F}_{\kappa}^{(n)}(t | x) = \frac{1}{\kappa} \sum_{i \in \mathcal{K}_{n,\kappa}(x)} 1_{(Y_i \leq t)},$$

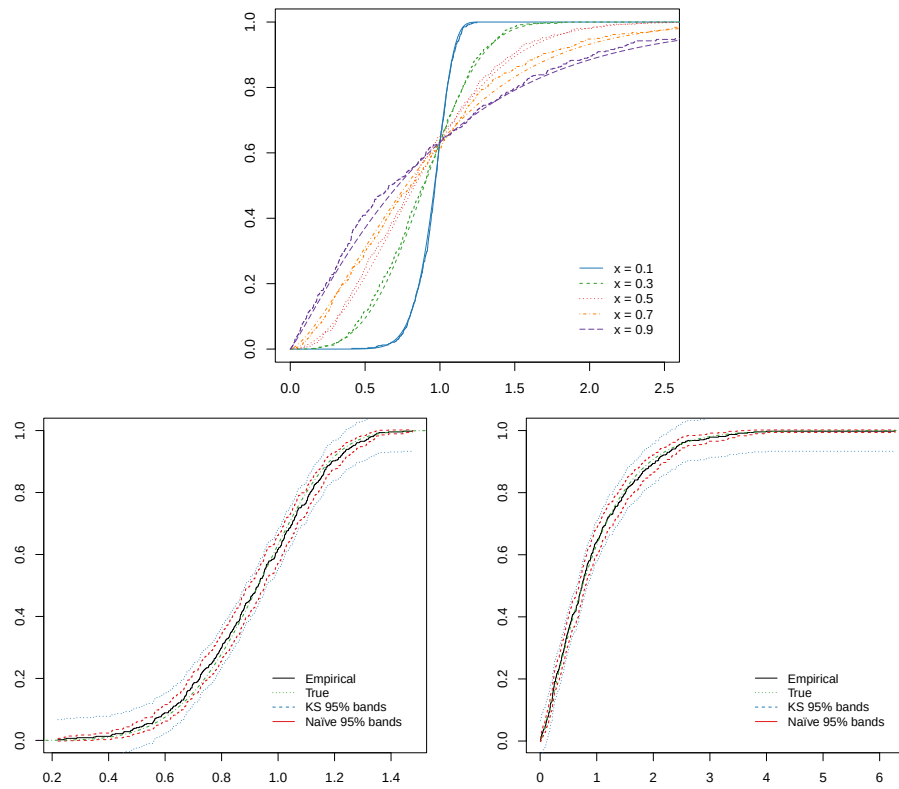


Figure 2.2: Nonparametric distributional regression: conditional distribution functions $y \mapsto F(y|x)$ for various x -values (top), and Kolmogorov–Smirnov and naïve confidence bands for $x = 0.2$ and $x = 0.8$ (bottom).

where $\mathcal{K}_{n,\kappa}(x)$ denotes the indices of the κ nearest covariates X_i to x .

The next result mirrors Theorem 2.8 and is obtained by verifying the conditions of Theorem 1.21 and Corollary 1.2 for the indicator class.

Theorem 2.9 (Pathwise weak convergence of the k -NN cdf) *Let x be such that $f(x) > 0$, with f continuous at x . Let $\kappa = \kappa_n$ satisfy $\kappa \rightarrow \infty$ and $\kappa^{3/2}/n \rightarrow 0$. Assume:*

- (i) *For every $t \in \mathbb{R}$, the map $z \mapsto F(t | z)$ is continuous at x .*
- (ii) *There exist constants $\tau_x, L_x > 0$ such that*

$$\sup_{t \in \mathbb{R}} |F(t | z) - F(t | x)| \leq L_x \|z - x\|, \quad z \in B(x, \tau_x).$$

Then

$$\sqrt{\kappa}(\mathbb{F}_{\kappa}^{(n)}(\cdot | x) - F(\cdot | x)) \xrightarrow{d} \mathbb{B} \circ F(\cdot | x)$$

in $\ell^\infty([0, \infty))$, where $\mathbb{B}$ is a standard Brownian bridge.

Proof. Set

$$\mathcal{F} = \{f_t : t \in \mathbb{R}, f_t(y) = 1_{(-\infty, t]}(y)\}.$$

Let $F \equiv 1$ be the envelope of $\mathcal{F}$. We verify (1.10)–(1.13) in Theorem 1.21. First, for fixed Q on $\mathbb{R}$ and $\varepsilon \in (0, 1)$, the same bracketing construction as in the proof of Theorem 1.22 (class of half-line indicators) gives

$$N_{[]}(\varepsilon \|F\|_{L^2(Q)}, \mathcal{F}, L^2(Q)) \leq 2/\varepsilon^2 + 1.$$

The bound is distribution-free, hence valid for all $Q = \mu_{x,u}$ with $|u| \leq \delta$. Hence, for every $\delta_n \rightarrow 0$,

$$\sup_{|u| \leq \delta} J(\mathcal{F}, L^2(\mu_{x,u}), \delta_n) \leq \int_0^{\delta_n} \sqrt{\log(2/\varepsilon^2 + 1)} \, d\varepsilon \rightarrow 0,$$

which is exactly (1.10). Second, the local Lindeberg condition is immediate, since

$$\sup_{|u| \leq \delta} \mu_{x,u}(F^2 1_{(F > M)}) = 0, \quad M > 1.$$

Thus (1.11) holds. Third, we verify (1.12). Fix $\eta > 0$. For each $u \in [0, \delta]$, by the same bracketing bound there exists a finite partition $-\infty = t_0^{(u)} < t_1^{(u)} < \dots < t_{m_u}^{(u)} = \infty$ with

$$\mu_{x,u}((t_{j-1}^{(u)}, t_j^{(u)}]) \leq \eta^2/4, \quad j = 1, \dots, m_u.$$

For fixed $(a, b]$, the map $v \mapsto \mu_{x,v}((a, b])$ is continuous on $[0, \delta]$; therefore each partition above remains bounded by $\eta^2/2$ on some neighbourhood U_u of u . By

compactness, choose $u_1, \dots, u_M$ with $[0, \delta] \subset \bigcup_{m=1}^M U_{u_m}$ and merge all corresponding partition endpoints into one grid $-\infty = s_0 < s_1 < \dots < s_K = \infty$. Then for every $v \in [0, \delta]$ and every $r = 1, \dots, K$,

$$\mu_{x,v}((s_{r-1}, s_r]) \leq \eta^2/2.$$

Hence, if $t \in (s_{r-1}, s_r]$,

$$\rho_\delta(f_t, f_{s_{r-1}})^2 = \sup_{|v| \leq \delta} \mu_{x,v}((s_{r-1}, t]) \leq \sup_{|v| \leq \delta} \mu_{x,v}((s_{r-1}, s_r]) \leq \eta^2/2.$$

Therefore

$$\sup_{f \in \mathcal{F}} \min_{0 \leq r \leq K} \rho_\delta(f, f_{s_r}) \leq \eta,$$

which is (1.12). Fourth, continuity of the covariance map is immediate, as for $s, t \in \mathbb{R}$,

$$z \mapsto (\mu_z(f_s f_t), \mu_z(f_t), \mu_z(F^2)) = (F(s \wedge t | z), F(t | z), 1),$$

is continuous at x by (i), so (1.13) holds. Finally, we check (1.14)–(1.15) from Corollary 1.2. For $\mathcal{F}$,

$$\sup_{f \in \mathcal{F}} |(\mu_z - \mu_x)(f)| = \sup_{t \in \mathbb{R}} |F(t | z) - F(t | x)| \leq L_x \|z - x\|$$

by (ii), which is (1.14). Also, here $d = 1$, so (1.15) reads $\kappa^{(d+2)/2}/n = \kappa^{3/2}/n \rightarrow 0$, which is assumed. Therefore Corollary 1.2 applies directly and yields the claim. $\square$

Remark 2.2 Condition (ii) is often verifiable directly. For instance, if $f_{Y|X}(\cdot | z)$ exists for z near x and

$$\sup_{|z-x| \leq \tau} \int_{\mathbb{R}} \left| \frac{\partial}{\partial z} f_{Y|X}(y | z) \right| dy < \infty,$$

then $\sup_t |F(t | z) - F(t | x)| \leq L_x |z - x|$ for some $L_x < \infty$ by the mean value theorem. $\circ$

Remark 2.3 Compared with Theorem 2.8, the normalisation changes from $\sqrt{n\hbar}$ to $\sqrt{\kappa}$ and the variance factor $\int K^2/f(x)$ disappears. This reflects the density-adaptive localisation of k -NN weighting.

The proof of the k -NN distributional regression weak convergence, Theorem 2.9, might seem more straightforward and less layered than its kernel smoothing counterpart, Theorem 2.8, but this is only because much of the k -NN tailoring is done behind the scenes in Theorem 1.21 and Corollary 1.2. $\circ$

2.5 Notes and Comments

The kernel estimators used in this chapter trace back to early ideas independently developed by [Rosenblatt \[1956\]](#) and [Parzen \[1962\]](#). Their setup then became the template for modern treatments of kernel bias, variance, and asymptotic normality, with classic expositions including [Silverman \[2018\]](#), [Wand and Jones \[1994\]](#), [Scott \[2015\]](#). Regression smoothing via kernels emerged soon after, with the conditional mean estimator independently introduced by [Nadaraya \[1964\]](#) and [Watson \[1964\]](#), and the closely related product of the regression function and marginal density considered by [Priestley and Chao \[1972\]](#). As seen in this chapter, under mild conditions, these estimators are weakly consistent and, after centring and appropriate scaling, converge to Gaussian limits. Now, while their joint asymptotics follow from standard theorems such as the Cramér–Wold device and the Lindeberg–Feller CLT, the empirical process approach given here follows the techniques of [Shorack and Wellner \[2009\]](#) and [van der Vaart and Wellner \[1996\]](#). From a general perspective, the minimax benchmarks that motivate kernel choices and bandwidth rates were explored in [Stone \[1980\]](#) and, in the nonparametric regression setting, in [Stone \[1982\]](#). For regression smoothing beyond simple kernels, local polynomial methods provide bias reduction and better boundary behaviour; see [Fan and Gijbels \[2018\]](#). Broader theoretical surveys include [Györfi et al. \[2002\]](#), [Tsybakov \[2009\]](#), while [Wasserman \[2006\]](#) offers a compact and applied overview. Classical results on kernel-shape efficiency include [Epanechnikov \[1969\]](#). Practical issues, such as boundary bias on one-sided supports and the effects of kernel shape, are discussed throughout the above-cited works and in focused papers, for instance [Hall et al. \[1999\]](#).

After these core estimators, the chapter turns to distributional regression and bandwidth choice. Reliable plug-in rules and related data-driven criteria for density estimation are given by [Sheather and Jones \[1991\]](#) and for local least-squares regression by [Ruppert et al. \[1995\]](#); a general treatment is given in [Scott \[2015\]](#), including the multivariate case. Foundational predictive cross-validation papers include [Stone \[1974\]](#), [Allen \[1974\]](#), [Geisser \[1975\]](#), and density-oriented cross-validation smoothing is classically treated in [Bowman \[1984\]](#). For mixed continuous and discrete covariates, from an econometrics perspective, see [Li and Racine \[2007\]](#). In distributional regression built from a conditional empirical distribution, the pathwise weak-convergence arguments use bracketing and empirical-process tools of [van der Vaart and Wellner \[1996\]](#), [Shorack and Wellner \[2009\]](#), while almost sure convergence results for conditional empirical distribution functions were established by [Stute \[1986\]](#). Bandwidth choice is more delicate for conditional distribution estimators. It was shown by [Altman and Léger \[1995\]](#) that naïve leave-one-out cross-validation can behave unsatisfactorily. Empirical processes for k -nearest neighbours were considered in [Portier \[2025\]](#), extending classical nearest-neighbour roots in classification, density and regression such as [Cover and Hart \[1967\]](#), [Loftsgaarden and Quesenberry \[1965\]](#), [Stone \[1977\]](#); see also references therein. Although our emphasis here is theoretical, these smoothing ideas are central for practical work in insurance data exploration, ratemaking and claim size modelling, where flexible nonparametric structure is a valuable applied tool, see [Silverman \[2018\]](#), [Scott \[2015\]](#).

2.6 Exercises

1. The Epanechnikov kernel is given by

$$K(x) = \frac{3}{4}(1 - x^2),$$

for $x \in [-1, 1]$ and 0 elsewhere. Show that it satisfies the kernel requirements.

2. Recall the Priestly–Chao estimator $\mathfrak{m}^{(n)}(x) = (nh)^{-1} \sum_{i=1}^n Y_i K(h^{-1}(x - X_i))$. The mean squared error of $\mathfrak{m}^{(n)}(x)$ can be decomposed into a sum of the variance and the squared bias as follows:

$$\mathbb{E}[(\mathfrak{m}^{(n)}(x) - m(x))^2] = \mathbb{E}[(\mathfrak{m}^{(n)}(x) - \mathbb{E}[\mathfrak{m}^{(n)}(x)])^2] + (\mathbb{E}[\mathfrak{m}^{(n)}(x)] - m(x))^2,$$

where we examined the latter two quantities in Theorem 2.3. Plugging these quantities into the mean squared error, we obtain that

$$\begin{aligned} \mathbb{E}[(\mathfrak{m}^{(n)}(x) - m(x))^2] &= \left(\frac{h^2}{2} m''(x) \int u^2 K(u) \, du \right)^2 + o(h^4) \\ &\quad + \frac{\mathbb{E}[Y^2 | X = x] f(x)}{nh} \int K^2(u) \, du + O(1/n). \end{aligned}$$

For large n , define the asymptotic mean squared error as

$$\begin{aligned} \text{AMSE}(\mathfrak{m}^{(n)}(x)) &= \left(\frac{h^2}{2} m''(x) \int u^2 K(u) \, du \right)^2 \\ &\quad + \frac{\mathbb{E}[Y^2 | X = x] f(x)}{nh} \int K^2(u) \, du. \end{aligned}$$

- (a) For fixed x , show that the optimal bandwidth that minimises the $\text{AMSE}(\mathfrak{m}^{(n)}(x))$ is given by

$$\hat{h}(x) = \left(\frac{\mathbb{E}[Y^2 | X = x] f(x) \int K^2(u) \, du}{(m''(x))^2 (\int u^2 K(u) \, du)^2} \right)^{\frac{1}{5}} n^{-\frac{1}{5}}.$$

- (b) Show that the optimal bandwidth that minimises $\int \text{AMSE}(\mathfrak{m}^{(n)}(x)) \, dx$, is given by

$$\hat{h} = \left(\frac{\mathbb{E}[Y^2] \int K^2(u) \, du}{\int (m''(x))^2 \, dx (\int u^2 K(u) \, du)^2} \right)^{\frac{1}{5}} n^{-\frac{1}{5}}.$$

- (c) Using the above optimal bandwidths, determine the decay rate, as a function of n , of $\text{AMSE}(\mathfrak{m}^{(n)}(x))$ and $\int \text{AMSE}(\mathfrak{m}^{(n)}(x)) \, dx$, and compare it with maximum likelihood estimation in a parametric setting.

3. Recall the definition of the cross-validation risk estimator:

$$\hat{J}(h) = \int \mathfrak{f}^{(n)}(x)^2 \, dx - \frac{2}{n} \sum_{i=1}^n \mathfrak{f}_{-i}^{(n)}(X_i).$$

Let $K_{(2)}$ denote the convolution of K with itself:

$$K_{(2)}(v) = \int_{-\infty}^{\infty} K(u) K(v - u) \, du.$$

- (a) Show that $\int \mathbb{F}^{(n)}(x)^2 dx = 1/(n^2 h) \sum_{i=1}^n \sum_{j=1}^n K_{(2)}(h^{-1}(X_j - X_i))$.
- (b) If we use the Gaussian kernel $K(u) = e^{-u^2/2}/\sqrt{2\pi}$, then $K_{(2)}(u) = 1/(2\sqrt{\pi})e^{-u^2/4}$. Show that in this case

$$\begin{aligned} \hat{J}(h) &= \frac{1}{n^2 h} \sum_{i=1}^n \sum_{j=1}^n 1/(2\sqrt{\pi}) e^{-\left(\frac{X_i - X_j}{h}\right)^2/4} \\ &\quad + \frac{2}{\sqrt{2\pi} h(n-1)} - \frac{2}{n(n-1)h} \sum_{i=1}^n \sum_{j=1}^n 1/\sqrt{2\pi} e^{-\left(\frac{X_i - X_j}{h}\right)^2/2}. \end{aligned}$$

4. Go through the details of the proof of Theorem 2.8.
5. Let $F(\cdot | x)$ be the conditional distribution function of Y given $X = x$, and let

$$q_\alpha(x) = F^{\leftarrow}(\alpha | x), \quad \alpha \in (0, 1),$$

denote the conditional quantile. Define the conditional empirical quantile

$$\hat{q}_{n,\alpha}(x) = \mathbb{F}^{(n)\leftarrow}(\alpha | x).$$

Assume the conditions of Theorem 2.8, and assume that $F(\cdot | x)$ has a density $f(\cdot | x)$ that is continuous and strictly positive at $q_\alpha(x)$. Show that the inverse map is Hadamard differentiable at $F(\cdot | x)$ and use the functional δ -method to show that, for fixed $\alpha \in (0, 1)$,

$$\sqrt{nh}(\hat{q}_{n,\alpha}(x) - q_\alpha(x)) \xrightarrow{d} N\left(0, \frac{\int K^2(u) du}{f(x)} \frac{\alpha(1-\alpha)}{f(q_\alpha(x) | x)^2}\right).$$

6. Let $\sigma^2(x) = \text{Var}(Y_1 | X_1 = x)$ and define

$$\mathfrak{m}_2^{(n)}(x) = \frac{1}{nh} \sum_{i=1}^n Y_i^2 K(h^{-1}(x - X_i)), \quad \hat{\sigma}^2(x) = \frac{\mathfrak{m}_2^{(n)}(x)}{\mathbb{F}^{(n)}(x)} - (\mathfrak{e}^{(n)}(x))^2.$$

Assume the conditions of Theorem 2.4 and that $\mathbb{E}[Y_1^4 | X_1 = x] < \infty$. Derive the leading bias of $\hat{\sigma}^2(x)$ and show that

$$\sqrt{nh}(\hat{\sigma}^2(x) - \sigma^2(x)) \xrightarrow{d} N(0, \tau^2(x)),$$

for a variance $\tau^2(x)$ that you should express in terms of conditional moments of Y given $X = x$.

7. Assume X has support $[0, 1]$ and the kernel K has support on $[-1, 1]$. Consider the conditional empirical distribution function $\mathbb{F}^{(n)}(t | x)$ at the boundary point $x = 0$. Proceed in two stages, first for the naive estimator and then for the reflection correction.
- (a) Show that $\mathbb{E}[\mathbb{F}^{(n)}(t | 0)]$ has a non-vanishing boundary bias, and identify its leading term.
- (b) Define a reflection-based estimator $\mathbb{F}_R^{(n)}(t | 0)$ by adding reflected covariates $-X_i$ with the same responses. Show that the leading boundary bias vanishes and give the resulting order of the bias.

- (c) Discuss a condition on the joint distribution of (X, Y) at $x = 0$ under which the reflection estimator achieves an $O(h^2)$ bias rate.
- 8. State and prove a functional central limit theorem for the conditional mean excess function.
- 9. State and prove a functional central limit theorem for the conditional cumulative hazard $\Lambda^{(n)}(\cdot|x)$, without product integration, suitably adapted from the first chapter.

Practical exercises

1. Consider the following simulation study:
 - (a) Simulate 1000 observations of a standard normal random variable. Implement the Parzen–Rosenblatt density estimator using the Epanechnikov kernel. Use the following bandwidths: 0.1, 0.3, 0.5, 0.7, 1.0.
 - (b) Let $X_1 \sim \text{Unif}(0, 1)$, $X_2 \sim \text{Unif}(1, 2)$ and $p \sim \text{Bernoulli}(0.8)$. Define

$$Y = pX_1 + (1 - p)X_2.$$

Use the same bandwidths.

- (c) Implement κ -nearest neighbours in both cases, for $\kappa = 10, 20, 50, 100, 200$.
2. We wish to investigate when it is appropriate to use kernel estimation.
 - (a) Simulate 1000 observations from a standard normal random variable. Estimate the density via kernel estimation and with the MLE. Plot the densities together with the theoretical density, and assess the fit overall and in the tail.
 - (b) Do the same experiment, but simulate now from a Pareto distribution with shape parameter 2 and threshold 1.
 - (c) Let $X_1 \sim \text{Unif}(0, 1)$, $X_2 \sim \text{Unif}(1, 2)$ and $p \sim \text{Bernoulli}(0.8)$. Define

$$Y = pX_1 + (1 - p)X_2$$

Estimate the density with kernels and assess the fit.

- (d) Repeat the three experiments with a different kernel using the R function `density()`. Assess the influence of the kernel choice and indicate when it matters.
- (e) Repeat the three experiments with a κ -nearest neighbours estimator with κ of your choice. Comment on the differences.
3. Consider the dataset `frecomfire`, from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#). Compute $\hat{h} = \text{argmin}_h \hat{J}(h)$ from Exercise 3 for the covariate `ClaimCost2007` using the first

$$n = 100, 500, 2000, 3000, 4500, 7000, 9613$$

observations, and plot $\hat{h}$ as a function of n .

4. Consider the dataset `brvehins1a` from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#), which is a Brazilian vehicle insurance dataset (by policy) with risk features. In this exercise, the response variable is `ClaimAmountTotColl` (denoted Y), which is the claim sizes caused by total collisions, given the covariate, sum insured `SumInsAvg` (denoted X).
 - (a) Load the data, select the variables `SumInsAvg` and `ClaimAmountTotColl`, and filter the data such that `ClaimAmountTotColl` $\in (0, M)$, with $M = 200,000$.
 - (b) Transform `SumInsAvg` = X to uniform variables by applying the empirical distribution function $\mathbb{F}_X^{(n)}(\cdot)$ on X .
 - (c) Let n be the number of observations in the dataset after filtering, and define a bandwidth $h_n = 3n^{-1/5}/\log \log n$. Let $x = 0.7$. Consider the grid of indices from 0 to M with step size 100. Compute the empirical distribution function $\mathbb{F}_Y^{(n)}(\cdot)$ and the conditional empirical distribution function, as per Definition 2.3, $\mathbb{F}^{(n)}(\cdot | 0.7)$ using the Gaussian kernel.
 - (d) We now compare the 95% naive confidence bands with the pathwise confidence bands. Plot $\mathbb{F}^{(n)}(\cdot | 0.7)$ with the 95% pointwise confidence bands arising from Theorem 2.4, and add the pathwise 95% confidence bands that arise from the limiting distribution of the quantity $\sqrt{nh_n} \sup_{t \in (0, M)} |\mathbb{F}^{(n)}(t | 0.7) - F(t | 0.7)|$, using Theorem 2.8.
Hint: The 95% quantile of the Kolmogorov distribution is $\text{KS}^{-1}(1 - 0.05) \approx 1.36$.

Chapter 3

Heavy-tailed distributions

Large claims and catastrophic losses are very common in non-life insurance, and they are, by definition, driven by tail behaviour. In this chapter, we first clarify the main heavy-tail classes and their relationships. We then introduce extreme value theory, develop estimation tools for regularly varying tails, and close with conditional extremes and their limit theory.

3.1 Definitions of heavy-tailed distributions

There are several definitions of heavy-tailed risk. We start by setting terminology and the main tail classes, since the statistical tools later in the chapter depend on these distinctions. In probability theory, a random variable Z is loosely said to have a heavy-tailed distribution if it lacks a moment generating function in some neighbourhood of the origin. More precisely, define the moment generating function of Z as $M_Z(t) = \mathbb{E}[e^{tZ}]$, $t \in \mathbb{R}$. A distribution is *heavy-tailed* if there exists no $\varepsilon > 0$ such that $M_Z(t) < \infty$ for every $t \in (-\varepsilon, \varepsilon)$. Intuitively, this definition indicates that the tails of the distribution decay more slowly than those of an exponential distribution, and can be equivalently stated, for positive random variables, in terms of the survival function $\bar{F} = 1 - F$ of Z as

$$\lim_{z \rightarrow \infty} \exp(tz) \bar{F}(z) = \infty, \quad \text{for all } t > 0. \quad (3.1)$$

A more specific class of heavy-tailed distributions is the class of long-tailed distributions. A distribution is called *long-tailed* if for all $x \geq 0$, the following limit holds:

$$\lim_{z \rightarrow \infty} \frac{\mathbb{P}(Z > z + x)}{\mathbb{P}(Z > z)} = 1.$$

This property means that the tail of the distribution is asymptotically invariant under translations, implying that a large value is not significantly affected by adding any fixed amount x . Perhaps surprisingly, long-tailed distributions are always heavy-tailed. The converse is generally not true.

Regularly varying distributions are another important and widely used subclass of heavy-tailed distributions. A distribution F or random variable Z is *regularly varying* if its survival function is a regularly varying function at infinity, i.e. there exists a function ℓ , slowly varying at infinity, and an index $1/\gamma > 0$ such that:

$$\mathbb{P}(Z > z) = z^{-1/\gamma} \ell(z).$$

Here, ℓ is said to be slowly varying if for every $t > 0$, $\lim_{z \rightarrow \infty} \ell(tz)/\ell(z) = 1$. Regularly varying distributions have heavy tails, as can be directly seen from (3.1) if ℓ is increasing. The general case can be shown using the fact that ℓ cannot asymptotically decrease faster than any power; this is made precise below. Many important distributions such as Pareto and stable distributions are regularly varying.

Finally, subexponential distributions are characterised by the property that the sum of two independent random variables with the same distribution has a tail that is asymptotically equivalent to the tail of the maximum of the two. Formally, a distribution is *subexponential* if:

$$\lim_{z \rightarrow \infty} \frac{\mathbb{P}(Z_1 + Z_2 > z)}{2\mathbb{P}(Z > z)} = 1, \quad (3.2)$$

where Z_1 and Z_2 are independent copies of Z . Subexponential distributions are a superset of regularly varying distributions and capture the idea that large sums are governed by their largest term. Indeed if $M_2 = Z_1 \vee Z_2$ then $\mathbb{P}(M_2 > z) = 1 - F(z)^2 = 1 - (1 - \bar{F}(z))^2 \approx 2\bar{F}(z)$ for large z , and the concept is equivalent to the analogous property for n copies of Z , and replacing 2 with n in (3.2). This is also known as the *single big event principle*.

The relationships between these classes can be summarised as follows: regularly varying distributions are subexponential, subexponential distributions are long-tailed, and long-tailed distributions are heavy-tailed. In this chapter, we focus on regularly varying distributions, which are arguably the most of the above definitions in statistical terms.

3.2 Introduction to extreme value theory

We now connect tail classes to asymptotic behaviour of maxima, which motivates much of the estimation theory that follows. A standard introductory reference for extreme value theory is [Embrechts et al. \[1997\]](#).

3.2.1 General max-domains of attraction

Before delving into regularly varying distributions, we first present the Fisher–Tippett theorem, which describes the limiting behaviour of the maxima of iid random variables. This theorem introduces the Generalised Extreme Value (GEV) distribution.

Theorem 3.1 (Fisher–Tippett) *Let (Z_n) be a sequence of iid random variables with common distribution function F . If there exist sequences of constants $a_n >$*

0 and $b_n \in \mathbb{R}$ such that the distribution of the normalised maximum converges to a non-degenerate distribution function H , that is,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\max(Z_1, Z_2, \dots, Z_n) - b_n}{a_n} \leq z \right) = H(z), \quad (3.3)$$

then H belongs to one of the three types of distributions, jointly parametrised by the Generalised Extreme Value (GEV) distribution:

$$H(z) = \exp \left(- \left(1 + \gamma \frac{z - \mu}{\sigma} \right)^{-1/\gamma} \right),$$

defined on the set $\{z : 1 + \gamma \frac{z - \mu}{\sigma} > 0\}$, where $\mu \in \mathbb{R}$, $\sigma > 0$, and $\gamma \in \mathbb{R}$. The $\gamma = 0$ expression can be understood as a limit.

The GEV distribution encompasses three families: the Gumbel, Fréchet, and Weibull distributions, which correspond to $\gamma = 0$, $\gamma > 0$, and $\gamma < 0$, respectively.

A distribution F is said to belong to the Gumbel, Fréchet, and Weibull *max-domain of attraction* if (3.3) holds with $\gamma = 0$, $\gamma > 0$, and $\gamma < 0$, respectively. We have the following:

1. All distributions in the Weibull max-domain of attraction have finite right-endpoints, and thus are not heavy-tailed at all.
2. Distributions in the Gumbel max-domain of attraction include light-tailed ones such as the exponential or normal distributions, but also heavy-tailed ones such as the Weibull or lognormal distributions.
3. For a distribution in the Fréchet max-domain of attraction, its tails are heavier than in the other two max-domains of attraction. In fact, we may characterise them in terms of regularly varying functions as follows.

The next theorem makes this characterisation precise.

Theorem 3.2 (Fréchet max-domain and regular variation) *A distribution function F belongs to the Fréchet max-domain of attraction if and only if its survival function is regularly varying with index $1/\gamma > 0$.*

This result shows that regularly varying distributions form a significantly large class within the heavy-tailed distributions and that polynomial decay of tails is not an arbitrary choice, but rather one of three ways that maxima behave asymptotically. A similar result is as follows, in terms of the Generalised Pareto Distribution (GPD).

Definition 3.1 (Generalised Pareto Distribution (GPD)) The Generalised Pareto Distribution is given by the cumulative distribution function:

$$G(y) = \begin{cases} 1 - \left(1 + \gamma \frac{y - \mu}{\sigma} \right)^{-1/\gamma} & \text{if } \gamma \neq 0, \\ 1 - \exp \left(-\frac{y - \mu}{\sigma} \right) & \text{if } \gamma = 0, \end{cases} \quad (3.4)$$

for $y \geq \mu$ when $\gamma \geq 0$, and $\mu \leq y \leq \mu - \sigma/\gamma$ when $\gamma < 0$, where $\mu \in \mathbb{R}$ is the location parameter, $\sigma > 0$ is the scale parameter, and $\gamma \in \mathbb{R}$ is the shape parameter.

Theorem 3.3 (Pickands–Balkema–de Haan) *Let Z be a random variable with distribution function F . Then F is in the max-domain of attraction of some extreme value distribution if and only if the exceedances over high thresholds converge to a Generalised Pareto Distribution (GPD). Formally, if and only if there exists a positive measurable function $a(u)$ such that*

$$\lim_{u \rightarrow x_F} \mathbb{P}((Z - u)/a(u) \leq z \mid Z > u) = G(z),$$

where G is the GPD with $\mu = 0$ and $\sigma = 1$, and where $x_F = \sup\{z : F(z) < 1\}$ is the right endpoint of F .

Thus, if the maxima of blocks of some data follow approximately a GEV distribution, then the exceedances over a high threshold of the same data are approximately GPD distributed, with the same γ . This relationship highlights that both the GPD and GEV describe different aspects of the same underlying extremal behaviour. The unification of all max-domains of attraction under the GEV/GPD framework is convenient for statistical estimation when data are bundled into *block-maxima* or treated as *peaks-over-thresholds*. In that case, maximum likelihood estimation of H and G is a viable option.

3.2.2 Results for regularly varying distributions

We survey some of the main properties of regularly varying distributions. These results are also helpful before moving to nonparametric estimation of $\gamma > 0$, the key parameter for this class of heavy-tailed distributions, by Theorem 3.2. A standard reference is [Bingham et al. \[1987\]](#). The order is deliberate. Uniform convergence gives the baseline tail approximation, Potter bounds provide robust inequalities, and the representation and Karamata results provide structural and integral tools used later in the asymptotic theory.

Theorem 3.4 (Uniform convergence property of regularly varying functions) *Let F be a regularly varying distribution with index $1/\gamma$, i.e. $\bar{F}(z) = z^{-1/\gamma}\ell(z)$ where ℓ is slowly varying. Then for any fixed $a > 0$, the following convergence holds:*

$$\lim_{z \rightarrow \infty} \sup_{t \in [a, \infty)} \left| \frac{\bar{F}(tz)}{\bar{F}(z)} - t^{-1/\gamma} \right| = 0. \quad (3.5)$$

The first version of Potter's bounds below deals with the survival function directly, and shows, in particular, that the oscillation of the slowly varying component can be bounded by any power.

Theorem 3.5 (Potter's bounds, first version) *Let F be a regularly varying distribution with index $1/\gamma$. For any $\varepsilon \in (0, 1/\gamma)$ there exist positive finite constants such that*

$$\begin{aligned} \bar{F}(z) &\leq cst \, z^{-1/\gamma+\varepsilon}, & z > 0, \\ \bar{F}(z) &\geq cst \, z^{-1/\gamma-\varepsilon}, & z > \varepsilon. \end{aligned}$$

Proof. We rewrite the tail in terms of the slowly varying function ℓ and bound it uniformly above and below. Set $\ell(z) = z^{1/\gamma}\bar{F}(z)$, which is a slowly varying

function. Let $\varepsilon \in (0, 1/\gamma)$, and note that $z \mapsto z^{-\varepsilon}\ell(z)$ is a regularly varying function with tail index $-\varepsilon$. We use that $\lim_{z \rightarrow \infty} z^{-\varepsilon}\ell(z) = 0$. In particular, there exists a b' such that for all $b \geq b'$ it holds that $\sup_{z \geq b} z^{-\varepsilon}\ell(z) < \infty$. In addition we note that $z^{-\varepsilon}\ell(z) = z^{-\varepsilon+1/\gamma}\bar{F}(z)$ is bounded on the compact interval $[1, b']$ i.e

$$\sup_{z \in [1, b']} z^{-\varepsilon}\ell(z) < \infty.$$

For $z \in (0, 1)$ we see that $z^{-\varepsilon}\ell(z) = z^{1/\gamma-\varepsilon}\bar{F}(z) \leq 1$. Consequently $z \mapsto z^{-\varepsilon}\ell(z)$ is uniformly bounded on $(0, \infty)$ and thus for all $z > 0$,

$$\bar{F}(z) = z^{-1/\gamma+\varepsilon} z^{-\varepsilon}\ell(z) \leq z^{-1/\gamma+\varepsilon} \sup_{x>0} x^{-\varepsilon}\ell(x) = \text{cst } z^{-1/\gamma+\varepsilon}.$$

This proves the first statement.

For the second statement, note that $z \mapsto z^{\varepsilon}\ell(z)$ is regularly varying with tail index ε . We use that $\lim_{z \rightarrow \infty} z^{\varepsilon}\ell(z) = \infty$. Thus for any $z > \varepsilon$, it holds that $z \mapsto z^{\varepsilon}\ell(z) = z^{\varepsilon+1/\gamma}\bar{F}(z)$ is strictly positive and diverging to infinity. Consequently $\inf_{x \geq \varepsilon} x^{\varepsilon}\ell(x) > 0$ and hence

$$\bar{F}(z) = z^{-1/\gamma-\varepsilon} z^{\varepsilon}\ell(z) \geq z^{-1/\gamma-\varepsilon} \inf_{x \geq \varepsilon} x^{\varepsilon}\ell(x) = \text{cst } z^{-1/\gamma-\varepsilon}.$$

□

The second version of Potter's bounds provides a similar control, but for the ratio of tails, which can be used to quantify the quality of the regular variation property.

Theorem 3.6 (Potter's bounds, second version) *Let F be a regularly varying distribution with index $1/\gamma$. For any $\varepsilon > 0$, there exists a finite positive constant such that for all $x \geq 1$ and $y > 0$,*

$$\begin{aligned} \frac{\bar{F}(x/y)}{\bar{F}(x)} &\leq \text{cst } (y^{1/\gamma-\varepsilon} \vee y^{1/\gamma+\varepsilon}), \\ &= \text{cst } y^{1/\gamma-\varepsilon} 1_{(y<1)} + \text{cst } y^{1/\gamma+\varepsilon} 1_{(y \geq 1)}. \end{aligned}$$

It turns out that all regularly varying distribution functions can be represented in a convenient exponential form, which can in particular help identify the regular variation, and prove limits of specific parametric classes.

Theorem 3.7 (Representation for regularly varying distributions) *A distribution function F is regularly varying if and only if it can be represented in the form*

$$\bar{F}(z) = c(z) \exp \left(\int_{z_0}^z \frac{\delta(u)}{u} du \right), \quad z \geq z_0$$

for some $z_0 > 0$, where $c(z) \rightarrow c_0 \in (0, \infty)$ and $\delta(z) \rightarrow -1/\gamma$ as $z \rightarrow \infty$.

The following result, as a special case, helps with the transfer of the regular variation property from densities to tails.

Theorem 3.8 (Karamata's Theorem) *Let $f(z) = z^{-1/\gamma-1}\ell(z)/\gamma$ be a regularly varying function at infinity with index $-1/\gamma-1$, and let ℓ be locally bounded in $[z_0, \infty)$. Then for $\gamma > 0$ we have*

$$\int_z^\infty f(t) dt \sim z^{-1/\gamma}\ell(z) \quad \text{as } z \rightarrow \infty.$$

In a nutshell, the first three results help with the taming of regular variation for limit theorems. From the second it follows that any regularly varying distribution is heavy-tailed. The last two results, apart from being theoretically significant, concretely help identify regularly varying distributions.

Example 3.1 (Some regularly varying distributions) Let $\gamma > 0$.

1. *Pareto distribution.* Define

$$F_P(x) = 1 - x^{-1/\gamma}, \quad x \geq 1, \quad F_P(x) = 0, \quad x < 1.$$

Then $\bar{F}_P(x) = x^{-1/\gamma}$ for $x \geq 1$, so F_P is regularly varying with index $1/\gamma$ and slowly varying function $\ell(x) \equiv 1$.

2. *Shifted Pareto (US-Pareto).* Define

$$F_{SP}(x) = 1 - (x+1)^{-1/\gamma}, \quad x \geq 0, \quad F_{SP}(x) = 0, \quad x < 0.$$

Then $\bar{F}_{SP}(x) = (x+1)^{-1/\gamma} = x^{-1/\gamma}\ell(x)$ with

$$\ell(x) = (1+x^{-1})^{-1/\gamma}, \quad x > 0.$$

Since $\ell(x) \rightarrow 1$ as $x \rightarrow \infty$, the function ℓ is slowly varying and hence F_{SP} is regularly varying with index $1/\gamma$.

3. *Student t distribution.* Define for the parameter $\nu > 0$ (degrees of freedom) the density

$$f_\nu(x) = c_\nu(1+x^2/\nu)^{-(\nu+1)/2}, \quad x \in \mathbb{R}, \quad c_\nu = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}.$$

Then, as $x \rightarrow \infty$,

$$f_\nu(x) = x^{-(\nu+1)}\ell(x), \quad \ell(x) = c_\nu x^{\nu+1}(1+x^2/\nu)^{-(\nu+1)/2}.$$

Since $\ell(x) \rightarrow c_\nu \nu^{(\nu+1)/2}$, the function ℓ is slowly varying. By Karamata's Theorem 3.8, the tail satisfies

$$\bar{F}_\nu(x) = \int_x^\infty f_\nu(t) dt \sim \frac{1}{\nu} x^{-\nu}\ell(x),$$

and therefore F_ν is regularly varying with index $1/\gamma = \nu$.

4. *Fréchet distribution.* Define

$$F_{Fr}(x) = \exp(-x^{-1/\gamma}), \quad x > 0.$$

Then $\bar{F}_{Fr}(x) = 1 - \exp(-x^{-1/\gamma}) = x^{-1/\gamma}\ell(x)$ with

$$\ell(x) = x^{1/\gamma}(1 - \exp(-x^{-1/\gamma})).$$

Since $\exp(-y) = 1 - y + o(y)$ as $y \downarrow 0$, we have $\ell(x) \rightarrow 1$ as $x \rightarrow \infty$, and hence ℓ is slowly varying. Thus F_{Fr} is regularly varying with index $1/\gamma$.



A common heavy-tailed distribution in insurance is the lognormal distribution, which in fact is not regularly varying.

Example 3.2 (Heavy tailed distribution which is not regularly varying) Let $Z = \exp(\sigma W)$ with $W \sim N(0, 1)$ and $\sigma > 0$, so that Z follows a *lognormal distribution*. For any $t > 0$,

$$\mathbb{E}[e^{tZ}] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp(te^{\sigma x} - x^2/2) dx = \infty,$$

since for x large enough one has $te^{\sigma x} \geq x^2$, and thus the integrand is bounded below by $\exp(x^2/2)$ on a tail interval, making the integral diverge. Hence Z is heavy-tailed. Writing Φ for the standard normal distribution function, we have by Mill's ratio that

$$\bar{F}_Z(x) = 1 - \Phi(\log x / \sigma) \sim \frac{\sigma}{\log x} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(\log x)^2}{2\sigma^2}\right),$$

so for any $t > 1$,

$$\frac{\bar{F}_Z(tx)}{\bar{F}_Z(x)} \sim \frac{\log x}{\log(tx)} \exp\left(-\frac{\log t}{\sigma^2} \log x - \frac{(\log t)^2}{2\sigma^2}\right) \rightarrow 0,$$

which is incompatible with regular variation. Therefore the lognormal distribution is heavy-tailed but not regularly varying. ◇

3.3 Statistics for regularly varying distributions

We now pivot from the general probabilistic theory to estimation, focusing on tools tailored to regularly varying tails. For exploratory analysis, QQ plots and the mean excess function are two valuable tools for investigating heavy-tailed behaviour in data. The former allows for a visual assessment of how well the tail of a distribution matches a theoretical heavy-tailed model, with deviations suggesting heavy tails, which may be mathematically investigated with empirical quantile processes. The mean excess function does not need a candidate distribution and instead reveals heavy-tailed patterns, though it can be quite volatile for large thresholds (as per the variance in Theorem 1.11). Despite these useful tools, the Hill estimator, defined below, is considered the gold standard for visualising and quantifying heavy-tailed behaviour.

In this section, we let $u_n = F^{\leftarrow}(1 - k/n)$ be a sequence of thresholds where $k = k_n$ is an intermediate sequence satisfying $k_n \rightarrow \infty$ and $k_n/n \rightarrow 0$. Note that in this case $n(1 - F(u_n)) = k$. Large k is necessary for having sufficient data to perform estimation, while small k/n indicates the need to focus solely on the tail.

The Hill estimator is motivated by the following three facts: a) regularly varying distributions are Pareto in the tail; b) the logarithm of a Pareto is an exponential with rate $1/\gamma$; c) the mean can be estimated by a sample average.

Definition 3.2 (Hill estimator) Let (Z_n) be a sequence of positive random variables with order statistics $(Z_{n,j})$, the latter non-decreasing in j for fixed n . The Hill estimator is given by

$$\gamma_{n,k} = \frac{1}{k} \sum_{i=1}^k \log(Z_{n,n-i+1}/Z_{n,n-k}).$$

3.3.1 Tail processes

Tail processes are stochastic processes that are defined in the far tail of a sample. Their analysis, and in particular establishing their convergence to a Pareto distribution, is a fundamental task for distributions in the Fréchet max-domain of attraction. In essence, this convergence allows us to dissect the $z^{-1/\gamma}$ part from the $\ell(z)$ part for regularly varying survival functions, which in turn makes estimating γ easier.

Definition 3.3 (Tail empirical distribution function) Define the tail empirical function and its expectation as

$$\tilde{T}_n(s) = \frac{1}{k} \sum_{i=1}^n 1_{(Z_i > u_n s)}, \quad T_n(s) = \mathbb{E}[\tilde{T}_n(s)] = \frac{1 - F(u_n s)}{1 - F(u_n)}, \quad s > 0.$$

Define also $T(s) = s^{-1/\gamma}$.

Note that, for $s \geq 1$, $\tilde{T}_n(s)$ estimates the conditional tail probability $\mathbb{P}(Z > u_n s \mid Z > u_n)$, so the Pareto limit corresponds to asymptotic scale-invariance of exceedances.

Theorem 3.9 (Consistency of tail empirical function) *We have $\tilde{T}_n(s) \xrightarrow{\mathbb{P}} T(s)$ uniformly on intervals bounded away from zero. Moreover $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$.*

Proof. We decompose into a stochastic term and a bias term, then use Dini's theorem for uniformity. Consider the decomposition $\tilde{T}_n(s) - T(s) = (\tilde{T}_n(s) - T_n(s)) + (T_n(s) - T(s)) = A_n + B_n$. By definition of regular variation, $B_n \rightarrow 0$. Then note that A_n has mean zero and

$$\text{Var}(A_n) = \frac{1}{n^2 \bar{F}(u_n)^2} n F(su_n) \bar{F}(su_n) = \frac{F(su_n) \bar{F}(su_n)}{k \bar{F}(u_n)} \rightarrow 0,$$

which implies that $A_n \xrightarrow{\mathbb{P}} 0$, and thus $\tilde{T}_n(s) \xrightarrow{\mathbb{P}} T(s)$ for fixed s . By Dini's Theorem 1.15, the convergence is uniform on bounded intervals. Finally, observe that for any $\varepsilon > 0$, if $Z_{n,n-k}/u_n > (1 + \varepsilon)$, then necessarily at least k terms are non-zero in the sum of $\tilde{T}_n$, or in other words $\tilde{T}_n(1 + \varepsilon) > 1$. Then by the previous result,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(Z_{n,n-k}/u_n > 1 + \varepsilon) \leq \lim_{n \rightarrow \infty} \mathbb{P}(\tilde{T}_n(1 + \varepsilon) > 1) = 0,$$

and similarly $\limsup_{n \rightarrow \infty} \mathbb{P}(Z_{n,n-k}/u_n < 1 - \varepsilon) = 0$. □

The $k/n \rightarrow 0$ requirement is essential; a fixed threshold breaks the limit, as the next example shows.

Example 3.3 (Failure without intermediate thresholds) Let the survival function be

$$\bar{F}(z) = z^{-1/\gamma} \ell(z), \quad z \geq e, \quad \ell(z) = \frac{1}{1 + \log z}.$$

Suppose the threshold is fixed, e.g. $u_n \equiv e$, so that $k/n = \bar{F}(e) > 0$ and u_n does not diverge. Then

$$T_n(s) = \frac{\bar{F}(u_n s)}{\bar{F}(u_n)} = s^{-1/\gamma} \frac{1 + \log u_n}{1 + \log(u_n s)} = s^{-1/\gamma} \frac{1 + \log e}{1 + \log(es)} \neq s^{-1/\gamma},$$

so we still obtain $\tilde{T}_n(s) \xrightarrow{\mathbb{P}} T_n(s)$ but no longer $T_n(s) \not\xrightarrow{\mathbb{P}} T(s)$. This highlights the importance of the assumption $k/n \rightarrow 0$. $\diamond$

Definition 3.4 (Tail empirical process) Define for a sequence u_n of thresholds the tail empirical process as

$$\tilde{\mathbb{T}}_n(s) = \sqrt{k}(\tilde{T}_n(s) - T_n(s)), \quad \mathbb{T}_n(s) = \sqrt{k}(\tilde{T}_n(s) - T(s)), \quad s > 0.$$

Theorem 3.10 (Weak convergence of the tail empirical process) *On compacts bounded away from zero, $\tilde{\mathbb{T}}_n \xrightarrow{d} \mathbb{T}$ where $\mathbb{T} = \mathbb{W} \circ T$ and $\mathbb{W}$ is a standard Brownian motion.*

Proof. Let us find the limit covariance structure. Define

$$Z_{ni}(s) = k^{-1/2} \{1_{(Z_i > u_n s)} - \mathbb{P}(Z_i > u_n s)\},$$

so that $\tilde{\mathbb{T}}_n = \sum_{i=1}^n Z_{ni}$. We have:

$$\begin{aligned} n \text{Cov}(Z_{ni}(s), Z_{ni}(t)) &= \frac{\mathbb{P}(Z_1 > u_n(s \vee t)) - \mathbb{P}(Z_1 > u_n s) \mathbb{P}(Z_1 > u_n t)}{1 - F(u_n)} \\ &\rightarrow (s \vee t)^{-1/\gamma} - 0. \end{aligned}$$

To prove functional convergence, we make use of the Bracketing CLT for changing functions, Theorem 1.20. Consider the bounded interval $I = [a, b]$ with $a > 0$, and define

$$\mathcal{F}_n = \{f_{n,t} : [0, \infty) \rightarrow [0, \infty) \text{ s.t. } f_{n,t}(z) = \bar{F}(u_n)^{-1/2} 1_{[u_n t, \infty)}(z) \text{ for } t \in [a, b]\}.$$

Clearly the envelope is given by $F_n(Z_1) = \bar{F}(u_n)^{-1/2} 1_{(Z_1 > u_n a)}$. It satisfies by regular variation

$$\mathbb{E}[F_n(Z_1)^2] = \frac{\bar{F}(u_n a)}{\bar{F}(u_n)} = O(1),$$

showing condition (1.6). We also have by regular variation and $k \rightarrow \infty$ that

$$\begin{aligned} \mathbb{E}[F_n(Z_1)^2 1_{(F_n(Z_1) > c\sqrt{n})}] &\leq \frac{1}{c\sqrt{n}} \mathbb{E}[F_n(Z_1)^3] = \frac{1}{c(n\bar{F}(u_n))^{1/2}} \frac{\bar{F}(u_n a)}{\bar{F}(u_n)} \\ &\rightarrow 0 \cdot a^{-1/\gamma} / c = 0. \end{aligned}$$

This shows (1.7). The bracketing number is calculated as follows. Consider for $j = 1, \dots, m$ brackets on the form $l_j(y) = \bar{F}(u_n)^{-1/2} 1_{[u_n t_j, \infty)}(y)$, and similarly $u_j(y) = \bar{F}(u_n)^{-1/2} 1_{[u_n t_{j-1}, \infty)}(y)$ for sequences on the form $a = t_0 < t_1 < \dots < t_m = b$. Let $\mathcal{F}_j^n = \{f \in \mathcal{F}_n : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F}_n = \bigcup_{j=1}^m \mathcal{F}_j^n$. Let $\varepsilon > 0$ be given, and pick a sequence for $m \leq \lceil 2/(\varepsilon^2/2) \rceil$ such that

$$T(t_{j-1}) - T(t_j) \leq \varepsilon^2 a^{-1/\gamma} / 2 \quad \text{for } 0 \leq j \leq m,$$

which is possible because T is a continuous and bounded function on $[a, b]$. Then this sequence covers all of $\mathcal{F}_n$ with brackets satisfying

$$\begin{aligned} \|u_j(Z_1) - l_j(Z_1)\|_{L^2} &= \frac{\mathbb{P}(u_n t_j \geq Z_1 > u_n t_{j-1})^{1/2}}{(\bar{F}(u_n))^{1/2}} \\ &= (T(t_{j-1}) - T(t_j))^{1/2} (1 + o(1)) \\ &= (T(t_{j-1}) - T(t_j))^{1/2} a^{1/(2\gamma)} \|F_n(Z_1)\|_{L^2} (1 + o(1)) \\ &\leq \varepsilon \|F_n(Z_1)\|_{L^2}, \end{aligned}$$

where we have used regular variation twice. Consequently for some constant C ,

$$N_{[]}(\varepsilon \|F_n(Z_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})) \leq m \leq \lceil C/\varepsilon^2 \rceil.$$

Integration now shows (1.9). Note that the above $o(1)$ term is independent of the partition by (3.5). In particular, the partition is independent of n and (1.8) does not need to be verified. $\square$

We now define another key process which plays an important role when finding a suitable estimator for the deterministic and unobservable threshold u_n .

Definition 3.5 (Empirical quantile process) The empirical quantile process is defined by

$$\tilde{Q}_n(t) = \tilde{T}_n^{\leftarrow}(\lfloor kt \rfloor / k) = \frac{Z_{n, n - \lfloor kt \rfloor}}{u_n}, \quad t \in (0, n/k).$$

Setting $Q(t) = t^{-\gamma}$ we define the associated empirical process as

$$\mathbb{Q}_n(t) = \sqrt{k}(\tilde{Q}_n(t) - Q(t)), \quad t \in (0, n/k).$$

Define $B_n(s_0) = \sup_{t \geq s_0} |T_n(t) - T(t)|$, which converges uniformly to 0 for each $s_0 > 0$.

Theorem 3.11 (Joint convergence of tail empirical and quantile processes) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$. Let $b > 1$. Then*

$$(\mathbb{T}_n, \mathbb{Q}_n) \xrightarrow{d} (\mathbb{T}, -Q' \cdot \mathbb{T} \circ Q) = (\mathbb{W} \circ T, -Q' \mathbb{W})$$

on $\ell^\infty([s_0, \infty)) \times \ell^\infty((T(b), T(s_0)])$. In particular,

$$(\mathbb{T}_n, \sqrt{k}(Z_{n, n-k}/u_n - 1)) \xrightarrow{d} (\mathbb{T}, \gamma \cdot \mathbb{T}(1)) = (\mathbb{W} \circ T, \gamma \cdot \mathbb{W}(1)).$$

Proof. We combine the tail empirical CLT with the functional δ -method for the inverse map. We may write, on $[s_0, \infty)$, by assumption, $\mathbb{T}_n = \tilde{\mathbb{T}}_n + o(1)$, and the latter converges to $\mathbb{T}$ in $\ell^\infty([s_0, \infty))$. Then the desired joint convergence follows from an application of the functional δ -method, Theorem 1.17, applied with the inverse mapping (also called Vervaat's Lemma in the literature), as per Lemma 1.10, by recalling that for Q a differentiable function, $Q' = 1/T' \circ Q$. $\square$

We may now define a tail empirical process which is observable from data, and which also has tractable asymptotic properties.

Definition 3.6 (Tail empirical distribution function with random levels) Define the tail empirical distribution function with random levels and its empirical process as

$$\hat{T}_n(s) = \frac{1}{k} \sum_{i=1}^n 1_{(Z_i > Z_{n,n-k} s)}, \quad \hat{\mathbb{T}}_n(s) = \sqrt{k}(\hat{T}_n(s) - T(s)), \quad s > 0.$$

Theorem 3.12 (Weak convergence of tail empirical process with random levels) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$. Then $\hat{\mathbb{T}}_n \xrightarrow{d} \mathbb{T} - T \cdot \mathbb{T}(1) = \mathbb{B} \circ T$, on $\ell^\infty([s_0, \infty))$, where $\mathbb{B}$ is a standard Brownian bridge.*

Proof. We expand around the random level $Z_{n,n-k}$ and show each term is controlled by earlier limits. We simply write

$$\begin{aligned} \hat{\mathbb{T}}_n(s) &= \sqrt{k}(\tilde{T}_n(sZ_{n,n-k}/u_n) - T_n(sZ_{n,n-k}/u_n)) \\ &\quad + \sqrt{k}(T_n(sZ_{n,n-k}/u_n) - T(sZ_{n,n-k}/u_n)) \\ &\quad + \sqrt{k}(T(sZ_{n,n-k}/u_n) - T(s)). \end{aligned}$$

Since by Theorem 3.9 we have $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, the first term has pathwise convergence to $\mathbb{T}$ by Theorem 3.10, Slutsky's lemma, Lemma 1.9, and the continuous mapping theorem, Theorem 1.14. Again by $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$ we get

$$\mathbb{P}(|T_n(sZ_{n,n-k}/u_n) - T(sZ_{n,n-k}/u_n)| \leq B_n(s_0)) \rightarrow 1,$$

for any $s \geq s_0$, so by assumption the second term becomes pathwise $o_{\mathbb{P}}(1)$.

Finally, we may pull out $T(s)$ from the last term, and then apply the δ -method, Theorem 1.10, to the convergence of Theorem 3.11 and the transform $y \mapsto T(y)$ with derivative $T'(y) = -y^{-1/\gamma-1}/\gamma$ to obtain pathwise convergence of the last term (jointly with the first), to $s^{-1/\gamma}(-1/\gamma) \cdot \gamma \mathbb{T}(1) = -T(s)\mathbb{T}(1)$. $\square$

3.3.2 The Hill estimator

We now introduce the Hill estimator and establish its basic asymptotic properties. It is the canonical tail index estimator and admits a simple integral representation in terms of the tail empirical function. Indeed, notice that we may re-express

$$\gamma_{n,k} = \int_1^\infty \hat{T}_n(t)/t \, dt.$$

In practice, for a fixed value of n , the value of k must be chosen with care. Taking k too small yields high variance, while taking k too large introduces bias from non-Pareto tail behaviour. A common diagnostic is the Hill plot, which plots $\gamma_{n,k}$ as a function of the smallest k -values. Then one should look for a stable region in k where the estimator is relatively flat, and take the largest k in that region.

Theorem 3.13 (Weak consistency of the Hill estimator) *We have that $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$.*

Proof. We write

$$\begin{aligned} \gamma_{n,k} &= \int_1^\infty \frac{\widehat{T}_n(t)}{t} dt = \int_1^\infty \frac{\widetilde{T}_n(tZ_{n,n-k}/u_n)}{t} dt \\ &= \int_{Z_{n,n-k}/u_n}^\infty \frac{\widetilde{T}_n(t)}{t} dt = \int_{Z_{n,n-k}/u_n}^1 \frac{\widetilde{T}_n(t)}{t} dt + \int_1^\infty \frac{\widetilde{T}_n(t)}{t} dt \end{aligned} \quad (3.6)$$

Since $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, the first integral is $o_{\mathbb{P}}(1)$. By consistency of the tail empirical function on bounded intervals, Theorem 3.9, for any $m > 0$,

$$\gamma_{nm} := \int_1^m \frac{\widetilde{T}_n(t)}{t} dt \xrightarrow{\mathbb{P}} \int_1^m \frac{T(t)}{t} dt =: \gamma_m$$

and $\lim_{m \rightarrow \infty} \gamma_m = \gamma$. Thus, by the approximation theorem for random elements, the proof is complete by Theorem 1.5 if we can show that $\gamma_{n,k} - \gamma_{nm}$ vanishes as $n \rightarrow \infty$ first and then $m \rightarrow \infty$, i.e. for $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\widetilde{T}_n(t)}{t} dt > \eta \right) = 0.$$

Note that by Potter's bounds, Proposition 3.6, we have for any $\varepsilon \in (0, 1/\gamma)$

$$\mathbb{E} \int_m^\infty \frac{\widetilde{T}_n(t)}{t} dt = \int_m^\infty \frac{1 - F(u_n t)}{t(1 - F(u_n))} dt \leq \text{cst} \int_m^\infty t^{-1/\gamma-1+\varepsilon} dt = O(m^{-1/\gamma+\varepsilon}).$$

The above expression is vanishing, so an application of Markov's inequality (equation (1.1) with φ the identity function) establishes the desired result. $\square$

Theorem 3.14 (Weak convergence of the Hill estimator) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$ and*

$$\lim_{n \rightarrow \infty} \sqrt{k} \int_1^\infty \frac{T_n(t) - T(t)}{t} dt = 0. \quad (3.7)$$

Then $\sqrt{k}(\gamma_{n,k} - \gamma) \xrightarrow{d} N(0, \gamma^2)$.

Proof. By the decomposition (3.6) we get

$$\begin{aligned} &\sqrt{k}(\gamma_{n,k} - \gamma) \\ &= \int_1^\infty \frac{\widetilde{T}_n(t)}{t} dt + \sqrt{k} \int_1^\infty \frac{T_n(t) - T(t)}{t} dt + \sqrt{k} \int_{Z_{n,n-k}/u_n}^1 \frac{\widetilde{T}_n(t)}{t} dt. \end{aligned} \quad (3.8)$$

The middle term vanishes by assumption. Consider the last term. Since $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$ we may assume w.l.o.g. that for a given $\varepsilon \in (0, 1)$ we have $Z_{n,n-k}/u_n \in (1 - \varepsilon, 1 + \varepsilon)$ for n large enough. Then by consistency of the tail empirical function and the last limit result of Theorem 3.11,

$$\begin{aligned} & \sqrt{k} \left| \int_{Z_{n,n-k}/u_n}^1 \frac{\tilde{T}_n(t)}{t} dt - \int_{Z_{n,n-k}/u_n}^1 \frac{T(t)}{t} dt \right| \\ & \leq \frac{\sqrt{k}}{1 - \varepsilon} \sup_{t \in (1 - \varepsilon, 1 + \varepsilon)} |\tilde{T}_n(t) - T(t)| |Z_{n,n-k}/u_n - 1| = o_{\mathbb{P}}(1). \end{aligned}$$

Thus the asymptotic of the last term is equivalent to that of

$$\sqrt{k} \int_{Z_{n,n-k}/u_n}^1 T(t)/t dt,$$

which by the δ -method, Theorem 1.10, applied to the function $x \mapsto \int_x^1 \frac{T(t)}{t} dt$, has the same limit as

$$\sqrt{k}(Z_{n,n-k}/u_n - 1)\{-T(\zeta_n)/\zeta_n\}$$

for some ζ_n between $Z_{n,n-k}/u_n$ and 1. It remains to note that the term in curly brackets converges in probability to -1 . Next consider the first term in (3.8), where we use a similar argument as in the proof of consistency. Namely, we know from Theorem 3.10 that $\int_1^m \tilde{\mathbb{T}}_n(t)/t dt \xrightarrow{d} \int_1^m \mathbb{T}(t)/t dt$ and that as $m \rightarrow \infty$, $\int_m^\infty \mathbb{T}(t)/t dt \xrightarrow{\mathbb{P}} 0$ (it is centred and has variance going to zero). Thus the proof is complete by Theorem 1.5 if we can show that for every $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{\mathbb{T}}_n(t)}{t} dt > \eta \right) = 0.$$

It suffices by Markov's inequality to show the corresponding limit property for the variance, which is given as follows:

$$\begin{aligned} \text{Var} \left(\int_m^\infty \frac{\tilde{\mathbb{T}}_n(t)}{t} dt \right) &= \frac{1}{k} \text{Var} \left(\int_m^\infty \frac{\sum_{i=1}^n \{1_{(Z_i > u_n t)} - \mathbb{P}(Z_i > u_n t)\}}{t} dt \right) \\ &= \frac{n}{k} \text{Var} \left(\int_m^{(Z_1/u_n) \vee m} \frac{1}{t} dt \right) = \frac{1}{\bar{F}(u_n)} \text{Var} \left(\log_+(Z_1/(m \cdot u_n)) \right) \\ &\leq \frac{\mathbb{E}[\log_+(Z_1/(m \cdot u_n))^2]}{\bar{F}(u_n)} = \frac{2 \int_{mu_n}^\infty \log(t/(mu_n)) \bar{F}(t) \frac{1}{t} dt}{\bar{F}(u_n)} \\ &= \frac{2 \int_1^\infty \log(t) \bar{F}(mu_n t) \frac{1}{t} dt}{\bar{F}(u_n)} \leq m^{-1/\gamma+\varepsilon} \cdot 2 \int_1^\infty \log(t) t^{-1/\gamma+\varepsilon-1} dt \end{aligned}$$

where we have used Potter's bounds, Theorem 3.6, in the last step with $\varepsilon \in (0, 1/\gamma)$. The final quantity does not depend on n and furthermore vanishes as $m \rightarrow \infty$, as desired. Finally, we get that the weak limit of $\sqrt{k}(\gamma_{n,k} - \gamma)$ has the following distributional representation

$$\int_1^\infty \frac{\mathbb{W}(T(t))}{t} dt - \gamma \mathbb{W}(1) = \int_1^\infty \frac{\mathbb{B}(T(t))}{t} dt = \gamma \int_0^1 \frac{\mathbb{B}(t)}{t} dt,$$

which has variance given by $\gamma^2 \int_0^1 \int_0^1 (s^{-1} \wedge t^{-1} - 1) ds dt = \gamma^2$. $\square$

To verify the bias condition in practice, it is useful to have simple tail-expansion criteria.

Proposition 3.1 (Class of distributions with reasonable bias) *Let F be a continuous distribution function on $\mathbb{R}$ and assume that there exist $c > 0$ and $\beta > 1/\gamma > 0$ such that*

$$\sup_{t \geq 1} t^\beta |\bar{F}(t) - ct^{-1/\gamma}| < \infty.$$

Let $u_n = F^{\leftarrow}(1 - k/n)$ so that $n\bar{F}(u_n) = k$, and define $T_n(t) = k^{-1}n\bar{F}(u_nt)$. If $k = o(n^\rho)$ with $\rho = 2(\beta\gamma - 1)/(2\beta\gamma - 1)$, then (3.7) holds.

Proof. For $t \geq 1$,

$$\begin{aligned} |T_n(t) - t^{-1/\gamma}| &\leq k^{-1}n|\bar{F}(u_nt) - c(u_nt)^{-1/\gamma}| + k^{-1}n|cu_n^{-1/\gamma} - \bar{F}(u_n)| t^{-1/\gamma} \\ &\leq \text{cst } k^{-1}nu_n^{-\beta}(t^{-\beta} + t^{-1/\gamma}) \leq \text{cst } k^{-1}nu_n^{-\beta}t^{-1/\gamma}, \end{aligned}$$

since $\beta > 1/\gamma$ and $t \geq 1$. The tail expansion implies $u_n = O((n/k)^\gamma)$. Therefore

$$\begin{aligned} \sqrt{k} \int_1^\infty \frac{|T_n(t) - t^{-1/\gamma}|}{t} dt &\leq \text{cst } \sqrt{k} k^{-1}nu_n^{-\beta} \int_1^\infty t^{-1/\gamma-1} dt \\ &\leq \text{cst } k^{-1/2+\beta\gamma} n^{1-\beta\gamma} = \text{cst } \frac{k^{(2\beta\gamma-1)/2}}{n^{\beta\gamma-1}}. \end{aligned}$$

If $k = o(n^\rho)$ with $\rho = 2(\beta\gamma - 1)/(2\beta\gamma - 1)$, the last term tends to 0, which yields (3.7). $\square$

The next example shows that the bias condition can be quite restrictive even for regularly varying tails with mild slowly varying components.

Example 3.4 (Big bias example) Consider the Pareto-type distribution with a heavy slowly varying component:

$$\bar{F}(z) = z^{-1/\gamma} \log z, \quad z \geq e.$$

Here, the slowly varying function is $\ell(z) = \log z$. Let u_n be the threshold associated with the sample fraction k , such that $k \sim n\bar{F}(u_n)$. We analyse the behaviour of the tail process:

$$T_n(t) = t^{-1/\gamma} \frac{\ell(u_nt)}{\ell(u_n)} = t^{-1/\gamma} \frac{\log(u_nt)}{\log u_n}.$$

Consequently

$$\int_1^\infty \frac{T_n(t) - T(t)}{t} dt = \int_1^\infty t^{-1/\gamma-1} \frac{\log t}{\log u_n} dt = \frac{1}{\log u_n} \int_1^\infty t^{-1/\gamma-1} \log t dt,$$

The integral $\int_1^\infty t^{-1/\gamma-1} \log t dt$ evaluates to γ^2 . Thus, the deterministic bias is exactly $\gamma^2/\log u_n$. The bias condition (3.7) then requires the vanishing of

$$\sqrt{k} \frac{\gamma^2}{\log u_n} \sim \frac{\gamma^2 \sqrt{k}}{\gamma \log(n/k)} \sim \gamma \frac{\sqrt{k}}{\log n}.$$

This means that k cannot be taken as a power of n , and the estimator has a strong systematic bias; for instance for a sample size of $n = 100,000$ we require $k \ll \log(n)^2 = 25$.

The pathology is even more severe if $\ell(z) = \exp((\log z)^\eta)$ for $0 < \eta < 1$. A first-order Taylor expansion yields:

$$\sqrt{k} \int_1^\infty \frac{T_n(t) - T(t)}{t} dt \sim \gamma \eta \frac{\sqrt{k}}{(\log n)^{1-\eta}}.$$

For asymptotic normality to hold, we require $k \ll (\log n)^{2(1-\eta)}$. If $\eta = 0.95$, then $k \ll (\log n)^{0.1}$. Even with $n = 10^{80}$ (number of atoms in the universe) we require $k \ll 1.684731$. So while $k \rightarrow \infty$ technically exists, one cannot obtain this regime in practice. $\diamond$

3.3.3 Probability and quantile estimation

Small tail probability and high quantile estimators for regularly varying distributions are motivated by a two-step procedure. First, a high quantile is estimated using the available data, providing a baseline from which further extrapolation can be made. In the second step, the tail index is used as the leading coefficient for extrapolation beyond the observed data. The Hill estimator is employed to estimate the latter index. This combination of direct high quantile estimation and tail index extrapolation sometimes goes under the name of *Weissman estimation*.

Focusing first on high quantiles, and writing in this subsection $Q(p) = F^\leftarrow(p)$ for the distribution quantile, inverting the Pareto tail approximation gives the *quantile estimator* of $Q(p)$

$$\hat{Q}(p) = ((1-p)n/k)^{-\gamma_{n,k}} Z_{n,n-k}, \quad p \in [1-k/n, 1).$$

The consistency of the ingredients already follows from Theorems 3.9 and 3.13. For statistical inference one needs a finer statement, and the extra work is to control the deterministic extrapolation bias at the level where the probability $1-p_n$ is evaluated. The next theorem gives a convenient condition directly in terms of $\bar{F}(u_n t)/\bar{F}(u_n)$.

Theorem 3.15 (Asymptotic normality of the Weissman quantile estimator) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k}B_n(s_0) \rightarrow 0$ and (3.7) holds. Let (p_n) be a sequence with $c_n := n(1-p_n)/k \rightarrow 0$. Assume also that F is continuous, that $\log(1/c_n) = o(\sqrt{k})$, and that*

$$\lim_{n \rightarrow \infty} \frac{\sqrt{k}}{\log(1/c_n)} \sup_{t \geq 1} \left| t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1 \right| = 0. \quad (3.9)$$

Then

$$\frac{\sqrt{k}}{\log(1/c_n)} \left(\frac{\hat{Q}(p_n)}{Q(p_n)} - 1 \right) \xrightarrow{d} N(0, \gamma^2).$$

Proof. Let $t_n = Q(p_n)/u_n$. Since $n(1-p_n)/k = c_n \rightarrow 0$, we have $1-p_n < k/n$ for n large enough and therefore $Q(p_n) \geq u_n$, so $t_n \geq 1$ eventually. By continuity

of F and the definition of u_n ,

$$\frac{\bar{F}(u_n t_n)}{\bar{F}(u_n)} = \frac{\bar{F}(Q(p_n))}{\bar{F}(u_n)} = c_n.$$

Now write

$$\frac{\hat{Q}(p_n)}{Q(p_n)} - 1 = A_n B_n C_n - 1,$$

where

$$A_n = \frac{Z_{n,n-k}}{u_n}, \quad B_n = c_n^{-(\gamma_{n,k}-\gamma)}, \quad C_n = \frac{c_n^{-\gamma} u_n}{Q(p_n)} = \frac{c_n^{-\gamma}}{t_n}.$$

Accordingly,

$$A_n B_n C_n - 1 = (A_n - 1) B_n C_n + (B_n - 1) C_n + (C_n - 1).$$

We show that the first and third terms are negligible after scaling by $\sqrt{k}/\log(1/c_n)$, while the second term provides the required asymptotic normality.

By Theorem 3.9, $A_n \xrightarrow{\mathbb{P}} 1$, and by Theorem 3.14, $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$. Moreover, since $\sqrt{k}(\gamma_{n,k} - \gamma) = O_{\mathbb{P}}(1)$ and $\log(1/c_n) = o(\sqrt{k})$, we get

$$(\gamma_{n,k} - \gamma) \log(1/c_n) \xrightarrow{\mathbb{P}} 0,$$

so $B_n \xrightarrow{\mathbb{P}} 1$. To analyse C_n , define

$$r_n(t) = t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1, \quad t \geq 1.$$

Then we get

$$c_n = t_n^{-1/\gamma} (1 + r_n(t_n)).$$

Therefore

$$C_n = \frac{c_n^{-\gamma}}{t_n} = (1 + r_n(t_n))^{-\gamma}.$$

By (3.9) and $\log(1/c_n) = o(\sqrt{k})$ we have $\sup_{t \geq 1} |r_n(t)| \rightarrow 0$, hence $C_n \rightarrow 1$. In addition, for n large enough, $\sup_{t \geq 1} |r_n(t)| \leq 1/2$, and then the mean value theorem gives

$$|C_n - 1| = |(1 + r_n(t_n))^{-\gamma} - 1| \leq \text{cst } |r_n(t_n)| \leq \text{cst } \sup_{t \geq 1} |r_n(t)|.$$

Consequently, we even get

$$\frac{\sqrt{k}}{\log(1/c_n)} (C_n - 1) \xrightarrow{\mathbb{P}} 0, \tag{3.10}$$

which deals with the third term. The first term is negligible because Theorem 3.11 yields $\sqrt{k}(A_n - 1) = O_{\mathbb{P}}(1)$, while $\log(1/c_n) \rightarrow \infty$ and $B_n C_n \xrightarrow{\mathbb{P}} 1$. Finally, for the middle term, by Taylor expansion of the exponential we obtain

$$\frac{\sqrt{k}}{\log(1/c_n)} (B_n - 1) C_n = \sqrt{k}(\gamma_{n,k} - \gamma)(1 + o_{\mathbb{P}}(1)) \xrightarrow{d} N(0, \gamma^2)$$

by Theorem 3.14. The result now follows from Slutsky's theorem. $\square$

The rate condition (3.9) is abstract on purpose, and it studies exactly the extrapolation error that is required for the Weissman step. It is always satisfied if one imposes second-order conditions found in standard literature. The next example shows that it can be verified explicitly for oscillating slowly varying functions (which do not obey the classic second order conditions), and that one can choose concrete sequences (k_n) and (p_n) satisfying all requirements.

Example 3.5 (Oscillating slowly varying tail and admissible sequences) Fix $\gamma > 0$ and $A_0 \in \mathbb{R}$. For sufficiently large z , let

$$\bar{F}(z) = z^{-1/\gamma} \ell(z), \quad \ell(z) = 1 + \frac{A_0 \cos(\log z)}{\log z}.$$

Since $\ell(z) \rightarrow 1$ and $\ell'(z) = O((z \log z)^{-1})$, the derivative of $z^{-1/\gamma} \ell(z)$ is negative for z large enough. After modifying the distribution on a bounded interval (and, if needed, starting the tail representation at a larger point), this defines a distribution with regularly varying tail index γ . The factor ℓ is slowly varying, but the oscillation prevents the usual monotone second-order parametrisations from holding. Nevertheless, the condition in Theorem 3.15 is easy to verify.

Indeed, for $t \geq 1$,

$$t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1 = \frac{\ell(u_n t)}{\ell(u_n)} - 1.$$

Since $\log(u_n t) \geq \log u_n$ and $\ell(u_n) \rightarrow 1$, we have uniformly in $t \geq 1$ that

$$|\ell(u_n t) - \ell(u_n)| \leq \frac{2|A_0|}{\log u_n},$$

and therefore

$$\sup_{t \geq 1} \left| t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1 \right| = O(1/\log u_n).$$

Because $k/n = \bar{F}(u_n) = u_n^{-1/\gamma} \ell(u_n)$ and $\ell(u_n) \rightarrow 1$, we also have that $\log u_n$ is asymptotically equivalent to $\log(n/k)$. Hence the rate condition follows if we can prescribe

$$\frac{\sqrt{k}}{\log(n/k) \log(1/c_n)} \rightarrow 0.$$

As a concrete choice, take

$$k_n = \lfloor (\log n)^{3/2} \rfloor, \quad c_n = \exp\{-(\log n)^{1/4}\}, \quad p_n = 1 - \frac{k_n}{n} c_n.$$

Then $k_n \rightarrow \infty$, $k_n/n \rightarrow 0$, and $n(1 - p_n)/k_n = c_n \rightarrow 0$. Moreover,

$$\log(1/c_n) = (\log n)^{1/4} = o(\sqrt{k_n}).$$

Finally,

$$\frac{\sqrt{k_n}}{\log(n/k_n) \log(1/c_n)} \sim \frac{(\log n)^{3/4}}{(\log n)(\log n)^{1/4}} = (\log n)^{-1/2} \rightarrow 0,$$

so the speed conditions of Theorem 3.15 are satisfied. It remains to check the Hill assumptions. Fix $s_0 \in (0, 1)$. For $t \geq s_0$ and n large enough, the same argument as above gives

$$\left| \frac{\ell(u_n t)}{\ell(u_n)} - 1 \right| \leq \frac{\text{cst}}{\log u_n},$$

and therefore

$$B_n(s_0) = \sup_{t \geq s_0} |T_n(t) - T(t)| \leq \text{cst } s_0^{-1/\gamma} \frac{1}{\log u_n} = O(1/\log u_n).$$

Hence $\sqrt{k_n} B_n(s_0) \rightarrow 0$. Likewise,

$$\left| \int_1^\infty \frac{T_n(t) - T(t)}{t} dt \right| \leq \frac{\text{cst}}{\log u_n} \int_1^\infty t^{-1/\gamma-1} dt = O(1/\log u_n),$$

so $\sqrt{k_n} \int_1^\infty (T_n(t) - T(t)) t^{-1} dt \rightarrow 0$, i.e. condition (3.7) holds. Thus the assumptions of Theorem 3.15 are satisfied. $\diamond$

The same extrapolation principle can be written in probability form. The corresponding Weissman tail probability estimator is

$$\hat{\bar{F}}(z) = \frac{k}{n} \left(\frac{z}{Z_{n,n-k}} \right)^{-1/\gamma_{n,k}}, \quad z > Z_{n,n-k}.$$

When the extrapolation point is chosen as $z_n = Q(p_n)$, the same rate $c_n = n(1-p_n)/k$ appears as in Theorem 3.15, and the proof may be carried out in a similar fashion.

Theorem 3.16 (Asymptotic normality of the Weissman probability estimator) *Let $k = k_n$ grow slowly enough such that for some $s_0 \in (0, 1)$, $\sqrt{k} B_n(s_0) \rightarrow 0$ and (3.7) holds. Let (p_n) be a sequence with $c_n := n(1-p_n)/k \rightarrow 0$. Assume also that F is continuous, that $\log(1/c_n) = o(\sqrt{k})$, and that (3.9) holds. Then*

$$\frac{\gamma \sqrt{k}}{\log(1/c_n)} \left(\frac{\hat{\bar{F}}(Q(p_n))}{1-p_n} - 1 \right) \xrightarrow{d} N(0, \gamma^2).$$

Proof. Let $t_n = Q(p_n)/u_n$. Since $c_n = n(1-p_n)/k \rightarrow 0$, we have $1-p_n < k/n$ for n large enough and therefore $Q(p_n) \geq u_n$, so $t_n \geq 1$ eventually. By continuity of F and the definition of u_n ,

$$\frac{\bar{F}(u_n t_n)}{\bar{F}(u_n)} = \frac{\bar{F}(Q(p_n))}{\bar{F}(u_n)} = c_n.$$

Now write

$$\frac{\hat{\bar{F}}(Q(p_n))}{1-p_n} = A_n B_n C_n,$$

where

$$A_n = \left(\frac{Z_{n,n-k}}{u_n} \right)^{1/\gamma}, \quad B_n = \left(\frac{Z_{n,n-k}}{Q(p_n)} \right)^{1/\gamma_{n,k}-1/\gamma},$$

$$C_n = \frac{(Q(p_n)/u_n)^{-1/\gamma}}{c_n} = \frac{t_n^{-1/\gamma}}{c_n}.$$

Accordingly,

$$A_n B_n C_n - 1 = (A_n - 1) B_n C_n + (B_n - 1) C_n + (C_n - 1).$$

We now use the same decomposition strategy as in Theorem 3.15: the first and third terms are negligible after scaling by $\gamma\sqrt{k}/\log(1/c_n)$, while the second term carries the asymptotic normality.

By Theorem 3.9, $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, hence $A_n \xrightarrow{\mathbb{P}} 1$. By Theorem 3.14, $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$ and $\sqrt{k}(\gamma_{n,k} - \gamma) = O_{\mathbb{P}}(1)$. To analyse C_n , define

$$r_n(t) = t^{1/\gamma} \frac{\bar{F}(u_n t)}{\bar{F}(u_n)} - 1, \quad t \geq 1.$$

Then $T_n(t) = t^{-1/\gamma}(1 + r_n(t))$, and at $t = t_n$ the identity above gives

$$C_n = \frac{t_n^{-1/\gamma}}{c_n} = (1 + r_n(t_n))^{-1}.$$

Exactly as in the proof of Theorem 3.15, condition (3.9) implies $\sup_{t \geq 1} |r_n(t)| \rightarrow 0$, hence $C_n \rightarrow 1$, and the mean value theorem yields $|C_n - 1| \leq \text{cst} \sup_{t \geq 1} |r_n(t)|$. Thus

$$\frac{\gamma\sqrt{k}}{\log(1/c_n)}(C_n - 1) \xrightarrow{\mathbb{P}} 0, \quad (3.11)$$

which deals with the third term. Since $C_n = t_n^{-1/\gamma}/c_n \rightarrow 1$, we also get

$$\frac{\log t_n}{\gamma \log(1/c_n)} = 1 + \frac{\log C_n}{\log(1/c_n)} \xrightarrow{\mathbb{P}} 1.$$

As $\log(Z_{n,n-k}/u_n) = o_{\mathbb{P}}(1)$ and $\log(1/c_n) \rightarrow \infty$, it follows that

$$\frac{\log(Q(p_n)/Z_{n,n-k})}{\gamma \log(1/c_n)} = \frac{\log t_n - \log(Z_{n,n-k}/u_n)}{\gamma \log(1/c_n)} \xrightarrow{\mathbb{P}} 1. \quad (3.12)$$

In particular, $\log(Q(p_n)/Z_{n,n-k}) = O_{\mathbb{P}}(\log(1/c_n))$, so $B_n \xrightarrow{\mathbb{P}} 1$ because $\gamma_{n,k} \xrightarrow{\mathbb{P}} \gamma$ and $(\gamma_{n,k} - \gamma) \log(1/c_n) \xrightarrow{\mathbb{P}} 0$. The first-term estimate is also identical in structure to the quantile case: Theorem 3.11 and the δ -method yield $\sqrt{k}(A_n - 1) = O_{\mathbb{P}}(1)$, and since $\log(1/c_n) \rightarrow \infty$ with $B_n C_n \xrightarrow{\mathbb{P}} 1$, we get

$$\frac{\gamma\sqrt{k}}{\log(1/c_n)}(A_n - 1) B_n C_n \xrightarrow{\mathbb{P}} 0,$$

which deals with the first term. Finally, for the middle term, by Taylor expansion, and since $C_n \xrightarrow{\mathbb{P}} 1$, we obtain

$$\begin{aligned} \frac{\gamma\sqrt{k}}{\log(1/c_n)}(B_n - 1)C_n &= \gamma\sqrt{k} \left(\frac{1}{\gamma} - \frac{1}{\gamma_{n,k}} \right) \frac{\log(Q(p_n)/Z_{n,n-k})}{\log(1/c_n)} (1 + o_{\mathbb{P}}(1)) \\ &= \sqrt{k}(\gamma_{n,k} - \gamma) \frac{\gamma}{\gamma_{n,k}} \frac{\log(Q(p_n)/Z_{n,n-k})}{\gamma \log(1/c_n)} (1 + o_{\mathbb{P}}(1)) \\ &\xrightarrow{d} N(0, \gamma^2), \end{aligned}$$

by Theorem 3.14, (3.12), and Slutsky's theorem. Combining the three terms completes the proof. $\square$

Numeric example 3.1 (Weissman versus empirical tail-probability estimation) To compare extrapolation with a non-extrapolating benchmark at a fixed sample size, consider the Burr distribution with distribution function

$$F(z) = 1 - (1 + (z/\theta)^\eta)^{-\beta}, \quad z \geq 0, \quad (3.13)$$

where $\theta, \eta, \beta > 0$. In this example we take $\theta = 1$ and $\eta = \beta = \sqrt{2}$, so the extreme-value index is $\gamma = (\beta\eta)^{-1} = 1/2$.

We fix $n = 5000$ and simulate iid samples from (3.13). For each sample, each intermediate value k , and each tail level z , we compare the Weissman estimator

$$\hat{\bar{F}}(z) = \frac{k}{n} \left(\frac{z}{Z_{n,n-k}} \right)^{-1/\gamma_{n,k}}, \quad z > Z_{n,n-k},$$

with the empirical tail estimator

$$1 - \mathbb{F}^{(n)}(z) = \frac{1}{n} \sum_{i=1}^n 1_{(Z_i > z)}.$$

To study the effect of extrapolation, we evaluate both estimators against the true tail probability $\bar{F}(z)$ and report the relative root mean squared error

$$\text{RRMSE}(\hat{p}(z)) = (\mathbb{E}[\{\hat{p}(z)/\bar{F}(z) - 1\}^2])^{1/2},$$

approximated by Monte Carlo averages. Figure 3.1 is based on 600 repetitions. The left panel shows the RRMSE as a function of z for three representative values of k , together with the empirical benchmark. The right panel shows the RRMSE as a function of k for three fixed tail levels z ; for each colour, the dashed horizontal line is the empirical estimator and the solid curve is the Weissman estimator. The comparison illustrates the usual bias-variance trade-off in the choice of k , and also the gain from extrapolation once z is far into the tail. $\diamond$

3.4 Other estimators in extremes

There exist other famous estimators for γ when there is no restriction on the max-domain of attraction. We briefly survey three of them below, and provide, mostly without proofs, their asymptotic distributions. This gives a broader context for comparing the Hill estimator in practice.

We begin by introducing the Pickands estimator, which is simply defined, but has the drawback of being based on only three distinct points, as opposed to using an entire set of top order statistics.

Definition 3.7 (Pickands estimator) The Pickands estimator for the index $\gamma \in \mathbb{R}$ is given by:

$$\gamma_{n,k}^P = \frac{1}{\log 2} \log \left(\frac{Z_{n,n-k+1} - Z_{n,n-2k+1}}{Z_{n,n-2k+1} - Z_{n,n-4k+1}} \right).$$

Theorem 3.17 (Consistency and asymptotic normality of Pickands estimator) Let (Z_n) be an iid sequence satisfying (3.3) for some $\gamma \in \mathbb{R}$. We have that

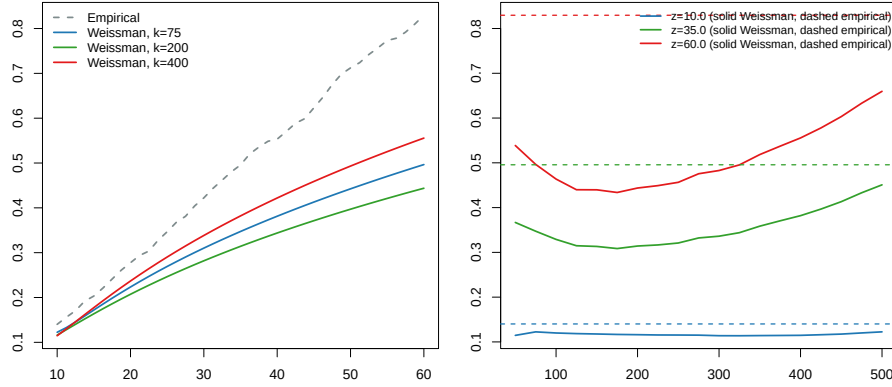


Figure 3.1: Finite-sample comparison of the Weissman tail-probability estimator and the empirical tail estimator under the Burr model (3.13) with $(\theta, \eta, \beta) = (1, \sqrt{2}, \sqrt{2})$, based on $n = 5000$ and 600 Monte Carlo repetitions. Left panel: relative root mean squared error as a function of the tail level z for $k \in \{75, 200, 400\}$. Right panel: relative root mean squared error as a function of k for three fixed tail levels z . In both panels, solid curves denote Weissman estimates and dashed curves denote empirical tail estimates.

$\gamma_{n,k}^P \xrightarrow{\mathbb{P}} \gamma$. Furthermore, if $k/\log \log n \rightarrow \infty$ then $\gamma_{n,k}^P \xrightarrow{a.s.} \gamma$. Finally, under suitable conditions on F and k provided in Dekkers and de Haan [1989],

$$\sqrt{k}(\gamma_{n,k}^P - \gamma) \xrightarrow{d} N\left(0, \gamma^2 \frac{2^{2\gamma+1} + 1}{(2(2^\gamma - 1) \log 2)^2}\right).$$

Next, we define the maximum likelihood estimators applied to peaks over thresholds, which is a parametric estimator, relying on the GPD.

Definition 3.8 (Peaks over thresholds (POT) maximum likelihood estimator (MLE)) The MLE estimator for excesses (Y_n) over the thresholds u for the index $\gamma \in \mathbb{R}$ is given by the argument that minimises the likelihood constructed using densities from the GPD (3.4), with $\mu = 0$ and (γ, σ) as parameters. We denote the resulting estimators $(\gamma_{n,u}, \sigma_{n,u})$.

Theorem 3.18 (Asymptotic normality of the GPD MLE estimator) Let (Z_n) be an iid sequence satisfying (3.3) for some $\gamma > -1/2$. Let k be the number of exceedances over the threshold u , and denote by $Y_1, \dots, Y_k$ the sequence of excesses over u . Then, under suitable conditions on F and the speed at which $k \rightarrow \infty$ and $k/n \rightarrow 0$ provided in Smith [1987], we get

$$k^{1/2}(\gamma_{k,u} - \gamma, \sigma_{k,u}/\sigma - 1) \xrightarrow{d} N(0, \Sigma(\gamma)),$$

where

$$\Sigma(\gamma) = (1 + \gamma) \begin{pmatrix} 1 + \gamma & 1 \\ 1 & 2 \end{pmatrix}.$$

Finally, we present an estimator which is very similar to the Hill estimator in that it uses the top k (scaled) order statistics. It is worth noting that it is known in advance that $\gamma > 0$, then the Hill estimator has a lower variance and consequently is preferable.

Definition 3.9 (Moment Estimator) Define

$$M_n^{(r)} = \frac{1}{k} \sum_{i=1}^k \log(Z_{n,n-i+1}/Z_{n,n-k})^r, \quad r = 1, 2.$$

The Moment estimator for the index $\gamma \in \mathbb{R}$ is given by:

$$\gamma_{n,k}^M := M_n^{(1)} + 1 - \frac{1}{2} \left\{ 1 - \frac{(M_n^{(1)})^2}{M_n^{(2)}} \right\}^{-1}.$$

The following result, for regularly varying tails, shows that the moment estimator is indeed a consistent estimator.

Proposition 3.2 (Weak consistency of the moment estimator) *We have that*
 $\gamma_{n,k}^M \xrightarrow{\mathbb{P}} \gamma$.

Proof. We only show the $\gamma > 0$ case. Since the first moment is the Hill estimator, we focus on the second moment. First note

$$M_n^{(2)} = - \int_1^\infty \log(t)^2 \hat{T}_n(dt) = 2 \int_1^\infty \frac{\log(t)}{t} \hat{T}_n(t) dt$$

By following the same technique as in the consistency of the Hill estimator we get

$$\begin{aligned} M_n^{(2)} &= 2 \int_{Z_{n,n-k}/u_n}^1 \frac{\log(t)}{t} \tilde{T}_n(t) dt + 2 \int_1^\infty \frac{\log(t)}{t} \tilde{T}_n(t) dt \\ &\quad + 2 \log(Z_{n,n-k}/u_n) \int_{Z_{n,n-k}/u_n}^\infty \frac{1}{t} \tilde{T}_n(t) dt. \end{aligned}$$

By consistency of the Hill estimator and $Z_{n,n-k}/u_n \xrightarrow{\mathbb{P}} 1$, i.e. Theorem 3.13 and Theorem 3.9, we have that the last term vanishes in probability. Using the same argument as in Theorem 3.9, the first term is bounded in probability. By uniform consistency of the tail empirical function on bounded intervals, Theorem 3.9, then for any $m > 0$,

$$\gamma_{nm} := \int_1^m \frac{\tilde{T}_n(t) \log(t)}{t} dt \xrightarrow{\mathbb{P}} \int_1^m \frac{T(t) \log(t)}{t} dt =: \gamma_m$$

and $\lim_{m \rightarrow \infty} \gamma_m = \gamma^2$. Thus, we have shown the result if for $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{T}_n(t) \log(t)}{t} dt > \eta \right) = 0.$$

For large enough m , we can pick any $1/\gamma > \delta > 0$ such that $\log(x) < x^\delta$ for $x > m$. Furthermore, we can pick a $\varepsilon \in (0, 1/\gamma - \delta)$.

$$\begin{aligned} \mathbb{E} \int_m^\infty \frac{\log(t) \tilde{T}_n(t)}{t} dt &= \int_m^\infty \log(t) \frac{1 - F(u_n t)}{t(1 - F(u_n))} dt \\ &\leq \text{cst} \int_m^\infty t^{-1/\gamma-1+\delta+\varepsilon} dt = O(m^{-1/\gamma+\delta+\varepsilon}) \end{aligned}$$

We get $M_n^{(2)} \xrightarrow{\mathbb{P}} 2\gamma^2$ by Markov's inequality. The result follows. $\square$

The general result, for all max-domains of attraction, including asymptotic normality, is given below.

Theorem 3.19 (Consistency and asymptotic normality of the moment estimator) *Let (Z_n) be an iid sequence satisfying (3.3) for some $\gamma \in \mathbb{R}$ and having positive right-endpoint. Under suitable conditions on F and k provided in Dekkers et al. [1989], then $\gamma_{n,k}^M \xrightarrow{\text{a.s.}} \gamma$, and*

$$\sqrt{k}(\gamma_{n,k}^M - \gamma) \xrightarrow{d} N(0, \sigma^2(\gamma)),$$

where

$$\sigma^2(\gamma) = \begin{cases} 1 + \gamma^2, & \gamma \geq 0, \\ (1 - \gamma)^2(1 - 2\gamma) \left\{ 4 - 8\frac{1-2\gamma}{1-3\gamma} + \frac{(5-11\gamma)(1-2\gamma)}{(1-3\gamma)(1-4\gamma)} \right\}, & \gamma < 0. \end{cases}$$

The standard way of plotting all the above estimators is as a function of k , the number of top-order statistics in question. We look for stability, ideally a prolonged region of small k -values where the estimator is constant. Automatic techniques for selecting such a region exist, but we instead propose to always plot the estimator and select the optimal k from visual inspection.

Numeric example 3.2 (Tail-index estimators across models) In Figure 3.2 we illustrate how plots from Definitions 3.2, 3.8, 3.9 and 3.7 look in practice. There, we have simulated two regularly varying distributions with $\gamma = 1/2$: a standard Pareto distribution and a Burr distribution as in (3.13), with $\theta = 1$ and $\eta = \beta = \sqrt{2}$. We also provide a $\gamma = 0$ case corresponding to a one-sided Normal distribution (constructed by taking the absolute value of a Normal random variable). Apart from the Pickands estimator's poor behaviour, which follows from the variance in Theorem 3.17, we observe that there is no overall best estimator, highlighting the difficulty of accurate inference for γ in finite samples.

The especially flat behaviour of the exact Pareto is not a surprise, since in that case $B_n(s_0) = \sup_{t \geq s_0} |T_n(t) - T(t)|$ is exactly zero, which is a bias term appearing in both Theorems 3.11 and 3.14. $\diamond$

3.5 Conditional estimation of regularly varying tails

Covariates are an important part of non-life insurance modelling, and it is sometimes desirable to have covariate-dependent tail behaviour. In this section we

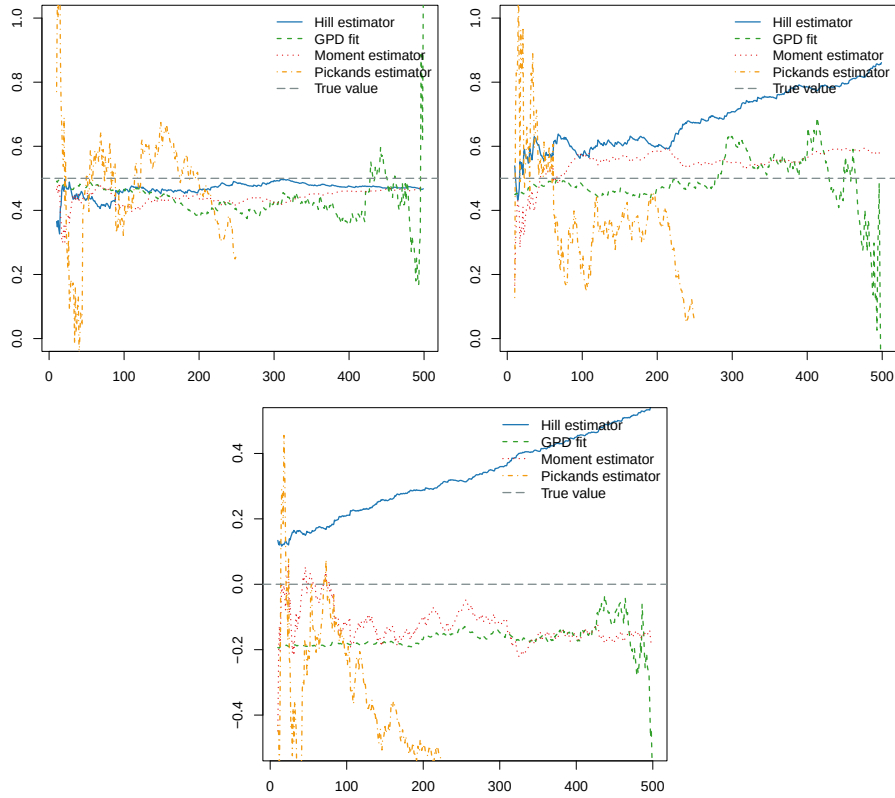


Figure 3.2: Estimation of γ through Definitions 3.2, 3.8, 3.9 and 3.7. Curves show $\hat{\gamma}$ against the number k of upper order statistics, for $n = 1000$ observations from the Pareto (top left), Burr (top right) and one-sided Normal (bottom) distributions.

introduce estimators for the conditional tail function and for the conditional tail index, for a sample where the covariate influences the shape of the tail. The mathematical goal is to let the tail index vary with covariates, while still preserving the tractable local regular variation structure at a given conditioning value x .

3.5.1 Conditional regular variation

Let (X_n, Y_n) be a bivariate iid sequence where (Y_n) is a positive process of interest and (X_n) is a process of covariates. Let the distribution function of Y_1 be denoted by F , the density of X_1 be denoted by g , which is assumed twice continuously differentiable, and let the distribution function of Y_1 given $X_1 = x$ be denoted by F^x .

Let $x \in \mathbb{R}$ with $g(x) > 0$ be given. The key assumption is that the conditional distribution of Y_1 , given $X_1 = x$, is regularly varying with tail index $\gamma(x) > 0$, and that this holds in a neighbourhood around x , that is

$$\bar{F}^u(y) = \mathbb{P}(Y_1 > y | X_1 = u) = L^u(y) y^{-\frac{1}{\gamma(u)}}, \quad y > 0, \quad (3.14)$$

for all u in an open set $U \subseteq \mathbb{R}$ containing x , and where L^u is slowly varying at infinity, that is $L^u(tv)/L^u(v) \rightarrow 1$ as $v \rightarrow \infty$ for any $t > 0$.

Let K be a kernel, supported on $[-1, 1]$, let (h_n) be the corresponding bandwidth sequence, and let (u_n^x) be a scaling sequence diverging to infinity.

The conditional survival function $\bar{F}^x$ is estimated with the Nadaraya–Watson pathwise estimator as

$$\bar{F}_n^x(y) = \sum_{j=1}^n \frac{1_{(Y_j > y)} K\left(\frac{x-X_j}{h_n}\right)}{\sum_{j=1}^n K\left(\frac{x-X_j}{h_n}\right)}, \quad y > 0. \quad (3.15)$$

Definition 3.10 (Conditional tail empirical distribution function) Let the corresponding tail empirical distribution function, for a deterministic threshold, be given by

$$\tilde{T}_n^x(s) = \frac{\bar{F}_n^x(su_n^x)}{\bar{F}_n^x(u_n^x)} = \frac{\frac{1}{nh_n \bar{F}_n^x(u_n^x)} \sum_{j=1}^n K\left(\frac{x-X_j}{h_n}\right) 1_{(Y_j > su_n^x)}}{\frac{1}{nh_n} \sum_{j=1}^n K\left(\frac{x-X_j}{h_n}\right)} =: \frac{\tilde{V}_n^x(s)}{g_n(x)}, \quad (3.16)$$

for $s > 0$. The target of $\tilde{T}_n^x$ is the conditional tail function $s \mapsto T^x(s) = s^{-1/\gamma(x)}$.

The above convergence, both in probability and weakly, is shown under suitable conditions below. However, as in the unconditional case, the deterministic thresholds u_n^x are not observable. Below we provide asymptotic results also for when these thresholds are estimated from data.

3.5.2 Conditional tail function and conditional Hill estimators

To transfer the tail function behaviour to the conditional tail index, the quantity of interest, the following key identity is analogous to the unconditional case:

$$\gamma(x) = \int_1^\infty \frac{T^x(s)}{s} ds.$$

We assume again that an *intermediate sequence* is given, that is a sequence $k_n \leq n$ such that $k_n \rightarrow \infty$, $k_n/n \rightarrow 0$. Let $u_n^x = F^{\leftarrow, x}(1 - k_n/n)$ be an unobserved sequence of deterministic thresholds, diverging to infinity, where $F^{\leftarrow, x}$ denotes the generalised inverse of F^x . We may and shall estimate this sequence by its empirical counterpart $\{q_n^x, n \in \mathbb{N}\}$ given by

$$q_n^x = F_n^{\leftarrow, x}(1 - k_n/n).$$

The conditional tail function is thus estimated by

$$\hat{T}_n^x(s) = \frac{\bar{F}_n^x(sq_n^x)}{\bar{F}^x(u_n^x)} = \tilde{T}_n^x(sq_n^x/u_n^x),$$

which no longer depends on the sequence u_n^x , as can be seen from the definition of $\tilde{T}_n^x$ and from the fact that $\bar{F}^x(u_n^x) = k_n/n$.

Definition 3.11 (Conditional Hill estimator) The conditional Hill estimator is given by

$$\gamma_{n,k}(x) = \int_1^\infty \frac{\hat{T}_n^x(s)}{s} ds = \frac{n}{k_n} \frac{\sum_{j=1}^n \log_+(\frac{Y_j}{q_n^x}) K(\frac{x-X_j}{h_n})}{\sum_{j=1}^n K(\frac{x-X_j}{h_n})}.$$

It is worth noticing that the effective sample size used in this estimator is proportional to $h_n k_n$ instead of k_n , which is also the rate for weak convergence. Consequently, the parameter k_n has a different interpretation than in the unconditional case. In particular, $\gamma_{n,k}(x)$ can be plotted for the entire range $k_n \in \{2, 3, \dots, n\}$.

Numeric example 3.3 Before delving into the mathematical treatment of our estimators, we provide an example of a conditional Hill plot. In Figure 3.3 we have taken (Y_j) to be the (absolute value of) the negative log-returns of the West Texas Intermediate (WTI) grade crude oil spot prices from 2007–2024. As a covariate (X_j) , we take the Crude Oil Volatility Index (CBOE OVX), transformed to have uniform marginals. For large x , we clearly observe a different, heavier tail regime. In contrast, for medium and small x , the tail regime is similar and lighter. The assumption that (Y_n, X_n) is an iid sequence is not fulfilled for this data. However, the conditional Hill estimator for stationary time-series also possesses large sample asymptotics, which happen to agree with the iid case under suitably lax assumptions; see [Bladt et al. \[2026\]](#). $\diamond$

3.6 Limit theorems for conditional extremes

In this section we provide consistency and weak convergence results for the estimators of the conditional tail function and conditional tail index under suitable

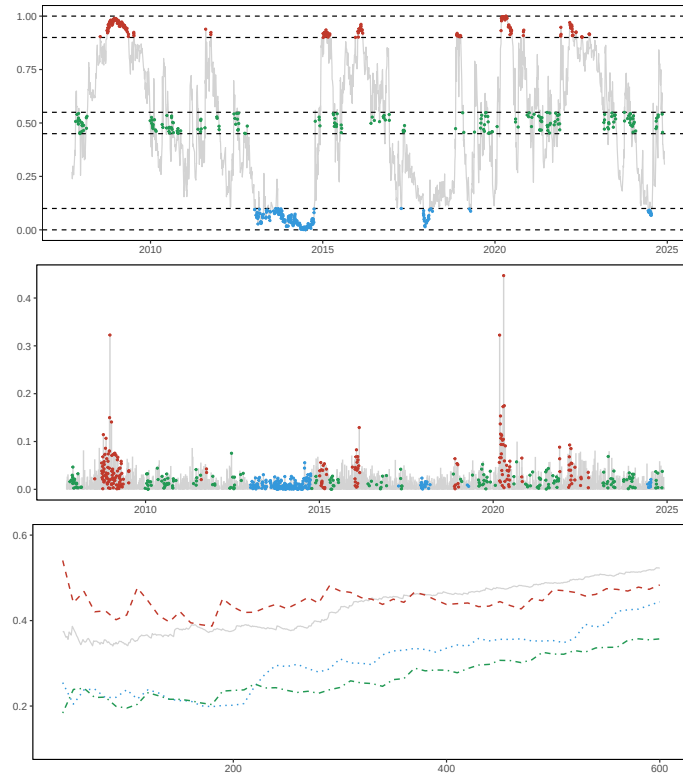


Figure 3.3: Top panel: covariate process (X_j) given by the Crude Oil Volatility Index (CBOE OVX), as a function of calendar year. The effective sample is calculated using a uniform kernel centred at $x = 0.05, 0.50, 0.95$ (highlighted in red, green and blue, respectively) and with $h_n = 0.05$. Middle panel: target process (Y_j) given by the absolute value of the negative log-returns of the West Texas Intermediate (WTI) grade crude oil spot prices with highlighted subsamples, as a function of calendar year. Bottom panel: conditional Hill estimates $\gamma_{n,k}(x)$ in dotted blue, dashdotted green and dashed red, respectively, as a function of k ; in solid grey we have the unconditional Hill estimator of the entire sample.

conditions. Furthermore, second order conditions are formulated in a very general and abstract form. It can be shown that broad classes of regularly varying tails satisfy them.

3.6.1 Consistency

A fundamental underlying result is the uniform consistency of the conditional tail empirical function, for deterministic thresholds. This provides the backbone for all subsequent asymptotic statements in the conditional setting.

Condition 3.1 The conditions for consistency are as follows.

3.1.1 The map $z \mapsto 1/\gamma(z)$ is c_γ -Lipschitz continuous,

3.1.2 It holds that $\lim_{n \rightarrow \infty} h_n \log(u_n^x) = 0$,

3.1.3 There exists $y_0 \in \mathbb{R}$ and $c_L > 0$ such that

$$\sup_{y \geq y_0} \left| \frac{\log(L^x(y)) - \log(L^u(y))}{\log(y)} \right| \leq c_L |x - u|,$$

uniformly in $x, u \in U$,

3.1.4 It holds that $\lim_{n \rightarrow \infty} nh_n \bar{F}^x(u_n^x) = \infty$,

We begin by proving some preliminary results related to the process $\tilde{V}_n^x$.

Lemma 3.1 (Mean of the tail process) *Assume conditions 3.1.1-3.1.3. Then*

$$\mathbb{E} \tilde{V}_n^x(s) \rightarrow s^{-1/\gamma(x)} g(x), \quad s > 0.$$

Proof. If $g(x) > 0$ it holds that

$$\begin{aligned} \mathbb{E} \tilde{V}_n^x(s) &= \\ g(x) \frac{\bar{F}^x(su_n^x)}{\bar{F}^x(u_n^x)} \int (su_n^x)^{-(1/\gamma(x-uh_n)-1/\gamma(x))} \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \frac{g(x-h_nu)}{g(x)} K(u) \, du. \end{aligned}$$

The expression before the integral converges to the desired limit, and hence it remains to show that the integral converges to 1. We consider each of the three n -dependent factors in the integral

$$\begin{aligned} (su_n^x)^{-(1/\gamma(x-uh_n)-1/\gamma(x))} &= \exp(-(1/\gamma(x-uh_n)-1/\gamma(x)) \log(su_n^x)) \\ &\leq \exp(c_\gamma h_n |u| \log(su_n^x)) = \exp(c_\gamma |u| h_n \log(u_n^x)) \exp(c_\gamma h_n |u| \log(s)) \rightarrow 1, \end{aligned}$$

and similarly

$$(su_n^x)^{-(1/\gamma(x-uh_n)-1/\gamma(x))} \geq \exp(-c_\gamma |u| h_n \log(u_n^x)) \exp(-c_\gamma h_n |u| \log(s)) \rightarrow 1,$$

uniformly in $u \in [-1, 1]$, using condition 3.1.1 and $h_n \log(u_n^x) \rightarrow 0$. Now, condition 3.1.3 implies

$$\exp(-\log(su_n^x) h_n c_L |u|) \leq \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \leq \exp(\log(su_n^x) h_n c_L |u|)$$

and both sides converge to 1, uniformly in $u \in [-1, 1]$. The density g is assumed twice continuously differentiable which implies Lipschitz continuity, so that

$$1 - \frac{c_g |u| h_n}{g(x)} \leq \frac{g(x-h_nu)}{g(x)} \leq 1 + \frac{c_g |u| h_n}{g(x)}$$

and both sides converge to 1, uniformly in $u \in [-1, 1]$. By the dominated convergence theorem we get $\mathbb{E} \tilde{V}_n^x(s) \rightarrow g(x) s^{-1/\gamma(x)}$. If $g(x) = 0$, similar arguments, using Lipschitz continuity of g , yield the result. $\square$

The following lemma underpins several of the asymptotic arguments to follow.

Lemma 3.2 (Kernel tail expectation) *Assume conditions 3.1.1-3.1.3. For any $t, q > 0$ it holds that*

$$\frac{\mathbb{E}[1_{(Y_1 > u_n^x t)} K^q(\frac{x-X_1}{h_n})]}{h_n \bar{F}^x(u_n^x)} \rightarrow t^{-1/\gamma(x)} g(x) \int K^q(u) du.$$

Proof. Calculations similar to those of Lemma 3.1 yield the result. $\square$

Theorem 3.20 (Consistency of the tail empirical process) *Assume that Condition 3.1 holds. Then*

$$\tilde{T}_n^x(s) \xrightarrow{\mathbb{P}} T^x(s) = s^{-1/\gamma(x)}, \quad s > 0, \quad (3.17)$$

the convergence being uniform on compacts bounded away from zero.

Proof. We have $g_n(x) \xrightarrow{\mathbb{P}} g(x)$, and hence, using the relation $\tilde{T}_n^x(s) = \tilde{V}_n^x(s)/g_n(x)$, it remains to show that $\tilde{V}_n^x(s) \xrightarrow{\mathbb{P}} g(x)s^{-1/\gamma(x)}$. Independence and Lemma 3.2 yield that

$$\mathbb{E}\tilde{V}_n^x(s) = \frac{\mathbb{E}[1_{(Y > u_n^x s)} K(\frac{x-X}{h_n})]}{h_n \bar{F}^x(u_n^x)} \rightarrow s^{-1/\gamma(x)} g(x),$$

and

$$\text{Var}(\tilde{V}_n^x(s)) = \frac{1}{nh_n \bar{F}^x(u_n^x)} \left[\frac{\mathbb{E}[1_{(Y_1 > u_n^x s)} K^2(\frac{x-X_1}{h_n})]}{h_n \bar{F}^x(u_n^x)} - \frac{\mathbb{E}[1_{(Y_1 > u_n^x s)} K(\frac{x-X_1}{h_n})]^2}{h_n \bar{F}^x(u_n^x)} \right],$$

which vanishes by Lemma 3.2. This finishes the proof, by Dini's Theorem 1.15. $\square$

As a first application we get that consistency of the conditional tail empirical function implies that the intermediate empirical quantile q_n^x is equivalent to the deterministic u_n^x . This is of practical importance, since the former is estimable from data. Consistency of the conditional Hill estimator then directly follows.

Proposition 3.3 (Consistency of conditional quantiles) *If (3.17) holds, then $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$.*

Proof. Let $\varepsilon > 0$ be given and assume that $q_n^x/u_n^x > 1 + \varepsilon$ for all $n \in \mathbb{N}$. Since $\bar{F}_n^x$ is decreasing we have

$$\tilde{T}_n^x(1 + \varepsilon) = \frac{\bar{F}_n^x((1 + \varepsilon)u_n^x)}{\bar{F}^x(u_n^x)} \geq \frac{\bar{F}_n^x(q_n^x)}{\bar{F}^x(u_n^x)} = \frac{\bar{F}_n^x(F_n^{\leftarrow, x}(1 - k_n/n))}{\bar{F}^x(F^{\leftarrow, x}(1 - k_n/n))} = 1.$$

In particular it holds that

$$\mathbb{P}(q_n^x/u_n^x > 1 + \varepsilon) \leq \mathbb{P}(\tilde{T}_n^x(1 + \varepsilon) \geq 1).$$

Since (3.17) holds, then $\tilde{T}_n^x(1 + \varepsilon) \xrightarrow{\mathbb{P}} (1 + \varepsilon)^{-1/\gamma(x)}$ and thus

$$\limsup_{n \rightarrow \infty} \mathbb{P}(q_n^x/u_n^x > 1 + \varepsilon) \leq \lim_{n \rightarrow \infty} \mathbb{P}(\tilde{T}_n^x(1 + \varepsilon) \geq 1) = 0.$$

Using similar arguments we obtain that $\limsup_{n \rightarrow \infty} \mathbb{P}(q_n^x/u_n^x < 1 - \varepsilon) = 0$. $\square$

Theorem 3.21 (Consistency of the conditional Hill estimator) *Assume conditions 3.1.1-3.1.3, and (3.17). Then $\gamma_{n,k}(x) \xrightarrow{\mathbb{P}} \gamma(x)$.*

Proof. We consider the following decomposition

$$\gamma_{n,k}(x) = \int_{q_n^x/u_n^x}^1 \frac{\tilde{T}_n^x(s)}{s} ds + \int_1^\infty \frac{\tilde{T}_n^x(s)}{s} ds.$$

Proposition 3.3 yields that $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$ and thus the first integral is $o_{\mathbb{P}}(1)$. The convergence (3.17) is uniform on compacts of $[1, \infty)$ and hence for all $m > 1$ we have that

$$\int_1^m \frac{\tilde{T}_n^x(s)}{s} ds \xrightarrow{\mathbb{P}} \int_1^m \frac{T^x(s)}{s} ds, \quad \text{as } n \rightarrow \infty.$$

Since $\lim_{m \rightarrow \infty} \int_1^m (T^x(s)/s) ds = \gamma(x)$, consistency of $\gamma_{n,k}$ follows by Theorem 1.5 if for every $\eta > 0$

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{T}_n^x(s)}{s} ds > \eta \right) = 0,$$

Now, since

$$\lim_{m \rightarrow \infty} \int_m^\infty \frac{\tilde{T}_n^x(s)}{s} ds = \frac{g(x)}{g_n(x)} \lim_{m \rightarrow \infty} \int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds,$$

and $g(x)/g_n(x) \xrightarrow{\mathbb{P}} 1$, it suffices to show that for every $\eta > 0$,

$$\lim_{m \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds > \eta \right) = 0.$$

Fix $\eta > 0$. We get

$$\begin{aligned} \mathbb{P} \left(\int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds > \eta \right) &\leq \frac{1}{\eta} \int_m^\infty \frac{\mathbb{E} \tilde{V}_n^x(s)}{g(x)s} ds \\ &= \frac{1}{\eta} \int_m^\infty \frac{\bar{F}^x(su_n^x)}{s \bar{F}^x(u_n^x)} \int (su_n^x)^{-(1/\gamma(x) - uh_n) - 1/\gamma(x)} \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \frac{g(x - h_n u)}{g(x)} K(u) du ds \\ &\leq \frac{\text{cst}}{\eta} \int_m^\infty \frac{\bar{F}^x(su_n^x)}{s \bar{F}^x(u_n^x)} \sup_{u \in [-1, 1]} \left((su_n^x)^{-(1/\gamma(x) - uh_n) - 1/\gamma(x)} \frac{L^{x-uh_n}(su_n^x)}{L^x(su_n^x)} \frac{g(x - h_n u)}{g(x)} \right) ds \\ &\leq \frac{\text{cst}}{\eta} \int_m^\infty \frac{\bar{F}^x(su_n^x)}{s \bar{F}^x(u_n^x)} \sup_{u \in [-1, 1]} \left(\frac{g(x - h_n u)}{g(x)} \right) s^{(c_\gamma + c_L)h_n} (u_n^x)^{(c_\gamma + c_L)h_n} ds \\ &\leq \frac{\text{cst}}{\eta} \sup_{u \in [-1, 1]} \left(\frac{g(x - h_n u)}{g(x)} \right) (u_n^x)^{(c_\gamma + c_L)h_n} \int_m^\infty s^{-1/\gamma(x) + \varepsilon - 1 + (c_\gamma + c_L)h_n} ds, \end{aligned}$$

for $\varepsilon \in (0, 1/\gamma(x))$, using Potter's bound. Let $\varepsilon' \in (0, 1/\gamma(x) - \varepsilon)$ be given. It is possible to find n' such that $(c_\gamma + c_L)h_n \leq \varepsilon'$ for all $n \geq n'$. Hence for $n \geq n'$ we have that the integrand $s \mapsto s^{-1/\gamma(x) + \varepsilon - 1 + (c_\gamma + c_L)h_n}$ is integrable and consequently

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_m^\infty \frac{\tilde{V}_n^x(s)}{g(x)s} ds > \eta \right) = O(m^{-1/\gamma(x) + \varepsilon + \varepsilon'}),$$

which vanishes as $m \rightarrow \infty$. $\square$

3.6.2 Central limit theorems

We now consider the univariate conditional tail empirical process for deterministic thresholds

$$\tilde{T}_n^x(s) = \sqrt{k_n h_n}(\tilde{T}_n^x(s) - T^x(s)), \quad s > 0,$$

for which a functional central limit is established. Notice that the speed of convergence is determined by the effective sample size $nh_n \bar{F}^x(u_n^x) = h_n k_n$, which is slower than in the unconditional case, which is simply k_n .

For $s_0 \in (0, 1)$, define

$$B_n^x(s_0) = \sup_{t \geq s_0} |\mathbb{E} \tilde{V}_n^x(t) - t^{-1/\gamma(x)} g(x)|.$$

Condition 3.2 The conditions for normality are as follows.

3.2.1 There exists $s_0 \in (0, 1)$, such that $\sqrt{k_n h_n} B_n^x(s_0) \rightarrow 0$,

3.2.2 It holds that

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n} \int_1^\infty \frac{\mathbb{E} \tilde{V}_n^x(t) - t^{-1/\gamma(x)} g(x)}{t} dt = 0.$$

Define the auxiliary empirical process

$$\tilde{V}_n^x(s) = \sqrt{k_n h_n}(\tilde{V}_n^x(s) - \mathbb{E}[\tilde{V}_n^x(s)]), \quad s > 0,$$

Lemma 3.3 (Covariance of the tail process) *Assume Condition 3.1. Then for all $s, t > 0$*

$$\lim_{n \rightarrow \infty} \text{Cov}(\tilde{V}_n^x(s), \tilde{V}_n^x(t)) = (s \vee t)^{-1/\gamma(x)} g(x) \int K^2(u) du. \quad (3.18)$$

Proof. Independence and identical distribution yield

$$\begin{aligned} \text{Cov}(\tilde{V}_n^x(s), \tilde{V}_n^x(t)) &= \frac{1}{h_n \bar{F}^x(u_n^x)} \mathbb{E} \left[1_{(Y_1 > (s \vee t) u_n^x)} K^2 \left(\frac{x - X_1}{h_n} \right) \right] \\ &\quad - h_n \bar{F}^x(u_n^x) \frac{\mathbb{E} \left[K \left(\frac{x - X_1}{h_n} \right) 1_{(Y_1 > s u_n^x)} \right]}{h_n \bar{F}^x(u_n^x)} \frac{\mathbb{E} \left[K \left(\frac{x - X_1}{h_n} \right) 1_{(Y_1 > t u_n^x)} \right]}{h_n \bar{F}^x(u_n^x)} \end{aligned}$$

The limit of the first term follows from Lemma 3.2:

$$\lim_{n \rightarrow \infty} \frac{1}{h_n \bar{F}^x(u_n^x)} \mathbb{E} \left[1_{(Y_1 > (s \vee t) u_n^x)} K^2 \left(\frac{x - X_1}{h_n} \right) \right] = (s \vee t)^{-1/\gamma(x)} g(x) \int K^2(u) du,$$

which is the asymptotic covariance. The second term disappears asymptotically due to Lemma 3.2. This finishes the proof. $\square$

With the covariance limit in place, we next establish finite-dimensional Gaussian limits and then lift this to functional weak convergence.

Theorem 3.22 (Finite-dimensional convergence) *Assume Conditions 3.1. Then the finite-dimensional distributions of $\tilde{\mathbb{V}}_n^x$ converge to those of a Gaussian process $\mathbb{V}^x$ with covariance function defined in (3.18).*

Proof. Fix $s > 0$. We only show the result for one marginal, but the general case follows by the Cramér–Wold device. Note first that $\mathbb{V}_n^x$ can be written as

$$\tilde{\mathbb{V}}_n^x(s) = \sum_{j=1}^n \frac{1_{(Y_j > su_n^x)} K\left(\frac{x-X_j}{h_n}\right) - \mathbb{E}\left[1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right)\right]}{\sqrt{k_n h_n}} =: \sum_{j=1}^n Z_{j,n}(s)$$

where the summands are mean zero iid random variables. In the calculations below we show that the conditions of the Lindeberg CLT are fulfilled. By Lemma 3.3 it only remains to show Lindeberg’s condition. Let $c > 0$ be given. Then we have that

$$\begin{aligned} \sum_{j=1}^n \int_{(|Z_{j,n}(s)| > c)} Z_{j,n}^2(s) \, d\mathbb{P} &= n \mathbb{E}\left[Z_{1,n}^2(s) 1_{(|Z_{1,n}(s)| > c)}\right] = \\ &= \frac{n}{k_n h_n} \mathbb{E}\left[\left(1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right) + \mathbb{E}\left[1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right)\right]\right)^2 1_{(|Z_{1,n}(s)| > c)}\right] \\ &\quad - \frac{2n}{k_n h_n} \mathbb{E}\left[K\left(\frac{x-X_1}{h_n}\right) 1_{(Y_1 > su_n^x)} \mathbb{E}\left[1_{(Y_1 > su_n^x)} K\left(\frac{x-X_1}{h_n}\right)\right] 1_{(|Z_{1,n}(s)| > c)}\right]. \end{aligned}$$

By two applications of the Cauchy–Schwarz inequality, and invoking Lemma 3.2 and noticing that $1_{(|Z_{1,n}(s)| > c)}$ converges almost surely to zero, the above term vanishes. $\square$

Theorem 3.23 (Functional CLT for the tail process) *Assume Condition 3.1. Then $\tilde{\mathbb{V}}_n^x$ converges weakly to the centred Gaussian process $\mathbb{V}^x$ in $\ell^\infty((0, \infty))$, and the process $\mathbb{V}^x$ admits an almost surely continuous version.*

Proof. Lemma 3.3 yields that the covariance structure of $\tilde{\mathbb{V}}_n^x$ converges to that of the proposed Gaussian process. Define

$$Z_{ni}(s) = (k_n h_n)^{-1/2} \{1_{(Y_i > u_n^x s)} K(h_n^{-1}(X_i - x)) - \mathbb{E}[1_{(Y_i > u_n^x s)} K(h_n^{-1}(X_i - x))]\},$$

so that $\tilde{\mathbb{V}}_n^x = \sum_{i=1}^n Z_{ni}$. To prove functional convergence, we make use of the Bracketing CLT for changing functions, Theorem 1.20. Consider the bounded interval $I = [a, b]$ with $a > 0$, and define

$$\begin{aligned} \mathcal{F}_n &= \{f_{n,t} : [0, \infty) \times \mathbb{R} \rightarrow [0, \infty) \\ \text{s.t. } f_{n,t}(y, z) &= (h_n \bar{F}(u_n^x))^{-1/2} 1_{[u_n^x t, \infty)}(y) K(h_n^{-1}(z - x)) \text{ for } t \in [a, b]\}. \end{aligned}$$

Clearly the envelope is given by $F_n(Y_1, X_1) = (h_n \bar{F}(u_n^x))^{-1/2} 1_{(Y_1 > u_n^x a)} K(h_n^{-1}(X_1 - x))$. It satisfies by Lemma 3.2 that $\mathbb{E}[F_n(Y_1, X_1)^2] = O(1)$, showing condition (1.6). We also have that by the same argument as in the Lindeberg CLT, Theorem 3.22, when proving finite-dimensional distributions that

$$\mathbb{E}[F_n(Y_1, X_1)^2 1_{(F_n(Y_1, X_1) > c\sqrt{n})}] \rightarrow 0.$$

This shows (1.7). The bracketing number is calculated as follows. Consider for $j = 1, \dots, m$ lower brackets on the form

$$l_j(y, z) = (h_n \bar{F}(u_n^x))^{-1/2} 1_{[u_n^x t_j, \infty)}(y) K(h_n^{-1}(z - x)),$$

and similarly for the upper brackets

$$u_j(y, z) = (h_n \bar{F}(u_n^x))^{-1/2} 1_{[u_n^x t_{j-1}, \infty)}(y) K(h_n^{-1}(z - x)),$$

for sequences on the form $a = t_0 < t_1 < \dots < t_m = b$. Let $\mathcal{F}_j^n = \{f \in \mathcal{F}_n : l_j \leq f \leq u_j\}$. Then clearly we have the partition $\mathcal{F}_n = \bigcup_{j=1}^m \mathcal{F}_j^n$. Let $\varepsilon > 0$ be given, and pick a sequence for $m \leq \lceil 2/(\varepsilon^2/2) \rceil$ such that

$$T^x(t_{j-1}) - T^x(t_j) \leq \varepsilon^2 a^{-1/\gamma(x)} / (2g(x)) \quad \text{for } 0 \leq j \leq m,$$

which is possible because T is a continuous and bounded function on $[a, b]$. Then this sequence covers all of $\mathcal{F}_n$ with brackets satisfying

$$\begin{aligned} & \|u_j(Y_1, X_1) - l_j(Y_1, X_1)\|_{L^2} \\ &= \frac{\mathbb{E} \left[1_{(u_n^x t_j \geq Y_1 > u_n^x t_{j-1})} K^2 \left(\frac{x - X_1}{h_n} \right) \right]^{1/2}}{(h_n \bar{F}(u_n^x))^{1/2}} \\ &= (g(x) \int K^2(u) \, du)^{1/2} (t_{j-1}^{-1/\gamma(x)} - t_j^{-1/\gamma(x)})^{1/2} (1 + o(1)) \\ &\leq \varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \end{aligned}$$

Consequently for some constant C ,

$$N_{[\cdot]}(\varepsilon \|F_n(Y_1, X_1)\|_{L^2}, \mathcal{F}_n, L^2(\mathbb{P})) \leq m \leq \lceil C/\varepsilon^2 \rceil.$$

Integration now shows (1.9). Note that the above $o(1)$ term is independent of the partition by Theorem 3.20. In particular, the partition is independent of n and (1.8) does not need to be verified. $\square$

Proposition 3.4 (Weak convergence of the conditional tail empirical process)
Assume that conditions 3.1 and 3.2.1 hold. Then it holds that $\tilde{\mathbb{T}}_n^x$ converges weakly to a centred Gaussian process $\mathbb{T}^x$ in $\ell^\infty([s_0, \infty))$, with covariance function

$$\Gamma(s, t) = \frac{(s \vee t)^{-1/\gamma(x)}}{g(x)} \int K^2(u) \, du, \quad (3.19)$$

and the process $\mathbb{T}^x$ has an almost surely continuous version.

Proof. Consider the following decomposition

$$\begin{aligned} \tilde{\mathbb{T}}_n^x(s) &= \sqrt{k_n h_n} (g(x) - g_n(x)) \frac{\tilde{T}_n^x(s)}{g(x)} + \frac{1}{g(x)} \tilde{\mathbb{V}}_n^x(s) \\ &\quad + \sqrt{k_n h_n} \frac{1}{g(x)} (\mathbb{E} \tilde{\mathbb{V}}_n^x(s) - g(x) s^{-1/\gamma(x)}). \end{aligned}$$

The absolute value of the last term vanishes, since it is bounded by $\sqrt{k_n h_n} B_n^x(s_0) \rightarrow 0$, so normality of the Parzen–Rosenblatt kernel density estimator, and Theorem 3.23 imply that

$$\tilde{T}_n^x(s) = \sqrt{k_n h_n} (g(x) - g_n(x)) \frac{\tilde{T}_n^x(s)}{g(x)} + \frac{1}{g(x)} \tilde{V}_n^x(s) + o(1) \xrightarrow{d} \frac{1}{g(x)} \mathbb{V}^x(s).$$

□

Recall the definitions $u_n^x = F_n^{\leftarrow, x}(1 - k_n/n)$ and $q_n^x = F_n^{\leftarrow, x}(1 - k_n/n)$. Then $n\bar{F}^x(u_n^x) = k_n$. Define the conditional tail empirical quantile function $\tilde{Q}_n^x$ on $(0, k_n/n)$ by

$$\tilde{Q}_n^x(t) = \tilde{T}_n^{\leftarrow, x}(\lfloor k_n t \rfloor / k_n) = \frac{q_{\lfloor k_n t \rfloor}}{u_n^x},$$

and let $Q^x(t) = t^{-\gamma(x)}$, $t > 0$. From here the argument is map-based. We first establish weak convergence of the tail process, then invert to the quantile process, and finally evaluate at random levels for the Hill estimator.

Theorem 3.24 (Joint CLT for tail and quantile processes) *Assume that conditions 3.1 and 3.2.1 hold. Then*

$$\sqrt{h_n k_n} (\tilde{T}_n^x - T^x, \tilde{Q}_n^x - Q^x) \xrightarrow{d} (\mathbb{T}^x, -(Q^x)' \cdot \mathbb{T}^x \circ Q^x),$$

in $\ell^\infty([s_0, \infty)) \times \ell^\infty((0, s_0^{-\gamma(x)}])$. Consequently

$$\sqrt{h_n k_n} (q_n^x / u_n^x - 1) \xrightarrow{d} \gamma(x) \mathbb{T}^x(1) \stackrel{d}{=} N(0, \frac{\gamma(x)^2}{g(x)} \int K^2(u) \, du),$$

jointly with the convergence of $\sqrt{h_n k_n} (\tilde{T}_n^x - T^x)$ to $\mathbb{T}^x$.

Proof. Note that $[s_0, \infty) \ni s \mapsto T^x(s)$ is surjective on $(0, s_0^{-\gamma(x)}]$, continuous and decreasing with negative derivative. As the processes $\tilde{T}_n^x$ have non-increasing sample paths for each $n \geq 1$, an application of the functional δ -method, Theorem 1.17, applied with the inverse mapping, as per Lemma 1.10, yields that

$$\sqrt{h_n k_n} (\tilde{T}_n^x - T^x, \tilde{T}_n^{\leftarrow, x} - T^{\leftarrow, x}) \xrightarrow{d} (\mathbb{T}^x, -(T^{\leftarrow, x})' \cdot \mathbb{T}^x \circ T^{\leftarrow, x}).$$

This concludes the proof. □

The sequence (u_n^x) is unobserved and we may estimate it with (q_n^x) . Let

$$\hat{\mathbb{T}}_n^x(s) = \sqrt{k_n h_n} (\hat{T}_n^x(s) - T^x(s)),$$

which we refer to as the tail empirical process with random levels.

Theorem 3.25 (Weak convergence of the tail empirical process with random levels) *Assume that conditions 3.1 and 3.2.1 hold. Then*

$$\hat{\mathbb{T}}_n^x \xrightarrow{d} \mathbb{T}^x - T^x \times \mathbb{T}^x(1),$$

in $\ell^\infty([s_0, \infty))$.

Proof. Decompose $\widehat{\mathbb{T}}_n^x$ into the following components

$$\widehat{\mathbb{T}}_n^x = \widetilde{\mathbb{T}}_n^x(sq_n^x/u_n^x) + \sqrt{k_n h_n}(T^x(sq_n^x/u_n^x) - T^x(s)).$$

By Theorem 3.20, the tail consistency (3.17) holds, while Proposition 3.3 yields that $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$, and the continuous mapping theorem thus yields that the first term above converges to $\mathbb{T}^x$. As for the second term, Theorem 3.24 and the delta-method yields that

$$\begin{aligned} \sqrt{k_n h_n}(T^x(sq_n^x/u_n^x) - T^x(s)) &= s^{-1/\gamma(x)} \sqrt{k_n h_n}(T^x(q_n^x/u_n^x) - T^x(1)) \\ &\xrightarrow{d} s^{-1/\gamma(x)}(T^x)'(1) \cdot \gamma(x)\mathbb{T}^x(1) = -T^x(s)\mathbb{T}^x(1), \end{aligned}$$

jointly with $\widetilde{\mathbb{T}}_n^x(sq_n^x/u_n^x)$. The argument can be transferred unchanged to path-wise weak convergence. $\square$

It turns out that Theorem 3.25 is not required to establish convergence of the conditional Hill estimator. However, it is an interesting result in its own right, since the empirical tail process can be used to measure how closely the top order statistics behave like a strict Pareto tail.

Theorem 3.26 (Asymptotic normality of the conditional Hill estimator) *Assume that Condition 3.1 and Condition 3.2 hold. Then*

$$\sqrt{k_n h_n}(\gamma_{n,k}(x) - \gamma(x)) \xrightarrow{d} N(0, \frac{\gamma(x)^2}{g(x)} \int K^2(u) du).$$

Proof. We consider the decomposition

$$\sqrt{k_n h_n}(\gamma_{n,k}(x) - \gamma(x)) = \sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{\widetilde{T}_n^x(s)}{s} ds + \int_1^\infty \frac{\widetilde{\mathbb{T}}_n^x(s)}{s} ds.$$

We examine the first term. Since $q_n^x/u_n^x \xrightarrow{\mathbb{P}} 1$, we may assume without loss of generality that for given $\varepsilon \in (0, 1)$, there exists $N \in \mathbb{N}$ such that for $n \geq N$ it holds that $q_n^x/u_n^x \in (1 - \varepsilon, 1 + \varepsilon)$. Thus for $n \geq N$ we have that

$$\begin{aligned} &\sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{\widetilde{T}_n^x(s) - T^x(s)}{s} ds \\ &\leq \frac{\sqrt{k_n h_n}}{1 - \varepsilon} |q_n^x/u_n^x - 1| \sup_{t \in (1 - \varepsilon, 1 + \varepsilon)} |\widetilde{T}_n^x(t) - T^x(t)|. \end{aligned}$$

Moreover, Theorem 3.24 and the continuous mapping theorem yield that

$$\sqrt{k_n h_n} |q_n^x/u_n^x - 1| \xrightarrow{d} \gamma(x) |\mathbb{T}^x(1)|,$$

while Theorem 3.20 yields $\sup_{t \in (1 - \varepsilon, 1 + \varepsilon)} |\widetilde{T}_n^x(t) - T^x(t)| \xrightarrow{\mathbb{P}} 0$ and thus

$$\sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{\widetilde{T}_n^x(s)}{s} ds = \sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{T^x(s)}{s} ds + o_{\mathbb{P}}(1).$$

Applying the delta-method with $x \mapsto \int_x^1 \frac{T^x(s)}{s} ds$, on the sequence (q_n^x/u_n^x) we obtain

$$\sqrt{k_n h_n} \int_{q_n^x/u_n^x}^1 \frac{T^x(s)}{s} ds \xrightarrow{d} -\gamma(x) \mathbb{T}^x(1).$$

This provides the asymptotics of the first term. We proceed to the second term

$$\begin{aligned} \int_1^\infty \frac{\tilde{T}_n^x(s)}{s} ds &= \sqrt{k_n h_n} \frac{1}{g(x)} (g(x) - g_n(x)) \int_1^\infty \frac{\tilde{T}_n^x(s)}{s} ds + \frac{1}{g(x)} \int_1^\infty \frac{\tilde{V}_n^x(s)}{s} ds \\ &\quad + \sqrt{k_n h_n} \frac{1}{g(x)} \int_1^\infty \frac{\mathbb{E} \tilde{V}_n^x(s) - g(x) s^{-1/\gamma(x)}}{s} ds. \end{aligned}$$

The last term is negligible due to condition 3.2.2. In the proof of Theorem 3.21 we showed $\int_1^\infty \tilde{T}_n^x(s)/s ds \xrightarrow{\mathbb{P}} \gamma(x)$. Combined with the asymptotic normality of the Parzen–Rosenblatt kernel density estimator we have that the first term also vanishes. Finally we consider $g(x)^{-1} \int_1^\infty \tilde{V}_n^x(s)/s ds$. We have that

$$\begin{aligned} \text{Var} \left(\int_m^\infty \tilde{V}_n^x(s)/s ds \right) &= \frac{1}{h_n \bar{F}^x(u_n^x)} \text{Var} \left(K \left(\frac{x - X_0}{h_n} \right) \int_m^\infty 1_{(Y_0 > s u_n)} \frac{1}{s} ds \right) \\ &= \frac{1}{h_n \bar{F}^x(u_n^x)} \text{Var} \left(K \left(\frac{x - X_0}{h_n} \right) \log \left(\frac{(Y_1/u_n^x) \vee m}{m} \right) \right) \\ &\leq \frac{1}{h_n \bar{F}^x(u_n^x)} \mathbb{E} \left[K^2 \left(\frac{x - X_0}{h_n} \right) \log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \right]. \end{aligned}$$

Since the kernel K has support $[-1, 1]$, we have that $1_{((z-x)/h_n \leq 1)} = 1_{(z \leq x+h_n)}$ and hence we have the further bound

$$\begin{aligned} &\frac{1}{h_n \bar{F}^x(u_n^x)} \int K^2 \left(\frac{x-z}{h_n} \right) \mathbb{E} \left[\log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \middle| X_0 = z \right] g(z) dz \\ &\leq \sup_{u \in [-h_n+x, x+h_n]} \left(\frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+^2 \left(\frac{Y_1}{u_n^x m} \right) \middle| X_0 = u \right] \right) \\ &\quad \times \frac{1}{h_n} \int K^2 \left(\frac{x-z}{h_n} \right) g(z) \frac{\bar{F}^z(u_n^x)}{\bar{F}^x(u_n^x)} dz, \end{aligned}$$

where it holds that

$$\begin{aligned} &\frac{1}{h_n} \int K^2 \left(\frac{x-z}{h_n} \right) g(z) \frac{\bar{F}^z(u_n^x)}{\bar{F}^x(u_n^x)} dz \\ &\leq \frac{1}{h_n} \int K^2 \left(\frac{x-z}{h_n} \right) g(z) (u_n^x)^{-(c_\gamma+c_L)|x-z|} dz \\ &\leq (u_n^x)^{(c_\gamma+c_L)h_n} \int K^2(w) g(x-wh_n) dw = O(1), \end{aligned}$$

and

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \sup_{u \in [-h_n + x, x + h_n]} \left(\frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = u \right] \right) \\
& \leq \lim_{n \rightarrow \infty} \left(\frac{1}{\bar{F}^x(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = x \right] \right) \\
& \quad + \lim_{n \rightarrow \infty} \sup_{u \in [-h_n + x, x + h_n]} \left| \frac{1}{\bar{F}^u(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = u \right] \right. \\
& \quad \quad \left. - \frac{1}{\bar{F}^x(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = x \right] \right| \\
& = \int_m^\infty \log^2(t) 1/\gamma(x) t^{-1/\gamma(x)-1} dt + 0,
\end{aligned}$$

where we have used Potter's bound to invoke the dominated convergence theorem for the first limit, and the term in the difference can be bounded using Potter's bound and applying twice the Lipschitz conditions on L^x and $1/\gamma(x)$ as was done in Theorem 3.21.

Thus, taking limit as $m \rightarrow \infty$ establishes the required asymptotic normality by Theorem 1.5. By construction, $\mathbb{T}^x(\cdot) = \mathbb{V}^x(\cdot)/g(x)$, and thus we have the representation

$$\sqrt{k_n h_n} (\gamma_{n,k}(x) - \gamma(x)) \xrightarrow{d} \frac{1}{g(x)} \int_1^\infty \frac{\mathbb{V}^x(s)}{s} ds - \frac{\gamma(x)}{g(x)} \mathbb{V}^x(1),$$

which has variance $\gamma(x)^2 \frac{1}{g(x)} \int K^2(u) du$. $\square$

Remark 3.1 The bias of the conditional Hill estimator vanishes due to condition 3.2.2. If this condition is relaxed to

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n} \int_1^\infty \frac{\mathbb{E} \tilde{V}_n^x(t) - t^{-1/\gamma(x)} g(x)}{t} dt = B \in \mathbb{R},$$

the asymptotic variance in Theorem 3.26 remains similar, while the bias becomes $B/g(x)$.

A theoretical bandwidth choice can be made as a function of k_n and n . For instance, $h_n = \sqrt{\log(k_n)/n}$ falls into the framework of conditions 3.1.2 and 3.1.4. It can also be seen to satisfy equation (3.21), related to second order behaviour, for a suitable (k_n) sequence. $\circ$

3.7 Notes and Comments

The terminology heavy-tailed, long-tailed, subexponential, and regularly varying goes back to classical asymptotic analysis and renewal theory, with modern probabilistic treatments given in monographs on regular variation and in general texts on regular variation, extremes and heavy tails such as [Bingham et al. \[1987\]](#), [Resnick \[1987, 2007\]](#), [Foss et al. \[2011\]](#). The detailed relationships between different definitions of heavy tails are standard and can be found with careful proofs in these references. For statistical practice, the insurance and finance perspective is well treated in [Embrechts et al. \[1997\]](#), which also treats modelling choices, diagnostics, a practical statistical section, and risk implications. The latter reference complements the more measure-theoretic treatments in [Resnick \[1987, 2007\]](#).

The classical probabilistic treatment of extreme value theory goes from the extremal types limit theorem in seminal works of Fisher–Tippett and Gnedenko [[Fisher and Tippett, 1928](#), [Gnedenko, 1943](#)] to the Generalised Extreme Value family, and from there to characterisations of max-domains of attraction; see [Leadbetter et al. \[1983\]](#) for a stationary-sequence viewpoint and [de Haan \[1970\]](#), [de Haan and Ferreira \[2006\]](#) for systematic treatments. The equivalence between membership in the Fréchet max-domain and regular variation of the tail is an important result connecting distributional limits of maxima to tail-regularity assumptions. The threshold-exceedance analogue, namely the Pickands–Balkema–de Haan limit, justifies modelling large exceedances by the Generalised Pareto family and leads to peaks-over-threshold methods, see [Balkema and de Haan \[1974\]](#), [III \[1975\]](#), [de Haan and Ferreira \[2006\]](#), [Embrechts et al. \[1997\]](#).

Regarding the manipulation of regularly varying tails, three tools recur throughout the chapter: uniform convergence of tail ratios on ranges bounded away from the origin; Potter-type bounds; and integral or representation theorems (Karamata-type results and de Haan’s representation), for which we refer to [Bingham et al. \[1987\]](#), [Resnick \[1987\]](#). The former reference also provides a very detailed account on properties of regularly varying functions, beyond the special case of distribution functions or quantile functions.

The statistical part of univariate extreme value estimation centres on tail-index and extreme-quantile estimation. In this simple context, the Hill estimator remains a benchmark for estimating the tail index, with consistency and asymptotic normality under standard intermediate-sequence and second-order conditions given in [Hill \[1975\]](#), [de Haan and Ferreira \[2006\]](#). The tail empirical and quantile process limits, specifically weak convergence to Brownian motion under a tail-driven time change, and the induced Gaussian limits for functionals such as Hill, are standard in empirical process approaches to extremes and are well shown in [Resnick \[1987\]](#), with the present treatment mirroring the slightly adapted approach and notation of [Kulik and Soulier \[2020\]](#), but specialising to the independent case. The block-maxima and threshold-based statistical counterparts of the Hill estimator include maximum likelihood for the Generalised Pareto model and alternative tail-index estimators such as the Pickands and the moment estimators; asymptotics and variance expressions for these estimators can be found in [III \[1975\]](#), [Smith \[1987\]](#), [Dekkers et al. \[1989\]](#). Extrapolation of

extreme quantiles via Weissman-type estimators is classical and investigated in various methodological papers and also in the general monographs cited above, see [Weissman \[1978\]](#). Finally, conditional tail estimation has been considered in various settings, where we notably mention [Gardes and Stupfler \[2013\]](#) and the time-series version [Bladt et al. \[2026\]](#) upon which the present exposition is based, again specialised to the independent case. In particular, the final consistency and central-limit-theorem sections for conditional extremes make explicit how these conditional estimators involve the same empirical-process structure as their simpler unconditional counterparts.

3.8 Exercises

1. Show that regularly varying distributions are subexponential, that subexponential distributions are long-tailed, and that long-tailed distributions are heavy-tailed.
2. Provide a counterexample for each reversed inclusion in the previous exercise.
3. Prove that $\gamma = \int_1^\infty \frac{T(s)}{s} ds$. Relate the sum version and integral version of the Hill estimator.
4. Let Z have Weibull distribution with survival function $\bar{F}(x) = \exp(-x^a)$, $x \geq 0$, for some $a > 0$.
 - (a) Show that the moment generating function is finite for some $t > 0$ if and only if $a \geq 1$, and conclude that the Weibull distribution is heavy-tailed if and only if $a < 1$.
 - (b) Show that $\bar{F}(tx)/\bar{F}(x) \rightarrow 0$ for any $t > 1$, and hence that the Weibull distribution is not regularly varying for any $a > 0$.
5. The distribution function of a Burr-distributed random variable is given by

$$F(x) = 1 - \left(1 + \left(\frac{x}{\theta}\right)^\eta\right)^{-\beta}, \quad x \geq 0, \theta > 0, \eta > 0, \beta > 0.$$

Show that the tail is regularly varying and find the corresponding index.

6. Discuss in detail the proof of Theorem 3.5 (Potter's bounds, first version). Prove Theorem 3.6 (Potter's bounds, second version).
7. Prove the following result, known as Breiman's lemma.

Lemma 3.4 (Breiman's Lemma) *Let Z and W be independent non-negative random variables, such that Z is regularly varying with tail index $1/\gamma > 0$ and there exists $\varepsilon > 0$ such that $\mathbb{E}[W^{1/\gamma+\varepsilon}] < \infty$. Then ZW is also regularly varying with index $1/\gamma$ and*

$$\lim_{x \rightarrow \infty} \frac{\mathbb{P}(ZW > x)}{\mathbb{P}(Z > x)} = \mathbb{E}[W^{1/\gamma}].$$

8. Recall that the weak limit of the Hill estimator has the following distributional representation:

$$\int_1^\infty \frac{\mathbb{W}(T(t))}{t} dt - \gamma \mathbb{W}(1) = \int_1^\infty \frac{\mathbb{B}(T(t))}{t} dt = \gamma \int_0^1 \frac{\mathbb{B}(t)}{t} dt.$$

Show that it has variance given by $\gamma^2 \int_0^1 \int_0^1 (s^{-1} \wedge t^{-1} - 1) ds dt = \gamma^2$.

9. Go through the details of the proof of consistency of the moment estimator in the $\gamma > 0$ case, Proposition 3.2.

10. Let F be a continuous distribution function on $\mathbb{R}$ and assume that there exist $c > 0$, $\beta > 0$ and $\gamma > 0$ such that

$$\sup_{t \geq 1} t^\beta |\overline{F}(t) - c \log(t) t^{-1/\gamma}| < \infty. \quad (9.6.3)$$

Deduce that $k = o(\log^2(n))$ is a suitable choice for (3.7) to hold.

11. Prove Lemma 3.2.
12. Recall that the weak limit of the conditional Hill estimator has the following distributional representation:

$$\sqrt{k_n h_n} (\hat{\gamma}_n(x) - \gamma(x)) \xrightarrow{d} \frac{1}{g(x)} \int_1^\infty \frac{\mathbb{V}^x(s)}{s} ds - \frac{\gamma(x)}{g(x)} \mathbb{V}^x(1),$$

Show that it has variance given by $\gamma(x)^2 \frac{1}{g(x)} \int K^2(u) du$.

13. Provide the details for the following limit, used in the proof of Theorem 3.26:

$$\lim_{n \rightarrow \infty} \sup_{u \in [-h_n + x, x + h_n]} \left| \frac{1}{\overline{F}^u(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = u \right] - \frac{1}{\overline{F}^x(u_n^x)} \mathbb{E} \left[\log_+ \left(\frac{Y_1}{u_n^x m} \right)^2 \middle| X_0 = x \right] \right| \rightarrow 0.$$

14. Assume Condition 3.1, that F^x is a continuous distribution and that there exist $c_x, 1/\gamma(x) > 0$, and $\beta_x > 1/\gamma(x)$ such that

$$\sup_{t \geq 1} t^{\beta_x} |\overline{F}^x(t) - c_x t^{-1/\gamma(x)}| < \infty. \quad (3.20)$$

In this exercise we wish to show that if there exist sequences (k_n) and (h_n) such that

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n^3} \log(n/k_n) = \lim_{n \rightarrow \infty} \sqrt{h_n k_n}^{\gamma(x)\beta_x - \frac{1}{2}} n^{1-\gamma(x)\beta_x} = 0, \quad (3.21)$$

then Condition 3.2 holds.

- (a) Argue that (3.20) implies that $u_n^x = O((n/k)^{\gamma(x)})$, and show that the following decomposition holds

$$\begin{aligned} & \mathbb{E}[\tilde{V}_n^x(t)] - t^{-1/\gamma(x)} g(x) \\ &= \int K(u) t^{-1/\gamma(x)} (g(x - h_n u) - g(x)) du \\ &+ \int K(u) \left(\frac{\overline{F}^{x-uh_n}(tu_n^x)}{\overline{F}^x(u_n^x)} - \frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} \right) g(x - h_n u) du \\ &+ \int K(u) \left(\frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} - t^{-1/\gamma(x)} \right) g(x - h_n u) du. \end{aligned}$$

- (b) Show that

$$\int K(u) t^{-1/\gamma(x)} (g(x - h_n u) - g(x)) du \leq O(t^{-1/\gamma(x)} h_n).$$

Hint: Use that g is assumed twice differentiable.

(c) Show that

$$\begin{aligned} & \left(\frac{\overline{F}^{x-uh_n}(tu_n^x)}{\overline{F}^x(u_n^x)} - \frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} \right) \\ & \leq \text{cst } t^{-1/\gamma(x)+\varepsilon} \log(tu_n^x) |u| h_n \left(1 + \log(tu_n^x) |u| h_n (1 + (tu_n^x)^{(c_\gamma+c_L)|u| h_n})/2 \right), \end{aligned}$$

and argue that the leading term is of order

$$O \left(t^{-1/\gamma(x)+\varepsilon} \log(tn/k_n) h_n \right),$$

for $\varepsilon \in (0, 1/\gamma(x))$.

Hint: Use that $|e^z - 1| \leq |z| + (1 + e^{|z|}) z^2/2$ along with the calculations in Lemma 3.1. You may use that Potter's bound still holds in the conditional setting.

(d) Show that

$$\left| \frac{\overline{F}^x(tu_n^x)}{\overline{F}^x(u_n^x)} - t^{-1/\gamma(x)} \right| \leq O(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)} t^{-1/\gamma(x)}).$$

Hint: Use the assumption in (3.20) and follow the same strategy as in the unconditional case.

(e) Argue that all three O -terms are uniformly bounded for $t \geq s_0$ for some $s_0 \in (0, 1)$ and conclude that condition 3.2.1 holds.

(f) By using the calculations from the previous subexercises, show that

$$\begin{aligned} & \int_1^\infty \frac{\mathbb{E}[\tilde{V}_n^x(t)] - t^{-1/\gamma(x)} g(x)}{t} dt \\ & \leq O(h_n) \int_1^\infty t^{-1/\gamma(x)-1} dt + O(k_n^{\beta_x \gamma(x)-1} n^{1-\beta_x \gamma(x)}) \int_1^\infty t^{-1/\gamma(x)-1} dt \\ & + O(h_n) \int_1^\infty t^{-1/\gamma(x)+\varepsilon-1} \log(tu_n^x) \left(1 + \log(tu_n^x) h_n (1 + (tu_n^x)^{(c_\gamma+c_L)h_n|u|})/2 \right) dt. \end{aligned}$$

(g) Prove that

$$O(h_n) \int_1^\infty t^{-1/\gamma(x)+\varepsilon-1} \log(tu_n^x) \left(1 + \log(tu_n^x) h_n (1 + (tu_n^x)^{(c_\gamma+c_L)h_n|u|})/2 \right) dt$$

is bounded for all n , and conclude that condition 3.2.2 also holds.

Hint: First show that $(u_n^x)^{h_n} = O(1)$ and that there exists n' such that for all $n \geq n'$ then $(c_\gamma + c_L)h_n \leq \varepsilon' \in (0, 1/\gamma(x) - \varepsilon)$, then analyse each sub-integral separately.

15. Assume Condition 3.1, that F^x is a continuous distribution and that there exist $c_x, 1/\gamma(x) > 0$, and $\beta_x > 1/\gamma(x)$ such that

$$\sup_{t \geq 1} t^{\beta_x} \left| \overline{F}^x(t) - c_x \log(t) t^{-1/\gamma(x)} \right| < \infty. \quad (3.22)$$

Show that if there exist sequences (k_n) and (h_n) such that

$$\lim_{n \rightarrow \infty} \sqrt{k_n h_n^3} \log(n/k_n) = \lim_{n \rightarrow \infty} \frac{\sqrt{k_n h_n}}{\log(n/k_n)} = 0, \quad (3.23)$$

then Condition 3.2 holds.

16. Assume that F^x is continuous and that conditions 3.1.1, 3.1.3, and condition (3.20) hold. Show that there exist sequences (k_n) , (h_n) such that all the conditions for consistency and normality are satisfied.
17. Assume that F^x is continuous and that conditions 3.1.1, 3.1.3, and condition (3.22) hold. Show that there exist sequences (k_n) , (h_n) such that all the conditions for consistency and normality are satisfied.

Practical exercises

1. Simulate 1000 observations of a Burr-distributed random variable with $\beta = 1$, $\eta = 2$ and $\theta = 1$ and of a standard normal random variable.
 - (a) Implement the Hill estimator as a function of k .
 - (b) Make a Hill plot of the Burr data. Also make one for a normally distributed sample. Estimate the tail index by selecting a suitable k .
 - (c) Plot the empirical mean excess functions. Qualify their behaviour based on their theoretical counterparts.
 - (d) Estimate the 90%, 99%, 99.99% quantile of the Burr-distributed data by using Section 3.3.3. Compare with the true quantiles.
 - (e) Estimate the 90%, 99%, 99.99% quantile of the Burr-distributed data based on the empirical distribution function. Compare the result with the previous subexercise.
2. Consider now the dataset `beaonre` from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#), which contains claim sizes caused by fire losses for which a reinsurance company Aon is liable. Consider the column `ClaimCost`.
 - (a) Construct the Hill estimator $\gamma_{n,k}$ as a function of the threshold k and use it to construct the Hill plot. Based on the plot, provide an estimated value for the tail index γ .
 - (b) Randomly sample 300 observations from `ClaimCost` and construct the Hill estimator, call it $\hat{\gamma}_{n,k}$. Repeat this experiment 100 times (sampling with replacement), and plot a histogram of the 100 samples of $\hat{\gamma}_{n,k}$ for $k = n/5$. Compare with the theoretical limit distribution.
3. Consider the datasets `freMTPL2sev` and `freMTPL2freq` from the R package `CASdatasets` [Dutang and Charpentier \[2024\]](#). The first dataset contains policy ID and claim size, the second contains policy ID along with covariates and claim number. We are interested in the conditional tail behaviour of the claim sizes (`ClaimAmount`) given driver age (`DrvAge`). The first task is to join the two datasets. Transform the covariate into

uniform random variables and consider the uniform kernel on $[-1, 1]$. We now implement the conditional Hill estimator of the claim severities (Y_n) given the chosen covariate (X_n) .

- (a) Recall that $q_n^x = (\mathbb{F}^{(n)})^\leftarrow(1 - k_n/n|x)$. In **R**, implement a function q_n^x that, for given $x \in \mathbb{R}$, returns a vector corresponding to each k_n , where (k_n) is chosen to live on an equidistant grid **K_grid** of 100 points on $[0.01n, n/2]$.
- (b) In **R**, implement the conditional Hill estimator: Construct an equidistant grid **X_grid** of 101 points on $[0, 1]$. Your function should return the value of $\hat{\gamma}_n(x)$ for each pair of (x, k_n) on $(\mathbf{X_grid}) \times (\mathbf{K_grid})$.
- (c) Consider $(x_1, \dots, x_5, x_{97}, \dots, x_{101}) \in (\mathbf{X_grid})$, which are the five smallest and five largest values on the x grid. For each of these values, plot the conditional Hill estimator as a function of k_n . Add corresponding 95% confidence bands arising from (3.26).
- (d) Consider $(k_1, \dots, k_5, k_{97}, \dots, k_{101}) \in (\mathbf{K_grid})$, which are the five smallest and five largest values on the k grid. For each of these values, plot the conditional Hill estimator as a function of x . Assess whether it makes sense to plot the corresponding 95% confidence bands that arise from (3.26), and justify your conclusion.
- (e) Construct a 3D plot of $\hat{\gamma}_n(x)$ on $(\mathbf{X_grid}) \times (\mathbf{K_grid})$.

Chapter 4

Parametric models

In statistical modelling, one often seeks an *optimal model* that best reflects the true data-generating process. Depending on the criterion function, optimality can take many different forms. The principle of maximum likelihood estimation is among the most widespread procedures of this kind and is often motivated by identifying the parameter values that make the observed data most probable. In fact, the procedure can also be derived from a minimisation criterion from the empirical distribution function.

To make this connection precise, introduce the Kullback–Leibler (KL) divergence, a measure of dissimilarity between two probability distributions. For two distributions P and Q , the KL divergence is defined as

$$D_{\text{KL}}(P \parallel Q) = \int p(z) \log \frac{p(z)}{q(z)} \, dz,$$

where p and q are the density functions of P and Q , respectively. Plugging in the empirical distribution function $\mathbb{F}^{(n)}$ and a parametric family of distributions $\{F_\theta\}$, we obtain heuristically that

$$\begin{aligned} \operatorname{argmin}_{\theta} D_{\text{KL}}(\mathbb{F}^{(n)} \parallel F_\theta) &= \operatorname{argmax}_{\theta} \int \log(f_\theta(z)) \, d\mathbb{F}^{(n)}(z) \\ &= \operatorname{argmax}_{\theta} \frac{1}{n} \sum_{i=1}^n \log f_\theta(Z_i). \end{aligned} \quad (4.1)$$

The right-hand side marks the starting point of our discussion, which is also shown schematically in Figure 4.1. We first develop the general M-estimation and likelihood framework, and then specialise it to regression models commonly used in actuarial practice.

4.1 Fundamental models

We begin with the general estimation setup that underlies maximum likelihood and related criteria. Assume we are interested in a (possibly high-dimensional) parameter $\theta \in \Theta$, where Θ is an abstract parameter space, with an associated

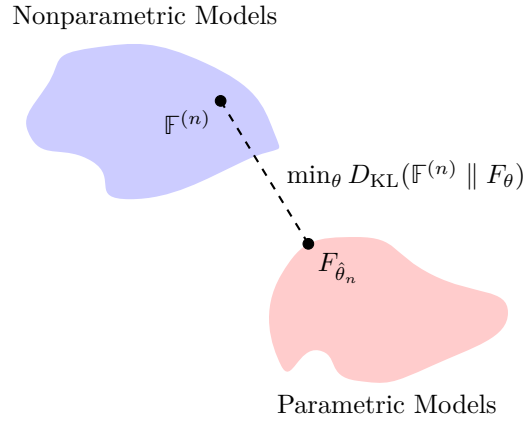


Figure 4.1: A schematic representation of the maximum likelihood estimate as a minimum Kullback–Leibler distribution.

distance d , related to the observations (Z_n) . The classical parametric approach is to consider estimators that solve maximisation problems. The archetype of these optimisation problems – and our main focus – is maximum likelihood estimation, the right-hand side of (4.1), though initially we present the theory for a slightly larger class of problems since proofs of consistency are still straightforward in this broader framework.

4.1.1 M-estimators

We begin with the general M -estimation framework. The following definition can be explored for general sequences of random variables, and hence we do not yet make any independence assumptions.

Definition 4.1 (M-estimators) Let $\hat{\theta}_n$ be a sequence of estimators arising as maximisers to the criterion function

$$\theta \mapsto M_n(\theta) = \frac{1}{n} \sum_{i=1}^n m(Z_i, \theta),$$

where $m(\cdot, \theta)$ are real-valued parametric functions. Then the estimators $\hat{\theta}_n$ are called *M-estimators*. Equivalently, when derivatives $\frac{d}{d\theta} m(\cdot, \theta) = \psi_{\theta}(\cdot)$ exist, we may obtain $\hat{\theta}_n$ as the zero of the function

$$\theta \mapsto \Psi_n(\theta) = \frac{1}{n} \sum_{i=1}^n \psi_{\theta}(Z_i).$$

The above derivatives play a smaller role in modern computation given efficient derivative-free optimisation routines. In the following theorem, notice the slight relaxation to near-maximisers, and the uniform convergence result which even in the simple iid case is not immediate from Theorem 1.2. The result also does not make use of the summation form of $M_n(\theta)$.

Theorem 4.1 (Consistency of M-estimators, near-maximisers) *Let M be a deterministic function with maximiser θ_0 . Assume*

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{\mathbb{P}} 0, \quad (4.2)$$

$$\sup_{\theta: d(\theta, \theta_0) \geq \varepsilon} M(\theta) < M(\theta_0), \quad \text{for every } \varepsilon > 0. \quad (4.3)$$

Then $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$ provided that

$$M_n(\hat{\theta}_n) \geq M_n(\theta_0) - o_{\mathbb{P}}(1). \quad (4.4)$$

Proof. We compare objective values and use separation to translate closeness in M_n into closeness of $\hat{\theta}_n$. By the uniform convergence assumption, (4.2), we get in particular $M_n(\theta_0) = M(\theta_0) + o_{\mathbb{P}}(1)$. Thus, by (4.4)

$$M_n(\hat{\theta}_n) \geq M(\theta_0) - o_{\mathbb{P}}(1).$$

Consequently we may evaluate

$$\begin{aligned} M(\theta_0) - M(\hat{\theta}_n) &\leq \{M_n(\hat{\theta}_n) + o_{\mathbb{P}}(1)\} - M(\hat{\theta}_n) \\ &\leq \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| + o_{\mathbb{P}}(1) \xrightarrow{\mathbb{P}} 0. \end{aligned} \quad (4.5)$$

Now, by the separation of (4.3), for every $\varepsilon > 0$ there exists $\eta > 0$ with

$$M(\theta) < M(\theta_0) - \eta, \quad \text{whenever } d(\theta, \theta_0) \geq \varepsilon.$$

Consequently

$$\{d(\hat{\theta}_n, \theta_0) \geq \varepsilon\} \subseteq \{M(\hat{\theta}_n) < M(\theta_0) - \eta\}.$$

and the latter has vanishing probability, by (4.5). $\square$

Condition (4.4) allows for approximate maximisers, which is the relevant notion when $\hat{\theta}_n$ is computed numerically with a stopping tolerance. The separation condition (4.3) is an identifiability requirement that prevents flat ridges of near-maximisers away from θ_0 . All conditions except the first in Theorem 4.1 are guaranteed if M is continuous, θ_0 is a unique maximiser of M , and Θ is compact. The real challenge is to show the uniform law of large numbers. Lemma 4.1 below provides some useful conditions under which such a result holds.

Lemma 4.1 (Uniform law of large numbers) *Let (Z_n) be iid random variables, $M(\theta) = \mathbb{E}[m(Z, \theta)]$ and $M_n(\theta) = n^{-1} \sum_{i=1}^n m(Z_i, \theta)$. Assume Θ is compact, $\theta \mapsto m(z, \theta)$ is continuous in $\theta \in \Theta$, there exists a dominating function $d(z)$ with $|m(z, \theta)| \leq d(z)$ for all $\theta \in \Theta$ and $\mathbb{E}[d(Z)] < \infty$. Then $M(\theta)$ is continuous and*

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{a.s.} 0.$$

Proof. We cover Θ by finitely many small balls and apply the strong law of large numbers on each to obtain uniform convergence. Let $\varepsilon > 0$ be given. Define

$$u(z, \theta, \delta) = \sup_{\gamma: |\gamma - \theta| \leq \delta} |m(z, \gamma) - m(z, \theta)|.$$

By almost sure continuity, for any fixed θ one has $\lim_{\delta \rightarrow 0} u(Z, \theta, \delta) = 0$. By the triangle inequality, $u(Z, \theta, \delta) \leq 2d(Z)$. Thus, by the dominated convergence theorem we have $\mathbb{E}[u(Z, \theta, \delta)] \leq \varepsilon$ whenever δ is small enough, say $\delta \leq \delta_0(\theta)$ for some $\delta_0(\theta) > 0$. Let $B(\theta)$ be an open ball with radius $\delta_0(\theta)$ and centre θ . Clearly $\Theta \subseteq \bigcup_{\theta \in \Theta} B(\theta)$, so the balls are an open covering of Θ . By compactness, one can find a finite open covering $B_k = B(\theta_k)$ ($k = 1, \dots, K$). Let $\delta_k = \delta_0(\theta_k)$ and $\mu_k = \mathbb{E}[u(Z, \theta_k, \delta_k)]$, and observe that by definition $\mu_k \leq \varepsilon$. Furthermore, for $\theta \in B_k$ we have

$$|M(\theta) - M(\theta_k)| \leq \mathbb{E}[|m(Z, \theta) - m(Z, \theta_k)|] \leq \mathbb{E}[u(Z, \theta, \delta_k)] \leq \varepsilon.$$

We shall use this bound repeatedly. Now again for $\theta \in B_k$, note

$$|M_n(\theta) - M(\theta)| \leq |M_n(\theta) - M_n(\theta_k)| + |M_n(\theta_k) - M(\theta_k)| + |M(\theta_k) - M(\theta)|.$$

We now consider each of the three terms separately. For the first term, observe by the triangle inequality

$$\begin{aligned} |M_n(\theta) - M_n(\theta_k)| &\leq \left(\frac{1}{n} \sum_{i=1}^n |m(Z_i, \theta) - m(Z_i, \theta_k)| - \mu_k \right) + \mu_k \\ &\leq \left(\frac{1}{n} \sum_{i=1}^n u(Z_i, \theta_k, \delta_k) - \mu_k \right) + \mu_k. \end{aligned}$$

The first part is, on a set of arbitrarily large probability, less than ε for $n \geq N_1(k, \varepsilon)$ by Theorem 1.2 (note that N_1 does not depend on θ), and the second part satisfies $\mu_k \leq \varepsilon$ by an earlier bound. Moving on to the second term, we note that by Theorem 1.2, it is almost surely smaller than ε for $n \geq N_2(k, \varepsilon)$. The third and final term is seen to be bounded by ε using one of the earlier bounds. Thus, for $n \geq \max\{N_1(k, \varepsilon), N_2(k, \varepsilon), k = 1, \dots, K\}$, it holds that

$$|M_n(\theta) - M(\theta)| \leq 4\varepsilon$$

almost surely. Since the right-hand side does not depend on θ , the inequality still holds if one takes the supremum over $\theta \in \Theta$ on the left-hand side. Since ε was arbitrary, we conclude that

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{a.s.} 0,$$

as required. □

Once the consistency of $\hat{\theta}_n$ is established, one can look at other asymptotic properties of the estimator. Under some regularity conditions, one may show that M-estimators converge at $\sqrt{n}$ rate to a Gaussian limit. In practice the requirement that the gradient satisfies a central limit theorem can usually be handled with standard tools such as Theorem 1.6.

Theorem 4.2 (Asymptotic normality of M-estimators) *Assume that $\hat{\theta}_n$ is an M-estimator with criterion function $M_n(\theta)$, and that:*

1. $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

2. $M_n(\theta)$ is twice continuously differentiable in an open subset $\mathcal{N}$ of Θ containing θ_0 .
3. $\sqrt{n} \nabla M_n(\theta_0) \xrightarrow{d} N(0, \Sigma)$.
4. There exists a function $H(\theta)$ which is continuous and nonsingular at θ_0 such that $\sup_{\theta \in \mathcal{N}} \|\nabla^2 M_n(\theta) - H(\theta)\| \xrightarrow{\mathbb{P}} 0$ for some norm.

Then, with $H_0 = H(\theta_0)$, we get $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, H_0^{-1} \Sigma H_0^{-1})$.

Proof. We linearise the score around θ_0 and control the Hessian uniformly. Without loss of generality, assume that $\mathcal{N}$ is an open ball with centre θ_0 and radius $r > 0$ contained in Θ . Let $\hat{1} = 1_{(\hat{\theta}_n \in \mathcal{N})}$, and note that $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$ implies $\hat{1} \xrightarrow{\mathbb{P}} 1$. Since $\hat{\theta}_n$ maximizes M_n , we have $\hat{1} \nabla M_n(\hat{\theta}_n) = 0$. We now want to use the mean value theorem to make an expansion of $\nabla M_n(\hat{\theta}_n)$. Let $(\nabla M_n(\theta))_i = \frac{\partial}{\partial \theta_i} M_n(\theta)$ be the i -th coordinate of $\nabla M_n(\theta)$. Then one can define a function

$$g_{n,i}(t) = (\nabla M_n((1-t)\theta_0 + t\hat{\theta}_n))_i, \quad t \in [0, 1].$$

Since $t \mapsto \hat{1} g_{n,i}(t)$ is a real-valued differentiable function, the mean-value theorem implies

$$\hat{1} g'_{n,i}(c_{n,i}) = \hat{1} (\nabla M_n(\hat{\theta}_n))_i - \hat{1} (\nabla M_n(\theta_0))_i$$

for some value $c_{n,i} \in (0, 1)$. Note that by the chain rule,

$$g'_{n,i}(t) = \nabla(\nabla M_n((1-t)\theta_0 + t\hat{\theta}_n))_i \cdot (\hat{\theta}_n - \theta_0)$$

and $\nabla(\nabla M_n(\theta))_i = (\frac{\partial^2}{\partial \theta_1 \partial \theta_i} M_n(\theta), \dots, \frac{\partial^2}{\partial \theta_p \partial \theta_i} M_n(\theta))^T$, which one may recognise as the i -th column of the Hessian $\nabla^2 M_n(\theta)$, or alternatively the i -th row of $(\nabla^2 M_n(\theta))^T$. In total, we obtain for any $i = 1, \dots, p$ that

$$0 = \hat{1} (\nabla M_n(\hat{\theta}_n))_i = \hat{1} (\nabla M_n(\theta_0))_i + \hat{1} (\nabla^2 M_n(\bar{\theta}_{n,i}))_i^T \cdot (\hat{\theta}_n - \theta_0)$$

for $\bar{\theta}_{n,i} = (1 - c_{n,i})\theta_0 + c_{n,i}\hat{\theta}_n$ when $\hat{1} = 1$ and being equal to θ_0 otherwise (for convenience). Then clearly $\bar{\theta}_{n,i} \xrightarrow{\mathbb{P}} \theta_0$. Let $\bar{H}_n$ be the matrix with i -th row being $(\nabla^2 M_n(\bar{\theta}_{n,i}))_i^T$ and $\bar{H}$ be the matrix with i -th row being $H(\bar{\theta}_{n,i})_i^T$. The main technical challenge going forward is that different rows of $\bar{H}_n$ and $\bar{H}$ are evaluated in different values of θ , meaning that there is not necessarily any $\bar{\theta}$ satisfying $\bar{H}_n = (\nabla^2 M_n(\bar{\theta}))^T$ and/or $\bar{H} = H(\bar{\theta})^T$. Note that we can rewrite

$$0 = \hat{1} \nabla M_n(\theta_0) + \hat{1} \bar{H}_n (\hat{\theta}_n - \theta_0).$$

Observe that

$$\|\bar{H}_n - H_0^T\| \leq \|\bar{H}_n - \bar{H}\| + \|\bar{H} - H_0^T\|.$$

Recall that all matrix norms are equivalent. Let $\|\cdot\|_1$ be the maximum absolute column sum and $\|\cdot\|_\infty$ be the maximum absolute row sum. For the first term, we get for some positive constants β_1, β_2 that

$$\begin{aligned} \|\bar{H}_n - \bar{H}\| &\leq \beta_1 \|\bar{H}_n - \bar{H}\|_\infty \\ &\leq \beta_1 \sup_{\theta \in \Theta} \|(\nabla^2 M_n(\theta))^T - H(\theta)^T\|_\infty \\ &= \beta_1 \sup_{\theta \in \Theta} \|\nabla^2 M_n(\theta) - H(\theta)\|_1 \leq \beta_2 \sup_{\theta \in \Theta} \|\nabla^2 M_n(\theta) - H(\theta)\| \xrightarrow{\mathbb{P}} 0 \end{aligned}$$

where the convergence follows by the last assumption. For the second term, using the norm $\|\cdot\|_{L^1}$ which is the absolute sum of the matrix entries, one can see that $H(\theta)$ is continuous at θ_0 if and only if every entry is continuous at θ_0 . Thus, for some positive constant β_3 we get

$$\|\bar{H} - H_0^T\| \leq \beta_3 \|\bar{H} - H_0^T\|_{L^1} \xrightarrow{\mathbb{P}} 0$$

where the convergence follows by using continuity of the entries of $H(\theta)$ at θ_0 , the continuous mapping Theorem 1.4, and that $\bar{\theta}_{n,i} \xrightarrow{\mathbb{P}} \theta_0$. Hence we conclude that $\bar{H}_n \xrightarrow{\mathbb{P}} H_0^T$. Now introduce $\bar{1} = 1_{(\hat{\theta}_n \in \mathcal{N}, \bar{H}_n \text{ nonsingular})} = \hat{1} \cdot 1_{(\bar{H}_n \text{ nonsingular})}$. Using that the determinant is a continuous function on the metric space of matrices (it is a polynomial of the entries), we see that $\bar{1} \xrightarrow{\mathbb{P}} 1$ since

$$\begin{aligned} (\|1_{(\bar{H}_n \text{ nonsingular})} - 1\| \geq \varepsilon) &= (\bar{H}_n \text{ singular}) \\ &\subseteq (\|\det(\bar{H}_n) - \det(H^T)\| \geq \|\det(H_0^T)\|), \end{aligned}$$

which has vanishing probability by the continuous mapping Theorem 1.4. Now since

$$0 = \bar{1} \nabla M_n(\theta_0) + \bar{1} \bar{H}_n(\hat{\theta}_n - \theta_0)$$

we get that

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\bar{1} \bar{H}_n^{-1} \sqrt{n} \nabla M_n(\theta_0) + (1 - \bar{1}) \sqrt{n}(\hat{\theta}_n - \theta_0).$$

Then since $\bar{1} \bar{H}_n^{-1} \xrightarrow{\mathbb{P}} (H_0^T)^{-1}$ by continuity of matrix inversion and the nonsingularity assumption, and $\bar{1} \xrightarrow{\mathbb{P}} 1$, we get by Slutsky's Theorem 1.3 that

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} (H_0^T)^{-1} N(0, \Sigma) = N(0, (H_0^T)^{-1} \Sigma H_0^{-1}).$$

Note finally that

$$\begin{aligned} \|H_0 - H_0^T\| &\leq \beta_3 \|H_0 - H_0^T\|_{L^1} \\ &\leq \beta_3 (\|\nabla^2 M_n(\theta_0) - H_0\|_{L^1} + \|\nabla^2 M_n(\theta_0) - H_0^T\|_{L^1}) \\ &= 2\beta_3 \|\nabla^2 M_n(\theta_0) - H_0\|_{L^1} \xrightarrow{\mathbb{P}} 0 \end{aligned}$$

where we in the last equality used that the L^1 -norm for matrices is invariant under transposition, and that $(\nabla^2 M_n(\theta_0))^T = \nabla^2 M_n(\theta_0)$ since the mixed partial derivatives are equal by Schwarz's theorem, due to the continuity of the second partial derivatives. Thus $H_0 = H_0^T$ and the result follows. $\square$

A nonlinear least squares model provides a simple instance of the general asymptotic normality result. Such a simple example serves as a template and can naturally be adapted to other specifications.

Example 4.1 (Nonlinear least squares) Suppose that we observe random iid copies (Z_n, X_n) of the random vector (Z, X) from a regression model

$$Z = \tilde{m}(X, \theta_0) + e, \quad \mathbb{E}[e | X] = 0.$$

Here $\tilde{m}(\cdot, \theta)$ is any parametric family of regression functions, for instance, we may take $\tilde{m}(x, \theta) = \theta_1 + \theta_2 x + \theta_3 e^{\theta_4 x}$, and we aim at estimating the unknown

vector θ_0 . We have assumed that the independent variables (X_n) are a random sample to fit the example in the iid notation, but the analysis could also be carried out conditionally, for a fixed conditioning value x . The least squares estimator that maximises

$$M_n(\theta) = -\frac{1}{n} \sum_{i=1}^n (Z_i - \tilde{m}(X_i, \theta))^2$$

is an M-estimator for $m(z, x, \theta) = -(z - \tilde{m}(x, \theta))^2$. It should be expected to converge to the maximiser of the limit criterion function

$$M(\theta) = \mathbb{E}[m(Z, X, \theta)] = -\mathbb{E}[(\tilde{m}(X, \theta_0) - \tilde{m}(X, \theta))^2] - \mathbb{E}[\epsilon^2].$$

Thus the least squares estimator should be consistent if θ_0 is identifiable from the model, in the sense that $\theta \neq \theta_0$ implies that $\tilde{m}(X, \theta) \neq \tilde{m}(X, \theta_0)$ with positive probability. For sufficiently regular regression models, we have by expansion

$$M(\theta) \approx -\mathbb{E}[(\theta - \theta_0)^\top \nabla \tilde{m}(X, \theta_0)]^2 - \mathbb{E}[\epsilon^2].$$

Thus using notation from Theorem 4.2 we get

$$H_0 = H(\theta_0) = \nabla^2 M(\theta_0) = 2\mathbb{E}[\nabla \tilde{m}(X, \theta_0) \nabla \tilde{m}(X, \theta_0)^\top]$$

and since

$$\nabla m(Z, X, \theta_0) = -2(Z - \tilde{m}(X, \theta_0)) \nabla \tilde{m}(X, \theta_0),$$

which has mean zero by the tower property, then

$$\begin{aligned} \Sigma &= \mathbb{E}[\nabla m(Z, X, \theta_0) \nabla m(Z, X, \theta_0)^\top] = 4\mathbb{E}[(Z - \tilde{m}(X, \theta_0))^2 \nabla \tilde{m}(X, \theta_0) \nabla \tilde{m}(X, \theta_0)^\top] \\ &= 4\mathbb{E}[\epsilon^2 \nabla \tilde{m}(X, \theta_0) \nabla \tilde{m}(X, \theta_0)^\top]. \end{aligned}$$

For instance, if ϵ and X are independent, then this leads to the asymptotic covariance matrix

$$H_0^{-1} \Sigma H_0^{-1} = H_0^{-1} (2 \cdot H \mathbb{E}[\epsilon^2]) H_0^{-1} = 2 \text{Var}(\epsilon) H_0^{-1}.$$

Figure 4.2 illustrates the CLT for the nonlinear least squares M-estimator, for the simpler case $\tilde{m}(x, \theta) = \theta_1 e^{\theta_2 x}$ with uniformly distributed covariates and centred beta errors (non-symmetric). The centred and scaled estimator converges to a normal distribution. $\diamond$

4.1.2 Maximum likelihood estimation

We now concentrate on the case where $\theta \in \Theta$ parametrises the distribution functions of the iid random variables (Z_n) . In particular, we assume that there exists $\theta_0 \in \Theta$ such that $F(\cdot | \theta_0)$ is the true distribution of Z_1 , and that the distributions $F(\cdot | \theta)$ admit densities $f(\cdot | \theta)$ with respect to Lebesgue measure (for continuous variables) or the counting measure (for discrete variables), though for the sake of exposition we only present the former.

We also now assume that

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(Z_i | \theta),$$

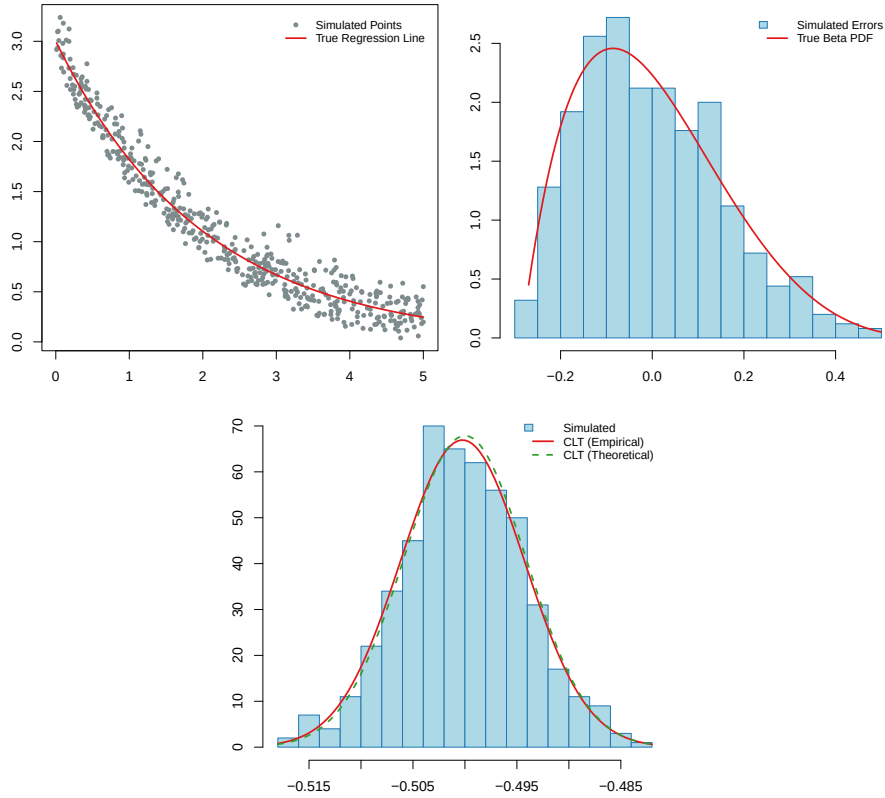


Figure 4.2: Nonlinear least squares M-estimation from a beta regression model. We consider $n = 500$ observations from the model $\tilde{m}(x, \theta) = \theta_1 e^{\theta_2 x}$, shown as response–covariate scatter with true regression curve (top left), with centred beta errors (top right). The assumptions imply that the CLT from Theorem 4.2 is applicable for the estimator component $\hat{\theta}_2$ (bottom).

and that it satisfies a uniform law of large numbers convergence in θ (see Lemma 4.1 for regularity conditions for this convergence to hold), with limit

$$M(\theta) = \int \log f(z|\theta) f(z|\theta_0) \, dz.$$

For intuition behind the following definition, we refer to (4.1).

Definition 4.2 (Likelihood, MLE, and Fisher information) Let (Z_n) be iid copies of the random variable Z . We define the *likelihood function* and *log-likelihood function* by

$$\begin{aligned} L_n(\theta) &= \prod_{i=1}^n f(Z_i|\theta), \\ \ell_n(\theta) &= \log L_n(\theta) = \sum_{i=1}^n \ell(Z_i|\theta) = \sum_{i=1}^n \log f(Z_i|\theta) = nM_n(\theta), \end{aligned}$$

the *maximum likelihood estimator* (MLE) by

$$\hat{\theta}_n = \operatorname{argmax}_{\theta \in \Theta} \ell_n(\theta) = \operatorname{argmin}_{\theta \in \Theta} \{-\ell_n(\theta)\},$$

and the *Fisher information* by

$$I(\theta_0) = \mathbb{E}_{\theta_0} [\ell'(Z|\theta_0)^2] \equiv \mathbb{E}_{\theta_0} \left[\left(\frac{\partial}{\partial \theta} \log f(Z|\theta) \Big|_{\theta=\theta_0} \right)^2 \right],$$

where $\mathbb{E}_{\theta_0}$ describes taking expectation of a function of Z under the distribution $F(\cdot|\theta_0)$.

Several regularity conditions are required in the theory of MLEs, mainly to be able to exchange integration with derivatives, and to guarantee the existence of the Fisher information. Additional regularity conditions require the parameter space Θ to be compact and θ_0 to be identifiable and in the interior of Θ . Below, we simply refer to these conditions as *standard regularity conditions*, which are listed in detail in the more abstract Theorem 4.2.

Lemma 4.2 (Computing the Fisher Information) *Assume standard regularity conditions from Theorem 4.2. Then we have that*

$$\mathbb{E}_{\theta_0} [\ell''(Z|\theta_0)] = -I(\theta_0).$$

Proof. We differentiate under the integral and simplify using the normalisation of the density. We simply compute using the quotient rule

$$\begin{aligned} \mathbb{E}_{\theta_0} [\ell''(Z|\theta_0)] &= \int \left(\frac{f''(z|\theta_0)}{f(z|\theta_0)} - \left(\frac{f'(z|\theta_0)}{f(z|\theta_0)} \right)^2 \right) f(z|\theta_0) \, dz \\ &= 0 - I(\theta_0), \end{aligned}$$

where we have taken the double derivative outside of the integral sign, used $\int f(x|\theta_0) \, dx = 1$, and the definition of I . $\square$

The result below is actually a direct consequence of Theorem 4.2. However, we restate it due to its historic significance and widespread usage. We also provide a conceptual proof which gives much more concrete insight than that of Theorem 4.2.

Theorem 4.3 (Asymptotic normality of MLEs, unidimensional case) *Assume standard regularity conditions from Theorem 4.2. Then we have*

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}).$$

Conceptual proof. We expand the score around θ_0 and apply Theorem 1.2 and Theorem 1.7 to the score and observed information. Although one may simply adapt Theorem 4.2, it is quite simple to give a direct and illuminating proof.

Since $\hat{\theta}_n$ maximises M_n , then $M'_n(\hat{\theta}_n) = 0$. On the other hand, by the mean value theorem, there exists $\tilde{\theta}_n$ between $\hat{\theta}_n$ and θ_0 with

$$0 = M'_n(\hat{\theta}_n) = M'_n(\theta_0) + M''_n(\tilde{\theta}_n)(\hat{\theta}_n - \theta_0).$$

Rearranging, and recalling that $M'(\theta_0) = 0$, we get

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\frac{\sqrt{n}\{M'_n(\theta_0) - M'(\theta_0)\}}{M''_n(\tilde{\theta}_n)}.$$

Now notice that

1. By a uniform strengthening of Theorem 1.2, the denominator converges to $\mathbb{E}_{\theta_0}[\ell''(Z | \theta_0)]$ (θ_n is clearly also consistent), which by Lemma 4.2 is equal to $-I(\theta_0)$.
2. By the Laplace's CLT, Theorem 1.6, the numerator is asymptotically distributed as $N(0, \text{Var}_{\theta_0}(\ell'(Z | \theta_0)))$.

Now, by definition of θ_0 ,

$$\mathbb{E}[\ell'(Z | \theta_0)] = \int \frac{f'(z | \theta_0)}{f(z | \theta_0)} f(z | \theta_0) dz = 0,$$

so that $\text{Var}_{\theta_0}(\ell'(Z | \theta_0)) = I(\theta_0)$. Thus the result follows by Slutsky's Lemma 1.3. $\square$

When the parameter lies on the support boundary, the normal limit can fail, as the uniform distribution example below illustrates.

Example 4.2 (Uniform model: asymptotic normality fails) Let (Z_n) be iid $\text{Unif}(0, \theta)$ with density $f(z | \theta) = \theta^{-1}1_{(0 < z < \theta)}$. The log-likelihood is

$$\ell_n(\theta) = -n \log \theta + \sum_{i=1}^n \log 1_{(Z_i < \theta)},$$

so the MLE is seen to be $\hat{\theta}_n = Z_{n:n} = \max_{1 \leq i \leq n} Z_i$. Then for $x > 0$,

$$\mathbb{P}(n(\theta - \hat{\theta}_n) \leq x) = \mathbb{P}\left(Z_{n:n} \geq \theta - \frac{x}{n}\right) = 1 - \left(1 - \frac{x}{n\theta}\right)^n \rightarrow 1 - e^{-x/\theta}.$$

Hence $n(\theta - \hat{\theta}_n) \xrightarrow{d} \text{Exp}(1/\theta)$ and so $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{\mathbb{P}} 0$. Consequently, the normal limit in Theorem 4.3 fails. The regularity assumptions break down here because the θ is the support boundary. $\diamond$

4.1.3 Actuarial considerations

Consider an iid sample (Z_n) drawn from a distribution $F(\cdot | \theta)$. In many practical scenarios, the observed data may be subject to truncation or censoring. First, we define left-truncation and right-censoring.

Left-truncation occurs when observations below a certain threshold are not included in the sample. In insurance, *left-truncation is related to deductibles*. A deductible is an amount below which the insurer does not cover any loss. Therefore, only claims exceeding the deductible are observed and included in the analysis. If L_i is the deductible, the observed claims Z_i are such that $Z_i \geq L_i$.

Definition 4.3 (Left-truncation) Assume that Z_i is only observed if $Z_i \geq L_i$, where L_i is the truncation point, so that the *left-truncated data* can be represented as:

$$Z_i | Z_i \geq L_i$$

The conditional probability density function of Z_i given that it is observed (i.e. $Z_i \geq L_i$) is given by

$$f_{Z_i | Z_i \geq L_i}(z | \theta) = \frac{f(z | \theta)}{\mathbb{P}(Z_i \geq L_i | \theta)} = \frac{f(z | \theta)}{\bar{F}(L_i | \theta)},$$

for $z \geq L_i$, where $f(\cdot | \theta)$ is the density function of Z_i .

On the other hand, right-censoring occurs when the value of an observation is only known to exceed a certain threshold. Right-censoring occurs in insurance when the value of a claim or a lifetime is not fully observed. For example, a claim might still be open (not fully settled) at the end of the observation period, or an individual might still be alive. In these cases, we know that the claim amount or lifetime exceeds a certain threshold, but we do not (yet) know the exact value.

Definition 4.4 (Right-censoring) Assume we observe either Z_i , or we know that Z_i exceeds a random censoring mechanism C_i , so the observed *right-censored data* can be represented as:

$$W_i = \min(Z_i, C_i), \quad \delta_i = 1_{(Z_i \leq C_i)}$$

where δ_i is an indicator variable that equals 1 if Z_i is observed and 0 if it is censored.

In short, these two mechanisms play different roles. Left-truncation specifies which observations enter the sample, while right-censoring specifies how much is observed after entry.

We assume noninformative truncation and censoring: (Z_i) is independent of (L_i, C_i) and $\mathbb{P}(C_i \geq L_i) = 1$. Under this assumption, we treat the L_i, C_i as fixed, and consequently one is formally dealing with a *partial likelihood*, as we are not interested in estimating the distribution of the censoring and truncation mechanisms. Under this assumption the conditional density of Z_i given $Z_i \geq L_i$ and L_i is $f(z | \theta) / \bar{F}(L_i | \theta)$, and the censoring contribution factors as in (4.7).

The independence is used to justify these conditional likelihoods and to keep the log-likelihood contributions iid; if Z_i and (L_i, C_i) are dependent, the selection mechanism must be modelled explicitly and the formulas below no longer apply.

Consequently, the log-likelihoods require adaptation in both cases, and are given, respectively, by:

$$\ell_n(\theta) = \sum_{i=1}^n \{ \log f(Z_i | \theta) - \log \bar{F}(L_i | \theta) \}, \quad (4.6)$$

and

$$\ell_n(\theta) = \sum_{i=1}^n \{ \delta_i \log f(Z_i | \theta) + (1 - \delta_i) \log \bar{F}(C_i | \theta) \}. \quad (4.7)$$

They can be derived in several ways. Perhaps the most intuitive way is considering a likelihood as “the probability of observing the dataset at hand”. For a formal derivation, we direct the reader to Section ??, of which the survival model is a special case. Consistency and asymptotic normality results for the resulting maximum partial likelihood estimates then follow from Theorem ?? and Theorem ??, respectively.

Example 4.3 (Left-truncation with right-censoring: Gompertz scale) Let (Z_n) be random variables following the Gompertz distribution with known shape $\eta > 0$ and unknown scale $b > 0$, that is with

$$\bar{F}(z | b) = \exp \left(-\frac{b}{\eta}(e^{\eta z} - 1) \right), \quad f(z | b) = b e^{\eta z} \exp \left(-\frac{b}{\eta}(e^{\eta z} - 1) \right).$$

Assume independent left-truncation at L_i and right-censoring with $W_i = \min(Z_i, C_i)$, $\delta_i = 1_{(Z_i \leq C_i)}$, and with $W_i \geq L_i$. The log-likelihood up to constants in b is

$$\ell_n(b) = \left(\sum_{i=1}^n \delta_i \right) \log b + \eta \sum_{i=1}^n \delta_i W_i - \frac{b}{\eta} \sum_{i=1}^n (e^{\eta W_i} - 1) + \frac{b}{\eta} \sum_{i=1}^n (e^{\eta L_i} - 1),$$

which yields the explicit MLE

$$\hat{b}_n = \frac{\eta \sum_{i=1}^n \delta_i}{\sum_{i=1}^n (e^{\eta W_i} - e^{\eta L_i})},$$

for $\sum_{i=1}^n \delta_i > 0$. ◇

A Weibull scale model yields an analogous explicit MLE under truncation and censoring.

Example 4.4 (Left-truncation with right-censoring: Weibull scale) Let (Z_n) be random variables following the Weibull distribution with known shape $k > 0$ and unknown scale $\lambda > 0$, that is with

$$\bar{F}(z | \lambda) = e^{-(z/\lambda)^k}, \quad f(z | \lambda) = \frac{k}{\lambda} \left(\frac{z}{\lambda} \right)^{k-1} e^{-(z/\lambda)^k}.$$

Again, assume independent left-truncation at L_i and right-censoring with $W_i = \min(Z_i, C_i)$, $\delta_i = 1_{(Z_i \leq C_i)}$, and with $W_i \geq L_i$. The log-likelihood up to constants in λ is

$$\ell_n(\lambda) = -k \left(\sum_{i=1}^n \delta_i \right) \log \lambda - \sum_{i=1}^n \left(\frac{W_i}{\lambda} \right)^k + \sum_{i=1}^n \left(\frac{L_i}{\lambda} \right)^k,$$

which gives the explicit MLE

$$\hat{\lambda}_n = \left(\frac{\sum_{i=1}^n W_i^k - \sum_{i=1}^n L_i^k}{\sum_{i=1}^n \delta_i} \right)^{1/k},$$

for $\sum_{i=1}^n \delta_i > 0$. The $k = 1$ case corresponds to the exponential distribution, for which the formula yields an interpretable adaptation of the usual maximum likelihood estimate, the sample mean. $\diamond$

Remark 4.1 (Simulation of truncated or censored random variables) The inversion idea gives direct simulation from conditional distributions, corresponding to truncation. Let Z have cdf F and let $a < b$ with $F(b) > F(a)$. Then the conditional law of Z given $a < Z \leq b$ has cdf

$$F_{(a,b]}(z) = \frac{F(z) - F(a)}{F(b) - F(a)}, \quad a < z \leq b,$$

and can therefore be generated by

$$Z_{a,b} = F^{-1}(F(a) + U(F(b) - F(a))), \quad U \sim \text{Unif}(0, 1).$$

Likewise, with $\bar{F} = 1 - F$, the overshoot law of $Z - a$ given $Z > a$ is generated by

$$Z_a^+ = F^{-1}(F(a) + U\bar{F}(a)) - a.$$

This avoids throwing away samples which fall below the threshold, which corresponds to naively simulating an overshoot. Censoring is easily handled through simulation of Z and the censoring mechanism C , and constructing $W = \min\{Z, C\}$, $\delta = 1_{(Z \leq C)}$. $\circ$

4.2 Matrix models

Matrix models provide a framework for constructing distributions through the lens of functional calculus. By representing distributions as the time until absorption in a Markovian system, we gain access to analytical tools that simplify statistical problems. This section details the progression from standard phase-type (PH) distributions to more flexible inhomogeneous (IPH) and discrete (DPH) variants, with a focus on flexible and computationally efficient models.

4.2.1 Phase-type distributions

Phase-type (PH) distributions form a versatile parametric class on $\mathbb{R}_+$ obtained as absorption times of finite-state Markov jump processes. Their flexibility

is a key selling point, since by enlarging the state space one can approximate essentially any positive distribution arbitrarily well, while retaining explicit likelihoods and a computationally tractable EM algorithm. We present the basic theory; the estimation for fully observed (up to a censoring time) jump processes is presented in Chapter ??.

Let $Z = (Z_t)_{t \geq 0}$ be a continuous-time Markov chain on the finite state space $\{1, \dots, p, p+1\}$, where states $1, \dots, p$ are transient and $p+1$ is absorbing. Let

$$Q = \begin{pmatrix} S & \mathbf{s} \\ 0 & 0 \end{pmatrix}$$

be the generator matrix of the Markov process, with subgenerator $S \in \mathbb{R}^{p \times p}$, $S_{jk} \geq 0$ for $j \neq k$, and $\mathbf{s} = -S\mathbf{1}$ the exit-rate vector (here $\mathbf{1}$ is the column vector of ones). The entries s_j represent the instantaneous rate of absorption from state j to state $p+1$. Note that each row of S sums to $-s_j$, so the rows of Q sum to zero as per the generator requirement. Let the initial distribution on the transient states be the row vector α , $\alpha\mathbf{1} = 1$, and define the absorption time

$$\tau = \inf\{t > 0 : Z_t = p+1\}.$$

We refer to τ as the *absorption time*. We use the row/column convention throughout, and denote the matrix exponential by

$$e^{tS} = \sum_{m=0}^{\infty} \frac{t^m}{m!} S^m,$$

which is always well-defined for any matrix S and solves $\frac{d}{dt}e^{tS} = e^{tS}S$ with $e^{0 \cdot S} = \text{Id}$.

Definition 4.5 (Phase-type distribution) The distribution of τ is a *phase-type distribution* with representation (α, S) , written $\tau \sim \text{PH}(\alpha, S)$.

Remark 4.2 (Simulation of PH observations) To simulate PH data, one simulates the underlying homogeneous Markov jump process with generator Q until absorption and saves the absorption time. This is exactly the homogeneous Markov jump-process simulation setup in Section ??; see in particular the methods of Proposition ?? and Proposition ??. $\circ$

Since the chain is Markov with generator Q , the transient transition matrix is e^{tS} . Consequently, the transient state distribution at time t is αe^{tS} , and probabilities and densities of τ can be written in closed form. This is the main reason PH models are analytically convenient and why they are often used as flexible parametric models for positive data. However, the representation (α, S) is not unique; different pairs can yield the same distribution. In applied work the dimension p and the structure of S are part of the modelling choice, balancing flexibility and parsimony. Restricting to structured classes such as Coxian or upper-triangular S often improves interpretability and in some special cases even reduces non-identifiability.

We now collect the standard analytic formulas that follow from the Markov structure. They are the basic tools for likelihood inference and for understanding how distributional properties are linked to the generator.

Proposition 4.1 (Basic formulas) *Let $\tau \sim PH(\alpha, S)$. Then for $t \geq 0$,*

$$\begin{aligned}\bar{F}(t) &= \mathbb{P}(\tau > t) = \alpha e^{tS} \mathbf{1}, \\ f(t) &= \alpha e^{tS} \mathbf{s}, \\ \mathbb{E}[e^{-u\tau}] &= \alpha(uI - S)^{-1} \mathbf{s}, \quad u > 0, \\ \mathbb{E}[\tau^k] &= k! \alpha(-S)^{-k} \mathbf{1}, \quad k \in \mathbb{N}.\end{aligned}$$

Proof. The transient state distribution is αe^{tS} , so the survival probability is the probability of not yet being absorbed, $\bar{F}(t) = \alpha e^{tS} \mathbf{1}$. Differentiating gives the density $f(t) = \alpha e^{tS} \mathbf{s}$. The Laplace transform follows from

$$\mathbb{E}[e^{-u\tau}] = \int_0^\infty e^{-ut} f(t) dt = \alpha \left(\int_0^\infty e^{t(S-uI)} dt \right) \mathbf{s} = \alpha(uI - S)^{-1} \mathbf{s},$$

and the moment formula follows by repeated differentiation at $u = 0$. $\square$

The explicit expressions show how moments and Laplace transforms reduce to matrix algebra in terms of S . In particular, the mean and higher moments exist whenever $-S$ is invertible, which is automatic for a transient subgenerator. The hazard function can be non-monotone and the density can be multimodal, reflecting the flexibility of the PH class.

The class of *matrix-exponential* (ME) distributions generalises PH by allowing an arbitrary representation $f(t) = \alpha e^{tS} \mathbf{s}$ without imposing the subgenerator constraints on S . In this chapter we remain within PH since it guarantees a probabilistic interpretation and a direct Markov construction.

Proposition 4.2 (Closure properties) *The PH class is closed under several natural operations that correspond to finite-state Markov constructions. Let $\tau_r \sim PH(\alpha_r, S_r)$, $r = 1, \dots, m$, be independent and write $\mathbf{s}_r = -S_r \mathbf{1}$.*

1. Finite mixtures. *Let C be independent of $(\tau_1, \dots, \tau_m)$ with $\mathbb{P}(C = r) = p_r$, $\sum_{r=1}^m p_r = 1$, and set $\tau = \tau_C$. Then τ is PH with*

$$\alpha^* = (p_1 \alpha_1, \dots, p_m \alpha_m), \quad S^* = \text{diag}(S_1, \dots, S_m).$$

2. Positive linear combinations. *For constants $c_1, \dots, c_m > 0$, the sum*

$$\tau^* = \sum_{r=1}^m c_r \tau_r$$

is PH. One convenient representation is

$$\alpha^* = (\alpha_1, 0, \dots, 0),$$

with block upper-bidiagonal generator

$$S^* = \begin{pmatrix} S_1/c_1 & (\mathbf{s}_1/c_1)\alpha_2 & 0 & \cdots & 0 \\ 0 & S_2/c_2 & (\mathbf{s}_2/c_2)\alpha_3 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & S_{m-1}/c_{m-1} & (\mathbf{s}_{m-1}/c_{m-1})\alpha_m \\ 0 & \cdots & \cdots & 0 & S_m/c_m \end{pmatrix}.$$

For $m = 2$ and $c_1 = c_2 = 1$, this is the usual convolution closure.

3. Order statistics. Let $\tau_{1:m} \leq \dots \leq \tau_{m:m}$ be the order statistics of $(\tau_1, \dots, \tau_m)$. Then $\tau_{k:m}$ is PH for each $k = 1, \dots, m$. In particular, both

$$\min(\tau_1, \dots, \tau_m) = \tau_{1:m} \quad \text{and} \quad \max(\tau_1, \dots, \tau_m) = \tau_{m:m}$$

are PH. For the minimum, one has the explicit representation

$$\alpha^{\min} = \alpha_1 \otimes \dots \otimes \alpha_m, \quad S^{\min} = S_1 \oplus \dots \oplus S_m,$$

where $\otimes$ denotes the Kronecker product and $\oplus$ the Kronecker sum ($A \oplus B := A \otimes I + I \otimes B$, with identity matrices of matching sizes).

Proof. For mixtures, the chain starts in block r with probability p_r and then evolves according to S_r , yielding the block-diagonal representation.

For linear combinations, scaling by c_r replaces S_r by S_r/c_r , and the block upper-bidiagonal construction runs components sequentially: after absorption from block r , the chain restarts in block $r+1$ with initial law α_{r+1} . The resulting absorption time is $\sum_{r=1}^m c_r \tau_r$.

For order statistics, consider the product chain of the m independent PH chains (including their absorbing states). For fixed k , declare absorbing all product states where at least k coordinates are absorbed, and transient otherwise. The first hitting time of this absorbing set is exactly $\tau_{k:m}$; since the state space is finite, it is PH. Taking $k = 1$ and $k = m$ gives minimum and maximum. For $k = 1$, all coordinates must stay transient up to time t , which gives the Kronecker representation above:

$$\mathbb{P}(\tau_{1:m} > t) = \prod_{r=1}^m \alpha_r e^{t S_r} \mathbf{1} = (\alpha_1 \otimes \dots \otimes \alpha_m) e^{t(S_1 \oplus \dots \oplus S_m)} \mathbf{1}.$$

□

Remark 4.3 (Special cases and density) By increasing the dimension, PH distributions are dense in the set of distributions on $\mathbb{R}_+$ under weak convergence; hence they can approximate essentially any positive distribution while retaining a parametric structure; see [Bladt and Nielsen \[2017\]](#) for a detailed account. ○

Example 4.5 (Structured phase-type subclasses) Let $\tau \sim \text{PH}(\alpha, S)$ with p transient states. The following choices of (α, S) yield standard families and the matrix expression $f(t) = \alpha e^{tS} \mathbf{s}$ collapses to familiar densities.

1. *Exponential.* With $p = 1$, $\alpha = (1)$, and $S = (-\lambda)$, one obtains

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0,$$

the unique memoryless PH law.

2. *Erlang (Gamma with integer shape).* With $\alpha = (1, 0, \dots, 0)$ and S upper-bidiagonal having diagonal entries $-\lambda$ and super-diagonal entries λ , the absorption time is $\text{Erlang}(p, \lambda)$ with density

$$f(t) = \frac{\lambda^p t^{p-1}}{(p-1)!} e^{-\lambda t}, \quad t \geq 0.$$

This is Gamma with shape p , hence only integer shapes are attained.

3. *Hypoexponential*. With $\alpha = (1, 0, \dots, 0)$ and S upper-bidiagonal having diagonal entries $-\lambda_1, \dots, -\lambda_p$ and super-diagonal entries $\lambda_1, \dots, \lambda_{p-1}$, the absorption time is the sum of independent exponentials with rates $\lambda_1, \dots, \lambda_p$ (a convolution model in which every sample passes through all p phases). If the rates are distinct, the density is

$$f(t) = \sum_{j=1}^p c_j e^{-\lambda_j t}, \quad c_j = \lambda_j \prod_{k \neq j} \frac{\lambda_k}{\lambda_k - \lambda_j}.$$

When some rates coincide, the density is a polynomial times $e^{-\lambda t}$, reducing to Erlang in the equal-rate case.

4. *Hyperexponential*. With $\alpha = (\alpha_1, \dots, \alpha_p)$ and $S = \text{diag}(-\lambda_1, \dots, -\lambda_p)$, one obtains the finite exponential mixture

$$f(t) = \sum_{j=1}^p \alpha_j \lambda_j e^{-\lambda_j t}, \quad \bar{F}(t) = \sum_{j=1}^p \alpha_j e^{-\lambda_j t}.$$

This class can exhibit much larger variance than a single exponential with the same mean. It is a mixture model, since each observation comes from one exponential component, in contrast to the hypoexponential case where each observation is a sum of p exponential phases.

5. *Coxian*. With $\alpha = (1, 0, \dots, 0)$ and S upper-bidiagonal having diagonal entries $-\lambda_j$ and super-diagonal entries $\lambda_j q_j$ with $q_j \in [0, 1]$ for $j = 1, \dots, p-1$ and $q_p = 0$, the chain moves from phase j to $j+1$ with probability q_j and is absorbed with probability $1 - q_j$. Writing f_j for the hypoexponential density based on the first j rates, one has

$$f(t) = \sum_{j=1}^p \left(\prod_{r=1}^{j-1} q_r \right) (1 - q_j) f_j(t).$$

Thus the Coxian model is a mixture over the number of completed phases, and when $q_j = 1$ for $j = 1, \dots, p-1$ the distribution reduces to the hypoexponential family (Erlang in the equal-rate case). If instead $\alpha = (\alpha_1, \dots, \alpha_p)$, the distribution is usually called *Generalised Coxian*, with the density being adapted accordingly.

◇

Remark 4.4 (Exponential tails) In all of the above subclasses the survival function is a finite linear combination of exponentials, hence the tail decays exponentially. In fact this holds for every PH distribution: since $\bar{F}(t) = \alpha e^{tS} \mathbf{1}$ and the spectrum of a transient subgenerator S lies strictly in the left half-plane, there exist constants $C, c > 0$ such that $\bar{F}(t) \leq C e^{-ct}$ for all $t \geq 0$. The dominant decay rate is governed by $-\max\{\Re(\lambda) : \lambda \in \text{spec}(S)\}$, possibly up to a polynomial factor when the leading eigenvalue has higher multiplicity. ◻

Likelihoods with censoring

Because f and $\bar{F}$ are explicit in terms of S , likelihood contributions for uncensored and censored observations follow directly from the general formulas

in (4.6)–(4.7). The resulting expressions are tractable and remain to some extent within the earlier framework of this chapter. Let (τ_n) be iid PH with representation (α, S) . The uncensored log-likelihood is

$$\ell_n(\alpha, S) = \sum_{i=1}^n \log(\alpha e^{\tau_i S} \mathbf{s}).$$

With right-censoring we observe (W_i, δ_i) as in Definition 4.2 and (4.7), and the log-likelihood becomes

$$\ell_n(\alpha, S) = \sum_{i=1}^n \{ \delta_i \log(\alpha e^{W_i S} \mathbf{s}) + (1 - \delta_i) \log(\alpha e^{W_i S} \mathbf{1}) \}.$$

For interval-censored data $(L_i, R_i]$ one obtains

$$\ell_n(\alpha, S) = \sum_{i=1}^n \log(\alpha e^{L_i S} (I - e^{(R_i - L_i) S}) \mathbf{1}).$$

Direct maximisation of $\ell_n(\alpha, S)$ is typically challenging because S enters through matrix exponentials, both for the objective function and for its associated gradients. A standard and stable alternative is the expectation-maximisation (EM) algorithm, which treats the unobserved Markov path as missing data.

Expectation-Maximisation (EM) estimation

The EM algorithm becomes a natural method once we regard the underlying Markov paths as missing data. If the full paths were observed, the likelihood factorises into the initial state probabilities, the transition intensities along the paths, and the exponential sojourn contributions. This leads to a small set of sufficient statistics.

In a nutshell, EM starts from an initial guess $(\alpha^{(0)}, S^{(0)})$ and alternates two steps: in the E-step, compute the conditional expectations of the complete-data sufficient statistics under the current parameter values; in the M-step, maximise the expected complete-data log-likelihood, which here yields explicit updates for (α, S) . Repeating these steps gives a non-decreasing observed-data log-likelihood and, in practice, one stops when successive improvements are negligible; see [Dempster et al. \[1977\]](#) for the general method, and the notes at the end of the chapter for the algorithm adapter to PH distributions presented below.

For observation i , let $Z^{(i)} = (Z_t^{(i)})_{t \geq 0}$ denote the latent path with absorption time τ_i , and define

$$B_{ij} = 1_{(Z_0^{(i)}=j)}, \quad N_{ijk} = \#\{t \leq \tau_i : Z_{t-}^{(i)} = j, Z_t^{(i)} = k\}, \quad V_{ij} = \int_0^{\tau_i} 1_{(Z_t^{(i)}=j)} dt.$$

Write the sample totals

$$B_j^+ = \sum_{i=1}^n B_{ij}, \quad N_{jk}^+ = \sum_{i=1}^n N_{ijk}, \quad V_j^+ = \sum_{i=1}^n V_{ij}.$$

The complete-data log-likelihood for the full sample is given by (see Section ?? for a general derivation)

$$\ell_c(\alpha, S) = \sum_{j=1}^p B_j^+ \log \alpha_j + \sum_{j \neq k} N_{jk}^+ \log S_{jk} + \sum_{j=1}^p N_{j,p+1}^+ \log s_j + \sum_{j=1}^p V_j^+ S_{jj},$$

where s_j denotes the j th component of $\mathbf{s}$. Hence the M-step has the explicit updates

$$\hat{\alpha}_j = \frac{\mathbb{E}[B_j^+]}{n}, \quad \hat{S}_{jk} = \frac{\mathbb{E}[N_{jk}^+]}{\mathbb{E}[V_j^+]} \quad (j \neq k), \quad \hat{s}_j = \frac{\mathbb{E}[N_{j,p+1}^+]}{\mathbb{E}[V_j^+]}, \quad \hat{S}_{jj} = - \sum_{k \neq j} \hat{S}_{jk} - \hat{s}_j.$$

These updates preserve the generator constraints by construction: off-diagonal rates remain nonnegative and rows sum to zero.

To describe the E-step it suffices to give the conditional expectations for one generic observation; for notational simplicity we suppress the observation index i in the formulas below. For an uncensored observation $W = w$,

$$f(w) = \alpha e^{wS} \mathbf{s},$$

and for a right-censored observation $W = w$,

$$\bar{F}(w) = \alpha e^{wS} \mathbf{1}.$$

Conditional expectations are then given by

$$\mathbb{E}[B_j | W = w, \delta = 1] = \frac{\alpha_j (e^{wS} \mathbf{s})_j}{f(w)}, \quad \mathbb{E}[B_j | W = w, \delta = 0] = \frac{\alpha_j (e^{wS} \mathbf{1})_j}{\bar{F}(w)},$$

and, for $j \neq k$,

$$\begin{aligned} \mathbb{E}[N_{jk} | W = w, \delta = 1] &= \frac{1}{f(w)} \int_0^w (\alpha e^{sS})_j S_{jk} (e^{(w-s)S} \mathbf{s})_k \, ds, \\ \mathbb{E}[V_j | W = w, \delta = 1] &= \frac{1}{f(w)} \int_0^w (\alpha e^{sS})_j (e^{(w-s)S} \mathbf{s})_j \, ds, \\ \mathbb{E}[N_{j,p+1} | W = w, \delta = 1] &= \frac{1}{f(w)} (\alpha e^{wS})_j s_j, \end{aligned}$$

and, for right-censoring,

$$\begin{aligned} \mathbb{E}[N_{jk} | W = w, \delta = 0] &= \frac{1}{\bar{F}(w)} \int_0^w (\alpha e^{sS})_j S_{jk} (e^{(w-s)S} \mathbf{1})_k \, ds, \\ \mathbb{E}[V_j | W = w, \delta = 0] &= \frac{1}{\bar{F}(w)} \int_0^w (\alpha e^{sS})_j (e^{(w-s)S} \mathbf{1})_j \, ds, \\ \mathbb{E}[N_{j,p+1} | W = w, \delta = 0] &= 0. \end{aligned}$$

Summing these quantities over $i = 1, \dots, n$ yields the expected sufficient statistics needed in the M-step.

Similar expressions are obtained for interval censoring by conditioning on $\{L < \tau \leq R\}$ and integrating the uncensored formulas over $w \in (L, R]$. The EM

algorithm alternates the E-step and the closed-form M-step until convergence, and in practice, initialisation and the chosen representation (α, S) influence the resulting fit. It is therefore recommended to try different starting settings to fit the same data. The dimension of the matrix is usually taken as a hyperparameter which may be fixed, or adaptively chosen based on some in-sample or predictive criterion.

Before specialising to structured subclasses, it is useful to recall briefly how the matrix terms in the E-step are evaluated in practice.

Remark 4.5 (Numerical evaluation of E-step matrix terms) The E-step requires repeated evaluation of e^{tS} and matrix-exponential integrals of the form

$$\mathcal{I}_A(w) = \int_0^w e^{uS} A e^{(w-u)S} du,$$

where A is a fixed matrix determined by the sufficient statistic under consideration. A standard identity embeds this integral into one matrix exponential:

$$\exp \left(w \begin{pmatrix} S & A \\ 0 & S \end{pmatrix} \right) = \begin{pmatrix} e^{wS} & \mathcal{I}_A(w) \\ 0 & e^{wS} \end{pmatrix},$$

so $\mathcal{I}_A(w)$ is obtained from the upper-right block [Loan, 1978].

For the exponential itself, three classical routes are commonly used in this context. One can solve the linear ODE $M'(t) = M(t)S$ with $M(0) = \text{Id}$ (or the row-vector ODE $y'(t) = y(t)S$ with $y(0) = \alpha$), for example by a fourth-order Runge–Kutta integrator. A second route is uniformisation (randomisation): for $\nu \geq \max_j(-S_{jj})$ and $P = \text{Id} + S/\nu$,

$$e^{tS} = e^{-\nu t} \sum_{n=0}^{\infty} \frac{(\nu t)^n}{n!} P^n,$$

which rewrites the transient probabilities as a Poisson-weighted series of matrix powers. The gain is numerical rather than symbolic. Compared with the raw Taylor expansion $\sum_{k \geq 0} (tS)^k/k!$, the terms P^n are nonnegative and bounded (for sub-stochastic P), and the scalar weights are Poisson probabilities; this substantially reduces cancellation and yields stable partial sums for probability vectors. Truncating at N gives an error controlled by a Poisson tail, so N can be selected directly from Poisson quantiles.

A third route, widely used in numerical linear algebra software, is scaling and squaring with Padé approximants Higham [2005]. For a matrix B , the diagonal $[m/m]$ Padé approximant is

$$r_m(B) = \left(\sum_{k=0}^m c_k B^k \right) \left(\sum_{k=0}^m c_k (-B)^k \right)^{-1}, \quad c_k = \frac{(2m-k)!m!}{(2m)!(m-k)!k!},$$

which matches the Taylor series of e^B through order $2m$; equivalently, for small $\|B\|$,

$$e^B - r_m(B) = \mathcal{O}(\|B\|^{2m+1}).$$

Hence scaling and squaring uses the approximation

$$e^{tS} \approx \left(r_m(2^{-s}tS) \right)^{2^s},$$

with s chosen so that $\|2^{-s}tS\|$ is small before squaring back. Combined with the block-exponential identity above, this yields a practical way to evaluate the EM integrals. $\circ$

With these numerical considerations in place, we now turn to a subclass where the same EM framework simplifies substantially. The Coxian structure yields explicit recursions and avoids matrix exponentials in the E-step.

Example 4.6 (Coxian EM without matrix exponentials) Consider the Coxian subclass from Example 4.5. For uncensored observations (τ_n) , the EM algorithm above simplifies substantially because the upper-bidiagonal structure implies that most quantities are finite mixtures of exponentials.

Define, for $i = 1, \dots, p$ and $t \geq 0$,

$$L_i(t) = (\alpha e^{tS})_i, \quad R_i(t) = (e^{tS} \mathbf{s})_i.$$

Here $L_i(t)$ is the probability of being in phase i at time t , and $R_i(t)$ is the density of eventual absorption exactly t time units after entering phase i . In particular, the Coxian density is

$$f(t) = R_1(t).$$

The Markov structure yields the recursions

$$\begin{aligned} L_1(t) &= e^{-\lambda_1 t}, \\ L_i(t) &= q_{i-1} \lambda_{i-1} \int_0^t L_{i-1}(u) e^{-\lambda_i(t-u)} du, \quad i = 2, \dots, p, \end{aligned}$$

and, backward from the last phase,

$$\begin{aligned} R_p(t) &= \lambda_p e^{-\lambda_p t}, \\ R_i(t) &= (1 - q_i) \lambda_i e^{-\lambda_i t} + q_i \lambda_i \int_0^t e^{-\lambda_i u} R_{i+1}(t-u) du, \quad i = 1, \dots, p-1. \end{aligned}$$

Substituting these identities into the uncensored E-step formulas above (with $\alpha = (1, 0, \dots, 0)$ and the Coxian form of S and $\mathbf{s}$), we obtain for one observation $\tau = w$:

$$\begin{aligned} \mathbb{E}[V_i \mid \tau = w] &= \frac{1}{f(w)} \int_0^w L_i(u) R_i(w-u) du, \\ \mathbb{E}[N_{i,p+1} \mid \tau = w] &= \frac{L_i(w)(1 - q_i) \lambda_i}{f(w)}, \\ \mathbb{E}[N_{i,i+1} \mid \tau = w] &= \frac{q_i \lambda_i}{f(w)} \int_0^w L_i(u) R_{i+1}(w-u) du, \quad i = 1, \dots, p-1. \end{aligned}$$

Summing over the sample gives $\mathbb{E}[V_i^+]$, $\mathbb{E}[N_{i,p+1}^+]$, and $\mathbb{E}[N_{i,i+1}^+]$, after which the general phase-type M-step reduces to explicit Coxian updates:

$$\hat{\lambda}_i = \frac{\mathbb{E}[N_{i,i+1}^+] + \mathbb{E}[N_{i,p+1}^+]}{\mathbb{E}[V_i^+]}, \quad i = 1, \dots, p-1, \quad \hat{\lambda}_p = \frac{\mathbb{E}[N_{p,p+1}^+]}{\mathbb{E}[V_p^+]},$$

and

$$\hat{q}_i = \frac{\mathbb{E}[N_{i,i+1}^+]}{\mathbb{E}[N_{i,i+1}^+] + \mathbb{E}[N_{i,p+1}^+]}, \quad i = 1, \dots, p-1, \quad \hat{q}_p = 0.$$

Since the initial distribution is fixed at $\alpha = (1, 0, \dots, 0)$, there is no α -update.

The computational gain comes from the fact that each recursion preserves the class of finite exponential mixtures. Consequently, the E-step integrals are finite sums of convolutions of exponential terms, each of which is explicit:

$$\int_0^w e^{-ru} e^{-s(w-u)} du = \begin{cases} \frac{e^{-rw} - e^{-sw}}{s - r}, & r \neq s, \\ we^{-rw}, & r = s. \end{cases}$$

Thus, one obtains a Coxian-specific EM algorithm that avoids numerical matrix exponentials while remaining consistent with the general PH EM formulas. In the terminology of Example 4.5, the same idea extends to the *Generalised Coxian* case (arbitrary α): the backward quantities R_i are unchanged, the forward quantities $(\alpha e^{tS})_i$ remain finite exponential mixtures, and α is updated by the general PH formula $\hat{\alpha}_j = \mathbb{E}[B_j^+]/n$. $\diamond$

In the Coxian case, the E-step is already explicit, but obtained through finite-mixture convolution recursions. For a mixture of Erlang components, the same E-step quantities simplify further: those recursions disappear and the relevant integrals reduce directly to closed-form Beta-integral expressions.

Example 4.7 (Erlang-mixture EM without matrix exponentials) An analogous simplification is available for mixtures of Erlang components. Let component $k \in \{1, \dots, K\}$ have m_k transient phases in series and common rate λ_k , and let the initial distribution place mass π_k on the first phase of component k (with $\sum_{k=1}^K \pi_k = 1$). This gives a PH model with $p = \sum_{k=1}^K m_k$ transient states, indexed by (k, j) with $j = 1, \dots, m_k$.

For $t \geq 0$, the forward and backward quantities are explicit:

$$L_{k,j}(t) = \pi_k e^{-\lambda_k t} \frac{(\lambda_k t)^{j-1}}{(j-1)!}, \quad R_{k,j}(t) = \lambda_k e^{-\lambda_k t} \frac{(\lambda_k t)^{m_k-j}}{(m_k-j)!}.$$

Hence the component density and total density are

$$f_k(t) = \pi_k \lambda_k e^{-\lambda_k t} \frac{(\lambda_k t)^{m_k-1}}{(m_k-1)!}, \quad f(t) = \sum_{k=1}^K f_k(t).$$

Substituting into the uncensored E-step formulas for one observation $\tau = w$,

the integrals reduce to Beta integrals:

$$\begin{aligned}\mathbb{E}[V_{k,j} \mid \tau = w] &= \frac{1}{f(w)} \int_0^w L_{k,j}(u) R_{k,j}(w-u) \, du \\ &= \frac{w}{m_k} \frac{f_k(w)}{f(w)}, \\ \mathbb{E}[N_{k,j,j+1} \mid \tau = w] &= \frac{\lambda_k}{f(w)} \int_0^w L_{k,j}(u) R_{k,j+1}(w-u) \, du \\ &= \frac{f_k(w)}{f(w)}, \quad j = 1, \dots, m_k - 1, \\ \mathbb{E}[N_{k,m_k,p+1} \mid \tau = w] &= \frac{L_{k,m_k}(w) \lambda_k}{f(w)} = \frac{f_k(w)}{f(w)}.\end{aligned}$$

Thus, conditional on component k , expected sojourn time is spread uniformly over its m_k phases, while each within-component jump count equals the same posterior component weight.

Define

$$\omega_{k,i} := \frac{f_k(\tau_i)}{f(\tau_i)}, \quad i = 1, \dots, n.$$

Summing over the sample yields

$$\begin{aligned}\mathbb{E}[V_{k,j}^+] &= \sum_{i=1}^n \frac{\tau_i}{m_k} \omega_{k,i}, \quad \mathbb{E}[N_{k,j,j+1}^+] = \sum_{i=1}^n \omega_{k,i}, \quad j = 1, \dots, m_k - 1, \\ \mathbb{E}[N_{k,m_k,p+1}^+] &= \sum_{i=1}^n \omega_{k,i}.\end{aligned}$$

The equality of the two count totals is expected: once component k is selected, each sample path performs every internal jump and the final jump to absorption exactly once. Since only the first phase of each component can be initial, $\pi_k = \alpha_{k,1}$ and the PH update for α gives

$$\hat{\pi}_k = \frac{1}{n} \sum_{i=1}^n \omega_{k,i}.$$

Pooling transition counts and sojourn times across the m_k phases of component k , the PH M-step for rates becomes

$$\hat{\lambda}_k = \frac{\sum_{j=1}^{m_k-1} \mathbb{E}[N_{k,j,j+1}^+] + \mathbb{E}[N_{k,m_k,p+1}^+]}{\sum_{j=1}^{m_k} \mathbb{E}[V_{k,j}^+]} = \frac{m_k \sum_{i=1}^n \omega_{k,i}}{\sum_{i=1}^n \tau_i \omega_{k,i}}.$$

Therefore the general PH EM algorithm collapses to the familiar finite-mixture EM updates for component weights and Erlang rates, with no direct matrix exponentials. $\diamond$

When all Erlang shapes are equal to one, the Beta-integral evaluations above collapse to trivial one-phase identities. This yields the hyperexponential case, where the same PH EM bookkeeping reduces to the standard exponential-mixture EM updates.

Example 4.8 (Mixture-of-exponentials EM) The simplification is even stronger in the hyperexponential case (a finite mixture of exponentials). Here

$$S = \text{diag}(-\lambda_1, \dots, -\lambda_p), \quad \mathbf{s} = (\lambda_1, \dots, \lambda_p)^\top, \quad \alpha = (\alpha_1, \dots, \alpha_p).$$

Since phases are mutually exclusive, there are no transient-to-transient jumps, so $N_{jk} = 0$ for $j \neq k$. For $t \geq 0$,

$$L_j(t) = \alpha_j e^{-\lambda_j t}, \quad R_j(t) = \lambda_j e^{-\lambda_j t},$$

and therefore

$$f_j(t) = \alpha_j \lambda_j e^{-\lambda_j t}, \quad f(t) = \sum_{j=1}^p f_j(t).$$

For one uncensored observation $\tau = w$, the E-step formulas reduce to

$$\begin{aligned} \mathbb{E}[V_j \mid \tau = w] &= \frac{1}{f(w)} \int_0^w \alpha_j e^{-\lambda_j u} \lambda_j e^{-\lambda_j (w-u)} du = \frac{w f_j(w)}{f(w)}, \\ \mathbb{E}[N_{j,p+1} \mid \tau = w] &= \frac{(\alpha e^{wS})_j s_j}{f(w)} = \frac{f_j(w)}{f(w)}, \\ \mathbb{E}[B_j \mid \tau = w] &= \frac{f_j(w)}{f(w)}. \end{aligned}$$

Define posterior component weights

$$\omega_{j,i} := \frac{f_j(\tau_i)}{f(\tau_i)}, \quad i = 1, \dots, n.$$

Then

$$\mathbb{E}[V_j^+] = \sum_{i=1}^n \tau_i \omega_{j,i}, \quad \mathbb{E}[N_{j,p+1}^+] = \sum_{i=1}^n \omega_{j,i}, \quad \mathbb{E}[B_j^+] = \sum_{i=1}^n \omega_{j,i}.$$

Hence the PH M-step becomes

$$\hat{\alpha}_j = \frac{1}{n} \sum_{i=1}^n \omega_{j,i}, \quad \hat{\lambda}_j = \frac{\sum_{i=1}^n \omega_{j,i}}{\sum_{i=1}^n \tau_i \omega_{j,i}}.$$

This is a well-known EM algorithm for finite exponential mixtures, in this case obtained as a direct special case of the general PH EM framework. Computationally, the E-step is linear in p per observation and requires no matrix exponentiation. $\diamond$

4.2.2 Inhomogeneous phase-type distributions

Inhomogeneous phase-type (IPH) distributions extend PH laws through deterministic time changes. They retain the matrix-analytic structure of the PH model while allowing substantially more flexible tail behaviour, including heavy-tailed models. This makes them useful when a homogeneous PH fit captures the body of the distribution well but enforces tails that are too light.

We encode the inhomogeneity through a function $g_\eta(\cdot)$, where η is a finite-dimensional parameter vector. Let $\tau_0 \sim \text{PH}(\alpha, S)$ and let $g_\eta : [0, \infty) \rightarrow [0, \infty)$ be measurable with cumulative transform

$$G_\eta(t) = \int_0^t g_\eta(u) \, du, \quad t \geq 0.$$

Assume for simplicity (although not needed) G_η is strictly increasing, $G_\eta(0) = 0$, and $G_\eta(t) \rightarrow \infty$ as $t \rightarrow \infty$. Define the (generalised) inverse

$$G_\eta^{-1}(x) = \inf\{t \geq 0 : G_\eta(t) \geq x\},$$

and set $\tau = G_\eta^{-1}(\tau_0)$.

Definition 4.6 (Inhomogeneous phase-type distribution) The distribution of τ is an *inhomogeneous phase-type distribution* with representation (α, S, g_η) , written $\tau \sim \text{IPH}(\alpha, S, g_\eta)$.

The representation is again non-unique: even for fixed g_η the PH representation (α, S) is not unique, and if the transform family contains a scale parameter then this can be confounded with a global scaling of S . In applications one therefore fixes a parametrisation or imposes a normalisation for identifiability.

Proposition 4.3 (Basic formulas for IPH) *Let $\tau \sim \text{IPH}(\alpha, S, g_\eta)$ and G_η be as above. Then for $t \geq 0$,*

$$\begin{aligned} \bar{F}(t) &= \mathbb{P}(\tau > t) = \alpha e^{G_\eta(t)S} \mathbf{1}, \\ f(t) &= g_\eta(t) \alpha e^{G_\eta(t)S} \mathbf{s}, \\ \mathbb{E}[e^{-u\tau}] &= \int_0^\infty e^{-uG_\eta^{-1}(x)} \alpha e^{xS} \mathbf{s} \, dx, \quad u > 0, \\ \mathbb{E}[\tau^k] &= \int_0^\infty (G_\eta^{-1}(x))^k \alpha e^{xS} \mathbf{s} \, dx, \quad k \in \mathbb{N}, \end{aligned}$$

whenever G_η is differentiable at t and the corresponding integrals are finite.

Proof. By construction, $\tau > t$ if and only if $\tau_0 > G_\eta(t)$, hence

$$\mathbb{P}(\tau > t) = \mathbb{P}(\tau_0 > G_\eta(t)) = \alpha e^{G_\eta(t)S} \mathbf{1}.$$

Differentiating the survival function and using $f(t) = -\frac{d}{dt} \bar{F}(t)$, where G_η is differentiable, gives

$$f(t) = g_\eta(t) \alpha e^{G_\eta(t)S} \mathbf{s}.$$

Since $\tau = G_\eta^{-1}(\tau_0)$ and τ_0 has PH density $x \mapsto \alpha e^{xS} \mathbf{s}$, for any measurable φ with finite expectation,

$$\mathbb{E}[\varphi(\tau)] = \mathbb{E}[\varphi(G_\eta^{-1}(\tau_0))] = \int_0^\infty \varphi(G_\eta^{-1}(x)) \alpha e^{xS} \mathbf{s} \, dx.$$

Choosing $\varphi(t) = e^{-ut}$ and $\varphi(t) = t^k$ gives the transform and moment formulas. $\square$

Thus IPH models inherit the PH matrix formulas after composition with G_η . In particular, moments and Laplace transforms of $G_\eta(\tau)$ are exactly the PH expressions from Proposition 4.1, whereas those of τ are one-dimensional integrals involving G_η^{-1} .

Remark 4.6 (Mixtures and order statistics under a common transform) Fix g_η and let $\tau_r \sim \text{IPH}(\alpha_r, S_r, g_\eta)$, $r = 1, \dots, m$, be independent. Finite mixtures are preserved: if C is independent with $\mathbb{P}(C = r) = p_r$, then τ_C is IPH with the same transform g_η and block-diagonal PH part as in Proposition 4.2. Order statistics are also preserved under the same transform: writing $\tau_{1:m} \leq \dots \leq \tau_{m:m}$ and $U_r := G_\eta(\tau_r)$, we have $U_r \sim \text{PH}(\alpha_r, S_r)$ and $U_{k:m}$ is PH by Proposition 4.2; hence

$$\tau_{k:m} = G_\eta^{-1}(U_{k:m}),$$

so $\tau_{k:m}$ is IPH with transform g_η for each k (in particular, minimum and maximum). In contrast, convolutions are generally not preserved in the original time scale. $\square$

Example 4.9 (Transform-based IPH subclasses) Let $\tau \sim \text{IPH}(\alpha, S, g_\eta)$ with baseline PH representation (α, S) and $\mathbf{s} = -S\mathbf{1}$. The following choices of G_η yield useful structured families. These are often called *matrix distributions*: the scalar exponential baseline in the corresponding classical family is replaced by a PH baseline (α, S) , while the same deterministic transform is retained. In each case, if $p = 1$, $\alpha = (1)$ and $S = (-\lambda)$, the scalar model reduces to a classical named distribution. For log-based transforms, an equivalent functional form is often used:

$$a(t)^S := \exp((\log a(t))S), \quad a(t) > 0,$$

which is standard in the IPH literature Albrecher and Bladt [2019], Bladt and Nielsen [2017].

1. *Matrix-Weibull (power transform)*. Let $G_\eta(t) = t^\beta$ with $\beta > 0$, so $g_\eta(t) = \beta t^{\beta-1}$. Then

$$\bar{F}(t) = \alpha e^{t^\beta S} \mathbf{1}, \quad f(t) = \beta t^{\beta-1} \alpha e^{t^\beta S} \mathbf{s}.$$

This is the PH analogue of the Weibull family, combining a Weibull-type time deformation with a flexible PH body. For $p = 1$ it is the standard Weibull distribution (with shape β and scale determined by λ).

2. *Matrix-Pareto/Lomax (log transform)*. Let $G_\eta(t) = \log(1 + t/\theta)$ with $\theta > 0$, so $g_\eta(t) = 1/(\theta + t)$. Then

$$\bar{F}(t) = \alpha(1 + t/\theta)^S \mathbf{1} = \alpha e^{(\log(1+t/\theta))S} \mathbf{1}, \quad f(t) = \frac{1}{\theta} \alpha(1 + t/\theta)^{S-I} \mathbf{s}.$$

The logarithmic transform converts the exponential PH tail into a polynomial-type tail and is therefore a standard route to heavy-tailed IPH models. For $p = 1$ one obtains $\bar{F}(t) = (1 + t/\theta)^{-\lambda}$, the Lomax tail.

3. *Matrix-Burr / matrix log-logistic (log-power transform)*. Let $G_\eta(t) = \log(1 + (t/\theta)^\beta)$ with $\theta, \beta > 0$, so

$$g_\eta(t) = \frac{\beta t^{\beta-1}}{\theta^\beta + t^\beta}.$$

Equivalently,

$$\bar{F}(t) = \alpha(1 + (t/\theta)^\beta)^S \mathbf{1}, \quad f(t) = \frac{\beta}{\theta} \left(\frac{t}{\theta}\right)^{\beta-1} \alpha(1 + (t/\theta)^\beta)^{S-I} \mathbf{s}.$$

This family interpolates between flexible body behaviour from the PH part and Burr/log-logistic-type tails and shapes induced by the transform. For $p = 1$ it yields the Burr type XII family; the log-logistic law is recovered as a special case.

4. *Matrix-Gompertz (exponential transform)*. Let $G_\eta(t) = \gamma^{-1}(e^{\gamma t} - 1)$ with $\gamma > 0$, so $g_\eta(t) = e^{\gamma t}$. Then

$$\bar{F}(t) = \alpha e^{((e^{\gamma t} - 1)/\gamma)S} \mathbf{1}, \quad f(t) = e^{\gamma t} \alpha e^{((e^{\gamma t} - 1)/\gamma)S} \mathbf{s}.$$

This is a matrix analogue of the Gompertz law, useful when one wants a Gompertz-type trend but more shape flexibility than a one-parameter baseline allows. For $p = 1$ one obtains the classical Gompertz distribution.

◇

Remark 4.7 (Tail behaviour for IPH) The tail behaviour of an IPH law is obtained by composing the PH tail with G_η . Indeed,

$$\bar{F}(t) = \alpha e^{G_\eta(t)S} \mathbf{1}.$$

Hence Remark 4.4 implies that there exist constants $C, c > 0$ such that

$$\bar{F}(t) \leq C e^{-c G_\eta(t)}, \quad t \geq 0.$$

The dominant decay rate is governed by the spectral abscissa of S composed with the growth of G_η , up to the same polynomial (now in $G_\eta(t)$) correction that appears in the PH case when the leading eigenvalue is multiple. In particular, power transforms give Weibull-type tails, while logarithmic transforms can generate polynomial (heavy) tails, and so forth. ○

Likelihoods with censoring

Because IPH laws are deterministic transforms of PH laws, likelihood contributions are obtained by composing the PH formulas with G_η and, for uncensored observations, adding the Jacobian term $\log g_\eta$. Let (τ_n) be iid $\text{IPH}(\alpha, S, g_\eta)$.

For uncensored observations the log-likelihood is

$$\ell_n(\alpha, S, \eta) = \sum_{i=1}^n \left\{ \log g_\eta(\tau_i) + \log \left(\alpha e^{G_\eta(\tau_i)S} \mathbf{s} \right) \right\}.$$

With right-censoring we observe (W_i, δ_i) and obtain

$$\ell_n(\alpha, S, \eta) = \sum_{i=1}^n [\delta_i \{ \log g_\eta(W_i) + \log (\alpha e^{G_\eta(W_i)S} \mathbf{s}) \} + (1 - \delta_i) \log (\alpha e^{G_\eta(W_i)S} \mathbf{1})].$$

For interval-censored data $(L_i, R_i]$ (with G_η nondecreasing) one has

$$\ell_n(\alpha, S, \eta) = \sum_{i=1}^n \log (\alpha e^{G_\eta(L_i)S} (I - e^{(G_\eta(R_i) - G_\eta(L_i))S}) \mathbf{1}).$$

One can similarly obtain a truncated version of the likelihood, however the statistical analysis does not simplify as easily as for the other cases, and hence is omitted below.

In any of the cases, direct maximisation is more demanding than in the PH case because the transform parameters η enter nonlinearly in both the matrix exponential arguments and (for uncensored data) the Jacobian term. Nevertheless, the PH structure still yields an efficient EM-type block algorithm.

Generalised Expectation-Maximisation (GEM) estimation

When η is fixed, estimation reduces exactly to PH estimation on transformed times: define $U_i = G_\eta(\tau_i)$ (or $\tilde{U}_i = G_\eta(W_i)$ for censored data), apply the PH likelihood and E-step formulas to the transformed sample, and update (α, S) by the PH M-step above.

When η is unknown but parametric, one typically alternates between PH-EM updates and a numerical optimisation of η ; this is naturally viewed as a generalised EM algorithm, or a GEM algorithm, and is implemented in the R package `matrixdist` [Bladt et al., 2025]. A typical iteration is:

1. Given $(\alpha^{(m)}, S^{(m)}, \eta^{(m)})$, transform the observations and censoring bounds using $G_{\eta^{(m)}}$.
2. Perform an E-step and M-step for the PH parameters on the transformed sample, obtaining $(\alpha^{(m+1)}, S^{(m+1)})$.
3. Update the transform parameters by numerical maximisation of the observed-data IPH log-likelihood with (α, S) fixed, for example by a quasi-Newton method:

$$\eta^{(m+1)} = \operatorname{argmax}_{\eta} \ell_n(\alpha^{(m+1)}, S^{(m+1)}, \eta).$$

Each iteration is designed and guaranteed to increase (or at least not decrease) the observed likelihood when the optimisation step is carried out sufficiently accurately. In practice, one may use a full maximisation in Step 3 or a single optimisation step, and one may alternate this with one or several PH-EM iterations before updating η , to obtain speed gains.

A practical issue is scale confounding: if $c > 0$, then replacing S by cS and G_η by $G_{\eta/c}$ yields the same IPH distribution. When the transform family

contains an explicit scale parameter, one should therefore fix one scale (either in g_η or in the PH part) to ensure identifiability. Parameter constraints for η (such as positivity) are usually handled directly by reparametrisation in the numerical optimisation step.

To illustrate the role of inhomogeneity in a simple light-tailed target, we first fit homogeneous and inhomogeneous PH models to truncated-normal data.

Numeric example 4.1 (Parsimony from inhomogeneity) We generate a sample of size $n = 1000$ from a normal law, and then truncate it on $\mathbb{R}_+$. We fit four models by EM with 3000 iterations: a homogeneous PH model with general structure and $p = 4$, a homogeneous mixed-Erlang model, and two IPH models with a $p = 2$ general PH baseline combined with Weibull and Gompertz transforms. The fitted densities and log-likelihood paths are shown in Figure 4.3. The inhomogeneous models dominate in this experiment despite the smaller baseline dimension, with final log-likelihoods. Thus deterministic time-change flexibility can deliver a more parsimonious representation than increasing the homogeneous PH dimension alone. $\diamond$

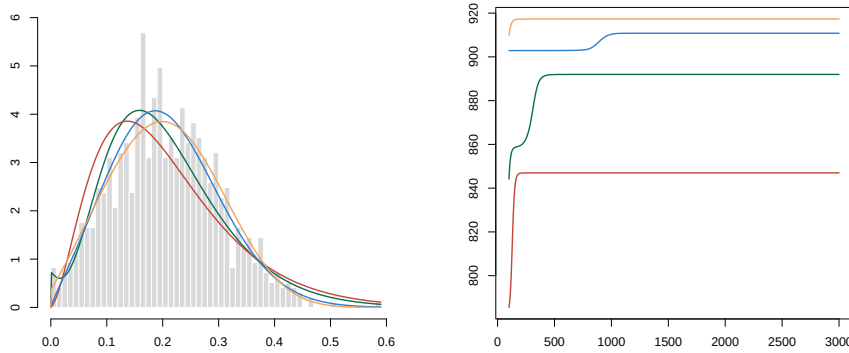


Figure 4.3: Truncated-normal study. Left: empirical histogram (grey) with fitted densities from PH general $p = 4$ (green), mixed Erlang (red), IPH with Weibull transform (blue), and IPH with Gompertz transform (orange). Right: EM log-likelihood trajectories as a function of EM iteration, for the same four models in the same colours.

For a heavy-tailed global fit, we next fix the transform family to Pareto and vary only the PH dimension, so both body fit and tail index are data-driven. Note that no threshold selection is required.

Numeric example 4.2 (Matrix-Pareto body and tail fitting) We generate a right-censored heavy-tailed sample by taking $n = 1000$ observations from the product of a standard Pareto and a Beta variable, and censoring with an independent Pareto variable. We then fit four IPH Pareto models with general PH baseline dimensions $p = 1, 2, 3, 4$ by GEM. Under the Pareto transform from Exam-

ple 4.9, the tail index is explicit through the fitted subgenerator:

$$\hat{\alpha}_n = 1/\hat{\gamma}_n = -\max\{\Re(\lambda) : \lambda \in \text{spec}(\hat{S}_n)\} > 0.$$

Hence positivity of the tail index is automatic in the model class, while the actual value remains data-adaptive through $\hat{S}_n$. Log-likelihood and tail index evolution are reported in Figure 4.4. Increasing p improves fit monotonically in this run, while final tail estimates remain positive and relatively close to their true value. The point here is that one can refine the fit by adding phases without having to change the transform family, which facilitates modelling choices. If the objective is prediction, model choice should follow the Chapter 6 principles based on out-of-sample performance. $\diamond$

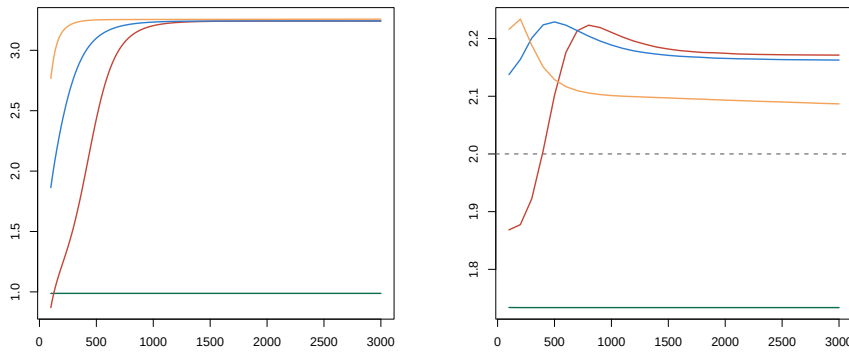


Figure 4.4: Matrix-Pareto study with PH dimensions $p = 1, 2, 3, 4$. Left: EM log-likelihood paths as a function of EM iteration, for $p = 1$ (green), $p = 2$ (red), $p = 3$ (blue), and $p = 4$ (orange). Right: corresponding tail-index paths $\hat{\alpha}_n$, with the dashed grey reference line at the true $\alpha = 2$.

Remark 4.8 (Regression and other extensions) Regression extensions typically link covariates to the initial vector, the transition-rate structure, or the time-change parameters and are estimated by likelihood-based or EM-type methods; see the notes for more details. Discrete-time PH distributions (DPH) are obtained by considering discrete-time Markov chains and are treated in the next subsection. Multivariate PH distributions arise by letting a Markov chain collect rewards in different states and inherit matrix-analytic structure that generalises the univariate case. For general inhomogeneous Markov jump processes beyond deterministic time changes, estimation of absorption-time distributions is treated in [Ahmad et al. \[2023\]](#). Heavy-tailed PH/IPH modelling from a unified perspective is developed in [Bladt and Yslas \[2022a\]](#). A broad and computationally oriented treatment, including PH/IPH fitting under censoring and related matrix-distribution models, is available in the R package `matrixdist` [[Bladt et al., 2025](#)]. $\circ$

4.2.3 Discrete phase-type distributions

Discrete phase-type (DPH) distributions are the discrete-time counterpart of PH laws. They are obtained as absorption times of finite-state *discrete*-time Markov chains and mirror the continuous theory with matrix powers replacing matrix exponentials.

Let $(Z_n)_{n \geq 0}$ be a discrete-time Markov chain on $\{1, \dots, p, p+1\}$ with transient states $1, \dots, p$ and absorbing state $p+1$. Write the transition matrix given by

$$P = \begin{pmatrix} T & \mathbf{t} \\ 0 & 1 \end{pmatrix},$$

where $T \in \mathbb{R}^{p \times p}$ is sub-stochastic with nonnegative entries and row sums at most one, and $\mathbf{t} = \mathbf{1} - T\mathbf{1}$ is the exit-probability vector. Let α be the initial distribution on the transient states and define the absorption time

$$\tau = \inf\{n \geq 1 : Z_n = p+1\}.$$

Definition 4.7 (Discrete phase-type distribution) The distribution of τ is a *discrete phase-type distribution* with representation (α, T) , written $\tau \sim \text{DPH}(\alpha, T)$.

Remark 4.9 (Simulation of DPH observations) Simulation is immediate from the embedded discrete-time chain. Draw Z_0 from α . If at step m the chain is in transient state j , draw the next state from

$$(T_{j1}, \dots, T_{jp}, t_j), \quad t_j = 1 - \sum_{k=1}^p T_{jk},$$

and stop at the first hit of the absorbing state $p+1$. The resulting step count is one DPH observation τ . $\circ$

Similarly to their continuous counterparts, DPH distributions enjoy explicit formulas for their basic characteristics in terms of matrix expressions.

Proposition 4.4 (Basic formulas for DPH) *Let $\tau \sim \text{DPH}(\alpha, T)$. Then for $n \in \mathbb{N}$,*

$$\begin{aligned} \mathbb{P}(\tau > n) &= \alpha T^n \mathbf{1}, \\ \mathbb{P}(\tau = n) &= \alpha T^{n-1} \mathbf{t}, \\ \mathbb{E}[z^\tau] &= z \alpha (I - zT)^{-1} \mathbf{t}, \quad |z| < \rho(T)^{-1}, \\ \mathbb{E}[\tau] &= \alpha (I - T)^{-1} \mathbf{1}. \end{aligned}$$

The discrete hazard is $\mathbb{P}(\tau = n | \tau \geq n) = \frac{\alpha T^{n-1} \mathbf{t}}{\alpha T^{n-1} \mathbf{1}}$.

Proof. The probability of still being transient after n steps is $\alpha T^n \mathbf{1}$, hence $\mathbb{P}(\tau = n) = \mathbb{P}(\tau > n-1) - \mathbb{P}(\tau > n) = \alpha T^{n-1} \mathbf{t}$. The probability generating function follows from the geometric series

$$\sum_{n \geq 1} z^n \alpha T^{n-1} \mathbf{t} = z \alpha (I - zT)^{-1} \mathbf{t},$$

and the mean is obtained by differentiating at $z = 1$. $\square$

Proposition 4.5 (Closure properties for DPH) *The DPH class is closed under several natural operations that preserve discrete time. Let $\tau_r \sim \text{DPH}(\alpha_r, T_r)$, $r = 1, \dots, m$, be independent and write $\mathbf{t}_r = \mathbf{1} - T_r \mathbf{1}$.*

1. Finite mixtures. *Let C be independent of $(\tau_1, \dots, \tau_m)$ with $\mathbb{P}(C = r) = p_r$, $\sum_{r=1}^m p_r = 1$, and set $\tau = \tau_C$. Then τ is DPH with*

$$\alpha^* = (p_1 \alpha_1, \dots, p_m \alpha_m), \quad T^* = \text{diag}(T_1, \dots, T_m).$$

2. Positive integer linear combinations. *For integers $c_1, \dots, c_m \in \mathbb{N}$, the sum*

$$\tau^* = \sum_{r=1}^m c_r \tau_r$$

is DPH. For $m = 2$ and $c_1 = c_2 = 1$ (convolution), one convenient representation is

$$\alpha^* = (\alpha_1, 0), \quad T^* = \begin{pmatrix} T_1 & \mathbf{t}_1 \alpha_2 \\ 0 & T_2 \end{pmatrix}.$$

3. Order statistics. *Let $\tau_{1:m} \leq \dots \leq \tau_{m:m}$ be the order statistics of $(\tau_1, \dots, \tau_m)$. Then $\tau_{k:m}$ is DPH for each $k = 1, \dots, m$. In particular, both*

$$\min(\tau_1, \dots, \tau_m) = \tau_{1:m} \quad \text{and} \quad \max(\tau_1, \dots, \tau_m) = \tau_{m:m}$$

are DPH. For the minimum, one has the explicit representation

$$\alpha^{\min} = \alpha_1 \otimes \dots \otimes \alpha_m, \quad T^{\min} = T_1 \otimes \dots \otimes T_m.$$

Here $\otimes$ denotes the Kronecker product.

Proof. The mixture statement follows by starting in block r with probability p_r and then evolving with transition matrix T_r .

For integer linear combinations, first note that if $\tau \sim \text{DPH}(\alpha, T)$ and $c \in \mathbb{N}$, then $c\tau$ is DPH by replacing each one-step transition of the original chain with a deterministic chain of c micro-steps. Then $\sum_{r=1}^m c_r \tau_r$ is obtained by repeated convolution of independent DPH variables. The displayed block matrix is the standard sequential construction for $\tau_1 + \tau_2$.

For order statistics, consider the product chain of the m independent DPH chains (including absorbing states). For fixed k , declare absorbing all product states with at least k absorbed coordinates, and transient otherwise. The first hitting time of this absorbing set is exactly $\tau_{k:m}$, hence DPH. Taking $k = 1$ and $k = m$ gives minimum and maximum. For $k = 1$,

$$\mathbb{P}(\tau_{1:m} > n) = \prod_{r=1}^m \alpha_r T_r^n \mathbf{1} = (\alpha_1 \otimes \dots \otimes \alpha_m)(T_1 \otimes \dots \otimes T_m)^n \mathbf{1},$$

which yields the minimum representation above. $\square$

Remark 4.10 (Denseness) By increasing the number of phases, DPH distributions are dense in the set of probability distributions on $\mathbb{N}$ under weak convergence, and hence can approximate arbitrary count distributions while retaining a useful Markov representation. $\circ$

Example 4.10 (Discrete phase-type subclasses) Let $\tau \sim \text{DPH}(\alpha, T)$ with p transient states. The following choices of (α, T) yield standard families and mirror the continuous PH subclasses.

1. *Geometric.* With $p = 1$, $\alpha = (1)$ and $T = (1 - q)$ for $q \in (0, 1)$, one obtains

$$\mathbb{P}(\tau = n) = q(1 - q)^{n-1}, \quad n \geq 1,$$

the memoryless DPH distribution.

2. *Discrete Erlang (negative binomial).* With $\alpha = (1, 0, \dots, 0)$ and T upper-bidiagonal with diagonal entries $1 - q$ and super-diagonal entries q , the absorption time is the sum of p iid geometric(q) variables, hence negative binomial with

$$\mathbb{P}(\tau = n) = \binom{n-1}{p-1} q^p (1 - q)^{n-p}, \quad n \geq p.$$

3. *Discrete hypoexponential.* With $\alpha = (1, 0, \dots, 0)$ and T upper-bidiagonal with diagonal entries $1 - q_1, \dots, 1 - q_p$ and super-diagonal entries $q_1, \dots, q_{p-1}$, the absorption time is the sum of independent geometric variables with success probabilities $q_1, \dots, q_p$. If the q_j are distinct, the pmf admits the representation

$$\mathbb{P}(\tau = n) = \sum_{j=1}^p c_j (1 - q_j)^{n-1}, \quad c_j = q_j \prod_{k \neq j} \frac{q_k}{q_k - q_j}.$$

4. *Discrete hyperexponential.* With $\alpha = (\alpha_1, \dots, \alpha_p)$ and $T = \text{diag}(1 - q_1, \dots, 1 - q_p)$, one obtains the finite mixture of geometric laws

$$\mathbb{P}(\tau = n) = \sum_{j=1}^p \alpha_j q_j (1 - q_j)^{n-1}, \quad n \geq 1.$$

5. *Discrete Coxian.* With $\alpha = (1, 0, \dots, 0)$ and T upper-bidiagonal having diagonal entries $1 - \lambda_j$ and super-diagonal entries $\lambda_j q_j$ with $\lambda_j \in (0, 1)$ and $q_j \in [0, 1]$, the chain remains in phase j with probability $1 - \lambda_j$, moves to $j + 1$ with probability $\lambda_j q_j$, and is absorbed with probability $\lambda_j (1 - q_j)$. Writing g_j for the discrete hypoexponential pmf based on the first j phases, one has

$$\mathbb{P}(\tau = n) = \sum_{j=1}^p \left(\prod_{r=1}^{j-1} q_r \right) (1 - q_j) g_j(n).$$

Thus the Coxian model is a mixture over the number of completed phases, and when $q_j = 1$ for $j = 1, \dots, p - 1$ it reduces to the discrete hypoexponential family (negative binomial in the equal- λ_j case).



Remark 4.11 (Geometric tails) Every DPH distribution has geometric tail decay. Indeed, $\mathbb{P}(\tau > n) = \alpha T^n \mathbf{1}$ with a transient sub-stochastic matrix T , so the tail decreases exponentially at a rate governed by the dominant eigenvalue modulus of T (which is strictly below one). In practical terms, for large n the tail behaves like a geometric sequence, possibly with a mild polynomial correction when dominant eigenvalues are repeated. $\circ$

EM estimation for DPH

Let $\tau_1, \dots, \tau_n$ be iid DPH with representation (α, T) , and for observation i let $Z^{(i)} = (Z_m^{(i)})_{m \geq 0}$ denote the latent discrete-time path with absorption time τ_i . The complete-data sufficient statistics mirror the continuous case:

$$B_{ij} = 1_{(Z_0^{(i)}=j)}, \quad N_{ijk} = \#\{m < \tau_i : Z_m^{(i)} = j, Z_{m+1}^{(i)} = k\}, \quad V_{ij} = \sum_{m=0}^{\tau_i-1} 1_{(Z_m^{(i)}=j)}.$$

For absorption jumps define

$$N_{ij,p+1} = \#\{m < \tau_i : Z_m^{(i)} = j, Z_{m+1}^{(i)} = p+1\},$$

and write sample totals

$$B_j^+ = \sum_{i=1}^n B_{ij}, \quad N_{jk}^+ = \sum_{i=1}^n N_{ijk}, \quad V_j^+ = \sum_{i=1}^n V_{ij}, \quad N_{j,p+1}^+ = \sum_{i=1}^n N_{ij,p+1}.$$

The complete-data log-likelihood for the full sample is

$$\ell_c(\alpha, T) = \sum_{j=1}^p B_j^+ \log \alpha_j + \sum_{j,k=1}^p N_{jk}^+ \log T_{jk} + \sum_{j=1}^p N_{j,p+1}^+ \log t_j,$$

where $t_j = 1 - \sum_{k=1}^p T_{jk}$. The M-step therefore updates

$$\hat{\alpha}_j = \frac{\mathbb{E}[B_j^+]}{n}, \quad \hat{T}_{jk} = \frac{\mathbb{E}[N_{jk}^+]}{\mathbb{E}[V_j^+]}, \quad \hat{t}_j = \frac{\mathbb{E}[N_{j,p+1}^+]}{\mathbb{E}[V_j^+]},$$

preserving row-stochastic constraints. To describe the E-step it suffices to give conditional expectations for one generic observation. Thus, suppress the index i and write $\tau = w \in \mathbb{N}$. Then

$$f(w) = \mathbb{P}(\tau = w) = \alpha T^{w-1} \mathbf{t},$$

and

$$\begin{aligned} \mathbb{E}[B_j | \tau = w] &= \frac{\alpha_j (T^{w-1} \mathbf{t})_j}{f(w)}, \\ \mathbb{E}[N_{jk} | \tau = w] &= \frac{1}{f(w)} \sum_{u=0}^{w-2} (\alpha T^u)_j T_{jk} (T^{w-2-u} \mathbf{t})_k, \\ \mathbb{E}[V_j | \tau = w] &= \frac{1}{f(w)} \sum_{u=0}^{w-1} (\alpha T^u)_j (T^{w-1-u} \mathbf{t})_j, \\ \mathbb{E}[N_{j,p+1} | \tau = w] &= \frac{(\alpha T^{w-1})_j t_j}{f(w)}. \end{aligned}$$

Summing these conditional expectations over $i = 1, \dots, n$ yields $\mathbb{E}[B_j^+]$, $\mathbb{E}[N_{jk}^+]$, $\mathbb{E}[V_j^+]$, and $\mathbb{E}[N_{j,p+1}^+]$ for the M-step.

Remark 4.12 (Numerical evaluation of DPH E-step sums) The required sums can be evaluated efficiently by precomputing powers of T or by forward recursion. A discrete analogue of the Van Loan block identity is

$$\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}^n = \begin{pmatrix} A^n & \sum_{u=0}^{n-1} A^u B C^{n-1-u} \\ 0 & C^n \end{pmatrix}, \quad n \geq 1,$$

for conformable matrices A, B, C . This allows finite convolution sums to be obtained from the upper-right block of one block-matrix power. Also, since we are dealing with discrete variables, ties can be used for numerical gains, as the contribution of tied observations can be handled using weights in the corresponding sums. $\circ$

Remark 4.13 (Incorporating exposure via continuous-time embedding) When counts are given with exposure e_i , one can embed the model into a continuous PH law by positing a latent continuous absorption time τ and setting $N = \lfloor e\tau \rfloor$. Then observing (N_i, e_i) is equivalent to interval censoring of τ on $[a_i, b_i]$ with

$$a_i = \frac{N_i}{e_i}, \quad b_i = \frac{N_i + 1}{e_i},$$

so (N_i, e_i) can be fitted by a continuous PH model using interval-censored likelihoods for τ , and then projecting to integers. The resulting distribution is a DPH with transition matrix $T = \exp(S)$. $\circ$

Numeric example 4.3 (Coxian DPH fit versus inflated count models) To illustrate the flexibility of discrete phase-type models on count data, we simulate a large sample from a deliberately heterogeneous distribution on $\{0, 1, 2, \dots\}$. Concretely, let J be a latent component indicator with

$$\mathbb{P}(J = j) = \frac{1}{5}, \quad j \in \{1, 2, 3, 4, 5\},$$

and define N conditionally on J by

$$N = \begin{cases} x_0, & J \in \{1, 2\}, \\ \text{Poisson}(1), & J = 3, \\ \text{NB}(\text{size} = 4, \mu = 2), & J = 4, \\ \text{Geom}(1/2), & J = 5, \end{cases}$$

with conditional independence across observations. The sample size is $n = 5,000,000$. The inflation point is $x_0 = 0$ in the first experiment and $x_0 = 1$ in the second. The geometric distribution is taken on $\{0, 1, 2, \dots\}$. We compare three fitted models: an inflated Poisson model, an inflated negative-binomial model (with inflation at the same atom x_0), and a Coxian DPH model with three transient phases, corresponding to five free parameters in the chosen parametrisation. The latter is fitted by 10,000 iterations of the EM algorithm through the R package `matrixdist`, taking approximately 0.13 seconds on a standard machine (Apple M1 Max, 32GB RAM).

Since the DPH support in Proposition 4.4 is $\mathbb{N} = \{1, 2, \dots\}$, we fit the Coxian model to the shifted variable $\tau = N + 1$ and then transform back to obtain fitted probabilities for N . For the inflated Poisson and inflated negative-binomial models we use direct maximum likelihood on the aggregated counts, and we compare all three models by log-likelihood, AIC and BIC. The numerical results in Table 4.1 show that the Coxian DPH model attains the largest log-likelihood and the smallest AIC and BIC in both experiments; the improvement over the inflated negative-binomial fit is modest but systematic, while the inflated Poisson fit is clearly less accurate. A practical advantage of the DPH specification is that no inflation point must be selected in advance, since the fitted Coxian structure can accommodate local excess mass without committing a priori to inflation at 0, 1, or any other specific count. $\diamond$

$x_0 = 0$ (zero inflation)				
Model	df	LogLik	AIC	BIC
Coxian DPH	5	-6100287	12200584	12200651
Inflated Poisson	2	-6179162	12358327	12358354
Inflated NB	3	-6100489	12200985	12201025
$x_0 = 1$ (one inflation)				
Model	df	LogLik	AIC	BIC
Coxian DPH	5	-6307452	12614914	12614981
Inflated Poisson	2	-6484523	12969051	12969077
Inflated NB	3	-6307493	12614992	12615032

Table 4.1: Likelihood and information-criterion comparison for the two simulated inflation scenarios. The Coxian DPH model uses three transient phases (five free parameters).

4.3 Regression models

We now specialise the general likelihood framework to regression settings. This section presents the most commonly used parametric regression models in actuarial science and some extensions to the case where the covariate space is possibly high-dimensional, as is also common in insurance datasets. Methods that are not likelihood-based can be adapted to the conditional case using the techniques of M-estimation above, but their analysis depends on the specific model.

4.3.1 Generalised Linear Models

This section provides a succinct overview of Generalised Linear Models (GLMs), a highly influential parametric regression model used extensively in, but not

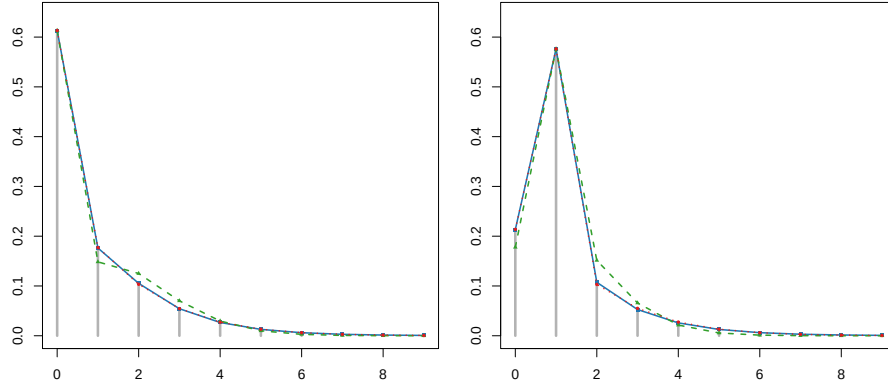


Figure 4.5: Empirical probabilities and fitted models for the simulated count mixture, with inflation at $x_0 = 0$ (left) and $x_0 = 1$ (right). Grey vertical lines show empirical proportions $\mathbb{P}(N = k)$. Blue, green and red curves (with square, triangle and circle markers) represent the Coxian DPH, inflated Poisson and inflated negative-binomial fits, respectively. Both panels are truncated at a high quantile.

exclusively for, insurance applications. They extend ordinary least squares regression to situations where the response is binomial, Poisson, gamma, or any exponential family.

Definition 4.8 (Exponential Dispersion Family) The *Exponential Dispersion Families* (EDFs) are a class of probability distributions for a random variable Z on a sample space $\mathcal{Z} \subseteq \mathbb{R}$. Their density or mass function $f_Z(z|\theta, \phi)$ with respect to some σ -finite measure μ can be expressed in the form:

$$f_Z(z|\theta, \phi) = \exp \left\{ \frac{z\theta - b(\theta)}{\phi} + c(z, \phi) \right\}, \quad (4.8)$$

where $z \in \mathcal{Z}$, the *canonical parameter* θ belongs to the *canonical parameter space* $\Theta \subseteq \mathbb{R}$, and the *dispersion parameter* ϕ belongs to the *dispersion parameter space* $\Phi \subseteq \mathbb{R}^+$. The functions $b : \Theta \rightarrow \mathbb{R}$ and $c : \mathcal{Z} \times \Phi \rightarrow \mathbb{R}$ are known functions.

The mean and variance of a random variable Z in the exponential dispersion family are conveniently given by:

$$\mathbb{E}[Z] = b'(\theta), \quad \text{Var}(Z) = b''(\theta)\phi.$$

Some of the most important examples used in actuarial practice are encompassed within the above definition. The form (4.8) yields the following EDF representations.

Example 4.11 (Common exponential dispersion families) We list canonical parametrisations in (4.8) together with common actuarial uses.

1. *Normal.* For $Z \sim N(\mu, \sigma^2)$ on $\mathbb{R}$ (Lebesgue measure), one has $\theta = \mu$, $\phi = \sigma^2$, $b(\theta) = \theta^2/2$, and $c(z, \phi) = -z^2/(2\phi) - \frac{1}{2} \log(2\pi\phi)$. Then $b'(\theta) = \mu$ and $b''(\theta) = 1$. This model is used for approximately symmetric, light-tailed continuous responses and for residual modelling.

2. *Bernoulli*. For $Z \sim \text{Bernoulli}(p)$ on $\{0, 1\}$ (counting measure), one has $\theta = \log(p/(1-p))$, $\phi = 1$, $b(\theta) = \log(1 + e^\theta)$, and $c(z, \phi) = 0$. Then $b'(\theta) = p$ and $b''(\theta) = p(1-p)$. This model is used for binary outcomes such as claim/no-claim indicators.
3. *Poisson*. For $Z \sim \text{Poisson}(\lambda)$ on $\{0, 1, 2, \dots\}$ (counting measure), one has $\theta = \log \lambda$, $\phi = 1$, $b(\theta) = e^\theta$, and $c(z, \phi) = -\log(z!)$. Then $b'(\theta) = \lambda$ and $b''(\theta) = \lambda$. This model is used for claim counts and event frequencies, typically with exposure offsets.
4. *Gamma*. For $Z \sim \text{Gamma}(\alpha, \beta)$ on $(0, \infty)$ (Lebesgue measure) with shape α and rate β , set $\phi = 1/\alpha$ and $\theta = -\beta/\alpha$. Then $b(\theta) = -\log(-\theta)$ and

$$c(z, \phi) = -\frac{\log \phi}{\phi} + \left(\frac{1}{\phi} - 1\right) \log z - \log \Gamma(1/\phi).$$

In this parameterisation $b'(\theta) = \alpha/\beta$ and $b''(\theta) = \alpha^2/\beta^2$, so $\mathbb{E}[Z] = \alpha/\beta$ and $\text{Var}(Z) = \alpha/\beta^2$. This model is used for positive, right-skewed severities and costs.

◇

We now incorporate covariates. The classical linear model assumes that the response variable Z is normally distributed with mean $\mu = X^\top \beta$ and constant variance σ^2 . However, this assumption is not suitable for many types of data, such as binary or count data. Thus, GLMs extend linear models to accommodate response variables in regression models that follow distributions from the exponential dispersion family.

Definition 4.9 (Generalised Linear Model) A Generalised Linear Model specifies a conditional response distribution from the exponential dispersion family (4.8) with mean μ_i , a linear predictor $\eta_i = X_i^\top \beta$ (the *score*), and a link function g such that $g(\mu_i) = \eta_i$, where g is monotone, continuously differentiable, and invertible.

Notice from the definition that the link function implies the relationship

$$b'(\theta) = \mathbb{E}[Z_i | X_i] = \mu_i = g^{-1}(X_i^\top \beta),$$

which is often the defining equation of a GLM. Now, if we apply g on both sides we obtain

$$(g \circ b')(\theta) = X_i^\top \beta,$$

and when g is chosen in a way that $g \circ b'$ is the identity map, we say that the link function is the *canonical link function*. Such a choice is mathematically convenient and has advantageous statistical properties, but is otherwise just as arbitrary as any other choice, and there is no correct recipe as to how to choose the link function in general. In practice, investigating the data can give indications on the relationship between covariates and distribution, but very often a choice is made without exploration.

The estimation of GLMs can be done by maximum likelihood estimation as per Definition 4.2, yielding an MLE $\hat{\beta}_n$. However, for computational reasons,

it is typically done using the Iteratively Reweighted Least Squares (IRLS) algorithm. This algorithm is an extension of a weighted least squares routine and is used to find the MLE of the parameters. In a nutshell, at each iteration the algorithm updates the estimates of the parameters by solving a weighted least squares problem where the weights are updated based on the current estimates; then the process continues until convergence. An important caveat is that the dispersion parameter ϕ is typically estimated separately, often using goodness-of-fit statistics such as the Pearson χ^2 statistic or the deviance.

The following results assume that ϕ is known. The asymptotics do not change if it is instead consistently estimated.

Lemma 4.3 (Consistency of the GLM MLE) *Assume the link is canonical, so $\theta_i(\beta) = X_i^\top \beta$. Let $\mathcal{B} \subset \mathbb{R}^q$ be a compact set and $\beta_0 \in \text{int}(\mathcal{B})$. Assume:*

1. *The mapping $\beta \mapsto \theta_i(\beta)$ is continuous for all X_i , and $b(\cdot)$ is strictly convex.*
2. *The regressors X are such that $\mathbb{E}[XX^\top]$ is positive definite.*
3. *There exists $d(z, x)$ such that $\sup_{\beta \in \mathcal{B}} |\ell(z, x | \beta)| \leq d(z, x)$ with $\mathbb{E}[d(Z_1, X_1)] < \infty$.*
4. *$\hat{\beta}_n$ satisfies $M_n(\hat{\beta}_n) \geq M_n(\beta_0) - o_{\mathbb{P}}(1)$.*

Then $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$.

Proof. Let $M(\beta) = \mathbb{E}[\ell_1(\beta)]$. By Assumption 3 and Lemma 4.1, $\sup_{\beta \in \mathcal{B}} |M_n(\beta) - M(\beta)| \xrightarrow{\mathbb{P}} 0$. Since $\beta \mapsto \ell(z, x | \beta)$ is continuous, $M(\beta)$ is continuous on the compact set $\mathcal{B}$.

To show β_0 is the unique maximiser of $M(\beta)$, note that under the canonical link,

$$\ell_1(\beta) = Z_1 X_1^\top \beta - b(X_1^\top \beta).$$

Since b is strictly convex, the map $\beta \mapsto \ell_1(\beta)$ is concave for every (Z_1, X_1) , hence $M(\beta) = \mathbb{E}[\ell_1(\beta)]$ is concave on $\mathcal{B}$. Moreover, by the property of the EDF,

$$\nabla M(\beta) = \mathbb{E}[(b'(X_1^\top \beta_0) - b'(X_1^\top \beta))X_1],$$

so $\nabla M(\beta_0) = 0$. The strict convexity of b , together with the positive definiteness of $\mathbb{E}[XX^\top]$, implies that M is strictly concave and therefore has the unique maximiser β_0 . Given the uniform convergence of M_n to M , the uniqueness of the maximiser β_0 , and the near-maximiser condition, Assumption 4, it follows by Theorem 4.1 that $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$. $\square$

With the consistency of the estimator provided by Lemma 4.3, we now characterise its asymptotic distribution. We use the specific properties of the GLM likelihood to verify the regularity conditions required to appeal to the general theory of M-estimation.

Theorem 4.4 (Asymptotic normality of GLM MLEs) *Assume Z_i follows, conditionally on X_i , an EDF law with log-likelihood contribution up to an additive constant:*

$$\ell_i(\beta) = Z_i\theta_i(\beta) - b(\theta_i(\beta)), \quad \text{where } \theta_i(\beta) = h(X_i^\top \beta),$$

with $h = (b')^{-1} \circ g^{-1}$. Let $M_n(\beta) = \frac{1}{n} \sum_{i=1}^n \ell_i(\beta)$ for $\beta \in \mathcal{B} \subset \mathbb{R}^q$, with true parameter $\beta_0 \in \text{int}(\mathcal{B})$. Suppose that:

1. The estimator is consistent, $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$, and satisfies $\mathbb{P}(\nabla M_n(\hat{\beta}_n) = 0) \rightarrow 1$.
2. The functions b and h are twice continuously differentiable.
3. There exists an open neighbourhood $\mathcal{N}$ of β_0 and a function $d(z, x)$ such that $\sup_{\beta \in \mathcal{N}} \|\nabla^2 \ell(z, x | \beta)\| \leq d(z, x)$ with $\mathbb{E}[d(Z_1, X_1)] < \infty$.
4. The Fisher information matrix

$$I(\beta_0) = \mathbb{E}\left[b''(\theta_1(\beta_0))(h'(X_1^\top \beta_0))^2 X_1 X_1^\top\right]$$

is nonsingular.

Then

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, I(\beta_0)^{-1}).$$

Proof. By the properties of the EDF, the conditional moments are $\mathbb{E}[Z_1 | X_1] = b'(\theta_1(\beta_0))$ and $\text{Var}(Z_1 | X_1) = b''(\theta_1(\beta_0))$. The score for a single observation is

$$\nabla \ell_1(\beta) = (Z_1 - b'(\theta_1(\beta)))h'(X_1^\top \beta)X_1.$$

Evaluating at β_0 and taking the conditional expectation yields $\mathbb{E}[\nabla \ell_1(\beta_0) | X_1] = 0$, so $\mathbb{E}[\nabla \ell_1(\beta_0)] = 0$. Its variance is exactly the Fisher information:

$$\Sigma = \text{Var}(\nabla \ell_1(\beta_0)) = \mathbb{E}\left[\text{Var}(Z_1 | X_1)(h'(X_1^\top \beta_0))^2 X_1 X_1^\top\right] = I(\beta_0).$$

Thus, by Theorem 1.6, $\sqrt{n}\nabla M_n(\beta_0) \xrightarrow{d} N(0, I(\beta_0))$.

The Hessian of the log-likelihood is

$$\nabla^2 \ell_1(\beta) = (Z_1 - b'(\theta_1(\beta)))h''(X_1^\top \beta)X_1 X_1^\top - b''(\theta_1(\beta))(h'(X_1^\top \beta))^2 X_1 X_1^\top.$$

Evaluating at β_0 , the conditional expectation of the first term vanishes since $\mathbb{E}[Z_1 - b'(\theta_1(\beta_0)) | X_1] = 0$. Taking the full expectation over the second term yields

$$H(\beta_0) = \mathbb{E}[\nabla^2 \ell_1(\beta_0)] = -I(\beta_0).$$

By Assumptions 2 and 3, together with Lemma 4.1, the Hessian satisfies the uniform convergence requirements. Since $\mathbb{E}[\nabla \ell_1(\beta_0)] = 0$, $\Sigma = I(\beta_0)$, and $H(\beta_0) = -I(\beta_0)$ is invertible by Assumption 4, all conditions of Theorem 4.2 are satisfied. Therefore,

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, H(\beta_0)^{-1}\Sigma H(\beta_0)^{-1}) = N(0, I(\beta_0)^{-1}).$$

□

Remark 4.14 (Conditional formulation and dispersion) Although the model is specified conditionally through $f_{Z|X}$, the asymptotics are taken under the joint law of (Z_i, X_i) (random design), so the M-estimation results apply directly to $M_n(\beta)$. For fixed-design asymptotics, one conditions on the observed covariates and imposes non-degeneracy conditions on the design sequence. The theorem is stated for a known dispersion ϕ ; if ϕ is replaced by a consistent estimator $\hat{\phi}_n$, the asymptotic distribution of $\sqrt{n}(\hat{\beta}_n - \beta_0)$ remains unchanged. $\square$

GLM specifications, as per Definition 4.9, may have probabilistic, covariate and link function choices yielding different results. Model selection is typically guided by likelihood-based criteria and diagnostics. Two widely used information criteria are the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), defined by

$$AIC = -2\ell_n + 2p, \quad BIC = -2\ell_n + \log(n) \cdot p,$$

where p is the number of parameters. AIC tends to favour more flexible fits, while BIC imposes a stronger penalty and therefore prefers parsimonious models; both require comparisons to be made on the same dataset. For nested models, likelihood ratio tests (ANOVA) compare the log-likelihood difference to a chi-squared reference distribution. Further validation is usually done with residual plots and out-of-sample performance measures. Indeed, the latter criterion has become standard, and hence in the statistical learning chapter we study predictive power as a general-purpose performance criterion in regression modelling.

4.3.2 Actuarial considerations

We now provide some practical considerations for the estimation of GLMs in actuarial applications. Some of the advice provided here is specific to the model, though others are generalisable to arbitrary regression models.

The most common GLM specifications in insurance correspond to the EDF examples above, but the regression setting adds the link and linear predictor structure. Using canonical link functions $g(\mu) = (b')^{-1}(\mu)$ and the linear predictor $\eta_i = x_i^\top \beta$, these models take the following forms and interpretations.

Example 4.12 (Canonical GLM specifications) 1. *Normal (linear regression)*.

With $Z_i \sim N(\mu_i, \phi)$ and $g(\mu) = \mu$, one has $\mu_i = \eta_i$ and constant variance. This is the classical linear model for continuous responses where additive effects are natural and residuals are approximately symmetric.

2. *Bernoulli (logistic regression)*. With $Z_i \sim \text{Bernoulli}(\mu_i)$ and $g(\mu) = \log(\mu/(1-\mu))$, one has

$$\log\left(\frac{\mu_i}{1-\mu_i}\right) = \eta_i,$$

with μ_i interpreted as the conditional success probability. The logit link ensures fitted probabilities remain in $(0, 1)$ and makes covariate effects multiplicative on the odds scale.

3. *Poisson regression.* With $Z_i \sim \text{Poisson}(\mu_i)$ and $g(\mu) = \log \mu$, one has $\mu_i = \exp(\eta_i)$ and $\text{Var}(Z_i | x_i) = \mu_i$. This is the standard model for claim counts or other event frequencies, especially with exposure offsets.
4. *Gamma regression.* With $Z_i \sim \text{Gamma}(\alpha, \beta_i)$ (fixed shape) and $g(\mu) = -1/\mu$, one has

$$-\frac{1}{\mu_i} = \eta_i,$$

and $\text{Var}(Z_i | x_i) = \phi \mu_i^2$ with $\phi = 1/\alpha$. This is a standard choice for positive severities where variability grows roughly quadratically with the mean.

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Choice of distribution and link function

The member of the exponential dispersion family is usually selected by the *type of data*: continuous, positive, discrete or binary. All members are light-tailed, so heavy-tailed data needs to be addressed with other models. A common workaround is to *transform the data* (for instance with the logarithm), though one should keep in mind that the model undergoes a distributional change; for instance transforming heavy-tailed data with the logarithm and fitting a linear regression effectively specifies a log-normal model, while fitting a Gamma regression specifies a Pareto-type tail model. *Splicing* is also common, where the top observations are modelled with a heavy-tailed distribution and the smaller ones with a GLM. To “paste” the two models together, either logistic regression is applied (to exceedance probabilities), or right-truncation needs to be considered.

As mentioned before, there is no set way of choosing the link function. Some pricing actuaries may prefer to set premiums according to multiplicative factors, from which a log link for both frequency and severity produce such results. Indeed, in that case the *pure premium* is given by

$$\mathbb{E}[Z_i N_i | X = x] = \mathbb{E}[Z_i | X = x] \mathbb{E}[N_i | X = x] = \exp(x_i^\top \beta^Z) \exp(x_i^\top \beta^N),$$

where Z_i and N_i represent the severity and frequency of a claim, respectively. The latter are assumed to be conditionally independent in most practical applications, despite the condition not always being satisfied by the data.

Choice of covariates

Covariates can be: continuous, ordinal, categorical, or binary. Several feature-engineering practices aim at improving the functional form of covariates through transformations so that the linearity of the score is more easily satisfied. This includes: a) transforming continuous components with real-valued functions; b) transformation of ordinal and even categorical components into real-valued covariates. The opposite procedure is to bundle continuous covariates into risk classes, though somewhat less common; c) interactions of components beyond an additive structure; d) avoidance of collinearity.

Offsets for different exposure periods

A common approach to dealing with different exposure periods for claims is to include such information in the covariate space. We recommend only doing so when no alternative is available. Luckily, for GLMs an alternative is available, which is to use an *offset* (or constant) in the linear predictor. Thus we consider the relationship

$$\mu_i = g^{-1}(\log(E_i) + X_i^\top \beta),$$

where E_i is an observation-specific constant, corresponding to exposure. In principle, the offset could be absorbed in a linear fashion into the features, though it is important to note that its coefficient should not be fitted, but identically set to 1. The most common instance where an offset is necessary is in Poisson models with canonical link functions, where we obtain

$$\mu_i = E_i \exp(X_i^\top \beta),$$

so that exposure acts multiplicatively (proportionally in time) with the weights E_i .

4.3.3 Regularisation

In actuarial applications, covariate spaces often include many rating factors, interactions, and sparse levels. In such settings the unpenalised GLM MLE can have high variance and unstable coefficients, especially under strong collinearity. Regularisation (or shrinkage) addresses this problem by introducing a controlled bias that reduces estimation variance and improves out-of-sample performance.

Write the conditional log-likelihood for a GLM as

$$\ell_n(\beta) = \sum_{i=1}^n \log f_{Z|X}(Z_i | X_i, \beta, \phi), \quad \eta_i = X_i^\top \beta,$$

and define the penalised estimator by

$$\hat{\beta}_\lambda = \underset{\beta}{\operatorname{argmax}} \{\ell_n(\beta) - \lambda P(\beta)\},$$

where P is a penalty and $\lambda \geq 0$ is a tuning parameter. In practice the intercept is excluded from P , and continuous covariates are centred and scaled so that shrinkage is comparable across units. Categorical factors are typically expanded into indicator variables; when entire factors should enter or leave jointly, block penalties are often preferable to componentwise L^1 shrinkage. Offsets remain fixed (coefficient equal to one) and are therefore not penalised. For convex penalties, and for GLM losses with convex negative log-likelihood, the optimisation problem is convex.

The tuning parameter is usually selected along a regularisation path. A standard practice is to evaluate a grid of λ values and select by cross-validation using out-of-sample log-likelihood or predictive deviance. In portfolio data with temporal structure, random folds may leak information across time, so blocked or rolling validation splits are preferable. A common practical choice is the largest λ whose validation performance remains within one standard error of the empirical minimum, thereby favouring parsimony.

Ridge, lasso, and elastic net

The three most common convex penalties can be presented in a single family:

$$\hat{\beta}_{\lambda,\alpha} = \operatorname{argmax}_{\beta} \left\{ \ell_n(\beta) - \lambda \sum_{j=1}^p \left[(1-\alpha) \frac{\beta_j^2}{2} + \alpha |\beta_j| \right] \right\}, \quad \alpha \in [0, 1].$$

The case $\alpha = 0$ gives ridge regression,

$$\hat{\beta}^r = \operatorname{argmax}_{\beta} \left\{ \ell_n(\beta) - \lambda \frac{1}{2} \sum_{j=1}^p \beta_j^2 \right\},$$

the case $\alpha = 1$ gives lasso,

$$\hat{\beta}^l = \operatorname{argmax}_{\beta} \left\{ \ell_n(\beta) - \lambda \sum_{j=1}^p |\beta_j| \right\},$$

and $0 < \alpha < 1$ gives elastic net.

This unified form also clarifies geometry and behaviour. Ridge corresponds to an L^2 ball constraint, lasso to an L^1 ball constraint, and elastic net to an interpolation between the two; equivalently,

$$\begin{aligned} \hat{\beta}^r &= \operatorname{argmax}_{\beta} \ell_n(\beta) \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq t, \\ \hat{\beta}^l &= \operatorname{argmax}_{\beta} \ell_n(\beta) \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq t, \end{aligned}$$

for a suitable one-to-one mapping between t and λ . The corner structure of the L^1 constraint explains exact zeros under lasso, while the smooth L^2 constraint explains why ridge shrinks continuously without variable deletion.

Practically, ridge is often preferred when predictive stability under collinearity is the main goal, whereas lasso is preferred when sparse variable screening is important. The elastic net is used when both sparsity and grouped shrinkage are desired in data with big collinearity. For lasso and elastic net, coordinate-descent algorithms are standard, while ridge can actually be solved by closed form in the Gaussian model and by Newton or penalised IRLS-type updates for general GLMs.

For inference, one should keep in mind that penalised estimators are biased by construction, and therefore the standard toolbox of uncertainty measures from the unpenalised likelihood are generally not valid after variable selection without additional adjustments. In practice, regularisation is best viewed first as a prediction and stabilisation tool, and coefficients do not deliver the same quality of statistical insight as the unpenalised counterparts

Regularisation also connects naturally to the next section, which is classical in actuarial science. Indeed, both regularisation and credibility estimators are shrinkage methods, but they shrink towards different targets and arise from different modelling principles.

4.3.4 Credibility theory

Credibility theory provides a principled form of shrinkage between individual experience and collective information. It is particularly useful when data are sparse for each risk but plentiful across a portfolio. The central object is the conditional mean of a risk, which we estimate by a linear combination of the individual experience and a portfolio-wide mean.

Collective risk model

Definition 4.10 (Collective risk model) Let Θ be an unobserved stochastic risk parameter. Conditional on Θ , let (Z_n) be independent with

$$\mathbb{E}[Z_i | \Theta] = \mu(\Theta), \quad \text{Var}(Z_i | \Theta) = \sigma^2(\Theta).$$

The target of credibility estimation is the conditional mean $\mu(\Theta)$.

Define the structural quantities

$$m = \mathbb{E}[\mu(\Theta)], \quad \text{EPV} = \mathbb{E}[\sigma^2(\Theta)], \quad \text{VHM} = \text{Var}(\mu(\Theta)),$$

where EPV is the *expected process variance* and VHM is the *variance of hypothetical means*. By the law of total variance,

$$\text{Var}(Z_1) = \text{EPV} + \text{VHM}.$$

Let $\bar{Z} = n^{-1} \sum_{i=1}^n Z_i$ denote the *individual experience*.

Bühlmann credibility

The classical Bühlmann model gives the best linear predictor of $\mu(\Theta)$ based on the individual experience $\bar{Z}$ and the collective mean m . The result below shows that the optimal shrinkage weight is determined by the balance between process noise (EPV) and portfolio heterogeneity (VHM).

Theorem 4.5 (Bühlmann credibility) *Among all linear estimators of the form $\hat{\mu} = a + b\bar{Z}$, the mean-square error*

$$\mathbb{E}[(\mu(\Theta) - \hat{\mu})^2]$$

is minimised by

$$\hat{\mu} = Z\bar{Z} + (1 - Z)m, \quad Z = \frac{n}{n + K}, \quad K = \frac{\text{EPV}}{\text{VHM}}.$$

The weight $Z \in [0, 1]$ is called the credibility factor.

Proof. Consider $\hat{\mu} = a + b\bar{Z}$ and minimise $\mathbb{E}[(\mu(\Theta) - a - b\bar{Z})^2]$. Since this objective is a convex quadratic in (a, b) , the minimiser is characterised by the first-order conditions

$$\mathbb{E}[\mu(\Theta) - a - b\bar{Z}] = 0, \quad \mathbb{E}[(\mu(\Theta) - a - b\bar{Z})\bar{Z}] = 0.$$

Since $\mathbb{E}[\bar{Z}] = m$, the first equation yields $a = (1 - b)m$. For the second, note that

$$\text{Cov}(\mu(\Theta), \bar{Z}) = \text{Cov}(\mu(\Theta), \mathbb{E}[\bar{Z} | \Theta]) = \text{Var}(\mu(\Theta)) = \text{VHM},$$

and

$$\text{Var}(\bar{Z}) = \text{Var}(\mathbb{E}[\bar{Z} | \Theta]) + \mathbb{E}[\text{Var}(\bar{Z} | \Theta)] = \text{VHM} + \frac{\text{EPV}}{n}.$$

Thus $b = \text{VHM}/(\text{VHM} + \text{EPV}/n) = n/(n + K)$ with $K = \text{EPV}/\text{VHM}$. Substituting $a = (1 - b)m$ gives the stated form. $\square$

The factor $Z = n/(n + K)$ increases with the amount of experience and decreases with process variability, so credibility moves smoothly from collective to individual information. In practice, however, exposures are rarely equal, which leads to the weighted extension below.

Bühlmann–Straub credibility

The Bühlmann model assumes equal information per observation. When observations have different exposures, we use weights. Assume that, conditional on Θ ,

$$\mathbb{E}[Z_i | \Theta] = \mu(\Theta), \quad \text{Var}(Z_i | \Theta) = \frac{\sigma^2(\Theta)}{w_i},$$

where $w_i > 0$ is the exposure of observation i and $W = \sum_{i=1}^n w_i$. Define the weighted mean $\bar{Z}_w = W^{-1} \sum_{i=1}^n w_i Z_i$.

Theorem 4.6 (Bühlmann–Straub credibility) *Among estimators of the form $\hat{\mu} = a + b\bar{Z}_w$, the mean-square error is minimised by*

$$\hat{\mu} = Z\bar{Z}_w + (1 - Z)m, \quad Z = \frac{W}{W + K}, \quad K = \frac{\text{EPV}}{\text{VHM}}.$$

Proof. The proof is identical to Theorem 4.5 after replacing $\bar{Z}$ by $\bar{Z}_w$ and n by W . Indeed,

$$\mathbb{E}[\bar{Z}_w | \Theta] = \mu(\Theta),$$

and, since the Z_i are conditionally independent,

$$\text{Var}(\bar{Z}_w | \Theta) = \frac{1}{W^2} \sum_{i=1}^n w_i^2 \frac{\sigma^2(\Theta)}{w_i} = \frac{\sigma^2(\Theta)}{W}.$$

Hence

$$\text{Var}(\bar{Z}_w) = \text{Var}(\mathbb{E}[\bar{Z}_w | \Theta]) + \mathbb{E}[\text{Var}(\bar{Z}_w | \Theta)] = \text{VHM} + \frac{\text{EPV}}{W},$$

which yields $Z = W/(W + K)$. $\square$

Hence the Bühlmann–Straub formula preserves the same shrinkage structure, with n replaced by total exposure W . This form is standard in non-life applications where contracts or observation windows have different effective volumes.

Exact credibility and examples

In a Bayesian formulation the optimal estimator of $\mu(\Theta)$ is the posterior mean $\mathbb{E}[\mu(\Theta) | Z_1, \dots, Z_n]$. When this posterior mean is linear in the sufficient statistic, the Bühlmann estimator is *exact*. This occurs for several classical conjugate pairs that are common in actuarial work.

Example 4.13 (Exact credibility in conjugate models) We illustrate how credibility weights arise in frequency, severity, and claim probability models.

1. *Poisson–Gamma*. If $Z_i | \Theta \sim \text{Poisson}(\Theta)$ and $\Theta \sim \text{Gamma}(\alpha, \beta)$, then $m = \alpha/\beta$, $\text{EPV} = \alpha/\beta$, $\text{VHM} = \alpha/\beta^2$, and $K = \beta$. The posterior mean is

$$\mathbb{E}[\Theta | Z_1, \dots, Z_n] = \frac{\alpha + \sum_{i=1}^n Z_i}{\beta + n} = \frac{n}{n + \beta} \bar{Z} + \frac{\beta}{n + \beta} m.$$

This model is a standard choice for claim frequency, where Θ represents an unobserved risk level and the gamma prior captures portfolio heterogeneity.

2. *Normal–Normal*. If $Z_i | \Theta \sim N(\Theta, \sigma^2)$ and $\Theta \sim N(m_0, \tau^2)$, then $\text{EPV} = \sigma^2$, $\text{VHM} = \tau^2$, and $K = \sigma^2/\tau^2$. The posterior mean is

$$\mathbb{E}[\Theta | Z_1, \dots, Z_n] = \frac{n\tau^2}{n\tau^2 + \sigma^2} \bar{Z} + \frac{\sigma^2}{n\tau^2 + \sigma^2} m_0.$$

This setting is natural for severity modelling with approximately normal noise and a population distribution for the underlying mean.

3. *Bernoulli–Beta*. If $Z_i | \Theta \sim \text{Bernoulli}(\Theta)$ and $\Theta \sim \text{Beta}(\alpha, \beta)$, then $m = \alpha/(\alpha + \beta)$, $\text{EPV} = \alpha\beta/[(\alpha + \beta)(\alpha + \beta + 1)]$, $\text{VHM} = \alpha\beta/[(\alpha + \beta)^2(\alpha + \beta + 1)]$, and $K = \alpha + \beta$. The posterior mean is

$$\mathbb{E}[\Theta | Z_1, \dots, Z_n] = \frac{\alpha + \sum_{i=1}^n Z_i}{\alpha + \beta + n} = \frac{n}{n + \alpha + \beta} \bar{Z} + \frac{\alpha + \beta}{n + \alpha + \beta} m.$$

This model applies to claim/no-claim indicators or policyholder lapse events, where Θ is the underlying probability for an individual risk.

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Estimating structural parameters

To implement credibility in practice, the structural quantities EPV and VHM must be estimated. Consider J independent risks, each with n observations $Z_{j1}, \dots, Z_{jn}$. Let

$$\bar{Z}_j = \frac{1}{n} \sum_{k=1}^n Z_{jk}, \quad S_j^2 = \frac{1}{n-1} \sum_{k=1}^n (Z_{jk} - \bar{Z}_j)^2.$$

Proposition 4.6 (Structural parameter estimates) *Assume the collective risk model holds for each risk and the number of observations per risk is common. Then*

$$\hat{m} = \frac{1}{J} \sum_{j=1}^J \bar{Z}_j, \quad \widehat{EPV} = \frac{1}{J} \sum_{j=1}^J S_j^2,$$

and

$$\widehat{VHM} = \frac{1}{J-1} \sum_{j=1}^J (\bar{Z}_j - \hat{m})^2 - \frac{\widehat{EPV}}{n}$$

are unbiased estimators of m , EPV , and VHM .

Proof. By the collective risk model, $\mathbb{E}[S_j^2] = EPV$ and $\mathbb{E}[\bar{Z}_j] = m$. Moreover,

$$\text{Var}(\bar{Z}_j) = VHM + \frac{EPV}{n}.$$

The usual unbiased sample variance across $\bar{Z}_1, \dots, \bar{Z}_J$ has expectation $\text{Var}(\bar{Z}_j)$, and therefore equals $VHM + EPV/n$ in expectation. Subtracting $\widehat{EPV}/n$ yields an unbiased estimator of VHM . $\square$

Remark 4.15 When exposures vary across observations, the Bühlmann–Straub model uses weighted means and weighted within-risk variances. The corresponding moment estimators are modifications of the above formulas with n replaced by total exposure and with weights in the sample variance; see the notes for further details. $\circ$

4.4 Notes and Comments

Large-sample statistical theory provides the foundation for the results on M-estimation developed in this chapter [van der Vaart, 1998], while a complementary rigorous treatment with econometric applications is given in Newey and McFadden [1994]. These works formalise the probabilistic conditions required for consistency and asymptotic normality. The maximum-likelihood parts of the chapter fit into this same asymptotic framework through score and information expansions. Broader introductions at the graduate level, such as Casella and Berger [2024], Lehmann [1991], Hogg et al. [2005], deal with related but mostly omitted topics such as sufficiency, efficiency, and Fisher information, which are relevant in modern theoretical statistics. The information-theoretic viewpoint, as introduced in Kullback and Leibler [1951], connects estimation to concepts of divergence and motivates likelihood-based methods, while the robustness literature in Huber [1981] shows how asymptotic properties adapt when regularity conditions are weakened.

Phase-type and matrix-exponential distributions are treated in a modern and systematic way in Bladt and Nielsen [2017], which also provides a statistical account, including likelihood-based and EM estimation. Historically, foundational developments include early phase-type renewal formulations by Neuts [1978] and structural characterisation results by O’Cinneide [1990]. The general EM framework is classical from Dempster et al. [1977], while the PH-specific EM developments are given in Asmussen et al. [1996] for complete observations and in Olsson [1996] for censored data. For the numerical steps that appear in PH-EM algorithms, the block-matrix identity used to compute integrals of matrix functions via one enlarged matrix exponential is due to Loan [1978]. Classical surveys of matrix-exponential computation, including ODE-based integration and Padé-based scaling-and-squaring viewpoints, are given in Moler and Loan [1978], with modern numerical analysis in Higham [2005]. In the stochastic-process literature, uniformisation (randomisation) traces back to Jensen [1953] and was developed computationally for transient Markov calculations in works such as Grassmann [1977]. For discrete phase-type distributions and their computational structure, a classical reference is Latouche and Ramaswami [1999].

Inhomogeneous phase-type distributions, based on time changes of homogeneous PH models, are developed in Albrecher and Bladt [2019]. A unified treatment of heavy-tailed PH/IPH models is given in Bladt and Yslas [2022a]. The multivariate phase-type class originates with Assaf et al. [1984] and has been extended in actuarial directions in Bladt [2023], Albrecher et al. [2023]. Recent actuarial developments include phase-type regression and mixture-of-experts models for claim frequencies (discrete outcomes) and severities (continuous outcomes) Bladt and Yslas [2023, 2022b], phase-type regression for severities Bladt [2022], and regression with matrix distributions in mortality modelling Albrecher et al. [2022]. Estimation of absorption time distributions for general inhomogeneous Markov jump processes is developed in Ahmad et al. [2023], extending the phase-type special case.

The generalised linear model (GLM) framework was introduced in Nelder and Wedderburn [1972] and developed systematically in McCullagh and Nelder [1989], including exponential-family formulations, canonical links, and asymp-

totic efficiency of maximum likelihood estimation. More applied discussions, including for logistic-regression, diagnostics, and broader implementation details, are found in [Hosmer and Lemeshow \[2000\]](#), [Hardin and Hilbe \[2018\]](#). From an actuarial perspective, the treatment of offsets for exposure periods and distribution-link choices is especially important for preserving interpretability. Model-selection principles draw on the information-criterion literature, in particular the Akaike Information Criterion [[Akaike, 1974](#)] and the Bayesian Information Criterion [[Schwarz, 1978](#)], both justified by large sample considerations. The extension of the regression methodology into high-dimensional and machine-learning contexts led to regularisation methods, beginning with ridge regression [[Hoerl and Kennard, 1970](#)], followed by the lasso [[Tibshirani, 1996](#)], and the elastic net [[Zou and Hastie, 2005](#)]. Their broader methodological connections are presented in [Hastie et al. \[2009\]](#), while asymptotic properties are not treated in this chapter, but are analysed rigorously for instance in [Fu and Knight \[2000\]](#).

Credibility theory is classical in insurance mathematics, and provides a shrinkage framework for insurance data with heterogeneous risk profiles. The foundational linear credibility model is due to [Bühlmann \[1967\]](#), and the exposure-weighted extension was developed by [Bühlmann and Straub \[1970\]](#). A comprehensive modern treatment is given in [Bühlmann and Gisler \[2005\]](#), while exact credibility for exponential families is characterised in [Jewell \[1974\]](#).

Regarding broader actuarial applications, these parametric methods also lead to domain-specific models depending on the data structure. Classical references such as [Bowers et al. \[1997\]](#), [Denuit et al. \[2005\]](#) are standard sources for parametric risk modelling, while survival analysis techniques useful for insurance applications are well developed in [Klein and Moeschberger \[2003\]](#). Recent work extends the statistical learning perspective to actuarial science, with [Wüthrich and Merz \[2023\]](#) providing an up-to-date treatment of machine learning for insurance, and the volumes [Denuit et al. \[2019a, 2020, 2019b\]](#) which treat GLMs and extensions, decision trees, and neural networks, respectively, all tailored to actuarial practice.

4.5 Exercises

1. Consider the sequence (Z_n) of iid exponentially distributed random variables with rate $\lambda > 0$. Then $Z \sim F$, where $F(z) = 1 - e^{-\lambda z}$, $z > 0$.
 - (a) Derive the MLE, $\hat{\lambda}_n$, and determine the asymptotic distribution of $\sqrt{n}(\hat{\lambda}_n - \lambda)$.
 - (b) Consider the sequence (Y_n) of iid Pareto-distributed random variables with scale parameter $\theta > 0$ and shape parameter $\alpha = 1/\gamma > 0$. The threshold θ has a natural meaning, so we treat it as given in this exercise and focus on estimating γ . Denote by $\hat{\gamma}_n$ the MLE of γ and argue why

$$\hat{\gamma}_n = \frac{1}{n} \sum_{i=1}^n (\log(Y_i) - \log(\theta)).$$

Hint: For $Y \sim \text{Pareto}(\theta, \alpha)$, it holds that $X = \log(Y) - \log(\theta) \sim \text{Gamma}(1, \alpha)$.

- (c) Determine the asymptotic distribution of $\sqrt{n}(\hat{\gamma}_n - \gamma)$ and compare with the asymptotic distribution of $\sqrt{k}(\hat{\gamma}_{n,k} - \gamma)$, where $\hat{\gamma}_{n,k}$ is the Hill estimator.
2. In this exercise we consider data that is either left-truncated or right-censored.
 - (a) Show that the log-likelihood for left-truncated data is given by (4.6), i.e.

$$\ell_n(\theta) = \sum_{i=1}^n \left\{ \log f_\theta(Z_i | \theta) - \log \bar{F}_\theta(L_i | \theta) \right\}.$$

- (b) Show that the log-likelihood for right-censored data is given by (4.7), i.e.

$$\ell_n(\theta) = \sum_{i=1}^n \left\{ \delta_i \log f_\theta(Z_i | \theta) + (1 - \delta_i) \log \bar{F}_\theta(C_i | \theta) \right\}.$$

Consider the cases when the sequence of censoring times (C_i) is stochastic and deterministic, respectively.

3. In this exercise we investigate what conditions on Ψ_n are sufficient to ensure that $\hat{\theta}_n$ is consistent. Two different sets of conditions can be used to show consistency, and we investigate both of them. To start, let Θ be a subset of the real line, let Ψ_n be random functions, and let Ψ be a fixed function of θ such that $\Psi_n(\theta) \rightarrow \Psi(\theta)$ in probability for every θ . Let θ_0 be a point such that $\Psi(\theta_0 - \varepsilon) < 0 < \Psi(\theta_0 + \varepsilon)$ for every $\varepsilon > 0$.
 - (a) Now assume that each map $\theta \mapsto \Psi_n(\theta)$ is continuous and has exactly one zero $\hat{\theta}_n$.

i. Show that

$$\mathbb{P}(\Psi_n(\theta_0 - \varepsilon) < 0 < \Psi_n(\theta_0 + \varepsilon)) \leq \mathbb{P}(\theta_0 - \varepsilon < \hat{\theta}_n < \theta_0 + \varepsilon)$$

for all $\varepsilon > 0$.

ii. By using (i), show that $\hat{\theta}_n$ is consistent.

(b) Now assume that $\theta \mapsto \Psi_n(\theta)$ is nondecreasing with $\Psi_n(\hat{\theta}_n) = o_{\mathbb{P}}(1)$.

i. Show that

$$\mathbb{P}(\Psi_n(\theta_0 - \varepsilon) < -\eta, \Psi_n(\theta_0 + \varepsilon) > \eta) \leq \mathbb{P}(\theta_0 - \varepsilon < \hat{\theta}_n < \theta_0 + \varepsilon) + o(1).$$

ii. Show that $\hat{\theta}_n$ is consistent.

(c) Let (X_n) be iid random variables with a unique median. Let $\Psi_n(\theta) = n^{-1} \sum_{i=1}^n \text{sign}(\theta - X_i)$, and let $\hat{\theta}_n$ be the sample median, i.e. $\hat{\theta}_n = X_{n, \lceil n/2 \rceil}$. Show that $\hat{\theta}_n$ is consistent.

4. Suppose that we observe a random iid sample (Z_i, X_i) from a regression model

$$Z = \tilde{m}(x, \theta_1^*, \theta_2^*) + e, \quad \mathbb{E}[e | X] = 0, \quad \mathbb{E}[e^2 | X] = 1$$

where $\tilde{m}(x, \theta_1, \theta_2) = \theta_1 x + e^{\theta_2 x}$, and we aim at estimating the parameters θ_1^* and θ_2^* . We assume that the independent variables (X_n) are a random sample. We further assume that $\Theta = [0, \tau]^2$ for some $\tau > 0$ and $\mathbb{E}[\sup_{\theta \in \Theta} |m(Z, X, \theta)|] < \infty$, where $m(z, x, \theta) = -(z - \theta_1 x - e^{\theta_2 x})^2$. We denote by $\hat{\theta}_n$ the estimator of (θ_1^*, θ_2^*) given by

$$\begin{aligned} \operatorname{argmax}_{(\theta_1, \theta_2) \in \Theta} M_n(\theta) &= \operatorname{argmax}_{(\theta_1, \theta_2) \in \Theta} \frac{1}{n} \sum_{i=1}^n m(Z_i, X_i, \theta) \\ &= \operatorname{argmax}_{(\theta_1, \theta_2) \in \Theta} \frac{1}{n} \sum_{i=1}^n -(Z_i - \theta_1 X_i - e^{\theta_2 X_i})^2 \end{aligned}$$

and we assume that θ^* uniquely maximises $M(\theta) = \mathbb{E}[m(Z, X, \theta)]$.

(a) Show that $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta^*$.

(b) First, assume that X has a finite moment generating function on a sufficiently large compact interval. Furthermore, assume

$$-\mathbb{E}[X^2] \mathbb{E}[a(\theta^*)] - \mathbb{E}[\exp(\theta_2^* X) X^2]^2 \neq 0,$$

where $a(\theta) = (Z_i - \theta_1 X_i - \exp(\theta_2 X_i)) \cdot \exp(\theta_2 X_i) X_i^2 - \exp(2\theta_2 X_i) X_i^2$. Use Theorem 4.2 to determine that $\sqrt{n}(\hat{\theta}_n - \theta^*)$ converges to a normal distribution.

5. We wish to investigate the conditions of Theorem 4.1.

(a) Find a sequence of fixed (nonrandom) functions $M_n: \mathbb{R} \rightarrow \mathbb{R}$ that converges pointwise to a limit M_0 and such that each M_n has a unique maximum at a point θ_n , but the sequence θ_n does not converge to θ_0 .

- (b) In the same setting, find a sequence of functions M_n that converges uniformly but still θ_n does not converge to θ_0 .
 - (c) Find a sequence of fixed (nonrandom) functions $M_n: \mathbb{R} \rightarrow \mathbb{R}$ that converges pointwise but not uniformly to a limit M_0 such that each M_n has a unique maximum at a point θ_n and the sequence θ_n converges to θ_0 .
6. Let (Z_n) be iid from a one-parameter exponential family with density

$$f_\theta(z) = \exp\{\theta T(z) - A(\theta)\}h(z), \quad \theta \in \Theta,$$

where A is twice continuously differentiable. Show that the MLE solves

$$\frac{1}{n} \sum_{i=1}^n T(Z_i) = A'(\theta),$$

and use the M-estimation theorem to derive the asymptotic distribution of the MLE. Express the asymptotic variance in terms of $A''(\theta_0)$.

7. Let (Z_n) be iid with a continuous density f such that $f(\theta_0) > 0$, where θ_0 is the median of Z_1 . Let

$$\hat{\theta}_n = \operatorname{argmin}_{\theta \in \mathbb{R}} \sum_{i=1}^n |Z_i - \theta|.$$

Show that $\hat{\theta}_n$ is asymptotically normal and determine its asymptotic variance using the M-estimation framework.

8. Phase-type representations are not unique. Construct two distinct pairs (α, S) (with different dimensions) that yield the same distribution, and verify directly from $f(t) = \alpha e^{tS} \mathbf{s}$ that the resulting densities coincide.
9. In the collective risk model, let $Z_1, \dots, Z_n$ be conditionally independent given Θ , with

$$\mu(\Theta) = \mathbb{E}[Z_1 | \Theta], \quad \sigma^2(\Theta) = \operatorname{Var}(Z_1 | \Theta),$$

and define

$$m = \mathbb{E}[\mu(\Theta)], \quad \operatorname{EPV} = \mathbb{E}[\sigma^2(\Theta)], \quad \operatorname{VHM} = \operatorname{Var}(\mu(\Theta)).$$

Among estimators of the form

$$\hat{\mu} = Z\bar{Z} + (1 - Z)m,$$

with credibility factor

$$Z = \frac{n}{n + K}, \quad K = \frac{\operatorname{EPV}}{\operatorname{VHM}},$$

answer the following:

- (a) For a target credibility level $z_0 \in (0, 1)$, derive the smallest sample size n such that $Z \geq z_0$.
- (b) In the Poisson–Gamma example $Z_i | \Theta \sim \operatorname{Poisson}(\Theta)$ and $\Theta \sim \operatorname{Gamma}(\alpha, \beta)$, explicitly compute m , EPV , VHM , and K , and write the resulting credibility estimator.
- (c) Explain how the required sample size from (a) changes when EPV is doubled and when VHM is doubled.

Practical exercises

1. Consider the `norauto` dataset from the R package `CASdatasets` [Dutang and Charpentier, 2024], which records Norwegian motor insurance claims. Model the number of claims (`NbClaim`) using a Poisson regression with covariates $x = (\text{Male}, \text{Young}, \text{DistLimit}, \text{GeoRegion})$ and offset `Expo`.

- (a) Fit a Poisson regression with an intercept and log-link function. Report the estimated coefficients.

Hint: Use the `glm()` function.

- (b) Fit a ridge regression (with intercept) across a sequence of shrinkage parameters λ using the R package `glmnet`. Plot the coefficient paths as a function of $\log(\lambda)$.

Hint: Pass a model matrix without an intercept to `glmnet()`, setting `alpha = 0` and `intercept = TRUE`. Leave λ unspecified to generate a plot-ready sequence.

- (c) Repeat (b) for lasso regression.

Hint: Use `alpha = 1`.

- (d) Perform 10-fold cross-validation to select the optimal λ for both the ridge and lasso models. Compare their optimal coefficients with the standard GLM coefficients from (a).

Hint: Use `cv.glmnet()` (which defaults to 10 folds). The optimal shrinkage parameter is stored as `lambda.min`.

2. In this exercise, simulate data from a Poisson GLM and fit it by coding the MLE yourself, then compare against a standard implementation. Let $n = 2000$, draw $X_{i1}, X_{i2} \sim N(0, 1)$ independently, set

$$\eta_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2}, \quad \mu_i = e^{\eta_i},$$

with $(\beta_0, \beta_1, \beta_2) = (0.2, 0.8, -0.5)$, and simulate $Y_i | X_i \sim \text{Poisson}(\mu_i)$.

- (a) Implement maximum likelihood estimation for β by coding a numerical optimiser (e.g. Newton–Raphson or `optim()`) for the Poisson log-likelihood.
- (b) Fit the same model with the `glm()` function. Compare estimated coefficients, log-likelihood values, and fitted means.

3. In this exercise, we implement Bühlmann credibility on simulated portfolio data.

- (a) For each $n \in \{2, 5, 10\}$, simulate a portfolio of $I = 2000$ policyholders with latent risk factors $\Theta_i \sim \text{Gamma}(\alpha, \beta)$, and conditionally iid annual claim counts

$$Z_{ij} | \Theta_i \sim \text{Poisson}(\Theta_i),$$

for $j = 1, \dots, n$.

- (b) Estimate m , EPV, and VHM by the method-of-moments estimators. Compute

$$K = \frac{\text{EPV}}{\text{VHM}}, \quad Z = \frac{n}{n + K}, \quad \hat{\mu}_i = Z \bar{Z}_i + (1 - Z)m.$$

- (c) Simulate one future claim count Z_i^* for each policyholder and compare test-set mean squared error of $\hat{\mu}_i$ against that with $\bar{Z}_i$ and with the collective premium m . Compare the three choices of n and comment on when credibility gives the largest gain.

Chapter 5

Claims reserving

An insurance claim is usually triggered by an event such as an accident. The time between the *accident date* and registration at the insurer is the *reporting delay*, and it can vary substantially across lines of business. For instance, if a vehicle accident occurs, the insured is often able to report the accident within a few weeks, while claims such as rot damage on a property or cancer induced by asbestos might take years or even decades before being noticed and hence reported.

As a consequence, legislation requires insurers to hold sufficient capital not only to cover claims that have been reported and settled, but also to cover all claims that have occurred in the period of coverage, including those that have not been reported or settled yet. The purpose of reserving is to estimate sufficient capital to cover unreported and unsettled claims, which make up the *outstanding loss liabilities*.

From a statistical perspective, the analysis of insurance claims is challenging. Claims data are inherently right-censored due to reporting delays, and their underlying distributions may vary over time because of yearly fluctuations (such as flood occurrences), calendar-time trends (such as inflation), or changes in reporting behaviour (for instance, through improved medical screening). Consequently, the right-censored likelihood methods developed in the previous chapter are not ideally suited for handling insurance claims data.

In this chapter, we introduce notation tailored to claims data and present techniques for estimating the outstanding loss liabilities. We start with the classical triangle framework and basic reserving methods, and then turn to more structured stochastic models used for uncertainty quantification. Throughout, the target quantities stay the same, and what changes from section to section is mainly how uncertainty around them is modelled.

5.1 Basic claims reserving

We first introduce the standard triangle notation and the main reserving quantities, and then build the estimators on that structure.

5.1.1 Introduction

Suppose an accident occurs at time T_1 within the coverage period (also known as the *risk period*). At time T_2 , which may or may not be in the risk period, the claim is reported to the insurer. Claims between the accident and reporting date are called *Incurred But Not Yet Reported* (IBNR). If the insured is eligible for coverage, claim payments are made at some point after the reporting date, until the settlement date, T_3 . Claims between reporting and settlement date are called *Reported But Not Settled* (RBNS).

In order to make statistical statements on the outstanding liabilities, the insurer constructs homogeneous groups of, e.g. insurance types, to which a law of large numbers can more realistically apply.

Claim triangles and reserves

The homogeneous groups (risk classes) are subdivided according to accident date, since claims that occur in the same accident period can often be triggered by the same underlying external risk factors, such as weather conditions, economic cycles, or pandemics.

Reserves, which are defined below, are in practice sometimes calculated monthly, but for the sake of discussion we consider determining the reserves for each accounting year. To this end we introduce the classical claims reserving notation for fixed $i \in \mathbb{N}$ and $j \in \mathbb{N}_0$ as

$$X_{i,j} = \text{all payments related to claims with accident year } i, \\ \text{paid in accounting year } i + j.$$

This notation motivates a further subdivision according to the payment delay, denoted by $j \in \mathbb{N}_0$ and called the *development year*.

We assume for simplicity that all claims are fully developed after $J \in \mathbb{N}$ development years, i.e. for all $i \in \mathbb{N}$, it holds that $X_{i,j} = 0$ for $j > J$. Note that J , the maximal settlement delay, varies between risk groups. For “short-tailed” claims such as comprehensive vehicle insurance, the maximal settlement delay is often much shorter than for “long-tailed” claims such as personal accident insurance or, liability insurance.

This notation allows for the visual representation of the claims that have occurred in accident year $1 \leq i \leq I$ (vertical axis), paid out in development year $0 \leq j \leq J$ (horizontal axis), and usually depicted by a triangle structure.

Notice that with this representation, claims for a given accounting year $i+j \in \{1, \dots, I+J\}$ are available on the diagonal. All available information, displayed in the top left trapezoid, is denoted by $\mathcal{D}_I$. More generally, we can fix $I \in \mathbb{N}$, and consider the information at time $t \geq I$, which is given by

$$\mathcal{D}_t = \{X_{i,j} : i+j \leq t, 1 \leq i \leq I, 0 \leq j \leq J\}.$$

The goal of reserving, and thus of this chapter, is to provide a best estimate of the outstanding loss liabilities (reserves), represented by $\mathcal{D}_t^e$, using the available information $\mathcal{D}_t$. We stress that estimated reserves include both IBNR and RBNS.

Accident year i	development year j					
	0	1	...	j	...	J
1	$X_{1,0}$	$X_{1,1}$		...		$X_{1,J}$
$\vdots$	$\vdots$	$\vdots$				$\vdots$
$I - J$	$X_{I-J,0}$	$X_{I-J,1}$		...		$X_{I-J,J}$
$\vdots$						
i	$\vdots$	observations $\mathcal{D}_I$				$\vdots$
$\vdots$						
$I - 1$						
I	$X_{I,0}$			...		

Figure 5.1: Visual representation of a claims reserving triangle.

Definition 5.1 (Claims reserves) At time $t \in \mathbb{N}$, the nominal best-estimate reserve of the outstanding loss liabilities, given the information $\mathcal{D}_t$, is defined by

$$\mathcal{R}^{(t)} = \sum_{i+j>t} \mathbb{E}[X_{i,j} | \mathcal{D}_t] = \sum_{i=t-J+1}^I \sum_{j=t-i+1}^J \mathbb{E}[X_{i,j} | \mathcal{D}_t],$$

for $t \geq I$.

In practice, one may additionally consider discounted reserves adjusted by a risk margin; however, this is not pursued here. Rather than the incremental payments $(X_{i,j})$, estimation routines can equivalently be given in terms of the *cumulative payments*, which are defined as

$$C_{i,j} = \sum_{k=0}^j X_{i,k}, \quad 1 \leq i \leq I, \quad 0 \leq j \leq J,$$

and where we call $C_{i,J}$ the *ultimate claim size* of accident year i .

5.1.2 Chain-ladder algorithm

As seen from the above trapezoid, our subdivision of claims yields an intuitive, albeit somewhat unusual, representation of claim payments, where time runs along three axes (horizontal, vertical, diagonal). For this reason, there are no immediate limit theorems that immediately provide limiting distributional results in a coherent way for all three dimensions, and for a chosen estimator.

Instead, classical reserving suggests a pragmatic approach based on heuristic extrapolation, rather than strict statistical modelling. Hence the name “chain-ladder algorithm”.

The chain-ladder rationale is as follows. Claims in accident years $1 \leq i \leq I$ develop similarly. For the cumulative claims we suggest

$$C_{i,j+1} = f_j C_{i,j},$$

for so-called *chain-ladder factors* $f_j > 0$.

Suppose that for accident year $i > t - J$, we have observed $C_{i,t-i}$ (which is on the diagonal). Then the chain-ladder rationale yields that the ultimate claim size of accident year i is given by $C_{i,J} = C_{i,t-i} \prod_{j=t-i}^{J-1} f_j$. Consequently, the ultimate claim size can be estimated as soon as we have obtained estimates of the (f_j) 's. In the simple setting where $t = I$, we simply compute

$$\hat{f}_j = \frac{\sum_{i=1}^{I-j-1} C_{i,j+1}}{\sum_{i=1}^{I-j-1} C_{i,j}},$$

which is a weighted average of individual development ratios $C_{i,j+1}/C_{i,j}$ with weights proportional to the cumulative exposure $C_{i,j}$. While in the slightly more general setting where $t \geq I$, we have

$$\hat{f}_j^{(t)} = \frac{\sum_{i=1}^{(t-j-1) \wedge I} C_{i,j+1}}{\sum_{i=1}^{(t-j-1) \wedge I} C_{i,j}} = \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sum_{i=1}^{(t-j-1) \wedge I} C_{i,j}} \frac{C_{i,j+1}}{C_{i,j}}. \quad (5.1)$$

Using this algorithm we obtain the chain-ladder estimate, at time $t \geq I$, of the ultimate claim size $C_{i,J}$, by

$$\hat{C}_{i,J}^{\text{CL}(t)} = C_{i,t-i} \prod_{j=t-i}^{J-1} \hat{f}_j^{(t)}, \quad t \geq I \geq i > t - J,$$

and the corresponding chain-ladder reserve

$$\mathcal{R}_i^{\text{CL}(t)} = \hat{C}_{i,J}^{\text{CL}(t)} - C_{i,t-i} = C_{i,t-i} \left(\prod_{j=t-i}^{J-1} \hat{f}_j^{(t)} - 1 \right).$$

Summing over all risk periods that are not fully developed yields the total chain-ladder reserve:

$$\mathcal{R}^{\text{CL}(t)} = \sum_{i=t-J+1}^I \mathcal{R}_i^{\text{CL}(t)}.$$

5.1.3 Bornhuetter–Ferguson algorithm

Another popular claims reserving algorithm is the Bornhuetter–Ferguson (BF) algorithm, which incorporates *prior information* regarding the expected ultimate claim size. More specifically, let $\hat{\mu}_i$ be some prior guess of $\mathbb{E}[C_{i,J}]$, and let $(\eta_j)_{j=0}^J$ be the proportions of $\hat{\mu}_i$ paid out at each development year.

The Bornhuetter–Ferguson rationale is as follows. Claims in accident years $1 \leq i \leq I$ develop similarly. For the incremental claims, we suggest

$$X_{i,j} = \eta_j \hat{\mu}_i,$$

for the *claim-development proportions* $(\eta_j)_{j=0}^J$, where $\sum_{j=0}^J \eta_j = 1$, and prior information $\hat{\mu}_i$.

The claim-development proportions are estimated using the chain-ladder factors; if we have estimated the factor (product of chain-ladder factors) to enlarge the last observed cumulative claim size, then we have an estimate of the proportion between the last observed cumulative claim size and the expected ultimate. For instance, suppose for accident year i that the last observed cumulative claim is $C_{i,t-i}$ and that $\prod_{j=t-i}^{J-1} \hat{f}_j^{(t)} = 1.25$. In this case, the chain-ladder algorithm yields the ultimate $\hat{C}_{i,J}^{\text{CL}(t)} = 1.25 \cdot C_{i,t-i}$, from which we might argue that the current claim size $C_{i,t-i}$ has only developed $1/1.25 = 80\%$. At time $t \geq I$, we formally let

$$\hat{\kappa}_j^{\text{CL}(t)} = \prod_{l=j}^{J-1} \frac{1}{\hat{f}_l^{(t)}} = \frac{\prod_{l=0}^{j-1} \hat{f}_l^{(t)}}{\prod_{l=0}^{J-1} \hat{f}_l^{(t)}}, \quad 0 \leq j \leq J, \quad \hat{\kappa}_J^{\text{CL}(t)} = 1,$$

which is related to claim-development proportions by $\hat{\eta}_0^{\text{CL}(t)} = \hat{\kappa}_0^{\text{CL}(t)}$ and $\hat{\eta}_j^{\text{CL}(t)} = \hat{\kappa}_j^{\text{CL}(t)} - \hat{\kappa}_{j-1}^{\text{CL}(t)}$ for $1 \leq j \leq J$.

The BF ultimate claim size estimate for accident year i , is then given by

$$\hat{C}_{i,J}^{\text{BF}(t)} = C_{i,t-i} + \left(\sum_{j=t-i+1}^J \hat{\eta}_j^{\text{CL}(t)} \right) \hat{\mu}_i = C_{i,t-i} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i,$$

with corresponding reserve

$$\mathcal{R}_i^{\text{BF}(t)} = \hat{C}_{i,J}^{\text{BF}(t)} - C_{i,t-i} = (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i.$$

Since the BF algorithm incorporates the chain-ladder factors, one might suspect the ultimate claim size estimates of the chain-ladder and of BF to be related. Indeed, we have that

$$\begin{aligned} \hat{C}_{i,J}^{\text{BF}(t)} &= C_{i,t-i} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i \\ &= \hat{\kappa}_{t-i}^{\text{CL}(t)} \left(C_{i,t-i} \frac{1}{\hat{\kappa}_{t-i}^{\text{CL}(t)}} \right) + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i \\ &= \hat{\kappa}_{t-i}^{\text{CL}(t)} \left(C_{i,t-i} \prod_{j=t-i}^{J-1} \hat{f}_j^{(t)} \right) + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i \\ &= \hat{\kappa}_{t-i}^{\text{CL}(t)} \hat{C}_{i,J}^{\text{CL}(t)} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i, \end{aligned} \tag{5.2}$$

from which we can see that the BF ultimate weighs the chain-ladder ultimate and the prior ultimate according to how developed the i -th accident year is. As t grows and $t-i$ approaches J , then $\hat{\kappa}_{t-i}^{\text{CL}(t)}$ approaches 1. This means that the more information available regarding accident year i , the more weight is put on the (data-driven) chain-ladder ultimate, as opposed to (data-independent) prior, and vice versa.

5.1.4 Stochastic claims reserving

Up until now, we have only discussed methods for estimating the nominal outstanding loss liabilities. Another important task is to quantify the uncertainty

of such estimated quantities, and for this purpose we need more than just algorithms. In this section, we consider claim reserving *models*. The objective is therefore unchanged, but we now add a probabilistic structure that yields prediction error measures.

As is discussed in the subsequent chapter, there are many candidate loss functions for computing prediction error; however, in this chapter we consider the mean squared error, since it can be calculated explicitly in many examples. Suppose at time $t \geq I$, for accident year i with $I \geq i > t - J$, we have estimated the ultimate claim size $C_{i,J}$ with some algorithm or model, denoted $\hat{C}_{i,J}^{(t)}$, which is assumed $\mathcal{D}_t$ -measurable. The mean squared error of accident year i , conditional on the available information at time t , is then given by

$$\mathbb{E} \left[(C_{i,J} - \hat{C}_{i,J}^{(t)})^2 \mid \mathcal{D}_t \right] = \text{Var} (C_{i,J} \mid \mathcal{D}_t) + \left(\mathbb{E} [C_{i,J} \mid \mathcal{D}_t] - \hat{C}_{i,J}^{(t)} \right)^2, \quad (5.3)$$

where the right-hand side is a bias-variance decomposition which is straightforward to verify. This error quantifies the total prediction error of the entire *run-off*, defined as the difference between the estimator and actual realisation of the ultimate.

Mack's chain-ladder model

Many reserving actuaries use the chain-ladder algorithm to determine reserves, since it is easy to implement, has the straightforward interpretation through linear regression with no intercept at every development year, and it has a robust performance. The first task is to introduce a simple model that reproduces, or embeds, the chain-ladder ultimates while allowing for error quantification. To this end, we introduce the celebrated Mack Model.

The model assumptions of Mack's CL model are as follows. The cumulative claim payments $(C_{i,j})_{j=0}^J$ are independent for each $1 \leq i \leq I$, and there exist positive constants (f_j) and (s_j) for $0 \leq j \leq J-1$ such that

$$\mathbb{E}[C_{i,j+1} \mid C_{i,j}, \dots, C_{i,0}] = f_j C_{i,j}, \quad (5.4)$$

$$\text{Var} (C_{i,j+1} \mid C_{i,j}, \dots, C_{i,0}) = s_j^2 C_{i,j}.$$

The first objective is to examine the conditional expectation and variance of the ultimate claim sizes. This is done in the following result.

Lemma 5.1 (Conditional mean and variance of the ultimate) *Assume the model assumptions of Mack's CL model. For $t \geq I \geq i > t - J$ it holds that*

$$\begin{aligned} \mathbb{E} [C_{i,J} \mid \mathcal{D}_t] &= C_{i,t-i} \prod_{j=t-i}^{J-1} f_j \\ \text{Var} (C_{i,J} \mid \mathcal{D}_t) &= \mathbb{E} [C_{i,J} \mid \mathcal{D}_t]^2 \sum_{j=t-i}^{J-1} \frac{s_j^2 / f_j^2}{\mathbb{E} [C_{i,j} \mid \mathcal{D}_t]} \end{aligned} \quad (5.5)$$

Proof. We iterate conditional expectations along development years and then apply the law of total variance. Repeated use of the tower property yields that

$$\begin{aligned}\mathbb{E}[C_{i,J} | \mathcal{D}_t] &= \mathbb{E}[\mathbb{E}[C_{i,J} | C_{i,J-1}, \dots, C_{i,0}] | \mathcal{D}_t] \\ &= \mathbb{E}[C_{i,J-1} | \mathcal{D}_t] f_{J-1} = C_{i,t-i} \prod_{j=t-i}^{J-1} f_j.\end{aligned}$$

Turning to the variance calculation,

$$\begin{aligned}\text{Var}(C_{i,J} | \mathcal{D}_t) &= \text{Var}(\mathbb{E}[C_{i,J} | C_{i,J-1}, \dots, C_{i,0}] | \mathcal{D}_t) \\ &\quad + \mathbb{E}[\text{Var}(C_{i,J} | C_{i,J-1}, \dots, C_{i,0}) | \mathcal{D}_t] \\ &= s_{J-1}^2 \mathbb{E}[C_{i,J-1} | \mathcal{D}_t] + f_{J-1}^2 \text{Var}(C_{i,J-1} | \mathcal{D}_t).\end{aligned}$$

We iteratively expand the term $\text{Var}(C_{i,J-1} | \mathcal{D}_t)$ on the right-hand side to obtain

$$\begin{aligned}\text{Var}(C_{i,J} | \mathcal{D}_t) &= s_{J-1}^2 \mathbb{E}[C_{i,J-1} | \mathcal{D}_t] \\ &\quad + f_{J-1}^2 \{s_{J-2}^2 \mathbb{E}[C_{i,J-2} | \mathcal{D}_t] + f_{J-2}^2 \text{Var}(C_{i,J-2} | \mathcal{D}_t)\} \\ &= \sum_{j=t-i}^{J-1} s_j^2 \mathbb{E}[C_{i,j} | \mathcal{D}_t] \prod_{l=j+1}^{J-1} f_l^2 \\ &= \sum_{j=t-i}^{J-1} s_j^2 \frac{(C_{i,t-i} \prod_{l=t-i}^{j-1} f_l)^2}{\mathbb{E}[C_{i,j} | \mathcal{D}_t]} \prod_{l=j+1}^{J-1} f_l^2 \\ &= \sum_{j=t-i}^{J-1} \frac{s_j^2}{f_j^2} \frac{(C_{i,t-i} \prod_{l=t-i}^{j-1} f_l)^2}{\mathbb{E}[C_{i,j} | \mathcal{D}_t]} \\ &= \mathbb{E}[C_{i,J} | \mathcal{D}_t]^2 \sum_{j=t-i}^{J-1} \frac{s_j^2 / f_j^2}{\mathbb{E}[C_{i,j} | \mathcal{D}_t]},\end{aligned}$$

as desired. $\square$

To predict and quantify uncertainty, we need to provide estimates of the chain-ladder factors f_j as well as the variance parameters s_j^2 . The chain-ladder factors are estimated as in equation (5.1), that is

$$\hat{f}_j^{(t)} = \frac{\sum_{i=1}^{(t-j-1) \wedge I} C_{i,j+1}}{\sum_{i=1}^{(t-j-1) \wedge I} C_{i,j}} = \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sum_{i=1}^{(t-j-1) \wedge I} C_{i,j}} \frac{C_{i,j+1}}{C_{i,j}},$$

while the variance parameters s_j^2 are estimated, for $j < t-2$, by

$$\hat{s}_j^2 = \frac{1}{((t-j-1) \wedge I) - 1} \sum_{l=1}^{(t-j-1) \wedge I} C_{l,j} \left(\frac{C_{l,j+1}}{C_{l,j}} - \hat{f}_j^{(t)} \right)^2, \quad (5.6)$$

which can be shown to be an unbiased estimator. When $t = I = J + 1$ we have observed a full triangle of observations and thus we are unable to estimate

s_{J-1}^2 as in (5.6), since only one observation of the last development period is available. In this case, Mack proposed to use the following amendment:

$$\hat{s}_{J-1}^2 = \min \left\{ \hat{s}_{J-3}^2, \frac{\hat{s}_{J-2}^4}{\hat{s}_{J-3}^2} \right\}.$$

We note from the above results that Mack's chain-ladder indeed reproduces the same ultimate claim size estimates for cumulative claims as the chain-ladder, that is Mack's chain-ladder claim size estimates are again

$$\hat{C}_{i,J}^{\text{CL}(t)} = C_{i,t-i} \prod_{j=t-i}^{J-1} \hat{f}_j^{(t)}, \quad t \geq I \geq i > t - J.$$

In addition, if we plug in the estimators of f_j and s_j into the variance expression (5.5), Mack's model suggests the following estimator of the conditional variance:

$$\widehat{\text{Var}}(C_{i,J} | \mathcal{D}_t) = \left(\hat{C}_{i,J}^{\text{CL}(t)} \right)^2 \sum_{j=t-i}^{J-1} \frac{\hat{s}_j^2 / (\hat{f}_j^{(t)})^2}{\hat{C}_{i,j}^{\text{CL}(t)}}, \quad (5.7)$$

which is a crucial component of the conditional mean squared error decomposition (5.3). We state the following lemma, which is used frequently in the subsections below.

Lemma 5.2 (Unbiasedness and optimality of CL factors) *Assume the model assumptions of Mack's CL model. The chain-ladder factor estimators $\hat{f}_j^{(t)}$ are unbiased, uncorrelated and optimal in the sense that*

$$\begin{aligned} & \text{Var}(\hat{f}_j^{(t)} | C_{(t-j-1) \wedge I, j}, \dots, C_{1,j}) \\ &= \min_{\gamma_i \in \mathbb{R}: \sum_{i=1}^{(t-j-1) \wedge I} \gamma_i = 1} \left\{ \text{Var} \left(\sum_{i=1}^{(t-j-1) \wedge I} \gamma_i \frac{C_{i,j+1}}{C_{i,j}} \mid C_{(t-j-1) \wedge I, j}, \dots, C_{1,j} \right) \right\}. \end{aligned}$$

The conditional variance is given by

$$\text{Var}(\hat{f}_j^{(t)} | C_{(t-j-1) \wedge I, j}, \dots, C_{1,j}) = \frac{s_j^2}{\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}}.$$

Proof. Left as an exercise to the reader. $\square$

In (5.7) we proposed an estimator of the conditional variance of the ultimate claim, which is the first term in the conditional mean squared error decomposition (5.3). It remains to propose a suitable estimator of the second term of (5.3), which is the squared bias.

Conditional mean squared error of Mack's model

The estimator of the conditional variance, given by (5.7), was simply obtained by replacing the unknown parameters with their corresponding estimators. When

examining the squared bias term, such an approach clearly fails, and we therefore take a slightly different route. Namely, we condition on a subset of $\mathcal{D}_t$ that is "truncated" at development year $0 \leq k \leq J$:

$$\mathcal{B}_{t,k} = \{C_{i,j} : i + j \leq t, 1 \leq i \leq I, 0 \leq j \leq k\} \subset \mathcal{D}_t.$$

This subset is natural to consider since

$$\mathbb{E}[C_{i,k+1} | \mathcal{B}_{t,k}] = \mathbb{E}[C_{i,k+1} | C_{i,k}, \dots, C_{i,0}] = f_k C_{i,k},$$

due to (5.4), and the assumption that claims are independent across accident years. In addition

$$\mathbb{E}[\hat{f}_k^{(t)} | \mathcal{B}_{t,k}] = \sum_{i=1}^{(t-k-1) \wedge I} \frac{\mathbb{E}[C_{i,k+1} | \mathcal{B}_{t,k}]}{\sum_{i=1}^{(t-k-1) \wedge I} C_{i,k}} = f_k.$$

Note that $\hat{f}_j^{(t)}$ is $\mathcal{B}_{t,j+1}$ -measurable for each j . To examine the squared bias term

$$(\mathbb{E}[C_{i,J} | \mathcal{D}_t] - \hat{C}_{i,J}^{(t)})^2 = C_{i,t-i}^2 \left(\prod_{j=t-i}^{J-1} f_j - \prod_{j=t-i}^{J-1} \hat{f}_j^{(t)} \right)^2, \quad (5.8)$$

we introduce the following notation

$$S_j^{(t)} = \hat{f}_{t-i}^{(t)} \cdots \hat{f}_{j-1}^{(t)} (f_j - \hat{f}_j^{(t)}) f_{j+1} \cdots f_{J-1}.$$

Then the parentheses on the right-hand side of (5.8) can be written as a telescoping sum

$$\left(\prod_{j=t-i}^{J-1} f_j - \prod_{j=t-i}^{J-1} \hat{f}_j^{(t)} \right)^2 = \left(\sum_{j=t-i}^{J-1} S_j^{(t)} \right)^2,$$

and in turn we observe that

$$(\mathbb{E}[C_{i,J} | \mathcal{D}_t] - \hat{C}_{i,J}^{(t)})^2 = C_{i,t-i}^2 \left(\sum_{j=t-i}^{J-1} (S_j^{(t)})^2 + 2 \sum_{j < l} S_j^{(t)} S_l^{(t)} \right), \quad (5.9)$$

where the latter sum is computed over $t-i \leq j < l \leq J-1$.

We approximate $(S_j^{(t)})^2$ with $\mathbb{E}[(S_j^{(t)})^2 | \mathcal{B}_{t,j}]$, for each j , and $S_j^{(t)} S_l^{(t)}$ with $\mathbb{E}[S_j^{(t)} S_l^{(t)} | \mathcal{B}_{t,l}]$, for each l . Note then that

$$\begin{aligned} \mathbb{E}[S_j^{(t)} | \mathcal{B}_{t,j}] &= \mathbb{E}[\hat{f}_{t-i}^{(t)} \cdots \hat{f}_{j-1}^{(t)} (f_j - \hat{f}_j^{(t)}) f_{j+1} \cdots f_{J-1} | \mathcal{B}_{t,j}] \\ &= \hat{f}_{t-i}^{(t)} \cdots \hat{f}_{j-1}^{(t)} (f_j - \mathbb{E}[\hat{f}_j^{(t)} | \mathcal{B}_{t,j}]) f_{j+1} \cdots f_{J-1} = 0. \end{aligned}$$

For $j < l$, $S_j^{(t)}$ is $\mathcal{B}_{t,l}$ measurable and hence

$$\mathbb{E}[S_j^{(t)} S_l^{(t)} | \mathcal{B}_{t,l}] = S_j^{(t)} \mathbb{E}[S_l^{(t)} | \mathcal{B}_{t,l}] = 0.$$

For this reason the last sum of (5.9) is simply estimated by zero. As for the corresponding first term, we compute

$$\begin{aligned}\mathbb{E}[(S_j^{(t)})^2 | \mathcal{B}_{t,j}] &= (\hat{f}_{t-i}^{(t)})^2 \dots (\hat{f}_{j-1}^{(t)})^2 \mathbb{E}[(f_j - \hat{f}_j^{(t)})^2 | \mathcal{B}_{t,j}] f_{j+1}^2 \dots f_{J-1}^2 \\ &= (\hat{f}_{t-i}^{(t)})^2 \dots (\hat{f}_{j-1}^{(t)})^2 \text{Var}(\hat{f}_j^{(t)} | C_{(t-j-1) \wedge I, j}, \dots, C_{1,j}) f_{j+1}^2 \dots f_{J-1}^2 \\ &= (\hat{f}_{t-i}^{(t)})^2 \dots (\hat{f}_{j-1}^{(t)})^2 \frac{s_j^2}{\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}} f_{j+1}^2 \dots f_{J-1}^2.\end{aligned}$$

Consequently, the modified version of the squared bias term (5.9) equals

$$\begin{aligned}C_{i,t-i}^2 \left(\sum_{j=t-i}^{J-1} \mathbb{E}[(S_j^{(t)})^2 | \mathcal{B}_{t,j}] + 2 \sum_{j < l} \mathbb{E}[S_j^{(t)} S_l^{(t)} | \mathcal{B}_{t,l}] \right) \\ = C_{i,t-i}^2 \sum_{j=t-i}^{J-1} (\hat{f}_{t-i}^{(t)})^2 \dots (\hat{f}_{j-1}^{(t)})^2 \frac{s_j^2}{\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}} f_{j+1}^2 \dots f_{J-1}^2,\end{aligned}$$

which we finally estimate with

$$\begin{aligned}C_{i,t-i}^2 \sum_{j=t-i}^{J-1} (\hat{f}_{t-i}^{(t)})^2 \dots (\hat{f}_{j-1}^{(t)})^2 \frac{\hat{s}_j^2}{\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}} (\hat{f}_{j+1}^{(t)})^2 \dots (\hat{f}_{J-1}^{(t)})^2 \\ = (\hat{C}_{i,J}^{\text{CL}(t)})^2 \sum_{j=t-i}^{J-1} \frac{\hat{s}_j^2 / (\hat{f}_j^{(t)})^2}{\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}}.\end{aligned}$$

We have now obtained, within Mack's chain-ladder model assumptions, estimators for both the conditional variance and the squared bias of the conditional mean squared error (5.3). Consequently, Mack's chain-ladder model suggests the following estimator for conditional mean squared error

$$\begin{aligned}\mathbb{E}[(C_{i,J} - \hat{C}_{i,J}^{(t)})^2 | \mathcal{D}_t] &= \text{Var}(C_{i,J} | \mathcal{D}_t) + (\mathbb{E}[C_{i,J} | \mathcal{D}_t] - \hat{C}_{i,J}^{(t)})^2 \\ &\approx (\hat{C}_{i,J}^{\text{CL}(t)})^2 \sum_{j=t-i}^{J-1} \left[\frac{\hat{s}_j^2 / (\hat{f}_j^{(t)})^2}{\hat{C}_{i,j}^{\text{CL}(t)}} + \frac{\hat{s}_j^2 / (\hat{f}_j^{(t)})^2}{\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}} \right].\end{aligned}$$

5.1.5 Tail development factors

Above, we assumed that all claims are fully developed after J development years. However, in practice, we might encounter claims that develop beyond a fixed year J . This phenomenon typically occurs in one of the following two cases.

1. Suppose the insurance company decides to sell a new product, such as cyber insurance or pregnancy insurance, and that this product has been available for 3-4 years. It is likely that the existing claims develop further, but the reserving actuary does not have data on the entire claim development, and thus has to rely on a different technique for estimating the ultimate claim sizes than those presented above.

2. Though less common than in earlier years, insurance companies sometimes offer full coverage on products with very “long-tailed” claim development. Examples are workers’ compensation, child insurance that covers until adulthood, and vehicle liability insurance. For such products, it can be difficult to correctly choose the maximal development year J .

In any case, additional reserves for claims that occur after the chosen J should be estimated. The most commonly used approach, that fits well into the chain-ladder setup, is to introduce a so-called *tail development factor* (or simply tail factor) τ_J , which is multiplied by the former (non-tail-corrected) ultimate claim size estimate to obtain

$$\hat{C}_{i,J+}^{\text{CL}(t)} = C_{i,t-i} \left(\prod_{j=t-i}^{J-1} \hat{f}_j^{(t)} \right) \hat{\tau}_J^{(t)} = \hat{C}_{i,J}^{\text{CL}(t)} \hat{\tau}_J^{(t)}, \quad t \geq I \geq i > t - J,$$

which is the estimated ultimate, but now corrected for development beyond J (hence the notation using $J+$). In practice, tail factors can be obtained by extrapolating a parametric trend in the late development years, for instance by fitting a log-linear trend to $(f_j - 1)$ or to cumulative development. The choice of extrapolation can materially affect the tail reserve and its variance, so it should be checked against domain knowledge. We stress that the setup where claims can occur after the maximal development period J contradicts the setup presented above, where all claims are fully developed at development year J . Thus, introducing a tail development factor is not a natural theoretical extension of the previous subsection, but instead a pragmatic adjustment to obtain sufficient reserves, which is crucial for the solvency of the insurance company. Below, we propose two common approaches for estimating the tail development factor.

Extrapolation of chain-ladder factors

The first approach is to fit a parametric survival function on the upper chain-ladder factors, from which we can extrapolate the claim development. The choice of parametric family is highly dependent on the product, respectively portfolio, considered, and in practice, one can fit several models and choose one according to some relevant score, such as mean squared error.

In this section, we present the simplest parametric extrapolation scheme, which assumes exponential decay of the chain-ladder factors and fitting an exponential tail thereafter. This is a relevant example for the first motivating example of this section, but not for the second example, since there we expect tails heavier than exponential.

Exponential decay corresponds to a linear regression on the log scale. As mentioned, the fitting is done on the upper chain-ladder factors $(\hat{f}_j)_{j=j_0}^{J-1}$, for appropriately chosen truncation index $0 \leq j_0 \leq J - 2$. More specifically, we fit a linear regression to $\log(\hat{f}_j^{(t)})$, with covariates $j_0 \leq j \leq J - 1$ using least squares, i.e. we obtain

$$(a^*, b^*) = \underset{(a,b)}{\operatorname{argmin}} \left\{ \sum_{j=j_0}^{J-1} (\log(\hat{f}_j^{(t)}) - (a + bj))^2 \right\}.$$

The fitted values (a^*, b^*) allow for extrapolation since we obtain chain-ladder factor estimates for $j \geq J$

$$\hat{f}_j^* = e^{a^* + b^* j} \vee 1 = e^{(a^* + b^* j) \vee 0}.$$

As the estimated factors should be positive and decreasing, we have that $b^* < 0$. The ultimate claim size of accident year i is then given by

$$\hat{C}_{i,J+}^{\text{CL}(t)} = C_{i,t-i} \left(\prod_{j=t-i}^{J-1} \hat{f}_j^{(t)} \right) \left(\prod_{j=J}^{\infty} \hat{f}_j^* \right) = \hat{C}_{i,J}^{\text{CL}(t)} \hat{\tau}_J^{(t)},$$

where we may use the approximation

$$\hat{\tau}_J^{(t)} = \prod_{j=J}^{\infty} \hat{f}_j^* = \exp \left\{ \sum_{j=J}^{\infty} ((a^* + b^* j) \vee 0) \right\} \approx \exp \left\{ -\frac{1}{2} (a^* + Jb^*) \left(\frac{a^*}{b^*} + J \right) \right\}.$$

The approximation above was done using the approximation of an arithmetic sequence. We now proceed to the second approach.

Tail development factors via incurred claims

In the introduction of this chapter we assumed that the insurance company pays the customer between the reporting date and settlement date, provided the customer is granted coverage. The process between the reporting date and settlement date can be specified as follows.

1. After reporting the claim, the insurance company assesses whether the customer is eligible for coverage. If so, a case worker or machine learning algorithm guesses or estimates the total amount of (future) payments and puts aside this amount, which is called the *case reserve*.
2. The payments from the company to the insured are withdrawn from the case reserve, which may need to be adjusted as further information on the accident is obtained.
3. The case reserve should eventually converge to zero, corresponding to the time when the claim is settled.

As above, we can aggregate a homogeneous portfolio, which yields a triangle of case reserves.

Definition 5.2 (Incurred cumulative claim sizes) Denote by $\text{CR}_{i,j}$ the cumulative case reserve of accident year i and development year j . The *incurred cumulative claim size*, denoted by $C_{i,j}^{\mathcal{I}}$, of accident year i and development year j , is the sum of the cumulative payment and case reserve, i.e.

$$C_{i,j}^{\mathcal{I}} = C_{i,j} + \text{CR}_{i,j}, \quad 1 \leq i \leq I, \quad 0 \leq j \leq J,$$

If we momentarily assume that there are no IBNR claims for accident year i , and that the case worker or machine learning algorithm is predicting the total

amount of payments appropriately, then it should hold that $C_{i,J} \approx C_{i,j}^{\mathcal{I}}$ for all j . Indeed, a sufficient case reserve was put aside to meet the outstanding liabilities to the customer, and thus, as j increases, the case reserve decreases according to the increase in $C_{i,j}$. In particular, once the claim is finally settled, and all case reserves have been paid out, then the cumulative claim size and the incurred cumulative claim size coincide.

However, by incorporating the incurred cumulative claim sizes, we might be able to utilise the information held in the case reserves to enhance the prediction of the ultimate claim size. We assume that after some development year $j_{\mathcal{I}}$, there are no more IBNR claims. Introduce the *incurred-paid ratio* of accident year i and development year $j \geq j_{\mathcal{I}}$

$$Q_{i,j} = \frac{C_{i,j}^{\mathcal{I}}}{C_{i,j}}.$$

This fraction provides information on the ratio between the amount that has been guessed or estimated to be paid out and the actual amount paid. If the claim is fully settled, this fraction, as mentioned above, should be 1. Below, we assume $I = t$. We then construct the following estimator for the tail development factor

$$\hat{\tau}_J^{(I)} = \frac{\sum_{i=1}^{I-J} C_{i,J}^{\mathcal{I}}}{\sum_{i=1}^{I-J} C_{i,J}} = \sum_{i=1}^{I-J} \frac{C_{i,J}}{\sum_{i=1}^{I-J} C_{i,J}} Q_{i,J}, \quad (5.10)$$

which yields the ultimate claim prediction at time I of

$$\hat{C}_{i,J+}^{\text{CL}(I)} = C_{i,I-i} \left(\prod_{j=I-i}^{J-1} \hat{f}_j^{(I)} \right) \hat{\tau}_J^{(I)}. \quad (5.11)$$

This estimation technique has noticeable weaknesses, of which we mention the important ones.

1. The method is based on the assumption that $C_{i,J}^{\mathcal{I}}$ of the last development period is the best estimate of the ultimate cumulative claim size; however, the case reserve is often determined by a case worker who does not have the reserving task in mind.
2. To reduce prediction error we require, as a rule of thumb, that the sums in (5.10) include at least a few terms, say we may desire that $I \geq J + 3$.

As in (5.6), we estimate the variance parameter of the tail development factor, when $I \geq J + 3$, by

$$\hat{s}_{J+}^2 = \frac{1}{I - J - 1} \sum_{i=1}^{I-J} C_{i,J} (Q_{i,J} - \hat{\tau}_J^{(I)})^2.$$

Claims after the maximal development period J contradict the model assumptions of Mack's model. If we nonetheless assume that the assumptions of Mack's model can be extended to include the tail development factor, we obtain the

following heuristic conditional mean squared error at $t = I > J + 1$ for accident year i :

$$\begin{aligned} & \mathbb{E}\left[(C_{i,J+} - \hat{C}_{i,J+}^{(I)})^2 \mid \mathcal{D}_I\right] \\ & \approx (\hat{C}_{i,J+}^{\text{CL}(I)})^2 \sum_{j=I-i}^{J-1} \left(\frac{\hat{s}_j^2 / (\hat{f}_j^{(I)})^2}{\hat{C}_{i,j}^{\text{CL}(I)}} + \frac{\hat{s}_j^2 / (\hat{f}_j^{(I)})^2}{\sum_{l=1}^{I-j-1} C_{l,j}} \right) \\ & \quad + (\hat{C}_{i,J+}^{\text{CL}(I)})^2 \left(\frac{\hat{s}_{J+}^2 / (\hat{\tau}_J^{(I)})^2}{\hat{C}_{i,J}^{\text{CL}(I)}} + \frac{\hat{s}_{J+}^2 / (\hat{\tau}_J^{(I)})^2}{\sum_{l=1}^{I-J} C_{l,J}} \right), \end{aligned}$$

where the first term corresponds to (5.6) and the second term is due to adding the tail development factor.

5.1.6 Reserving and pricing

One of the most important quantities in insurance is the *loss-ratio*, which provides, at time t , a snapshot of the profitability. The definition below is to be understood product-wise or according to each homogeneous portfolio constructed for reserving.

Definition 5.3 (Loss-ratio) Let π_i be the total premium collected in the i -th accident year, and let $C_{i,t-i}$ be the cumulative claim payment. The loss ratio of accident year i is given by

$$\text{LR}_i^{(t)} = \frac{C_{i,t-i} + \mathcal{R}_i^{(t)}}{\pi_i}, \quad (5.12)$$

where $\mathcal{R}_i^{(t)}$ is the best estimate of the reserve at time t .

Since reserving actuaries determine the reserves, it is not uncommon that they also determine the loss-ratios. As discussed, the estimation techniques above are done at an *aggregate level* by conducting the analysis on homogeneous portfolios of products.

Premiums are determined by pricing actuaries and are often set with micro-tariffing, i.e. using data on the *individual level*. The premium is thus estimated on a different data subset or format than that on which the reserves are being estimated. For this reason, the estimation techniques are fundamentally different, and thus communication between reserving and pricing sections is often more business-related than statistics-related.

5.2 Special claims reserving models

The chain-ladder algorithm is simple, performs robustly, and is based on a rather intuitive idea that allows for transparency. For this reason, the algorithm is widely used in practice by reserving actuaries. The motivation behind Mack's model was to introduce a stochastic setup that provides reserves corresponding to the chain-ladder reserves while allowing for quantification of the conditional

mean squared error. This section aims to explore two slightly more advanced stochastic setups that also provide the chain-ladder reserves while allowing for uncertainty quantification. As opposed to Mack's model, we consider only parametric models in this section. Since several models produce the same reserves, it is natural to compare their respective conditional mean squared error of the ultimate claim prediction.

The first task is to introduce the incremental claims as members of a certain exponential dispersion family and examine the conditional mean squared error. Then the parameters of that exponential dispersion family are specified to obtain the so-called over-dispersed Poisson model, which provides the chain-ladder reserves. Remarkably, this family can also be related to the Bornhuetter–Ferguson algorithm considered above. The second task is to propose a (parametric) Bayesian approach which provides exact and explicit conditional mean and mean squared error.

5.2.1 Cross-classified models

We consider incremental claims with exposure parameter μ_i for accident year $1 \leq i \leq I$ and development parameter η_j for development year $0 \leq j \leq J$. Assume that

$$\mathbb{E}[X_{i,j}] = \eta_j \mu_i,$$

which resembles the Bornhuetter–Ferguson idea, and can be seen as a generalisation thereof. The goal is to estimate the μ_i 's and η_j 's using maximum likelihood estimation.

Tweedie's model

Tweedie models provide a flexible exponential dispersion family that naturally fits cross-classified claims, interpolating between discrete and continuous behaviour. We first state the cross-classified assumptions and then embed them in the Tweedie likelihood. We begin by imposing a set of assumptions that are necessary but not sufficient to specify the Tweedie distribution, namely the assumption of a *cross-classified model*.

The Cross-classified model assumptions are as follows. Let $p \in [1, 2]$ be fixed. Assume that there exist positive constants $\eta_0, \dots, \eta_J$, $\mu_1, \dots, \mu_I$ and ϕ such that incremental claims $X_{i,j}$ are independent across i and j , non-negative and

$$\mathbb{E}[X_{i,j}] = \eta_j \mu_i, \quad \text{Var}(X_{i,j}) = \phi (\eta_j \mu_i)^p,$$

for all $1 \leq i \leq I$, and $0 \leq j \leq J$.

In order to have identifiable parameters (η_j) and (μ_i) , we always impose one of the following three constraints

$$(I) : \mu_1 = 1, \quad (II) : \eta_0 = 1 \quad \text{or} \quad (III) : \sum_{j=0}^J \eta_j = 1,$$

where the first two are often more convenient when considering GLMs and the latter is the expression of, and hence suitable for, the Bornhuetter–Ferguson method.

A model satisfying the cross-classified model assumption can be embedded in an exponential dispersion family with log-likelihood ℓ_n , conditional on the observations $\mathcal{D}_t$ (with $n = |\mathcal{D}_t|$), given by

$$\ell_n(\theta, \phi) = \sum_{i+j \leq t} \left\{ \frac{X_{i,j}\theta_{i,j} - b_p(\theta_{i,j})}{\phi} + \log c_p(X_{i,j}, \phi) \right\},$$

where

$$\begin{aligned} \mathbb{E}[X_{i,j}] &= b'_p(\theta_{i,j}) = \eta_j \mu_i, \\ \text{Var}(X_{i,j}) &= \phi b''_p(\theta_{i,j}) = \phi (\eta_j \mu_i)^p. \end{aligned}$$

The above specification is satisfied for the following pairs of functions, collectively specifying *Tweedie's model*:

$$\begin{cases} b_p(\theta_{i,j}) = \exp(\theta_{i,j}) & \theta_{i,j} = \log(\eta_j \mu_i) & \text{for } p = 1, \\ b_p(\theta_{i,j}) = \frac{1}{2-p} ((1-p)\theta_{i,j})^{\frac{2-p}{1-p}} & \theta_{i,j} = \frac{1}{1-p} (\eta_j \mu_i)^{1-p} & \text{for } p \in (1, 2), \\ b_p(\theta_{i,j}) = -\log(-\theta_{i,j}) & \theta_{i,j} = \frac{1}{1-p} (\eta_j \mu_i)^{1-p} & \text{for } p = 2. \end{cases}$$

The first case, where $p = 1$, corresponds to the over-dispersed Poisson model, the second case, $p \in (1, 2)$, to a compound Poisson with gamma claim sizes, and the last case, $p = 2$, corresponds to a gamma distribution. The maximum likelihood estimators of η_j and μ_i , at time $t \geq I$, are solutions to the following system of equations

$$\eta_j \sum_{i=1}^{(t-j) \wedge I} \mu_i^{2-p} = \sum_{i=1}^{(t-j) \wedge I} X_{i,j} \mu_i^{1-p} \quad \text{for } 0 \leq j \leq J, \quad (5.13)$$

$$\mu_i \sum_{j=0}^{(t-i) \wedge J} \eta_j^{2-p} = \sum_{j=0}^{(t-i) \wedge J} X_{i,j} \eta_j^{1-p} \quad \text{for } 1 \leq i \leq I, \quad (5.14)$$

using one of the constraints above. The system is solved numerically, e.g. with Fisher's scoring algorithm, and we denote the corresponding solutions by $\hat{\mu}_i^{\text{Tw}_p(t)}$ and $\hat{\eta}_j^{\text{Tw}_p(t)}$ for $1 \leq i \leq I$ and $0 \leq j \leq J$ respectively. The resulting best-estimate reserve and ultimate claim size, for fixed $p \in [1, 2]$ at time $t \geq I$ for accident year $I \geq i > t - J$, are respectively given by

$$\mathcal{R}_i^{\text{Tw}_p(t)} = \sum_{j=t-i+1}^J \hat{\eta}_j^{\text{Tw}_p(t)} \hat{\mu}_i^{\text{Tw}_p(t)}, \quad \hat{C}_{i,J}^{\text{Tw}_p(t)} = C_{i,t-i} + \mathcal{R}_i^{\text{Tw}_p(t)}.$$

The Poisson case highlights how the chain-ladder equations arise directly from likelihood.

Example 5.1 (Poisson likelihood and chain-ladder equations) Consider the special case $p = 1$ and impose the constraint (III), $\sum_{j=0}^J \eta_j = 1$. Then $X_{i,j}$

correspond to independent Poisson random variables with mean $\eta_j \mu_i$, and the conditional log-likelihood is

$$\ell_n(\mu, \eta) = \sum_{i+j \leq t} \left\{ X_{i,j} \log(\eta_j \mu_i) - \eta_j \mu_i - \log(X_{i,j}!) \right\}.$$

Differentiating yields the likelihood equations

$$\mu_i \sum_{j=0}^{(t-i) \wedge J} \eta_j = \sum_{j=0}^{(t-i) \wedge J} X_{i,j}, \quad \eta_j \sum_{i=1}^{(t-j) \wedge I} \mu_i = \sum_{i=1}^{(t-j) \wedge I} X_{i,j},$$

which are exactly (5.13)–(5.14) at $p = 1$. Under this normalisation, the solution admits the chain-ladder representation given in Theorem 5.1. $\diamond$

Conditional mean squared error of Tweedie's model

The conditional mean squared error decomposes according to (5.3) by

$$\mathbb{E} \left[(C_{i,J} - \hat{C}_{i,J}^{\text{Tw}_p(t)})^2 \mid \mathcal{D}_t \right] = \text{Var}(C_{i,J} \mid \mathcal{D}_t) + \left(\mathbb{E}[C_{i,J} \mid \mathcal{D}_t] - \hat{C}_{i,J}^{\text{Tw}_p(t)} \right)^2.$$

The first term, which is the variance term, is analysed using the independence and variance assumption of the cross-classified model

$$\begin{aligned} \text{Var}(C_{i,J} \mid \mathcal{D}_t) &= \text{Var} \left(\sum_{j=0}^{t-i} X_{i,j} + \sum_{j=t-i+1}^J X_{i,j} \mid \mathcal{D}_t \right) \\ &= \sum_{j=t-i+1}^J \text{Var}(X_{i,j} \mid \mathcal{D}_t) = \phi \sum_{j=t-i+1}^J (\eta_j \mu_i)^p. \end{aligned}$$

If $\hat{\phi}^{(t)}$ is an estimator of ϕ , we simply estimate the variance term by

$$\widehat{\text{Var}}(C_{i,J} \mid \mathcal{D}_t) = \hat{\phi}^{(t)} \sum_{j=t-i+1}^J (\hat{\eta}_j^{\text{Tw}_p(t)} \hat{\mu}_i^{\text{Tw}_p(t)})^p.$$

The squared bias term is cumbersome to handle, and hence we make some quite crude approximations in the derivation below. The ideal goal is to examine

$$\left(\mathbb{E}[C_{i,J} \mid \mathcal{D}_t] - \hat{C}_{i,J}^{\text{Tw}_p(t)} \right)^2 = \left(\mu_i \sum_{j=t-i+1}^J \eta_j - \hat{\mu}_i^{\text{Tw}_p(t)} \sum_{j=t-i+1}^J \hat{\eta}_j^{\text{Tw}_p(t)} \right)^2,$$

which is the squared bias for accident year i . Earlier in this chapter we derived the conditional mean squared error of each accident year. However, here we proceed to calculate the squared bias for all accident years that are not yet fully developed. To this end, define

$$\sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} \mid \mathcal{D}_t] = \sum_{i=t-J+1}^I \left(C_{i,t-i} + \mu_i \sum_{j=t-i+1}^J \eta_j \right) =: E^+(\tilde{\mu}, \tilde{\eta}),$$

where $\tilde{\mu} = (\mu_{t-J+1}, \dots, \mu_I)$ and $\tilde{\eta} = (\eta_{t-I+1}, \dots, \eta_J)$. Note that the first-order derivatives of E^+ with respect to each μ_i and η_j are given by

$$\begin{aligned}\frac{\partial}{\partial \mu_i} E^+ &= \sum_{j=t-i+1}^J \eta_j =: \frac{m_i^{(t)}}{\mu_i} \sum_{l=t-J+1}^I \mathbb{E}[C_{l,J} | \mathcal{D}_t], \\ \frac{\partial}{\partial \eta_j} E^+ &= \sum_{i=t-j+1}^I \mu_i =: \frac{g_j^{(t)}}{\eta_j} \sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{D}_t].\end{aligned}$$

The idea is to use that $E^+((\hat{\mu}_{t-J+1}^{\text{Tw}_p(t)}, \dots, \hat{\mu}_I^{\text{Tw}_p(t)}), (\hat{\eta}_{t-I+1}^{\text{Tw}_p(t)}, \dots, \hat{\eta}_J^{\text{Tw}_p(t)})) = E^+(\tilde{\mu}^{\text{Tw}_p(t)}, \tilde{\eta}^{\text{Tw}_p(t)}) = \sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_p(t)}$ and to make a first-order expansion of the latter around the true parameter value (μ, η) , disregarding all higher order terms. Thus, we get

$$\begin{aligned}\sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_p(t)} &= E^+(\tilde{\mu}^{\text{Tw}_p(t)}, \tilde{\eta}^{\text{Tw}_p(t)}) \\ &\approx E^+(\tilde{\mu}, \tilde{\eta}) + (\nabla_{\tilde{\mu}} E^+(\tilde{\mu}, \tilde{\eta}), \nabla_{\tilde{\eta}} E^+(\tilde{\mu}, \tilde{\eta})) (\tilde{\mu}^{\text{Tw}_p(t)} - \tilde{\mu}, \tilde{\eta}^{\text{Tw}_p(t)} - \tilde{\eta})^\top \\ &= \sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{D}_t] + \sum_{i=t-J+1}^I \left(\frac{\partial}{\partial \mu_i} E^+(\tilde{\mu}, \tilde{\eta}) \right) (\hat{\mu}_i^{\text{Tw}_p(t)} - \mu_i) \\ &\quad + \sum_{j=t-I+1}^J \left(\frac{\partial}{\partial \eta_j} E^+(\tilde{\mu}, \tilde{\eta}) \right) (\hat{\eta}_j^{\text{Tw}_p(t)} - \eta_j).\end{aligned}$$

In particular, we propose that

$$\begin{aligned}&\frac{(\sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{D}_t] - \sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_p(t)})^2}{(\sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{D}_t])^2} \\ &\approx \left(\sum_{i=t-J+1}^I m_i^{(t)} \frac{(\mu_i - \hat{\mu}_i^{\text{Tw}_p(t)})}{\mu_i} + \sum_{j=t-I+1}^J g_j^{(t)} \frac{(\eta_j - \hat{\eta}_j^{\text{Tw}_p(t)})}{\eta_j} \right)^2,\end{aligned}$$

where we used that $m_i^{(t)} = (\mu_i \sum_{j=t-i+1}^J \eta_j) / \sum_{l=t-J+1}^I \mathbb{E}[C_{l,J} | \mathcal{D}_t]$ and $g_j^{(t)} = (\eta_j \sum_{i=t-j+1}^I \mu_i) / \sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{D}_t]$.

The right-hand side has to be estimated. The idea of the following approximation is to rewrite the latter expression into a matrix expression where we utilise the asymptotic normality of MLEs, see for instance Theorem 4.4.

Fix ϕ and choose normalisation $\eta_0 = 1$. The Fisher information matrix at the vector $\nu = (\mu_1, \dots, \mu_I, \eta_1, \dots, \eta_J)$ is then given by

$$I_t(\nu, \phi) = \left(-\mathbb{E} \left[\frac{\partial^2}{\partial \nu_r \partial \nu_s} \ell_n(\nu, \phi) \right] \right)_{r,s=1, \dots, I+J},$$

with entries

$$\begin{aligned}
I_t(\nu, \phi)_{r,s} &= 1_{(r=s)} \frac{\mu_i^{-p}}{\phi} \sum_{j=0}^{(t-i) \wedge J} \eta_j^{2-p}, \quad r, s \in \{1, \dots, I\}, \quad i = r, \\
I_t(\nu, \phi)_{r,s} &= 1_{(r=s)} \frac{\eta_j^{-p}}{\phi} \sum_{i=1}^{(t-j) \wedge I} \mu_i^{2-p}, \quad r, s \in \{I+1, \dots, I+J\}, \quad j = r - I, \\
I_t(\nu, \phi)_{r,s} &= 1_{(i+j \leq t)} \frac{1}{\phi} \eta_j^{1-p} \mu_i^{1-p}, \quad r \in \{I+1, \dots, I+J\}, s \in \{1, \dots, I\}, \\
&\quad j = r - I, \quad i = s,
\end{aligned}$$

where the first and second lines correspond to the Hessian at $\mu = (\mu_1, \dots, \mu_I)$ and $\eta = (\eta_1, \dots, \eta_J)$, respectively, while the third line corresponds to all cross-derivatives of (μ_i, η_j) pairs. According to Theorem 4.4, under regularity conditions, we have that $\text{Cov}(\hat{\nu}^{\text{Tw}_p(t)}) \rightarrow I_t(\nu, \phi)^{-1}$, for

$$\hat{\nu}^{\text{Tw}_p(t)} = (\hat{\mu}_1^{\text{Tw}_p(t)}, \dots, \hat{\mu}_I^{\text{Tw}_p(t)}, \hat{\eta}_1^{\text{Tw}_p(t)}, \dots, \hat{\eta}_J^{\text{Tw}_p(t)}).$$

Thus if we define

$$\begin{aligned}
\hat{\delta}^{\text{Tw}_p(t)} &= \left(\frac{\mu_1 - \hat{\mu}_1^{\text{Tw}_p(t)}}{\mu_1}, \dots, \frac{\mu_I - \hat{\mu}_I^{\text{Tw}_p(t)}}{\mu_I}, \frac{\eta_1 - \hat{\eta}_1^{\text{Tw}_p(t)}}{\eta_1}, \dots, \frac{\eta_J - \hat{\eta}_J^{\text{Tw}_p(t)}}{\eta_J} \right)^\top, \\
w^{(t)} &= (0, \dots, 0, m_{t-J+1}^{(t)}, \dots, m_I^{(t)}, 0, \dots, 0, g_{t-I+1}^{(t)}, \dots, g_J^{(t)})^\top,
\end{aligned}$$

we can write the above first-order Taylor expansion as

$$\begin{aligned}
&\frac{(\sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{D}_t] - \sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_p(t)})^2}{(\sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{D}_t])^2} \\
&\approx ((w^{(t)})^\top \hat{\delta}^{\text{Tw}_p(t)})^2 = (w^{(t)})^\top \hat{\delta}^{\text{Tw}_p(t)} (\hat{\delta}^{\text{Tw}_p(t)})^\top w^{(t)}.
\end{aligned}$$

Assuming we have estimated ϕ with $\hat{\phi}^{(t)}$, we propose to estimate the quadratic form by the inverse Fisher information matrix as follows

$$\hat{\delta}^{\text{Tw}_p(t)} (\hat{\delta}^{\text{Tw}_p(t)})^\top \approx \text{diag}(\hat{\nu}^{\text{Tw}_p(t)})^{-1} I_t(\hat{\nu}^{\text{Tw}_p(t)}, \hat{\phi}^{(t)})^{-1} \text{diag}(\hat{\nu}^{\text{Tw}_p(t)})^{-1} =: \hat{\mathcal{I}}_t^{(p)}.$$

Note that the latter approximation suggests replacing each matrix entry of the left-hand side by its expected value, as well as replacing the denominators of $\hat{\delta}^{\text{Tw}_p(t)}$ by the corresponding estimators.

Consequently, the above calculations provide an estimate for the squared bias term. In conclusion, we approximate the conditional mean squared error by

$$\begin{aligned}
&\mathbb{E} \left[\left(\sum_{i=t-J+1}^I C_{i,J} - \sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_p(t)} \right)^2 \middle| \mathcal{D}_t \right] \\
&\approx \hat{\phi}^{(t)} \sum_{i=t-J+1}^I \sum_{j=t-i+1}^J (\hat{\eta}_j^{\text{Tw}_p(t)} \hat{\mu}_i^{\text{Tw}_p(t)})^p + \left(\sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_p(t)} \right)^2 (\hat{w}^{(t)})^\top \hat{\mathcal{I}}_t^{(p)} \hat{w}^{(t)},
\end{aligned}$$

where $\hat{w}^{(t)}$ corresponds to $w^{(t)}$ with parameters replaced by their estimates

$$\hat{m}_i^{(t)} = \frac{\hat{\mu}_i^{\text{Tw}_p(t)} \sum_{j=t-i+1}^J \hat{\eta}_j^{\text{Tw}_p(t)}}{\sum_{l=t-J+1}^I \hat{C}_{l,J}^{\text{Tw}_p(t)}} \quad \text{and} \quad \hat{g}_j^{(t)} = \frac{\hat{\eta}_j^{\text{Tw}_p(t)} \sum_{i=t-j+1}^I \hat{\mu}_i^{\text{Tw}_p(t)}}{\sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_p(t)}}.$$

The parameter ϕ is usually estimated with the Pearson residuals

$$\hat{\phi}^{(t)} = \frac{1}{|\mathcal{D}_t| - (I + J)} \sum_{i+j \leq t} \frac{(X_{i,j} - \hat{\eta}_j^{\text{Tw}_p(t)} \hat{\mu}_i^{\text{Tw}_p(t)})^2}{(\hat{\eta}_j^{\text{Tw}_p(t)} \hat{\mu}_i^{\text{Tw}_p(t)})^p}.$$

Another common approach is to apply the bootstrap to obtain the right-hand side of the first-order Taylor approximation. To obtain a closed-form benchmark, we now specialise to $p = 1$.

Over-dispersed Poisson model

The case $p = 1$ is especially useful because it yields an explicit likelihood-based route to chain-ladder. In this case, the system of equations (5.13)-(5.14) can be solved analytically. This model is called the over-dispersed Poisson model and, remarkably, it provides the chain-ladder reserves. The following theorem makes this equivalence explicit.

Theorem 5.1 (ODP yields chain-ladder) *Assume the Tweedie model assumptions with $p = 1$ and normalisation $\sum_{j=0}^J \eta_j = 1$. Then the MLEs of the system of equations (5.13)-(5.14) are given by*

$$\hat{C}_{i,J}^{\text{Tw}_1(t)} = \hat{\mu}_i^{\text{Tw}_1(t)} = \hat{C}_{i,J}^{\text{CL}(t)}, \quad \hat{\eta}_j^{\text{Tw}_1(t)} = \hat{\eta}_j^{\text{CL}(t)} = \prod_{k=j}^{J-1} \frac{1}{\hat{f}_k^{(t)}} \left(1 - \frac{1}{\hat{f}_{j-1}^{(t)}}\right),$$

for $1 \leq i \leq I$ and $1 \leq j \leq J$. Additionally $\hat{\eta}_0^{\text{Tw}_1(t)} = \hat{\eta}_0^{\text{CL}(t)} = \prod_{k=0}^{J-1} 1/\hat{f}_k^{(t)}$, and the nominal best-estimate reserve for accident year i is given by

$$\mathcal{R}_i^{\text{Tw}_1(t)} = \sum_{j=t-i+1}^J \hat{\eta}_j^{\text{Tw}_1(t)} \hat{\mu}_i^{\text{Tw}_1(t)} = \mathcal{R}_i^{\text{CL}(t)}$$

Proof. The proof is left as an exercise for the reader. $\square$

As the over-dispersed Poisson model is a special case of Tweedie's model, we immediately get the conditional mean squared error estimate

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{i=t-J+1}^I C_{i,J} - \sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_1(t)} \right)^2 \middle| \mathcal{D}_t \right] \\ & \approx \hat{\phi}^{(t)} \sum_{i=t-J+1}^I \sum_{j=t-i+1}^J \hat{\eta}_j^{\text{Tw}_1(t)} \hat{\mu}_i^{\text{Tw}_1(t)} + \left(\sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{Tw}_1(t)} \right)^2 (\hat{w}^{(t)})^\top \hat{\mathcal{I}}_t^{(1)} \hat{w}^{(t)}, \end{aligned}$$

where the first term on the right-hand side is the approximated variance and the second term is the approximated squared bias. In the over-dispersed Poisson model, we have used a different normalisation than the one used in the derivation of the mean squared error in the general Tweedie model. This derivation can, however, be replicated with the normalisation $\sum_{j=0}^J \eta_j = 1$, which justifies reusing the mean squared error approximation.

Revisiting the Bornhuetter–Ferguson method

Recall that if we, somewhat artificially, use the chain-ladder ultimate claim size estimate as prior in the Bornhuetter–Ferguson method, the ultimates of both algorithms agree, according to (5.2).

Now, the ultimate claim size estimate in the over-dispersed Poisson model equals the chain-ladder ultimate claim size estimate, which allows for the following comparison

$$\begin{aligned}\hat{C}_{i,J}^{\text{CL}(t)} &= C_{i,t-i} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)})\hat{\mu}_i^{\text{Tw}_1(t)} \\ \hat{C}_{i,J}^{\text{BF}(t)} &= C_{i,t-i} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)})\hat{\mu}_i,\end{aligned}$$

where $\hat{\mu}_i$ is an exogenously given prior. This motivates utilising the framework and assumptions of cross-classified models to provide an estimate of the conditional mean squared error of the Bornhuetter–Ferguson ultimate, which we currently have only viewed as an algorithm without any underlying model.

This change in viewpoint, however, requires some adaptation. In the over-dispersed Poisson model, we enlarge the underlying filtration to accommodate the given priors, i.e. we consider

$$\mathcal{F}_t = \sigma(\mathcal{D}_t, \hat{\mu}_1, \dots, \hat{\mu}_I).$$

In addition we assume that

$$\{X_{i,j} : 1 \leq i \leq I, 0 \leq j \leq J\} \perp\!\!\!\perp \{\hat{\mu}_1, \dots, \hat{\mu}_I\}. \quad (5.15)$$

From a practical point of view, the independence assumption (5.15) could be potentially problematic, as expert information may be based on data. Indeed, an approach that could be used to pick the prior for accident year i is to take it equal to a weighted average of loss ratios of recent accident years that are all developed more than a given percentage (e.g. 80%) and multiply that average with the premium of accident year i . The loss ratios of former accident years are given as a linear transformation of former claims and are thus highly dependent on the latter. With this caveat in mind, we proceed with our calculations.

To examine conditional mean squared error, we follow the derivations from the Tweedie model, however using now the parametrisation $\sum_{j=0}^J \eta_j = 1$ and specifying $p = 1$. Consequently, the conditional variance is estimated by

$$\widehat{\text{Var}}^{\text{BF}}(C_{i,J} | \mathcal{F}_t) = \hat{\phi}^{(t)} \sum_{j=t-i+1}^J \hat{\eta}_j^{\text{Tw}_1(t)} \hat{\mu}_i.$$

The first-order Taylor expansion above is repeated with $\hat{\mu}_i^{\text{Tw}_1(t)}$ replaced by $\hat{\mu}_i$ and yields

$$\begin{aligned}& \frac{(\sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{F}_t] - \sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{BF}(t)})^2}{(\sum_{i=t-J+1}^I \mathbb{E}[C_{i,J} | \mathcal{F}_t])^2} \\ & \approx \left(\sum_{i=t-J+1}^I m_i^{(t)} \frac{(\mu_i - \hat{\mu}_i)}{\mu_i} + \sum_{j=t-I+1}^J g_j^{(t)} \frac{(\eta_j - \hat{\eta}_j^{\text{Tw}_1(t)})}{\eta_j} \right)^2.\end{aligned}$$

The corresponding quadratic expression related to the inverse Fisher information is adapted by setting

$$\widehat{\mathcal{I}}_t^{BF} = \begin{pmatrix} \mathcal{I}_\mu & 0 \\ 0 & \widehat{\mathcal{I}}_\eta \end{pmatrix}$$

where $\widehat{\mathcal{I}}_\eta = (\widehat{\mathcal{I}}_t^{(1)})_{j,j' \in (I+1, \dots, I+J)}$ is the bottom right matrix of $\widehat{\mathcal{I}}_t^{(1)}$ corresponding to the covariance matrix of $(\eta_j - \hat{\eta}_j^{\text{Tw}_1(t)})/\eta_j$ for $1 \leq j \leq J$, while $\mathcal{I}_\mu$ estimates the covariance matrix of $(\mu_i - \hat{\mu}_i)/\mu_i$ for $1 \leq i \leq I$. The latter matrix should be estimated exogenously by those providing the prior estimates $(\hat{\mu}_i)$, which may or may not be the reserving actuary themselves.

Consequently, the corresponding conditional mean squared error of the Bornhuetter–Ferguson algorithm (under over-dispersed Poisson model assumptions) is given by

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{i=t-J+1}^I C_{i,J} - \sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{BF}(t)} \right)^2 \middle| \mathcal{F}_t \right] \\ & \approx \hat{\phi}^{(t)} \sum_{i=t-J+1}^I \sum_{j=t-i+1}^J \hat{\eta}_j^{\text{Tw}_1(t)} \hat{\mu}_i + \left(\sum_{i=t-J+1}^I \hat{C}_{i,J}^{\text{BF}(t)} \right)^2 (\hat{w}^{(t)})^\top \widehat{\mathcal{I}}_t^{BF} \hat{w}^{(t)}. \end{aligned}$$

5.2.2 Introduction to Bayesian chain-ladder models

In the above section we introduced stochastic claim models where the distribution of the incremental claims were embedded in a Tweedie family, and we used the asymptotic properties derived earlier in this chapter to estimate the conditional mean squared error of the ultimate claims.

In this section, we step in another direction, not only away from the Tweedie family, but also away from the frequentist approach. Here we take a Bayesian approach which, in the gamma-gamma case below, gives rise to explicit analytical expressions for the conditional mean squared error. The essential change is that development factors are treated as random quantities and updated through posterior distributions. For a more systematic treatment of Bayesian methods, we refer to Chapter ??.

The Gamma-gamma Bayesian chain-ladder model assumptions are as follows. Let $\sigma_j > 0$ be given constants for $0 \leq j \leq J-1$.

1. The components Θ_j of $\Theta = (\Theta_0, \dots, \Theta_{J-1})$ (called the *prior*) are independent and $\text{Gamma}(\gamma_j, f_j(\gamma_j - 1))$ -distributed with given hyper prior parameters $f_j > 0$ and $\gamma_j > 1$.
2. Conditionally on Θ , the sequence of cumulative claims in different accident years $\{(C_{i,j})_{0 \leq j \leq J}, 1 \leq i \leq I\}$ are independent Markov processes with conditional distributions

$$C_{i,j+1} \mid C_{i,j}, \Theta \sim \text{Gamma}(C_{i,j}\sigma_j^{-2}, \Theta_j\sigma_j^{-2}),$$

for all $1 \leq i \leq I$ and $0 \leq j \leq J-1$. This automatically specifies the *likelihood*.

3. The random variables Θ and $C_{1,0}, \dots, C_{I,0}$ are independent and each $C_{i,0}$ is almost surely strictly positive.

The above model has some resemblance to Mack's model, which makes their comparison quite natural. As in Mack's model, we examine the conditional expected value and variance of $C_{i,j+1}$, where the conditioning is not only on former claims but also the outcome of the prior Θ .

$$\mathbb{E}[C_{i,j+1} | C_{i,j}, \Theta] = \frac{C_{i,j} \sigma_j^{-2}}{\Theta_j \sigma_j^{-2}} = \Theta_j^{-1} C_{i,j},$$

where Θ_j^{-1} is inverse gamma distributed $\text{IG}(\gamma_j, f_j(\gamma_j - 1))$, which implies that $\mathbb{E}[\Theta_j^{-1}] = f_j(\gamma_j - 1)/(\gamma_j - 1) = f_j$. We see that the hyper-prior f_j corresponds to the prior guess of the chain-ladder factor of development period j . As for the conditional variance, it holds that

$$\text{Var}(C_{i,j+1} | C_{i,j}, \Theta) = \frac{C_{i,j} \sigma_j^{-2}}{(\Theta_j \sigma_j^{-2})^2} = \sigma_j^2 \Theta_j^{-2} C_{i,j} =: \sigma_j^2(\Theta_j) C_{i,j}.$$

Here $\sigma_j^2(\Theta_j)$ plays the role of s_j^2 in Mack's model. Note that if the parameters $\Theta = \theta$ were deterministic, we would have an explicit distributional embedding of Mack's model, which is otherwise distribution free.

One of the main advantages of the Bayesian approach is that we update the parameters of the distribution of Θ as we get more data. In particular, we can derive the so-called posterior distribution of each Θ_j given certain data $\mathcal{D}_t$.

Lemma 5.3 (Posterior of chain-ladder factors) *Assume the gamma-gamma Bayesian chain-ladder assumptions. Then $\Theta_0, \dots, \Theta_{J-1}$ are, conditionally given $\mathcal{D}_t$, independent and distributed as*

$$\Theta_j | \mathcal{D}_t \sim \text{Gamma}\left(\gamma_j + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sigma_j^2}, f_j(\gamma_j - 1) + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j+1}}{\sigma_j^2}\right) \quad (5.16)$$

Proof. The conditional likelihood of the parameters Θ given observations $\mathcal{D}_t$ is simply given by

$$\begin{aligned} \pi(\theta | \mathcal{D}_t) &= \frac{\pi_\Theta(\theta) \ell_{(C_{i,j})_{i+j \leq t}}(\mathcal{D}_t | \theta)}{\int \pi_\Theta(\theta) \ell_{(C_{i,j})_{i+j \leq t}}(\mathcal{D}_t | \theta) d\theta} \\ &\propto \pi_\Theta(\theta) \ell_{(C_{i,j})_{i+j \leq t}}(\mathcal{D}_t | \theta) \\ &= \prod_{j=0}^{J-1} \pi_{\Theta_j}(\theta_j) \prod_{j=0}^{J-1} \prod_{i=1}^{(t-j-1) \wedge I} \ell_{C_{i,j+1}}(C_{i,j+1} | C_{i,j}, \theta) g_{C_{i,0}}(C_{i,0}) \\ &= \prod_{j=0}^{J-1} \frac{(f_j(\gamma_j - 1))^{\gamma_j}}{\Gamma(\gamma_j)} \theta_j^{\gamma_j - 1} e^{-\theta_j f_j(\gamma_j - 1)} \\ &\times \prod_{j=0}^{J-1} \prod_{i=1}^{(t-j-1) \wedge I} \left(\frac{(\theta_j / \sigma_j^2)^{C_{i,j} / \sigma_j^2}}{\Gamma(C_{i,j} / \sigma_j^2)} C_{i,j+1}^{C_{i,j} / \sigma_j^2 - 1} e^{-(\theta_j / \sigma_j^2) C_{i,j+1}} \right) \ell_{C_{i,0}}(C_{i,0}) \\ &\propto \prod_{j=0}^{J-1} \theta_j^{\gamma_j + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sigma_j^2} - 1} e^{-\theta_j (f_j(\gamma_j - 1) + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j+1}}{\sigma_j^2})}, \end{aligned}$$

which we readily recognise as being proportional to the density of a random variable distributed according to (5.16). $\square$

The above derivation relies on Bayes' rule, which yields the posterior distribution being proportional to the density of the prior multiplied by the likelihood of observations given Θ . Note that both Θ_j and the conditional likelihoods of $C_{i,j+1} | C_{i,j}, \Theta$ are gamma distributed, and so the above Lemma yields that the posterior distribution of each Θ_j is also gamma-distributed. When the prior and posterior distributions belong to the same family of distributions (for the chosen likelihood) we say the prior and posterior distributions are *conjugate*.

Corollary 5.1 (Posterior BCL factors) *Assume the gamma-gamma Bayesian chain-ladder assumptions and that $t > j + 1$. Then the posterior Bayesian chain-ladder factors are given by*

$$\hat{f}_j^{BCL(t)} = \mathbb{E}[\Theta_j^{-1} | \mathcal{D}_t] = \omega_j^{(t)} \hat{f}_j^{(t)} + (1 - \omega_j^{(t)}) f_j,$$

where $\hat{f}_j^{(t)}$ are the chain-ladder factors and

$$\omega_j^{(t)} = \frac{\sum_{i=1}^{(t-j-1) \wedge I} C_{i,j}}{\sigma_j^2(\gamma_j - 1) + \sum_{i=1}^{(t-j-1) \wedge I} C_{i,j}} \in (0, 1).$$

Proof. Similarly to above we have that $\Theta_j^{-1} | \mathcal{D}_t$ is inverse gamma distributed with parameters as in (5.16). Consequently

$$\begin{aligned} \mathbb{E}[\Theta_j^{-1} | \mathcal{D}_t] &= \frac{1}{\gamma_j - 1 + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sigma_j^2}} \left(f_j(\gamma_j - 1) + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j+1}}{\sigma_j^2} \right) \\ &= \frac{\gamma_j - 1}{\gamma_j - 1 + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sigma_j^2}} f_j + \frac{\sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sigma_j^2}}{\gamma_j - 1 + \sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sigma_j^2}} \frac{\sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j+1}}{\sigma_j^2}}{\sum_{i=1}^{(t-j-1) \wedge I} \frac{C_{i,j}}{\sigma_j^2}}. \end{aligned}$$

$\square$

The posterior Bayesian chain-ladder factor is thus a credibility-weighted average between the prior f_j and the chain-ladder estimate $\hat{f}_j^{(t)}$, and Corollary 5.1 explains how the chosen value of γ_j determines the weight on the prior f_j and the data-driven estimate $\hat{f}_j^{(t)}$. In particular, γ_j should in some sense be interpreted as the degree of information included in the prior distribution of Θ_j .

Letting $\gamma_j \rightarrow 1$ implies $\omega_j^{(t)} \rightarrow 1$, which means that all weight is put on $\hat{f}_j^{(t)}$ and should reflect the situation in which no information is included in the prior. Similarly, when $\gamma_j \rightarrow \infty$ all weight is put on the prior f_j . Another consequence of Lemma 5.3 is the following result. We can now express the Bayesian chain-ladder predictor explicitly.

Theorem 5.2 (Bayesian chain-ladder predictor) *Assume the gamma-gamma Bayesian chain-ladder assumptions. Then the Bayesian chain-ladder predictor of $C_{i,j}$, for $t \geq I \geq i > t - j \geq t - J$, is given by*

$$C_{i,j}^{BCL(t)} = \mathbb{E}[C_{i,j} | \mathcal{D}_t] = C_{i,t-i} \prod_{l=t-i}^{j-1} \hat{f}_l^{BCL(t)}.$$

Proof. The gamma-gamma Bayesian chain-ladder assumptions yield that

$$\begin{aligned}
C_{i,j}^{\text{BCL}(t)} &= \mathbb{E}[C_{i,j} \mid \mathcal{D}_t] = \mathbb{E}[\mathbb{E}[C_{i,j} \mid \mathcal{D}_t, \Theta] \mid \mathcal{D}_t] \\
&= \mathbb{E}[\mathbb{E}[C_{i,j} \mid C_{i,0}, \dots, C_{i,j-1}, \Theta] \mid \mathcal{D}_t] = \mathbb{E}[\mathbb{E}[C_{i,j} \mid C_{i,j-1}, \Theta] \mid \mathcal{D}_t] \\
&= \mathbb{E}[\Theta_{j-1}^{-1} C_{i,j-1} \mid \mathcal{D}_t] = \dots = \mathbb{E}[\Theta_{j-1}^{-1} \dots \Theta_{t-i}^{-1} C_{i,t-i} \mid \mathcal{D}_t] \\
&= C_{i,t-i} \prod_{l=t-i}^{j-1} \mathbb{E}[\Theta_l^{-1} \mid \mathcal{D}_t] = C_{i,t-i} \prod_{l=t-i}^{j-1} \hat{f}_l^{\text{BCL}(t)}
\end{aligned}$$

□

Following our remark above it follows that $C_{i,j}^{\text{BCL}(t)} \rightarrow C_{i,j}^{\text{CL}(t)}$ when $\gamma_j \rightarrow 1$, for $I-i \leq j \leq J-1$. Consequently, the gamma-gamma Bayesian chain-ladder model allows, as an edge case, for exact replication of the chain-ladder reserves. We now proceed to examine the mean squared error.

Conditional mean squared error of the gamma-gamma Bayesian chain-ladder model

The closed form of the conditional mean shows that the bias is zero. Consequently, the mean squared error of the ultimate of accident year i in this model is (subject to existence) given by the conditional variance, which we can split by conditioning on Θ as follows

$$\begin{aligned}
\mathbb{E}\left[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 \mid \mathcal{D}_t\right] &= \text{Var}(C_{i,J} \mid \mathcal{D}_t) + (\mathbb{E}[C_{i,J} \mid \mathcal{D}_t] - C_{i,J}^{\text{BCL}(t)})^2 \\
&= \text{Var}(C_{i,J} \mid \mathcal{D}_t) \\
&= \text{Var}(\mathbb{E}[C_{i,J} \mid \Theta, \mathcal{D}_t] \mid \mathcal{D}_t) + \mathbb{E}[\text{Var}(C_{i,J} \mid \Theta, \mathcal{D}_t) \mid \mathcal{D}_t],
\end{aligned}$$

where the first term is the conditional variance of the *parameter volatility* and the latter term is the expected *process variance*. Our next task is to calculate the parameter volatility and process variance.

Define for $0 \leq j \leq J-1$ and $t > j+1$ the term

$$\Psi_j^{(t)} = \frac{\sigma_j^2}{\sigma_j^2(\gamma_j - 2) + \sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}}.$$

The next theorem gives a closed-form expression for the conditional mean squared error.

Theorem 5.3 (BCL conditional mean squared error) *Assume the gamma-gamma Bayesian chain-ladder assumptions. Then the gamma-gamma Bayesian chain-ladder estimator satisfies for $t \geq I \geq i > t-J$*

$$\begin{aligned}
\mathbb{E}\left[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 \mid \mathcal{D}_t\right] &= C_{i,J}^{\text{BCL}(t)} \sum_{j=t-i}^{J-1} \sigma_j^2 \prod_{n=j}^{J-1} \left(\hat{f}_n^{\text{BCL}(t)} (1 + \Psi_n^{(t)})\right) \\
&\quad + (C_{i,J}^{\text{BCL}(t)})^2 \left(-1 + \prod_{j=t-i}^{J-1} (1 + \Psi_j^{(t)})\right),
\end{aligned}$$

assuming that $\gamma_j + \sum_{l=1}^{(t-j-1) \wedge I} (C_{l,j}/\sigma_j^2) > 2$ for all $t-i \leq j \leq J-1$. When the latter is not satisfied, the second moment is infinite.

Proof. Left as an exercise for the reader. $\square$

The above theorem gives an exact expression of the conditional mean squared error. This is in contrast to all the other stochastic claims reserving models we have seen so far; they all relied on some approximation, and the actual discrepancy can be difficult to quantify in practice.

Now, the dissection of the above expressions into parameter estimation uncertainty and process variance uncertainty can be done in many ways. Below, we propose a seemingly natural split, but the reader should bear in mind that the corresponding terms that arise can influence each other.

As discussed above, when letting $\gamma_j \rightarrow \infty$ for all components, all weight of the development factor is placed on the prior, which causes the remaining randomness to be concentrated on the realised cumulative claims. In this case $\Psi_j^{(t)} \rightarrow 0$ and

$$\lim_{\gamma \rightarrow \infty} \frac{\mathbb{E}[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 | \mathcal{D}_t]}{(C_{i,J}^{\text{BCL}(t)})^2} = \frac{\sigma_{t-i}^2}{C_{i,t-i}} + \sum_{j=t-i+1}^{J-1} \frac{\sigma_j^2}{C_{i,j}^{\text{BCL}(t)}},$$

which motivates the following split

$$\begin{aligned} \frac{\mathbb{E}[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 | \mathcal{D}_t]}{(C_{i,J}^{\text{BCL}(t)})^2} &= \frac{\sigma_{t-i}^2}{C_{i,t-i}} \prod_{j=t-i}^{J-1} (1 + \Psi_j^{(t)}) \\ &+ \sum_{j=t-i+1}^{J-1} \frac{\sigma_j^2}{C_{i,j}^{\text{BCL}(t)}} \prod_{n=j}^{J-1} (1 + \Psi_n^{(t)}) + \left(-1 + \prod_{j=t-i}^{J-1} (1 + \Psi_j^{(t)}) \right) \\ &=: \Gamma_{i,t} + \Delta_{i,t}, \end{aligned}$$

where $\Delta_{i,t} = \prod_{j=t-i}^{J-1} (1 + \Psi_j^{(t)}) - 1$. The above findings can be summarised in concise notation as follows

$$\mathbb{E}[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 | \mathcal{D}_t] = (C_{i,J}^{\text{BCL}(t)})^2 (\Gamma_{i,t} + \Delta_{i,t}),$$

where $\Gamma_{i,t}$ corresponds to the process variance and $\Delta_{i,t}$ corresponds to parameter estimation uncertainty.

We proceed to examine the degenerate on data case more closely, where $\gamma_j \rightarrow 1$. In this case, the assumption

$$\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j} > \sigma_j^2, \quad \text{for } t \geq I \text{ and all } 0 \leq j \leq J-1,$$

is required, which implies finiteness of the mean squared error in Theorem 5.3. Under this requirement, it holds that

$$\lim_{\gamma \rightarrow 1} \Psi_j^{(t)} = \frac{\sigma_j^2}{-\sigma_j^2 + \sum_{l=1}^{(t-j-1) \wedge I} C_{l,j}} =: \bar{\Psi}_j^{(t)} > 0,$$

and similarly to above, we may define

$$\begin{aligned}\bar{\Gamma}_{i,t} &= \frac{\sigma_{t-i}^2}{C_{i,t-i}} \prod_{j=t-i}^{J-1} (1 + \bar{\Psi}_j^{(t)}) + \sum_{j=t-i+1}^{J-1} \frac{\sigma_j^2}{C_{i,j}^{\text{BCL}(t)}} \prod_{n=j}^{J-1} (1 + \bar{\Psi}_n^{(t)}) \\ \bar{\Delta}_{i,t} &= \prod_{j=t-i}^{J-1} (1 + \bar{\Psi}_j^{(t)}) - 1.\end{aligned}$$

Hence when $\gamma_j \rightarrow 1$,

$$\mathbb{E}\left[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 \mid \mathcal{D}_t\right] = (C_{i,J}^{\text{BCL}(t)})^2 (\bar{\Gamma}_{i,t} + \bar{\Delta}_{i,t}).$$

This concludes the gamma-gamma Bayesian chain-ladder. In the next section, we combine the ideas of Bayesian and cross-classified models, which allows for increased stochastic complexity.

5.2.3 Bayesian cross-classified models

The Bornhuetter-Ferguson algorithm uses priors (μ_i) to determine the ultimate claim sizes. In a Bayesian setting, a prior is not a given constant, but a random variable with some underlying distribution. A natural approach for extending the Bornhuetter-Ferguson algorithm to a Bayesian model is via the over-dispersed Poisson model.

The Bayesian over-dispersed Poisson model assumptions are as follows.

1. The components of $\Theta = (\mu_1, \dots, \mu_I, \eta_0, \dots, \eta_J, \phi)$ are independent and strictly positive with

$$\mu_i \sim \text{Gamma}(a_i, a_i/m_i), \quad \eta_j \sim \text{Gamma}(b, b/g_j),$$

where $a_i, m_i, b, g_j > 0$.

2. Conditionally on Θ , the incremental claims $X_{i,j}$ are independent. Let $Z_{i,j} = X_{i,j}/\phi$ and assume

$$Z_{i,j} \sim \text{Pois}(\eta_j \mu_i / \phi),$$

for all $1 \leq i \leq I$ and $0 \leq j \leq J$.

Despite the very natural specification, the posterior distribution $\pi(\theta \mid \mathcal{D}_t)$ is unfortunately not tractable and thus one has to rely on numerical methods for this estimation. While methods from variational inference offer a direct approach, reserving literature has suggested using Markov Chain Monte Carlo; we refer to Chapter ?? for general details.

5.3 Notes and Comments

Deterministic chain-ladder remains the baseline against which stochastic refinements are interpreted; perhaps not for its performance, but certainly for its popularity, simplicity, and widespread use in the insurance industry. Stochastic versions are then a natural starting point. In particular, [Mack \[1993\]](#) provided a distribution-free treatment with explicit uncertainty quantification, and [Mack \[1994\]](#) provided a stochastic model underlying the chain-ladder method. Classical second-moment analysis of outstanding claims reserves is given in [Taylor and Ashe \[1983\]](#). The Bornhuetter–Ferguson model [[Bornhuetter and Ferguson, 1972](#)] then combines prior information with observed claims development, effectively establishing a link between the traditional chain-ladder approach and the common actuarial judgement.

The chapter then turns to tail development and reserving extensions. Standard actuarial treatments are given in [Taylor \[2000\]](#), [Wüthrich and Merz \[2012\]](#). Recursive standard-error calculations with explicit inclusion of a tail factor were given by [Mack \[1999\]](#). Estimation of the tail using incurred-paid information in the paid-incurred chain framework was developed by [Merz and Wüthrich \[2013\]](#), offering an alternative way of extending development patterns beyond the observed data. For practical market guidance, see also the CAS Tail Factor Working Party report [[CAS Tail Factor Working Party, 2013](#)].

Regarding special models, the over-dispersed Poisson formulation for chain-ladder settings was developed in [Renshaw and Verrall \[1998\]](#) and unified for practical stochastic reserving in [England and Verrall \[2002\]](#); see also the chain-ladder maximum-likelihood formulation in [Verrall \[1991\]](#). This also connects with the Tweedie viewpoint used in cross-classified models, where variance is modelled as a power of the mean [[England and Verrall, 2002](#)]. The quantification of parameter uncertainty in the Bornhuetter–Ferguson model was addressed in [Alai et al. \[2009, 2010\]](#). Further developments within the Bayesian framework, including the Gamma–Gamma Bayesian chain-ladder model, are presented in [Merz and Wüthrich \[2014\]](#) and [Wüthrich and Merz \[2015\]](#). Finally, for cross-classified reserving models, the two-way ANOVA approach goes back to [Kremer \[1982\]](#). Bayesian chain-ladder and related log-linear reserving formulations are discussed in [Taylor \[2014\]](#), [Ntzoufras and Dellaportas \[2002\]](#).

5.4 Exercises

1. Recall that the ultimate claim size of accident year $i > t - J$ in the Bornhuetter–Ferguson algorithm can be decomposed as follows:

$$\hat{C}_{i,J}^{\text{BF}(t)} = \hat{\kappa}_{t-i}^{\text{CL}(t)} \hat{C}_{i,J}^{\text{CL}(t)} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i \quad (5.17)$$

Here $\hat{\mu}_i$ is some prior estimate of the ultimate claim size for the i -th accident year. The Benktander algorithm corresponds to the Bornhuetter–Ferguson algorithm, where we replace $\hat{\mu}_i$ with the ultimate of the Bornhuetter–Ferguson algorithm, i.e.

$$\hat{C}_{i,J}^{\text{Benk}(t)} = \hat{\kappa}_{t-i}^{\text{CL}(t)} \hat{C}_{i,J}^{\text{CL}(t)} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{C}_{i,J}^{\text{BF}(t)}.$$

- (a) Show that

$$\hat{C}_{i,J}^{\text{Benk}(t)} = (1 - (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)})^2) \hat{C}_{i,J}^{\text{CL}(t)} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)})^2 \hat{\mu}_i. \quad (5.18)$$

- (b) Repeat this process once again by plugging in the ultimate from (5.18) into the prior $\hat{\mu}_i$ in (5.17). Argue that iterating this process eventually results in the chain-ladder ultimate.

2. Define the *claims development result* by

$$\text{CDR}_{i,t+1} = \mathbb{E}[C_{i,J} | \mathcal{D}_t] - \mathbb{E}[C_{i,J} | \mathcal{D}_{t+1}]. \quad (5.19)$$

The claims development result quantifies the change in the predicted ultimate as a consequence of observing an additional development year.

- (a) Assume that $C_{i,J}$ has finite first moment for $t \geq I \geq i > t - J$. Show that

$$\mathbb{E}[\text{CDR}_{i,t+1} | \mathcal{D}_t] = 0.$$

- (b) Assume that $C_{i,J}$ has finite second moment given $\mathcal{D}_t$, for $t \geq I \geq i > t - J$. Show that $\text{CDR}_{i,t+1}, \dots, \text{CDR}_{i,i+J}$ are uncorrelated and that

$$\mathbb{E}[(C_{i,J} - \mathbb{E}[C_{i,J} | \mathcal{D}_t])^2 | \mathcal{D}_t] = \sum_{s=t+1}^{i+J} \text{Var}(\text{CDR}_{i,s} | \mathcal{D}_t).$$

Explain how this result is to be understood.

- (c) Since we are frequently interested in the conditional mean squared error of $\hat{C}_{i,J}^{(t)}$, we now also study it for the claims development result. Show that

$$\mathbb{E}[(\text{CDR}_{i,t+1})^2 | \mathcal{D}_t] = \text{Var}(\mathbb{E}[C_{i,J} | \mathcal{D}_{t+1}] | \mathcal{D}_t)$$

- (d) For $0 \leq k \leq i + J - t - 1$, show that

$$\begin{aligned} & \mathbb{E}[\mathbb{E}[(\text{CDR}_{i,t+k+1})^2 | \mathcal{D}_{t+k}] | \mathcal{D}_t] \\ &= \text{Var}(\mathbb{E}[C_{i,J} | \mathcal{D}_{t+k+1}] | \mathcal{D}_t) - \text{Var}(\mathbb{E}[C_{i,J} | \mathcal{D}_{t+k}] | \mathcal{D}_t) \end{aligned}$$

3. In this exercise we prove Lemma 5.2.

- (a) Show that the chain-ladder factor estimators $\hat{f}_j^{(t)}$ are unbiased.
- (b) Show that the chain-ladder factor estimators $\hat{f}_j^{(t)}$ are uncorrelated.
- (c) Prove the following lemma:

Lemma 5.4 *Assume that $P_1, \dots, P_H$ are independent, unbiased estimators for μ with variances $\sigma_1^2, \dots, \sigma_H^2$. Then the minimum variance linear combination of $P_1, \dots, P_H$ is given by*

$$G = \frac{\sum_{h=1}^H \frac{P_h}{\sigma_h^2}}{\sum_{h=1}^H \frac{1}{\sigma_h^2}},$$

with

$$\text{Var}(G) = \left(\sum_{h=1}^H \frac{1}{\sigma_h^2} \right)^{-1}.$$

- (d) Use the lemma to conclude that the chain-ladder factor estimators $\hat{f}_j^{(t)}$ are optimal in the sense that

$$\begin{aligned} & \text{Var}(\hat{f}_j^{(t)} \mid C_{(t-j-1) \wedge I, j}, \dots, C_{1, j}) \\ &= \min_{\gamma_i \in \mathbb{R}: \sum_{i=1}^{(t-j-1) \wedge I} \gamma_i = 1} \left\{ \text{Var} \left(\sum_{i=1}^{(t-j-1) \wedge I} \gamma_i \frac{C_{i, j+1}}{C_{i, j}} \mid C_{(t-j-1) \wedge I, j}, \dots, C_{1, j} \right) \right\}. \end{aligned}$$

and the conditional variance is given by

$$\text{Var}(\hat{f}_j^{(t)} \mid C_{(t-j-1) \wedge I, j}, \dots, C_{1, j}) = \frac{s_j^2}{\sum_{l=1}^{(t-j-1) \wedge I} C_{l, j}}.$$

4. In this exercise, we prove Theorem 5.1.

- (a) Show that when $i + J \leq t$ we have

$$C_{i, J} = \hat{C}_{i, J}^{\text{Tw}_1(t)} = \hat{\mu}_i^{\text{Tw}_1(t)} = \hat{C}_{i, J}^{\text{CL}(t)}$$

- (b) Show that for $t \geq I \geq i > t - J$

$$\hat{C}_{i, J}^{\text{Tw}_1(t)} = \hat{\mu}_i^{\text{Tw}_1(t)} \sum_{j=0}^J \hat{\eta}_j^{\text{Tw}_1(t)} = \hat{\mu}_i^{\text{Tw}_1(t)},$$

and that

$$\hat{C}_{i, J}^{\text{Tw}_1(t)} = C_{i, t-i} \frac{\sum_{j=0}^{t-i-1} \hat{\eta}_j^{\text{Tw}_1(t)}}{\sum_{j=0}^{t-i} \hat{\eta}_j^{\text{Tw}_1(t)}} \cdots \frac{\sum_{j=0}^J \hat{\eta}_j^{\text{Tw}_1(t)}}{\sum_{j=0}^{J-1} \hat{\eta}_j^{\text{Tw}_1(t)}}$$

(c) Show that for $t \geq I \geq i > t - J$

$$\hat{C}_{i,J}^{\text{Tw}_1(t)} = \hat{\mu}_i^{\text{Tw}_1(t)} = \hat{C}_{i,J}^{\text{CL}(t)}$$

Hint: Use that under the assumptions of Theorem 5.1 we have on $\mathcal{D}_t$ for $t \geq I$ and all $0 \leq j \leq J$

$$\sum_{i=1}^{(t-j) \wedge I} C_{i,j} = \sum_{i=1}^{(t-j) \wedge I} \hat{\mu}_i^{\text{Tw}_1(t)} \sum_{k=0}^j \hat{\eta}_k^{\text{Tw}_1(t)}.$$

(d) Show that

$$\begin{aligned} \hat{\eta}_0^{\text{Tw}_1(t)} &= \prod_{k=0}^{J-1} \frac{1}{\hat{f}_k^{(t)}}, \\ \hat{\eta}_j^{\text{Tw}_1(t)} &= \prod_{k=j}^{J-1} \frac{1}{\hat{f}_k^{(t)}} \left(1 - \frac{1}{\hat{f}_{j-1}^{(t)}} \right), \quad 1 \leq j \leq J. \end{aligned}$$

5. In this exercise, we prove Theorem 5.3 and discuss the implementation of the gamma-gamma Bayesian chain-ladder method. Recall first that, under the gamma-gamma Bayesian chain-ladder assumptions,

$$\mathbb{E} \left[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 | \mathcal{D}_t \right] = \text{Var}(\mathbb{E}[C_{i,J} | \Theta, \mathcal{D}_t] | \mathcal{D}_t) + \mathbb{E}[\text{Var}(C_{i,J} | \Theta, \mathcal{D}_t) | \mathcal{D}_t].$$

(a) Assume $i > t - J$. Show that

$$\text{Var}(C_{i,J} | \Theta, \mathcal{D}_t) = \Theta_{J-1}^{-2} \left(C_{i,t-i} \sigma_{J-1}^2 \prod_{j=t-i}^{J-2} \Theta_j^{-1} + \text{Var}(C_{i,J-1} | \Theta, \mathcal{D}_t) \right).$$

Then follow the derivations of (5.5) to show that

$$\begin{aligned} &\text{Var}(C_{i,J} | \Theta, \mathcal{D}_t) \\ &= C_{i,t-i} \left(\sigma_{t-i}^2 \Theta_{t-i}^{-2} \prod_{n=t-i+1}^{J-1} \Theta_n^{-2} + \sum_{j=t-i+1}^{J-1} \sigma_j^2 \Theta_j^{-2} \prod_{l=t-i}^{j-1} \Theta_l^{-1} \prod_{n=j+1}^{J-1} \Theta_n^{-2} \right). \end{aligned}$$

(b) Taking the conditional expected value, conditioning on $\mathcal{D}_t$, of $\text{Var}(C_{i,J} | \Theta, \mathcal{D}_t)$ yields the expected process variance. Show that

$$\mathbb{E}[\text{Var}(C_{i,J} | \Theta, \mathcal{D}_t) | \mathcal{D}_t] = C_{i,J}^{\text{BCL}(t)} \sum_{j=t-i}^{J-1} \sigma_j^2 \prod_{n=j}^{J-1} \left(\hat{f}_n^{\text{BCL}(t)} (1 + \Psi_n^{(t)}) \right).$$

(c) Show that

$$\text{Var}(\mathbb{E}[C_{i,J} | \Theta, \mathcal{D}_t] | \mathcal{D}_t) = (C_{i,J}^{\text{BCL}(t)})^2 \left(-1 + \prod_{j=t-i}^{J-1} (1 + \Psi_j^{(t)}) \right),$$

and argue that this proves Theorem 5.3.

- (d) The conditional mean squared error in the current setup can be related to Mack's model. Recall that in one of the degenerate cases, it is assumed that

$$\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j} > \sigma_j^2, \quad \text{for } t \geq I \text{ and all } 0 \leq j \leq J-1.$$

We now strengthen this assumption for $t \geq I$ to

$$\sum_{l=1}^{(t-j-1) \wedge I} C_{l,j} \geq \sum_{l=1}^{I-j-1} C_{l,j} \gg \sigma_j^2, \quad (5.20)$$

i.e. σ_j^2 is much smaller than j 'th column sum of the cumulative trapezoid at time I . Use (5.20) to show that

$$\mathbb{E}[(C_{i,J} - C_{i,J}^{\text{BCL}(t)})^2 | \mathcal{D}_t] \approx (C_{i,J}^{\text{BCL}(t)})^2 \left(\sum_{j=t-i}^{J-1} \frac{\sigma_j^2}{C_{i,j}^{\text{BCL}(t)}} + \sum_{j=t-i}^{J-1} \Psi_j^{(t)} \right), \quad (5.21)$$

for any $\gamma_j > 1$ (subject to the existence of the posterior second moment). Argue why the right-hand side is a lower bound for the left-hand side.

- (e) To relate the gamma-gamma BCL model to Mack's model, we consider the assumption $\Theta_j^{-1} \approx \hat{f}_j^{(t)}$. Discuss whether this assumption is reasonable in the current degenerate context. Argue that then $\sigma_j^2 \approx (\hat{s}_j^2 / (\hat{f}_j^{(t)})^2)$, and consequently that the conditional mean squared error can be approximated by Mack's conditional mean squared error.
- (f) Assume $\Theta_j^{-1} \approx \hat{f}_j^{(t)}$. Using the dataset MW2014 from the R package **ChainLadder**, examine whether the assumption (5.20) is satisfied.
- (g) Use (5.21) to estimate the conditional mean squared error of the gamma-gamma BCL model. Compare with that of Mack's model.
6. Recall the Bornhuetter–Ferguson ultimate

$$\hat{C}_{i,J}^{\text{BF}(t)} = \hat{\kappa}_{t-i}^{\text{CL}(t)} \hat{C}_{i,J}^{\text{CL}(t)} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i.$$

Show that $\hat{C}_{i,J}^{\text{BF}(t)}$ is a convex combination of the chain-ladder ultimate and the prior $\hat{\mu}_i$, and that it is monotone in the credibility factor $\hat{\kappa}_{t-i}^{\text{CL}(t)}$. Interpret the edge cases $\hat{\kappa}_{t-i}^{\text{CL}(t)} \rightarrow 0$ and $\hat{\kappa}_{t-i}^{\text{CL}(t)} \rightarrow 1$.

7. Under Mack's model, define the standardised development residuals by

$$R_{i,j} = \frac{\sqrt{C_{i,j}}(C_{i,j+1}/C_{i,j} - \hat{f}_j^{(t)})}{\hat{s}_j^{(t)}}, \quad 1 \leq i \leq (t-j-1) \wedge I, \quad 0 \leq j \leq J-1.$$

Show that, conditionally on $\mathcal{D}_t$, the residuals satisfy $\mathbb{E}[R_{i,j} | \mathcal{D}_t] = 0$ and $\text{Var}(R_{i,j} | \mathcal{D}_t) \approx 1$, and argue when they are approximately uncorrelated across accident years for fixed j .

8. For $t \geq I$, recall

$$\hat{C}_{i,J}^{\text{BF}(t)} = \hat{\kappa}_{t-i}^{\text{CL}(t)} \hat{C}_{i,J}^{\text{CL}(t)} + (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) \hat{\mu}_i.$$

Define the total Bornhuetter–Ferguson reserve

$$\mathcal{R}^{\text{BF}(t)} = \sum_{i=t-J+1}^I (\hat{C}_{i,J}^{\text{BF}(t)} - C_{i,t-i}).$$

For deterministic perturbations (d_i) and $\varepsilon \in \mathbb{R}$, set $\hat{\mu}_i(\varepsilon) = \hat{\mu}_i + \varepsilon d_i$ and let $\mathcal{R}^{\text{BF}(t)}(\varepsilon)$ be the reserve computed with $\hat{\mu}_i(\varepsilon)$.

(a) Show that

$$\mathcal{R}^{\text{BF}(t)}(\varepsilon) = \sum_{i=t-J+1}^I (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) (\hat{\mu}_i + \varepsilon d_i).$$

(b) Derive

$$\left. \frac{\partial}{\partial \varepsilon} \mathcal{R}^{\text{BF}(t)}(\varepsilon) \right|_{\varepsilon=0} = \sum_{i=t-J+1}^I (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}) d_i.$$

(c) If $|d_i| \leq d$ for all i , show that

$$|\mathcal{R}^{\text{BF}(t)}(\varepsilon) - \mathcal{R}^{\text{BF}(t)}(0)| \leq |\varepsilon| d \sum_{i=t-J+1}^I (1 - \hat{\kappa}_{t-i}^{\text{CL}(t)}),$$

and further comment on which accident years dominate prior-sensitivity.

9. Consider a Bayesian multiplicative incremental model

$$X_{i,j} \mid \mu_i, \eta_j \sim \text{Poisson}(\eta_j \mu_i), \quad i + j \leq t,$$

with independent priors

$$\mu_i \sim \text{Gamma}(a_\mu, b_\mu), \quad \eta_j \sim \text{Gamma}(a_\eta, b_\eta),$$

where $i = 1, \dots, I$ and $j = 1, \dots, J$, and identifiability constraint $\eta_0 = 1$.

- (a) Derive the full conditional distribution of each μ_i given the remaining parameters and the data.
- (b) For $j \geq 1$, derive the full conditional distribution of η_j .
- (c) Give the mean of one future incremental cell $X_{i,j}^*$ with $i + j > t$, conditional on the data and posterior draws of (μ_i, η_j) . Derive by integration the corresponding mean when only conditioning on the data.

Practical exercises

1. In R we consider the claim triangle dataset `austri1autoBI7895` from the R package `CASdatasets` [Dutang and Charpentier, 2024].

- (a) Consider the subset `paid`, which is the claim triangle of incremental claims. Implement the chain-ladder algorithm, and provide the estimated total ultimate and reserve.

Hint: In this dataset we have $t = I = J + 1$ i.e. a true triangle. Be careful when indexing in R.

- (b) Use the calculations from subexercise (a) to compute $\hat{\kappa}_{I-i}^{\text{CL}(t)}$ for all i where $\hat{\kappa}_{I-i}^{\text{CL}(t)}$ is well-defined. Consider the following prior

$$\hat{\mu}_i = \begin{cases} \hat{C}_{i,J}^{\text{CL}(t)}, & \text{for } \hat{\kappa}_{I-i}^{\text{CL}(t)} > 0.8 \\ \text{expo}_i \cdot 0.37, & \text{for } \hat{\kappa}_{I-i}^{\text{CL}(t)} < 0.8 \end{cases} \quad 1 \leq i \leq I.$$

Implement the Bornhuetter–Ferguson algorithm, and provide the estimated ultimates for each accident year.

- (c) Under the assumptions of Mack’s chain-ladder, compute for each accident year the estimator of the conditional variance, $\widehat{\text{Var}}(C_{i,J}|\mathcal{D}_t)$ given by (5.7). Compute the proposed estimator of the squared bias of the conditional mean squared error. Provide for each accident year an estimate of the conditional mean squared error of the estimated ultimate claim sizes.
2. The purpose of this exercise is to estimate a tail development factor. In R, we consider the claim triangle dataset `usmedclaim` from the R package `CASdatasets` [Dutang and Charpentier, 2024]. For a suitable threshold j_0 , implement the extrapolation using exponential decay of the chain-ladder factors. Provide, for each accident year i , an estimate for the ultimate cumulative claim size with and without estimated tail development factor.
 3. In this exercise, we consider the Tweedie model. Consider the data `austri1autoBI7895` from the R package `CASdatasets` [Dutang and Charpentier, 2024].
 - (a) Use the R function `glmReserve()` from the R package `ChainLadder` to determine the total (sum of) conditional mean squared errors in the over-dispersed Poisson model. Compare with the conditional mean squared error of Mack’s model.

Hint: If the cumulative claim payments are stored as `triangle` then the over-dispersed Poisson model can be fitted via `glmReserve(triangle, var.power = 1, mse.method = "formula")`. Mack’s model may be fitted using the function `MackChainLadder()`.
 - (b) Plot the `glmReserve()` object from the previous subexercise to illustrate the development of payments for different accident years. Assess whether payments in different accident years develop similarly. Use the variable `exposure` to correct the triangle in a suitable way, redefine the model and construct the plot again. Comment on the results.

- (c) As we are dealing with an exponential dispersion family, it is possible to calculate the AIC for different choices of p . As this is rather complicated (at least analytically), we can compare approximated likelihoods for different p 's. Run the code `tweedieReserve(triangle = triangle, p.optim = TRUE, p.check = c(1,1.1,1.2,1.3,1.4,1.5,1.6,1.7,1.8,1.9,2), design.type = c(1,1,0), rereserving = FALSE, bootstrap = 0)`. Select the model (in terms of p), fit it, and compare the resulting conditional mean squared error with that of the over-dispersed Poisson and Mack's model. Repeat with the exposure corrected triangle.
- (d) Error can also be quantified with bootstrapping. Run `glmReserve(triangle, var.power = p, mse.method = "bootstrap", nsim = b)` for the choices $b = 10, 25, 50, 100, 200, 500, 1000, 2000$, and extract the total (sum) conditional mean squared error for each b . Plot the error as a function of b , identify where the curve stabilises, report the corresponding error, and compare it with the results from the former subexercises.
4. In this exercise, we implement a Bayesian multiplicative incremental reserving model and compare it with chain-ladder.
- (a) Use the incremental triangle `paid` from the data `austri1autoBI7895` from the R package `CASdatasets` [Dutang and Charpentier, 2024] and fit

$$X_{i,j} \mid \mu_i, \eta_j \sim \text{Poisson}(\eta_j \mu_i), \quad i + j \leq t,$$

with identifiability constraint $\eta_0 = 1$. For a standard triangle, take $t = I + 1$.

- (b) Choose independent Gamma priors for μ_i and for η_j with $j \geq 1$ (keeping $\eta_0 = 1$ fixed), and draw posterior samples, either by Monte Carlo integration, or by implementing a Gibbs sampler as in Section ??, using the conditional distributions.
- (c) For each posterior draw, simulate the future cells with $i + j > t$ and compute total reserve

$$R^* = \sum_{i+j>t} X_{i,j}^*.$$

Report posterior mean and a 95% credible interval for R^* , compare with the chain-ladder reserve, and repeat under a second prior choice to assess prior sensitivity.

Chapter 6

Foundations of statistical learning

This chapter provides a short overview of the main concepts underpinning statistical learning. These are conceptually distinct from those of classical statistical inference, although we do not pursue that distinction further here. Statistical learning methods are increasingly relevant in actuarial science and insurance, where they play an important role in prediction, classification, and the analysis of complex insurance data. We begin with model assessment and selection, move to model averaging, and conclude with a survey of widely used learning models.

6.1 Model assessment and selection

Model selection and assessment are central in machine learning, particularly for real-world problems such as those encountered in insurance. The primary goal in this context is to build a model that can predict outcomes accurately. This section covers the fundamental principles of selecting the best model and evaluating its performance.

Prediction as the main objective

The focus in machine learning applications, especially in predictive modelling for insurance, is on the ability of the model to *predict* future events. This is the driving factor behind model selection. To ensure that the chosen model performs well on new, unseen data, a reliable measure of predictive performance is necessary. Two points are worth noting:

1. It is crucial that this measure of predictive performance is robust, and not dependent on the specific dataset used for training the model. If the evaluation metric is data-driven, it can lead to overfitting, i.e. the model performs well on the training data but poorly on new data.

2. The decision-making process in selecting a model must be carried out with caution. Poor decisions at this stage can lead to choosing a model that appears to perform well during training but fails in practical applications.

While goodness-of-fit, i.e. how well a model captures the data it was trained on, is an important component in statistics, it is a secondary concern compared to predictive performance in machine learning. Nonetheless, a model that does not fit the training data at all is likely to be subject to substantial model error.

One advantage of focusing on prediction over goodness-of-fit is the flexibility it provides in choosing modelling techniques. More specifically, it allows for the use of non-statistical methods (which may sometimes seem ad hoc) alongside rigorous statistical methods. The ultimate goal is to find the model that offers the best predictive accuracy, regardless of the technique used to develop it.

Assessing prediction error

Once a model has been selected based on its predictive performance, it is crucial to assess the prediction error. This step quantifies performance on unseen data and provides an estimate of the generalisation error (which does not follow from the model selection phase). Accurate assessment of prediction error is essential for understanding the reliability of the model, especially in non-life insurance lines.

Consider a set of inputs denoted by X and a target variable Y . The goal is to predict Y using a model $\hat{f}(X)$ that has been estimated from a *training set* $\mathcal{T}$. To evaluate the model's predictive accuracy, we use a general *loss function*, represented by $L(Y, \hat{f}(X))$. For instance, depending on the objective, one may use:

- L^p loss, defined as $L(u, v) = |u - v|^p$, with $p \in \mathbb{N}$. The case $p = 1$ is more robust to outliers, while $p = 2$ gives the familiar squared-error loss known from linear models. Minimising the expected squared loss yields the conditional mean, while minimising the absolute loss yields the conditional median.
- Negative log-likelihood, defined for a single observation as

$$L(Y, \hat{f}(X)) = -\log \mathbb{P}_{\theta(X)}(Y),$$

where $\theta(X)$ denotes a probabilistic model for Y given X . This loss is widely used when the model outputs a full conditional distribution, for instance in classification problems. However, one can also use such a function even when the model is not probabilistic.

- Classification error, defined as

$$L(u, v) = \begin{cases} 1 & \text{if } u \neq v, \\ 0 & \text{if } u = v. \end{cases}$$

This loss is used when the prediction itself is a class label. Minimising the expected loss yields the most probable class, that is, the so-called Bayes classifier.

- Quantile loss, defined for $\tau \in (0, 1)$ as

$$L_\tau(u, v) = \begin{cases} \tau(u - v), & u \geq v, \\ (\tau - 1)(u - v), & u < v. \end{cases}$$

This loss is used when the aim is to predict a conditional quantile rather than a conditional mean, and is particularly useful when underprediction and overprediction have different consequences. Minimising the expected loss yields the conditional τ -quantile.

6.1.1 Types of errors in predictive models

The first step is to distinguish carefully between the error of a *fitted model* and the error of a *learning procedure*. This is easy to blur because the predictor $\hat{f}$ is itself random before the training sample is observed. The distinction matters throughout this chapter, for instance cross-validation and bootstrap methods mainly target average predictive performance of a procedure, a genuinely independent test sample is the natural tool for assessing the specific fitted model that is to be selected.

Definition 6.1 (Test error, or generalisation error) The *test error* or *generalisation error* measures how well the model performs on new, unseen data. It is defined as:

$$\text{Err}_{\mathcal{T}} = \mathbb{E}[L(Y, \hat{f}(X)) \mid \mathcal{T}]$$

Here, $\mathcal{T}$ represents the training set, and the expectation $\mathbb{E}$ is taken over a the distribution of a sample (X, Y) from the population, conditionally on the training set.

The quantity $\text{Err}_{\mathcal{T}}$ is specific to the observed training set $\mathcal{T}$, and the expectation is taken with respect to a fresh draw from the target population (or population of interest). It is therefore the most relevant target for final model assessment of the fitted predictor, but it is also the hardest to estimate accurately from a finite dataset. In practice, a test set only provides, at best and upon fancy procedures, a noisy empirical and averaged approximation to $\text{Err}_{\mathcal{T}}$.

Definition 6.2 (Expected test error, or expected prediction error) The *expected test error* or *expected prediction error* is another critical measure, which averages over all possible training sets:

$$\text{Err} = \mathbb{E}[L(Y, \hat{f}(X))]$$

This expected error averages over both sources of randomness, the future observation and the training sample, and is therefore the natural target when comparing *procedures* (model classes, tuning rules, fitting pipelines) rather than one specific realised fit. It is also the quantity most naturally estimated by resampling methods such as cross-validation and bootstrap.

Definition 6.3 (Training error) The *training error* is the error calculated on the same data that was used to train the model:

$$\overline{\text{err}} = \frac{1}{n} \sum_{i=1}^n L(Y_i, \hat{f}(X_i)), \quad \{(X_i, Y_i), i = 1, \dots, n\} = \mathcal{T}.$$

Training error is an in-sample quantity. It is useful for fitting and optimisation, but it is usually optimistic as an estimate of predictive performance because the same observations were used to construct $\hat{f}$. In particular, training error typically decreases with model complexity even in situations where test error eventually increases as variance begins to dominate bias. The expected gap between test error and training error is often called *optimism*, and covariance penalties below make this notion explicit in probabilistic models. For this reason, training error is not suitable, by itself, for model assessment.

These three errors correspond to three distinct tasks: fitting (training error), model selection and tuning (typically expected prediction error), and final model assessment (conditional test error). Keeping this separation explicit prevents several common forms of model error misjudgement and makes the role of each resampling method much clearer. We still emphasise that most methods centre around evaluating the expected prediction error, since a proper estimate of the generalisation error is often unavailable, not least due to sample size constraints.

Example 6.1 Assume that $Y = f(X) + \varepsilon$, where ε is a conditionally independent, zero-mean error term with conditional variance σ_ε^2 . We calculate Err conditional on the input for this simple model.

$$\begin{aligned} \text{Err}(x_0) &= \mathbb{E}[(Y - \hat{f}(x_0))^2 | X = x_0] \\ &= \mathbb{E}[(f(x_0) + \varepsilon - \hat{f}(x_0))^2 | X = x_0] \\ &= \sigma_\varepsilon^2 + \mathbb{E}[(f(x_0) - \hat{f}(x_0))^2 | X = x_0] \\ &= \sigma_\varepsilon^2 + \mathbb{E}[f(x_0) - \hat{f}(x_0) | X = x_0]^2 + \mathbb{E}[(\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)])^2 | X = x_0] \\ &= \sigma_\varepsilon^2 + \text{Bias}(\hat{f}(x_0))^2 + \text{Var}(\hat{f}(x_0)) \\ &= \text{Irreducible error} + \text{Bias}^2 + \text{Variance}. \end{aligned}$$

As a rule of thumb, more complex models *decrease the bias but increase the variance*. This is known as the *bias-variance tradeoff*. $\diamond$

Numeric example 6.1 (Train-test error versus model complexity) Consider now a simulation study where

$$X \sim N(0, 1), \quad Y = 0.5X + \sin(4X) + \varepsilon, \quad \varepsilon \sim N(0, 0.4^2). \quad (6.1)$$

We fit additive models (anticipated from the sequel, with natural splines) with $5 \leq p \leq 50$ degrees of freedom to $n = 100$ simulated points and compute the training error, as well as the test error on 100 independent data points from the same model. The experiment is repeated 200 times, and the resulting curves, as a function of p , are provided in Figure 6.1. We observe that training error always decreases, but test error attains a sweet spot between 10 and 20 degrees of freedom. The bias-variance tradeoff is also apparent, with the spread of the errors increasing with p . $\diamond$

Estimating the test error

If an independent sample from the target population is available, then the empirical loss on that sample is the cleanest estimator of predictive performance.

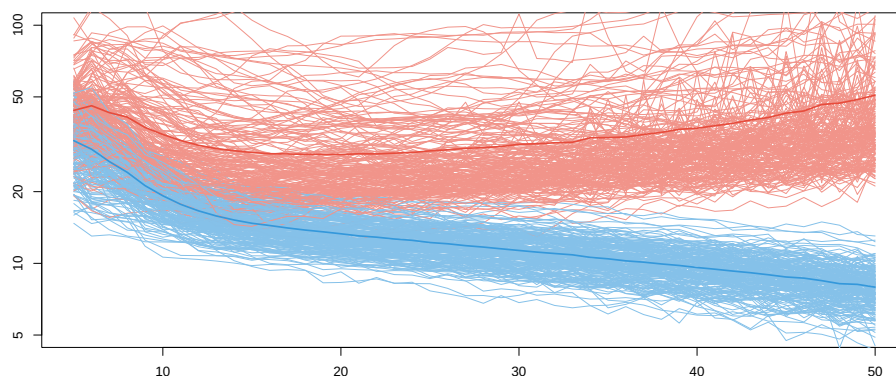


Figure 6.1: Test (light red) and training errors (light blue) as a function of degrees of freedom, together with their averages (red and blue, respectively).

In most applications, however, the available data must be used both to construct and to evaluate models, so data splitting is not merely a computational convenience but part of the statistical design of the analysis.

To separate model selection from final model assessment, it is often preferable to split the data into *three distinct sets*:

1. Training set: used to fit the model. This is where the model learns the relationships between inputs X and outputs Y .
2. Validation set: used to select the model by estimating the prediction error for different models. This process is known as *model selection*.
3. Test set: used to estimate the generalisation error, providing an unbiased evaluation of the model performance. This is crucial for *model assessment*.

A crucial point is that the test set must remain untouched throughout model development, since repeatedly checking test-set performance and making design choices accordingly turns the test set into another validation set and reintroduces selection bias. When the sample size is too small for a stable three-way split, resampling methods such as cross-validation can replace the validation set, but if a final external test sample exists it should still be reserved for the last step. In dependent-data settings (time-series, grouped risks, repeated observations on the same unit), the split itself must respect the dependence structure; otherwise the resulting error estimate can be seriously biased even if the later calculations are formally correct.

6.1.2 Cross-validation

Cross-validation (CV) is a widely used method in machine learning for estimating the expected prediction error of a model or learning procedure. It is

best interpreted as a repeated train/validation experiment carried out internally within the available sample. As such, it is primarily a tool for estimating average predictive performance, and not the conditional test error of the final fitted model.

In K -fold CV, each fit is trained on approximately $(1 - 1/K)n$ observations and validated on the remaining data, or fold. This means the resulting estimate is usually slightly pessimistic for the predictor that is later refit on all n observations. The choice of K induces a bias-variance tradeoff, with small K giving higher bias and lower computational cost, while large K reduces bias but may have higher variance.

Definition of K -fold cross-validation

Let $\mathcal{T} = (X_i, Y_i)_{i=1}^n$ and let $\kappa : \{1, \dots, n\} \rightarrow \{1, \dots, K\}$ denote a fold assignment. In practice, folds are typically chosen to have nearly equal sizes, and one may impose additional constraints such as stratification or grouping.

For K -fold cross-validation, the procedure is:

1. For each fold k , fit the model on the data excluding the observations with $\kappa(i) = k$. Denote the resulting predictor by $\hat{f}^{-k}$.
2. Predict each observation using only the predictor trained without that observation.
3. Average the validation losses:

$$\text{CV}(\hat{f}) = \frac{1}{n} \sum_{i=1}^n L(Y_i, \hat{f}^{-\kappa(i)}(X_i))$$

Equivalently,

$$\text{CV}(\hat{f}) = \frac{1}{n} \sum_{k=1}^K \sum_{i: \kappa(i)=k} L(Y_i, \hat{f}^{-k}(X_i)),$$

which makes clear that each observation appears exactly once as validation data.

When $K = n$ we recover *leave-one-out cross-validation* (LOOCV), which was already encountered to select bandwidths for some kernel methods, see Definitions 2.4 and 2.5. In many applications, $K = 5$ or 10 gives a practical compromise, and repeated CV (with several random fold assignments) can reduce Monte Carlo variability due to a particular split.

Cross-validation with tuning parameters

Suppose now that the model depends on a tuning parameter α (possibly vector-valued), such as a penalty level, tree depth, number of neighbours, or number of boosting iterations (unfamiliar concepts are described in subsequent sections in detail, when dealing with concrete models). Cross-validation should then be used to estimate predictive performance *as a function of α* .

For a fixed value of α , the CV estimate is:

$$\text{CV}(\hat{f}, \alpha) = \frac{1}{n} \sum_{i=1}^n L(Y_i, \hat{f}^{-\kappa(i)}(X_i, \alpha)).$$

Evaluating this expression over a grid or search set of candidate values yields a CV curve (or surface) as a function of α . One then selects

$$\hat{\alpha} \in \underset{\alpha}{\operatorname{argmin}} \text{CV}(\hat{f}, \alpha),$$

possibly with an additional stability rule when several candidates perform similarly.

After choosing $\hat{\alpha}$, the model is typically refit on all data available for fitting (e.g. the training and validation sets combined, or the full sample in a pure CV approach, when data is scarce). It is important to distinguish:

- *model selection*, which chooses $\hat{\alpha}$ using CV; and
- *model assessment*, which estimates the performance of the entire procedure including the tuning step.

If one reports the minimum value of the same CV curve used to choose $\hat{\alpha}$, the reported error is usually slightly optimistic because α has been selected adaptively. A separate test set or nested cross-validation is needed when an approximately unbiased assessment of the full selection procedure is required.

6.1.3 Bootstrap

The bootstrap is a general-purpose method in statistics used to estimate the accuracy of a model, especially when the theoretical distribution of the estimator is complex or unknown. The method involves *resampling with replacement* from the original dataset and is particularly effective for estimating the expected prediction error. However, it is important to note that the bootstrap method *works poorly for estimating the conditional test error*.

One advantage is its simplicity and versatility, as it is straightforward to implement and can be applied to a wide range of problems. It does not rely heavily on assumptions about the underlying distribution, making it applicable to different types of data and models. However, the bootstrap can be computationally heavy, particularly for large datasets or complex models.

Definition of bootstrap for B replications

We formalise the standard resampling scheme and fix notation for later error estimators. The bootstrap replaces the unknown population distribution by the empirical distribution of the observed sample and then repeats the statistical analysis under that empirical law. As before, let $\mathcal{T} = \{Z_1, \dots, Z_n\}$ denote the original sample (with $Z_i = (X_i, Y_i)$ in predictive problems).

1. For each replication $b = 1, \dots, B$, draw a bootstrap sample $\mathcal{T}^b = (Z_1^{*b}, \dots, Z_n^{*b})$ of size n *with replacement* from $\mathcal{T}$.

2. Compute the statistic or fitted model of interest on each bootstrap sample.
3. Use the empirical distribution of the B bootstrap replicates to estimate variability, bias, quantiles, or prediction error.

Because sampling is with replacement, each bootstrap sample contains duplicates and omits some original observations. The expected fraction of distinct observations in a size- n bootstrap sample is approximately $1 - e^{-1} \approx 0.632$, a fact that later explains out-of-bag error estimators and the “.632” bootstrap correction.

The number of replications B controls Monte Carlo variability, with larger B stabilising the numerical approximation to the bootstrap distribution, but does not fix model misspecification or poor resampling design. One may for instance use the R package `bootstrap` for implementation.

Example 6.2 If we are interested in a specific statistic $S(\mathcal{T})$ computed from the dataset, we can use the bootstrap samples to estimate the variance of this statistic. Indeed, the variance of $S(\mathcal{T})$ across the B bootstrap samples can be estimated using the following formula:

$$\widehat{\text{Var}}(S(\mathcal{T})) = \frac{1}{B-1} \sum_{b=1}^B (S(\mathcal{T}^b) - \bar{S})^2,$$

where $\bar{S} = \frac{1}{B} \sum_{b=1}^B S(\mathcal{T}^b)$ is the mean of the statistic across all bootstrap samples. $\diamond$

More generally, the bootstrap allows us to assess the variability or uncertainty of a statistic, which is not limited to variance; it can be used to construct confidence intervals, bias estimates, and any other statistical measures.

How not to use the bootstrap for estimating prediction error

As mentioned, the bootstrap can also be applied to estimate expected prediction error. A straightforward, but usually flawed, estimator is:

$$\widehat{\text{Err}}_{\text{boot}} = \frac{1}{B} \frac{1}{n} \sum_{b=1}^B \sum_{i=1}^n L(Y_i, \hat{f}^{*b}(X_i)),$$

where $\hat{f}^{*b}$ is the predictor fitted on the b -th bootstrap sample. The problem is that this estimator evaluates many observations using models that were trained on those same observations, often several times. The training and evaluation roles are therefore not separated.

This induces the same type of optimism as training error: bootstrap resampling has been used, but the dependence between fitting and evaluation has not been broken. To estimate predictive error, each observation should be evaluated only on bootstrap models for which that observation was not used in fitting.

Correct approach: leave-one-out bootstrap estimator

Indeed, a more reliable method for estimating prediction error is the leave-one-out bootstrap estimator (also called an out-of-bag bootstrap estimator), defined as:

$$\widehat{\text{Err}}^{(1)} = \frac{1}{n} \sum_{i=1}^n \frac{1}{|C^{-i}|} \sum_{b \in C^{-i}} L(Y_i, \hat{f}^{*b}(X_i)),$$

where C^{-i} is the set of indices of the bootstrap samples that do *not* contain the i -th observation. An observation which is not in a particular bootstrap sample is sometimes called an *out-of-bag* (OOB) observation for that sample.

This approach breaks the most direct dependence by evaluating observation i only on models that did not use i for fitting. It is therefore much closer in spirit to cross-validation than the naive bootstrap estimator above.

Since

$$\mathbb{P}(i \notin \mathcal{T}^{*b}) = \left(1 - \frac{1}{n}\right)^n \approx e^{-1},$$

each observation has on average about $0.368B$ out-of-bag predictions available. This gives a practical rule of thumb for choosing B if OOB error estimation is the main goal.

A practical issue remains: for finite B , some observations may have $C^{-i} = \emptyset$. This becomes rare when B is large, but can occur in short bootstrap runs. One may then increase B or omit such terms from the average, while reporting the effective number of observations contributing to the estimate.

A curious bootstrap estimator: the “632” estimator

The leave-one-out bootstrap estimator can itself be unstable or biased for highly adaptive fitting rules. A classical compromise is the “632” bootstrap estimator, which blends the training error and the OOB bootstrap error using weights motivated by bootstrap inclusion probabilities. From the calculation above, $\mathbb{P}(i \notin \mathcal{T}^{*b}) \approx e^{-1}$, so the corresponding inclusion probability is approximately $1 - e^{-1} \approx 0.632$. The “632” estimator is therefore defined by

$$\widehat{\text{Err}}^{(.632)} = 0.368 \overline{\text{err}} + 0.632 \widehat{\text{Err}}^{(1)}.$$

The idea is to compromise between two biased estimators, on the one hand the training error (often too small), and on the other the OOB bootstrap estimate (which can be too large in some settings). This can work well in practice, but it is not universally preferable. In severe overfitting settings the training-error component can still induce optimism. Many variants appear in the literature, although their analysis is deeper than the simple heuristic provided above.

6.1.4 The mathematics behind the bootstrap

We now formalise the bootstrap resampling mechanism and some of its distributional properties. The key point is to view bootstrap quantities as random

functionals of the empirical measure. The aim is to provide basic results which shed light on how bootstrap routines can be analysed mathematically.

Defining bootstrap samples

The classical bootstrap defined above goes back to Efron [1979], and works by resampling n times with replacement from the original sample $X_1, \dots, X_n$ and assigning probability $1/n$ to each observation. A more formal construction of the bootstrap sample is as follows. Let $\tau_1^{(n)}, \dots, \tau_n^{(n)}$ be discrete uniform random variables on the set $\{1, \dots, n\}$ and independent of $X_1, \dots, X_n$. We may then define the *bootstrap sample* $X_1^*, \dots, X_n^*$ by

$$X_i^* := X_{\tau_i^{(n)}}.$$

In general, we use the superscript $*$ to indicate that a quantity is constructed via bootstrapping, distinguishing it from the non-bootstrapped version. The following results establish some basic properties of bootstrap samples.

Proposition 6.1 (Conditional distribution) *The following holds.*

1. The $X_1^*, \dots, X_n^*$ are conditionally independent given $X_1, \dots, X_n$.
2. The conditional distribution of X_j^* given $X_1, \dots, X_n$ is the empirical measure $\mathbb{P}^{(n)}$ given by

$$\mathbb{P}^{(n)}(\cdot) = \frac{1}{n} \sum_{i=1}^n 1_{(X_i \in \cdot)}$$

Having access to the conditional distribution then allows for explicit calculation of resampling probabilities, which turn, unexpectedly, to be uniform.

Corollary 6.1 (Uniformity of bootstrap resamples) *For any $j_1, \dots, j_n \in \{1, \dots, n\}$,*

$$\mathbb{P}((X_1^*, \dots, X_n^*) = (X_{j_1}, \dots, X_{j_n}) \mid X_1, \dots, X_n) = \frac{1}{n^n}.$$

More generally, one can compute integrals of functionals of the original sample and a bootstrap sample, by averaging across the original uniform random variables.

Lemma 6.1 (Computing conditional expectations) *Let $Z := f(X_1, \dots, X_n, X_1^*, \dots, X_n^*)$ with f a measurable function. Assuming that Z is integrable, we have*

$$\mathbb{E}[Z \mid X_1, \dots, X_n] = \mathbb{E}_{(\tau_1^{(n)}, \dots, \tau_n^{(n)})}[Z], \quad \mathbb{P}\text{-a.s.}$$

where the subscript indicates integrating with respect to (the image measures of) the variables $\tau_1^{(n)}, \dots, \tau_n^{(n)}$ only.

Proof. We verify the defining property of conditional expectation by checking the identity against sets in $\sigma(X_1, \dots, X_n)$. Using Fubini's theorem,

$$\mathbb{E}_{(\tau_1^{(n)}, \dots, \tau_n^{(n)})}[Z] = \int \cdots \int f(X_1, \dots, X_n, X_{k_1}, \dots, X_{k_n}) \, d\mu(k_1) \cdots d\mu(k_n)$$

where $\mu := \tau_1^{(n)}(\mathbb{P})$. Clearly, $\mathbb{E}_{(\tau_1^{(n)}, \dots, \tau_n^{(n)})}[Z]$ is $\sigma(X_1, \dots, X_n)$ -measurable. Let $A \in \sigma(X_1, \dots, X_n)$. We may assume that A is of the form $A = (X_1, \dots, X_n)^{-1}(B)$ for some measurable subset $B \subseteq \mathbb{R}^n$. Then 1_A only depends on $X_1, \dots, X_n$. We may write $Z = g(X_1, \dots, X_n, \tau_1^{(n)}, \dots, \tau_n^{(n)})$ for some function g . Letting $h_A := 1_A g$ and $\nu := X_1(\mathbb{P})$, we again get by Fubini and the change of variable formula that

$$\begin{aligned} \mathbb{E}[1_A Z] &= \int \cdots \int h_A(x_1, \dots, x_n, k_1, \dots, k_n) \, d\mu(k_1) \cdots d\mu(k_n) \, d\nu(x_1) \cdots d\nu(x_n) \\ &= \int \cdots \int h_A(X_1, \dots, X_n, k_1, \dots, k_n) \, d\mu(k_1) \cdots d\mu(k_n) \, d\mathbb{P} \\ &= \int \mathbb{E}_{(\tau_1^{(n)}, \dots, \tau_n^{(n)})}[h_A(X_1, \dots, X_n, \tau_1^{(n)}, \dots, \tau_n^{(n)})] \, d\mathbb{P} \\ &= \mathbb{E}[1_A \mathbb{E}_{(\tau_1^{(n)}, \dots, \tau_n^{(n)})}[Z]]. \end{aligned}$$

This completes the proof. $\square$

The multinomial distribution

When establishing statistical properties of estimators based on bootstrap samples, it is convenient to work with a representation based on the *multinomial distribution*, which generalises the binomial distribution to more than two outcomes. Suppose we have k disjoint outcomes of an experiment, and let $I_j = 1$ if outcome j occurs and zero otherwise for $j = 1, \dots, k$. Let $p_j = \mathbb{P}(I_j = 1)$. Since the outcomes are disjoint, $I_1 + \cdots + I_k = 1$ and $p_1 + \cdots + p_k = 1$. Let

$$I = (I_1, \dots, I_k).$$

Suppose we have n iid trials $I^{(1)}, \dots, I^{(n)}$. We then say that

$$M = \sum_{i=1}^n I^{(i)}$$

is multinomially distributed with n trials and probabilities $(p_1, \dots, p_k)$. In the following, we write $M = (M_{n1}, \dots, M_{nk})$ where $M_{nj} = \sum_{i=1}^n I_j^{(i)}$. It is clear from the construction that $M_{n1} + \cdots + M_{nk} = n$. Also,

$$M_{nj} \sim \text{Bin}(n, p_j) \quad \text{and} \quad M_{nj} + M_{nj'} \sim \text{Bin}(n, p_j + p_{j'}), \quad j \neq j'.$$

Further properties of the multinomial distribution are collected in the following proposition.

Proposition 6.2 (Properties of the multinomial distribution) *Let $M = (M_{n1}, \dots, M_{nk})$ be multinomial with n trials and probabilities $(p_1, \dots, p_k)$. Then the following hold.*

1. *The support of M is $\{(m_1, \dots, m_k) \in \mathbb{N}_0^k : m_1 + \cdots + m_k = n\}$ and the density is given by*

$$\mathbb{P}(M = (m_1, \dots, m_k)) = \frac{n!}{m_1! \cdots m_k!} \prod_{i=1}^k p_i^{m_i}.$$

2. Let $\{1, \dots, k\} = A \cup B$ be a partition. The conditional distribution of $\{M_{ni} : i \in A\}$ given $\{M_{nj} : j \in B\}$ is again multinomial with $n - \sum_{j \in B} M_{nj}$ trials and probabilities

$$\frac{p_i}{\sum_{j \in A} p_j}, \quad i \in A.$$

Proof. We count permutations of outcomes to obtain the multinomial mass function. We only prove the first assertion. The second is left as an exercise for the reader. It is clear that the support is as specified. Now let $(m_1, \dots, m_k)$ have non-negative integer entries summing to n . The probability of m_1 outcomes of type 1, m_2 outcomes of type 2 and so on is $p_1^{m_1} \dots p_k^{m_k}$ by independence. We have $\binom{n}{m_1}$ possible permutations where m_1 outcomes of type 1 occur. Given one such permutation, we have $\binom{n-m_1}{m_2}$ permutations giving rise to m_2 outcomes of type 2 and so on. Therefore, the total number of permutations is

$$\binom{n}{m_1} \binom{n-m_1}{m_2} \dots \binom{n-m_1-\dots-m_{k-1}}{m_k} = \frac{n!}{m_1! \dots m_k!}.$$

This shows that the density is given as stated. $\square$

The second assertion is particularly useful when mixed moments of multinomial entries appear.

Bootstrap estimators

We now state a general template for bootstrap estimators and indicate how their fluctuations are analysed by comparison with the original estimator. A useful abstract viewpoint is the plug-in principle: if an estimator can be written as a functional of the empirical measure, say $S_n = T(\mathbb{P}^{(n)})$, then the bootstrap analogue is $S_n^* = T(\mathbb{P}^{(n)*})$, where $\mathbb{P}^{(n)*}$ is the empirical measure of the bootstrap sample. The bootstrap is then used to approximate the distribution of S_n by the conditional distribution of S_n^* given the data.

In many concrete problems, this abstract representation reduces to sums or smooth functions of sums, so it is enough to understand how bootstrap resampling acts on empirical sums. Many estimators based on a sample $X_1, \dots, X_n$ can either be expressed in the form

$$\sum_{i=1}^n f(X_i)$$

for a suitable measurable function f or be written as a function of estimators of this form. For the corresponding bootstrap estimator

$$\sum_{i=1}^n f(X_i^*),$$

the variables $X_1^*, \dots, X_n^*$ are conditionally iid given the data, but unconditional calculations are often cumbersome because the resampling distribution itself

depends on the sample. A common technique is to use limit results for the original estimator and then analyse the difference

$$\sum_{i=1}^n f(X_i^*) - \sum_{i=1}^n f(X_i),$$

which is often easier to control conditionally on $X_1, \dots, X_n$. To do so, it is convenient to use a representation in terms of multinomial counts, which makes explicit how often each original observation is reused in the bootstrap sample. We start by noting that we may write

$$X_i^* = \sum_{k=1}^n X_k 1_{(\tau_i^{(n)}=k)}.$$

From this representation, we have for every measurable f that

$$\begin{aligned} \sum_{i=1}^n f(X_i^*) &= \sum_{i=1}^n \sum_{k=1}^n f(X_k) 1_{(\tau_i^{(n)}=k)} = \sum_{k=1}^n \sum_{i=1}^n f(X_k) 1_{(\tau_i^{(n)}=k)} \\ &= \sum_{k=1}^n f(X_k) \sum_{i=1}^n 1_{(\tau_i^{(n)}=k)} = \sum_{k=1}^n f(X_k) M_{nk} \end{aligned}$$

where $M_{nk} := \sum_{i=1}^n 1_{(\tau_i^{(n)}=k)}$ so that $(M_{n1}, \dots, M_{nn})$ is multinomial with n trials and probabilities $(1/n, \dots, 1/n)$. Since $(\tau_1^{(n)}, \dots, \tau_n^{(n)})$ is independent of $(X_1, \dots, X_n)$, it follows that $(M_{n1}, \dots, M_{nn})$ is also independent of $(X_1, \dots, X_n)$.

We now specialise the general representation to bootstrap mean and variance estimators. Let $Z_1, \dots, Z_n$ be iid with mean $\mu = \mathbb{E}[Z_1]$ and variance $\sigma^2 = \text{Var}(Z_1)$. Define

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n Z_i, \quad \hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (Z_i - \hat{\mu}_n)^2,$$

and let $Z_1^*, \dots, Z_n^*$ be the bootstrap sample with associated estimators

$$\hat{\mu}_n^* = \frac{1}{n} \sum_{i=1}^n Z_i^*, \quad (\hat{\sigma}_n^2)^* = \frac{1}{n} \sum_{i=1}^n (Z_i^* - \hat{\mu}_n^*)^2.$$

The multinomial representation gives the key quadratic bound.

Lemma 6.2 (Multinomial quadratic bound) *Let $(M_{n1}, \dots, M_{nn})$ be multinomial with n trials and probabilities $(1/n, \dots, 1/n)$, independent of $Z_1, \dots, Z_n$, and assume $\mathbb{E}[Z_1^2] < \infty$. Then*

$$\mathbb{E} \left[\left(\sum_{i=1}^n (M_{ni} - 1) Z_i \right)^2 \right] = O(n), \quad \text{as } n \rightarrow \infty.$$

Proof. We compute

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{i=1}^n (M_{ni} - 1) Z_i \right)^2 \right] \\ &= \sum_{i=1}^n \mathbb{E}[(M_{ni} - 1)^2] \mathbb{E}[Z_i^2] + \sum_{i \neq j} \mathbb{E}[(M_{ni} - 1)(M_{nj} - 1)] \mathbb{E}[Z_i Z_j] \\ &= n \mathbb{E}[(M_{n1} - 1)^2] \mathbb{E}[Z_1^2] + n(n-1) \mathbb{E}[(M_{n1} - 1)(M_{n2} - 1)] \mathbb{E}[Z_1]^2. \end{aligned}$$

Since $M_{n1} \sim \text{Bin}(n, 1/n)$, we have

$$\mathbb{E}[(M_{n1} - 1)^2] = \text{Var}(M_{n1}) = 1 - \frac{1}{n}.$$

For the mixed term, Proposition 6.2 gives that $M_{n1} | M_{n2}$ is binomial with parameters $n - M_{n2}$ and $1/(n - 1)$, hence

$$\begin{aligned} \mathbb{E}[(M_{n1} - 1)(M_{n2} - 1)] &= \mathbb{E}[(M_{n2} - 1) \mathbb{E}[(M_{n1} - 1) | M_{n2}]] \\ &= \mathbb{E} \left[(M_{n2} - 1) \left(\frac{n - M_{n2}}{n - 1} - 1 \right) \right] = -\frac{1}{n}. \end{aligned}$$

Therefore,

$$\mathbb{E} \left[\left(\sum_{i=1}^n (M_{ni} - 1) Z_i \right)^2 \right] = (n - 1) \text{Var}(Z_1),$$

which proves the claim. $\square$

The previous lemma can be practically used to assert consistency of the bootstrap plug-in estimators.

Proposition 6.3 (Consistency of bootstrap mean and variance) *Assume $\mathbb{E}[Z_1^2] < \infty$. Then*

$$\hat{\mu}_n^* \xrightarrow{\mathbb{P}} \mu.$$

If, in addition, $\mathbb{E}[Z_1^4] < \infty$, then

$$(\hat{\sigma}_n^2)^* \xrightarrow{\mathbb{P}} \sigma^2.$$

Proof. By Theorem 1.2, we in particular get $\hat{\mu}_n \xrightarrow{\mathbb{P}} \mu$. Hence it is enough to show $\hat{\mu}_n^* - \hat{\mu}_n \xrightarrow{\mathbb{P}} 0$. Using the multinomial representation,

$$\hat{\mu}_n^* - \hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n (M_{ni} - 1) Z_i,$$

and therefore by Lemma 6.2,

$$\mathbb{E}[(\hat{\mu}_n^* - \hat{\mu}_n)^2] = \frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n (M_{ni} - 1) Z_i \right)^2 \right] = O(1/n),$$

which implies $\hat{\mu}_n^* - \hat{\mu}_n \xrightarrow{\mathbb{P}} 0$.

For the variance, $\hat{\sigma}_n^2 \xrightarrow{\mathbb{P}} \sigma^2$, so it remains to control

$$(\hat{\sigma}_n^2)^* - \hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (M_{ni} - 1) Z_i^2 - ((\hat{\mu}_n^*)^2 - \hat{\mu}_n^2).$$

The first term converges to 0 in probability by the same argument as for $\hat{\mu}_n^* - \hat{\mu}_n$, now applied to Z_i^2 (using $\mathbb{E}[Z_1^4] < \infty$). The second term converges to 0 in probability because $\hat{\mu}_n^* - \hat{\mu}_n \xrightarrow{\mathbb{P}} 0$ and both $\hat{\mu}_n^*$ and $\hat{\mu}_n$ convergent. This proves $(\hat{\sigma}_n^2)^* \xrightarrow{\mathbb{P}} \sigma^2$. $\square$

The same decomposition yields the natural $\sqrt{n}$ -scale bounds. In fact, normality can be established, through the proof techniques are not elementary and beyond the scope of this section.

Proposition 6.4 (Bootstrap rate of convergence) *Assume $\mathbb{E}[Z_1^2] < \infty$. Then*

$$\sqrt{n}(\hat{\mu}_n^* - \hat{\mu}_n) = O_{\mathbb{P}}(1).$$

If, in addition, $\mathbb{E}[Z_1^4] < \infty$, then

$$\sqrt{n}((\hat{\sigma}_n^2)^* - \hat{\sigma}_n^2) = O_{\mathbb{P}}(1).$$

Proof. By Lemma 6.2,

$$\mathbb{E}[n(\hat{\mu}_n^* - \hat{\mu}_n)^2] = \frac{1}{n} \mathbb{E}\left[\left(\sum_{i=1}^n (M_{ni} - 1) Z_i\right)^2\right] = O(1),$$

which gives $\sqrt{n}(\hat{\mu}_n^* - \hat{\mu}_n) = O_{\mathbb{P}}(1)$.

For the variance, use

$$\sqrt{n}((\hat{\sigma}_n^2)^* - \hat{\sigma}_n^2) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (M_{ni} - 1) Z_i^2 - \sqrt{n}((\hat{\mu}_n^*)^2 - \hat{\mu}_n^2).$$

The first term is $O_{\mathbb{P}}(1)$ by Lemma 6.2 applied to Z_i^2 . For the second term,

$$\sqrt{n}|(\hat{\mu}_n^*)^2 - \hat{\mu}_n^2| = |\hat{\mu}_n^* + \hat{\mu}_n| \sqrt{n}|\hat{\mu}_n^* - \hat{\mu}_n| = O_{\mathbb{P}}(1),$$

since $\hat{\mu}_n^*, \hat{\mu}_n = O_{\mathbb{P}}(1)$ and the first assertion gives $\sqrt{n}(\hat{\mu}_n^* - \hat{\mu}_n) = O_{\mathbb{P}}(1)$, and the absolute value is a continuous mapping. Hence the whole expression is $O_{\mathbb{P}}(1)$. $\square$

6.1.5 Covariance penalties

Cross-validation and bootstrap methods work well for arbitrary models because they treat the fitted predictor as a black box. However, if the model is probabilistic, additional structure can often be exploited to obtain less noisy estimates of prediction error. Covariance penalty methods do this by quantifying the optimism of the training error directly.

The guiding idea is simple: fitted values that react strongly to the observed responses tend to fit the training sample too well, and this excess fit can be

measured through a covariance term. This perspective underlies several classical criteria, including Mallows's C_p and AIC.

The covariance penalty approach actually originates from a regression framework. Let us assume initially that the model in question is of the form of an n -dimensional probabilistic model where

$$Y \sim (\mu, \sigma^2 \text{Id}),$$

that is, the Y_i , $i = 1, \dots, n$, are uncorrelated with mean μ_i and variance σ^2 (which needs to be estimated too). Note that we do not assume normality or any other structure. Assume also that a regression rule $r(\cdot)$ has been produced, and is given by

$$\hat{\mu} = r(Y).$$

Now assume that Y_0 is an independent copy of Y , and we want to know how accurate $\hat{\mu}(Y)$ is at predicting Y_0 . For squared error loss, we define

$$\text{Err}_{\mathcal{T}} = \frac{1}{n} \sum_{i=1}^n \text{Err}_i = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(Y_{0i} - \hat{\mu}_i)^2 | \hat{\mu}_i].$$

Define also $\text{err}_i = (Y_i - \hat{\mu}_i)^2$.

Proposition 6.5 (Covariance penalties) *We have that*

$$\mathbb{E}[\text{Err}_i] = \mathbb{E}[\text{err}_i] + 2 \text{Cov}(\hat{\mu}_i, Y_i).$$

Proof. We expand squared errors around the conditional mean and compare the resulting terms. It is not hard to verify that

$$(Y_i - \hat{\mu}_i)^2 = (Y_i - \mu_i)^2 - 2(\hat{\mu}_i - \mu_i)(Y_i - \mu_i) + (\hat{\mu}_i - \mu_i)^2,$$

and similarly

$$(Y_{0i} - \hat{\mu}_i)^2 = (Y_{0i} - \mu_i)^2 - 2(\hat{\mu}_i - \mu_i)(Y_{0i} - \mu_i) + (\hat{\mu}_i - \mu_i)^2.$$

Taking expectations on both sides gives

$$\mathbb{E}[\text{err}_i] = \sigma^2 - 2 \text{Cov}(\hat{\mu}_i, Y_i) + \mathbb{E}[(\hat{\mu}_i - \mu_i)^2],$$

and since Y_{0i} is independent of $\hat{\mu}_i$

$$\mathbb{E}[\text{Err}_i] = \sigma^2 + \mathbb{E}[(\hat{\mu}_i - \mu_i)^2].$$

Then taking the difference between the two expressions yields the result. $\square$

The above still holds if σ^2 varies with i . The significance of Proposition 6.5 is that it provides a way of estimating the expected prediction error from the training error, provided we can estimate the covariance term. Summing up over observations we obtain

$$\widehat{\text{Err}} = \overline{\text{err}} + \frac{2}{n} \sum_{i=1}^n \widehat{\text{Cov}}(\hat{\mu}_i, Y_i) = \frac{1}{n} \sum_{i=1}^n \text{err}_i + \frac{2}{n} \sum_{i=1}^n \widehat{\text{Cov}}(\hat{\mu}_i, Y_i). \quad (6.2)$$

The covariance term can be interpreted as the *optimism* of the training error, that is, the expected gap between training and test performance. For linear smoothers it reduces to a trace penalty (see Mallows's C_p below), which provides a direct complexity adjustment and motivates the notion of effective degrees of freedom.

In practice, one must be clear about what procedure is being penalised. If model tuning or variable selection is part of the fitting rule, then the covariance is taken with respect to that full adaptive procedure. Covariance penalties can be less noisy than resampling estimates, but they are also more model-dependent, and hence more sensitive to misspecification. The only unknown is thus $\widehat{\text{Cov}}(\hat{\mu}_i, Y_i)$, which is model dependent. Below we provide some examples.

1. *Mallows's C_p estimate of prediction loss*: for a linear model $\hat{\mu} = c + MY$ for a known $n \times n$ matrix M we have $\widehat{\text{Cov}}(\hat{\mu}, Y) = \sigma^2 M$, and in particular $\widehat{\text{Cov}}(\hat{\mu}_i, Y_i) = \sigma^2 M_{ii}$. This yields

$$\widehat{\text{Err}} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\mu}_i)^2 + \frac{2\sigma^2}{n} \text{tr}(M).$$

2. For ordinary regression, $M = X(X^\top X)^{-1}X^\top$ and $\text{tr}(M) = p$, the number of predictors. Then

$$\widehat{\text{Err}} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\mu}_i)^2 + \frac{2\sigma^2 p}{n}.$$

The above formula, equated with (6.2) provides a definition of degrees of freedom in terms of covariance penalties:

$$\text{df} = \frac{1}{\sigma^2} \sum_{i=1}^n \widehat{\text{Cov}}(\hat{\mu}_i, Y_i).$$

Thus, larger degrees of freedom provide flexibility, but at the cost of a covariance penalty for the generalisation error.

3. We may use the *parametric bootstrap* to provide an estimate for each of the covariance terms. The idea is to resample the training data, fitting the same parametric model, yielding $\hat{\mu}_i^b$ replicates and responses Y_i^b . Then

$$\widehat{\text{Cov}}(\hat{\mu}_i, Y_i) = \frac{1}{B} \sum_{b=1}^B (\hat{\mu}_i^b - \tilde{\mu}_i)(Y_i^b - \tilde{Y}_i),$$

where

$$\tilde{\mu}_i = \frac{1}{B} \sum_{b=1}^B \hat{\mu}_i^b, \quad \tilde{Y}_i = \frac{1}{B} \sum_{b=1}^B Y_i^b.$$

Now, covariance penalties may also be applied in situations where the loss function is not squared error. Notable extensions are the following:

1. For *classification* we get $\text{err}_i = 1_{(y_i \neq \hat{\mu}_i)}$, and we clearly have that $\text{Err}_i = \mathbb{P}(y_{0i} \neq \hat{\mu}_i | \hat{\mu}_i)$. Then the statement of Proposition 6.5 holds verbatim, yielding

$$\widehat{\text{Err}} = \frac{1}{n} \sum_{i=1}^n 1_{(Y_i \neq \hat{\mu}_i)} + \frac{2}{n} \sum_{i=1}^n \widehat{\text{Cov}}(\hat{\mu}_i, Y_i).$$

But notice also that

$$\text{Cov}(\hat{\mu}_i, Y_i) = \mu_i(1 - \mu_i) [\mathbb{P}(\hat{\mu}_i = 1 | Y_i = 1) - \mathbb{P}(\hat{\mu}_i = 1 | Y_i = 0)],$$

where $\mu_i = \mathbb{P}(Y_i = 1)$. This again shows that more accurate models give rise to larger penalties.

2. For members of the exponential dispersion family, with deviance error (two times the Kullback–Leibler divergence), it can be shown that the penalty is well approximated by $2p/n$, where p is the degrees of freedom of the corresponding GLM. Then we get

$$\widehat{\text{Err}} = \frac{1}{n} \{ -2 \log f_{\hat{\mu}}(Y) + 2p \} + \text{cst.}$$

The term in curly brackets is defined as the *Akaike information criterion* (AIC), which actually extends beyond GLMs as a model selection criterion.

6.2 Model averaging

Once predictive performance is understood, it is natural to ask whether combining models can further stabilise or improve accuracy. Model averaging is not one method, but rather a family of constructions in which several predictors are combined by averaging, weighted averaging, or selection over resampled fits. The main statistical motivation is often variance reduction, although bias may also improve when different models capture different aspects of the signal.

These gains are not automatic. Averaging highly correlated predictors gives little benefit, and averaging poor predictors merely propagates their bias. What matters is the joint behaviour of the prediction errors and the loss used to combine them. We now discuss three standard examples.

6.2.1 Bagging: bootstrap aggregating

Bagging, short for *bootstrap aggregating*, uses the bootstrap not to estimate uncertainty, but to construct a new predictor by averaging many refits of the same learning rule on bootstrap samples. It is best viewed as a bootstrap smoothing operation, whereby a possibly unstable predictor is replaced by its average under the empirical bootstrap law.

Bagging is especially effective for unstable learners, where small perturbations of the training data produce large changes in the fitted predictor. Decision trees are the canonical example. For very stable linear procedures, bagging often has a much smaller effect.

It is implemented as follows:

1. Start with a training dataset $\mathcal{T} = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ and a predictive model $\hat{f}$ that has been fitted to this data.
2. Generate B bootstrap samples from $\mathcal{T}$, where each bootstrap sample $\mathcal{T}^b$ is obtained by sampling with replacement from $\mathcal{T}$.
3. For each bootstrap sample, we fit the *same model* $\hat{f}^b$. This includes all possible adaptive selection techniques.
4. The *bagging estimate* is then defined as the average of the predictions from all B bootstrap models:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^b(x).$$

We have that, almost surely, as $B \rightarrow \infty$,

$$\hat{f}_{\text{bag}}(x) \rightarrow \mathbb{E}_{\mathbb{P}^{(n)}}[\hat{f}^*(x)],$$

where $\mathbb{P}^{(n)}$ is the empirical cumulative distribution function (ecdf) of the training data $\mathcal{T}$, each $\mathcal{T}^* = \{(X_1^*, Y_1^*), \dots, (X_n^*, Y_n^*)\}$ is a bootstrap sample where $(X_i^*, Y_i^*) \sim \mathbb{P}^{(n)}$, and $\hat{f}^*$ is the model fitted on $\mathcal{T}^*$.

One of the key advantages of bagging is variance reduction, often at the cost of a small increase in bias. If the bootstrap replicates $\hat{f}^b(x)$ are highly correlated, the gain is limited; if they are diverse and individually noisy, averaging can improve predictive accuracy substantially.

The bootstrap replicates also provide more than a mean, and one may inspect empirical quantiles or other summaries of $\{\hat{f}^b(x)\}_{b=1}^B$. It is important to note that bagging changes the predictor only when the fitting rule is nonlinear or adaptive; for many linear smoothers the effect is small or close to zero.

Numeric example 6.2 (Bagging and bootstrap intervals) In Figure 6.2 we revisit the setup of equation (6.1), for a chosen model with 10 degrees of freedom. We generate $B = 1000$ bootstrap samples. Averaging the resulting fitted curves yields a bagging predictive model, while the same bootstrap sample of fits may also be used to construct confidence intervals for the chosen predictive model. Luckily, in this case, the theory of linear models assures that theoretical intervals are available, and these agree closely with the bootstrap intervals. The example is therefore useful in separating two roles of the bootstrap, namely, predictor stabilisation through averaging and uncertainty quantification through resampling variability.

In more complicated scenarios, theoretical confidence bounds are usually unavailable, whereas the bootstrap construction remains essentially unchanged.

◇

6.2.2 Stacking several models

Stacking addresses a different setting from bagging. Instead of repeatedly refitting the same learning rule, we combine several different candidate predictors

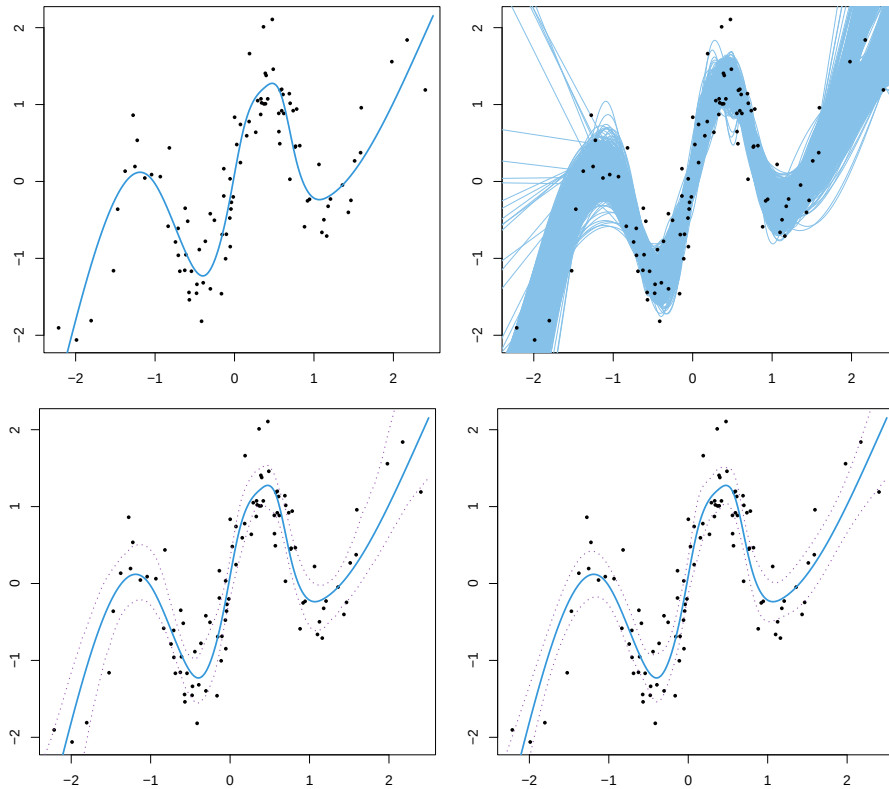


Figure 6.2: Fitted linear model $\hat{f}(x)$ (top left) to $n = 100$ simulated data points (x_i, y_i) , and $B = 1000$ bootstrap samples $(\hat{f}^b)$ (top right). The latter are then used to construct 95% confidence intervals over x (bottom left), which follow closely the theoretical confidence bounds (bottom right).

for the same task. The goal is to exploit complementary strengths of models that make different approximation errors.

Rather than selecting a single “best” model, stacking learns a meta-predictor based on the base-model predictions. In the simplest case, this meta-predictor is a weighted linear combination.

Thus, suppose we have M different models, each trained on the same training dataset $\mathcal{T}$. These models provide predictions $\hat{f}_1(x), \hat{f}_2(x), \dots, \hat{f}_M(x)$ for a given input x . We seek weights $w = (w_1, w_2, \dots, w_M)$ that minimise the expected squared error between the true outcome Y and the weighted average of the predictions. Thus, define

$$\hat{w} = \operatorname{argmin}_w \mathbb{E} \left[\left(Y - \sum_{m=1}^M w_m \hat{f}_m(X) \right)^2 \right].$$

The theoretical solution to this problem is the population linear projection of Y onto the span of $(\hat{f}_1(X), \dots, \hat{f}_M(X))$. This is not computable because the population law is unknown.

Nonetheless, this theoretical framework guarantees that the expected squared error of the combined model is at least as good as the error of any individual model:

$$\mathbb{E} \left[\left(Y - \sum_{m=1}^M \hat{w}_m \hat{f}_m(X) \right)^2 \right] \leq \mathbb{E} \left[\left(Y - \hat{f}_m(X) \right)^2 \right], \quad \forall m.$$

In practice, the population is not available, so the weights w must be estimated from the observed data. A crucial warning is that estimating w from in-sample predictions $\hat{f}_m(X_i)$ can severely overfit and often gives an unfair advantage to more complex base learners.

A more robust method for model averaging is the *stacking estimator*, which uses out-of-sample predictions to fit the meta-learner:

1. Weights are estimated using a cross-validation approach. Specifically, for the i -th data point, the predictions $\hat{f}_m^{-i}(X_i)$ are obtained from models trained without the i -th observation (or, more generally, without the fold containing i).
2. The stacking weights are then chosen to minimise the squared error on these out-of-sample predictions:

$$\hat{w}^{\text{st}} = \operatorname{argmin}_w \sum_{i=1}^n \left[Y_i - \sum_{m=1}^M w_m \hat{f}_m^{-i}(X_i) \right]^2.$$

3. The final model is then given by the weighted sum:

$$\sum_{m=1}^M \hat{w}_m^{\text{st}} \hat{f}_m(x).$$

The final model resulting from stacking is less interpretable than individual models but often leads to better predictive performance, especially when the

base models capture different aspects of the data. In practice one often imposes constraints such as $w_m \geq 0$ and $\sum_{m=1}^M w_m = 1$, or uses regularisation on w , to stabilise the meta-fit.

More generally, one may use $(\hat{f}_1^{-i}(X_i), \dots, \hat{f}_M^{-i}(X_i))$ as predictors and fit any meta-learner under an appropriate loss function. This is often called *super learning*. The same leakage warning applies regarding the meta-learner having to be trained on out-of-sample base predictions if one wants an honest assessment of its performance.

6.2.3 Bumping

Bumping is a bootstrap-based model search strategy for unstable optimisation problems, where the fitting algorithm may get trapped in local minima or depend strongly on influential observations. In contrast with bagging, bumping does not average bootstrap fits; it generates many candidate fits and then selects one according to a performance criterion.

This can be useful when bootstrap perturbations force the fitting algorithm to explore different regions of parameter space. The price is that the final selection can itself be noisy and may overfit if the selection criterion is too optimistic.

A basic bumping construction is:

1. Given a training dataset $\mathcal{T}$, generate B bootstrapped samples, often including the original training sample among the candidates.
2. For each bootstrapped sample b , fit a predictive model $\hat{f}^b$, including any adaptive selection algorithm used in the original procedure.
3. Select the candidate that minimises a chosen criterion (here, squared error on the original training data). Formally, the best model $\hat{b}$ is selected as:

$$\hat{b} = \operatorname{argmin}_b \sum_{i=1}^n (Y_i - \hat{f}^b(X_i))^2,$$

where the summation runs over all observations in the original training dataset.

4. The final prediction model is then $\hat{f}^{\hat{b}}(x)$.

Compared with bagging, bumping can produce a stronger correction because it replaces averaging by selection. This may help when some bootstrap perturbations uncover substantially better local optima, but it can also increase variance. In practice, using a validation or OOB criterion instead of pure training error may improve robustness when computationally feasible.

6.3 Machine learning models

Next, we survey several commonly used learning models to illustrate how the assessment and averaging ideas play out in practice. In this section, we explore

several commonly used machine learning models, expanding on our prior discussions of kernel methods, k -nearest neighbours, Generalised Linear Models (GLMs), and regularised (shrinkage) models. We delve into additive models, generalised additive models, decision trees, random forests, and boosting techniques, each offering unique strengths for predictive modelling. While these models are versatile and widely applicable, it is important to note that this chapter is not comprehensive by any means, and certain key areas are omitted, such as classification methods, unsupervised learning (which plays a vital role in creating risk classes in insurance), and neural networks, which, though powerful, are often more complex and challenging to tune compared to the methods presented here.

6.3.1 Additive models

In many modelling scenarios, the assumption that the relationship between the explanatory variables and the target variable is linear can be restrictive. While linear models have their advantages, such as interpretability and simplicity, they may not capture the underlying complexities in the data. This section discusses moving beyond linear models by using nonparametric transformations of explanatory variables to enhance model flexibility.

The expectation of the target variable given the explanatory variables is generally expressed as:

$$\mathbb{E}[Y | X] = f(X).$$

Linear models can be seen as the first-order Taylor expansion of this relationship, offering a straightforward and interpretable approach. They are also useful in scenarios with a small number of explanatory variables p and a relatively large sample size n . The main idea here is to retain the same formula as a starting point but treat transformations of explanatory variables more systematically.

Consider explanatory variables $X \in \mathbb{R}^p$. Each explanatory variable X can be transformed using a function $h_m(X)$, referred to as the m -th transformation or *basis function*. These basis functions map the explanatory variables from $\mathbb{R}^p$ to $\mathbb{R}$, with $m = 1, \dots, M$. It is important to note that in general $M \neq p$.

The model we consider is of the form:

$$f(X) = \sum_{m=1}^M \beta_m h_m(X),$$

where β_m are the coefficients to be estimated. This model is referred to as a *linear basis expansion*, where the function $f(X)$ is expressed as a linear combination of the basis functions. Once these basis functions are determined, the fitting of the model follows standard routines, such as least squares or other optimisation techniques.

Examples of basis functions

Basis functions provide a general language for flexible modelling. The key point is that the model is linear in the coefficients β_m , even when it is highly nonlinear

in the original covariates. This separates two tasks: choosing a useful dictionary of transformations h_m , and estimating or regularising the coefficients in that dictionary.

If we choose $h_m(X) = X_m$ for $m = 1, \dots, p$, we recover the standard linear model. Choosing $h_m(X) = X_j^2$, X_j^3 , or $X_j X_k$ leads to polynomial and interaction expansions. These enrich the approximation space, but the number of candidate terms grows quickly with the degree d (roughly $O(p^d)$), so naive high-order polynomial expansions become impractical in moderate or high dimensions.

Other common transformations include nonlinear one-dimensional maps such as $h_m(X) = \log(X_j)$ or $h_m(X) = \sqrt{X_j}$, as well as geometric features such as $h_m(X) = \|X\|$. In some applications, it is useful to discretise a covariate into intervals; for example,

$$h_m(X) = 1_{(X_k \in [L_m, U_m])},$$

which yields a piecewise constant contribution from X_k . Such indicator bases connect directly to histogram estimators and tree-based models.

It is useful to distinguish between *global* and *local* bases. Global bases (e.g. low-order polynomials) are parsimonious but can extrapolate poorly and may be sensitive to misspecification. Local bases (e.g. splines, wavelets, indicator bins) provide greater local adaptivity, but require tuning of knot locations, resolution, or regularisation.

In general, basis functions are a *device* for constructing flexible approximations to $f(X)$. When the resulting collection of candidate functions (a *dictionary*) is large, often much larger than the sample size, one needs explicit complexity control, as follows:

1. *Restriction*: preselect a subset of M_j basis functions for each additive component of the model. The resulting model is of the form:

$$f(X) = \sum_{j=1}^p f_j(X_j) = \sum_{j=1}^p \sum_{m=1}^{M_j} \beta_{jm} h_{jm}(X_j).$$

2. *Selection methods*: adaptively identify and select the basis functions that contribute most significantly to the model fit. Techniques such as regression trees and boosting are examples of selection methods (to be defined below).
3. *Regularisation methods*: regularisation techniques like ridge regression and lasso are used to shrink the coefficients, thereby reducing model complexity.

Third-order polynomials and splines

When constructing splines, different continuity restrictions can be imposed on f , such as no continuity, continuity at the first derivative, or continuity at both the first and second derivatives. Enforcing second-order smoothness yields a

function known as a *cubic spline*. The basis functions for cubic splines can be expressed as:

$$\begin{aligned} h_1(X) &= 1, & h_2(X) &= X, & h_3(X) &= X^2, \\ h_4(X) &= X^3, & h_5(X) &= (X - \xi_1)_+^3, & h_6(X) &= (X - \xi_2)_+^3. \end{aligned}$$

Similar expressions can be derived for additional splitting points ξ_j , also referred to as *knots*. When accounting for constraints, such as those imposed by continuity requirements, the total number of parameters is reduced. For instance, with four knots and second-order continuity, the total number of parameters is $3 \times 4 - 2 \times 3 = 6$ (four degrees of freedom in each sub-interval, two removed from each centre knot, and one at each boundary knot to enforce linearity).

Definition 6.4 (Order- M spline) An *order- M spline* with K knots at $\xi_1, \dots, \xi_K$ is a piecewise polynomial of order M with continuous derivatives of order $M - 2$ at each knot. The basis functions are given by:

$$\begin{aligned} h_j(X) &= X^{j-1}, & j &= 1, \dots, M, \\ h_{M+\ell}(X) &= (X - \xi_\ell)_+^{M-1}, & \ell &= 1, \dots, K. \end{aligned}$$

For example, a cubic spline has order $M = 4$ (not 3). In **R**, the functions `splines::bs()` and `splines::ns()` implement spline bases and use different yet equivalent parameterisations for efficient fitting, as direct computation of powers of large numbers can be computationally challenging.

Numeric example 6.3 (Spline fits with increasing order) For instance, in Figure 6.3 we consider simulated data with $n = 200$ and

$$X \sim U(-2.5, 2.5), \quad Y = X + \sin(4X) + \varepsilon, \quad \varepsilon \sim N(0, 0.4^2). \quad (6.3)$$

We then fit additive models corresponding to $M = 1, 2, 3, 4$, with knots at

$$(-5\pi/8, -3\pi/8, -\pi/8, \pi/8, 3\pi/8, 5\pi/8).$$

Across the four panels we keep both the simulated sample and the knot locations fixed, and vary only the spline order M , so that the figure isolates the effect of the smoothness constraints. The placement of the knots is strategic, and a feature that other models below do not require.

The progression makes clear how increasing the spline order removes visible kinks at the knots and produces smoother transitions, at the cost of imposing stronger local regularity. $\diamond$

One drawback of cubic splines is that they tend to behave erratically outside the range of the observed data, making them unsuitable for extrapolation. *Natural cubic splines* address this issue by imposing the constraint that the function should be linear beyond the boundary knots, thereby adding two extra constraints to the model.

Definition 6.5 (Natural cubic spline) A natural cubic spline with K knots has K basis functions, which can be expressed as:

$$N_1(X) = 1, \quad N_2(X) = X, \quad N_{k+2}(X) = d_k(X) - d_{K-1}(X),$$

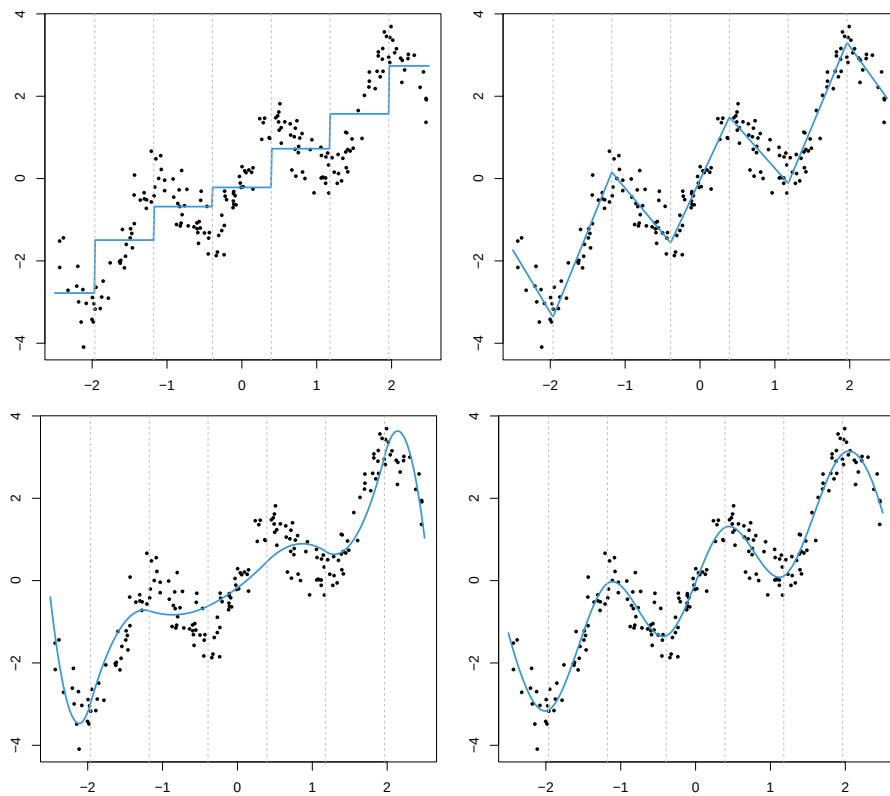


Figure 6.3: Additive models based on piecewise constant (top left), piecewise linear (top right), piecewise quadratic (bottom left), and piecewise cubic (bottom right) spline fits, illustrating increased smoothness at each knot as the polynomial degree increases.

where

$$d_k(X) = \frac{(X - \xi_k)_+^3 - (X - \xi_K)_+^3}{\xi_K - \xi_k}, \quad k = 1, \dots, K-1.$$

The R function `splines::ns()` implements these natural cubic splines, mitigating poor extrapolation.

6.3.2 Generalised Additive Models

Generalised Additive Models (GAMs) extend the flexibility of linear models by allowing nonlinear effects while retaining an additive structure. This additive decomposition is the key modelling compromise, since it captures rich one-dimensional nonlinearities without attempting a fully unrestricted multivariate nonparametric fit.

A GAM has the following form:

$$\mathbb{E}[Y | X_1, \dots, X_p] = \alpha + \sum_{j=1}^p f_j(X_j),$$

where each function f_i represents a smooth, possibly non-linear, effect of the predictor X_i .

The decomposition is not unique unless one imposes identifiability constraints. A standard choice is

$$\sum_{r=1}^n f_j(X_{rj}) = 0, \quad j = 1, \dots, p,$$

so that the intercept α absorbs the overall mean level.

If the functions f_i are represented using basis expansions, the model reduces to a penalised regression problem. In modern implementations, GAMs are therefore typically fitted using finite-dimensional spline bases together with smoothness penalties. A link-function formulation,

$$g(\mathbb{E}[Y | X_1, \dots, X_p]) = \alpha + \sum_{i=1}^p f_i(X_i),$$

extends GAMs to settings previously covered by Generalised Linear Models (GLMs). Other extensions include semiparametric models and additive models with selected interaction terms.

Fitting Generalised Additive Models

Fully general f_j functions are computationally unrealistic, so constraints and penalisation are necessary to fit a GAM effectively. In the Gaussian case, a standard approach is to impose smoothness through a roughness penalty and minimise the penalised residual sum of squares (PRSS):

$$\text{PRSS}(\alpha, f_1, \dots, f_p) = \sum_{i=1}^n \left(Y_i - \alpha - \sum_{j=1}^p f_j(X_{ij}) \right)^2 + \sum_{j=1}^p \lambda_j \int (f_j''(t_j))^2 dt_j, \quad (6.4)$$

where $\lambda_j \geq 0$ are smoothing parameters controlling the smoothness of each f_j . The minimiser can be shown to be a cubic spline with knots at the observed values X_{ij} (or, equivalently, a penalised spline fit in a finite-dimensional basis representation).

This formulation makes the bias-variance tradeoff explicit. Large λ_j shrink the corresponding component towards a low-curvature function (high bias, low variance), while small λ_j permit more local adaptation (low bias, higher variance). In practice, λ_j is often re-expressed in terms of effective degrees of freedom, which are easier to interpret.

Smoothing parameters may be selected by cross-validation, generalised cross-validation (GCV), or related criteria, depending on the modelling assumptions and software implementation.

The *backfitting algorithm* is a classical method for finding the minimiser of (6.4). It alternates between smoothing partial residuals for one component while holding the remaining components fixed. This can be interpreted as block coordinate descent in function space (or in coefficient space after basis expansion). The algorithm is as follows:

1. Initialise: Set $\hat{\alpha} = \frac{1}{n} \sum_{i=1}^n Y_i$, and $\hat{f}_j \equiv 0$ for all j .
2. Iterate for each $j = 1, \dots, p$:

$$\hat{f}_j = \mathcal{S}_j \left[\{y_i - \hat{\alpha} - \sum_{k \neq j} \hat{f}_k(X_{ik})\}_{i=1, \dots, n} \right],$$

where $\mathcal{S}_j$ is a smoother (e.g. a cubic spline) applied to the residuals.

3. Centre the function: adjust $\hat{f}_j$ by subtracting its mean to enforce the identifiability constraint $\sum_{i=1}^n \hat{f}_j(X_{ij}) = 0$:

$$\hat{f}_j \leftarrow \hat{f}_j - \frac{1}{n} \sum_{i=1}^n \hat{f}_j(X_{ij}).$$

4. Repeat steps 2 and 3 until the change in the functions $\hat{f}_j$ is below a pre-specified tolerance level.

This process continues until convergence, resulting in a set of functions $\hat{f}_j$ that together form the fitted GAM. Under standard but technical conditions on the smoothing operators (for example linear smoothers with suitable contraction properties), backfitting converges to the unique penalised least-squares solution.

The resulting component functions are often easier to interpret than a large list of basis coefficients, since each $\hat{f}_j$ can be plotted directly as a smooth effect. However, uncertainty quantification and inference must account for smoothing-parameter selection when those parameters are estimated from the same data.

6.3.3 Decision Trees

Decision trees are a powerful and intuitive method for classification and regression tasks. The core idea behind decision trees is to partition the feature space

into distinct regions and fit a simple model, such as a constant value, to each of these sub-regions. This approach is particularly useful when the relationship between the predictors and the response is complex and nonlinear.

We typically focus on binary splits, where each split divides the feature space into two distinct halves. This makes the resulting regions easy to interpret, although the overall model can nonetheless become less interpretable as the number of splits increases. The sub-regions then get split again, recursively until a stopping rule is applied. This splitting method partitions the space in rectangle-like regions, as is depicted in Figure 6.4.

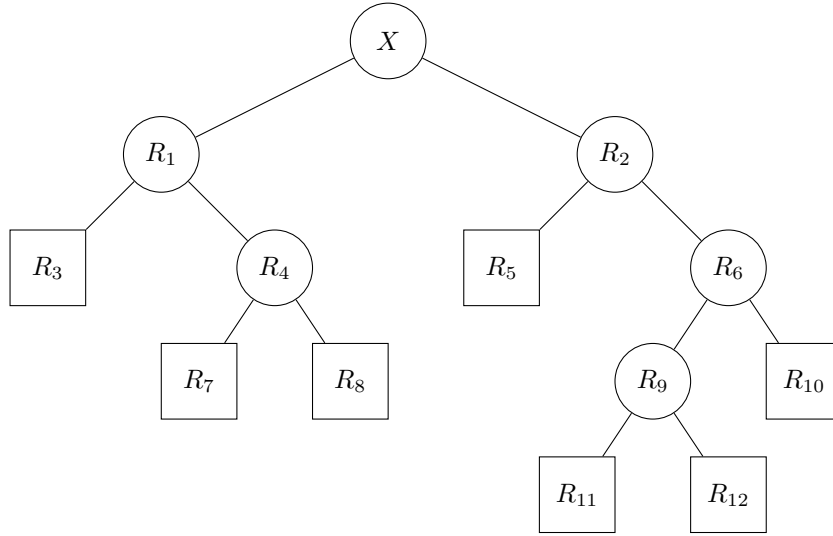


Figure 6.4: A simple decision tree. Here $X = \{X_1, \dots, X_n\}$ denotes all the available feature data, and R_k are elements of partitions of the space. For instance, we have $R_1 \cap R_2 = \emptyset$ and $R_1 \cup R_2 = X$, in line with space partitioning. The terminal nodes, or leaves, are depicted in rectangles and are then relabelled to correspond to $R_1, \dots, R_M$, with M the number of terminal nodes. Here we have $M = 7$.

A canonical example of a prediction model in a decision tree is one that uses the mean of the observations in each terminal node (or leaf) to make predictions. This can be expressed as:

$$\hat{f}(X) = \sum_{m=1}^M c_m 1_{((X_1, \dots, X_p) \in R_m)},$$

where c_m represents the mean value in region R_m , and M is the number of regions.

Growing a decision tree

Decision-tree fitting can be viewed as empirical risk minimisation over a class of piecewise-constant functions whose regions are generated recursively by binary splits. In principle one would like to optimise over all tree structures and

all terminal-node values simultaneously, but that problem is combinatorially difficult. Practical algorithms therefore use greedy recursive splitting.

Given a set of regions $R_1, \dots, R_M$, we might want to model the response as a constant within each region:

$$f(x) = \sum_{m=1}^M c_m 1_{(x \in R_m)}.$$

The constants c_m are determined by minimising the sum of squared errors (SSE) within each region. Specifically, for each region R_m , and considering squared error loss, the optimal c_m is the average of the response values in that region:

$$\hat{c}_m = \frac{\sum_{x_i \in R_m} Y_i}{\#\{X_i \in R_m\}}.$$

If the partition is fixed, estimation of the constants is straightforward. The hard part is selecting the partition. Including partition selection directly in the SSE minimisation is computationally intensive, so a feasible alternative is the greedy recursive splitting algorithm:

1. *Initial split*: begin with the entire dataset and consider a split on each predictor variable j at each possible split point s . For continuous predictors, the candidate split points are typically chosen from sorted observed values (or their midpoints). The split produces two regions:

$$R_1(j, s) = \{X \mid X_j \leq s\}, \quad R_2(j, s) = \{X \mid X_j > s\}.$$

2. *Local optimisation*: for each candidate split, compute the resulting within-node loss and choose the best split. In the regression case with squared error, we solve:

$$\min_{j,s} \left[\min_{c_1} \sum_{X_i \in R_1(j,s)} (Y_i - c_1)^2 + \min_{c_2} \sum_{X_i \in R_2(j,s)} (Y_i - c_2)^2 \right].$$

Here, the inner minimisations are solved by setting c_1 and c_2 to the sample means of the observations in R_1 and R_2 , respectively. In classification, the same scheme is used with a different impurity criterion such as the Gini index or cross-entropy.

3. *Recursion*: if the stopping rule (discussed next) is not satisfied, repeat the same procedure in each daughter node.

Because the algorithm is greedy, the resulting tree is generally not globally optimal for a given number of leaves. This is one reason why a pruning stage is typically essential.

Stopping rule and pruning

Choosing when to stop growing the tree is crucial. If we stop too late, the tree may overfit by tracking noise; if we stop too early, it may underfit and miss important structure.

There are two conceptually distinct strategies:

- *Pre-pruning* (early stopping): stop growth according to rules such as minimum node size, maximum depth, or minimum impurity reduction.
- *Post-pruning*: first grow a large tree and then prune it back using a complexity penalty.

In practice, post-pruning is often more stable because it lets the algorithm explore a richer family of candidate splits before the final complexity choice is made.

The standard post-pruning method is *cost-complexity pruning*. It balances fit and complexity through the penalised criterion

$$C_\alpha(T) = \sum_{m=1}^{|T|} N_m Q_m(T) + \alpha |T|,$$

where $|T|$ is the number of terminal nodes in the tree, N_m is the number of observations in region R_m , and $Q_m(T)$ is the mean squared error within region R_m (or the corresponding impurity in classification).

Pruning then proceeds through the following two steps:

1. *Weakest-link pruning*: starting from a large tree T_0 , construct a nested sequence of subtrees and, for each value of the complexity parameter α , identify the subtree $T_\alpha \subseteq T_0$ that minimises $C_\alpha(T)$.
2. *Model selection*: use validation or cross-validation to select the optimal value of α . The final tree is $T_{\hat{\alpha}}$, where $\hat{\alpha}$ is chosen to minimise the estimated prediction error.

As with any tuning procedure, the same validation criterion used to choose $\hat{\alpha}$ should not automatically be interpreted as an unbiased estimate of final predictive performance. A separate assessment step is required if one wants an approximately unbiased estimate of the full tree-building procedure.

Numeric example 6.4 (Tree partitions, overfitting, and pruning) To visualise how stopping rules and pruning affect the geometry of a decision tree, we consider a simple regression experiment on the unit square. We simulate $n = 1000$ iid covariates $X = (X_1, X_2)$ with independent coordinates $X_1, X_2 \sim \text{Unif}(0, 1)$ and generate responses from

$$Y = f(X) + \varepsilon, \quad f(X) = \left(X_1 - \frac{1}{2}\right)^2 + \left(X_2 - \frac{1}{2}\right)^2, \quad \varepsilon \sim N(0, 0.05^2).$$

The regression function is a symmetric paraboloid centred at $(1/2, 1/2)$, so its level sets in the (X_1, X_2) -plane are concentric curves. This makes it easy to see, in a single picture, both the lack of smoothness of tree predictors and the effect of model complexity on the partition $\{R_1, \dots, R_M\}$.

Figure 6.5 compares three trees fitted with `rpart()`, using the same prediction rule $\hat{f}(x) = \sum_{m=1}^M c_m 1_{(x \in R_m)}$ discussed above. The first tree is deliberately shallow, enforcing an early stopping rule through a restricted depth (and a minimum leaf size). The second tree is grown much more aggressively, with a very

small complexity penalty, so that many local fluctuations of the noisy sample are reflected in the partition. The third tree starts from the deep tree and is then pruned by cost-complexity pruning, with the tuning parameter selected by cross-validation as in the construction of $T_{\hat{\alpha}}$.

In the partition plots, black line segments show the boundaries of the fitted regions, while grey dashed curves show level sets of the true regression function f . The contrast is instructive: the shallow tree underfits by using only a few large rectangles, the deep tree overfits by creating many small regions, and the pruned tree recovers a more stable compromise. The example also illustrates why decision trees are often effective as local approximators of a signal, but can struggle to represent smooth structure without averaging. Further general issues are explained below. $\diamond$

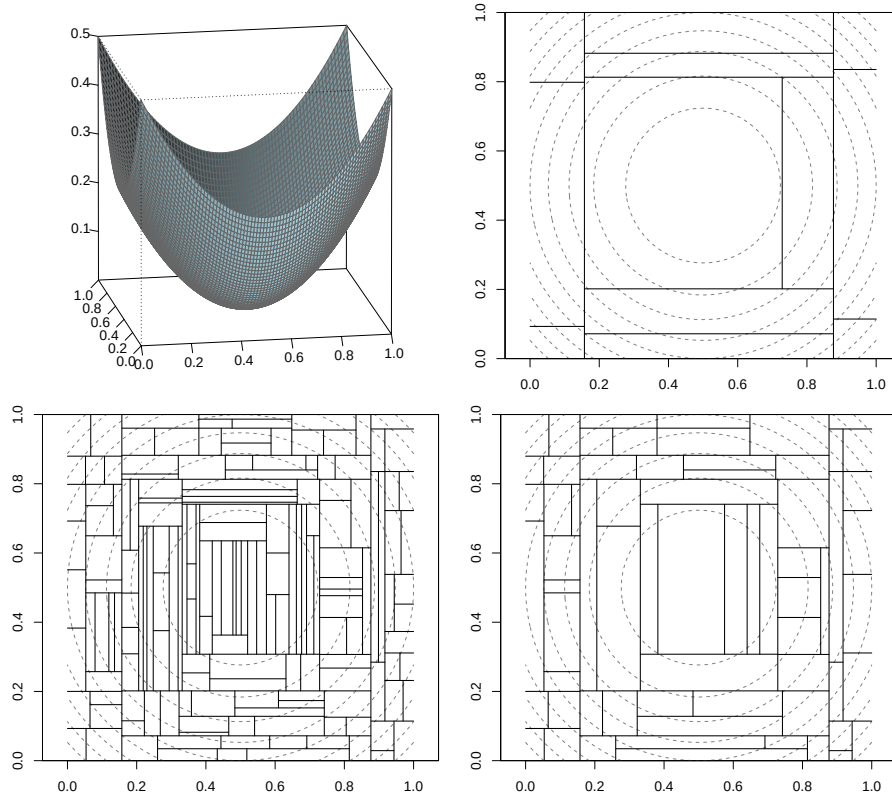


Figure 6.5: Top left: the true regression surface $f(x_1, x_2) = (x_1 - 1/2)^2 + (x_2 - 1/2)^2$ on $[0, 1]^2$. Top right, bottom left, and bottom right: partitions induced by a shallow tree, a deep tree, and a cost-complexity pruned tree, respectively. In the partition panels, black lines are tree split boundaries and grey dashed curves are level sets of the true regression function projected onto the covariate plane.

Issues with decision trees

While decision trees offer an intuitive approach to modelling, they come with several structural challenges that can materially affect performance and reliability.

1. *Unordered categorical predictors*: large numbers of unordered categorical predictors present a critical problem for decision trees. For a categorical variable with q possible values, the number of possible binary partitions is $2^q - 1$ (up to symmetry). This very large search space can cause the tree to overfit and can bias split selection towards high-cardinality variables.
2. *High variance and instability*: small changes in the training data can lead to very different tree structures, split points, and predictions. Bagging and random forests reduce this variance by averaging many trees, but then the transparency of a single tree is largely lost.
3. *Lack of smoothness*: since trees partition the feature space into distinct regions and assign constant predictions within those regions, the predictions exhibit sharp jumps. This can be problematic when the underlying signal is smooth or when continuity is required by the application.
4. *Axis-aligned splits and extrapolation*: standard trees use axis-aligned splits. This can be inefficient for oblique decision boundaries or smooth interactions. In addition, trees extrapolate poorly outside the observed covariate range because predictions are inherited from boundary leaves.

6.3.4 Random forests

The high variance associated with decision trees requires techniques to stabilise and improve their predictions. One such approach is *bagging*, which involves generating multiple bootstrapped samples of the training data, building a tree on each sample, and then averaging the predictions. However, while bagging reduces variance, it does not fully address the issue of correlation between the trees.

A powerful alternative is *boosting*, which we cover later on. For now, we focus on *random forests*, a modification of bagging designed to reduce correlation among the trees.

Random forests build a collection of de-correlated trees and then aggregate their predictions. This method provides similar performance to boosting but with simpler training and tuning processes. Random forests have become popular due to their robust performance and wide implementation across various statistical packages.

In the context of random forests, understanding correlation is crucial. If we had B iid samples, their average would have a variance of $\frac{\sigma^2}{B}$. However, if these samples are correlated, the variance of their average is given by:

$$\text{Variance} = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2,$$

where ρ is the correlation coefficient. This equation shows that a variance term remains in bagging due to sampling with replacement from the same population. Random forests aim to reduce this correlation while also minimising the increase in variance. The key idea is to randomly select a subset of input variables when growing each tree, which decorrelates the trees.

6.3.5 Algorithm for random forests

We summarise the standard construction to fix notation for later discussion. Random forests combine bagging with randomised split selection and are designed to reduce correlation among trees while preserving their low bias.

For regression, the random forest algorithm can be summarised as follows:

1. *Bootstrap sampling and tree building*: for each tree $b = 1, \dots, B$:
 - a) Draw a bootstrap sample of size n from the training data.
 - b) Grow a random forest tree T_b on the bootstrapped data:
 - i. Randomly select m out of the p variables.
 - ii. Choose the best variable and split point among the m selected variables according to the chosen impurity criterion.
 - iii. Split the data into two daughter nodes.
 - iv. Repeat the process for each daughter node until a minimal node size is reached, typically growing the trees deep and without pruning.
2. *Aggregation of trees*: after growing all B trees, the final random forest predictor is given by:

$$\hat{f}_{\text{rf}}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x).$$

Reducing the number of variables m used at each split decreases correlation between trees, which is the main source of variance that remains after bagging.

For classification, the aggregation step is usually majority vote (or averaging class probabilities, when available).

Typical tuning parameters include the number of trees B , the number of candidate variables per split m (often called `mtry`), and the minimum node size. Classical heuristics such as $m = p/3$ for regression are useful starting points but should not replace empirical tuning. We refer to the R package `randomForest`.

Out-of-bag (OOB) samples provide a useful mechanism for error estimation and model validation. For each observation in the training dataset $Z_i = (X_i, Y_i)$, the OOB prediction is calculated as the average prediction of all trees where Z_i did not appear in the bootstrap sample. The resulting OOB error is a built-in analogue of cross-validation and is often sufficiently accurate for tuning random-forest hyperparameters.

One of the advantages of random forests is that there is usually no need for pruning individual trees. Instead, one can allow to grow trees deep and let averaging control variance; tree depth can be treated as a hyperparameter. As B increases, the OOB error typically stabilises, reflecting diminishing Monte Carlo error from using finitely many trees.

Numeric example 6.5 (Random forests versus splines) We revisit the setup in (6.3), where we fit a single tree with maximum depth of 6, and a random forest with 400 trees with at most 30 nodes per tree. The result is given in the left panel of Figure 6.6. Additionally, in the right panel of the latter figure we consider the following sharper dependence structure, where the random forest shows advantage over natural cubic splines with 10 degrees of freedom:

$$X \sim U(-2.5, 2.5), \quad Y = \begin{cases} X^2 + \varepsilon & \text{if } X < -1, \\ -3X + 2 + \varepsilon & \text{if } -1 \leq X < 1, \\ 0.5X^3 + \varepsilon & \text{if } X \geq 1, \end{cases} \quad \varepsilon \sim N(0, 0.4^2).$$

◇

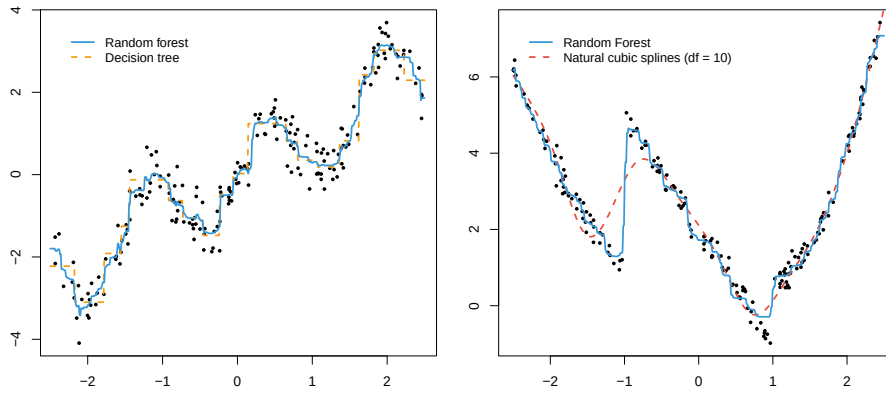


Figure 6.6: Fitted random forest and natural cubic splines for responses y versus covariate x , under smooth dependence (left) and sharp dependence (right). We see how the different scenarios favour different methods.

As with trees, a large number of uninformative variables can degrade the performance of random forests. It is also important to note that although increasing B does not lead to overfitting, excessive reduction of the node size can. Extremely small node sizes may capture noise in the data, leading to poor generalisation. Variable-importance diagnostics (including split-based and permutation-based variants, with OOB implementations for random forests) are discussed later in a more general interpretability section, to avoid introducing notation and definitions twice.

6.3.6 Some theory for decision trees and random forests

We now state a set of assumptions under which consistency can be established for a certain class of random forests. Since the tree partitions depend on the

data, explicit structural conditions are needed in order to obtain any asymptotic theory. We emphasise that these assumptions are introduced primarily for mathematical convenience and do not fit perfectly within the previous framework. Even so, the resulting analysis is instructive and reveals many parallels with kernel methods.

Notation and assumptions

A rigorous and direct statistical analysis of decision trees and random forests is notoriously difficult. The primary reason is that each tree makes a data-dependent partition of the covariate space, and that the procedure for creating the partition is very intricate. As a result, one needs tractable assumptions on both the data-generating process as well as the tree-growing procedure itself in order to derive limit results. We start by making the following mild assumptions on the data:

- (i) We have iid data $(X_1, Y_1), \dots, (X_n, Y_n)$ where the X_i are p -dimensional features and Y a one-dimensional response (we consider the regression setup where the goal is to estimate $\mathbb{E}[Y | X = x]$).
- (ii) Letting X denote a generic X_i , we assume that $X \in [0, 1]^p$ with continuous density f with respect to Lebesgue measure satisfying

$$0 < \inf_{x \in [0, 1]^p} f(x) \quad \text{and} \quad \sup_{x \in [0, 1]^p} f(x) < \infty.$$

- (iii) We assume that Y has second moment, $\mathbb{E}[Y^2]$, and we assume that $x \mapsto \mathbb{E}[Y^k | X = x]$ for $k = 1, 2$ are continuous. This implies in particular that $\mathbb{E}[Y^2 | X = x] \leq K$ for some constant K .

The intuition behind (ii) is that, as the tree partitions the feature space, we cannot make predictions in areas with sparse to non-existing data. This would occur if f was allowed to approach zero. Condition (iii) is purely out of theoretical convenience. We also remark that since decision trees are invariant under scaling of the data (not hard to see), the assumption that $X \in [0, 1]^p$ is equivalent to assuming that X resides in any closed and bounded box in $\mathbb{R}^p$. As mentioned before, the data-dependence in placing splits is one of the main difficulties when analysing decision trees. The following assumption, referred to as *honesty* in the literature (see e.g. [Wager and Athey \[2018\]](#)), elegantly circumvents this problem.

Assumption (honesty): We assume that the data $(X_1, Y_1), \dots, (X_n, Y_n)$ has been partitioned into two disjoint subsets $\mathcal{I}$ and $\mathcal{J}$, with $\mathcal{I}$ used to compute the estimate in the nodes and $\mathcal{J}$ used to place splits. We assume for simplicity that $\#\mathcal{I} = \lceil n/2 \rceil$ and $\#\mathcal{J} = \lfloor n/2 \rfloor$.

Now let us fix some notation. First of all, when growing a decision tree, one injects some randomness into the splitting procedure. In practice, this means choosing a subset of covariates to consider in each split. There may also be some randomness in how the splitting points are chosen. We model this using an auxiliary random parameter θ . Now let $n_{\mathcal{I}} := \#\mathcal{I}$ denote the

number of $\mathcal{I}$ -samples. By reordering the observations if necessary, we may assume $\mathcal{I} = \{(X_1, Y_1), \dots, (X_{n_{\mathcal{I}}}, Y_{n_{\mathcal{I}}})\}$. We denote by $L(x, \theta)$ the leaf that the prediction point $x \in [0, 1]^p$ falls into, and let

$$N_L(x, \theta) = \sum_{i=1}^{n_{\mathcal{I}}} 1_{(X_i \in L(x, \theta))}$$

denote the number of $\mathcal{I}$ -observations which fall into $L(x, \theta)$. We emphasise that the honesty assumption implies that $L(x, \theta)$ only depends on θ and $\mathcal{J}$, while $N_L(x, \theta)$ depends on the entire dataset as well as θ . We can now write the tree predictor as

$$T(x, \theta) = \sum_{i=1}^{n_{\mathcal{I}}} \frac{1_{(X_i \in L(x, \theta))}}{N_L(x, \theta)} Y_i.$$

It is convenient to introduce more notation related to the leaf $L(x, \theta)$. We write

$$L(x, \theta) = (a_{n,1}^x, b_{n,1}^x] \times \dots \times (a_{n,p}^x, b_{n,p}^x]$$

with, slightly abusing notation, $a_{n,j}^x := a_{n,j}^x(\theta, \mathcal{J}, s(x))$ and $b_{n,j}^x := b_{n,j}^x(\theta, \mathcal{J}, s(x))$ where $s(x)$ denotes the number of splits leading to the leaf $L(x, \theta)$. We furthermore let

$$\text{Vol}(L(x, \theta)) = \prod_{j=1}^p (b_{n,j}^x - a_{n,j}^x)$$

denotes the volume of the leaf $L(x, \theta)$. Choosing the conditions on the splitting procedure is more intricate. In the literature, it is typical to only consider a forest (sometimes assuming no Monte Carlo effects). The approach is then to assume that the forest has a form which imitates well-known classical estimators such as U -statistics. We take an entirely different approach and instead make some assumptions on how the leaves of the tree should behave. We assume the following properties of the splitting rule of the tree:

- (I) $b_{n,j}^x - a_{n,j}^x \rightarrow 0$, $\mathbb{P}$ -a.s. for all $j = 1, \dots, p$.
- (II) $n_{\mathcal{I}} \text{Vol}(L(x, \theta)) \rightarrow \infty$, $\mathbb{P}$ -a.s.

Note the resemblance of conditions (I) and (II) to the kernel setup from earlier. (I) ensures that the node volume shrinks to zero around x . It is natural to ask why it does not suffice to have $\text{Vol}(L(x, \theta)) \rightarrow 0$, $\mathbb{P}$ -a.s. The practical reason is that we want to prevent strange behaviour such as the leaves shrinking to a line, preventing the estimator from concentrating around x only. The mathematical reason will become clear in the proofs below.

Consistency

Proving consistency for decision trees in the present setup mirrors the approach for kernel estimators, but with some added conditioning arguments. In the kernel estimation setup, we first proved consistency of the Parzen–Rosenblatt density estimator and then the Priestly–Chao estimator of $m(x) = \mathbb{E}[Y | X = x]f(x)$. We define the decision tree density estimator

$$\hat{f}(x, \theta) := \sum_{i=1}^{n_{\mathcal{I}}} \frac{1_{(X_i \in L(x, \theta))}}{n_{\mathcal{I}} \text{Vol}(L(x, \theta))}.$$

We then have the following result.

Proposition 6.6 (Consistency of tree density estimator) *Under the assumptions presented above, it holds that*

$$\hat{f}(x, \theta) \xrightarrow{\mathbb{P}} f(x).$$

The proof relies on the following technical lemma, which allows us to use a conditioning argument. To state it, note that, as the data grows, we may think of the partitioning into $\mathcal{I}$ and $\mathcal{J}$ as happening dynamically in the sense that we fill up the $\mathcal{I}$ and $\mathcal{J}$ with observations in parallel. If we for a moment choose to not suppress the dependence on n in the notation and write $\mathcal{J}_n$ instead of $\mathcal{J}$, we may then define

$$\mathcal{J}_\infty := \bigcup_{n \in \mathbb{N}} \mathcal{J}_n.$$

Lemma 6.3 (Conditioning on $\mathcal{J}_\infty$) *Fix $\delta > 0$. It holds that*

$$\begin{aligned} \mathbb{P}(|\hat{f}(x, \theta) - \mathbb{E}[\hat{f}(x, \theta) | \theta, \mathcal{J}]| > \delta | \theta, \mathcal{J}) \\ = \mathbb{P}(|\hat{f}(x, \theta) - \mathbb{E}[\hat{f}(x, \theta) | \theta, \mathcal{J}]| > \delta | \theta, \mathcal{J}_\infty) \end{aligned}$$

and that

$$\begin{aligned} \mathbb{P}(|\hat{m}(x, \theta) - \mathbb{E}[\hat{m}(x, \theta) | \theta, \mathcal{J}]| > \delta | \theta, \mathcal{J}) \\ = \mathbb{P}(|\hat{m}(x, \theta) - \mathbb{E}[\hat{m}(x, \theta) | \theta, \mathcal{J}]| > \delta | \theta, \mathcal{J}_\infty). \end{aligned}$$

Proof. This follows immediately from the general fact that for a $\mathcal{D}$ -measurable random variable Z with $\mathbb{E}[|Z|] < \infty$, it holds that $\mathbb{E}[Z | \mathcal{G}] = \mathbb{E}[Z | \sigma(\mathcal{G}, \mathcal{H})]$ whenever $\mathcal{G} \subseteq \mathcal{D}$ and $\mathcal{D}$ is independent of $\mathcal{H}$. Now apply this fact to the σ -algebras $\mathcal{D} := \sigma(\theta, (X_1, Y_1), \dots, (X_n, Y_n))$, $\mathcal{G} := \sigma(\theta, \mathcal{J})$ and $\mathcal{H} := \sigma(\mathcal{J}_\infty \setminus \mathcal{J})$. $\square$

Proof of Proposition 6.6. We note that

$$\begin{aligned} \mathbb{E}[\hat{f}(x, \theta) | \theta, \mathcal{J}] &= \frac{1}{n_{\mathcal{I}} \text{Vol}(L(x, \theta))} \sum_{i=1}^{n_{\mathcal{I}}} \mathbb{E}[1_{(X_i \in L(x, \theta))} | \theta, \mathcal{J}] \\ &= \frac{1}{\text{Vol}(L(x, \theta))} \int_{L(x, \theta)} f(\tilde{x}) \, d\tilde{x}, \end{aligned}$$

and this converges in probability to $f(x)$ by condition (I)¹. Hence it suffices to show that $\hat{f}(x, \theta) - \mathbb{E}[\hat{f}(x, \theta) | \theta, \mathcal{J}] \xrightarrow{\mathbb{P}} 0$. By Markov's inequality, for any $\delta > 0$,

$$\mathbb{P}(|\hat{f}(x, \theta) - \mathbb{E}[\hat{f}(x, \theta) | \theta, \mathcal{J}]| > \delta | \theta, \mathcal{J}) \leq \frac{\text{Var}(\hat{f}(x, \theta) | \theta, \mathcal{J})}{\delta^2}$$

and

$$\text{Var}(\hat{f}(x, \theta) | \theta, \mathcal{J}) = \mathbb{E}[\hat{f}(x, \theta)^2 | \theta, \mathcal{J}] - \left(\frac{1}{\text{Vol}(L(x, \theta))} \int_{L(x, \theta)} f(\tilde{x}) \, d\tilde{x} \right)^2,$$

¹Due to continuity, every $x \in [0, 1]^d$ is a Lebesgue point of f . We refer to [Rudin \[1987\]](#) for background. See also the exercises below.

so we require to establish that $\mathbb{E}[\hat{f}(x, \theta)^2 | \theta, \mathcal{J}] \xrightarrow{\mathbb{P}} f(x)^2$. We compute

$$\begin{aligned} \mathbb{E}[\hat{f}(x, \theta)^2 | \theta, \mathcal{J}] &= \frac{1}{n_{\mathcal{I}}^2 \text{Vol}(L(x, \theta))^2} \sum_{i=1}^{n_{\mathcal{I}}} \sum_{j=1}^{n_{\mathcal{I}}} \mathbb{E}[1_{(X_i \in L(x, \theta))} 1_{(X_j \in L(x, \theta))} | \theta, \mathcal{J}] \\ &= \frac{1}{n_{\mathcal{I}}^2 \text{Vol}(L(x, \theta))^2} (n_{\mathcal{I}} \mathbb{P}(X_1 \in L(x, \theta) | \theta, \mathcal{J}) + n_{\mathcal{I}}(n_{\mathcal{I}} - 1) \mathbb{P}(X_1 \in L(x, \theta) | \theta, \mathcal{J})^2) \\ &= \frac{1}{n_{\mathcal{I}} \text{Vol}(L(x, \theta))} \frac{\int_{L(x, \theta)} f(\tilde{x}) \, d\tilde{x}}{\text{Vol}(L(x, \theta))} + \frac{n_{\mathcal{I}} - 1}{n_{\mathcal{I}}} \left(\frac{1}{\text{Vol}(L(x, \theta))} \int_{L(x, \theta)} f(\tilde{x}) \, d\tilde{x} \right)^2 \\ &\xrightarrow{\mathbb{P}} 0 \cdot f(x) + 1 \cdot f(x)^2 = f(x)^2 \end{aligned}$$

which shows that

$$\mathbb{P}(|\hat{f}(x, \theta) - \mathbb{E}[\hat{f}(x, \theta) | \theta, \mathcal{J}]| > \delta | \theta, \mathcal{J}) \xrightarrow{\mathbb{P}} 0.$$

By Lemma 6.3, we can replace $\theta, \mathcal{J}$ in the probability by $\theta, \mathcal{J}_{\infty}$. Take expectations and apply the tower property along with dominated convergence to conclude the desired statement. $\square$

We can now provide a full proof of consistency for the tree estimator.

Theorem 6.1 (Consistency of regression trees) *Under the assumptions above, we have*

$$T(x, \theta) \xrightarrow{\mathbb{P}} \mathbb{E}[Y | X = x].$$

Proof. Just like in the proof of Proposition 6.6, we have

$$\mathbb{E}[\hat{m}(x, \theta) | \theta, \mathcal{J}] = \frac{1}{\text{Vol}(L(x, \theta))} \int_{L(x, \theta)} m(\tilde{x}) \, d\tilde{x} \xrightarrow{\mathbb{P}} m(x) := \mathbb{E}[Y | X = x]f(x).$$

To see that $\hat{m}(x, \theta) - \mathbb{E}[\hat{m}(x, \theta) | \theta, \mathcal{J}] \xrightarrow{\mathbb{P}} 0$, we again apply a conditional Chebyshev inequality. We compute

$$\begin{aligned} \mathbb{E}[\hat{m}(x, \theta)^2 | \theta, \mathcal{J}] &= \frac{1}{n_{\mathcal{I}}^2 \text{Vol}(L(x, \theta))^2} \sum_{i=1}^{n_{\mathcal{I}}} \sum_{j=1}^{n_{\mathcal{I}}} \mathbb{E}[1_{(X_i \in L(x, \theta))} 1_{(X_j \in L(x, \theta))} Y_i Y_j | \theta, \mathcal{J}] \\ &= \frac{1}{n_{\mathcal{I}} \text{Vol}(L(x, \theta))} \frac{\int_{L(x, \theta)} \mathbb{E}[Y^2 | X = \tilde{x}] f(\tilde{x}) \, d\tilde{x}}{\text{Vol}(L(x, \theta))} \\ &\quad + \frac{n_{\mathcal{I}} - 1}{n_{\mathcal{I}}} \left(\frac{1}{\text{Vol}(L(x, \theta))} \int_{L(x, \theta)} m(\tilde{x}) \, d\tilde{x} \right)^2 \end{aligned}$$

which converges to $m(x)^2$ in probability, and so the conditional variance again converges to zero in probability. The rest of the argument is identical to the one for the density estimator. We have thus shown that $\hat{f}(x, \theta) \xrightarrow{\mathbb{P}} f(x)$ and $\hat{m}(x, \theta) \xrightarrow{\mathbb{P}} m(x)$. Now apply the continuous mapping theorem to complete the proof. $\square$

One may also extend the convergence in probability to L^p -convergence, see the exercises for details. In order to consider a whole forest, we need to extend these results to trees grown on bootstrapped datasets. In order to do

so, one needs to determine what is meant by bootstrapping in the honesty setup. If one considers a double bootstrap setup where the whole dataset is first randomly partitioned into $\mathcal{I}$ and $\mathcal{J}$ and then each subset is bootstrapped independently, it is possible to show (using some natural assumptions on the bootstrapping procedure) that the bootstrap tree is still weakly consistent. Letting $T_1^*(x, \theta_1), \dots, T_B^*(x, \theta_B)$ denote B bootstrap trees, it follows immediately that the random forest

$$\hat{f}_{rf}(x) = \frac{1}{B} \sum_{b=1}^B T_b^*(x, \theta_b)$$

inherits the asymptotic behaviour of the component trees, and in particular is also consistent.

6.3.7 Boosting

Boosting is a highly effective and widely used ensemble learning technique in machine learning. While it shares similarities with bagging, the two approaches are fundamentally different in their execution. The core idea of boosting revolves around building a strong predictor by *sequentially* combining multiple weak learners, i.e. models that are only slightly better than random guessing. These weak learners are fit sequentially to modified versions of the data, and their predictions are combined at the end to form a final, more accurate model.

Additive models and basis expansion

To understand boosting, it is useful to revisit additive expansions from an optimisation perspective. Boosting can be viewed as a strategy for building a large additive model sequentially, one component at a time, instead of solving a difficult joint optimisation problem over all basis functions simultaneously.

Consider the generic basis expansion

$$f(x) = \sum_{m=1}^M \beta_m b(x; \gamma_m),$$

where $b(x; \gamma_m)$ are basis functions with parameters γ_m , and β_m are the corresponding coefficients. Many machine learning models can be expressed in this framework:

- Kernel smoothing methods employ a mother function, like a Gaussian density, with γ_m controlling the shifts and scales.
- Regression splines utilise power basis functions, with γ_m determining the variables and knot positions.
- Decision trees aggregate binary basis functions, where γ_m controls split variables, split points at internal nodes, and predictions at terminal nodes.

- Neural networks use sigmoid functions as the basis.

The weak learner in boosting determines the class of admissible basis functions. For example, if the weak learner is a shallow tree, then each stage adds a tree basis element; if the weak learner is a spline or linear fit, boosting builds an additive expansion in that class.

In principle, and in full generality, one would ideally fit the additive model by minimising a loss function $L(y, f(x))$ jointly over all basis terms:

$$\min_{\{\beta_m, \gamma_m\}_{m=1}^M} \sum_{i=1}^n L\left(Y_i, \sum_{m=1}^M \beta_m b(X_i; \gamma_m)\right).$$

Directly optimising this problem is of course computationally expensive since the problem is nonconvex. A more tractable approach is available when one can solve (or approximately solve) a one-step subproblem quickly:

$$\min_{\beta, \gamma} \sum_{i=1}^n L(Y_i, \beta b(X_i; \gamma)).$$

Boosting exploits this reduction by repeatedly adding basis functions chosen to improve the current fit. The resulting procedure is a greedy algorithm, but often highly effective in high-dimensional and nonlinear problems.

Forward stagewise additive modelling

Forward stagewise additive modelling is a greedy iterative scheme that constructs the predictor one increment at a time. At each stage, it searches for the basis function that gives the largest improvement when added to the current predictor.

The procedure can be summarised as follows:

1. Start with an initial model: $f_0(x) = 0$.
2. For each step $m = 1, \dots, M$:
 - a) Compute the optimal parameters (β_m, γ_m) by minimising the loss function:

$$(\beta_m, \gamma_m) = \underset{\beta, \gamma}{\operatorname{argmin}} \sum_{i=1}^n L(Y_i, f_{m-1}(X_i) + \beta b(X_i; \gamma)).$$

- b) Update the model by adding the new basis function:

$$f_m(x) = f_{m-1}(x) + \beta_m b(x; \gamma_m).$$

This algorithm is greedy and does not generally recover the joint minimiser over all $\{(\beta_m, \gamma_m)\}_{m=1}^M$. Its strength is computational rather than exact optimisation.

When using squared error loss, the stagewise problem is equivalent to fitting a new regression model to the current residuals, as can be directly checked from the above algorithm. This residual-fitting interpretation is rather intuitive and, beyond their excellent performance, adds to the popularity of boosted trees in regression and classification applications.

Choice of loss function

The choice of loss function is central in boosting because it determines both the statistical target and the optimisation geometry. We again highlight that, although tree-based models are not probabilistically grounded models, and certainly not parametric, standard loss functions from regression settings can be used to evaluate them.

For instance, squared error loss is common, although not always appropriate. Squared loss targets the conditional mean and can be sensitive to outliers. Absolute loss targets the conditional median and is more robust. A third option is the Huber loss, which combines local quadratic behaviour with linear tails:

$$L(y, f(x)) = \begin{cases} [y - f(x)]^2 & \text{for } |y - f(x)| \leq \delta, \\ \delta(2|y - f(x)| - \delta) & \text{otherwise.} \end{cases}$$

Beyond regression, boosting is often used with classification losses such as logistic loss or exponential loss, and with asymmetric losses for quantile prediction. In insurance applications, the loss should be chosen to match the decision problem (pricing, reserving, triage, ranking, tail risk estimation, etc.), not merely computational convenience.

The choice of loss also affects the availability of efficient optimisation. Gradient boosting, discussed next, requires differentiability (or a suitable subgradient formulation) of the loss with respect to the fitted value.

Boosted trees

When choosing trees as weak learners, boosting constructs an additive expansion in tree functions. The boosted tree model is

$$f_M(x) = \sum_{m=1}^M T(x; \Theta_m),$$

where $T(x; \Theta_m)$ represents a decision tree with parameters Θ_m . At each iteration, one adds a tree chosen to reduce the current empirical loss:

$$\hat{\Theta}_m = \operatorname{argmin}_{\Theta_m} \sum_{i=1}^n L(Y_i, f_{m-1}(X_i) + T(X_i; \Theta_m)).$$

As with single-tree fitting, exact optimisation over tree structures is difficult, so practical algorithms use greedy tree construction at each stage. For squared error loss, the problem simplifies to fitting trees to residuals (or pseudo-residuals in the gradient-boosting formulation below).

The depth or number of terminal nodes of each tree controls the complexity of the basis functions being added. Very shallow trees (sometimes called stumps) target additive effects, whereas deeper trees may capture interactions to be introduced gradually over the boosting iterations. As with random forests, there is no set rule regarding the tree depth, and it is considered a hyperparameter of the model.

Gradient boosting

A key challenge in boosting is that simple residual-fitting arguments only apply directly to a few losses (such as squared error). Gradient boosting extends boosting to general differentiable loss functions by interpreting the procedure as steepest descent in function space and replacing residuals by negative gradients of the loss.

At iteration m , the current predictor f_{m-1} is updated in a direction that approximately decreases the empirical risk. Because we restrict updates to the span of weak learners (here trees), the negative gradient is first *projected* onto the weak-learner class by fitting a tree to the pseudo-residuals.

The gradient boosting algorithm proceeds as follows:

1. Initialise with a constant fit: $f_0(x) = \operatorname{argmin}_{\gamma} \sum_{i=1}^n L(Y_i, \gamma)$.
2. For each iteration $m = 1, \dots, M$:
 - a) Compute the gradient residuals:

$$r_{im} = -\left(\frac{\partial L(Y_i, f(X_i))}{\partial f(X_i)}\right)_{f=f_{m-1}}, \quad i = 1, \dots, n.$$

- b) Fit a regression tree to these residuals, yielding regions $R_{jm}, j = 1, \dots, J_m$.
 - c) Determine the optimal constants for each region:

$$\gamma_{jm} = \operatorname{argmin}_{\gamma} \sum_{x_i \in R_{jm}} L(Y_i, f_{m-1}(X_i) + \gamma), \quad j = 1, \dots, J_m.$$

- d) Update the model:

$$f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jm} \mathbf{1}_{(x \in R_{jm})}.$$

3. The final model is $\hat{f}(x) = f_M(x)$.

For squared error loss, we have $r_{im} = Y_i - f_{m-1}(X_i)$, so the pseudo-residuals coincide with ordinary residuals. For other losses, the pseudo-residuals encode how the current fit should move to reduce the empirical risk.

This algorithm is usually combined with shrinkage and early stopping (discussed below), which turn a powerful fitting scheme into a regularised estimator.

Practical considerations

Boosting is powerful but also easy to overfit if the stagewise procedure is allowed to run unchecked. In practice, boosting performance is governed by a small set of interacting hyperparameter choices:

- It is common to use a fixed number of terminal nodes $J_m = J$ for all trees, treating J as a hyperparameter that controls interaction complexity. Small J yields weak learners and often better generalisation, whereas larger J fits training data faster but increases overfitting risk.
- The number of boosting iterations M is a major regularisation parameter and is typically selected using a validation set or cross-validation (early stopping).
- Shrinkage (learning-rate regularisation) scales down the contribution of each tree:

$$f_m(x) = f_{m-1}(x) + \nu \cdot \sum_{j=1}^J \gamma_{jm} 1_{(x \in R_{jm})},$$

where $0 < \nu < 1$. Smaller ν usually requires larger M but often improves generalisation and stabilises the fitting path.

- Stochastic gradient boosting introduces randomness by selecting a random fraction η of the training data at each iteration to grow the next tree. This can reduce variance, improve computational efficiency, and act as an additional regulariser.
- Other practical choices (minimum node size, subsampling without replacement, column subsampling, monotonicity constraints in some implementations) can matter materially and should be tuned to the application.

Because these parameters interact strongly, a systematic tuning strategy is preferable to one-at-a-time adjustment, and it is also recommended to closely monitor the predictive performance.

Numeric example 6.6 (Random forests and boosting on the paraboloid example) We return to the paraboloid regression setup from Figure 6.5. The goal is to compare how averaging (random forests) and stagewise fitting (boosting) reconstruct the smooth signal that was difficult for a single tree to represent without rectangular artefacts.

For the random forest, we fit $\hat{f}_{\text{rf}}$ with a large number of trees ($B = 5000$) and a relatively large minimum node size, so that the individual trees remain

weak. For boosting, we fit a boosted tree model with shrinkage $\nu = 0.01$, and select the number of boosting iterations M by 5-fold cross-validation, in line with the discussion of validation-based tuning. In both cases we predict on a fine grid over $[0, 1]^2$ for comparison.

Figure 6.7 shows contour representations of the two fitted surfaces. Grey dashed curves are level sets of the true paraboloid, while black solid curves are level sets of the fitted predictor. Both methods recover the radial structure better than a single tree, though some remains of tree artefacts remain. The sample size is relatively small $n = 1000$, and the fitted contours improve as it grows (not shown). For both models, predictive performance improves at the cost of direct interpretability of individual splits. Below we explore interpretability of general models in more detail. $\diamond$

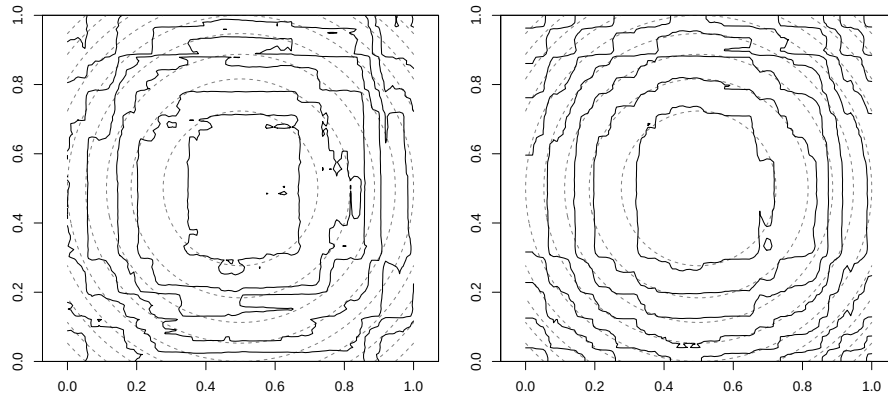


Figure 6.7: Contour comparison of fitted surfaces on the paraboloid regression example. Left: random forest predictor $\hat{f}_{\text{rf}}$ with $B = 5000$ trees and enlarged minimum node size. Right: boosted tree predictor $\hat{f}$ with shrinkage $\nu = 0.01$ and the number of iterations selected by 5-fold cross-validation. Grey dashed curves denote level sets of the true regression function, and black solid curves denote level sets of the fitted predictor.

Ensemble methods such as random forests and boosted trees are often among the strongest predictors in practice, but they are also harder to interpret than linear models or a single shallow tree. Interpretation is therefore usually post hoc and model-based, since one summarises what the fitted predictor does, rather than reading off a small set of coefficients or a single tree. The most common summaries are global in nature, in the sense that they describe average behaviour of the fitted predictor over a sample. They are often very useful for diagnostics and communication, but they are not causal by themselves and they can be misleading when predictors are strongly dependent or when the model is investigated far from the observed covariate support.

6.3.8 Variable importance

Variable importance aims to quantify how strongly each predictor contributes to the performance of the fitted model. In ensemble tree methods (such as

random forests and boosted trees) this is subtle, because a variable may be influential through main effects, through interactions, or by acting as a surrogate for another correlated predictor. Accordingly, variable importance should be read as a *model summary*, not a structural statement about the underlying data-generating mechanism.

One common approach is *split-based importance*. Every time a variable is used in a tree split, the split produces an improvement in the empirical criterion (for example a reduction in squared error, deviance, or another loss proxy). Summing these improvements over all splits and over all boosting iterations yields an importance score. More precisely, let s_m denote the number of internal nodes (splits) in the m -th boosted tree. For the ℓ -th split in tree m , let $\Delta_{\ell m} \geq 0$ denote the associated improvement in the split criterion, and let $v_{\ell m} \in \{1, \dots, p\}$ denote the index of the variable used in that split. Then one may define

$$I_j^{\text{split}} = \sum_{m=1}^M \sum_{\ell=1}^{s_m} \Delta_{\ell m} 1_{(v_{\ell m}=j)},$$

possibly followed by a normalisation such as $I_j^{\text{split}} / \sum_{k=1}^p I_k^{\text{split}}$. This measure is computationally cheap and directly tied to the fitting process, but it inherits known biases from tree-splitting procedures, including preference for variables with many available split points (continuous variables or high-cardinality factors) and sensitivity to the tree depth and impurity criterion used.

A second and often more directly predictive notion is *permutation importance*. Here it is convenient to reuse the loss function L introduced earlier. Let $\mathcal{V} = \{(X_i, Y_i)\}_{i=1}^{n_V}$ be a held-out evaluation sample (preferably validation or test data, not the training data), and let π be a random permutation of $\{1, \dots, n_V\}$. We construct a permuted version of the sample by replacing the j -th covariate column $(X_{1j}, \dots, X_{n_V j})$ by $(X_{\pi(1)j}, \dots, X_{\pi(n_V)j})$, while keeping all other covariates and the responses fixed. The permutation importance of variable j for model f may then be defined as

$$I_j^{\text{perm}} = \frac{1}{n_V} \sum_{i=1}^{n_V} L(Y_i, f(X_{i1}, \dots, X_{\pi(i)j}, \dots, X_{ip})) - \frac{1}{n_V} \sum_{i=1}^{n_V} L(Y_i, f(X_i)).$$

Large positive values indicate that destroying the information in X_j materially harms prediction, while values near zero indicate that the fitted predictor can largely compensate without X_j . In finite samples the estimate may even be slightly negative due to Monte Carlo noise or overfitting, so repeated permutations and reporting means and standard deviations can be informative.

Permutation importance is often preferred when the goal is to rank variables by predictive contribution, but it is not immune to interpretability problems. If predictors are strongly correlated, permuting one variable can create unrealistic combinations and can also understate importance because correlated variables may substitute for each other in the fitted predictor. In such settings, grouped permutation (permuting a block of correlated variables together) or conditional permutation schemes may be more meaningful. In all cases, “importance” is relative to the fitted model, the chosen loss function L , and the evaluation sample $\mathcal{V}$.

Another conceptually clean alternative is *drop-column importance* (also called leave-one-covariate-out importance). Here one refits the model without variable X_j , using the same fitting procedure (and, ideally, the same tuning protocol), and compares predictive error on the same held-out sample $\mathcal{V}$. If $\hat{f}^{(-j)}$ denotes the predictor refit without the j -th variable, one may define

$$I_j^{\text{drop}} = \frac{1}{n_V} \sum_{i=1}^{n_V} L(Y_i, \hat{f}^{(-j)}(X_{i,-j})) - \frac{1}{n_V} \sum_{i=1}^{n_V} L(Y_i, \hat{f}(X_i)),$$

where $X_{i,-j}$ denotes the feature vector with the j -th component removed. This quantity often aligns closely with the practical intuition regarding how much predictive performance is lost if this variable is unavailable. However, it is computationally expensive because it requires refitting the model p times (or more, if repeated or cross-validated). As with permutation importance, strong dependence among predictors complicates interpretation, and again a variable may have small drop-column importance because the remaining variables can partially replace it.

A further family of importance measures, common in modern machine learning software for tree ensembles, is based on Shapley-value attributions (often reported as SHAP values). Rooted in cooperative game theory, the exact attribution ϕ_{ij} assigned to variable j for observation i is calculated as the weighted average of its marginal contributions across all possible feature subsets. If P represents the set of all p features, this is formally defined as:

$$\phi_{ij} = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} [v_i(S \cup \{j\}) - v_i(S)].$$

Here, S is a subset of features excluding j , and $v_i(S)$ is the conditional expectation of the prediction of the model given only the features in S (that is, $\mathbb{E}[f(X) \mid X_S = x_{i,S}]$). Crucially, this does not involve refitting the model; rather, it evaluates the already-fitted model by integrating out the omitted features. In modern tree-ensemble software, this expectation is approximated either by simulating missing values via a reference background dataset (interventional SHAP) or by taking weighted averages across all possible decision paths in the trees (observational TreeSHAP). Using these local attributions, a global importance score is then typically calculated as:

$$I_j^{\text{SHAP}} = \frac{1}{n_V} \sum_{i=1}^{n_V} |\phi_{ij}|.$$

This framework has the advantage of mathematically aggregating local explanations into a single global ranking while retaining the possibility of observation-level inspection. However, because evaluating $v_i(S)$ requires the model to simulate the absence of certain features, it introduces additional modelling choices. Notably, the reference or background distribution used to fill in those missing features, and how predictor dependence is handled. Therefore, it should again be interpreted as a model-based summary rather than an intrinsic property of the data-generating mechanism.

6.3.9 Partial dependence

Partial dependence is one of the most widely used tools for visualising the effect of predictors in flexible machine learning models, including tree ensembles. It measures the average effect of a subset of predictor variables X_S on the fitted prediction, averaging over the remaining variables X_C . As such, it is a property of the fitted predictor f , not of the data-generating mechanism itself. Formally, it is defined by

$$f_S(X_S) = \int f(X_S, x) \, d(\mathbb{P} \circ X_C)(x),$$

where $\mathcal{S} \cup \mathcal{C} = \{1, 2, \dots, p\}$ is a partition of the predictor indices. In practice, the unknown marginal distribution of X_C is replaced by the empirical distribution of the observed sample, giving the estimator

$$\bar{f}_S(X_S) = \frac{1}{n} \sum_{i=1}^n f(X_S, x_{iC}).$$

For one-dimensional summaries ($\mathcal{S} = \{j\}$), one evaluates $\bar{f}_j(x_j)$ on a grid of values of x_j and plots the resulting curve; for two-dimensional summaries one obtains a surface or contour plot. Because additive constants are often uninformative visually, implementations frequently centre the partial dependence function by subtracting its sample mean over the plotting grid. In classification or count models, one should also be explicit about the scale on which the plot is drawn (probability scale, logit scale, log-link scale, etc.), since apparent effect size can differ substantially across scales.

This quantity captures the effect of varying X_S while averaging over the empirical distribution of the remaining variables. It is not the same as “ignoring” X_C , and it is not the same as the conditional expectation $\mathbb{E}[f(X_S, X_C) | X_S]$, which averages with respect to the *conditional* distribution of X_C given X_S . A useful perspective is that partial dependence is the average of *individual conditional expectation* (ICE) curves: for fixed observation i , the curve $x_j \mapsto f(x_j, x_{i,-j})$ describes how the fitted prediction changes when only X_j is varied and all other covariates are frozen at their observed values, and the PDP is the average of these curves over $i = 1, \dots, n$. Inspecting ICE curves alongside the PDP is often helpful, since the PDP may hide strong heterogeneity across observations.

The main caveat is predictor dependence. If X_S and X_C are strongly correlated, the combinations (X_S, x_{iC}) used in the empirical partial dependence estimator may correspond to unrealistic or even impossible covariate combinations, so the resulting plot may partly describe extrapolation behaviour rather than behaviour on the observed support. For that reason, partial dependence plots should ideally be read together with diagnostics of data density (for example rugs or marginal histograms) and, in strongly dependent settings, complemented by alternatives such as conditional partial dependence or accumulated local effects (ALE) plots, which are designed to reduce extrapolation artefacts.

It is also worth distinguishing *effect magnitude* from *predictive importance*. In practice, one sometimes summarises a one-variable partial dependence curve

by its range or variance over a plotting grid and uses this as an “importance” score. Such summaries can be useful descriptively, but they measure the amplitude of the fitted effect on the chosen scale, not the deterioration in predictive performance when information about that variable is removed or destroyed. A variable may therefore have a visually large partial dependence effect yet modest predictive importance (for example because its effect occurs in a sparse region or is largely redundant with other predictors), and conversely a variable may be predictively important through interactions while having a relatively flat marginal partial dependence curve.

Numeric example 6.7 (Variable importance and partial dependence with a hidden interaction) We consider a boosted-tree regression model with three independent covariates X_1, X_2, X_3 all distributed as $U(-1, 1)$ and response

$$Y = X_1^2 + \sin(5X_2) + X_1X_2 + \varepsilon, \quad \varepsilon \sim N(0, 1).$$

The variable X_3 carries no signal, while the term X_1X_2 introduces a genuine interaction between X_1 and X_2 . The sample size is $n = 10,000$. We fit a boosted-tree model with square error loss and select the number of boosting iterations by 5-fold cross-validation, in the spirit of the preceding subsections.

The results are shown in Figure 6.8, which combines the variable-importance summary with the one-dimensional partial dependence functions $\bar{f}_j$, $j = 1, 2, 3$. In the partial dependence panels, the coloured solid curve is the fitted partial dependence plot and the grey dashed curve is the theoretical one-dimensional partial dependence function of the data-generating regression surface, obtained by averaging over the remaining covariates as in the definition above.

In the present setup these theoretical curves can be computed explicitly. Since X_1 and X_2 are independent and symmetric on $(-1, 1)$, we have $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0$, $\mathbb{E}[X_1^2] = 1/3$, and $\mathbb{E}[\sin(5X_2)] = 0$. Hence

$$\bar{f}_1(x_1) = x_1^2, \quad \bar{f}_2(x_2) = \frac{1}{3} + \sin(5x_2), \quad \bar{f}_3(x_3) = \frac{1}{3}.$$

The variable-importance panel correctly identifies X_1 and X_2 as relevant and leaves X_3 essentially negligible. More interestingly, the interaction term X_1X_2 does not visibly appear in the one-dimensional partial dependence plots: because the covariates are independent and symmetric around zero, the cross-term averages to zero when constructing each $\bar{f}_j$. Hence the PDPs can look close to the marginal specifications (quadratic for X_1 , sinusoidal for X_2 , and flat for X_3) even though the underlying regression surface is not additive. This is exactly the point to keep in mind when interpreting PDPs, since they are averaged summaries of a fitted predictor, and not plots of the conditional distribution of Y given a single covariate. $\diamond$

Numeric example 6.8 (Bivariate partial dependence reveals the interaction) Using the same simulation design and fitted boosted-tree model as in the previous example, we now consider the two-variable partial dependence function for $\mathcal{S} = \{1, 2\}$. Since the response does not depend on X_3 , averaging over the remaining covariate leaves the full (X_1, X_2) surface unchanged:

$$\bar{f}_{\{1,2\}}(x_1, x_2) = x_1^2 + \sin(5x_2) + x_1x_2.$$

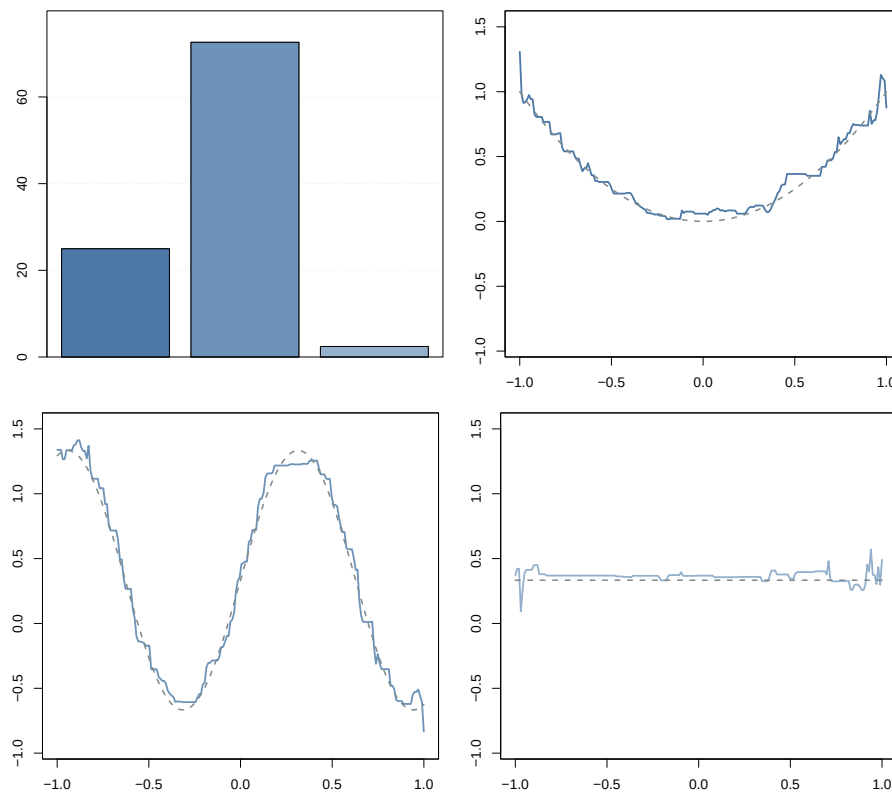


Figure 6.8: Boosted trees for the interaction model. Top left: variable importance for (X_1, X_2, X_3) (left to right). Top right, bottom left, and bottom right: one-dimensional partial dependence plots for X_1 , X_2 , and X_3 , respectively. Coloured solid curves are fitted partial dependence functions and grey dashed curves are the corresponding theoretical partial dependence functions. The three partial-dependence panels use the same vertical scale.

In contrast to the one-dimensional PDPs, the interaction term is therefore retained once both interacting variables are kept in the argument of the partial dependence function.

Figure 6.9 confirms this numerically. The left panel shows the theoretical bivariate partial dependence surface together with fitted bivariate PDP values from the boosted-tree model. The right panel compares level curves of the theoretical and fitted bivariate PDP surfaces. The close agreement of the contour geometry shows that the fitted model recovers the interaction structure once the PDP is evaluated jointly in (X_1, X_2) . $\diamond$

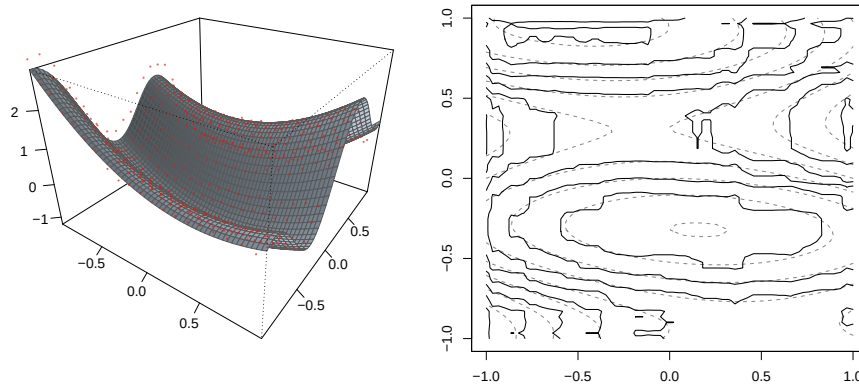


Figure 6.9: Bivariate partial dependence for (X_1, X_2) under the interaction model. Left: light-blue theoretical surface $x_1^2 + \sin(5x_2) + x_1x_2$ with overlaid fitted bivariate PDP values (dark points). Right: grey dashed theoretical level curves and black fitted level curves, drawn at common contour levels.

Interactions

In the present context, interactions quantify departures from additivity in the fitted predictor. For two variables X_j and X_k , a natural idea is to compare the two-variable partial dependence surface with the sum of the corresponding one-variable partial dependences. If the model is approximately additive in (X_j, X_k) , then little is lost by this decomposition; otherwise, the residual surface indicates interaction.

For two-way interactions, Friedman's H-statistic provides a normalised measure of this departure:

$$H_{jk}^2 = \frac{\sum_{i=1}^n [\bar{f}_{jk}(x_{ij}, x_{ik}) - \bar{f}_j(x_{ij}) - \bar{f}_k(x_{ik})]^2}{\sum_{i=1}^n \bar{f}_{jk}^2(x_{ij}, x_{ik})},$$

where $\bar{f}_{jk}(x_{ij}, x_{ik})$ is the estimated two-variable partial dependence and $\bar{f}_j(x_{ij})$, $\bar{f}_k(x_{ik})$ are the corresponding one-variable partial dependences (typically centred in implementation). Larger values indicate stronger departures from additivity.

As with partial dependence itself, interpretation of interaction diagnostics becomes more delicate when predictors are strongly dependent. Once again, apparent interactions can partly reflect extrapolation of the fitted model into sparsely observed regions of covariate space.

6.4 Notes and Comments

Statistical learning theory focuses on prediction and explanation. Classical sources such as Vapnik [1995] and Hastie et al. [2009] emphasise prediction accuracy as the main criterion, distinguishing it from traditional statistical inference. Indeed, modern treatments define the expected test error as a function of bias, variance, and irreducible noise, with model selection aiming to minimise this quantity rather than training error. Techniques such as cross-validation [Stone, 1974, Allen, 1974, Geisser, 1975] and the bootstrap [Efron, 1979] became fundamental tools for empirical error estimation, with refinements such as the Akaike Information Criterion [Akaike, 1974], the .632 bootstrap estimator [Efron, 1983], and covariance-penalty methods [Efron, 2004] providing analytical insight into bias correction and estimator variability; see Efron and Hastie, 2016 as the main reference of the chapter.

Model averaging and ensemble methods are the next conceptual step, motivated by the reduction of variance and the improvement of predictive stability. Bagging and random forests [Breiman, 1996, 2001] formalised bootstrap aggregation and randomised trees as powerful general-purpose learners, while boosting [Freund and Schapire, 1997, Friedman, 2001] introduced sequential additive fitting to reduce bias by using weak learners. Stacking [Wolpert, 1992] and related meta-learning strategies combine predictions using several heterogeneous models, and bumping was introduced by Tibshirani and Knight [1999]. The literature on mathematical theory for machine learning methods is vast. The one covered in this chapter is partially adapted based on modern ideas such as Wager and Athey [2018], with foundational consistency results for random forests given in Scornet et al. [2015], with recent extensions in Bladt and Lemvig [2026a].

Specific model classes have been developed in parallel and are equally vast and growing; we limit ourselves to mention some highlights aligned with the chapter. Additive-model foundations and GAM methodology are classically treated in Stone [1985], Hastie and Tibshirani [2017], Wood [2017]. Decision-tree construction and pruning are based on Breiman et al. [2017], Quinlan [1986]. In addition, the random-forest and boosting subsections place substantial emphasis on tuning choices such as tree depth, learning rate, and early stopping, which are well discussed already in Hastie et al. [2009]. For interpretability topics, including variable-importance, partial-dependence and accumulated-local-effects plots, and interaction diagnostics, see Strobl et al. [2007], Friedman [2001], Apley and Zhu [2020], Hooker [2007].

In applications, statistical learning has been widely adopted, not least in actuarial and insurance analytics. Texts such as Denuit et al. [2019a, 2020, 2019b] and Wüthrich and Merz [2023] integrate machine learning and predictive modelling with classical insurance mathematics, showing how statistical learning can complement traditional risk models. In Efron and Hastie [2016] and in the probabilistic perspective of Murphy [2012] the broader computational-statistical context is provided, within which actuarial applications can easily be situated. These foundational books, together with Bishop [2006], Hastie and Tibshirani [2017], Hastie et al. [2009], constitute important and foundational literature of statistical learning theory.

6.5 Exercises

1. Consider ordinary regression, $\hat{\mu} = My$, with $M = X(X^\top X)^{-1}X^\top$, where X is the model matrix of predictors. Suppose that $y \sim (\mu, \sigma^2 \text{Id})$.
 - (a) Show that $\text{Cov}(\hat{\mu}, y) = X(X^\top X)^{-1}X^\top \sigma^2$.
 - (b) The trace of a $p \times p$ matrix $A = (a_{i,j})$ is defined as the sum of diagonal elements:

$$\text{tr}(A) = \sum_{i=1}^p a_{ii}.$$

Show that

$$\widehat{\text{Err}} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2 + \frac{2\sigma^2 p}{n}.$$

2. Consider a linear fitting model $\hat{\mu} = Sy$, where S is some matrix function of the covariates. Let $\hat{f}^{-i}(x_i)$ denote the fit at x_i obtained by leaving out observation i . Show that

$$\hat{f}^{-i}(x_i) = \frac{\hat{f}(x_i) - S_{ii}y_i}{1 - S_{ii}}$$

and hence

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}^{-i}(x_i))^2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}(x_i)}{1 - S_{ii}} \right)^2.$$

It is often computationally lighter to estimate the trace of S rather than the individual S_{ii} 's. Thus we introduce the generalised cross-validation by

$$\text{GCV}(\hat{f}) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}(x_i)}{1 - \text{tr}(S)/n} \right)^2.$$

- (a) Assume that $y_i = \hat{f}(x_i) + \varepsilon_i$, where $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$ are independent noise terms. Using the approximation $1/(1-x)^2 \approx 1+2x$, show that

$$\text{GCV}(\hat{f}) \approx \overline{\text{err}}_{MSE} + \frac{2 \text{tr}(S)}{n} \sigma_\varepsilon^2.$$

- (b) Compute $\text{GCV}(\hat{f})$ when $\overline{\text{err}}$ is defined with twice the Kullback–Leibler divergence rather than the mean squared error and $\sigma_\varepsilon^2 = 1$.

Hint: $\text{tr}(S)$ equals the number of parameters.

3. Prove Proposition 6.1.
4. Prove the second part of Proposition 6.2.
5. In this exercise we derive smoothing splines. Suppose $n \geq 2$ and that g is the natural cubic spline interpolant to the pairs $\{x_i, y_i\}$, with $a < x_1 < \dots < x_n < b$. This is a natural spline with a knot at every x_i . Since this is an n -dimensional space of functions we can determine the coefficients that interpolate the sequence y_i exactly. Let $\tilde{g}$ be any other function on $[a, b]$ that interpolates the n pairs. We assume $\tilde{g}$ is at least twice continuously differentiable.

- (a) Let $h(x) = \tilde{g}(x) - g(x)$. Use that g is a natural cubic spline with integration by parts to show that

$$\int_a^b g''(x)h''(x) \, dx = 0.$$

- (b) Show that

$$\int_a^b \tilde{g}''(x)^2 \, dx \geq \int_a^b g''(x)^2 \, dx,$$

with equality if h is identically zero in $[a, b]$.

- (c) We consider the penalised least squares problem

$$\min_f \left[\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_a^b f''(t)^2 \, dt \right].$$

Use (b) to argue that the minimiser must be a cubic spline with knots at each x_i .

6. Go through the proofs of Proposition 6.6 and Theorem 6.1 and fill in the missing details. In particular, look at the following key steps.

- (a) Use that the density f is continuous to verify that

$$\frac{1}{\text{Vol}(L(x, \theta))} \int_{L(x, \theta)} f(\tilde{x}) \, d\tilde{x} \rightarrow f(x), \quad \mathbb{P}\text{-a.s.}, \quad \text{as } n \rightarrow \infty$$

- (b) In the proof of Theorem 6.1, we omitted the conditional variance calculation for $\hat{m}(x, \theta)$. Perform this calculation.

7. Consider the theoretical framework for decision trees from above. In this exercise, we prove the following result.

Theorem 6.2 *Under the assumptions presented in the chapter, it holds that*

$$T(x, \theta) \xrightarrow{L^1} \mathbb{E}[Y \mid X = x].$$

Moreover, strengthen the moment condition (iii) to $\mathbb{E}[|Y|^{p+\delta} \mid X = x] \leq K$ for some constants $p \geq 1$ and $\delta, K > 0$. Then it holds that

$$T(x, \theta) \xrightarrow{L^p} \mathbb{E}[Y \mid X = x].$$

Clearly, the second assertion of the theorem implies the first. Since we have already established weak convergence, it suffices to show that $\{|T(x, \theta)|^p\}$ is uniformly integrable by the Lebesgue–Vitali Theorem. We do this by verifying that $\sup_n \{\mathbb{E}[|T(x, \theta)|^{p+\delta}]\} < \infty$.

- (a) Let $m \in \{1, \dots, n_{\mathcal{I}}\}$. Verify that

$$\mathbb{E}[|T(x, \theta)|^{p+\delta} \mid N_L(x, \theta) = m] = \mathbb{E}\left[\left|\sum_{i=1}^m \frac{Y_i}{m}\right|^{p+\delta} \mid X_1, \dots, X_m \in L(x, \theta)\right].$$

(b) Use convexity of $x \mapsto |x|^{p+\delta}$ to show that

$$\mathbb{E}[|T(x, \theta)|^{p+\delta} \mid N_L(x, \theta) = m] \leq \mathbb{E}[|Y_1|^{p+\delta} \mid X_1 \in L(x, \theta)].$$

(c) Complete the proof of the theorem.

8. For large n , argue as $B \rightarrow \infty$ that the limit of the OOB error estimate for a random forest approaches its n -fold CV error estimate.
9. Let $\hat{f}_1, \hat{f}_2$ be two predictors of Y at a fixed covariate value x , and consider the stacked predictor

$$\hat{f}_w = w\hat{f}_1 + (1-w)\hat{f}_2, \quad w \in \mathbb{R}.$$

Assume squared loss and define $R(w) = \mathbb{E}[(Y - \hat{f}_w)^2]$.

- (a) Show that $R(w)$ is a quadratic polynomial in w and derive the minimiser

$$w^* = \frac{\mathbb{E}[(Y - \hat{f}_2)(\hat{f}_1 - \hat{f}_2)]}{\mathbb{E}[(\hat{f}_1 - \hat{f}_2)^2]},$$

whenever the denominator is positive.

- (b) Explain how this identifies optimal stacking weights in population.
- (c) Find the optimal stacking weights for a given sample of size n .

10. Assume

$$Y = m(X) + \varepsilon, \quad m(x) = \sum_{j=1}^p g_j(x_j), \quad \mathbb{E}[\varepsilon \mid X] = 0, \quad \mathbb{E}[g_j(X_j)] = 0,$$

and that all the p components of X are mutually independent.

- (a) Show that the partial dependence function for coordinate j satisfies

$$\bar{f}_j(x_j) = \mathbb{E}[m(x_j, X_{-j})] = g_j(x_j).$$

- (b) Let $\tilde{X}_j$ be an independent copy of X_j , independent of (Y, X) , and define permutation importance in population as

$$I_j^{\text{perm}} = \mathbb{E}[(Y - m(X_1, \dots, \tilde{X}_j, \dots, X_p))^2] - \mathbb{E}[(Y - m(X))^2].$$

Show that

$$I_j^{\text{perm}} = 2 \text{Var}(g_j(X_j)).$$

- (c) Discuss how both partial dependence and permutation importance change in the case where each function g_j depends on two components of x .

Practical exercises

1. This exercise explores the effect of bootstrap aggregating (bagging). Let $X \sim N(0, 1)$ and $Y = 1_{(X \leq 1)}e^X + 1_{(X > 1)}\log(X) + \varepsilon$, where $\varepsilon \sim N(0, (0.6)^2)$, independent of X and Y .
 - (a) Simulate $n = 500$ samples of X and the corresponding Y . Fit regression splines with 10 degrees of freedom. Plot the simulations and the fitted model.
Hint: You may fit the model using `lm(Y ~ splines::ns(X, df = 10))`.
 - (b) Perform bootstrap aggregation with $B = 1000$ on the simulated data, fitting regression splines with 10 degrees of freedom for each bootstrap sample, using the fit to predict on a suitable grid $[-2.5, 2.5]$. Add the prediction to the same plot.
 - (c) Compute the empirical 0.025 and 0.975 of your bootstrap prediction at each grid point of $[-2.5, 2.5]$. Compare with the confidence bands from your fitted regression splines.
Hint: The latter can be computed with `predict()`.
2. This exercise compares bagging between linear and non-linear models.
 - (a) Assume that we use the parametric bootstrap. Calculate the bias and variance of the bagging estimator. Compare it with the variance and bias of the original model.
 - (b) For the rest of the exercise, we are interested in the following model:

$$Y = 2 + 3X + 2\sin(X) + e$$

where e is t -distributed with 10 degrees of freedom. Now, start by simulating 500 observations to use as training data and 200 observations to use as test data.

- i. Fit a linear regression model to the training data. Perform parametric and nonparametric bagging. In both cases use $B = 100$. Calculate the MSE on the test data for all three models, report how bagging affects the error, and discuss the implications in light of (a).
 - ii. Fit a decision tree to the data using `rpart(Y ~ X)`. Perform nonparametric bagging with $B = 100$, calculate the MSE, and discuss the effect of bagging for this model in light of (a).
 - iii. A way to visualise the variance of the models is to plot each bootstrap estimate. Do this for the linear regression model and the decision tree. Compare the empirical variance and bias across the models, indicate which model benefits most from bagging, and state whether this aligns with the simulation results.
3. In this exercise, we implement trees and forests on simulated data. Let X, S be uniformly distributed on $[0, 1]$, let $\eta(x) = 0.3 + 2\phi_{\mu=0.2, \sigma=0.05}(x)$, where ϕ is the Gaussian density, and let $Y = S^{-\eta(X)} + 20 \cdot 1_{(X > 3/4)}$.

- (a) Simulate 1000 realisations of X and Y , corresponding to covariates and responses, respectively. Fit a decision tree of depth d , for $d = 2, 4, 6$. Plot predicted curves along with the simulated data.
 - (b) Implement a random forest consisting of 400 trees, each with a maximum of 15 nodes. Compare with the decision trees.
 - (c) Implement a gradient boosting model using the R package `gbm` with regularisation parameter $\nu = 0.1$, $J = 4$, squared error loss function and 400 trees. Compare with the preceding models.
4. In this exercise we implement M -order splines on simulated data. Let X be uniformly distributed on $[-2.5, 2.5]$ and let $Y = \sin(2X) + \varepsilon$, where $\varepsilon \sim N(0, (0.2)^2)$ is an independent noise term. We fix 3 knots

$$\xi_1 = -1.25, \quad \xi_2 = 0, \quad \xi_3 = 1.25.$$

- (a) Perform 1000 simulations of X and corresponding Y . Fit an M -order spline, using the above knots, of order $M = 1, 2, 3, 4$ on the simulations and compare. Repeat with $Z = \sin(2X) + 2.2 \mathbf{1}_{(X < 0)} \cos(2X) + \varepsilon$.

Hint: The R code can be implemented in various ways. Familiarise yourself with the functions `splines::bs()` and `splines::ns()`. A spline model of order M can be fitted with `lm(Y ~ splines::bs(X, degree = M, knots = knots_chosen))`, which can then be used to predict with `predict()`.

- (b) In this subexercise we quantify how training and test error behave for increasing order of M .
 - Perform 100 rounds of simulations, where each time you draw $n = 100$ realisations of X as above, with $Z = \sin(2X) + 2.2 \mathbf{1}_{(X < 0)} \cos(2X) + \varepsilon$ and $Y = \sin(2X) + \varepsilon$.
 - For each simulation, split the data 50/50 into train and test data.
 - For each order $M = 1, \dots, 15$, fit an M -order spline on the training data (of the current simulation) for $(X_{\text{train}}, Y_{\text{train}})$ and $(X_{\text{train}}, Z_{\text{train}})$, and save the training and test error for both fits.

Plot the training and test error for all simulations in the two models, and visually determine the order M where the *mean* test error for each model is minimal.

5. In this exercise, we implement cross-validated stacking and compare it with its single component.

- (a) Simulate $n = 3000$ observations with $X_1, \dots, X_5 \sim N(0, 1)$ iid and

$$Y = 2 + X_1 + 2 \sin(X_2) + X_3^2 + \varepsilon, \quad \varepsilon \sim N(0, 1).$$

- (b) Split the sample into 70% training and 30% test data. On the training set, fit three base models: a linear model, a regression tree, and a GAM with spline terms.
- (c) Construct 5-fold out-of-fold predictions for each base model on the training set, and then estimate stacking weights by constrained least squares with nonnegative weights summing to one.

- (d) Compare test-set mean squared errors of the stacked predictor and each base model. Report the estimated weights, and compute permutation importance for each covariate in the stacked predictor. Verify that X_4 and X_5 have little relative importance.

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